

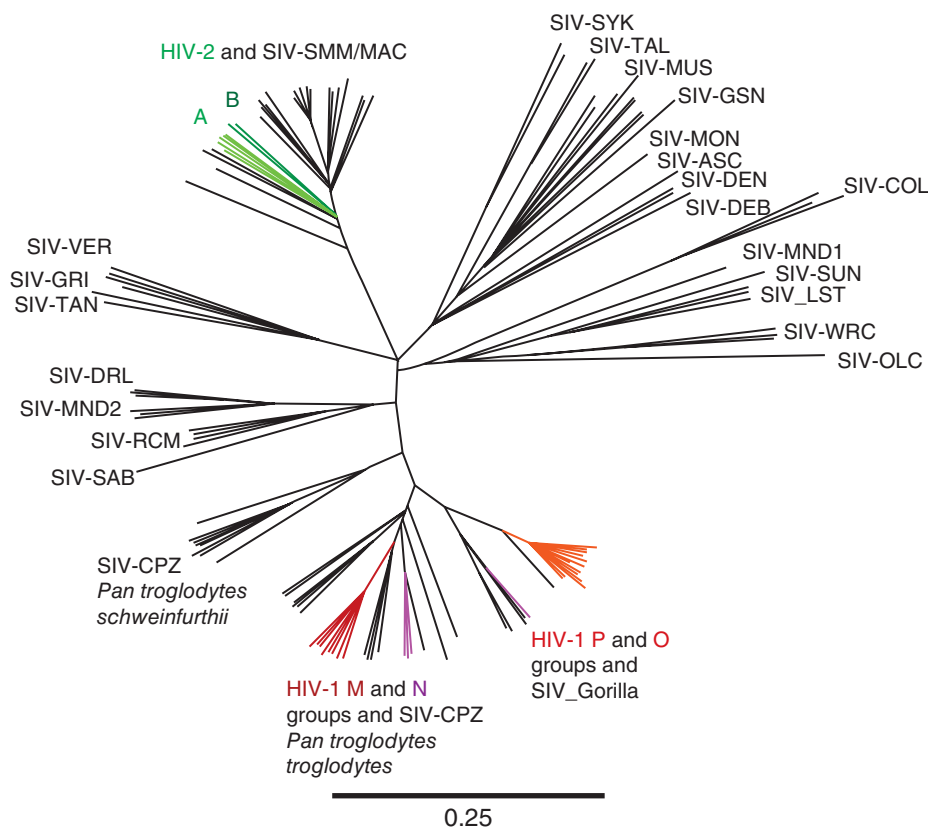


FIGURE 208-1 A phylogenetic tree based on the nearly complete genomes (*gag* through *nef* genes) of primate immunodeficiency viruses. The scale (0.25) indicates a 25% phylogenetically corrected genetic distance at the nucleotide level. Clades in color represent viruses (HIV-1, HIV-2) identified in humans after relatively recent transfers from chimpanzee, gorilla, and sooty mangabey reservoirs. (Prepared by Brian Foley, PhD, of the HIV Sequence Database, Theoretical Biology and Biophysics Group, Los Alamos National Laboratory; additional information at www.hiv.lanl.gov/content/sequence/HelpDocs/subtypes.html)

spread is also facilitated by accessory molecules such as the C-type lectin receptor *DC-SIGN* expressed on certain dendritic cells (DCs) that bind to the HIV gp120 envelope protein, allowing virus captured on DCs to spread to CD4⁺ T cells. Following binding of the envelope protein to the CD4 molecule associated with the above-mentioned conformational change in the viral envelope gp120, which facilitates binding to a co-receptor, fusion with the host cell membrane occurs as the newly exposed gp41 molecule penetrates the plasma membrane of the target cell and then folds upon itself to bring the virion and target cell together (Fig. 208-4). Following fusion, uncoating of the capsid protein shell is initiated—a step that facilitates reverse transcription and leads to formation of the preintegration complex, composed of viral RNA, enzymes, and accessory proteins and surrounded by capsid and matrix proteins (Fig. 208-3). All these post-fusion viral components constitute the HIV replication complex, including the outer capsid shell, which plays an integral role in supporting reverse transcription of viral RNA. As the preintegration complex traverses the cytoplasm to reach the nucleus, the viral reverse transcriptase enzyme catalyzes the reverse transcription of the genomic RNA into DNA, resulting in the formation of double-stranded HIV proviral DNA. At several steps of the replication cycle, the virus is vulnerable to various cellular factors that can block the progression of infection. The cytoplasmic tripartite motif-containing protein 5 α (TRIM5 α) is a host restriction factor that interacts with retroviral capsids, causing their premature disassembly and induction of innate immune responses. The apolipoprotein B mRNA editing enzyme (catalytic polypeptide-like 3 [APOBEC3]) family of cellular proteins also inhibits progression of virus infection after virus has entered the cell and prior to entering the nucleus. APOBEC3 proteins, which are incorporated into virions and released into the cytoplasm of a newly infected cell, bind to the single minus-strand DNA intermediate and deaminate viral cytidine, causing hypermutation of retroviral genomes. HIV has evolved a

powerful strategy to protect itself from APOBEC. The viral protein Vif targets APOBEC3 for proteasomal degradation. SAMHD1 is another post-entry host factor that prevents reverse transcription by depleting pools of deoxynucleotides (dNTPs). The type I interferon (IFN)-induced myxovirus resistance protein 2 (MX2) is another restriction factor associated with innate immunity that inhibits HIV-1 nuclear entry.

Following reverse transcription, the proviral DNA accesses the nuclear pore and is transferred from the cytoplasm to the nucleus, where it is integrated into a host cell chromosome through the action of another virally encoded enzyme, *integrase* (Fig. 208-3). HIV proviral DNA preferentially integrates into the host genomic DNA in regions of active transcription. This provirus may remain transcriptionally inactive (latent) or may manifest varying levels of gene expression, up to active transcription and production of virus.

Cellular activation plays an important role in the replication cycle of HIV and is critical to the pathogenesis of HIV disease (see “Pathophysiology and Pathogenesis,” below). Following initial binding, fusion, and internalization of the nucleic acid contents of virions into the target cell, incompletely reverse-transcribed DNA intermediates are labile in quiescent cells and do not integrate efficiently into the host cell genome unless cellular activation occurs shortly after infection. Furthermore, some degree of activation of the host cell is required for the initiation of transcription of the integrated proviral DNA into either genomic RNA or mRNA.

This latter process may not necessarily be associated with the detectable expression of the classic cell-surface markers of activation. This is reflected in the fact that cell-associated HIV RNA transcribed from competent or defective proviruses can be detected in infected resting CD4⁺ T cells. In this regard, activation of HIV expression from the latent state depends on the interaction of various cellular and viral factors. Following transcription, HIV mRNA is translated into proteins that undergo modification through glycosylation, myristoylation, phosphorylation, and cleavage. The viral particle is formed by the assembly of HIV proteins, enzymes, and genomic RNA at the plasma membrane of the cells. Budding of the progeny virion through the lipid bilayer of the host cell membrane is the point at which the core acquires its external envelope and where the host restriction factor tetherin can inhibit the release of budding particles. Tetherin is an IFN-induced type II transmembrane protein that interferes with virion detachment. The HIV accessory protein vpu counteracts this effect through direct interactions with tetherin. During or soon after budding, the virally encoded protease catalyzes the cleavage of the gag-pol precursor to yield the mature virion. Progression through the virus replication cycle is profoundly influenced by a variety of viral regulatory gene products. Likewise, each point in the replication cycle of HIV is a real or potential target for therapeutic intervention. Thus far, the reverse transcriptase, protease, and integrase enzymes as well as the process of virus-target cell binding and fusion have proved to be susceptible to pharmacologic disruption.

■ HIV GENOME

Figure 208-5 illustrates schematically the arrangement of the HIV genome. Like other retroviruses, HIV-1 has genes that encode the structural proteins of the virus: *gag* encodes the proteins that form the core of the virion (including p24 antigen); *pol* encodes the enzymes responsible for protease processing of viral proteins, reverse

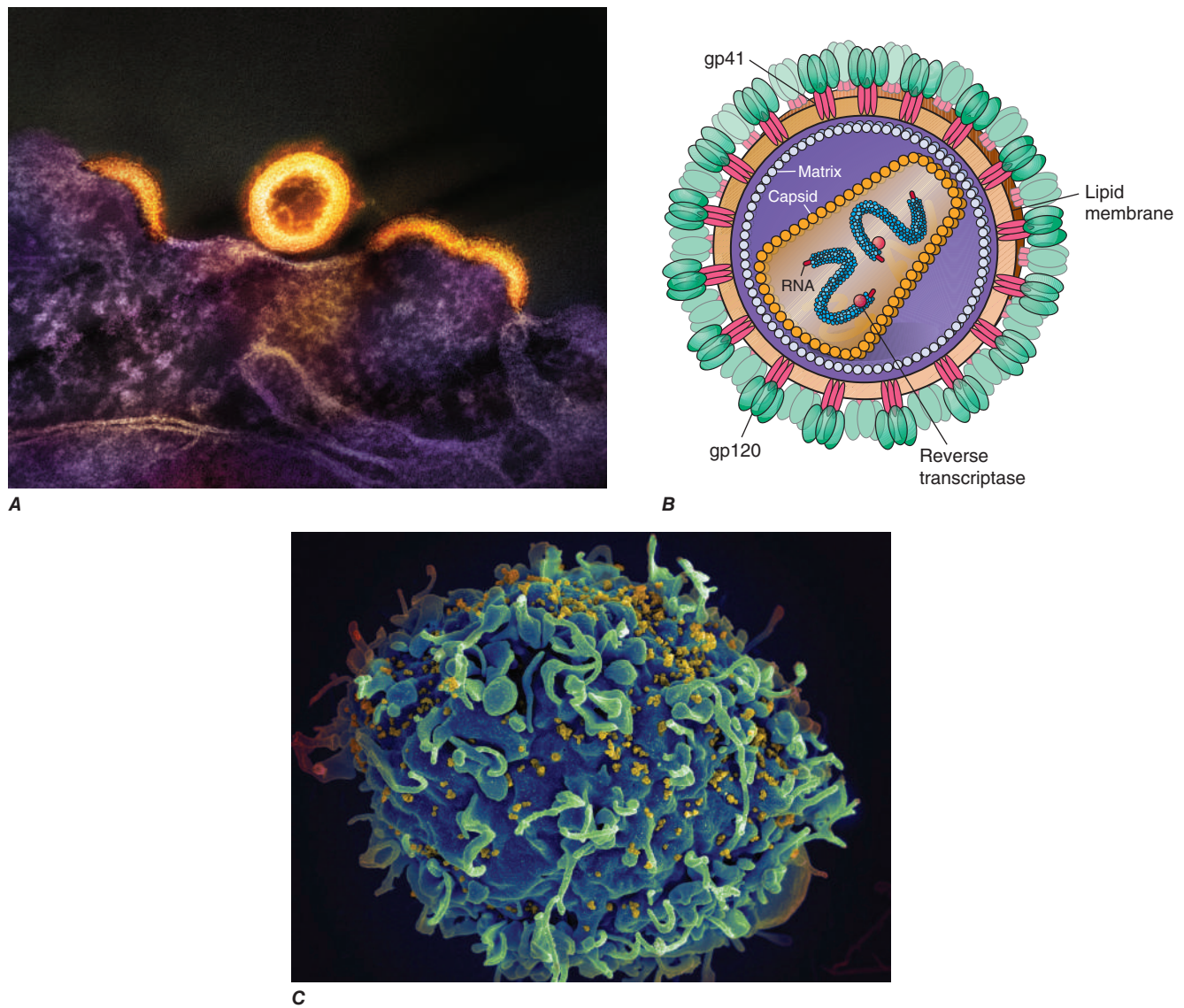


FIGURE 208-2 **A.** Transmission electron micrograph of HIV-1 virus particles (colored yellow/gold) replicating from an HIV-infected H9 T cell (purple). Budding virus particles that have not yet separated from the cell appear as semicircles. A separated, spherical immature particle is seen at the center of the image. (Image captured at the NIAID Integrated Research Facility [IRF] in Fort Detrick, Maryland. Courtesy of NIAID.) **B.** Structure of HIV-1, including the gp120 envelope, gp41 transmembrane components of the envelope, genomic RNA, enzyme reverse transcriptase, p18(17) inner membrane (matrix), and p24 core protein (capsid). (Courtesy of George V. Kelvin. Adapted from RC Gallo: *Sci Am* 256:46, 1987.) **C.** Scanning electron micrograph of a human H9 T cell (blue/green) infected with HIV-1 virus particles (yellow). (Image captured at NIAID Rocky Mountain Laboratories in Hamilton, Montana. Courtesy of NIAID.)

transcription, and integration; and *env* encodes the envelope glycoproteins. However, HIV-1 is more complex than other retroviruses, particularly those of the nonprimate group, in that it also contains at least six other regulatory genes (*tat*, *rev*, *nef*, *vif*, *vpr*, and *vpu*), which code for proteins involved in the modification of the host cell to enhance virus growth and the regulation of viral gene expression. Several of these proteins are thought to play a role in the pathogenesis of HIV disease; their various functions are listed in Fig. 208-5. Flanking these genes are long terminal repeats (LTRs), which contain regulatory elements involved in gene expression (Fig. 208-5). The major difference between the genomes of HIV-1 and HIV-2 is the fact that HIV-2 lacks the *vpu* gene and has a *vpx* gene not contained in HIV-1.

MOLECULAR HETEROGENEITY OF HIV-1

Molecular analyses of HIV isolates reveal varying levels of sequence diversity over all regions of the viral genome. For example, the degree of difference in the coding sequences of the viral envelope protein ranges from a few percent (very close, among isolates from the same infected individual) to more than 50% (extreme diversity, between isolates from the different groups of HIV-1: M, N, O, and P). The changes tend to cluster in hypervariable regions. HIV can evolve by several means, including simple base substitution, insertions and

deletions, recombination, and gain and loss of glycosylation sites. HIV sequence diversity arises directly from the limited fidelity of the reverse transcriptase, i.e., a tendency toward copying errors. The balance of immune pressure and functional constraints on proteins influences the regional level of variation within proteins. For example, envelope, which is exposed on the surface of the virion and is under immune selective pressure from both antibodies and cytolytic T lymphocytes, is extremely variable, with clusters of mutations in hypervariable domains. In contrast, reverse transcriptase, with important enzymatic functions, is relatively conserved in the absence of drug-induced selective pressure, particularly around the active site. The extraordinary variability of HIV-1 contrasts markedly with the relative stability of HTLV-1 and 2.

The four groups (M, N, O, and P) of HIV-1 are the result of four separate chimpanzee-to-human (or possibly gorilla-to-human for groups O and P) transfers. Group M (major), which is responsible for most infections in the world, has diversified into subtypes and inter-subtype recombinant forms, due to “sub-epidemics” within humans after one of those transfers.

Among primate lentiviruses, HIV-1 is most closely related to viruses isolated from chimpanzees and gorillas (Fig. 208-1). The chimpanzee subspecies *Pan troglodytes troglodytes* has been established to be a

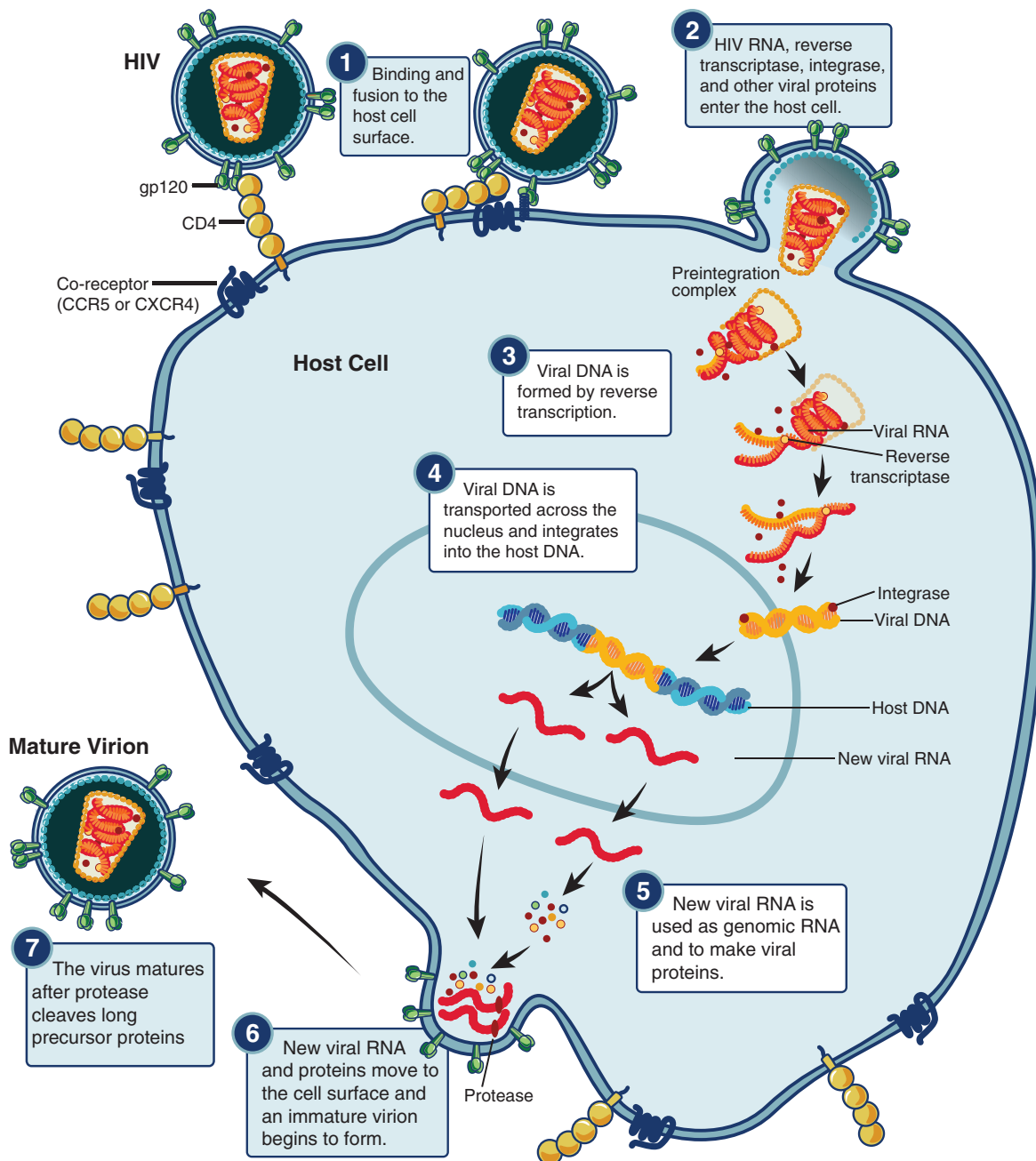


FIGURE 208-3 The replication cycle of HIV. See text for description. (From the National Institute of Allergy and Infectious Diseases.)

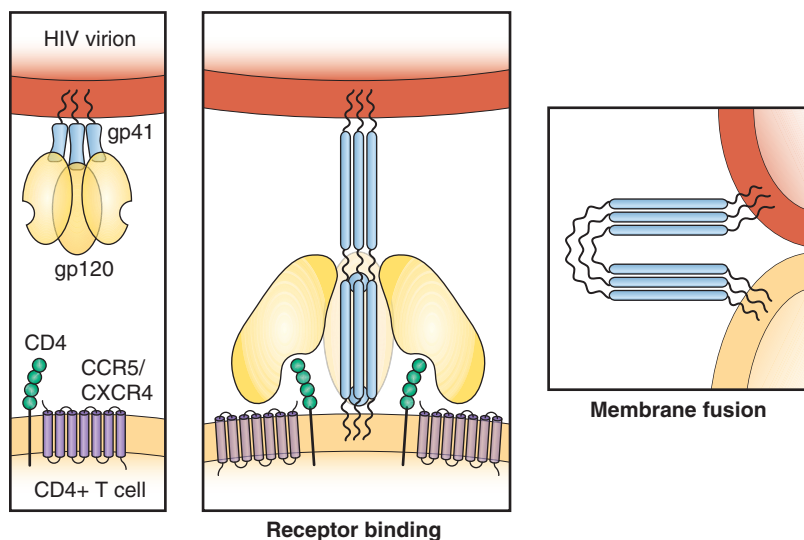


FIGURE 208-4 Binding and fusion of HIV-1 with its target cell. HIV-1 binds to its target cell via the CD4 molecule, leading to a conformational change in the gp120 molecule that allows it to bind to the co-receptor CCR5 (for R5-using viruses). The virus then firmly attaches to the host cell membrane in a coiled-spring fashion via the newly exposed gp41 molecule. Virus-cell fusion occurs as the transitional intermediate of gp41 undergoes further changes to form a hairpin structure that draws the two membranes into close proximity (see text for details). (Adapted from D Montefiori, JP Moore: HIV vaccines. *Magic of the occult?* Science 283:336, 1999.)

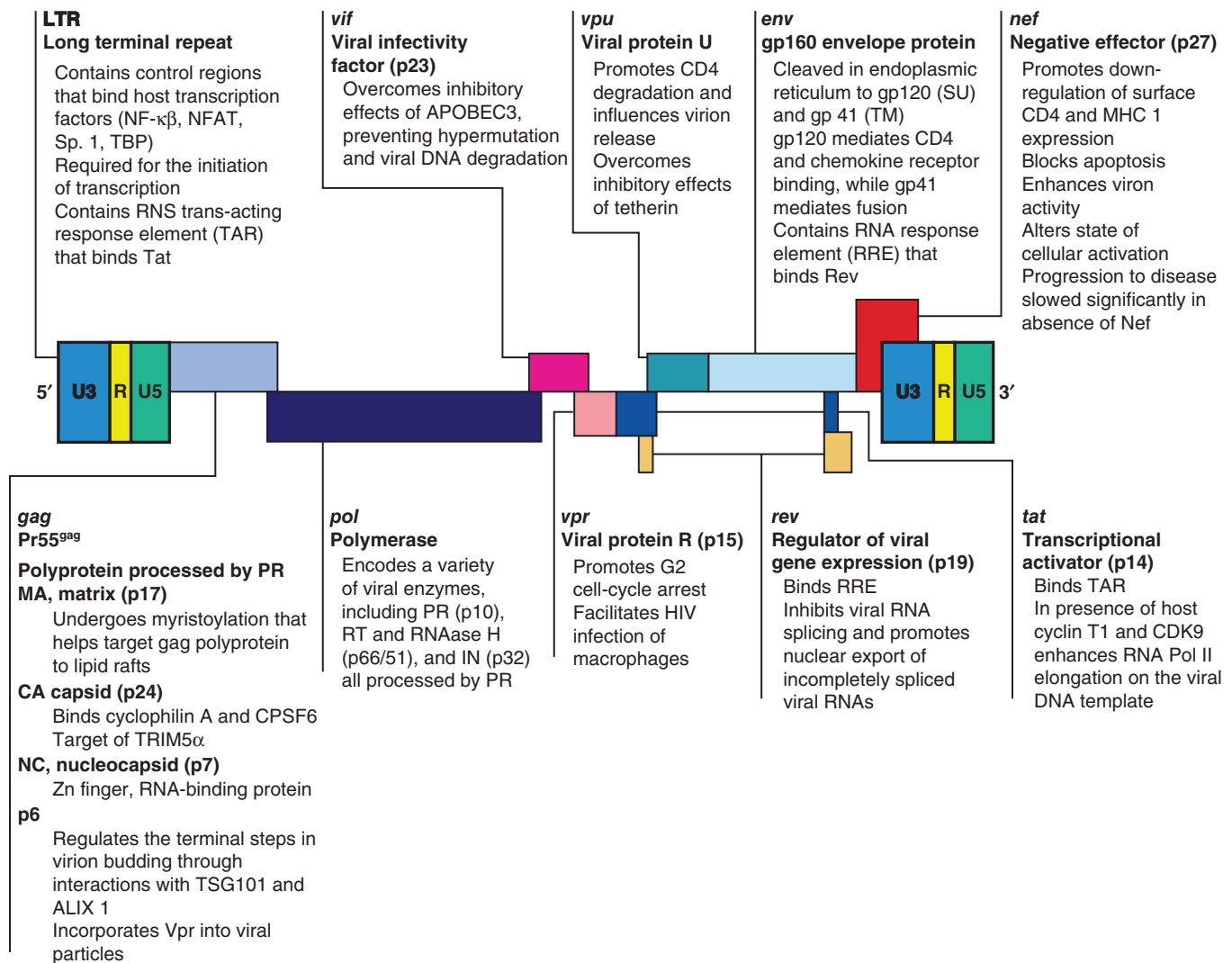


FIGURE 208-5 Organization of the genome of the HIV provirus together with a summary description of its 9 genes encoding 15 proteins. (Reproduced with permission from WC Greene et al: *Charting HIV's remarkable voyage through the cell: Basic science as a passport to future therapy.* *Nat Med* 8:673, 2002.)

natural reservoir of the HIV-1 M and N groups. The rare viruses of the HIV-1 O and P groups are most closely related to viruses found in Cameroonian gorillas. The M group comprises ten subtypes, or *clades*, designated A, B, C, D, F, G, H, J, K, and L, as well as more than 150 known circulating recombinant forms (CRFs) and numerous unique recombinant forms. Inter-subtype recombinants are generated by infection of an individual with two subtypes that then recombine and create a virus with a selective advantage. These CRFs range from highly prevalent forms—such as CRF01_AE, common in southeast Asia, and CRF02_AG in west and central Africa—to numerous CRFs that are relatively rare, either because they are of a more recent origin (newly recombined) or because they have not broken out into a major population. The subtypes and CRFs constitute the major lineages of the M group of HIV-1. HIV-1 M group subtype C dominates the global pandemic, and although there is speculation that it is more transmissible than other subtypes, solid data on variations in transmissibility between subtypes are lacking. Human population densities, access to prevention and treatment, prevalence of genital ulcers, iatrogenic transmissions, and other confounding host factors are all possible reasons why one subtype has spread more than another.

Figure 208-6 schematically diagrams the worldwide distribution of HIV-1 subtypes by region. Nine strains account for most new HIV infections globally: HIV-1 subtypes A, B, C, D, F, and G and three of the CRFs, CRF01_AE, CRF02_AG, and CRF07_BC. Subtype C viruses (of the M group) are by far the most common form, likely accounting for ~50% of infections worldwide. In sub-Saharan Africa, home to approximately two-thirds of all people living with HIV/AIDS, most infections

are caused by subtype C, with smaller proportions of infections caused by subtype A, subtype D, CRF02_AG, and other subtypes and recombinants. In South Africa, the country with the largest number of prevalent infections (7.7 million in 2023), 98% of all HIV-1 isolates sequenced are of subtype C. In Asia, HIV-1 isolates of the CRF01_AE lineage and subtypes B and C predominate. CRF01_AE accounts for most infections in south and southeast Asia, while ~97% of infections in India, home to an estimated 2.5 million people with HIV, are of subtype C (see “HIV Infection and AIDS Worldwide,” below). Subtype B viruses are overwhelmingly predominant in the United States, Canada, certain countries in South America, western Europe, and Australia. It is thought that, purely by chance, subtype B was seeded into the United States and Europe in the late 1970s, thereby establishing an overwhelming founder effect. Many countries have co-circulating viral subtypes that are giving rise to new CRFs. Sequence analyses of HIV-1 isolates from infected individuals indicate that recombination among viruses of different clades likely occurs when an individual is infected with viruses of more than one subtype, particularly in geographic areas where subtypes overlap, and more often in sub-epidemics driven by injection drug use than in those driven by sexual transmission.

The extraordinary diversity of HIV, reflected by the presence of multiple subtypes, circulating recombinant forms, and continuous viral evolution, has implications for possible differential rates of transmission, rates of disease progression, and the development of resistance to antiretroviral drugs. This diversity may also prove to be a formidable obstacle to HIV vaccine development, as a broadly useful vaccine would need to induce protective responses against a wide range of viral strains.

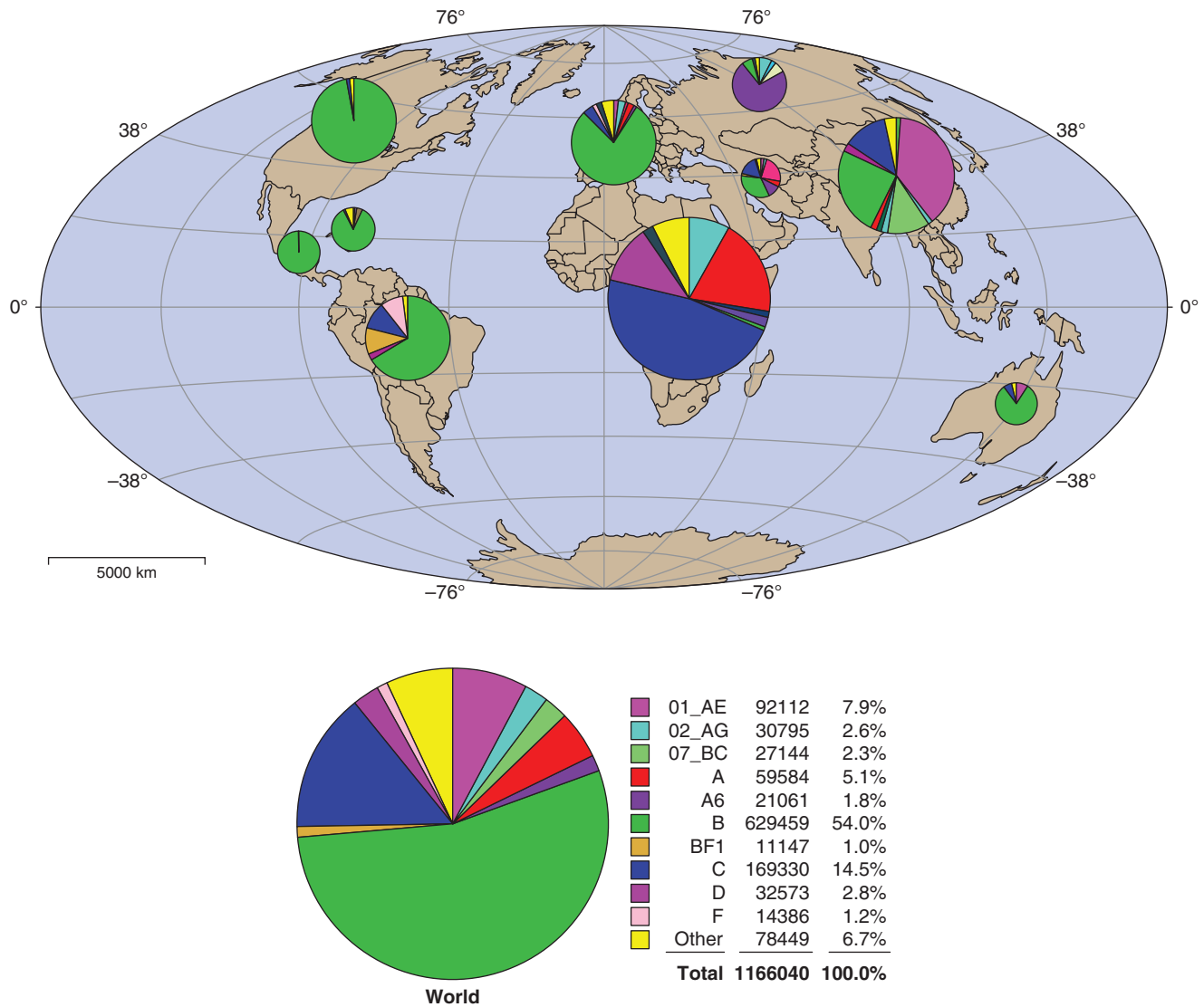


FIGURE 208-6 Global geographic distribution of HIV-1 subtypes and recombinant forms. Distributions derived from relative frequency of subtypes among >1.16 million HIV genomic sequences in the Los Alamos National Laboratory HIV Sequence Database. (Additional information available at www.hiv.lanl.gov/components/sequence/HIV/geo/geo.comp.)

TRANSMISSION

HIV is transmitted primarily by sexual contact (both heterosexual and male to male); by blood and blood products; and by infected mothers to infants intrapartum, perinatally, or via breast milk. After four decades of experience and observations, there is no evidence that HIV is transmitted by any other modality. Table 208-3 lists the estimated risk of HIV transmission for various types of exposures.

SEXUAL TRANSMISSION

HIV infection is predominantly a sexually transmitted infection (STI) worldwide. By far the most common mode of infection, particularly in developing countries, is heterosexual transmission, although in many western countries male-to-male sexual transmission dominates. Although a wide variety of factors including viral load and the presence of ulcerative genital diseases influence the efficiency of heterosexual transmission of HIV, such transmission is generally inefficient. A systemic review found a low per-act risk of heterosexual transmission in the absence of antiretrovirals: 0.04% for female-to-male transmission and 0.08% for male-to-female transmission during vaginal intercourse in the absence of antiretroviral therapy or condom use (Table 208-3). As discussed below, this risk approaches zero in settings where the infected partner is on effective antiretroviral therapy or when the uninfected partner is on a program of pre-exposure or postexposure prophylaxis (PrEP or PEP).

HIV has been demonstrated in seminal fluid both within infected mononuclear cells and in cell-free material. The virus appears to

TABLE 208-3 Estimated Per-Act Probability of Acquiring HIV from an Infected Source, By Exposure Act	
TYPE OF EXPOSURE	RISK PER 10,000 EXPOSURES
Parenteral	
Blood transfusion	9250
Needle-sharing during injection drug use	63
Percutaneous (needle-stick)	23
Sexual	
Receptive anal intercourse	138
Insertive anal intercourse	11
Receptive penile-vaginal intercourse	8
Insertive penile-vaginal intercourse	4
Receptive oral intercourse	Low
Insertive oral intercourse	Low
Other ^a	
Biting	Negligible
Spitting	Negligible
Throwing body fluids (including semen or saliva)	Negligible
Sharing sex toys	Negligible

^aHIV transmission through these exposure routes is technically possible but unlikely and not well documented.
Source: CDC, www.cdc.gov/hivpartners/php/riskandprevention/.

concentrate in the seminal fluid, particularly in situations where there are increased numbers of lymphocytes and monocytes in the fluid, as seen in genital inflammatory states such as urethritis and epididymitis, conditions closely associated with other STIs. The virus has also been demonstrated in cervical smears and vaginal fluid. There is an elevated risk of HIV transmission associated with unprotected receptive anal intercourse (URAI) among both men and women compared to the risk associated with unprotected receptive vaginal intercourse. Although data are limited, the per-act risk for HIV transmission via URAI has been estimated to be ~1.4% (Table 208-3). The risk of HIV acquisition associated with URAI is higher than that seen in penile–vaginal intercourse probably because only a thin, fragile rectal mucosal membrane separates the deposited semen from potentially susceptible cells in and beneath the mucosa, and microtrauma of the mucosal membrane has been associated with anal intercourse. Anal douching and sexual practices that traumatize the rectal mucosa also increase the likelihood of infection. It is likely that anal intercourse provides at least two modalities of infection: (1) direct inoculation into blood in cases of traumatic tears in the mucosa; and (2) infection of susceptible target cells, such as Langerhans cells, in the mucosal layer in the absence of trauma. Insertive anal intercourse also confers an increased risk of HIV acquisition compared with insertive vaginal intercourse in the receptive partner since the vaginal mucosa is several layers thicker than the rectal mucosa and less likely to be traumatized during intercourse. Nonetheless, the virus can be transmitted to either partner through vaginal intercourse. As noted in Table 208-3, male-to-female HIV transmission is more efficient than female-to-male transmission. The differences in reported transmission rates between men and women may be due in part to the prolonged exposure of the vaginal and cervical mucosa to infected seminal fluid; the endometrium also can be exposed to virus when semen enters through the cervical os. By comparison, the penis and urethral orifice of the uninfected male partner are only exposed relatively briefly to infected vaginal fluid.

Among various cofactors examined in studies of heterosexual HIV transmission, the presence of other STIs has been strongly associated with HIV transmission. In this regard, there is a close association between genital ulcerations and transmission, owing to both susceptibility to infection and infectivity. Infections with microorganisms such as *Treponema pallidum* (Chap. 187), *Haemophilus ducreyi* (Chap. 162), and herpes simplex virus (HSV; Chap. 197) are important causes of genital ulcerations linked to transmission of HIV. In addition, pathogens responsible for non-ulcerative inflammatory STIs such as those caused by *Chlamydia trachomatis* (Chap. 194), *Neisseria gonorrhoeae* (Chap. 161), and *Trichomonas vaginalis* (Chap. 236) also are associated with an increased risk of transmission of HIV infection. Bacterial vaginosis, an infection related to sexual behavior, but not strictly an STI, also may be linked to an increased risk of transmission of HIV infection. Multiple studies have suggested that treating STIs and genital tract syndromes may help decrease transmission of HIV. This effect is most prominent in populations in which the prevalence of HIV infection is relatively low. It is noteworthy that this principle may not apply to the treatment of HSV infections since it has been shown that even following anti-HSV therapy with resulting healing of HSV-related genital ulcers, HIV acquisition is not reduced. Biopsy studies revealed that the likely explanation is that HIV receptor–positive inflammatory cells persisted in the genital tissue despite the healing of ulcers, and so HIV-susceptible targets remained at the site.

The quantity of HIV-1 in plasma (viral load) is a primary determinant of the risk of HIV-1 transmission. In a cohort of heterosexual couples in Uganda discordant for HIV infection and not receiving antiretroviral therapy, the mean serum HIV RNA level was significantly higher among individuals with HIV whose partners seroconverted than among those whose partners did not seroconvert. In fact, transmission was rare when the infected partner had a plasma level of <1700 copies of HIV RNA per milliliter, even when genital ulcer disease was present (Fig. 208-7). The rate of HIV transmission per coital act was highest during the early stage of HIV infection when plasma HIV RNA levels were high and in advanced disease with high viral set points.

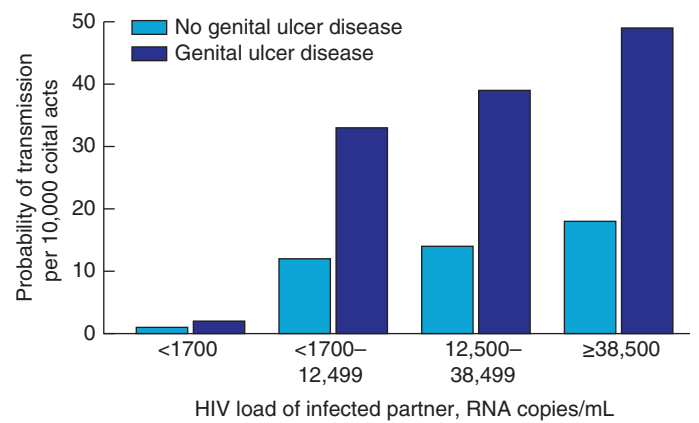


FIGURE 208-7 Probability of HIV transmission per coital act among monogamous, heterosexual, HIV-serodiscordant couples in Uganda. (From RH Gray et al: *Lancet* 357:1149, 2001.)

Antiretroviral therapy dramatically reduces plasma viremia in most people with HIV (see “Antiretroviral Therapy” and “HIV Prevention,” below) and is associated with a dramatic reduction in risk of transmission, an approach widely referred to as *treatment as prevention* or TasP. Multiple studies have demonstrated that if the viral load of a person with HIV is reduced by antiretroviral therapy to <20 copies/mL as measured by conventional commercial assays, there is essentially no chance of sexual transmission to the person’s sexual partner. This is true for heterosexuals as well as men who have sex with men, leading to the commonly used description of this phenomenon as “undetectable equals untransmittable” or “U = U.”

Multiple studies including large, randomized, controlled trials clearly have indicated that male circumcision is associated with a lower risk of acquisition of HIV infection for heterosexual men. Studies also suggest that circumcision is protective against HIV acquisition for men who have sex with men reporting mainly or only insertive sex. The benefit of circumcision may be due to increased susceptibility of uncircumcised men to ulcerative STIs, as well as to other factors such as microtrauma to the foreskin and glans penis. In addition, the highly vascularized inner layer of foreskin tissue contains a high density of Langerhans cells as well as increased numbers of CD4+ T cells, macrophages, and other cellular targets for HIV. Finally, the moist environment under the foreskin may promote the presence or persistence of microbial flora that, via inflammatory changes, may lead to even higher concentrations of target cells for HIV in the foreskin. In addition, randomized clinical trials have demonstrated that male circumcision also reduces HSV type 2, human papillomavirus virus (HPV), and genital ulcer disease in men as well as HPV, genital ulcer disease, bacterial vaginosis, and *Trichomonas vaginalis* infections among female partners of circumcised men. Thus, there may be an added indirect benefit of diminution of risk for HIV acquisition to the female sexual partners of circumcised men.

In some studies, the use of oral contraceptives was associated with an increase in incidence of HIV infection over and above that which might be expected by not using a condom for birth control. This phenomenon may be due to drug-induced changes in the cervical mucosa, rendering it more vulnerable to penetration by the virus. Adolescent girls might also be more susceptible to infection upon exposure due to the properties of an immature genital tract with increased cervical ectopy or exposed columnar epithelium.

Oral sex is a much less efficient mode of transmission of HIV than is anal intercourse or vaginal intercourse (Table 208-3). Multiple studies have reported that the incidence of transmission of infection by oral sex among couples discordant for HIV is extremely low. However, there have been well-documented reports of HIV transmission that likely resulted from fellatio or cunnilingus. Therefore, the assumption that oral sex is completely safe is not warranted.

The association of alcohol consumption and illicit drug use with unsafe sexual behavior, both homosexual and heterosexual, leads to an increased

risk of sexual transmission of HIV. Methamphetamine and other so-called “club drugs” such as 3,4-methylenedioxymethamphetamine (MDMA; also known as “ecstasy”), ketamine, gamma-hydroxybutyrate (GHB), and inhaled nitrites (known as “poppers”), sometimes taken in conjunction with PDE-5 inhibitors such as sildenafil (Viagra), tadalafil (Cialis), or vardenafil (Levitra), have been associated with risky sexual practices and increased risk of HIV infection, particularly among men who have sex with men.

■ TRANSMISSION THROUGH INJECTION DRUG USE

HIV can be transmitted to injection drug users (IDUs) who are exposed to HIV while sharing injection paraphernalia such as needles, syringes, the water in which drugs are mixed, or the cotton through which drugs are filtered. Parenteral transmission of HIV during injection drug use does not require IV puncture; subcutaneous (“skin popping”) or intramuscular (“muscling”) injections can transmit HIV as well, even though these behaviors are sometimes erroneously perceived as low risk. Among IDUs, the risk of HIV infection increases with the duration of injection drug use; the frequency of needle sharing; the number of partners with whom paraphernalia are shared; comorbid psychiatric conditions such as antisocial personality disorder; the use of cocaine in injectable form or smoked as “crack”; and the use of injection drugs in a geographic location with a high prevalence of HIV infection. As noted in Table 208-3, the per-act risk of transmission from injection drug use with a contaminated needle has been estimated to be approximately 0.6%.

■ TRANSMISSION BY TRANSFUSED BLOOD AND BLOOD PRODUCTS

HIV can be transmitted to individuals who receive HIV-contaminated blood transfusions, blood products, or transplanted tissue. The vast majority of HIV infections acquired via contaminated blood transfusions, blood components, or transplanted tissue in resource-rich countries occurred prior to the spring of 1985, when mandatory testing of donated blood for HIV-1 was initiated. It is estimated that >90% of individuals exposed to HIV-contaminated blood products become infected (Table 208-3). Transfusions of whole blood, packed red blood cells, platelets, leukocytes, and plasma are all capable of transmitting HIV infection. In contrast, hyperimmune gamma globulin, hepatitis B immune globulin, plasma-derived hepatitis B vaccine, and Rh₀ immune globulin have not been associated with transmission of HIV infection. The procedures involved in processing these products either inactivate or remove the virus.

Currently, in the United States and in most developed countries, the following measures have made the risk of transmission of HIV infection by transfused blood or blood products extremely small: the screening of blood donations for antibodies to HIV-1 and HIV-2 and determination of the presence of HIV nucleic acid usually in minipools of several specimens; the careful selection of potential blood donors with health history questionnaires to exclude individuals with risk behaviors; and opportunities for self-deferral and the screening out of HIV-negative individuals with serologic testing for infections that have shared risk factors with HIV, such as hepatitis B and C and syphilis. The chance of infection of a hemophiliac via clotting factor concentrates has essentially been eliminated because of standard screening of blood together with the added layer of safety resulting from heat treatment of the concentrates. It is currently estimated that the risk of infection with HIV in the United States via transfused screened blood is approximately 1 in 2 million units. Since nearly 21 million blood components are transfused in the United States each year, eliminating the risk of transfusion-related HIV transmission likely will not be possible. Transmission of HIV (both HIV-1 and HIV-2) by blood or blood products is still an ongoing threat in certain developing countries where routine screening of blood is not universally practiced. Furthermore, there have been reports in certain countries of sporadic breakdowns in routinely available screening procedures in which contaminated blood or plasma was transfused, resulting in small clusters of patients becoming infected.

■ OCCUPATIONAL TRANSMISSION OF HIV: HEALTH CARE WORKERS, LABORATORY WORKERS, AND THE HEALTH CARE SETTING

There is a small but definite occupational risk of HIV transmission to health care workers and laboratory personnel and potentially others who work with HIV-containing materials, particularly when sharp objects are used. More than 300,000 health care workers are stuck with needles or other sharp medical instruments in the United States each year. The global number of HIV infections among health care workers attributable to sharps injuries has been estimated to be 1000 cases (range, 200–5000) per year. In the United States, a total of 58 documented cases of occupational HIV transmission to health care workers, and 150 possible transmissions, have been reported by the CDC. Since 1999, only one confirmed case (a laboratory technician sustaining a needle puncture while working with a live HIV culture in 2008) has been reported.

Exposures that place a health care worker at potential risk of HIV infection are percutaneous injuries (e.g., a needle stick or cut with a sharp object) or contact of mucous membrane or nonintact skin (e.g., exposed skin that is chapped, abraded, or afflicted with dermatitis) with blood, tissue, or other potentially infectious body fluids. Large, multi-institutional studies have indicated that the risk of HIV transmission following skin puncture from a needle or a sharp object that was contaminated with blood from a person with documented HIV infection is ~0.23%, and after a mucous membrane exposure it is ~0.09% (see “HIV and the Health Care Worker,” below) if the injured and/or exposed person is not treated within 24 hours with antiretroviral drugs. The risk of hepatitis B virus (HBV) infection following a similar type of exposure is ~6–30% in nonimmune individuals. If a susceptible worker is exposed to HBV, postexposure prophylaxis with hepatitis B immune globulin and initiation of HBV vaccine is >90% effective in preventing HBV infection. The risk of HCV infection following percutaneous injury is ~1.8% ([Chap. 350](#)).

Rare HIV transmission after nonintact skin exposure has been documented. The average risk for transmission by this route has not been precisely determined; however, it is estimated to be less than the risk for mucous membrane exposure. Transmission of HIV through intact skin has not been documented. All health care workers experiencing a puncture wound or mucous membrane exposures involving blood from a patient with documented HIV infection should be treated prophylactically with combination antiretroviral therapy (ART). This practice, referred to as *postexposure prophylaxis* or PEP, has dramatically reduced the occurrence of puncture-related transmissions of HIV to health care workers.

In addition to blood and visibly bloody body fluids, semen and vaginal secretions also are considered potentially infectious; however, they have not been implicated in occupational transmission from patients to health care workers. The following fluids also are considered potentially infectious: cerebrospinal fluid, synovial fluid, pleural fluid, peritoneal fluid, pericardial fluid, and amniotic fluid. The risk for transmission after exposure to fluids or tissues other than HIV-infected blood has not been quantified, but it is probably considerably lower than the risk after blood exposures. Feces, nasal secretions, saliva, sputum, sweat, tears, urine, and vomitus are not considered potentially infectious for HIV unless they are visibly bloody. Rare cases of HIV transmission via human bites have been reported, but not in the setting of occupational exposure.

An increased risk for HIV infection following percutaneous exposures to HIV-infected blood is associated with exposures involving a relatively large quantity of blood, as in the case of a device visibly contaminated with the patient's blood, a procedure that involves a hollow-bore needle placed directly in a vein or artery, or a deep injury. Factors that might be associated with mucocutaneous transmission of HIV include exposure to an unusually large volume of blood and prolonged contact. In addition, the risk increases for exposures to blood from untreated patients with high levels of HIV in the blood. Since the beginning of the HIV epidemic, there have been rare instances where transmission of infection from a health care worker to patients seemed highly probable. Despite this small number of documented cases, the

risk of HIV transmission involving infected health care workers to patients is extremely low in developed countries—in fact, too low to be measured accurately. In this regard, several retrospective epidemiologic studies have been performed tracing thousands of patients of dentists, physicians, surgeons, obstetricians, and gynecologists with HIV, and no cases of HIV transmission that could be linked to the health care providers were identified other than the already identified documented cases.

Breaches in infection control and the reuse of contaminated syringes, failure to properly sterilize surgical instruments, and/or hemodialysis equipment also have resulted rarely in the transmission of HIV from patient to patient in hospitals, nursing homes, and outpatient settings. Finally, these very rare occurrences of transmission of HIV as well as HBV and HCV to and from health care workers in the workplace underscore the importance of the use of universal precautions when caring for all patients (see below and [Chap. 147](#)).

■ MOTHER-TO-CHILD TRANSMISSION OF HIV

HIV infection can be transmitted from an infected mother to her fetus during pregnancy, during delivery, or by breast-feeding. This remains a persistent form of transmission of HIV infection in certain developing countries. Virologic analyses of aborted fetuses indicate that HIV can be transmitted to the fetus during the first or second trimesters of pregnancy. However, maternal transmission occurs most commonly during birth. Two studies performed in Rwanda and the Democratic Republic of Congo (then called Zaire) indicated that among all mother-to-child transmissions of HIV, the relative proportions were 23–30% before birth, 50–65% during birth, and 12–20% via breast-feeding.

In the absence of antiretroviral therapy for the mother during pregnancy, labor, and delivery, and for the infant prophylactically following birth, the overall probability of transmission of HIV from mother to infant/fetus ranges from 15% to 25% in industrialized countries and from 25% to 35% in developing countries. These differences may relate to the adequacy of prenatal care as well as to the stage of HIV disease and the general health of the mother during pregnancy. Higher rates of transmission have been reported to be associated with many factors—the best documented of which is the presence of high maternal levels of plasma viremia, with the risk increasing linearly with the level of maternal plasma viremia. It is very unlikely that mother-to-child transmission will occur if the mother's level of plasma viremia is <1000 copies of HIV RNA/mL of blood and extremely unlikely if the level is <50 copies/mL. Increased mother-to-child transmission is also correlated with closer human leukocyte antigen (HLA) match between mother and child. A prolonged interval between membrane rupture and delivery is another well-documented risk factor for transmission of HIV. Other conditions that are potential risk factors, but that have not been consistently demonstrated, are the presence of chorioamnionitis at delivery; STIs during pregnancy; illicit drug use during pregnancy; cigarette smoking; preterm delivery; and obstetric procedures such as amniocentesis, amniocentesis, fetal scalp electrodes, and episiotomy. Today, the rate of mother-to-child transmission has fallen to <1% in pregnant women who are receiving ART for their HIV infection. Such treatment, combined with cesarean section delivery, has rendered mother-to-child transmission of HIV an extremely unusual event in the United States and other developed nations. In this regard, both the United States Public Health Service and the World Health Organization guidelines recommend that all pregnant women with HIV receive lifelong ART for the health of the mother (regardless of plasma HIV RNA copy number or CD4+ T cell count) as well as to prevent perinatal transmission.

Breast-feeding is an important modality of transmission of HIV infection in certain developing countries, particularly where untreated mothers continue to breast-feed for prolonged periods. The risk factors for mother-to-child transmission of HIV via breast-feeding by an untreated mother include detectable levels of HIV in breast milk, the presence of mastitis, low maternal CD4+ T cell counts, and maternal vitamin A deficiency. The risk of HIV infection via breast-feeding is highest in the early months of breast-feeding. In addition, exclusive breast-feeding has been reported to carry a lower risk of

HIV transmission than mixed feeding. In developed countries, breast-feeding of babies by a mother with HIV is contraindicated since alternative forms of adequate nutrition, i.e., formulas, are readily available. In developing countries, where breast-feeding may be essential for the overall health of the infant, the continuation of ART in the infected mother during the period of breastfeeding markedly diminishes the risk of transmission of HIV to the infant. In fact, treatment of a pregnant woman with ART should be provided for the benefit of the woman as much as for the prevention of mother-to-child transmission and should be continued beyond the pregnancy, for life.

■ TRANSMISSION OF HIV BY OTHER BODY FLUIDS

Although HIV can be isolated typically in low titers from saliva of a small proportion of infected individuals, there is no convincing evidence that saliva can transmit HIV infection, either through kissing or through other exposures, such as occupationally to health care workers. Saliva contains endogenous antiviral factors; among these factors, HIV-specific immunoglobulins of IgA, IgG, and IgM isotypes are detected readily in salivary secretions of infected individuals. It has been suggested that large glycoproteins such as mucins and thrombospondin 1 sequester HIV into aggregates for clearance by the host. In addition, multiple soluble salivary factors inhibit HIV to various degrees in vitro, probably by targeting host cell receptors rather than the virus itself. Perhaps the best studied of these, secretory leukocyte protease inhibitor (SLPI), blocks HIV infection in several cell culture systems, and it is found in saliva at levels that approximate those required for inhibition of HIV in vitro. In this regard, higher salivary levels of SLPI in breast-fed infants were associated with a decreased risk of HIV transmission through breast milk. It has also been suggested that submandibular saliva reduces HIV infectivity by stripping gp120 from the surface of virions, and that saliva-mediated disruption and lysis of HIV-infected cells occurs because of the hypotonicity of oral secretions. Transmission of HIV by a human bite can occur but is a rare event. Although virus can be detected, if not isolated, from virtually any body fluid, there is no evidence that HIV transmission can occur as a result of exposure to tears, sweat, or urine. However, there have been isolated cases of transmission of HIV infection by body fluids that may or may not have been contaminated with blood. Most of these situations occurred in the setting of a close relative providing intensive nursing care for a person with HIV without observing universal precautions, underscoring the importance of adhering to such precautions in the handling of body fluids and wastes from people with HIV.

EPIDEMIOLOGY

■ HIV INFECTION AND AIDS WORLDWIDE

HIV infection/AIDS is a global pandemic, with cases reported from virtually every country. In 2023, an estimated 39.9 million individuals were living with HIV infection, according to the Joint United Nations Programme on HIV/AIDS (UNAIDS). An estimated 95% of people living with HIV/AIDS reside in low- and middle-income countries; ~53% are female, and 1.4 million are children <15 years. The regional distribution of these cases is illustrated in [Fig. 208-8](#). The estimated number

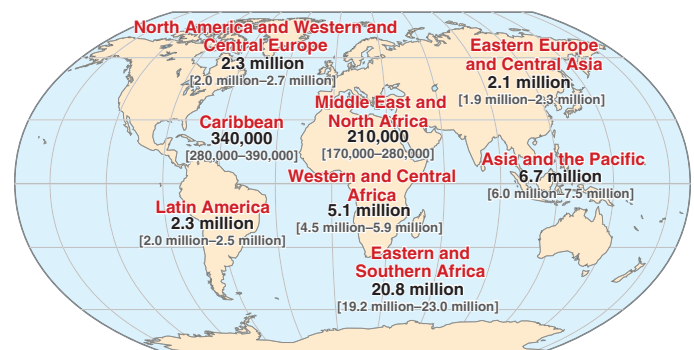


FIGURE 208-8 Estimated number of adults and children living with HIV infection as of December 2023. Total: 39.9 million (36.1 million–44.6 million). (From Joint United Nations Programme on HIV/AIDS [UNAIDS].)

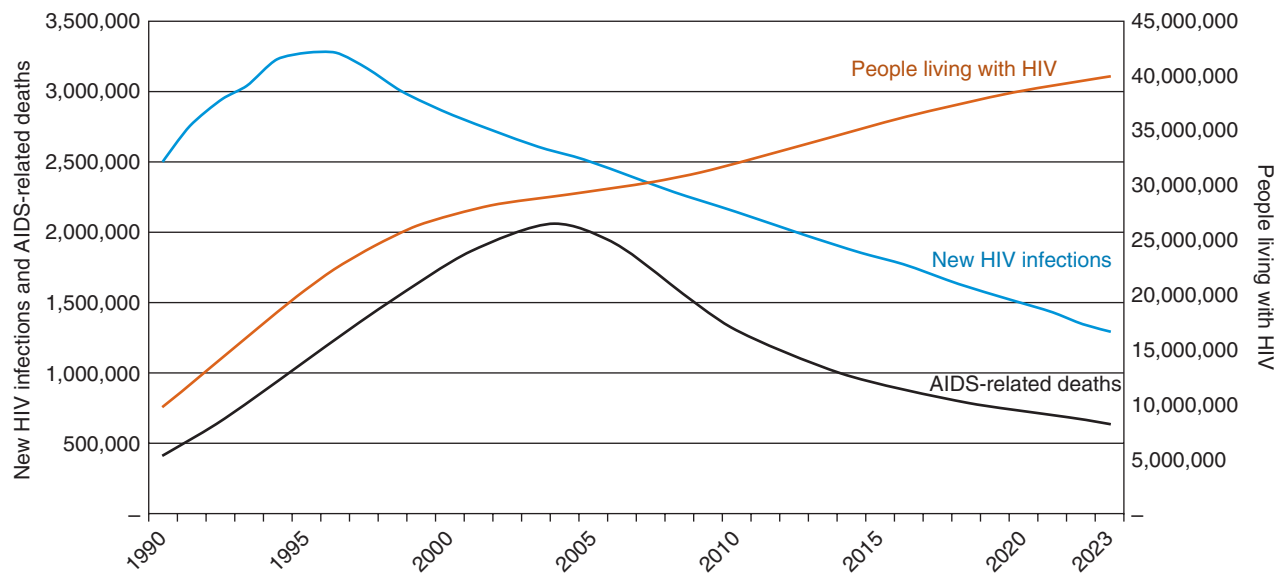


FIGURE 208-9 Global estimates of new HIV cases, AIDS-related deaths, and HIV prevalence, 1990–2023. (From UNAIDS.)

of people living with HIV—i.e., the global prevalence—has increased more than fourfold since 1990, reflecting the combined effects of continued high rates of new HIV infections and the life-prolonging impact of antiretroviral therapy (Fig. 208-9). The overall global prevalence of HIV infection among persons 15–49 years of age is ~0.7%, with rates varying widely by country and region as illustrated in Fig. 208-10. The WHO Eastern and Southern Africa region remains most severely affected, with more than 1 in 20 adults (5.7%) living with HIV and accounting for more than half of the people with HIV worldwide. An estimated 88.4 million people have become infected with HIV since the start of the pandemic.

In 2023, an estimated 1.3 million new cases of HIV infection occurred worldwide, including 120,000 among children <15 years; about one-third of new infections were among people age 15–24 years. The regional distribution of new HIV cases in 2023 is shown

in Fig. 208-11. Globally, members of certain high-risk populations are disproportionately affected by HIV infection, often because of factors such as marginalization, discrimination, and in some cases criminalization. Sex workers, people who inject drugs, transgender people, prisoners, gay men and other men who have sex with men, the clients of sex workers, and the sexual partners of these key populations accounted for 55% of all new HIV infections in 2022 (Fig. 208-12).

New HIV infections globally have fallen by 60% since their peak in 1995 (Fig. 208-9). In 2023, 1.3 million people were newly infected with HIV, compared with 3.3 million people in 1995. Reductions in global HIV incidence likely reflect progress with HIV prevention efforts and the increased provision to HIV-infected people of antiretroviral therapy, which makes them much less likely to transmit the virus to sexual partners. Between 2010 and 2023, new HIV infections declined by 39%, from 2.1 million to 1.3 million. During the same period a ~62%

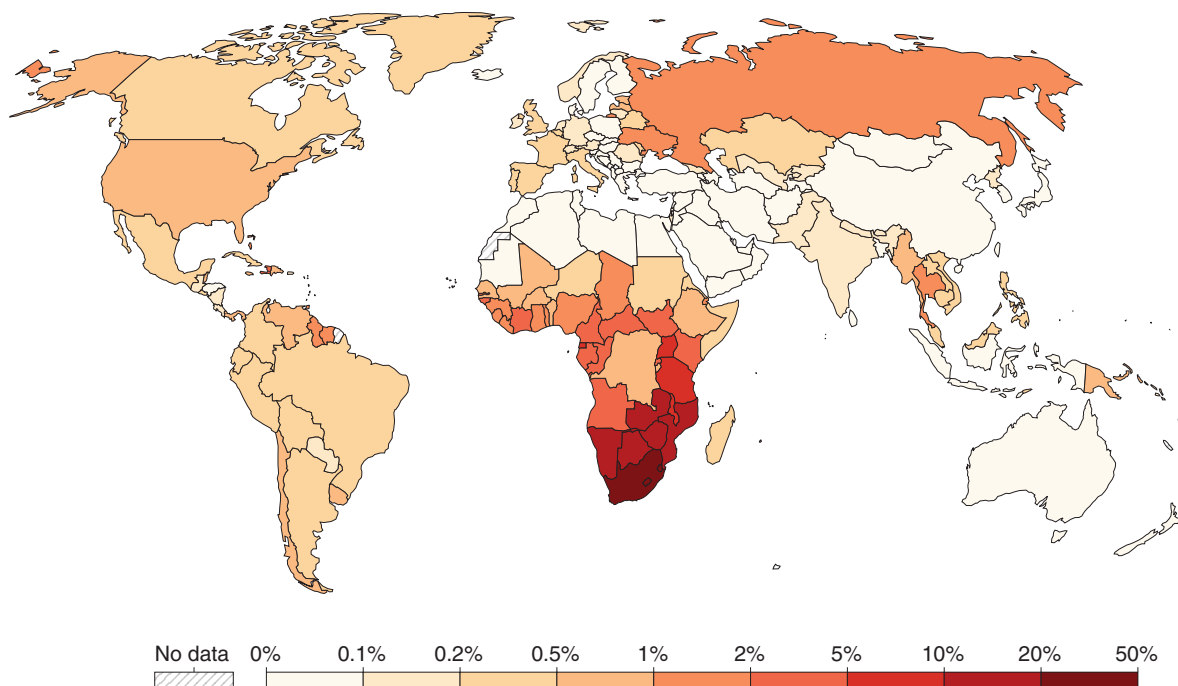


FIGURE 208-10 Adult HIV prevalence rates by country, 2021. Data are estimates for adults age 15–49 years. (Reproduced from IHME, *Global Burden of Disease with major processing by Our World in Data*; 2024.)

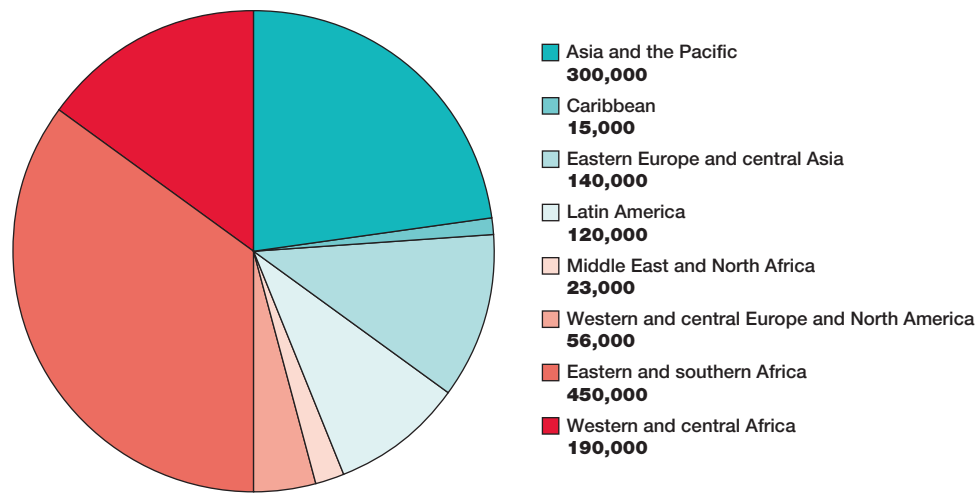


FIGURE 208-11 Distribution of new HIV infections, by region, 2023. (Reproduced with permission from UNAIDS.)

reduction in HIV infections among children <15 years was observed, from 300,000 in 2010 to 120,000 in 2023. This progress is due largely to the increasing availability of antiretroviral medications to prevent the transmission of HIV from mother to infant. An estimated 30.7 million people with HIV globally, 77% of all people living with HIV, were accessing antiretroviral therapy as of December 2023, up from 7.7 million people in 2010. Among pregnant women with HIV, 84% had access to antiretroviral medicines to prevent transmission of HIV to their child in 2023.

In 2023, global AIDS deaths totaled 630,000 (including 76,000 children <15 years), a 65% decrease since 2000 that coincides with a rapid expansion of access to antiretroviral therapy (Fig. 208-13). Since the beginning of the HIV pandemic, an estimated 42.3 million persons globally have died of an AIDS-related illness.

The HIV epidemic has occurred in “waves” in different regions of the world, each wave having somewhat different characteristics depending on the demographics of the country and region in question and the timing of the introduction of HIV into the population. Although the AIDS epidemic was first recognized in the United States and shortly thereafter in Western Europe, it very likely began in sub-Saharan Africa (see above), a region particularly devastated by the epidemic.

The 21 countries of the **Eastern and Southern Africa** region are home to about 7% of the world’s population but had 20.8 million people living with HIV in 2023, >50% of the global total (Fig. 208-8). HIV prevalence among adults age 15–49 years across the region is 5.7%. Of 21 countries in the region, 17 have generalized epidemics, that is, their national prevalence is >1%. In 6 countries in the region, >10% of

the adult population age 15–49 has HIV infection (Fig. 208-10). South Africa has the highest number of people living with HIV in the world (7.7 million); Eswatini (formerly known as Swaziland) has the highest adult HIV prevalence globally (25.1%). Heterosexual exposure is the primary mode of HIV transmission in most countries in this region, as is the case throughout sub-Saharan Africa. Women and girls age 15 years and older account for ~63% of all HIV infections in the region. Key populations, notably sex workers and clients of sex workers, accounted for 23% of new infections in the region in 2022.

Major progress in the HIV response has been made in the Eastern and Southern Africa region in recent years. The annual number of people acquiring HIV in the region fell by 59% between 2010 and 2023, from 1.1 million to 450,000. AIDS-related deaths decreased by 57% in that period, from 600,000 in 2010 to 260,000 in 2023. As of 2023, 93% of people living with HIV in the region knew their infections status; 83% of those living with HIV (17.4 million people) were on antiretroviral therapy, and 78% had suppressed viral loads.

The 25 countries of the **Western and Central Africa** region are home to 5.1 million people living with HIV, of whom 3.1 million are women age ≥15 years and 380,000 are children. HIV prevalence in most countries is relatively low compared with Eastern and Southern Africa. HIV prevalence among adults across the region overall stands at 1.2%. About 40% of new infections in the region in 2023 occurred in Nigeria, a large country with an HIV seroprevalence rate of 1.3%. As in Eastern and Southern Africa, heterosexual transmission accounts for most HIV transmission in West and Central Africa, with sex workers and their clients accounting for about one-fifth of new infections.

The **Middle East and North Africa** region has one of the lowest HIV prevalence rates in the world (<0.1%), although annual new infections more than doubled between 2010 and 2023, from 11,000 to 23,000. In 2023, an estimated 230,000 people were living with HIV in the 19 countries in the region. Cases are largely concentrated among IDUs, men who have sex with men, and sex workers and their clients.

In **Asia and the Pacific**, an estimated 6.7 million people were living with HIV at the end of 2023, with an overall adult prevalence rate of 0.2%. With 300,000 new infections in 2023, the region accounted for one-quarter of all new infections globally that year. The region saw a 13% decrease in new infections between 2010 and 2023, with reductions in Thailand and Vietnam offset by increases in Afghanistan, Bangladesh, Fiji, the Lao People’s Democratic Republic, Papua New Guinea, and the Philippines. In the same time frame, AIDS-related deaths in the region fell by 51% to 150,000. Among countries in this region, only Thailand has an adult seroprevalence rate that reaches 1%. However, the populations of many countries are so large that even low infection and seroprevalence rates result in large numbers of people living with HIV. In this regard, three populous countries—China, India, and Indonesia—account for around three-quarters of all people living with HIV in the region. Key populations (Fig. 208-11) and their

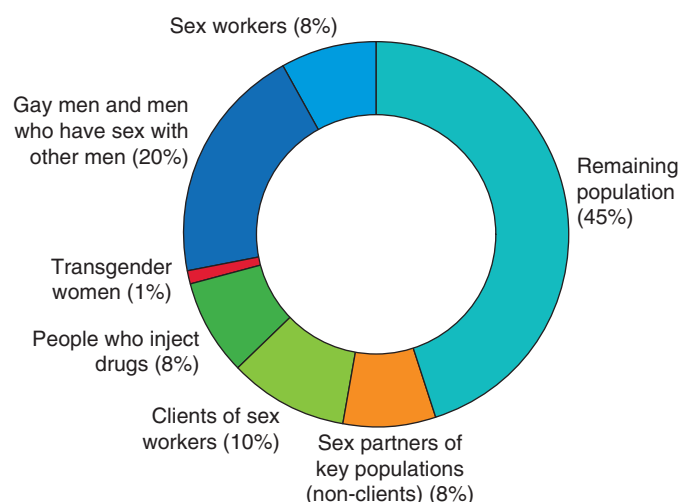


FIGURE 208-12 Global distribution of new HIV infections by population. Data for 2022. (From UNAIDS.)

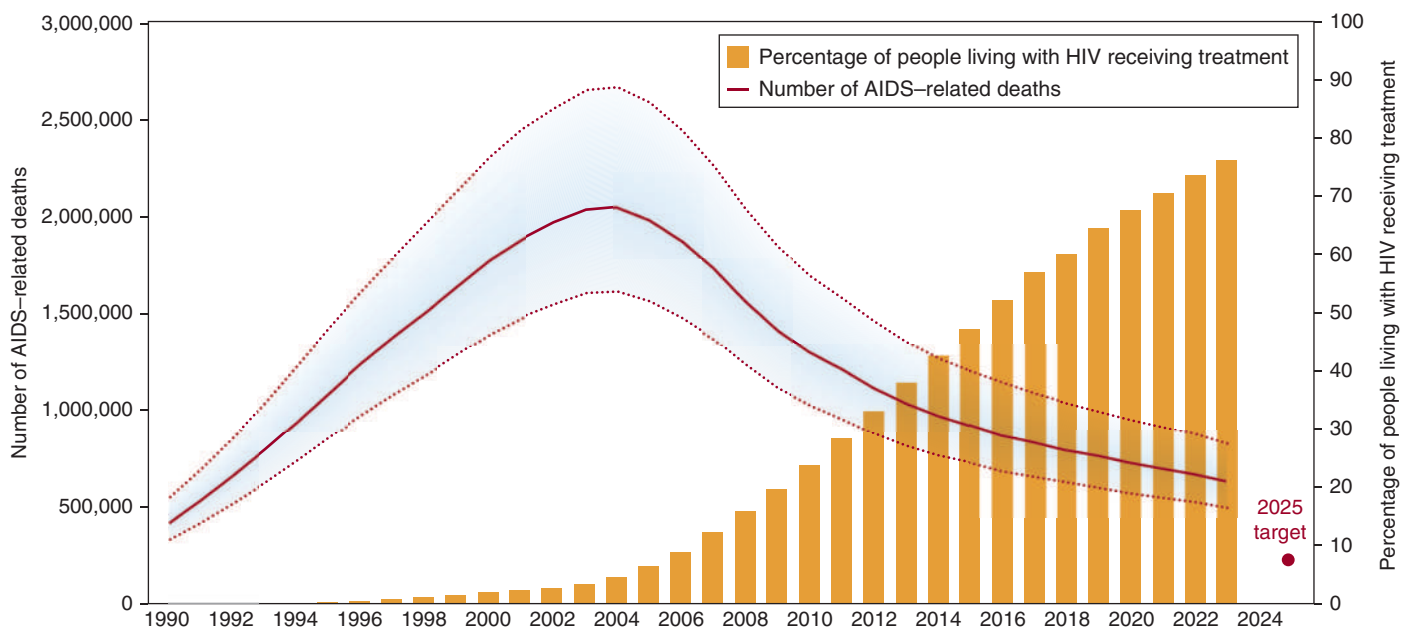


FIGURE 208-13 Global antiretroviral therapy coverage and number of AIDS-related deaths, 1990-2023. (Data from *The Urgency of Now: AIDS at a Crossroads*. Geneva: Joint United Nations Programme on HIV/AIDS; 2024.)

partners accounted for ~80% of new HIV infections in the region in 2022. Rising numbers of new infections among gay men and other men who have sex with men in this region are a major concern.

The HIV epidemic continues to expand in **Eastern Europe and Central Asia**, with a 20% increase in annual new HIV infections and 34% increase in AIDS deaths between 2010 and 2023. There are about 2.1 million people living with HIV in the region, where the epidemic has been driven by injection drug use but has an increasing proportion of new HIV infections transmitted sexually. Members of key populations and their sexual partners account for ~94% of new infections in the region. The Russian Federation, Ukraine, Kazakhstan, and Uzbekistan reported 92% of all registered cases in the region in 2023.

Approximately 2.3 million people were living with HIV/AIDS in **Latin America** at the end of 2023. The annual number of new infections in the region increased by 9% between 2010 and 2023 to 120,000. About two-thirds of new infections were among key populations and their sex partners, with increasing rates among men who have sex with men, sex workers, and transgender women. AIDS-related deaths declined by 28% to 30,000 from 2010 to 2023. Brazil is home to the largest number of HIV-infected persons (1,00,000) in the region.

In the **Caribbean**, an estimated 340,000 people were living with HIV in 2023, with 15,000 new infections that year. New infections have declined by 22% in the region since 2010. About 90% of new infections in the region in 2023 were in Cuba, the Dominican Republic, Haiti, and Jamaica. People from key populations and their sex partners accounted for almost half of new infections in the region in 2023.

Approximately 2.3 million people were living with HIV/AIDS in **North America and Western and Central Europe** in 2023. While modes of transmission vary greatly by country, HIV in the region disproportionately affects men who have sex with men. New infections in the region fell by 24% between 2010 and 2023 to 56,000, while AIDS-related deaths fell 34%, to 13,000. Key populations accounted for ~83% of new infections in the region in 2022.

HIV INFECTION IN THE UNITED STATES

As of 2022, an estimated 1.2 million individuals in the United States were living with HIV infection, ~13% of whom were unaware of their infection. Among people with diagnosed HIV in 2022, about 76% have received some HIV care, and about 65% have achieved viral suppression (see below).

Nearly 40% of people living with HIV in the United States are Black/African American and 26% are Hispanic/Latino; ~60% are men who have sex with men, according to CDC estimates. The HIV prevalence

rate among all individuals age 13 years or older in the United States is 0.4%. Approximately 1.4% of Black/African-American adults are living with HIV in the United States, more than any other racial/ethnic group.

The estimated annual number of new HIV infections in the United States has fallen by more than two-thirds since its height in the late 1980s of about 130,000 per year. CDC data indicate further progress in recent years, as overall incidence fell 12% from 2018 to 2022, from 36,200 new infections to 31,800. The estimated distribution of new infections in 2022, broken down by transmission category, is shown in **Fig. 208-14**.

In the United States, the burden of HIV infection is not evenly distributed across states and regions. In most areas of the country, HIV is concentrated in urban areas. In the southern United States, larger percentages of diagnoses are in smaller metropolitan and nonmetropolitan areas. HIV has disproportionately affected minority populations in the United States in both urban and rural areas. Among new HIV infections from 2018 to 2022 in the United States, 37% were among Blacks/African Americans, a group that constitutes only 12% of the U.S. population. Hispanics/Latinos, 18% of the U.S. population, accounted for 33% of new HIV infections. The rate of new HIV infections in 2022 by race/ethnicity in the United States is shown in **Fig. 208-15**.

Perinatal HIV transmission, from an HIV-infected mother to her baby, has declined significantly in the United States, largely due to the implementation of guidelines for the universal counseling and voluntary HIV testing of pregnant women and the use of antiretroviral

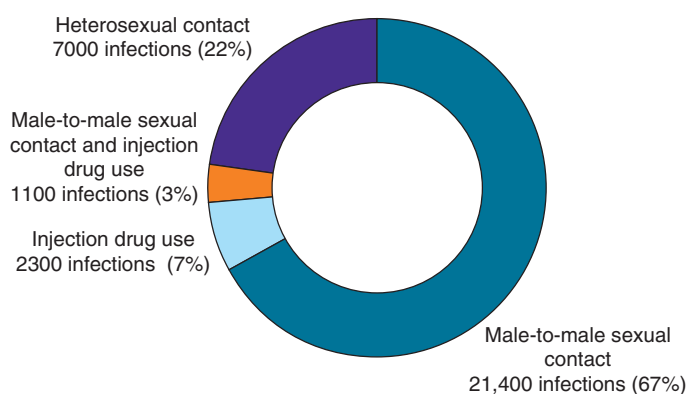


FIGURE 208-14 Estimated distribution of new HIV infections in the United States among people age ≥13 years, by transmission category, 2022. Total: 31,800. (From CDC: HIV Surveillance Supplemental Report 29 [No. 1], 2024.)

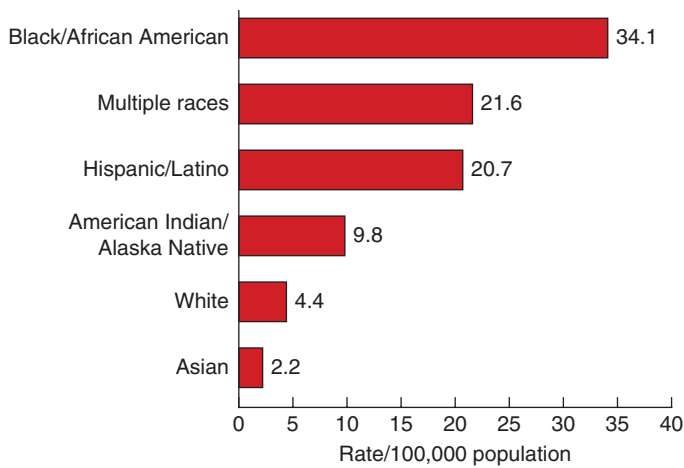


FIGURE 208-15 Estimated rate of new HIV infections in the United States among people age ≥ 13 years, by race/ethnicity (per 100,000 population), 2022. (From CDC: HIV Surveillance Supplemental Report 29 [No. 1], 2024.)

therapy for pregnant women and newborn infants to prevent infection. In 2022, 62 children were newly diagnosed with HIV infection in the United States, down from a peak of ~1750 in 1991.

The rate of HIV-related deaths in the United States rose steadily through the 1980s, peaked around 1994, and fell rapidly through 1997 (Fig. 208-16). Since then, the rate of death due to HIV disease continues to decline, decreasing from 2.6 deaths per 100,000 in 2010 to 1.3 deaths per 100,000 in 2022. This trend is likely due to several factors, including improved prophylaxis and treatment of opportunistic infections, growing experience among the health professions in caring for HIV-infected individuals, improved access to health care, and a decrease in new infections. However, the most influential factor clearly has been the increased use—and continued refinement—of antiretroviral therapy (ART), generally administered in a combination of two, three, or four agents. Despite much progress, HIV disease continues to remain among the leading causes of death among all persons age 25–44 years in the United States, ranking 13th in 2022. HIV disease as an underlying cause of death is particularly high among Black/African-American persons age 25–44 years, ranking as the 8th leading cause among Black males and the 12th leading cause among Black females in 2022.

PATHOPHYSIOLOGY AND PATHOGENESIS

The hallmark of HIV disease is a profound immunodeficiency resulting primarily from a progressive quantitative and qualitative deficiency of the subset of T lymphocytes referred to as *helper T cells* occurring

in a setting of aberrant immune activation creating a state of immunosuppression. The *helper* subset of T cells is defined phenotypically by the presence on its surface of the CD4 molecule (Chap. 360), which serves as the primary cellular receptor for HIV. A co-receptor also must be present together with CD4 for efficient binding, fusion, and entry of HIV-1 into its target cells (Figs. 208-3 and 208-4). HIV-1 uses two major co-receptors, CCR5 and CXCR4, for fusion and entry; these co-receptors are also the primary receptors for certain chemoattractant cytokines termed *chemokines* and belong to the seven-transmembrane-domain G protein-coupled family of receptors. Multiple mechanisms responsible for cellular depletion and/or immune dysfunction of CD4+ T cells have been demonstrated in vitro. These include direct infection and destruction of these cells by HIV, as well as indirect effects such as immune clearance of infected cells; cell death associated with aberrant immune activation and inflammation, including caspase 1-mediated pyroptosis prompted by tissue CD4+ T cells undergoing abortive/nonproductive HIV infection; and immune exhaustion due to persistent cellular activation with resulting cellular dysfunction. Patients with CD4+ T cell levels below certain thresholds are at high risk of developing a variety of opportunistic diseases, particularly the infections and neoplasms that are AIDS-defining illnesses. Some features of AIDS, such as Kaposi's sarcoma and certain neurologic abnormalities, cannot be explained completely by the immunodeficiency caused by HIV infection, since these complications may occur prior to the development of severe immunologic impairment.

The combination of viral pathogenic and immunopathogenic events that occur during the course of HIV disease from the moment of initial (primary) infection through the development of advanced-stage disease is complex and varied. It is important to appreciate that the pathogenic mechanisms of HIV disease are multifactorial and multiphasic and are different at different stages of the disease. Therefore, it is essential to consider the typical clinical course of an untreated individual with HIV to better appreciate these pathogenic events (Fig. 208-17).

■ EARLY EVENTS IN HIV INFECTION: PRIMARY INFECTION AND INITIAL DISSEMINATION OF VIRUS

Using rectal or vaginal mucosal transmission in nonhuman primates as a model, the earliest events (within hours) that occur following exposure of HIV to the mucosal surface determine whether an infection will be established or aborted as well as the subsequent course of events following infection. Although the mucosal barrier is relatively effective in limiting access of HIV to susceptible targets in the submucosal tissue, the virus can cross the barrier by transport on Langerhans cells, an epidermal type of DC, just beneath the surface or through microscopic rents in the mucosa. Significant disruptions in the mucosal barrier as seen in ulcerative genital disease facilitate viral entry and increase the

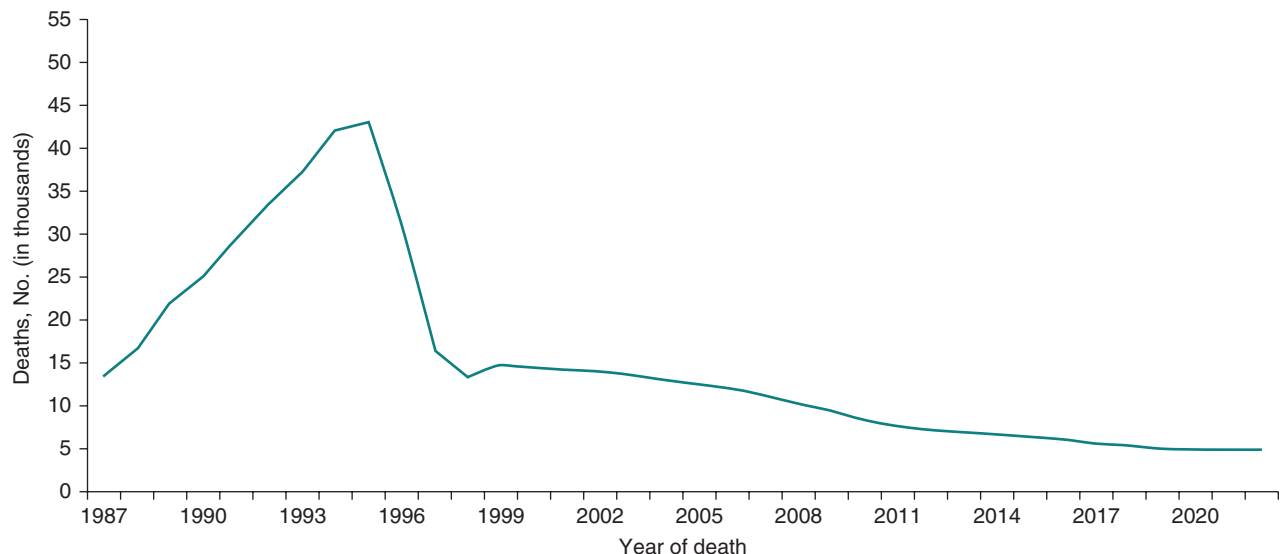


FIGURE 208-16 Trends in annual age-adjusted rates of death in the general population with HIV infection as the underlying cause, United States, 1987–2022. (From CDC.)

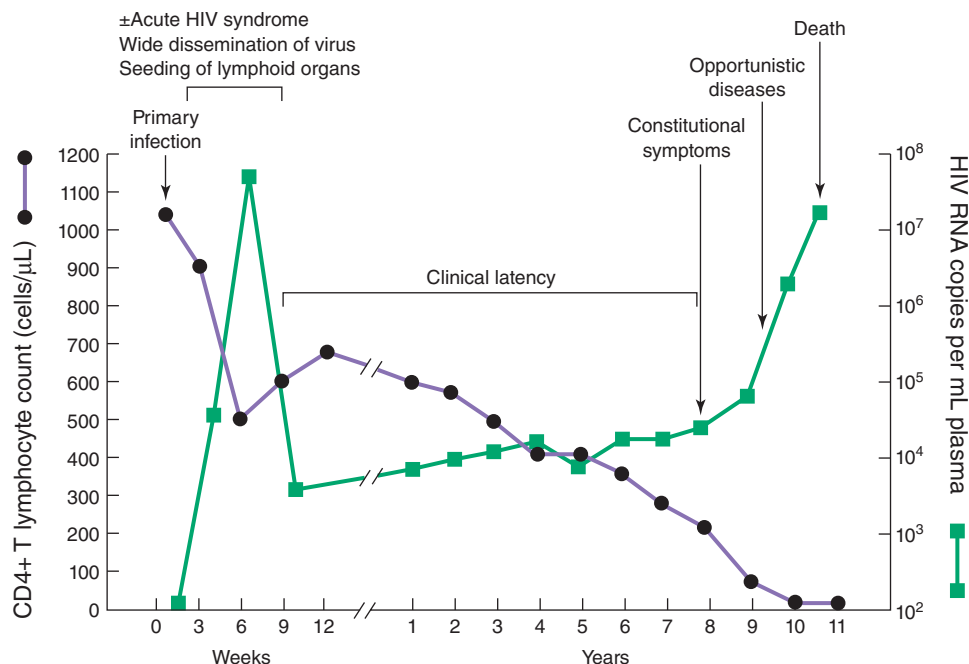


FIGURE 208-17 Typical course of an untreated HIV-infected individual. See text for detailed description. (From G Pantaleo, C Graziosi, AS Fauci: *The Immunopathogenesis of Human Immunodeficiency Virus Infection*. N Engl J Med 328:327, 1993. Copyright © 1993 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)

efficiency of infection. Viruses then seek susceptible targets, which are primarily CD4⁺ T cells that are spatially dispersed in the mucosa. This spatial dispersion of targets provides a significant obstacle to the establishment of infection. Such obstacles account for the low efficiency of sexual transmission of HIV (see “Sexual Transmission,” above). Both “partially” resting CD4⁺ T cells and activated CD4⁺ T cells serve as early amplifiers of infection. Resting CD4⁺ T cells are more abundant; however, activated CD4⁺ T cells support productive infection and thus generate larger amounts of virus. For infection to become established, the basic reproductive rate (R_0) must become equal to or greater than 1, i.e., each infected cell would infect at least one other cell. Once infection is established, the virus replicates in lymphoid cells in the mucosa, the submucosa, and to some extent the lymphoreticular tissues that drain the gut or genital tissues. For a variable period up to several days, the virus is typically not detected in the plasma. This period is

called the “eclipse” phase of infection. As more virus is produced within several days to weeks, it is disseminated, first to the draining lymph nodes and then to other lymphoid compartments where it has easy access to dense concentrations of CD4⁺ T cell targets, allowing for a burst of high-level plasma viremia that is readily detectable by currently available assays measuring viral RNA (Fig. 208-18). The gut-associated lymphoid tissue (GALT) is one target of HIV infection and a location where CD4⁺ T cells (primarily memory cells) are infected and depleted, both by direct viral effects and by activation-associated apoptosis. Once virus replication reaches this threshold and virus is widely disseminated, infection is firmly established throughout the lymphoid tissues of the body and persists for the life of the individual. It is important to point out that the efficiency of initial infection of susceptible cells may vary somewhat with the route of infection. Virus that enters directly into the bloodstream via infected blood or blood products (i.e., transfusions, use of contaminated needles for injection drugs, sharp-object injuries, maternal-

to-fetal transmission either intrapartum or perinatally, or sexual intercourse where there is enough trauma to cause bleeding) is likely first cleared from the circulation to the spleen and other lymphoid organs, where primary focal infections begin, followed by wider dissemination throughout other lymphoid tissues as described above.

It has been demonstrated that sexual transmission of HIV is the result of a single infectious event and that a viral genetic bottleneck exists for transmission with selective transmission of certain viruses. In this regard, certain characteristics of the HIV envelope glycoprotein have a major influence on transmission, at least in subtype A and C viruses. Transmitting viruses, often referred to as “founder viruses,” are typically a small fraction of the circulating viremia of the transmitting partner and are less-diverged viruses with signature sequences including shorter V1-V2 loop sequences and fewer predicted N-linked glycosylation sites relative to the major circulating variants. These

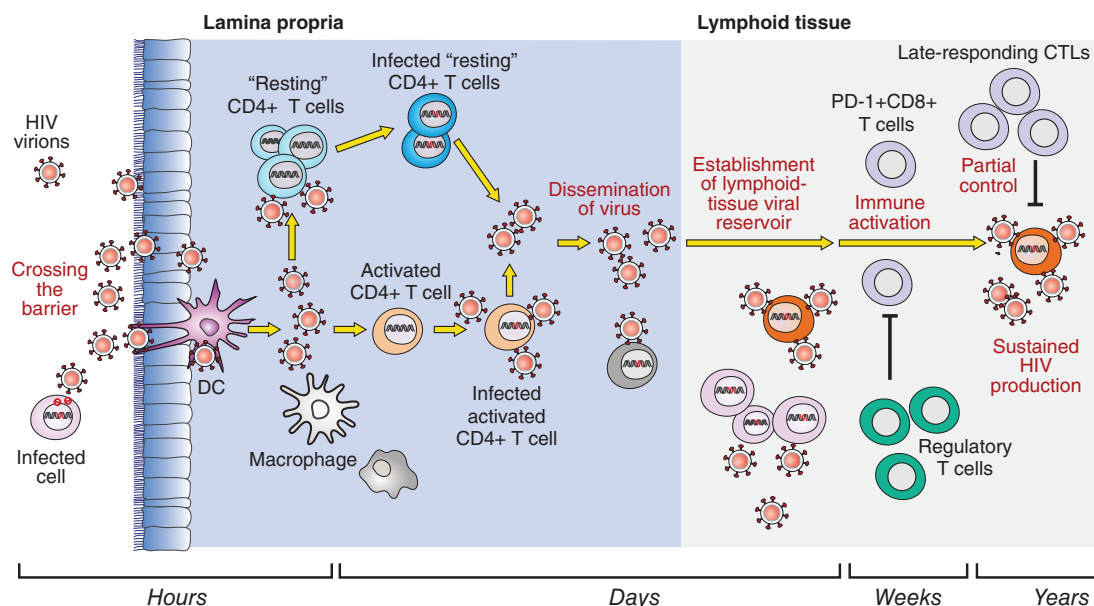


FIGURE 208-18 Summary of early events in HIV infection. See text for detailed description. CTLs, cytotoxic T lymphocytes; HIV, human immunodeficiency virus. (Adapted from AT Haase: *Nat Rev Immunol* 5:783, 2005.)

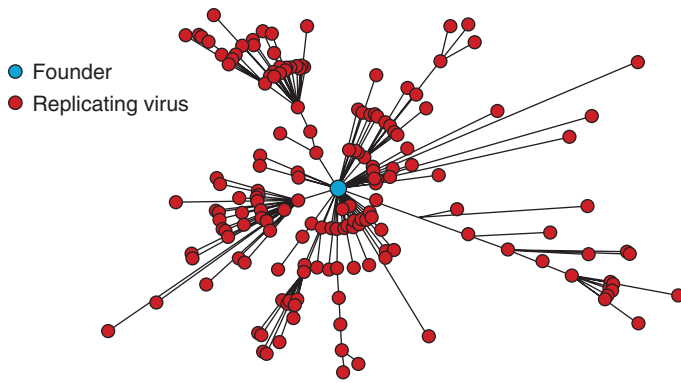


FIGURE 208-19 As HIV diverges from founder to chronically replicating virus, it accumulates N-linked glycosylation sites. See text for detailed description. (Adapted from CA Derdeyn et al: *Science* 303:2019, 2004; B Chohan et al: *J Virol* 79:6528, 2005; and BF Keele et al: *Proc Natl Acad Sci USA* 105:7552, 2008.)

viruses are almost exclusively R5 strains and are usually sensitive to neutralizing antibody. Once replication proceeds in the newly infected partner, the founder virus diverges and accumulates glycosylation sites, becoming progressively more resistant to neutralization (Fig. 208-19).

The acute burst of viremia and wide dissemination of virus in primary HIV infection may be associated with an *acute HIV syndrome*, which occurs to varying degrees in ~50% of individuals within 2 to 4 weeks of initial infection (see below). This syndrome is usually associated with high levels of plasma viremia reflected in millions of copies of HIV RNA per milliliter of plasma that can last for several weeks. Acute mononucleosis-like symptoms are well correlated with the presence of high levels of plasma viremia. Virtually all patients develop some degree of plasma viremia during primary infection, which contributes to virus dissemination throughout the lymphoid tissue, even though they may remain asymptomatic or not recall experiencing symptoms. The peak level of plasma viremia in primary HIV infection does not necessarily determine the rate of disease progression; however, the set point of the level of steady-state plasma viremia after ~1 year correlates with the rate of disease progression in the untreated patient and with immunologic and virologic aberrancies that may persist in the treated patient. The strikingly high levels of viremia observed in many patients during acute HIV infection is felt to be associated with a higher likelihood of transmission of the virus to others by a variety of routes including sexual transmission, shared needles and syringes, and mother-to-child transmission intrapartum, perinatally, or via breast milk.

■ ESTABLISHMENT OF CHRONIC INFECTION

Persistence of Virus Replication HIV infection is unique among human viral infections. Despite the robust cellular and humoral immune responses that are mounted following primary infection (see “Immune Response to HIV,” below), once infection has been established the virus succeeds in escaping complete immune-mediated clearance, paradoxically seems to thrive on immune activation, and is never eliminated completely from the body. Rather, a chronic infection develops and persists with varying degrees of continual virus replication in the untreated patient for a median of ~10 years before the patient becomes clinically ill (see “Advanced HIV Disease,” below). It is this establishment of a chronic, persistent infection that is the hallmark of HIV disease. Throughout the often-protracted course of chronic infection, virus replication can invariably be detected in untreated patients by widely available molecular assays that measure copies of virion-associated HIV RNA in plasma (copies per milliliter). Levels of virus vary greatly in most untreated patients, usually ranging from fewer than 50 to greater than a million copies of HIV RNA per milliliter of plasma. Studies using highly sensitive molecular techniques have demonstrated that even in treated patients in whom plasma viremia is suppressed to below detection limits of commercial assays (20–50 copies of HIV RNA per milliliter depending on assay kit manufacturer)

by ART, there is a continual low level of virion production in most infected patients. In other human viral infections (with some exceptions) if the host survives, the virus is completely cleared from the body and a state of immunity against subsequent infection develops. HIV infection very rarely kills the host during primary infection. Certain viruses, such as HSV (Chap. 197), are not completely cleared from the body after infection, but instead enter a latent state; in these cases, clinical latency is accompanied by microbiologic latency. This is not the case with HIV infection as described above. Chronicity associated with persistent virus replication can also be seen in certain cases of HBV and HCV infections (Chap. 352); however, in these infections the immune system is not a target of the virus.

Escape of HIV from Effective Immune System Control Inherent to the establishment of chronicity of HIV infection is the ability of the virus to evade adequate control and elimination by the cellular and humoral immune responses. There are several mechanisms whereby the virus accomplishes this evasion. Paramount among these is the establishment of a sustained level of replication associated with the generation of viral diversity via mutation and recombination. The selection of mutants that escape control by CD8⁺ cytolytic T lymphocytes (CTLs) is critical to the propagation and progression of HIV infection. The high rate of virus replication associated with inevitable mutations also contributes to the inability of antibody to neutralize and/or clear the autologous virus. Furthermore, for reasons that remain unclear, the humoral immune system does not readily produce classic neutralizing antibodies against the HIV envelope and does so only after years of persistent virus replication and after the infection is firmly established (see below). Extensive analyses of sequential HIV isolates and host responses have demonstrated that viral escape from B cell and CD8⁺ T cell responses occurs early after infection and allows the virus to continually evade effective immune responses. Virus-specific CD8⁺ CTLs expand greatly during primary HIV infection, and they likely represent the high-affinity responses that would be expected to be most efficient in eliminating virus-infected cells; however, with very rare exceptions, viral control is incomplete as viral replication persists at relatively high levels in the majority of individuals. In addition to viral escape from CTLs through high rates of mutation, it is thought that the initially strong immune response becomes qualitatively dysfunctional owing to the overwhelming immune activation associated with persistent viral replication, leading to immune “exhaustion” that affects both arms of adaptive immunity. Several studies have indicated that exhaustion of HIV-specific CD8⁺ T cells during prolonged immune activation is associated with upregulation of several inhibitory receptors, such as the programmed death (PD) 1 molecule (of the B7-CD28 family of molecules), T cell immunoreceptor with Ig and ITIM domains (TIGIT), T cell immunoglobulin and mucin domain-containing molecule 3 (Tim-3), and lymphocyte activating gene 3 (Lag-3), collectively referred to as *immune-checkpoint receptors*. Upregulation of these surface proteins restricts polyreactivity and proliferative capacity, functional attributes of CD8⁺ T cells that are essential for effective killing of pathogens. Another mechanism contributing to the evasion by HIV of immune system control is the downregulation of HLA class I molecules on the surface of HIV-infected cells by the viral proteins Nef, Tat, and Vpu, resulting in the lack of ability of CD8⁺ CTLs to recognize and kill infected target cells. Although this downregulation of HLA class I molecules would seem to favor elimination of HIV-infected cells by natural killer (NK) cells, this latter mechanism does not remove HIV-infected cells effectively (see below). Another potential means of escape of HIV-infected cells from elimination by CD8⁺ CTLs is the sequestration of infected cells in immunologically privileged sites such as the central nervous system (CNS), as well as the low frequency of virus-specific CD8⁺ CTLs in areas of lymphoid tissues, namely germinal centers, where HIV actively replicates.

The principal targets of neutralizing antibodies against HIV are the envelope proteins gp120 and gp41. HIV employs at least three mechanisms to evade neutralizing antibody responses: hypervariability in the primary sequence of the envelope, extensive glycosylation of the envelope, and conformational masking of neutralizing epitopes.

Several studies that have followed the evolution of the humoral immune response to HIV from the earliest points after primary infection indicate that the virus continually mutates to escape the emerging antibody response such that the sequential antibodies that are induced do not neutralize the currently autologous virus. *Broadly neutralizing antibodies* capable of neutralizing a wide range of primary HIV isolates in vitro occur in only about 20% of people with HIV, and, when they do occur, 2–3 years of infection with continual virus replication are generally required to drive the affinity maturation of the antibodies. Unfortunately, by the time these broadly neutralizing antibodies are formed, they are ineffective in containing the virus currently replicating in the patient. Persistent viremia also results in exhaustion of B cells like the exhaustion reported for CD8+ T cells, adding to the defects in the humoral response to HIV.

CD4+ T cell help is essential for the integrity of both humoral and cell-mediated antigen-specific immune responses. HIV preferentially infects activated CD4+ T cells including HIV-specific CD4+ T cells, and so this loss of viral-specific helper T cell responses has profoundly negative consequences for the immunologic control of HIV replication. Furthermore, this loss occurs early in the course of infection, and animal studies indicate that 40–70% of all memory CD4+ T cells in the GALT are eliminated during acute infection. During chronic HIV viremia, CD4+ T cells also exhibit evidence of exhaustion, reflected in upregulation of the cytotoxic T lymphocyte-associated antigen 4 (CTLA-4), also a member of the B7-CD28 family.

Finally, the escape of HIV from immune-mediated elimination during primary infection allows the formation of a pool of latently infected CD4+ T cells, referred to as the *viral reservoir*, which may not be recognized or completely eliminated by virus-specific CTLs or by ART (see below). Thus, despite a potent immune response and the marked downregulation of virus replication following primary HIV infection, HIV succeeds in establishing a state of chronic infection with a variable degree of persistent virus replication. During this period most patients make the clinical transition from acute primary infection to variable periods of clinical latency or smoldering disease activity (see below).

The HIV Reservoir: Obstacles to the Eradication of Virus

A pool of latently infected, resting CD4+ T cells that serves as at least one component of the persistent reservoir of virus exists in virtually all people with HIV, including those who are receiving ART. Such cells carry an integrated form of HIV DNA in the genome of the host and can remain in this state until an activation signal drives the expression of HIV transcripts. Only a small fraction of the latently infected cells in the viral reservoir contains replication-competent virus, with the overwhelming majority of cells containing defective proviruses incapable of a full replication cycle. However, upon activation of the reservoir variable degrees of sustained virus replication invariably occur. This form of latency is to be distinguished from preintegration latency, in which HIV enters a resting CD4+ T cell and, in the absence of an activation signal, reverse transcription of the HIV genome occurs to a certain extent but the resulting proviral DNA fails to integrate into the host genome. This period of preintegration latency may last hours to days, and if no activation signal is delivered to the cell, the proviral DNA loses its capacity to initiate a productive infection. If these cells do become activated prior to decay of the preintegration complex, reverse transcription proceeds to completion and the virus continues along its replication cycle (see above and Fig. 208-20). The pool of cells that are in the postintegration state of latency is established early during primary HIV infection. Despite the suppression of plasma viremia to <20–50 copies per milliliter by potent regimens of ART administered over several years, this pool of latently infected cells persists and can give rise to replication-competent virus upon cellular activation ex vivo. Modeling studies built on projections of decay curves have estimated that in such a setting of prolonged viral suppression, it would require many years to the entire life of the host for the pool of latently infected cells to be eliminated. This has not been documented to occur spontaneously in any patients very likely because the latent viral reservoir is long-lived and is continually replenished by the low levels of persistent virus replication that may remain below the limits

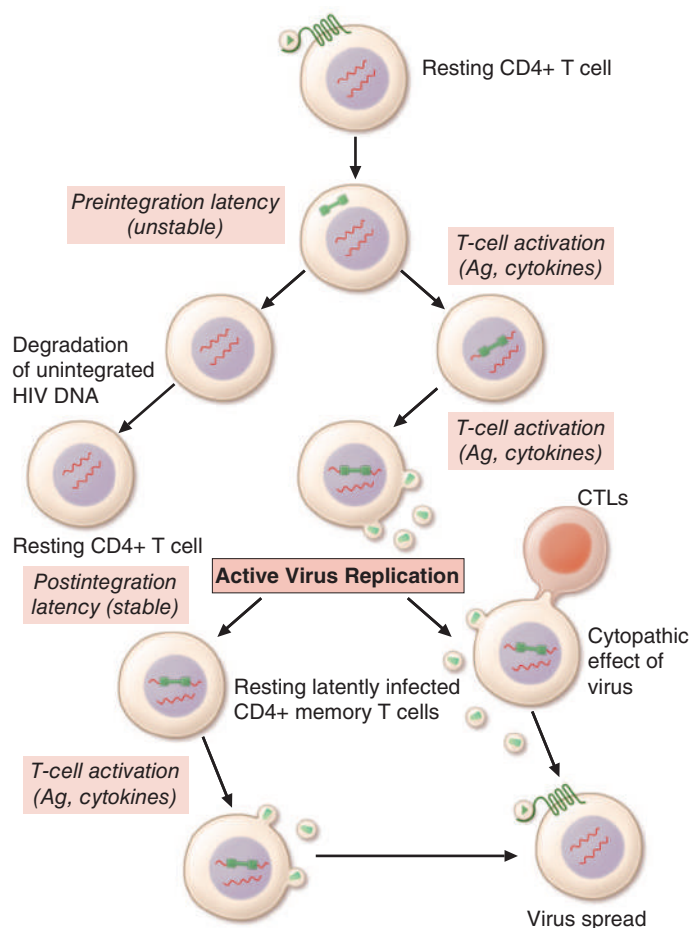


FIGURE 208-20 Generation of latently infected, resting CD4+ T cells in people with HIV. See text for details. Ag, antigen; CTLs, cytotoxic T lymphocytes. (Courtesy of TW Chun.)

of detection of current assays (see below) as well as by the expansion by proliferation of the pool of latently infected cells (Fig. 208-20), even in patients who for the most part are treated successfully. Reservoirs of HIV-infected cells, latent or otherwise, can exist in multiple compartments including the lymphoid tissue, peripheral blood, and CNS (likely in cells of the monocyte/macrophage lineage) as well as in other unidentified locations. Over the past several years attempts have been made to eliminate HIV in the latent viral reservoir using agents that activate resting CD4+ T cells and/or reinstate viral expression without systemic activation during the course of ART; however, such attempts, referred to as “shock and kill,” have thus far been unsuccessful. This persistent reservoir of infected cells remains a major obstacle to the goal of eradication of virus from infected individuals and hence a classic “cure,” despite the favorable clinical outcomes that have resulted from ART. Consequently, intense efforts are being directed toward investigating the feasibility of achieving ART-free HIV remission through passive transfer of long-acting broadly neutralizing antibodies and therapeutic agents that could enhance the host immune responses against the virus.

Viral Dynamics The dynamics of viral production and turnover have been quantified using mathematical modeling in the setting of the administration of reverse transcriptase and protease inhibitors to people with HIV in clinical studies. Treatment with these drugs resulted in a precipitous decline in the level of plasma viremia, which typically fell by well over 90% within 2 weeks. It was determined on the basis of modeling the kinetics of viral decline and the emergence of resistant mutants during therapy that 93–99% of the circulating virus originated from recently infected, rapidly turning over CD4+ T cells and that ~1–7% of circulating virus originated from longer-lived cells, likely monocytes/macrophages. A negligible amount of circulating virus originated from the pool of latently infected cells (Fig. 208-21).

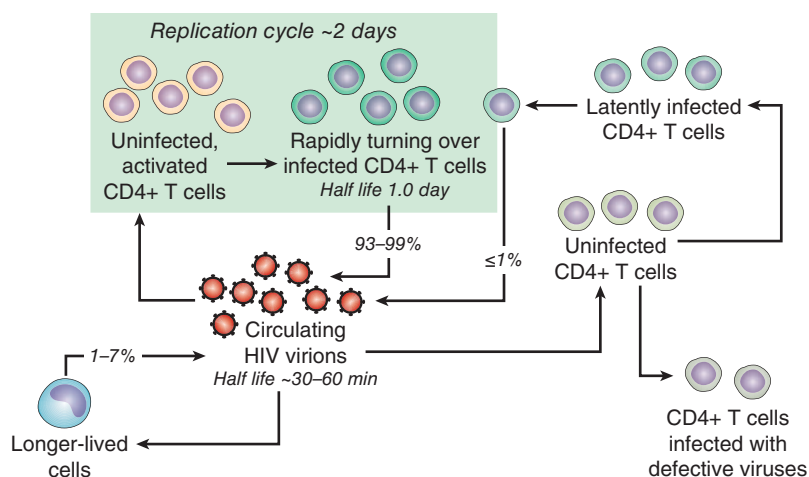


FIGURE 208-21 Dynamics of HIV infection in vivo. See text for detailed description. (Adapted from Perelson AS et al: HIV-1 dynamics in vivo: Virion clearance rate, infected cell life-span, and viral generation time. *Science* 271:1582, 1996.)

It was also determined that the half-life of a circulating virion was ~30–60 min and that of productively infected cells was 1 day. Given the relatively steady level of plasma viremia and of infected cells, it appears that extremely large amounts of virus ($\sim 10^{10}$ – 10^{11} virions) are produced and cleared from the circulation each day. In addition, data suggest that the minimal duration of the HIV-1 replication cycle in vivo is ~2 days. Other studies have demonstrated that the decrease in plasma viremia that results from treatment with ART correlates closely with a decrease in virus replication in lymph nodes, further confirming that lymphoid tissue is the main site of HIV replication and the main source of plasma viremia.

The level of steady-state viremia, called the viral *set point*, at ~1 year following acquisition of HIV infection has important prognostic implications for the progression of HIV disease in the untreated patient. It has been demonstrated that, as a group, untreated people with HIV who have a low set point at 6 months to 1 year following infection progress to AIDS much more slowly than do individuals whose set point is very high at that time (Fig. 208-22).

Clinical Latency versus Microbiologic Latency With the exception of certain long-term nonprogressors and “elite controllers” of HIV replication, the level of CD4+ T cells in the blood inevitably decreases progressively in viremic people with HIV in the absence of ART. The decline in CD4+ T cells may be gradual or abrupt, the latter usually reflecting a significant spike in the level of plasma viremia. Most patients are relatively asymptomatic while this progressive

decline is taking place (see below) and are often described as being in a state of *clinical latency*. However, this term is misleading; it does not mean disease latency, since progression, although slow in many cases and often without symptoms, is generally relentless as evidenced by readily detectable plasma viremia, during this period. Furthermore, clinical latency should not be confused with microbiologic latency since varying levels of virus replication inevitably occur during this period of clinical latency. Even in those rare patients, such as elite controllers, who have <50 copies of HIV RNA per milliliter in the absence of therapy, there is virtually always some degree of low-level ongoing virus replication.

■ ADVANCED HIV DISEASE

In untreated patients or in patients in whom therapy has not adequately controlled virus replication, after a variable period, usually measured in years, the CD4+ T cell count falls below a critical level (<200/ μ L) and the patient becomes highly susceptible to opportunistic disease (Fig. 208-17). For this reason, the CDC case definition of stage 3 (AIDS) includes all people with HIV >5 years of age with CD4+ T cell counts below this level (Table 208-2). Patients may experience constitutional signs and symptoms or may develop an opportunistic disease abruptly without any prior symptoms. The depletion of CD4+ T cells continues to be progressive and unrelenting in this phase. It is not uncommon for CD4+ T cell counts in the untreated patient to drop to as low as 10/ μ L or even to zero. In countries where ART as well as prophylaxis and treatment for opportunistic infections are readily accessible, survival is increased dramatically even in those patients with advanced HIV disease. In contrast, untreated patients who progress to this severest form of immunodeficiency usually succumb to opportunistic infections or neoplasms (see below).

■ LONG-TERM SURVIVORS, LONG-TERM NONPROGRESSORS, AND ELITE CONTROLLERS

It is important to distinguish between the terms *long-term survivor* and *long-term nonprogressor*. Long-term nonprogressors are by definition long-term survivors; however, the reverse is not always true. Predictions from one study that antedated the availability of effective ART estimated that ~13% of homosexual/bisexual men who were infected at an early age may remain free of clinical AIDS for >20 years. Many of these individuals may have gradually progressed in their degree of immune deficiency; however, they certainly survived for a considerable period. With the advent of effective ART, the survival of people with HIV has dramatically increased. Early in the AIDS pandemic, prior to the availability of antiretroviral therapy, if a patient presented with

a life-threatening opportunistic infection, the median survival was 26 weeks from the time of presentation. Currently, a 20-year-old individual with HIV who is appropriately treated with ART can expect to live at least 50 years according to mathematical model projections. In the face of ART, long-term survival is now commonplace. Definitions of long-term nonprogressors have varied considerably over the years, and so such individuals constitute a heterogeneous group. Long-term nonprogressors were first described in the 1990s. Originally, individuals were considered to be long-term nonprogressors if they had been infected with HIV for a long period (≥ 10 years), their CD4+ T cell counts were in the normal range, their plasma viremia remained relatively low (undetectable to several thousand copies of HIV RNA/mL plasma), and they remained clinically stable over years without receiving ART. Approximately 5–15% of people

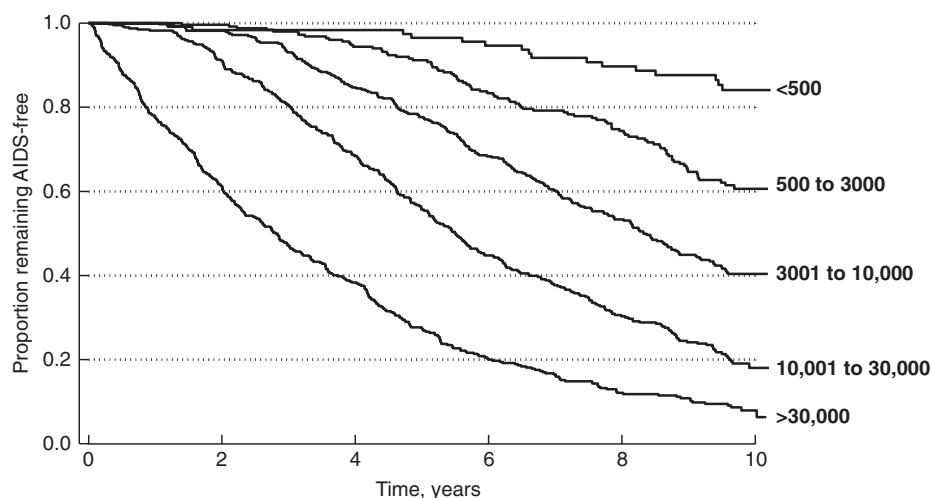


FIGURE 208-22 Relationship between levels of virus and rates of disease progression. Kaplan-Meier curves showing proportion of 1604 untreated patients remaining AIDS-free over 10 years, stratified by baseline HIV-1 RNA categories (copies per milliliter). (From Multicenter AIDS Cohort Study; JW Mellors, A Muñoz, JV Giorgi, JB Margolick, CJ Tassoni, P Gupta, LA Kingsley, JA Todd, AJ Saah, R Detels, JP Phair, CR Rinaldo, Jr.)

with HIV fell into this broader nonprogressor category. However, this group was rather heterogeneous and over time a significant proportion of these individuals progressed and ultimately required antiretroviral therapy. From this broader group, a much smaller subgroup of “elite” controllers was identified, and they constituted a fraction of 1% of people with HIV. These elite controllers, by definition, have extremely low levels of plasma viremia that is often undetectable by standard assays and normal CD4+ T cell counts. It is noteworthy that their HIV-specific immune response, especially HIV-specific CD8+ CTLs that can clear infected CD4+ T cells, is robust and distinctly superior to those of progressors with HIV infection. In this group of elite controllers, certain HLA class I haplotypes are overrepresented, particularly HLA-B57-01 and HLA-B27-05. Outside of the subgroup of elite controllers, multiple other genetic factors have been shown to be involved to a greater or lesser degree in the control of virus replication and thus in the rate of HIV disease progression (see “Genetic Factors in HIV-1 and AIDS Pathogenesis,” below).

■ LYMPHOID ORGANS AND HIV PATHOGENESIS

Regardless of the portal of entry of HIV, lymphoid tissues are the major anatomic sites for the establishment and propagation of HIV infection. Despite the use of measurements of plasma viremia to determine the level of disease activity, virus replication occurs mainly in lymphoid tissue and not in blood; indeed, the level of plasma viremia directly reflects virus production in lymphoid tissue.

Some patients experience progressive generalized lymphadenopathy early in the course of the infection; others experience varying degrees of transient lymphadenopathy. Lymphadenopathy reflects the cellular activation and immune response to the virus in the lymphoid tissue, which is generally characterized by follicular or germinal center hyperplasia. Lymphoid tissue involvement is a common denominator of virtually all patients with HIV infection, even those without easily detectable lymphadenopathy.

Examinations of lymph tissue and peripheral blood in patients and monkeys during various stages of HIV and simian immunodeficiency virus (SIV) infection, respectively, have led to substantial insight into the pathogenesis of HIV disease. In most of the original human studies, peripheral lymph nodes were the predominant sources for analyses into changes in lymphoid tissues associated with HIV and SIV infection, whereas more recent studies have expanded to include the GALT, where the earliest burst of virus replication occurs associated with marked depletion of CD4+ T cells. A variety of techniques, including sensitive molecular and imaging approaches to visualize virus and cells in location or suspension, have been employed to describe events associated with HIV disease. During acute HIV infection resulting from mucosal transmission, virus replication progressively amplifies from scattered lymphoid cells in the lamina propria of the gut to draining lymph nodes, leading to high levels of plasma viremia. The GALT plays a major role in the amplification of virus replication, and virus is disseminated from replication in the GALT to peripheral lymphoid tissues. A profound degree of cellular activation occurs within lymphoid tissues (see below) and is reflected in follicular or germinal center hyperplasia. At this time copious amounts of extracellular virions (both infectious and defective) are trapped on the processes of the follicular dendritic cells (FDCs) that form the stromal cell network in the light zones of lymph node germinal centers. Virions that have bound complement components on their surfaces attach to the surface of FDCs via interactions with complement receptors and likely via Fc receptors that bind to antibodies that are attached to the virions. The use of in situ hybridization techniques, including those that allow detection of viral RNA in the context of tissue architecture, has revealed that HIV is primarily expressed in CD4+ T cells of the paracortical area and, to a lesser extent, in specialized CD4+ T cells (see below) in light zones of germinal centers (Fig. 208-23). The persistence of trapped virus on the surface of FDC likely reflects both a long-lived viral reservoir and virus that is replaced by continual expression in nearby CD4+ T cells. The trapped virus, either as whole virion or shed envelope, also serves as a continual activator of CD4+ T cells, thus driving further virus replication.

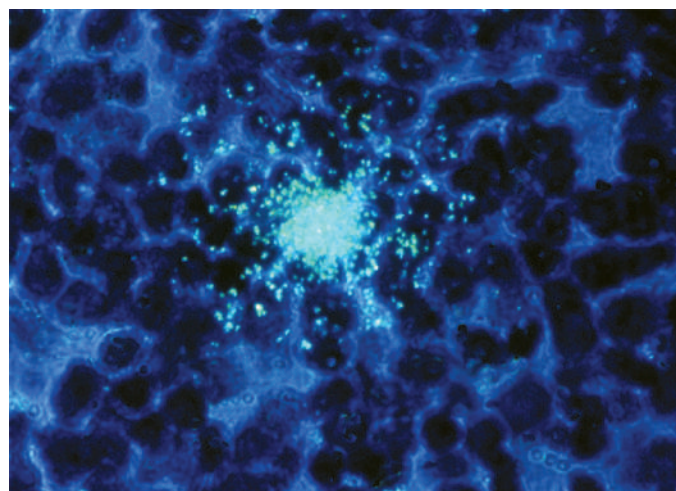


FIGURE 208-23 HIV in the lymph node of an HIV-infected individual. An individual cell infected with HIV shown expressing HIV RNA by in situ hybridization using a radiolabeled molecular probe. Original $\times 500$. (Reproduced with permission from G Pantaleo et al: HIV infection is active and progressive in lymphoid tissue during the clinically latent stage of disease. *Nature* 362:355, 1993.)

During the early stages of HIV disease, the architecture of lymphoid tissues is generally preserved and may even be hyperplastic owing to an increased presence of B cells and specialized CD4+ T cells called follicular helper CD4+ T cells (TF_H) in prominent germinal centers. Extracellular virions can be seen by electron microscopy attached to FDC processes. The trapping of antigen is a physiologically normal function for the FDCs, which present antigen to B cells and secrete factors such as CXCL13 that retain B and TF_H cells in the light zones of germinal centers. These FDC functions, along with stimulatory factors produced by TF_H cells, contribute to the generation of B cell memory. However, in the case of HIV, persistent cellular activation, resulting in a shift to secretion of proinflammatory cytokines such as interleukin (IL) 1 β , tumor necrosis factor (TNF) α , IFN- γ , and IL-6, can induce viral replication (see below) and diminish the effectiveness of the immune response against the virus. In addition, the CD4+ TF_H cells that are recruited into the germinal center to provide help to B cells in the generation of an HIV-specific immune response are highly susceptible to infection and may be an important component of the HIV reservoir. Thus, in HIV infection, a normal physiologic function of the immune system, i.e., the generation of an HIV-specific immune response that contributes to the clearance of virus, can also have deleterious consequences.

As HIV disease progresses, the architecture of lymphoid tissues begins to disrupt. Confocal microscopy reveals destruction of the fibroblastic reticular cell (FRC) and FDC networks in the T cell zone and B cell follicles/germinal centers, respectively. The mechanisms of destruction are not completely understood, but they are thought to be associated with collagen deposition causing fibrosis and a shift in the expression of certain cytokines, namely decreases in IL-7 and lymphotoxin α , which are critical to the maintenance of lymphoid tissues and their lymphocyte constituents, and increased levels of transforming growth factor (TGF) β . As the disease progresses to an advanced stage, there is complete disruption of the architecture of the lymphoid tissues, accompanied by dissolution of the FRC and FDC networks. At this point, the lymph nodes are “burnt out.” This destruction of lymphoid tissue compounds the immunodeficiency of HIV disease and contributes both to the inability to control HIV replication and to the inability to mount adequate immune responses against opportunistic pathogens and vaccination. The events from primary infection to the ultimate destruction of the immune system are illustrated in Fig. 208-24. In nonhuman primate studies and some human studies that have examined GALT following SIV or HIV infection, the basal level of cellular activation combined with virus-mediated activation leads to the rapid infection and elimination of an estimated 50–90% of CD4+ T cells in the gut.

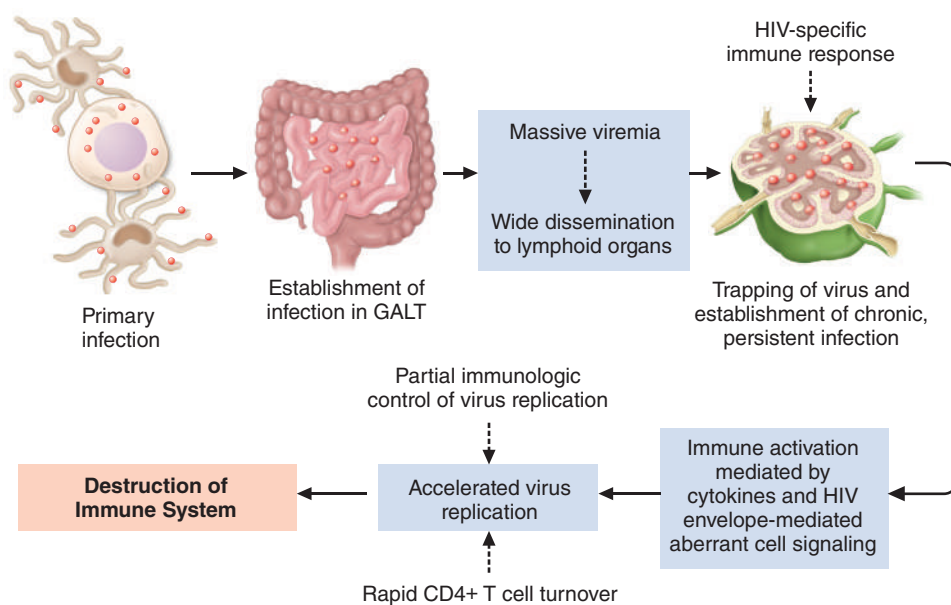


FIGURE 208-24 Events that transpire from primary HIV infection through the establishment of chronic persistent infection to the ultimate destruction of the immune system. See text for details. GALT, gut-associated lymphoid tissue.

THE ROLE OF IMMUNE ACTIVATION AND INFLAMMATION IN HIV PATHOGENESIS

Activation of the immune system and variable degrees of inflammation are essential components of any appropriate immune response to a foreign antigen. However, immune activation and inflammation, which are aberrant in certain individuals with HIV, play a critical role in the pathogenesis of HIV disease as well as other chronic conditions associated with HIV infection. Immune activation and inflammation in individuals with HIV contribute substantially to (1) the replication of HIV, (2) the induction of immune dysfunction, and (3) the increased incidence of chronic conditions such as premature cardiovascular disease (Table 208-4).

INDUCTION OF HIV REPLICATION BY ABERRANT IMMUNE ACTIVATION

The immune system is normally in a state of homeostasis, awaiting perturbation by foreign antigenic stimuli. Once the immune response deals with and clears the antigen, the system returns to relative quiescence (Chap. 360). This is generally not the case in HIV infection where, in the untreated patient, virus replication is invariably persistent with very few exceptions and as a result immune activation is persistent. HIV replicates most efficiently in activated CD4⁺ T cells; in HIV infection, chronic activation provides the cell substrates necessary for persistent virus replication throughout the course of HIV disease, particularly in the untreated patient. Even in certain patients receiving ART whose levels of plasma viremia are suppressed to <20 copies per milliliter, there are low but detectable degrees of virus replication that drives low-level persistent immune activation. In addition, immune activation may result from RNA transcription of the integrated DNA of defective proviruses. From a virologic standpoint, although quiescent CD4⁺ T cells can be infected with HIV, albeit inefficiently, reverse transcription,

integration, and virus spread are much more efficient in activated cells. Furthermore, cellular activation induces expression of virus in cells latently infected with HIV. In essence, immune activation and inflammation provide the engine that drives HIV replication. In addition to endogenous factors such as cytokines, multiple exogenous factors such as other microbes that induce cellular activation can enhance HIV replication and thus may play a role in HIV pathogenesis.

Co-infection with a range of viruses, such as HSV types 1 and 2, cytomegalovirus (CMV), human herpesvirus (HHV) 6, Epstein-Barr virus (EBV), HBV, HCV, adenovirus, and HTLV-1 have been shown to upregulate HIV expression. In addition, infestation with nematodes has been shown to be associated with a heightened state of immune activation that facilitates HIV replication; in certain studies, deworming of the infected host has resulted in a decrease in plasma viremia. Two diseases of great global health significance, malaria and tuberculosis

(TB), have been shown to increase HIV viral load in dually infected individuals. Globally, *Mycobacterium tuberculosis* is the most common opportunistic infection in people with HIV (Chap. 183). In addition to the fact that individuals with HIV are more likely to develop active TB after exposure and to reactivate latent TB, it has been demonstrated that active TB can accelerate the course of HIV infection. It has also been shown that levels of plasma viremia are greatly elevated in individuals with HIV who have active TB and who are not receiving ART, compared with pre-TB levels and levels of viremia after successful treatment of the active TB. The situation is similar in the interaction between HIV and malaria parasites (Chap. 231). Acute infection with *Plasmodium falciparum* of individuals with HIV increases viral load, and the increased viral load is reversed by effective treatment of malaria.

MICROBIAL TRANSLOCATION AND PERSISTENT IMMUNE ACTIVATION

One proposed mechanism of persistent immune activation involves the disruption of the mucosal barrier in the gut due to HIV replication in submucosal lymphoid tissue. As a result of this disruption, there is an increase in the products of bacteria, particularly lipopolysaccharide (LPS), that translocate from the bowel lumen through the damaged mucosa to the circulation, leading to persistent systemic immune activation and inflammation. This effect can persist even after the HIV viral load is brought to <20 copies/mL by ART. Other related factors that are thought to contribute to the pathogenesis of HIV include depletion in the GALT of IL-17-producing T cells, which are responsible for defense against extracellular bacteria and fungi, as well as alterations in gut microbiota and the metabolic pathways involved.

PERSISTENT IMMUNE ACTIVATION AND INFLAMMATION INDUCE IMMUNE DYSFUNCTION

The immune activated state in HIV infection is reflected by hyperactivation of B cells leading to hypergammaglobulinemia; increased lymphocyte turnover; activation of monocytes; expression of activation markers and immune checkpoint receptors on CD4⁺ and CD8⁺ T cells; increased activation-associated cellular apoptosis and pyroptosis; lymph node hyperplasia, particularly during the chronic phase prior to disease progression; increased secretion of proinflammatory cytokines, particularly IL-6 and type I interferons; elevated levels of high-sensitivity C-reactive protein (CRP), CXCL10, D-dimer, neopterin, β_2 -microglobulin, soluble (s) CD14, sTNFR, sCD27, sCD163, and sCD40L; and autoimmune phenomena (see “Autoimmune Phenomena,” below). Even in the absence of direct infection of a target cell, HIV envelope proteins can interact with cellular receptors (CD4 molecules and chemokine

TABLE 208-4 Conditions Associated with Persistent Immune Activation and Inflammation in Patients with HIV Infection

Accelerated aging syndrome
Bone fragility
Cancers
Cardiovascular disease
Diabetes
Kidney disease
Liver disease
Neurocognitive dysfunction

receptors) to deliver potent activation signals resulting in calcium flux, the phosphorylation of certain proteins involved in signal transduction, co-localization of cytoplasmic proteins including those involved in cell trafficking, immune dysfunction, and, under certain circumstances, apoptosis and pyroptosis. From an immunologic standpoint, chronic exposure of the immune system to a particular antigen over an extended period may ultimately lead to an inability to sustain an adequate immune response to the antigen in question. In many chronic viral infections, including HIV infection, persistent viremia is associated with “functional exhaustion” of virus-specific T cells, decreasing their capacity to proliferate and perform effector functions. It has been demonstrated that this phenomenon of immune exhaustion may be mediated, at least in part, by the upregulation of inhibitory receptors on HIV-specific T cells, such as PD-1, LAG-3, and Tim-3 that are shared by both CD4+ and CD8+ T cells, as well as CTLA-4 on CD4+ and 2B4 and CD160 on CD8+ T cells. Furthermore, the ability of the immune system to respond to a broad spectrum of non-HIV antigens may be compromised if immunocompetent bystander cells are maintained in a state of chronic activation.

The deleterious effects of chronic immune activation on the progression of HIV disease are well established. As in most conditions of persistent antigen exposure, the host maintains activation of antigen (HIV)-specific responses but must also prevent excessive activation and potential immune-mediated damage to tissues. Certain studies suggest that normal immunoregulatory mechanisms that act to keep hyperimmune activation in check, particularly CD4+, FoxP3+, and CD25+ regulatory T cells (T-regs), may be dysfunctional or depleted in the context of advanced HIV disease. One possibility is a role for the inhibitory receptor LAG-3 (see below), which is overexpressed on exhausted T cells and shown to inhibit the proliferation of T-regs.

Apoptosis Apoptosis is a form of programmed cell death that is a normal mechanism for the elimination of effete cells in organogenesis as well as in the cellular proliferation that occurs during a normal immune response (**Chap. 360**). Apoptosis can occur by intrinsic or extrinsic pathways, the latter of which is largely dependent on cellular activation, and in this regard the aberrant cellular activation associated with HIV disease is correlated with a heightened state of apoptosis. HIV can trigger activation-induced cell death through the upregulation of the death receptors, such as Fas/CD95, TNFR1, or TNF-related apoptosis-inducing ligand (TRAIL) receptors 1 and 2. Their corresponding ligands FasL, TNF, and TRAIL also are upregulated in HIV disease. HIV-induced stress and alterations in homeostasis also can trigger intrinsic apoptosis due to the downregulation of antiapoptotic proteins such as Bcl-2. Other mechanisms of HIV-induced cell death have been described, including autophagy, necrosis, necroptosis, and pyroptosis. The phenomenon of pyroptosis, an inflammatory form of cell death involving the upregulation of the proinflammatory enzyme caspase 1 and release of the proinflammatory cytokines IL-1 β and IL-18, has been linked to a bystander effect of HIV replication on depletion of CD4+ T cells (see “Pathophysiology and Pathogenesis,” above). The process of pyroptosis generates multimeric complexes called inflammasomes, which can also be activated by LPS. Certain viral gene products have been associated with enhanced susceptibility to apoptosis; these include Env, Tat, and Vpr. In contrast, Nef has been shown to possess antiapoptotic properties. The intensity of apoptosis correlates with the general state of activation of the immune system and not with the stage of disease or with viral burden. Multiple studies, including those examining lymphoid tissue, have demonstrated that the rate of apoptosis is elevated in HIV infection and that apoptosis is seen in “bystander” cells such as CD8+ T cells and B cells as well as in uninfected CD4+ T cells. It is likely that this bystander apoptosis of immunocompetent cells related to immune activation contributes to the general immunologic abnormalities in HIV disease.

MEDICAL CONDITIONS ASSOCIATED WITH PERSISTENT IMMUNE ACTIVATION AND INFLAMMATION IN HIV DISEASE It has become clear, as the survival of people with HIV has increased, that multiple, previously unrecognized medical complications are associated with HIV disease—and that these complications relate to chronic immune

activation and inflammation (Table 208-4). These complications can appear even after patients have experienced years of ART-induced adequate control of viral replication (plasma viremia <50 copies per milliliter of plasma) for several years. Other chronic conditions that have been reported include bone fragility, certain cancers, diabetes, kidney and liver disease, and neurocognitive dysfunction, thus presenting an overall picture of accelerated aging.

Autoimmune Phenomena Autoimmune phenomena are commonly observed in people with HIV, and they reflect, at least in part, chronic immune activation and the dysregulation of B and T cells. Although these phenomena usually occur in the absence of autoimmune disease, a wide spectrum of clinical manifestations that may be associated with autoimmunity have been described (see “Immunologic and Rheumatologic Diseases,” below). Autoimmune phenomena include antibodies against autoantigens expressed on intact lymphocytes and other cells, or against proteins released from dying cells. Antiplatelet and antierythrocyte antibodies have some clinical relevance in that they may contribute to thrombocytopenia and autoimmune hemolytic anemia, respectively, in HIV disease (see below). Antibodies to nuclear and cytoplasmic components of cells have been reported, as have antibodies to cardiolipin and phospholipids, as well as surface receptors, including CD4, and serum proteins. However, these manifestations are relatively low in the era of ART. Molecular mimicry, either from opportunistic pathogens or from HIV itself, also is a trigger or cofactor in autoimmunity. Antibodies against the HIV envelope proteins, especially gp41, often cross-react with host proteins; the best-known examples are antibodies directed against the membrane-proximal external region (MPER) of gp41 that also react with phospholipids and cardiolipin. The phenomenon of polyreactive HIV-specific antibodies may be beneficial to the host (see “Immune Response to HIV,” below).

The increased occurrence and/or exacerbation of certain autoimmune diseases have been reported in HIV infection; these diseases include psoriasis, idiopathic thrombocytopenic purpura, autoimmune hemolytic anemia, Graves’ disease, antiphospholipid syndrome, and primary biliary cirrhosis. Most of these manifestations were described prior to the advent of ART and have decreased in frequency since its widespread use. However, with increasing availability of ART, an *immune reconstitution inflammatory syndrome* (IRIS) has been increasingly observed in infected individuals, particularly those with low CD4+ T cell counts (see below). IRIS is an autoimmune-like phenomenon characterized by a paradoxical deterioration of clinical condition, which is usually compartmentalized to a particular organ system in individuals in whom ART has recently been initiated. It is associated with a decrease in viral load and at least partial recovery of immune competence, which is usually associated with increases in CD4+ T cell counts. The immunopathogenesis of this syndrome is felt to be related to an increase in immune response against the presence of residual antigens that are usually microbial and is most commonly seen with underlying mycobacterial (*Mycobacterium tuberculosis* [TB] or *avium* complex [MAC]), fungal (cryptococcal), and viral (CMV, HHV) infections. This syndrome is discussed in more detail below.

■ CYTOKINES AND OTHER SOLUBLE FACTORS IN HIV PATHOGENESIS

The immune system is homeostatically regulated by a complex network of immunoregulatory cytokines, which are pleiotropic and redundant and operate in an autocrine and paracrine manner. They are expressed continuously, even during periods of apparent quiescence of the immune system. On perturbation of the immune system by antigenic challenge, the expression of cytokines increases to varying degrees (**Chap. 360**). Cytokines that are important components of this immunoregulatory network are thought to play major roles in HIV disease, during both the early and chronic phases of infection. A potent proinflammatory “cytokine storm” is induced during the acute phase of HIV infection, likely a response of inflammatory cells to virus replicating at very high levels. Cytokines and chemokines that are induced during this early phase include the type I interferon IFN- α ,

IL-15, and CXCL10, followed by IL-6, IL-12, and TNF- α , and a delayed peak of the anti-inflammatory cytokine IL-10. Soluble factors of innate immunity also are induced shortly after infection, including neopterin and β -microglobulin. Several of these early-expressed cytokines and factors are not downregulated following the early phase of HIV infection, as seen in other self-resolving viral infections, and persist during the chronic phase of infection and contribute to maintaining high levels of immune activation. Among the cytokines and factors associated with early innate immune responses, they are intended to contain viral replication, although paradoxically most are potent inducers of HIV expression/replication because of their ability to induce immune activation that leads to enhanced viral production and an increase in readily available target cells for HIV (activated CD4⁺ T cells). The induction of IFN- α , one of the first cytokines induced during primary HIV infection and an important element of innate immune sensing, is thought to play a particularly important role in HIV pathogenesis by inducing a large number of IFN-associated genes that activate the immune system, alter the homeostasis of CD4⁺ T cells, and influence the virus variants that are selected during the HIV transmission bottleneck. Other cytokines that are elevated during the chronic phase of HIV infection and linked to immune activation include IFN- γ , the CC-chemokine RANTES (CCL5), macrophage inflammatory protein (MIP) 1 β (CCL4), and IL-18.

Multiple cytokines and soluble factors have been associated with HIV pathogenesis at various stages of disease, in various tissues or organs, and in the regulation of HIV replication. Plasma levels of IP-10 are predictive of disease progression, whereas the proinflammatory cytokine IL-6, marker of monocyte/macrophage activation soluble CD14 (sCD14), and coagulation marker D-dimer are associated with increased risk of all-cause mortality in people with HIV. In particular, IL-6, sCD14, and D-dimer are associated with increased risk of cardiovascular disease and other causes of death, even in individuals receiving ART. IL-18 has also been shown to play a role in the development of the HIV-associated lipodystrophy syndrome. Elevated levels of TNF- α and IL-6 have been demonstrated in plasma and cerebrospinal fluid (CSF), and increased expression of TNF- α , IL-1 β , IFN- γ , and IL-6 has been demonstrated in the lymph nodes of people with HIV prior to disease progression and a shift to TGF- β in advanced disease (see “Lymphoid Organs and HIV Pathogenesis,” above). RANTES (CCL5), MIP-1 α (CCL3), and MIP-1 β (CCL4) (Chap. 360) inhibit infection by and spread of R5 HIV-1 strains, while *stromal cell–derived factor* (SDF) 1 inhibits infection by and spread of X4 strains. The mechanisms whereby the CC-chemokines RANTES (CCL5), MIP-1 α (CCL3), and MIP-1 β (CCL4) inhibit infection of R5 strains of HIV, or SDF-1 blocks X4 strains of HIV, involve blocking of the binding of the virus to its co-receptors, the CC-chemokine receptor CCR5 and the CXC-chemokine receptor CXCR4, respectively. Other soluble factors that have not yet been fully characterized, such as soluble CD8 antiviral factor (CAF), also have been shown to suppress HIV replication, independent of co-receptor usage.

■ LYMPHOCYTE TURNOVER IN HIV INFECTION

The immune systems of patients with HIV infection are characterized by a profound increase in lymphocyte turnover that is immediately reduced with effective ART. Studies utilizing in vivo or in vitro labeling of lymphocytes in the S-phase of the cell cycle have demonstrated a tight correlation between the degree of lymphocyte turnover and plasma viremia. This increase in turnover is seen in CD4⁺ and CD8⁺ T lymphocytes as well as B lymphocytes and can be observed in peripheral blood and lymphoid tissue. Mathematical models derived from these data suggest that one can view the lymphoid pool as consisting of dynamically distinct subpopulations of cells that are differentially affected by HIV infection. A major consequence of HIV infection appears to be a shift in cells from a more quiescent pool to a pool with a higher turnover rate. It is likely that a consequence of a higher rate of turnover is a higher rate of cell death. It has been suggested that the more rapid decline in CD4⁺ compared with CD8⁺ T cells may be linked to alterations in inflammatory and homeostatic cytokines that cause increased activation-induced death without replenishment of CD4⁺ T cells. (See Table 208-5 for additional mechanisms of depletion.)

TABLE 208-5 Proposed Mechanisms of CD4⁺ T Cell Dysfunction and Depletion

DIRECT MECHANISMS	INDIRECT MECHANISMS
Loss of plasma membrane integrity due to viral budding	Aberrant intracellular signaling events
Accumulation of unintegrated viral DNA	Autoimmunity
Activation of DNA-dependent protein kinase during viral integration into host genome	
Interference with cellular RNA processing	Innocent bystander killing of viral antigen-coated cells
Intracellular gp120–CD4 autolysis events	Apoptosis, pyroptosis (caspase 1-associated inflammation), autophagy
Syncytia formation	Inhibition of lymphopoiesis from reduced survival cytokines and lymphoid tissue integrity
	Activation-induced cell death
	Elimination of HIV-infected cells by virus-specific immune responses

■ THE ROLE OF VIRAL RECEPTORS AND CO-RECEPTORS IN HIV PATHOGENESIS

CCR5 AND CXCR4 As mentioned above, HIV-1 utilizes two major co-receptors along with CD4 to bind to, fuse with, and enter target cells; these co-receptors are CCR5 and CXCR4, which are also receptors for certain endogenous chemokines. Strains of HIV that utilize CCR5 as a co-receptor are referred to as *R5 viruses*. Strains of HIV that utilize CXCR4 are referred to as *X4 viruses*. Many virus strains are *dual tropic* in that they utilize both CCR5 and CXCR4; these are referred to as *R5X4 viruses*.

The natural chemokine ligands for the major HIV co-receptors can readily block entry of HIV. For example, the CC-chemokines RANTES (CCL5), MIP-1 α (CCL3), and MIP-1 β (CCL4), which are the natural ligands for CCR5, block entry of R5 viruses, whereas SDF-1, the natural ligand for CXCR4, blocks entry of X4 viruses. The mechanism of inhibition of viral entry is a steric inhibition of binding that is not dependent on signal transduction (Fig. 208-25).

The transmitting virus is almost invariably an R5 virus that predominates during the early stages of HIV disease, although in the era of deep sequencing, more X4 variants have been detected in early disease than were previously reported. In the absence of ART or in therapeutic failures, there is a transition to a predominantly X4 virus in approximately half of individuals infected with subtype B virus. The transition is often preceded by dual R5X4 strains, and detection of X4 variants is associated with a relatively rapid decline in CD4⁺ T cell counts, increased HIV plasma viremia, and progression of disease. However, the other half of infected individuals progress in their disease while maintaining predominance of an R5 virus, and individuals infected with non-subtype B clades more rarely switch from CCR5 tropism to CXCR4 tropism than do those infected with subtype B. The reason for this difference is unclear.

The basis for the tropism of different envelope glycoproteins for either CCR5 or CXCR4 relates to the ability of the HIV envelope, including the third variable region (V3 loop) of gp120, to interact with these co-receptors. In this regard, binding of gp120 to CD4 induces a conformational change in gp120 that increases its affinity for the relevant co-receptor. Finally, R5 viruses are more efficient in infecting monocytes/macrophages and microglial cells of the brain (see “Neuropathogenesis in HIV Disease,” below).

THE INTEGRIN $\alpha 4\beta 7$ The integrin $\alpha 4\beta 7$ is an accessory receptor for HIV. It is not essential for the binding and infection of a CD4⁺ T cell by HIV; however, it likely plays an important role in the transmission of HIV at mucosal surfaces such as the genital tract and gut and contributes somewhat to the pathogenesis of HIV disease. The integrin $\alpha 4\beta 7$, which is the gut homing receptor for peripheral T cells, binds

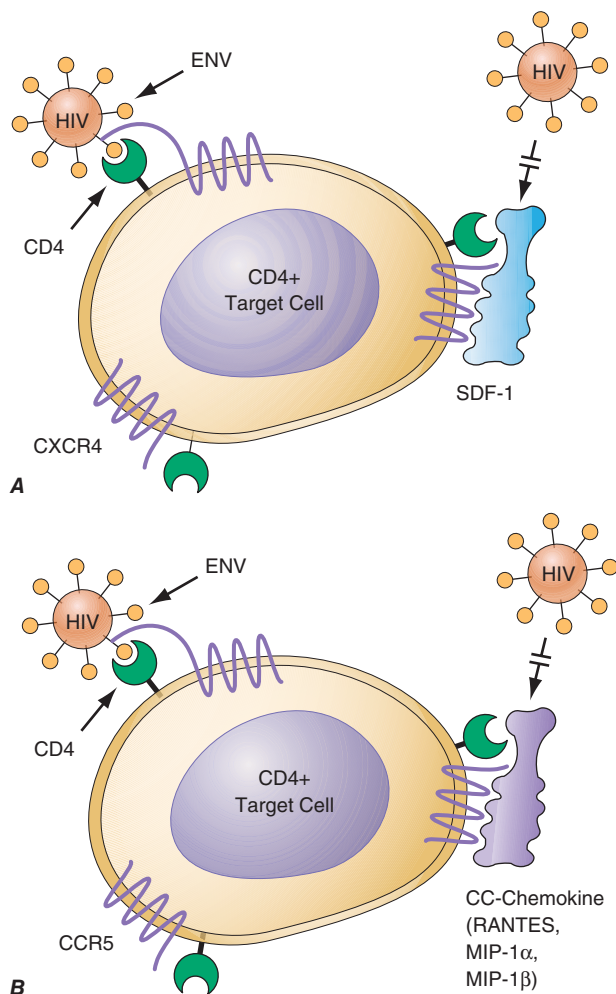


FIGURE 208-25 Model for the role of co-receptors CXCR4 and CCR5 in the efficient binding and entry of X4 (A) and R5 (B) strains of HIV-1, respectively, into CD4+ target cells. Blocking of this initial event in the virus life cycle can be accomplished by inhibition of binding to the co-receptor by the normal ligand for the receptor in question. The ligand for CXCR4 is stromal cell–derived factor (SDF-1); the ligands for CCR5 are RANTES, MIP-1 α , and MIP-1 β .

in its activated form to a specific tripeptide in the V2 loop of gp120, resulting in rapid activation of leukocyte function–associated antigen 1 (LFA-1), the central integrin in the establishment of virologic synapses, which facilitate efficient cell-to-cell spread of HIV. It has been demonstrated that $\alpha 4\beta 7^{\text{high}}$ CD4+ T cells are more susceptible to productive infection than are $\alpha 4\beta 7^{\text{low-neg}}$ CD4+ T cells because this cellular subset is enriched with metabolically active CD4+ T cells that are CCR5 $^{\text{high}}$. These cells are present in the mucosal surfaces of the gut and genital tract. Importantly, it has been demonstrated that the virus that is transmitted during sexual exposure binds much more efficiently to $\alpha 4\beta 7$ than does the virus that diversifies from the transmitting virus over time by mutation, particularly involving the accumulation of glycogens on the surface of the HIV envelope (see “Early Events in HIV Infection: Primary Infection and Initial Dissemination of Virus,” above).

■ CELLULAR TARGETS OF HIV

CD4+ T lymphocytes and to a lesser extent CD4+ cells of the myeloid lineage are the principal targets of HIV and are the only cells that can be productively infected with HIV. Circulating DCs have been reported to express low levels of CD4, although high expression of the restriction factor SAMHD1 in myeloid (mDC) and plasmacytoid (pDC) DCs limits HIV replication in these cells by depleting intracellular pools of dNTPs and directly degrading viral RNA. Epidermal Langerhans cells express CD4 and have been infected by HIV in vivo, although they too restrict replication by high expression of the host restriction factor langerin. As has been shown in vivo for DCs, FDCs, and B cells,

Langerhans cells are more likely to bind and transfer virus to activated CD4+ T cells than to be productively infected themselves.

Of potential clinical relevance is the demonstration that thymic precursor cells, which were assumed to be negative for CD3, CD4, and CD8 molecules, express low levels of CD4 and can be infected with HIV in vitro. In addition, human thymic epithelial cells transplanted into an immunodeficient mouse can be infected with HIV by direct inoculation of virus into the thymus. Since these cells may play a role in the normal regeneration of CD4+ T cells, it is possible that their infection and depletion contribute, at least in part, to the impaired ability of the CD4+ T cell pool to completely reconstitute itself in certain infected individuals in whom ART has suppressed plasma viremia to below the level of detection (see below). In addition, CD34+ monocyte precursor cells have been shown to be infected in vivo in patients with advanced HIV disease. It is likely that these cells express low levels of CD4, and therefore it is not essential to invoke CD4-independent mechanisms to explain the infection. The clinical relevance of this finding is unclear.

■ QUALITATIVE AND QUANTITATIVE ABNORMALITIES OF MONONUCLEAR CELLS

CD4+ T Cells The primary immunopathogenic lesion in HIV infection involves CD4+ T cells, and the range of CD4+ T cell abnormalities in advanced HIV infection is broad. The defects are both quantitative and qualitative and ultimately impact virtually every limb of the immune system, indicating the critical dependence of the integrity of the immune system on the inducer/helper function of CD4+ T cells. In advanced HIV disease, most of the observed immune defects can ultimately be explained by the quantitative depletion of CD4+ T cells. However, T cell dysfunction can be demonstrated in patients early in the course of infection, even when the CD4+ T cell count is in the low–normal range. The degree and spectrum of dysfunctions increase as the disease progresses, reflecting the range of CD4+ T cell functional heterogeneity, especially in lymphoid tissues. One of the first sites of intense HIV replication is in the GALT where CD4+ T_H17 cells reside; they are important for host defense against extracellular pathogens in the intestinal mucosa and help maintain the integrity of the gut epithelium. In HIV infection, they are depleted by direct and indirect effects of viral replication and cause loss of gut homeostasis and integrity, as well as a shift toward a T_H1 phenotype. Studies have shown that even after many years of ART, normalization of the CD4+ T cells in the GALT remains incomplete. In lymph nodes, HIV perturbs another important subset of the CD4+ helper T lineage, namely TF_H cells (see “Lymphoid Organs and HIV Pathogenesis,” above). TF_H cells, which are derived either directly from naïve CD4+ T cells or from other T_H precursors, migrate into B cell follicles during germinal center reactions and provide help to antigen-specific B cells through cell–cell interactions and secretion of cytokines to which B cells respond, the most important of which is IL-21. In addition, it has been shown that people with HIV who have broadly neutralizing antibodies have higher frequencies of memory TF_H CD4+ T cells. As with T_H17 cells, TF_H cells are highly susceptible to HIV infection. However, in contrast to T_H17 and most other CD4+ T cell subsets, the number of TF_H cells is increased in lymph nodes of people with HIV, especially those who are viremic. It is unclear whether this increase is helpful to responding B cells, although the likely outcome is that the increase in numbers is detrimental to the quality of the humoral immune response against HIV (see “Immune Response to HIV,” below). In addition, defects of central memory cells are a critical component of HIV immunopathogenesis. The progressive loss of antigen-specific CD4+ T cells has important implications for the control of HIV infection. In this regard, there is a correlation between the maintenance of HIV-specific CD4+ T cell proliferative responses and improved control of infection. Essentially every T cell function has been reported to be abnormal at some stage of HIV infection. Loss of polyfunctional HIV-specific CD4+ T cells, especially those that produce IL-2, occurs early in disease, whereas IFN- γ -producing CD4+ T cells are maintained longer and do not correlate with control of HIV viremia. Other abnormalities include impaired expression of IL-2 receptors,

defective IL-2 production, reduced expression of the IL-7 receptor (CD127), and a decreased proportion of CD4+ T cells that express CD28, a major co-stimulatory molecule necessary for the normal activation of T cells, which is also depleted due to aging. Cells lacking expression of CD28 do not respond normally to activation signals and may express markers of terminal activation including HLA-DR, CD38, and CD45RO. As mentioned above (“The Role of Immune Activation and Inflammation in HIV Pathogenesis”), a subset of CD4+ T cells referred to as *T regulatory cells*, or T-regs, may be involved in dampening aberrant immune activation that propagates HIV replication. The presence of these T-reg cells correlates with lower viral loads and higher CD4+/CD8+ T cell ratios. A loss of this T-reg capability with advanced disease may be detrimental to the control of virus replication.

It is difficult to explain completely the profound immunodeficiency noted in people with HIV solely based on direct infection and quantitative depletion of CD4+ T cells. This is particularly apparent during the early stages of HIV disease, when CD4+ T cell numbers may be only marginally decreased. In this regard, it is likely that CD4+ T cell dysfunction results from a combination of depletion of cells due to direct infection of the cell and multiple virus-related but indirect effects on the cell (Table 208-5). Several of these effects have been demonstrated *ex vivo* and/or by the analysis of cells isolated from the peripheral blood. Soluble viral proteins, particularly gp120, can bind with high affinity to the CD4 molecules on uninfected T cells and monocytes; in addition, virus and/or viral proteins can bind to DCs or FDCs. HIV-specific antibodies can recognize these bound molecules and potentially collaborate in the elimination of the cells by ADCC. HIV envelope glycoproteins gp120 and gp160 manifest high-affinity binding to the CD4 molecule as well as to various chemokine receptors. Intracellular signals transduced by gp120 through both CD4 and CCR5/CXCR4 have been associated with a number of immunopathogenic processes including anergy, apoptosis, and abnormalities of cell trafficking. The molecular mechanisms responsible for these abnormalities include dysregulation of the T cell receptor–phosphoinositide pathway, p56lck activation, phosphorylation of focal adhesion kinase, activation of the MAP kinase and ras signaling pathways, and downregulation of the co-stimulatory molecules CD40 ligand and CD80.

The inexorable decline in CD4+ T cell counts that occurs in most untreated people with HIV may result in part from the inability of the immune system to regenerate over an extended period of time the rapidly turning over CD4+ T cell pool efficiently enough to compensate for both HIV-mediated and naturally occurring attrition of cells. In this regard, the degree and duration of decline of CD4+ T cells at the time of initiation of therapy is an important predictor of the restoration of these cells. A person who maintains a very low CD4+ T cell count for a considerable period before the initiation of ART almost invariably has an incomplete reconstitution of such cells. At least two major mechanisms may contribute to the failure of the CD4+ T cell pool to reconstitute itself adequately over the course of HIV infection. The first is the destruction of lymphoid precursor cells, including thymic and bone marrow progenitor cells; the other is the gradual disruption of the lymphoid tissue architecture and microenvironment, which is essential for efficient regeneration of immunocompetent cells. Finally, during the advanced stages of CD4+ T lymphopenia, there are increased serum levels of the homeostatic cytokine IL-7. It was initially felt that this elevation was a homeostatic response to the lymphopenia; however, recent findings suggest that the increase in serum IL-7 was a result of reduced utilization of the cytokine related to the loss of cells expressing the IL-7 receptor, CD127, which serves as a normal physiological regulator of IL-7 production.

CD8+ T Cells A relative CD8+ T lymphocytosis is generally associated with high levels of HIV plasma viremia and likely reflects an immune response to the virus as well as dysregulated homeostasis associated with generalized immune activation. During the late stages of HIV infection, there may be a significant reduction in the numbers of CD8+ T cells despite the presence of high levels of viremia. HIV-specific CD8+ CTLs have been demonstrated in people with HIV early in the course of disease, and their emergence often coincides with a decrease

in plasma viremia—an observation that is a factor in the proposal that virus-specific CTLs can control HIV disease for a finite period of time in a certain percentage of infected individuals. However, emergence of HIV escape mutants that ultimately evade these HIV-specific CD8+ T cells has been described in most people with HIV who are not receiving ART. In addition, as the disease progresses, the functional capability of these cells gradually decreases, at least in part due to the persistent nature of HIV infection that causes functional exhaustion via the upregulation of inhibitory receptors such as PD-1, TIGIT, LAG-3, and TIM-3 on HIV-specific CD8+ T cells (see “The Role of Immune Activation and Inflammation in HIV Pathogenesis,” above). As chronic immune activation persists, there are also systemic effects on CD8+ T cells, such that as a population they assume an abnormal phenotype characterized by expression of activation markers such as co-expression of HLA-DR and CD38 with an absence of expression of the IL-2 receptor (CD25) and a reduced expression of the IL-7 receptor (CD127). In addition, CD8+ T cells lacking CD28 expression are increased in HIV disease, reflecting a skewed expansion of a less differentiated CD8+ T cell subset. This skewing of subsets is also associated with diminished polyfunctionality, a qualitative difference that distinguishes elite controllers from progressors. Elite controllers can also be distinguished from progressors by the maintenance in the former of a high proliferative capacity of their HIV-specific CD8+ T cells coupled to increases in perforin expression and elimination of infected targets, characteristics that are markedly diminished in advanced HIV disease. It has been reported that the phenotype of CD8+ T cells in people with HIV may be of prognostic significance. Those individuals whose CD8+ T cells developed a phenotype of HLA-DR+/CD38– following seroconversion had stabilization of their CD4+ T cell counts, whereas those whose CD8+ T cells developed a phenotype of HLA-DR+/CD38+ had a more aggressive course and a poorer prognosis. In addition to the defects in HIV-specific CD8+ CTLs, functional defects in other MHC-restricted CTLs, such as those directed against influenza and CMV, have been demonstrated. CD8+ T cells secrete a variety of soluble factors that inhibit HIV replication, including the CC-chemokines RANTES (CCL5), MIP-1 α (CCL3), and MIP-1 β (CCL4) and potentially several yet-unidentified factors. The presence of high levels of HIV viremia *in vivo* as well as exposure of CD8+ T cells *in vitro* to HIV envelope, both of which are associated with aberrant immune activation, have been shown to be associated with a variety of cellular functional abnormalities. Furthermore, since the integrity of CD8+ T cell function depends in part on adequate inductive signals from CD4+ T cells, the defect in CD8+ CTLs is likely compounded by the quantitative loss and qualitative dysfunction of CD4+ T cells.

B Cells The predominant defect in B cells from people with HIV is one of aberrant cellular activation, which is reflected by increased propensity to terminal differentiation and immunoglobulin secretion, as well as increased expression of markers of activation and exhaustion. As a result of activation and differentiation *in vivo* and induction of inhibitory pathways of regulation, B cells from HIV viremic patients manifest a decreased capacity to undergo cell signaling and mount a proliferative response *ex vivo*. B cells from people with HIV manifest enhanced spontaneous secretion of immunoglobulins *in vitro*, a process that reflects their highly differentiated state *in vivo*. There is also an increased incidence of EBV-related B cell lymphomas in people with HIV that are likely due to combined effects of defective T cell immune surveillance and increased B cell turnover that increases the risk of oncogenesis. Untransformed B cells cannot be infected with HIV, although HIV or its products can activate B cells directly. B cells from patients with high levels of viremia bind virions to their surface via the CD21 complement receptor. It is likely that *in vivo* activation of B cells by replication-competent or defective virus as well as viral products during the viremic state accounts at least in part for their activated phenotype. B cell subpopulations from people with HIV undergo a number of changes over the course of HIV disease, including the attrition of resting memory B cells and replacement with several aberrant memory and differentiated B cell subpopulations that collectively express reduced levels of CD21 and either increased expression of activation

markers or inhibitory receptors associated with functional exhaustion. The more activated and differentiated B cells are also responsible for increased secretion of immunoglobulins and increased susceptibility to Fas-mediated apoptosis. In more advanced disease, there is also the appearance of immature B cells associated with CD4⁺ T cell lymphopenia. Despite increased frequencies of germinal center B cells and CD4⁺ TF_H cells, both of which are required for effective humoral immunity, cognate B cell–CD4⁺ T cell interactions in lymphoid tissues are perturbed in people with HIV, especially those with persistent viremia. In vivo, the aberrant activated state of B cells manifests itself by hypergammaglobulinemia and by the presence of circulating immune complexes that bind the B cells and restrict their capacity to respond to further stimulation. People with HIV respond poorly to primary and secondary immunizations with protein and polysaccharide antigens. Using immunization with influenza vaccine, it has been demonstrated that there is a memory B cell defect in people with HIV, particularly those with high levels of HIV viremia. There is also evidence that responses to HIV and non-HIV antigens in infected individuals, especially those who remain viremic, are enriched in abnormal subsets of B cells that either are highly prone to apoptosis or show signs of functional exhaustion. Taken together, these B cell defects are likely responsible at least in part for the inadequate humoral response to HIV as well as to decreased response to vaccinations and the increase in certain bacterial infections seen in advanced HIV disease in adults. In addition, they likely contribute to the inadequacy of host defenses against bacterial infections that play a role in the increased morbidity and mortality of children with HIV. The absolute number of circulating B cells also may be depressed in HIV infection; this phenomenon likely reflects increased activation-induced apoptosis as well as a redistribution of cells out of the circulation and into the lymphoid tissue—phenomena that are associated with ongoing viral replication.

Monocytes/Macrophages Circulating monocytes are generally normal in number in people with HIV; however, there is evidence of increased activation within this lineage. The increased level of sCD14 and other biomarkers (see above) reported in people with HIV is an indirect marker of monocyte activation in vivo. Levels of sCD14 can remain elevated in individuals whose plasma viremia has been suppressed by ART for several years, an indicator of the residual immune activation and inflammation observed in HIV infection and effects on the monocyte/macrophage lineage. Multiple other abnormalities of circulating monocytes have been reported in people with HIV, many of which may be related directly or indirectly to aberrant in vivo immune activation. In this regard, increased levels of lipopolysaccharide (LPS) are found in the sera of people with HIV due, at least in part, to translocation across the gut mucosal barrier (see above). LPS is a highly inflammatory bacterial product that preferentially binds to macrophages through CD14 and Toll-like receptors, resulting in cellular activation. In the peripheral blood, expansion of monocytes that express the intermediate and nonclassical marker CD16 and markers of activation (HLA-DR) and stimulation (CD40 and CD86) has been described, especially in viremic individuals. Activated monocytes are also responsible for secretion of inflammatory cytokines and chemokines observed in HIV infection, including CXCL10, IL-1 β , and IL-6. Monocytes express the CD4 molecule and several co-receptors for HIV on their surface, and thus are potential targets of HIV infection. However, in vivo infection of circulating monocytes is difficult to demonstrate, although infection of tissue macrophages and macrophage-lineage cells in the brain (infiltrating macrophages or resident microglial cells) and lung (pulmonary alveolar macrophages) can be demonstrated easily. Tissue macrophages are an important source of HIV during the inflammatory response associated with opportunistic infections and can serve as persistent reservoirs of HIV infection, thus representing an obstacle to the eradication of HIV by antiretroviral drugs.

Dendritic and Langerhans Cells DCs and Langerhans cells are not productively infected with HIV, likely in part due to their expression of host restriction factors, including APOBEC3G and SAMHD1 (see above). However, they are thought to play an important role in the initiation of HIV infection by virtue of the ability of HIV to bind

to cell-surface C-type lectin receptors, particularly DC-SIGN (see above) and langerin. However, while langerin provides a host barrier for replication by trafficking HIV to acidic compartments for degradation, DC-SIGN retains HIV in early endosomal compartments. This allows efficient presentation of intact virus to CD4⁺ T cell targets that become infected; complexes of infected CD4⁺ T cells and DCs provide an optimal microenvironment for virus replication. Furthermore, pDCs secrete large amounts of IFN- α in response to viral infections and as such play an important role in innate sensing of HIV during early phase of infection. The numbers of circulating pDCs and mDCs are decreased in HIV infection through mechanisms that remain unclear, although several studies have shown increased lymphoid tissue recruitment of DCs associated with lymphoid hyperplasia and inflammation. The mDCs are also involved in the initiation of adaptive immunity in draining lymph nodes by presenting antigen to T cells and B cells, as well as by secreting cytokines such as IL-12, IL-15, and IL-18 that activate other immune cells, although these functions are perturbed in HIV infection.

Natural Killer Cells and Innate Lymphoid Cells NK cells represent the prototypical member of innate lymphoid cells (ILCs) that collectively provide tissue homeostasis and immunosurveillance against virus-infected cells, certain tumor cells, and allogeneic cells (Chap. 360). There are no convincing data that HIV productively infects NK cells in vivo; however, functional abnormalities in NK cells have been observed throughout the course of HIV disease, and the severity of these abnormalities increases as disease progresses. NK cells are part of the innate immune system and act by direct killing of infected cells and secretion of antiviral cytokines and chemokines. In early HIV infection there is an increase in the activation of NK cells, and the capacity to secrete IFN- γ is maintained, although they manifest reduced cytotoxic function as a result of altered maturation. During chronic HIV infection, both NK cell cytotoxicity and cytokine secretion become impaired. Given that HIV infection of target cells downregulates HLA-A and B, but not HLA-C and D molecules, this may explain in part the relative inability of NK cells to kill HIV-infected target cells. However, the NK cell impairments, especially in patients with high levels of virus replication, are associated with an expansion of an “anergic” CD56[−]/CD16⁺ NK cell subset. This abnormal subset of NK cells manifests an increased expression of inhibitory NK cell receptors (iNKR) and a substantial decrease in expression of natural cytotoxicity receptors (NCR) and shows a markedly impaired lytic activity. The overrepresentation of this abnormal subset of NK cells may explain in part the observed defects in NK cell function in people with HIV and likely begins to occur during primary infection. The relative expression of iNKR and NCR—as well as their ligands, which include HLA class I molecules—has an impact on the antiviral functions associated with NK cells, including direct killing and ADCC. Polymorphisms in iNKR and NCR alleles have been linked to HIV-1 disease outcomes, and there are indications that the early control of HIV may be mediated by cytotoxic NK cell-mediated responses. NK cells may also serve as sources of HIV-inhibitory soluble factors, including CC-chemokines such as MIP-1 α (CCL3), MIP-1 β (CCL4), and RANTES (CCL5). Finally, both inflammatory cytokines and alterations in the GALT of HIV infected individuals disrupt NK cells and other ILCs.

■ GENETIC FACTORS IN HIV-1 AND AIDS PATHOGENESIS

Candidate gene approaches and genome-wide association studies (GWAS) have identified polymorphisms in host genes that contribute to inter-individual variation in (1) the risk of acquiring HIV, (2) the steady-state levels of HIV that are established soon after infection (virologic set point), (3) the rate at which untreated HIV infection progresses to AIDS defined by a CD4⁺ T cell count that is lower than 200 cells/mm³ and/or development of AIDS-defining illnesses, (4) the level of immune reconstitution (e.g., CD4⁺ cell counts) achieved and risk of non-AIDS-associated diseases after initiation of virally suppressive antiretroviral therapy (ART), and (5) adverse reactions to antiretroviral agents. The key polymorphisms that influence these five outcomes are summarized in Table 208-6, and their identification has

TABLE 208-6 Host Genetic Factors Influencing HIV/AIDS Pathogenesis and Therapy Responses

GENE ^a	GENETIC VARIATION	MECHANISMS ^b	GENETIC ASSOCIATIONS ^c
Genes in MHC Locus			
HLA-B	<i>B*27</i> and <i>B*57</i>	Altered presentation of specific HIV antigens	Slower progression to AIDS; lower viral load
	<i>B*35</i>	Restriction of specific HIV peptide presentation	Faster progression to AIDS; higher viral load
	<i>HLA-Bw4</i>	Providing ligands for activating KIR	Slower progression to AIDS
	<i>B*57:01</i>	Altered presentation of specific HIV antigens (as above); possible abacavir-specific activation of cytokine-producing CD8+ T cells in carriers of this allele	Slower progression to AIDS; higher risk of abacavir-associated hypersensitivity
	<i>HLA-B*21M</i> allele	Enhanced HLA-E expression levels correlated with higher HLA-A expression and inhibition of NKG2A-expressing cells	Higher viral load, reduced CD4+ counts, and accelerated disease progression
	<i>B*57:03</i> bearing the rs2523608-A allele	Altered presentation of specific HIV antigens	Variant overexpressed in HIV-1 controllers of African descent
HLA class I allele	Homozygosity of <i>HLA</i> -class I alleles	Reduced repertoire for epitope recognition	Faster progression to AIDS; increased risk of mother-to-child transmission
	Shared donor–recipient <i>HLA</i> alleles	Preadaptation of HIV strains	Faster progression to AIDS
	Rare <i>HLA</i> alleles	Limited adaptation of HIV strains; less frequent escape mutants	Protection against HIV infection
HLA class II allele	<i>HLA-DRB1</i> alleles	Influence on protein specificity of CD4+ T cell responses to HIV Gag and Nef proteins	<i>HLA-DRB1*15:02</i> —lower viral load <i>HLA-DRB1*03:01</i> —higher viral load
	A1-B8-DR3-DQ2 (AH 8.1)	Increased proinflammatory responses; higher TNF- α production	Faster progression to AIDS
HLA-C	rs9264942-C allele (35 kb upstream of <i>HLA-C</i>) in linkage with rs67384697-Del	Increased expression of <i>HLA-C</i> by reducing binding of miRNA-148a	Decreased viral load set point
	rs5010528-G (1 kb upstream of <i>HLA-C</i>)	Unknown	Higher risk of developing nevirapine-associated hypersensitivity
<i>HCP5</i>	rs2395029-G	Linkage disequilibrium with <i>HLA-B*57:01</i>	Lower viral load and slower progression to AIDS
<i>MICA</i>	Noncoding SNV near <i>MICA</i> , rs4418214-T	May affect HLA class I peptide presentation—linkage with protective <i>HLA-B</i> alleles	Enriched in HIV-1 controllers
<i>PSORS1C3</i>	rs3131018-A	May affect HLA class I peptide presentation	Enriched in HIV-1 controllers
<i>ZNRD1</i>	rs9261174-C	Possible interference in processing of HIV transcripts; influence <i>ZNRD1</i> expression	Slower disease progression to AIDS
Chemokine Receptors			
<i>CCR5</i>	rs333: 32-bp deletion in the ORF (Δ 32) found in persons of European descent	Truncated <i>CCR5</i> protein; reduced co-receptor activity of R5 HIV strain	Δ 32/ Δ 32: <i>CCR5</i> -null state associated with resistance to acquiring HIV infection
			Δ 32/wild type: slower progression to AIDS; better CD4+ T cell recovery during ART
	Promoter SNVs, haplotypes (<i>HHA</i> to <i>HHG*2</i>)	Altered <i>CCR5</i> expression, e.g., <i>HHE</i> haplotype correlates with high <i>CCR5</i> expression	<i>HHE/HHE</i> : increased HIV susceptibility and faster progression to AIDS
	rs1015164 G→A (34 kb downstream from <i>CCR5</i> and close to <i>CCRL2</i>)	Increased expression of the lncRNA <i>RP11-24-11.2</i> , which corresponds to an antisense transcript that overlaps <i>CCR5</i> (<i>CCR5AS</i>); results in increased <i>CCR5</i> expression	rs1015164A allele associated with higher viral load
<i>CCR2</i>	rs1799864: SNV in ORF (64 V→I)	Linkage with polymorphisms in <i>CCR5</i> promoter	64I-bearing haplotype associated with delayed progression to AIDS
<i>CCRL2</i>	rs3204849: SNV in ORF (167 Y→F)	SNV in linkage with <i>CCR5</i> haplotype	167F associated with accelerated progression to AIDS and PCP
<i>CXCR6</i>	rs2234358: G→T in the 3' UTR	Trafficking of effector T cells and activation of NK T cells; minor HIV co-receptor	Prevalence of rs2234358-T lower in long-term nonprogressors and viremic controllers of African descent
<i>CX3CR1</i>	SNVs in ORF: rs3732379 (249 V→I) and rs3732378 (280 T→M)	Alleles bearing 249I and 280M reduce receptor expression and binding of fractalkine, the CX3CR1 ligand	249I and 280M associated with faster AIDS progression in persons of European descent
<i>DARC</i>	rs2814778: Promoter SNV (–46T→C) found in persons of African descent	–46C/C associated with absent <i>DARC</i> expression (Duffy null), low neutrophil counts, and altered circulating chemokine levels as well as HIV binding to RBCs and trans-infection of HIV-1	–46C/C: increased risk of acquiring HIV but slower HIV disease progression; Duffy null—associated low neutrophil trait associated with increased HIV risk
Chemokines			
<i>CCL3L</i> , <i>CCL4L</i>	Gene copy number of <i>CCL3L</i> and <i>CCL4L</i>	High numbers of <i>CCL3L</i> and <i>CCL4L</i> gene-containing segmental duplications correlated with high <i>CCL3L</i> and <i>CCL4L</i> levels	Gene copy number lower than population median associated with increased HIV-AIDS susceptibility and lower CD4+ T cell recovery during ART
<i>CCL5</i>	Promoter SNVs	Altered gene expression	Influenced HIV-AIDS susceptibility
<i>CCL2</i>	rs1024611: Promoter SNV (–2578 T→G)	–2578G allele: increased <i>CCL2</i> expression and monocyte recruitment	–2578G/G associated with increased risk of developing HIV-1—associated dementia and faster AIDS onset

(Continued)

TABLE 208-6 Host Genetic Factors Influencing HIV/AIDS Pathogenesis and Therapy Responses (Continued)

GENE ^a	GENETIC VARIATION	MECHANISMS ^b	GENETIC ASSOCIATIONS ^c
<i>CXCL12</i>	rs7919208: Promoter SNV (G→A)	rs7919208A creates a new transcription factor binding site associated with increased <i>CXCL12</i> expression	rs7919208A associated with higher susceptibility to HIV-related non-Hodgkin's lymphoma
Cytokines			
<i>IL-6</i>	rs1800795: Promoter SNV (−174 G→C)	−174G/G associated with increased IL-6 and CRP levels	−174G/G associated with high risk of KS development and variable recovery of CD4+ T cells during ART
<i>IL-7RA</i>	rs6897932: Coding SNV (244 T→I)	244 I/I associated with increased signal transduction and proliferation in response to IL-7	244 I/I associated with faster CD4+ T cell recovery during ART
<i>IL-10</i>	rs1800872: Promoter SNV (−592 C→A)	−592A associated with decreased IL-10 levels	−592A associated with increased HIV infection risk and AIDS progression rate
Drug-Metabolizing Enzyme Gene			
<i>CYP2B6</i>	Multiple variants (e.g., rs3745274 [516 G→T], i.e., <i>CYP2B6</i> *6)	<i>CYP2B6</i> variants influence enzyme activity	516T/T associated with higher risk of adverse reactions to efavirenz
Innate Immunity Genes			
<i>MBL</i>	Alleles defined by 3 coding SNVs	Low plasma concentration and structural variation of MBL protein	Slow progression to AIDS with heterozygosity for coding SNVs
	X allele (promoter SNV −221)	Decreased levels of MBL protein	Faster progression to AIDS with X/X genotype
<i>APOBEC3G</i>	rs8177832: ORF SNV (186 H→R)	Reduced anti-HIV-1 activity	186R associated with rapid AIDS progression in persons of African descent
<i>APOBEC3F</i>	Haplotype tagged by rs2076101 in ORF (231 I→V)	231V variant may influence Vif-mediated <i>APOBEC3F</i> degradation	231V associated with lower viral load, slower progression to AIDS and PCP
<i>TLR7</i>	rs179008: ORF SNV (32A→T) on Chr. X	Lower TLR7 mRNA translation efficiency and impaired TLR7-dependent IFN-α production	rs179008-T associated with lower viral load and cell-associated HIV-1 DNA in women
<i>PARD3B</i>	rs11884476 near exon 20: C→G	Direct interaction with HIV signaling through SMAD family of proteins	rs11884476-G associated with slower progression to AIDS
<i>IFNL4</i>	rs368234815: Frameshift mutation (TT→ΔG)	Polymorphism in <i>IFNL4</i> gene in linkage with a <i>IFNL3</i> variant; this haplotype influences <i>IFNL3</i> levels	rs368234815-ΔG associated with higher prevalence of AIDS-defining illnesses and potentially increased HIV-1 infection risk
	rs8099917: T→G	Unknown	rs8099917-G associated with higher susceptibility to KS
Others			
<i>ApoE</i>	E4 allele defined by two coding SNVs	ApoE is an HIV-1-inducible inhibitor of HIV-1 replication and infectivity in macrophages	E4/E4 associated with rapid AIDS progression and HIV-associated dementia
<i>ApoL1/MYH9</i>	Several risk haplotypes, including G1	Overexpression of the <i>ApoL1</i> kidney risk variants may increase kidney cell death	Increased risk for HIV-associated nephropathy
<i>RYR3</i>	rs2229116: ORF SNV (A→G)	Unknown; potential impact on calcium signaling and homeostasis	rs2229116-G associated with subclinical atherosclerosis during ART
<i>PROX1</i>	rs17762192-G 36 kb upstream of <i>PROX1</i>	Unknown; presumably due to its impact on <i>PROX1</i> expression, which is a negative regulator of IFN-γ	rs17762192-G associated with reduced rate of HIV disease progression
Gene–Gene Interaction			
<i>KIR+HLA</i>	<i>KIR3DS1</i> interaction with <i>HLA-Bw4-80Ile</i>	Altered NK cell activity required to eliminate HIV-infected cells	<i>KIR3DS1/HLA-Bw4-80Ile</i> associated with delayed AIDS onset
	<i>KIR2DL3</i> interaction with <i>HLA-C1</i>	Reduction of inhibitory KIR likely resulting in increased immune activation, impaired killing of latently infected cells, and a higher proviral burden	<i>HLA-C1* KIR2DL*</i> associated with better immune recovery during ART
	<i>KIR3DL1</i> 147V interaction with <i>HLA-B*57:01</i>	Variation in an immune NK cell receptor that binds <i>B*57:01</i> , modifying the protective effect of <i>B*57:01</i>	Increasing copy numbers of 47V associated with lower viral load in persons carrying <i>HLA-B*57</i>
<i>LILRB2+HLA</i>	<i>LILRB2</i> interaction with HLA class I	Regulation of dendritic cells by <i>LILRB2</i> -HLA engagement	Control of HIV-1
<i>CCL3L1+CCR5</i>	Low <i>CCL3L1</i> gene copies + detrimental <i>CCR5</i> genotypes	Low <i>CCL3L1</i> and high <i>CCR5</i> expression	Increased HIV/AIDS susceptibility and reduced immune reconstitution during ART

^aRepresentative genes and polymorphisms and ^bpossible mechanisms are listed. ^cSome of the associations are population specific and may display cohort-specific effects. Most of the associations were derived from persons of European descent.

Abbreviations: APOBEC, apolipoprotein B mRNA editing enzyme, catalytic polypeptide-like; ApoE, apolipoprotein E; ApoL1, apolipoprotein L1; ART, antiretroviral therapy; CCL, CC ligand; CCL3L, CCL3-like; CCR5, CC chemokine receptor 5; CCR5AS, CCR5 antisense RNA; CCRL2, CC chemokine receptor-like 2; CRP, C-reactive protein; CYP2B6, cytochrome P450 family 2 subfamily B member 6; CXCL12, chemokine (C-X-C motif) ligand 12; CXCR6, chemokine (C-X-C motif) receptor 6; CX3CR1, chemokine (C-X3-C motif) receptor 1; DARC, Duffy antigen receptor for chemokines; Del, deletion; HCP5, HLA class I histocompatibility antigen protein P5; HHE, human haplogroup E; HLA, human leukocyte antigen; IFN, interferon; IFNL3, interferon λ3 gene; IFNL4, interferon λ4 gene; IL, interleukin; IL-7RA, interleukin 7 receptor α; KIR, killer cell immunoglobulin-like receptors; KS, Kaposi's sarcoma; LILRB2, leukocyte immunoglobulin-like receptor B2; MBL, mannose-binding lectin; MHC, major histocompatibility complex; MICA, MHC class I polypeptide-related sequence A; MYH9, myosin heavy chain 9; NK, natural killer; ORF, open reading frame; PARD3B, par-3 family cell polarity regulator beta; PCP, *Pneumocystis jirovecii* pneumonia; PROX1, prospero homeobox 1; PSORS1C3, psoriasis susceptibility 1 candidate 3; RYR3, ryanodine receptor 3; SMAD, mothers against decapentaplegic homolog; SNV, single nucleotide variant; rs#, SNV identification number; TLR7, Toll-like receptor 7; TNF-α, tumor necrosis factor α; UTR, untranslated region; VL, viral load; ZNRD1, zinc ribbon domain containing 1; +, present; −, absent.

Sources: SK Ahuja, W He. Reviews for additional information; P An et al: Trends Genet 26:119, 2010; J Fellay: Antivir Ther 14:731, 2009; RA Kaslow et al: J Infect Dis 191:S68, 2005; D van Manen et al: Retrovirology 9:70, 2012; MP Martin et al: Immunol Rev 254:245, 2013; S Limou et al: Front Immunol 4:118, 2013; PJ McLaren et al: Curr Opin HIV AIDS 10:110, 2015; PJ McLaren et al: Proc Natl Acad Sci USA 112:14658, 2015; PJ McLaren, M Carrington: Nat Immunol 16:577, 2015; P An et al: PLoS Genet 12:e1005921, 2016; F Pereyra et al: Science 330:1551, 2010; I Bartha et al: PLoS Comput Biol 13:e1005339, 2017; S Kulkarni et al: Nat Immunol 20:824, 2019; S Le Clerc et al: Front Genet 10:799, 2019; V Kalidasan et al: Front Microbiol 11:46, 2020; SN Gingras et al: Hum Genet 139:865, 2020.

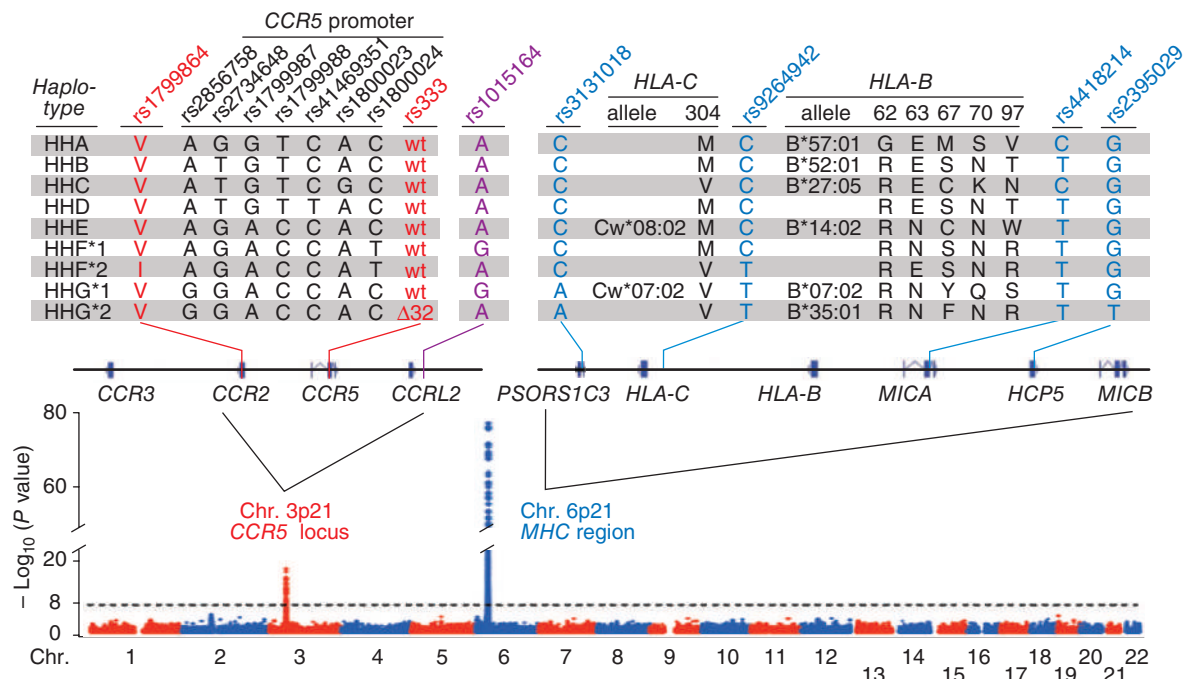


FIGURE 208-26 Schema depicting haplotypes within two regions that contribute significantly to HIV-AIDS susceptibility. *Top:* Haplotypes (left, *CCR5*; right, HLA alleles). *Bottom:* GWAS Manhattan plots schematized. Chr, chromosome. Horizontal dotted line: genome level significance threshold.

greatly advanced our understanding of the genes that influence HIV-AIDS pathogenesis and ART-associated immune reconstitution. Of particular interest are polymorphisms in two chromosomal regions, as they are associated with consistent effects on HIV acquisition, virologic set point, and/or rates of HIV disease progression: the region in chromosome 3 that includes the gene that encodes the HIV co-receptor *CC-chemokine receptor 5* (*CCR5*) and the *major histocompatibility locus* (*MHC*) in chromosome 6 (Fig. 208-26).

GENETICS OF CCR5: FROM BENCH TO BEDSIDE While the discovery of *CCR5* as a major co-receptor for cell entry of HIV-1 was established by in vitro studies, genetic association studies established its seminal role in HIV pathogenesis. Initial in vitro studies revealed that a 32-bp deletion ($\Delta 32$) in the coding region of *CCR5* contributes to resistance to *CCR5* using R5 strains of HIV. The *CCR5* $\Delta 32$ allele encodes a truncated protein that is not expressed on the cell surface. Congruently, genotype-phenotype association studies in large cohorts demonstrated that individuals homozygous for the *CCR5* $\Delta 32$ allele ($\Delta 32/\Delta 32$) lack *CCR5* surface expression and are highly resistant to acquiring HIV infection; heterozygosity for the *CCR5* $\Delta 32$ allele is associated with a lower risk of acquiring HIV.

The distribution of the *CCR5* $\Delta 32$ allele is population specific. Approximately 1% of individuals of European ancestry are homozygous for the *CCR5* $\Delta 32$ allele. Depending on the geographic region in Europe, up to 18% of individuals are heterozygous for the *CCR5* $\Delta 32$ allele. The *CCR5* $\Delta 32$ allele is rare in other populations. The evolutionary pressure that resulted in the emergence of the *CCR5* $\Delta 32$ allele in the European population remains unknown and has been speculated to be secondary to an ancestral pandemic, such as the plague.

Subsequent studies identified single nucleotide variants (SNVs) in the promoter (regulatory) region of *CCR5* that influence gene expression levels. Alleles bearing specific cassettes of linked polymorphisms (haplotypes) were identified and designated as human haplogroups A to G*2 (*HHA* to *HHG**2) (Fig. 208-26). The *CCR5* $\Delta 32$ polymorphism is found on the *HHG**2 haplotype. *CCR5* haplotypes A–D versus E–G*2 differ by bearing GT versus AC at polymorphic sites rs1799987 and rs1799988 (Fig. 208-26). *CCR5*-*HHA* haplotype represents the ancestral haplotype (found in chimpanzees) and is associated with lower *CCR5* gene expression, whereas the *CCR5*-*HHE* haplotype is associated with higher *CCR5* expression. Methylation of DNA is a common epigenetic signaling mechanism that cells use to lock genes in the “off”

position, and polymorphisms in *CCR5* haplotypes may mediate their effects by influencing DNA methylation levels in the *CCR5* locus. The *CCR5*-*HHE* and *CCR5*-*HHA* haplotypes are more sensitive and resistant, respectively, to T cell activation-induced demethylation of the *CCR5* locus.

In worldwide populations, *HHE* and *HHC* are prevalent haplotypes, whereas the ancestral *HHA* haplotype is more common in persons of African ancestry. The associations of *CCR5* haplotypes with HIV acquisition and/or HIV disease course are largely consistent with their effects on *CCR5* gene expression. For example, homozygosity for the *CCR5*-*HHE* haplotype is associated with an increased risk of acquiring HIV, progressing rapidly to AIDS, and reduced immune recovery during ART. The *HHA* haplotype is associated with slower disease progression in African populations and has been speculated to be a basis for why chimpanzees (who all carry the ancestral *CCR5*-*HHA* haplotype) naturally infected with SIV resist disease progression. The pairing of the *HHC* and *CCR5* $\Delta 32$ -bearing *HHG**2 haplotypes (*HHC*/*HHG**2 genotype) is associated with a lower risk of acquiring HIV infection and slower rate of HIV disease progression, whereas the pairing of the *HHE* haplotype with the *HHG**2 haplotype is associated with the opposite effects. The *CCR2*-64I–bearing *HHF**2 haplotype is associated with a slower HIV disease course.

Consistent with these genetic associations, polymorphisms in genes encoding ligands for *CCR5* have also been associated with variable HIV susceptibility and disease progression rates. Examples include copy number variations of *CCL3L1* and SNVs in *CCL5*. The sum of these studies established a pivotal role of *CCR5* and its ligands in HIV-AIDS pathogenesis and, potentially, immune recovery.

The discovery that the *CCR5* $\Delta 32/\Delta 32$ genotype is associated with strong resistance to HIV infection, and that uninfected persons bearing this genotype did not appear to have impaired immunity, led to the development of two kinds of novel therapies. First, it spurred the development of a new class of therapies approved by the U.S. Food and Drug Administration: entry inhibitors (e.g., maraviroc) that block the interaction of *CCR5* with the HIV envelope. Second, it led to the evaluation of novel experimental cellular therapies. A person with HIV and acute myelogenous leukemia was given an allogeneic stem cell transplantation from an HLA-compatible person whose cells lacked expression of *CCR5* due to the $\Delta 32/\Delta 32$ genotype. There was no evidence of HIV-1 infection for 13 years in the patient who underwent the transplant; the patient eventually died due to recurrence of leukemia.

1584 This observation provided a proof of concept for an HIV cure and led to the development of additional novel cellular therapies involving autologous transplantation of CD4+ T cells in which the *CCR5* gene is inactivated ex vivo using new gene editing procedures. Similar cellular strategies have had mixed success, mainly due to the latent viral reservoir in various tissues.

DISCOVERY OF HLA CLASS I ALLELES THAT ASSOCIATE WITH VIROLOGIC CONTROL OF HIV INFECTION There is a strong association between variations within the *HLA-B* gene with protective (e.g., *HLA-B*57* and *HLA-B*27* alleles) or detrimental (e.g., *HLA-B*35* allele) outcomes during HIV infection. Carriage of the *HLA-B*57* and/or *HLA-B*27* alleles is associated with slower disease progression. The beneficial effects of these alleles may relate in part to their associations with a lower virologic set point as well as to higher cell-mediated immunity in people with HIV. The protective effect of the *HLA-B*57* and *HLA-B*27* alleles on the HIV disease course is underscored by the finding that the prevalence of these alleles is higher among persons with long-term nonprogression and persons who control HIV replication spontaneously (elite controllers). In contrast, the *HLA-B*35* allele has been associated with faster progression to AIDS and higher viral load. The prevalence of the *HLA-B* alleles differs between populations. *HLA-B*57:01* in Europeans and *HLA-B*57:03* in persons of African descent are the protective alleles. In some populations (e.g., Japanese) where the *HLA-B*57/HLA-B*27* alleles are absent, *HLA-B*51* is associated with a protective phenotype.

Possession of the protective *HLA-B* alleles is associated with broader and stronger CD8+ T cell responses to HIV epitopes. The mechanisms underlying the differential effects of the *HLA-B* alleles on the course of HIV disease may relate to differences in the ability of antigen-presenting cells to present immunodominant HIV epitopes to T helper or cytotoxic T lymphocytes in the context of MHC-encoded molecules. This may result in differential immune responses that influence viral replication. In this regard, the *HLA-B* alleles that impact the course of HIV disease differ in their amino acid residues in the *HLA-B* peptide-binding groove; this difference may play a critical role in virologic control.

The *HLA-B*21M* allele does not influence *HLA-B* gene expression; however, it is in linkage with *HLA-B* haplotypes that are associated with higher *HLA-A* and *HLA-E* expression. Higher *HLA-A* levels associate with poorer control of HIV as well as higher viral load, reduced CD4+ T cell counts, and accelerated progression to AIDS. *HLA-E* is the ligand for natural killer (NK) cell NKG2A, an inhibitory receptor. Engagement of NKG2A with *HLA-E* inhibits NK cells that would normally be potent eliminators of virally infected cells. Thus, targeting NKG2A might provide a therapeutic avenue for HIV treatment.

Investigators have also examined the influence of extended *HLA* haplotypes (linked alleles) on the course of HIV disease. The extended *HLA* ancestral haplotype (AH) 8.1 is defined by the presence of *HLA-A1*, *HLA-B8*, and *HLA-DR3* alleles. AH 8.1 is the most common ancestral haplotype in persons of European descent (present in 10%) and is associated with multiple autoimmune diseases in HIV-seronegative persons. These associations of AH 8.1 are thought to be due to a genetically determined hyperresponsiveness characterized by high TNF- α production and lack of complement C4A. Strong epidemiologic data indicate that carriage of AH 8.1 in HIV-seropositive persons is associated with a rapid decline in the number of CD4+ T cells and faster progression to AIDS development. Gene-gene interactions between *HLA* alleles and other genes (e.g., killer cell immunoglobulin-like receptors) also may influence HIV disease progression rates.

POLYMORPHISMS IDENTIFIED BY GWAS THAT ASSOCIATE WITH VIROLOGIC CONTROL AND DISEASE PROGRESSION GWAS have not identified additional genetic variations that associate with the risk of HIV-1 acquisition, presumably due to a paucity of well-characterized risk cohorts in which level of exposure has been quantified. By contrast, large-scale GWAS have identified SNVs, especially in the MHC, that influence HIV viral load, including in a large group of individuals termed *HIV controllers* (including elite controllers) who spontaneously (without ART) control viral replication. GWAS in people with HIV

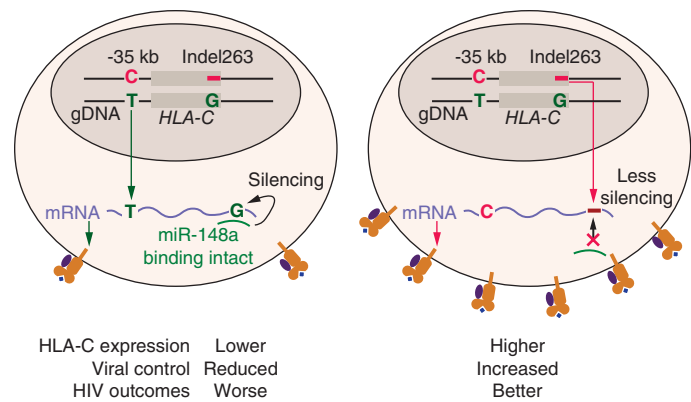


FIGURE 208-27 Linkage disequilibrium between two variants in the *HLA-C* locus and their influence on binding of miR-148a to the 3'-untranslated region (UTR). Altered binding of miR-148a associates with *HLA-C* protein expression levels and, in turn, viral control and HIV disease outcomes. Effects associated with T-G (left) and C-del (right) haplotypes are depicted. The C-deletion haplotype prevents binding of miR-148a to 3'-UTR of *HLA-C* (less silencing). Kb, kilobase.

of European ancestry identified four SNVs in genes in the *HLA* class I loci that associated with virologic control. These SNVs are within or in the vicinity of *PSORS1C3*, *HLA-C*, *MICA*, and *HCP5* genes (Fig. 208-26). As noted in this figure, the individual effects of these alleles are difficult to discern because of linkage disequilibrium. The protective effects of the SNVs in *HCP5* and *MICA* may relate to their linkage with known protective *HLA-B* alleles. The protective *HCP5* allele is in linkage disequilibrium with the *HLA-B*57:01* allele, and the protective *MICA* allele tags with the *HLA-B*57:01* and *HLA-B*27:05* alleles. The protective *HLA-C* SNV is associated with higher *HLA-C* expression, which has been associated with viral control and better HIV outcomes. This protective SNV (rs9264942; T→C) resides 35 kb upstream of the *HLA-C* gene and is in strong linkage disequilibrium with a 3'-UTR indel263 SNV (rs67384697; G→deletion), generating the T-G or C-deletion haplotypes (Fig. 208-27). miR-148a binds to the 3'-UTR region encompassing the rs67384697 SNV and silences *HLA-C* expression. Binding of miR-148a to the 3'-UTR is disrupted on the mRNA transcribed from the C-deletion haplotype; this disruption associates with less silencing of the mRNA and therefore higher *HLA-C* cell surface expression, which associates with better HIV disease outcomes (Fig. 208-27). Conversely, binding of miR-148a to the 3'-UTR is intact on the mRNA transcribed from the T-G haplotype; this binding associates with silencing of the mRNA and therefore lower *HLA-C* cell surface expression associates with worse HIV disease outcomes (Fig. 208-27). GWAS in persons of African descent have identified an SNV (rs2523608) that tags the *HLA-B*57:03* allele that is known to associate with HIV-1 control and a slower disease course. Together, these GWAS data underscore the importance of variations in *HLA* class I loci in control of viral replication.

A recent GWAS suggested that an SNV (rs1015164G→A) approximately 34 kb downstream of the *CCR5* loci is associated with a higher viral load set point (Fig. 208-26) and lower CD4+ T cell counts in therapy-naïve HIV-seropositive persons. rs1015164 maps to a lncRNA gene in proximity to the *CCRL2* gene (Fig. 208-26). The lncRNA is transcribed from the antisense strand of *CCR5* and was therefore named *CCR5AS*. The rs1015164A allele is associated with higher expression of *CCR5AS* in CD4+ T cells, which in turn was associated with increased levels of *CCR5* mRNA. Although the detrimental effect of the rs1015164A allele was suggested to be independent of the detrimental effects of the abovementioned *CCR5-HHE* haplotype, further investigation is warranted as the rs1015164A allele and *CCR5-HHE* haplotype are in a high degree of linkage disequilibrium.

Most GWAS studies have been performed in European populations, limiting generalizability to other populations. Additionally, GWAS are generally not suitable for identifying rare variants (<1% prevalence). Therefore, next-generation sequencing (NGS) approaches were suggested to identify these rare variants. However, a recent NGS study suggests that exonic variants with large effect sizes are unlikely to have

a major contribution to host control of HIV infection. Mathematical modeling revealed that variations in host genes may explain about 10% of the observed variability in HIV viral load, whereas viral genetic diversity may explain 29% of the variability.

GENETIC ASSOCIATIONS WITH SPECIFIC AIDS AND NON-AIDS CONDITIONS

Carotid artery disease Many of the non-AIDS events in HIV-seropositive individuals resemble those attributable to immune senescence and those found in the HIV-seronegative aging population. A functional SNV in the ryanodine receptor 3 (*RYR3*) gene was found to be associated with an increased risk of common carotid intima-media thickness (cIMT), which is a surrogate for subclinical atherosclerosis. Functional studies on *RYR3* and its isoforms demonstrate a major role of these receptors in modulating endothelial function and atherogenesis via calcium-signaling pathways, providing a biologically plausible mechanism by which the SNV in *RYR3* may associate with increased cIMT risk.

Kidney disease HIV-1-associated nephropathy (HIVAN) is a form of focal sclerosing glomerulonephritis caused by direct infection of kidney epithelial cells with HIV. HIVAN is more common in persons of African descent. There is evidence that polymorphisms in the *MYH9* gene and in the neighboring *APOL1* gene are a strong determinant of susceptibility to HIVAN in persons of African descent. The effect of carrying two *APOL1* risk alleles explains nearly 35% of HIVAN. Overexpression of the *APOL1* kidney risk variants may associate with increased kidney cell death.

HIV-associated neurocognitive disorder HIV-associated neurocognitive disorder (HAND) comprises a spectrum of neurocognitive deficits due to HIV infection. Variations in the apolipoprotein E (*ApoE*) gene have strong associations with Alzheimer's disease in the HIV-seronegative population. In HIV-seropositive persons, possession of the *E4/E4* genotype has been associated with dementia, peripheral neuropathy, and impairment in cognition as well as immediate and delayed verbal memory. Macrophage recruitment and activation play a central role in the development of many of the HAND syndromes. Variations in chemokines that play an influential role in macrophage activation and recruitment, namely *CCL2* (*MCP-1*) and *CCL3* (*MIP-1 α*), have been shown to influence the risk of developing HAND. Variations in mitochondrial genes also have been associated with a risk of AIDS and HAND. A GWAS identified a polymorphism in chromosome 14 in the T cell receptor α locus that may influence neurocognitive outcomes.

HIV-1-associated pneumocystis pneumonia Human Apobec3 cytidine deaminases are intrinsic resistance factors to HIV-1. However, HIV-1 encodes a viral infectivity factor (Vif) that degrades *APOBEC3* proteins. Association studies suggest a role of genetic variations in the *APOBEC3* family in HIV disease. A common haplotype derived from 6 SNVs in the *APOBEC3F* gene and tagged by a codon-changing variant is associated with a significantly lower viral load set point, slower rate of progression to AIDS, and delayed development of *Pneumocystis jirovecii* pneumonia (PCP). In addition, a coding SNV in the *CCRL2* gene is associated with accelerated progression to AIDS and rapid development of PCP.

HIV-related non-hodgkin lymphoma (NHL) The relative risk of developing NHL in HIV-seropositive persons is highly elevated compared with the general population. NHL represents approximately 34% of all identified cancers in HIV-seropositive persons. A recent GWAS identified a promoter SNV in the *CXCL12* gene that was associated with higher susceptibility to develop HIV-related NHL. The effect of this SNV is likely causal as it creates new transcription factor binding sites, impacting *CXCL12* expression.

ASSOCIATIONS WITH ART-RELATED ADVERSE EVENTS Abacavir, an effective antiretroviral agent, is associated with significant risk of hypersensitivity reactions (2–9% of cases). Interestingly, while the *HLA-B*57:01* allele is associated with a slower HIV disease course, possession of this allele is associated with a higher risk of abacavir-associated hypersensitivity, possibly due to the abacavir-specific activation of cytokine-producing CD8⁺ T cells only in *HLA-B*57:01*

carriers. Pharmacogenetic screening for the *HLA-B*57:01* allele is recommended before initiation of abacavir treatment.

The antiretroviral agent nevirapine is associated with hypersensitivity reactions in 6–10% of patients, including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). rs5010528G, a strong proxy for *HLA-C*04:01* carriage, was associated with high risk of SJS and TEN during nevirapine treatment. In addition, efavirenz was among the first antiretroviral agents to be co-formulated into single-pill regimens for mass rollout globally. Several genetic variants in the drug-metabolizing enzyme CYP2B6 have been associated with high efavirenz plasma concentrations and increased risk of adverse neuropsychiatric effects. For example, homozygosity for one such variant, rs3745274 T/T, increases the risk of adverse reactions to efavirenz up to fivefold, and this risk genotype is much more common in Africans (13.7%) than Europeans (5.6%).

■ NEUROPATHOGENESIS IN HIV DISEASE

While there has been a remarkable decrease in the incidence of the severe forms of HIV encephalopathy among those with access to treatment in the era of effective ART, people with HIV can still experience milder forms of neurocognitive impairment despite adequate ART. Factors that contribute to the neurocognitive decline include lack of complete control of HIV replication in the brain; production of HIV proteins that may be neurotoxic; low CD4⁺ T cell nadir; chronic immune activation; comorbidities such as drug abuse, microvascular disease, older age, and diabetes; and the potential for neurotoxicity of certain antiretroviral drugs. HIV has been demonstrated in the brain and CSF of infected individuals with and without neuropsychiatric abnormalities. As opposed to lymphoid tissues, there are no resident lymphocytes in the brain. The main cell types that are infected in the brain in vivo are the perivascular macrophages and the microglial cells, which can sometimes form syncytia resulting in multinucleated giant cells; low-level viral replication is also seen in perivascular astrocytes. It has been proposed that monocytes that have already been infected in the blood can migrate into the brain, where they then reside as macrophages, or macrophages can be directly infected while residing within the brain. The precise mechanisms whereby HIV enters the brain are unclear; however, they are thought to relate, at least in part, to the ability of virus-infected and immune-activated macrophages to induce adhesion molecules such as E-selectin and vascular cell adhesion molecule 1 (VCAM-1) on brain endothelium. Other studies have demonstrated that HIV gp120 enhances the expression of intercellular adhesion molecule 1 (ICAM-1) in glial cells and HIV Tat protein can disrupt the tight junctions of the brain endothelial cells to facilitate entry of HIV-infected cells into the CNS. Virus isolates from the brain are preferentially R5 strains as opposed to X4 strains; in this regard, people with HIV who are heterozygous for *CCR5- Δ 32* appear to be relatively protected against the development of HIV encephalopathy. Once HIV enters the brain due to pressures of the local environment, it evolves to develop distinct sequences in the env, tat, and LTR genes. These unique sequences have been associated with neurocognitive dysfunction; however, it is unclear if they are causal (see below).

People with HIV may manifest white matter lesions as well as neuronal loss. The white matter lesions are due to axonal injury and a disruption of the blood-brain barrier and not due to demyelination. Given the absence of evidence of HIV infection of neurons, HIV-mediated effects on neurons are thought to involve indirect pathways whereby viral proteins, particularly gp120 and Tat, trigger the release of endogenous neurotoxins from macrophages and to a lesser extent from astrocytes. In addition, it has been demonstrated that both HIV-1 Nef and Tat can induce chemotaxis of leukocytes, including monocytes, into the CNS. Neurotoxins can be released from monocytes as a consequence of infection and/or immune activation. Monocyte-derived neurotoxic factors have been reported to kill neurons via a variety of mechanisms including activation of the N-methyl-D-aspartate (NMDA) receptors and induction of oxidative stress. In addition, HIV gp120 shed by virus-infected monocytes could cause neurotoxicity by antagonizing the function of vasoactive intestinal peptide (VIP), by elevating intracellular calcium levels, and by decreasing neurotrophic factor levels in the

cerebral cortex. A variety of monocyte-derived cytokines can contribute directly or indirectly to the neurotoxic effects in HIV infection; these include TNF- α , IL-1, IL-6, TGF- β , IFN- γ , platelet-activating factor, and endothelin. Furthermore, among the CC-chemokines, elevated levels of monocyte chemotactic protein 1 (MCP-1 or CCL-2) in the brain and CSF have been shown to correlate best with the presence and degree of HIV encephalopathy in ART-naïve patients. In addition, infection and/or activation of monocyte-lineage cells can result in increased production of eicosanoids, quinolinic acid, nitric oxide, excitatory amino acids such as L-cysteine and glutamate, arachidonic acid, platelet-activating factor, free radicals, TNF- α , and TGF- β , which may contribute to neurotoxicity. Astrocytes may play diverse roles in HIV neuropathogenesis. Reactive gliosis or astrocytosis has been demonstrated in the brains of people with HIV, and TNF- α and IL-6 have been shown to induce astrocyte proliferation. In addition, astrocyte-derived IL-6 can induce HIV expression in infected cells in vitro. Furthermore, it has been suggested that astrocytes may downregulate macrophage-produced neurotoxins. Evidence of neuronal injury can be demonstrated by measuring neurofilament levels in CSF. Treatment with ART leads to improvement in neuropsychiatric manifestations and a decrease in these cytokine levels in CSF, suggesting that they are driven by the virus or by its products. However, even in patients on long-term ART, there may be evidence of persistently activated lymphocytes in the CSF. It is unclear if these lymphocytes may contribute to neuronal injury in the brain or are critical for controlling the CNS viral reservoir. However, some individuals may develop a subacute encephalitis due to an IRIS reaction (see below). This often occurs weeks or a few months after initiation of ART in individuals with low CD4+ T cell counts. It is thought that the recovery of CD4+ T cells causes a lymphocyte response to the CNS HIV reservoir. The contribution of host genetic factors to development of neuropsychiatric manifestations of HIV infection has not been well studied. However, evidence supports the role of several genetic factors including the E4 allele for apoE in an increased risk of HIV-associated neurocognitive disorders and peripheral neuropathy.

It has also been suggested that the CNS may serve as a relatively sequestered site for a reservoir of latently infected cells that might be a barrier for the eradication of virus by ART (see “The HIV Reservoir: Obstacles to the Eradication of Virus,” above).

■ PATHOGENESIS OF KAPOSI’S SARCOMA

There are at least four distinct epidemiologic forms of KS: (1) the classic form that occurs in older men of predominantly Mediterranean or eastern European Jewish backgrounds with no recognized contributing factors; (2) the equatorial African form that occurs in all ages, also without any recognized precipitating factors; (3) the form associated with organ transplantation and its attendant iatrogenic immunosuppressed state; and (4) the form associated with HIV-1 infection. In the latter two forms, KS is an opportunistic disease; in people with HIV, unlike typical opportunistic infections, its occurrence is not strictly related to the level of depression of CD4+ T cell counts. The pathogenesis of KS is complex; fundamentally, it is an angioproliferative disease that is not a true neoplastic sarcoma, at least not in its early stages. It is a manifestation of excessive proliferation of spindle cells that are believed to be of vascular origin and have features in common with endothelial and smooth-muscle cells. In HIV disease the development of KS is dependent on the interplay of a variety of factors including HIV-1 itself, human herpes virus 8 (HHV-8), immune activation, and cytokine secretion. Numerous epidemiologic and virologic studies have clearly linked HHV-8, which is also referred to as *Kaposi’s sarcoma-associated herpesvirus* (KSHV), to KS not only in people with HIV but also in individuals with the other forms of KS. HHV-8 is a γ -herpesvirus related to EBV and *herpesvirus saimiri*. It encodes a homologue to human IL-6 and, in addition to KS, has been implicated in the pathogenesis of body cavity lymphoma, multiple myeloma, and monoclonal gammopathy of undetermined significance. Sequences of HHV-8 are found universally in the lesions of KS, and patients with KS are virtually all seropositive for HHV-8. HHV-8 DNA sequences can be found in the B cells of 30–50% of patients with KS and 7% of patients with AIDS without clinically apparent KS.

Between 1% and 2% of eligible blood donors are positive for antibodies to HHV-8, while the prevalence of HHV-8 seropositivity in men with HIV is 30–35%. The prevalence of HHV-8 seropositivity in women with HIV is ~4%. This finding is reflective of the lower incidence of KS in women. It has been debated whether HHV-8 is the transforming agent in KS; the bulk of the cells in the tumor lesions of KS are not neoplastic cells. However, it has been demonstrated that endothelial cells can be transformed in vitro by HHV-8. In this regard, HHV-8 possesses genes, including homologues of the IL-8 receptor, Bcl-2, and cyclin D, which can potentially transform the host cell. Despite the complexity of the pathogenic events associated with the development of KS in people with HIV, HHV-8 is the etiologic agent of this disease. The initiation and/or propagation of KS requires an activated state and is mediated, at least in part, by cytokines. Multiple factors, including TNF- α , IL-1 β , IL-6, granulocyte-macrophage colony-stimulating factor (GM-CSF), basic fibroblast growth factor, and oncostatin M, function in an autocrine and paracrine manner to sustain the growth and chemotaxis of the KS spindle cells. In this regard, KSHV-derived IL-6 has been demonstrated to induce proliferation of lymphoma cells and to inhibit the cytostatic effects of IFN- α on KSHV-infected lymphoma cells.

IMMUNE RESPONSE TO HIV

As detailed above and below, following the initial burst of viremia during primary infection, people with HIV mount robust immune responses that in most cases substantially curtail the levels of plasma viremia and likely contribute to delaying the ultimate development of clinically apparent disease for a median of 10 years in untreated individuals. This immune response contains elements of both humoral and cell-mediated immunity involving both adaptive and innate immune responses (Table 208-7; Fig. 208-28). It is directed against multiple antigenic determinants of the HIV virion as well as against viral proteins expressed on the surface of infected cells. Ironically, those CD4+ T cells with T cell receptors specific for HIV are theoretically those CD4+ T cells most likely to be activated—and thus to serve as early targets for productive HIV infection and the cell death or dysfunction associated with infection. Thus, an early consequence of HIV infection is interference with and decrease of the helper T cell population needed to generate an effective immune response.

Although a great deal of investigation has been directed toward delineating and better understanding the components of this immune response, it remains unclear which immunologic effector mechanisms are most important in delaying progression of infection and which, if any, play a role in the pathogenesis of HIV disease. This lack of knowledge has also hampered the ability to develop an effective vaccine for HIV disease.

TABLE 208-7 Elements of the Immune Response to HIV	
Humoral immunity	
Neutralizing antibodies	
Type specific	
Group specific	
Broadly neutralizing	
Antibodies participating in antibody-dependent cellular cytotoxicity (ADCC)	
Protective	
Pathogenic (bystander killing)	
Enhancing antibodies	
Complement	
Cell-mediated immunity	
Helper CD4+ T lymphocytes	
Class I MHC–restricted cytotoxic CD8+ T lymphocytes	
CD8+ T cell–mediated inhibition (noncytolytic)	
ADCC	
Natural killer cells	

Abbreviation: MHC, major histocompatibility complex.

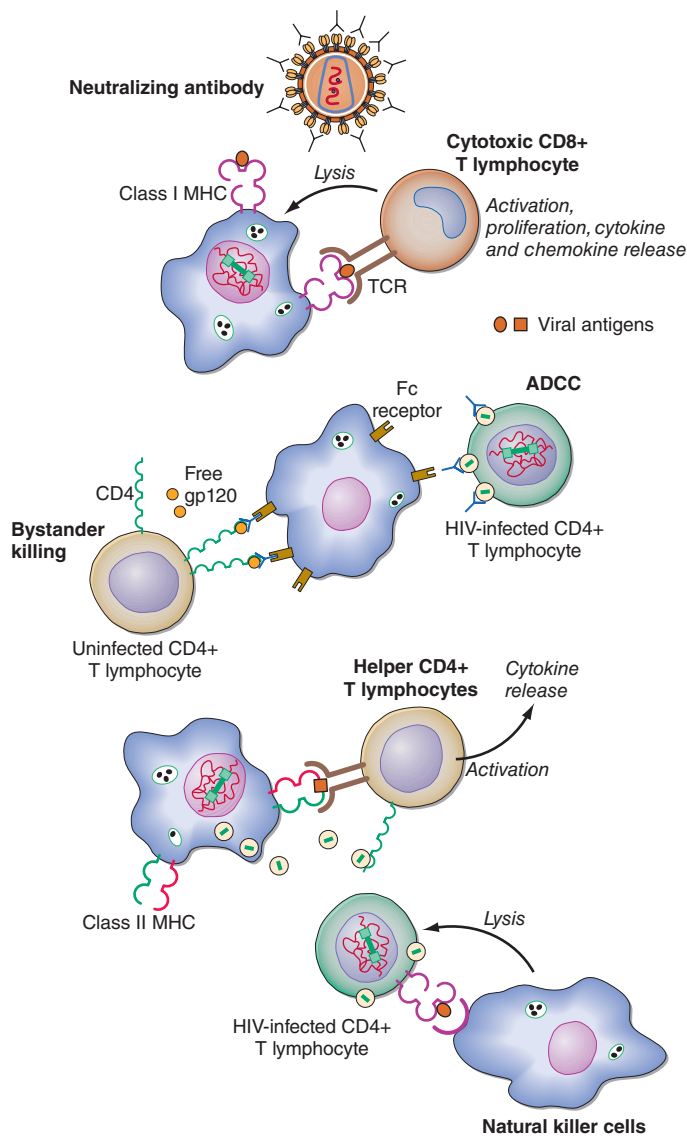


FIGURE 208-28 Schematic representation of the different immunologic effector mechanisms thought to be active in the setting of HIV infection. Detailed descriptions are given in the text. ADCC, antibody-dependent cellular cytotoxicity; MHC, major histocompatibility complex; TCR, T cell receptor.

■ HUMORAL IMMUNE RESPONSE

Antibodies to HIV usually appear within 3–6 weeks and almost invariably within 12 weeks of primary infection (Fig. 208-29); rare exceptions are in individuals who have defects in the ability to produce HIV-specific antibodies. Along with plasma levels of HIV RNA and p24 antigen, detection of these antibodies forms the basis of most diagnostic screening tests for HIV infection. The appearance of HIV-binding antibodies detected by ELISA and Western blot assays occurs prior to the appearance of neutralizing antibodies; the latter generally appear following the initial decreases in plasma viremia and are more closely related to the appearance of HIV-specific CD8⁺ T lymphocytes. The first antibodies detected are those directed against the immunodominant region of the envelope gp41, followed by the appearance of antibodies to the structural or gag protein p24 and the gag precursor p55. Antibodies to p24 gag are followed by the appearance of antibodies to the outer envelope glycoprotein (gp120), the gag protein p17, and the products of the *pol* gene (p31 and p66). In addition, one may see antibodies to the low-molecular-weight regulatory proteins encoded by the HIV genes *vpr*, *vpu*, *vif*, *rev*, *tat*, and *nef*. On rare occasions, levels of HIV-specific antibodies may decline during treatment of acute HIV infection; otherwise they remain persistently positive.

While antibodies to multiple antigens of HIV are produced, the precise functional significance of these different antibodies is unclear. The only viral proteins that elicit neutralizing antibodies are the envelope

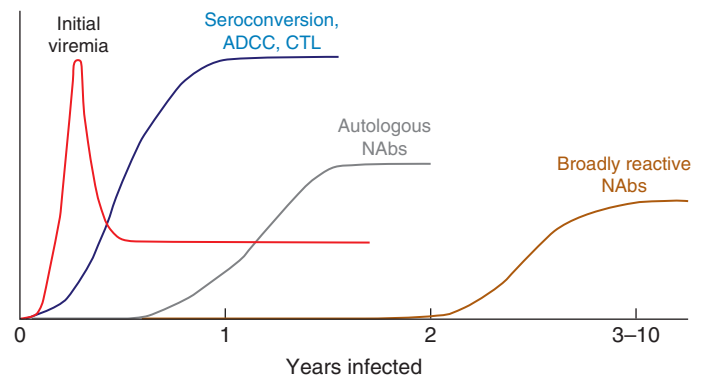


FIGURE 208-29 Relationship between initial HIV viremia and the development of antibodies to HIV. Within 3 to 6 weeks of initial HIV infection, nonneutralizing antibodies to HIV appear. These antibodies are capable of mediating antibody-dependent cellular cytotoxicity (ADCC). The decline in plasma viremia generally correlates with the appearance of cytotoxic T lymphocytes (CTL). After approximately 3 months, autologous neutralizing antibodies (NABs) capable of neutralizing prior circulating strains of HIV appear. After 2 or more years, broadly reactive NABs appear. (Reproduced with permission from *Annual Reviews, Inc. from The Role of Antibodies in HIV Vaccines*, JR Mascola and DC Montefiori. 28:413, 2010.)

proteins gp120 and gp41. Antibodies directed toward the envelope proteins of HIV have been characterized both as being protective and as possibly contributing to the pathogenesis of HIV disease. Among the protective antibodies are those that function to neutralize HIV directly and prevent the spread of infection to additional cells, as well as those that participate in ADCC. The first neutralizing antibodies are directed against the autologous infecting virus and appear after approximately 12 to 24 weeks of infection. Due to its high rate of mutation the virus is usually able to quickly escape these (and subsequent) neutralizing antibodies. One important mechanism of immune escape is the addition of N-linked glycosylation sites, forming a glycan shield that interferes with envelope recognition by these initial antibodies.

A number of broad and potent HIV-neutralizing envelope-specific antibodies have been isolated from people with HIV in studies designed to better understand the host response to HIV infection. Approximately 20% of patients develop antibodies capable of neutralizing highly diverse strains. These usually appear 2 or more years following infection in the face of continual viremia. These studies have revealed at least five major sites within the HIV envelope trimer that are able to elicit broadly neutralizing antibodies. These sites include antibodies directed toward the CD4 binding site (CD4bs) of gp120, those binding glycan-dependent epitopes in the V1/V2 region of gp120, those near the base of the V3 region of gp120, those binding to the gp120/gp41 bridge, and those binding to the membrane-proximal region of gp41 (Fig. 208-30). Several of these antibodies contain unique features including high levels of somatic hypermutation, selective germ-line gene usage (especially for CD4bs antibodies), and long

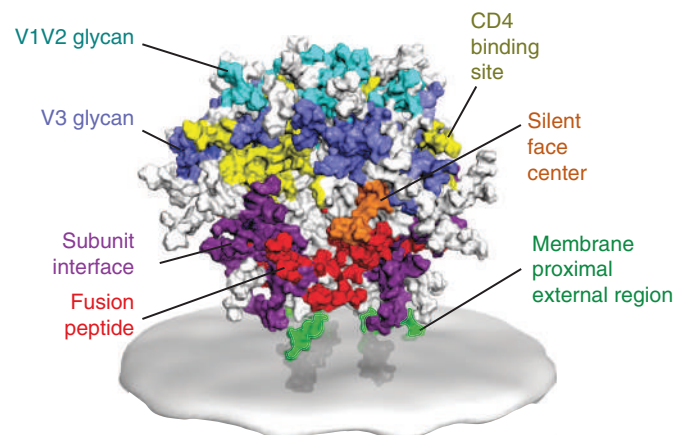


FIGURE 208-30 Known targets of broadly neutralizing antibodies against HIV-1. (Courtesy of J Stuckey, GY Chuang.)

heavy chain complementary determining regions (especially CDRH3). Of note, while these antibodies are broadly neutralizing in vitro, their precise in vivo significance is unclear and the patients from whom they were derived demonstrate evidence of ongoing viral replication unless treated with ART.

The other major class of protective antibodies are those that participate in ADCC, a form of cell-mediated immunity (**Chap. 360**) in which NK cells that bear Fc receptors bind specific anti-HIV antibodies that then juxtapose the NK cells to and destroy cells expressing HIV antigens. The levels of anti-envelope antibodies capable of mediating ADCC are highest in the earlier stages of HIV infection. Antibodies to both gp120 and gp41 have been shown to participate in ADCC-mediated killing of HIV-infected cells. In vitro, IL-2 can augment ADCC-mediated killing.

In addition to playing a role in host defense, HIV-specific antibodies have also been implicated in disease pathogenesis. Antibodies directed to gp41, when present in low titer, have been shown in vitro to be capable of facilitating infection of cells through an Fc receptor-mediated mechanism known as *antibody enhancement*. Thus, the same regions of the envelope protein of HIV that give rise to antibodies capable of mediating ADCC can also elicit the production of antibodies that can facilitate infection of cells in vitro. In addition, it has been postulated that anti-gp120 antibodies that participate in the ADCC killing of HIV-infected cells might also kill uninfected CD4+ T cells if the uninfected cells had bound free gp120, a phenomenon referred to as *bystander killing*.

One of the most primitive components of the humoral immune system is the complement system (**Chap. 360**). This element of innate immunity consists of ~30 proteins that are found circulating in blood or associated with cell membranes. While HIV alone is capable of directly activating the complement cascade, the resulting lysis is weak due to the presence of host cell regulatory proteins captured in the virion envelope during budding. It is possible that complement-opsonized HIV virions have increased infectivity in a manner analogous to antibody-mediated enhancement.

■ CELLULAR IMMUNE RESPONSE

T cell-mediated immunity plays a major role in host defense against most viral infections (**Chap. 360**) and is thought to be an important component of the host immune response to HIV. T cell immunity can be divided into two major categories: that mediated by *helper/inducer CD4+ T cells* and that mediated by *cytotoxic/immunoregulatory CD8+ T cells*.

HIV-specific CD4+ T cells can be detected in the majority of people with HIV via flow cytometry to measure intracellular cytokine production in response to MHC class II tetramers pulsed with HIV peptides or through lymphocyte proliferation assays utilizing HIV antigens such as p24. These cells likely play a critical role in the orchestration of the immune response to HIV by providing help to HIV-specific B cells and CD8+ T cells. They may also be capable of directly killing HIV-infected cells. HIV-specific CD4+ T cells may be preferential targets of HIV infection by HIV-infected antigen-presenting cells during the generation of an immune response to HIV (Fig. 208-28). However, they also are likely to undergo clonal expansions in response to HIV antigens and thus survive as a population of cells despite the virus. No clear correlations exist between levels of HIV-specific CD4+ T lymphocytes and plasma HIV RNA levels; however, in the setting of high viral loads, CD4+ T cell responses to HIV antigens appear to shift from one of proliferation and IL-2 production to one of IFN- γ production. Thus, while a reverse correlation exists between the level of p24-specific proliferation and levels of plasma HIV viremia, the nature of the causal relationship between these parameters is unclear.

MHC class I-restricted, HIV-specific CD8+ T cells have been identified in the peripheral blood of patients with HIV-1 infection. These cells include CTLs that produce perforins and granzyme, and T cells that can be induced by HIV antigens to express an array of cytokines such as IFN- γ , IL-2, MIP-1 β , and TNF- α . Multiple HIV antigens, including Gag, Env, Pol, Tat, Rev, and Nef, can elicit CD8+ T cell responses. CTLs have been identified in the peripheral blood of patients within weeks of HIV infection and prior to the appearance of plasma virus. The selective pressure they exert on the evolution of the population of circulating viruses reflects their potential role in

control of HIV infection. These CD8+ T lymphocytes, through their HIV-specific antigen receptors, bind to and cause the lytic destruction of target cells bearing autologous MHC class I molecules presenting HIV antigens. Two types of CTL activity can be demonstrated in the peripheral blood or lymph node mononuclear cells of people with HIV. The first type directly lyses appropriate target cells in culture without prior in vitro stimulation (*spontaneous CTL activity*). The other type of CTL activity reflects the *precursor frequency of CTLs* (CTLp); this type of CTL activity can be demonstrated by stimulation of CD8+ T cells in vitro with a mitogen such as phytohemagglutinin or anti-CD3 antibody to expand the population prior to detection.

In addition to CTLs, CD8+ T cells capable of being induced by HIV antigens to express cytokines such as IFN- γ also appear in the setting of HIV-1 infection. It is not clear whether these are the same or different effector pools compared with those cells mediating cytotoxicity; in addition, the relative roles of each in host defense against HIV are not fully understood. It does appear that these CD8+ T cells are driven to in vivo expansion by HIV antigen. There is a direct correlation between levels of CD8+ T cells capable of producing IFN- γ in response to HIV antigens and plasma levels of HIV-1 RNA. Thus, while these cells are clearly induced by HIV-1 infection, in most instances they are not able to effectively control infection. One exception may be a subset of patients who control viral replication in the absence of antiretroviral drugs and are referred to as *elite nonprogressors* (see “Long-Term Survivors, Long-Term Nonprogressors, and Elite Controllers,” above). The peripheral blood of these patients contains a population of CD8+ T cells that undergo substantial in vitro proliferation in response to HIV antigens and express perforin and granzyme.

At least three other forms of cell-mediated immunity to HIV have been described: non-cytolytic CD8+ T cell-mediated suppression of HIV replication, ADCC, and NK cell activity. *Non-cytolytic CD8+ T cell-mediated suppression of HIV replication* refers to the ability of CD8+ T cells from a patient with HIV to inhibit the replication of HIV in tissue culture without killing infected targets. There is no requirement for HLA compatibility between the CD8+ T cells and the HIV-infected cells. This effector mechanism is thus nonspecific and appears to be mediated by soluble factor(s) including the CC-chemokines RANTES (CCL5), MIP-1 α (CCL3), and MIP-1 β (CCL4). These CC-chemokines are potent suppressors of HIV replication and operate at least in part via blockade of the HIV co-receptor (CCR5) for R5 (macrophage-tropic) strains of HIV-1 (see above). ADCC, as described above in relation to humoral immunity, involves the killing of HIV-expressing cells by NK cells armed with specific antibodies directed against HIV antigens. Finally, *NK cells* alone have been shown to be capable of killing HIV-infected target cells in tissue culture. This primitive cytotoxic mechanism of host defense is directed toward non-specific surveillance for neoplastic transformation and viral infection through recognition of altered class I MHC molecules.

DIAGNOSIS AND LABORATORY MONITORING OF HIV INFECTION

The establishment of HIV as the causative agent of AIDS and related syndromes early in 1984 was followed by the rapid development of sensitive screening tests for HIV infection. By March 1985, blood donors in the United States were routinely screened for antibodies to HIV. In 1996, blood banks in the United States added the p24 antigen capture assay to the screening process to help identify the rare, infected individuals who were donating blood in the time (up to 3 months) between infection and the development of antibodies. In 2002, the ability to detect early infection with HIV was further enhanced by the licensure of nucleic acid testing (NAT) as a routine part of blood donor screening. These refinements decreased the interval between infection and detection (window period) from 22 days for antibody testing to 16 days with p24 antigen testing and subsequently to 12 days with NAT. The development of sensitive assays for monitoring levels of plasma viremia ushered in a new era of being able to monitor the progression of HIV disease more closely. Utilization of these tests, coupled with the measurement of levels of CD4+ T lymphocytes in peripheral blood, is an important component of the management of persons with HIV infection.

■ DIAGNOSIS OF HIV INFECTION

The CDC has recommended that screening for HIV infection be performed as a matter of routine health care. The diagnosis of HIV infection depends on the demonstration of antibodies to HIV and/or the direct detection of HIV or one of its components. As noted above, antibodies to HIV generally appear in the circulation 3–12 weeks following infection. In addition to laboratory-based screening tests, several home tests are available.

The standard blood screening tests for HIV infection are based on the detection of antibodies to HIV and/or the p24 antigen (see below) of HIV. A common laboratory-based platform is the ELISA, also referred to as an *enzyme immunoassay* (EIA). This solid-phase assay is an extremely good screening test with a sensitivity of >99.5%. Most diagnostic laboratories use commercial kits that contain antigens from both HIV-1 and HIV-2 and thus can detect antibodies to either. These kits use both natural and recombinant antigens and are continuously updated to increase their sensitivity to newly discovered species, such as group O viruses (Fig. 208-1). The fourth-generation EIA tests combine detection of antibodies to HIV-1 or HIV-2 with detection of the p24 antigen of HIV. EIA tests are generally scored as positive (highly reactive), negative (nonreactive), or indeterminate (partially reactive). While the EIA is an extremely sensitive test, it is not optimal with regard to specificity. This is particularly true in studies of low-risk individuals, such as volunteer blood donors. In this latter population, as few as 10% of EIA-positive individuals are subsequently confirmed to have HIV infection. Among the factors associated with false-positive EIA tests are antibodies to class II antigens (such as may be seen following pregnancy, blood transfusion, or transplantation), autoantibodies, hepatic disease, recent influenza vaccination, acute viral infections, and administration of an HIV vaccine. For these reasons, anyone suspected of having HIV infection based on a positive or inconclusive fourth-generation EIA result should have the result confirmed with a more specific assay such as an HIV-1- or HIV-2-specific antibody immunoassay or a plasma HIV RNA level. One can estimate whether an individual has a recent infection with HIV-1 by comparing the results on a standard EIA test that will score positive for all infected individuals with the results on an assay modified to be less sensitive (“detuned assay”) that will score positive for individuals with established HIV infection and negative for individuals with recent infection. In rare instances, an individual with HIV treated early in the course of infection may revert to a negative EIA. This does *not* indicate clearing of infection; rather, it signifies levels of ongoing exposure to virus or viral proteins insufficient to maintain a measurable antibody response. When these individuals have discontinued therapy, viruses and antibodies have reappeared.

CDC recommendations indicate that a positive fourth-generation assay confirmed by a second HIV-1- or HIV-2-specific immunoassay or a plasma HIV RNA level is adequate for diagnosis. The Western blot, which had previously been used for a confirmatory test, is no longer used for this purpose.

A guideline for the use of these serologic tests in attempting to make a diagnosis of HIV infection is depicted in Fig. 208-31. In patients in whom HIV infection is suspected, the appropriate initial test is a fourth-generation HIV-1/2 antigen antibody immunoassay. If the result is negative, unless there is strong reason to suspect early HIV infection (as in a patient exposed within the previous 3 months), the diagnosis is ruled out and retesting should be performed only as clinically indicated. If the immunoassay is indeterminate or positive, the test should be repeated. If the repeat is negative on two occasions, one can assume that the initial positive reading was due to a

technical error in the performance of the assay and that the patient is negative. If the repeat is indeterminate or positive, one should proceed to an HIV-1/HIV-2 antibody differentiation immunoassay such as the Bio-Rad Geenius. If testing is positive for HIV-1 and/or HIV-2 one may make a diagnosis of HIV-1 and/or HIV-2 infection. If the HIV-1/HIV-2 antibody testing is negative or indeterminate one should proceed to HIV-1 RNA testing with a nucleic acid test (NAT; see below). If the NAT is positive, in the presence of a negative antibody test, one can make a diagnosis of acute HIV-1 infection. If the NAT test is negative for HIV-1 one should consider additional NAT testing for HIV-2. One can conclude a false-positive fourth-generation test in the setting of repeated negative or indeterminate HIV-2/HIV-2 antibody tests in the setting of negative NAT tests.

In addition to these standard laboratory-based assays for detecting antibodies to HIV, a series of point-of-care tests can provide results in 1 to 60 minutes. While the sensitivity and specificity of these tests are generally quite high, it is generally recommended that any positive results be confirmed with standard laboratory testing. Currently in the United States there is one FDA-approved rapid test kit available for use at home (OraQuick). It provides results in approximately 20 minutes. There are several other tests for which the sample is obtained at home and mailed to the lab. A positive result with any of these tests should be followed with confirmatory laboratory testing by a health care professional.

A variety of laboratory tests are available for the direct detection of HIV or its components (Table 208-8). These tests may be of considerable help in making a diagnosis of HIV infection when the antibody determination assays are indeterminate. In addition, the tests detecting levels of HIV RNA can be used to determine prognosis and to assess the response to antiretroviral therapies. The simplest, least expensive, and most rarely used of the direct detection tests is the *p24 antigen capture assay*. This is an EIA-type assay in which the solid phase consists of antibodies to the p24 antigen of HIV. It detects the viral protein p24 in the blood of people with HIV where it exists either as free antigen or complexed to anti-p24 antibodies. It is currently part of the fourth-generation HIV-1/2 antigen antibody immunoassay test recommended for initial screening. Overall, ~30% of individuals with untreated HIV infection have detectable levels of free p24 antigen. This increases to ~50% when samples are treated with a weak acid to dissociate antigen-antibody complexes. Throughout the course of HIV infection, an equilibrium exists between p24 antigen and anti-p24 antibodies. During the first few weeks of infection, before an immune response develops, there is a brisk rise in p24 antigen levels. After the development of anti-p24 antibodies, these levels decline. Late in the course of infection, when

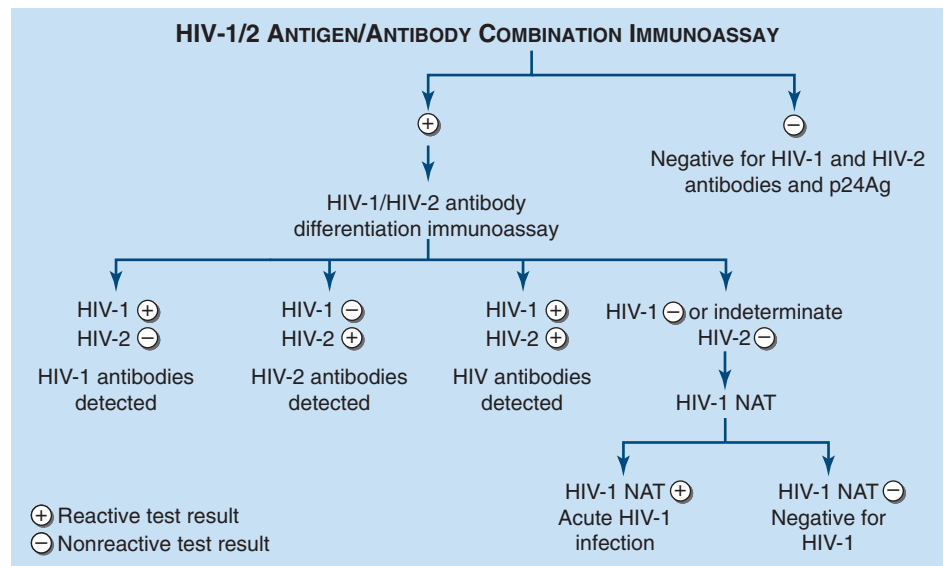


FIGURE 208-31 CDC algorithm for making a diagnosis of HIV infection using tests for antibody, antigen, and RNA. NAT, nucleic acid test. (Adapted from stacks.cdc.gov/view/cdc/23446.)

TABLE 208-8 Characteristics of Tests for Direct Detection of HIV

TEST	TECHNIQUE	SENSITIVITY ^a	COST/TEST ^b
Immune complex–dissociated p24 antigen capture assay	Measurement of levels of HIV-1 core (p24) protein in an EIA-based format following dissociation of antigen–antibody complexes by weak acid treatment	Positive in 50% of patients; detects down to 15 pg/mL of p24 protein	\$1–2
HIV RNA by PCR	Target amplification of HIV-1 RNA via reverse transcription followed by PCR	Reliable to 20 copies/mL of HIV RNA	\$75–150
HIV RNA by bDNA	Measurement of levels of particle-associated HIV RNA in a nucleic acid capture assay employing signal amplification	Reliable to 50 copies/mL of HIV RNA	\$75–150
HIV RNA by TMA	Target amplification of HIV-1 RNA via reverse transcription followed by T7 RNA polymerase	Reliable to 100 copies/mL of HIV RNA	\$225
HIV RNA by NASBA	Isothermal nucleic acid amplification with internal controls	Reliable to 80 copies/mL of HIV RNA	\$75–150

^aSensitivity figures refer to those approved by the U.S. Food and Drug Administration. ^bPrices may be lower in large-volume settings.
Abbreviations: bDNA, branched DNA; cDNA, complementary DNA; EIA, enzyme immunoassay; NASBA, nucleic acid sequence–based amplification; PCR, polymerase chain reaction; TMA, transcription-mediated amplification.

circulating levels of virus are high, p24 antigen levels also increase, particularly when detected by techniques involving dissociation of antigen-antibody complexes. The p24 antigen capture assay has its greatest use as a screening test for HIV infection in patients suspected of having the acute HIV syndrome (see below), as high levels of p24 antigen are present prior to the development of antibodies. Its use as a stand-alone test for routine blood donor screening for HIV infection has been replaced by the use of “fourth-generation” assays that combine antigen and antibody testing or NAT.

The ability to measure and monitor levels of HIV RNA in the plasma of patients with HIV infection has been of extraordinary value in furthering our understanding of the pathogenesis of HIV infection, in monitoring the response to ART, and in providing a diagnostic tool in settings where measurements of anti-HIV antibodies may be misleading, such as in acute infection and neonatal infection. In addition to the commercially available tests for measuring HIV RNA, DNA PCR assays also are employed by research laboratories for making a diagnosis of HIV infection by amplifying HIV proviral DNA from peripheral blood mononuclear cells. The commercially available RNA detection tests have a sensitivity of 20–50 copies of HIV RNA per milliliter of plasma. Research laboratory–based RNA assays can detect as few as one HIV RNA copy per milliliter, while the DNA PCR tests can detect proviral DNA at a frequency of one copy per 10,000–100,000 cells. Thus, these tests are extremely sensitive. One frequent consequence of a high degree of sensitivity is some loss of specificity, and false-positive results have been reported with each of these techniques. For this reason, a positive EIA with a positive HIV RNA assay can be considered the “gold standard” for a diagnosis of HIV infection, and the interpretation of other test results must be done with this in mind.

In the RT-PCR technique, following DNase treatment, a cDNA copy is made of the HIV RNA species present in plasma typically using tRNA_{Lys3} as a primary primer. This is the same host cell primer used during the reverse transcription stage of the viral life cycle (Fig. 208-32). Because HIV is an RNA virus, this will result in the production of DNA copies of the HIV genome in amounts proportional to the amount of HIV RNA present in plasma. This cDNA is then amplified and characterized using standard PCR techniques. In addition to being diagnostic and prognostic tools, RT-PCR and DNA-PCR also are useful for amplifying defined areas of the HIV genome for sequence analysis and have become an important technique for studies of sequence diversity and microbial resistance to antiretroviral agents. In patients with a positive or indeterminate EIA test and an indeterminate Western blot, and in patients in whom serologic testing may be unreliable (such as patients with hypogammaglobulinemia or advanced HIV disease), these tests for quantitating HIV RNA in plasma or detecting proviral DNA in peripheral blood mononuclear cells are valuable tools for making a diagnosis of HIV infection; however, they should be used for diagnosis only when standard serologic testing has failed to provide a definitive result.

LABORATORY MONITORING OF PATIENTS WITH HIV INFECTION

The integration of clinical and laboratory data is essential to optimal management of patients with HIV infection. The close relationship

between clinical manifestations of HIV infection and CD4+ T cell count has made measurement of CD4+ T cell numbers a routine part of the initial evaluation of people with HIV. The discovery of HIV as the cause of AIDS led to the development of sensitive tests that allow one to monitor the levels of HIV in the blood. Determinations of peripheral blood CD4+ T cell counts and measurements of the plasma levels of HIV RNA provide a powerful set of tools for determining prognosis and monitoring response to therapy.

CD4+ T Cell Counts The CD4+ T cell count is the laboratory test generally accepted as the best indicator of the immediate state of immunologic competence of the patient with HIV infection. This measurement has been shown to correlate very well with the level of immunologic competence. Patients with CD4+ T cell counts <200/μL are at high risk of disease from *P. jirovecii*, while patients with CD4+ T cell counts <50/μL are also at high risk of disease from CMV, mycobacteria of the *M. avium* complex (MAC), and/or *T. gondii* (Fig. 208-32). Once the CD4+ T cell count is <200/μL, patients should be placed on a regimen for *P. jirovecii* prophylaxis. Once the count is <50/μL, primary prophylaxis for MAC infection is indicated unless the patient is immediately started on ART. As with any laboratory measurement, one may wish to obtain two determinations prior to any significant changes in patient management based on CD4+ T cell count alone. Patients with HIV infection should have CD4+ T cell measurements performed at the time of diagnosis and every 3–6 months for the first 2 years thereafter. More frequent measurements should be made if a declining trend is noted. For patients who have been on ART for at least 2 years with HIV RNA levels persistently <50 copies/mL and CD4 counts 300–500/μL, monitoring may be decreased to every year. For those with CD4 counts >500/μL, the monitoring of the CD4 count is felt by many to be optional. There are a handful of clinical situations in which the CD4+ T cell count may be misleading. Patients with HTLV-1/HIV co-infection may have elevated CD4+ T cell counts that do not accurately reflect their degree of immune competence. In patients with hypersplenism or those who have undergone splenectomy, and in patients receiving medications that suppress the bone marrow such as IFN-α, the CD4+ T cell percentage may be a more reliable indication of immune function than the CD4+ T cell count. A CD4+ T cell percentage of 15 is comparable to a CD4+ T cell count of 200/μL.

HIV RNA Determinations Facilitated by highly sensitive techniques for the precise quantitation of small amounts of nucleic acids, the measurement of serum or plasma levels of HIV RNA has become an essential component in the monitoring of patients with HIV infection. As discussed in “Diagnosis of HIV Infection,” above, the most used technique is the RT-PCR assay. This assay generates data in the form of number of copies of HIV RNA per milliliter of serum or plasma, and commercial assays can reliably detect as few as 20 copies of HIV RNA per milliliter of plasma. Research-based assays can detect down to one copy per milliliter. While it is common practice to describe levels of HIV RNA below these cut-offs as “undetectable,” this is a term that should be avoided as it is imprecise and leaves the false impression that the level of virus is 0. By utilizing more sensitive,

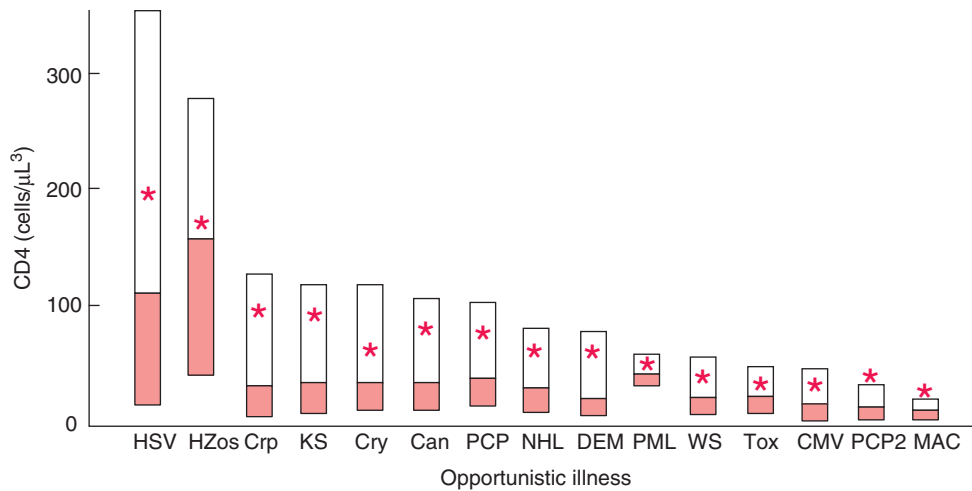


FIGURE 208-32 Relationship between CD4+ T cell counts and the development of opportunistic diseases. Boxplot of the median (line inside the box), first quartile (bottom of the box), third quartile (top of the box), and mean (asterisk) CD4+ lymphocyte count at the time of the development of opportunistic disease. Can, candida esophagitis; CMV, cytomegalovirus infection; Crp, cryptosporidiosis; Cry, cryptococcal meningitis; DEM, AIDS dementia complex; HSV, herpes simplex virus infection; HZos, herpes zoster; KS, Kaposi's sarcoma; MAC, *Mycobacterium avium* complex bacteremia; NHL, non-Hodgkin's lymphoma; PCP, primary *Pneumocystis jirovecii* pneumonia; PCP2, secondary *P. jirovecii* pneumonia; PML, progressive multifocal leukoencephalopathy; Tox, *Toxoplasma gondii* encephalitis; WS, wasting syndrome. (From *Annals of Internal Medicine*, RD Moore, RE Chaisson: *Natural History of Opportunistic Disease in an HIV-Infected Urban Clinical Cohort*. 124(7):633-642, 1996. Copyright © 1996 American College of Physicians. All Rights Reserved. Reprinted with the permission of American College of Physicians, Inc.)

nested PCR techniques and by studying tissue levels of virus as well as plasma levels, HIV RNA can be detected in virtually every patient with HIV infection. There are notable exceptions to this that involve patients who underwent cytoreductive therapy followed by transplant from CCR5Δ32 donors.

Measurements of changes in HIV RNA levels over time have been of great value in delineating the relationship between levels of virus and rates of disease progression (Fig. 208-22), the rates of viral turnover, the relationship between immune system activation and viral replication, and the time to development of drug resistance. HIV RNA measurements are greatly influenced by the state of activation of the immune system and may fluctuate greatly in the setting of secondary infections or immunization. For these reasons, decisions based on HIV RNA levels should never be made on a single determination. Measurements of plasma HIV RNA levels should be made at the time of HIV diagnosis. At the time of diagnosis, ART should be initiated, and HIV RNA levels monitored approximately every 4 weeks until the effectiveness of the therapeutic regimen is determined by the development of a new steady-state level of HIV RNA. In most instances of effective antiretroviral therapy, the plasma level of HIV RNA will drop to <50 copies/mL within 6 months of the initiation of treatment. During therapy, levels of HIV RNA should be monitored every 3–6 months to evaluate the continuing effectiveness of therapy.

HIV Resistance Testing The availability of multiple antiretroviral drugs as treatment options has generated a great deal of interest in the potential for measuring the sensitivity of an individual's HIV viral quasispecies to different antiretroviral agents. HIV resistance testing can be done through either genotypic or phenotypic measurements. In the genotypic assays, sequence analyses of the HIV genomes obtained from patients are compared with sequences of viruses with known antiretroviral resistance profiles. In the phenotypic assays, the *in vivo* growth of patient-derived viral isolates or genetically constructed pseudoviruses is compared with the growth of reference strains of the virus in the presence or absence of different antiretroviral drugs. These tests are quite good at identifying those antiretroviral agents that have been utilized in the past and suggesting agents that may be of future value in a given patient. Resistance testing is recommended at the time of initial diagnosis and, if therapy is not initiated at that time, at the time of initiation of ART. Drug resistance testing is also indicated in the setting of virologic failure and should be performed while the patient is still

on the failing regimen because of the propensity for the pool of HIV quasispecies to rapidly revert to wild-type in the absence of the selective pressures of ART. In the hands of experts, resistance testing enhances the short-term ability to decrease viral load by ~0.5 log compared with changing drugs merely based on drug history. In addition to the use of resistance testing to help in the selection of new drugs in patients with virologic failure, it may also be of value in selecting an initial regimen for treatment of therapy-naïve individuals. This is particularly true in geographic areas with a high level of background resistance. The patient needs to have an HIV-1 RNA level above 500–1000 copies/mL for an accurate resistance determination. Resistance assays lose their consistency at lower levels of plasma viremia.

Co-Receptor Tropism Assays

Following the licensure of maraviroc as the first CCR5 antagonist for the treatment of HIV infection (see below), it became necessary to be able

to determine whether a patient's virus was likely to respond to this treatment. Patients tend to have CCR5-tropic virus early in the course of infection, with a trend toward CXCR4 viruses later in disease. The antiretroviral agent maraviroc is effective only against CCR5-tropic viruses. Because the genotypic determinants of cellular tropism are poorly defined, a phenotypic assay is necessary to determine this property of HIV. The Trofile assay (Monogram Biosciences) is available to make this determination. This assay clones the envelope regions of the patient's virus into an indicator virus that is then used to infect target cells expressing either CCR5 or CXCR4 as their co-receptor. The assay takes weeks to perform and is expensive. Another, less costly option is to obtain a genotypic assay of the V3 region of HIV-1 and then employ a computer algorithm to predict viral tropism from the sequence. While this approach is less expensive than the classic phenotypic assay, there are fewer data to validate its predictive value.

Other Tests A variety of other laboratory tests have been studied as potential markers of HIV disease activity. Among these are quantitative culture of replication-competent HIV from plasma, peripheral blood mononuclear cells, or resting memory CD4+ T cells; circulating levels of β₂-microglobulin, soluble IL-2 receptor, IgA, acid-labile endogenous IFN, or TNF-α; and the presence or absence of activation markers such as CD38, HLA-DR, and PD-1 on CD4+ or CD8+ T cells. Nonspecific serologic markers of inflammation and/or coagulation such as IL-6, D-dimer, and sCD14 have been shown to have a high correlation with all-cause mortality (Table 208-9). While these measurements have value as markers of disease activity and help to increase our understanding of the pathogenesis of HIV disease, they do not currently play a major role in the monitoring of patients with HIV infection.

TABLE 208-9 Association between High-Sensitivity CRP, IL-6, and D-Dimer with All-Cause Mortality in Patients with HIV Infection

MARKER	UNADJUSTED		ADJUSTED	
	ODDS RATIO (FOURTH/FIRST)	P	ODDS RATIO (FOURTH/FIRST)	P
Hs-CRP	2.0	.05	2.8	.03
IL-6	8.3	<.0001	11.8	<.0001
D-dimer	12.4	<.0001	26.5	<.0001

Abbreviations: Hs-CRP, high-sensitivity C-reactive protein; IL-6, interleukin 6.

Source: From LH Kuller et al: *PLoS Med* 5:e203, 2008.

The clinical consequences of HIV infection encompass a spectrum ranging from an acute syndrome associated with primary infection to a prolonged asymptomatic state to advanced disease. It is best to regard HIV disease as beginning at the time of primary infection and progressing through various stages. As mentioned above, active virus replication and progressive immunologic impairment occur throughout the course of HIV infection in most patients. Except for the rare, true, “elite” virus controllers or long-term nonprogressors (see “Long-Term Survivors, Long-Term Nonprogressors, and Elite Controllers,” above), HIV disease in untreated patients inexorably progresses even during the clinically latent stage. Since the mid-1990s, ART has had a major impact on preventing and reversing the progression of disease over extended periods of time in a substantial proportion of adequately treated patients. Today, a person diagnosed with HIV infection and treated with ART has a close to normal life expectancy.

■ ACUTE HIV INFECTION

It is estimated that 50–70% of individuals with HIV infection experience an acute clinical syndrome ~3–6 weeks after primary infection (Fig. 208-33). Varying degrees of clinical severity have been reported, and although it has been suggested that symptomatic seroconversion leading to the seeking of medical attention indicates an increased risk for an accelerated course of disease, there does not appear to be a correlation between the level of the initial burst of viremia in acute HIV infection and the subsequent course of disease. The typical clinical findings in the acute HIV syndrome are listed in Table 208-10; they occur along with a burst of plasma viremia. It has been reported that several symptoms of the acute HIV syndrome (fever, skin rash, pharyngitis, and myalgia) occur less frequently in those infected by injection drug use compared with those infected by sexual contact. The syndrome is typical of an acute viral syndrome and has been likened to acute infectious mononucleosis. Symptoms usually persist for one to several weeks and gradually subside as an immune response to HIV develops and the levels of plasma viremia decrease. Opportunistic infections have been reported during this stage of infection, reflecting the immunodeficiency that results from reduced numbers of CD4+ T cells and likely also from the dysfunction of CD4+ T cells owing to viral protein and endogenous cytokine-induced perturbations of cells (Table 208-5) associated with the extremely high levels of plasma viremia. The Fiebig staging system has been used to describe the different stages of acute HIV infection, ranging from stage 1 (HIV RNA positive alone) to stage VI (HIV RNA and full Western blot positive). A number of immunologic abnormalities accompany the acute HIV syndrome, including multiphasic perturbations of the numbers of circulating lymphocyte subsets. The numbers of total lymphocytes and T cell subsets (CD4+ and CD8+) are initially reduced. An inversion of the CD4+/CD8+ T cell ratio occurs later because of a rise in the number

TABLE 208-10 Clinical Findings in the Acute HIV Syndrome	
General	Neurologic
Fever	Meningitis
Pharyngitis	Encephalitis
Lymphadenopathy	Peripheral neuropathy
Headache/retroorbital pain	Myelopathy
Arthralgias/myalgias	Dermatologic
Lethargy/malaise	Erythematous maculopapular rash
Anorexia/weight loss	Mucocutaneous ulceration
Nausea/vomiting/diarrhea	

Source: Reproduced with permission from B Tindall, DA Cooper: Primary HIV infection: Host responses and intervention strategies. AIDS 5:1, 1991.

of CD8+ T cells. In fact, there may be a selective and transient expansion of CD8+ T cell subsets, as determined by T cell receptor analysis (see above). The total circulating CD8+ T cell count may remain elevated or return to normal; however, CD4+ T cell levels usually remain somewhat depressed, although there may be a slight rebound toward normal. Lymphadenopathy occurs in ~70% of individuals with primary HIV infection. Most patients recover spontaneously from this syndrome, and many are left with only a mildly depressed CD4+ T cell count that remains stable for a variable period before beginning its progressive decline; in some individuals, the CD4+ T cell count returns to the normal range. Approximately 10% of patients manifest a fulminant course of immunologic and clinical deterioration after primary infection, even after the disappearance of initial symptoms. In most patients, primary infection with or without the acute syndrome is followed by a prolonged period of clinical latency or smoldering low disease activity.

■ THE ASYMPTOMATIC STAGE—CLINICAL LATENCY

Although the length of time from initial infection to the development of clinical disease varies greatly, the median time for untreated patients is ~10 years. As emphasized above, HIV disease with active virus replication is ongoing and progressive during this asymptomatic period. The rate of disease progression is directly correlated with HIV RNA levels. Patients with high levels of HIV RNA in plasma progress to symptomatic disease faster than patients with low levels of HIV RNA (Fig. 208-22). Some patients referred to as *long-term nonprogressors* show little if any decline in CD4+ T cell counts over extended periods of time. These patients generally have extremely low levels of HIV RNA; a subset, referred to as *elite nonprogressors*, exhibits HIV RNA levels <50 copies/mL. Certain other patients remain entirely asymptomatic even though their CD4+ T cell counts show a steady progressive decline to extremely low levels. In these patients, the appearance of an opportunistic disease may be the first manifestation of HIV infection. During the asymptomatic period of HIV infection, the average rate of CD4+ T cell decline is ~50/μL per year in an untreated patient. When the CD4+ T cell count falls to <200/μL, the resulting state of immunodeficiency is severe enough to place the patient at high risk for opportunistic infections and neoplasms and, hence, for clinically apparent disease.

■ SYMPTOMATIC DISEASE

Symptoms of HIV disease can appear at any time during the course of HIV infection. Generally, the spectrum of illnesses that one observes changes as the CD4+ T cell count declines. The more severe and life-threatening complications of HIV infection occur in patients with CD4+ T cell counts <200/μL. A diagnosis of AIDS is made in any individual age 6 years and older with HIV infection and a CD4+ T cell count <200/μL (stage 3, Table 208-2) and in anyone with HIV infection who develops one of the HIV-associated diseases considered to be indicative of a severe defect in cell-mediated immunity (Table 208-1). While the causative agents of the secondary infections are characteristically opportunistic organisms such as *P. jirovecii*, atypical mycobacteria, CMV, and other organisms that do not ordinarily cause disease in the absence of a compromised immune system, they also include several common bacterial and mycobacterial pathogens. Following the widespread use of ART and implementation of guidelines for the prevention of opportunistic infections (Table 208-11), the incidence of these

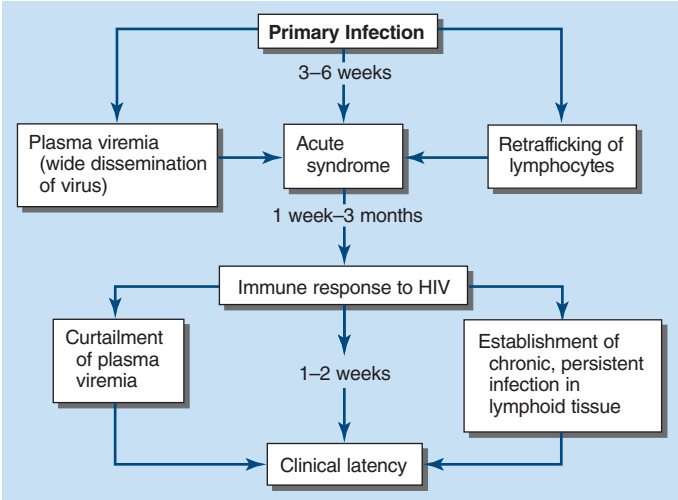


FIGURE 208-33 The acute HIV syndrome. See text for detailed description. (From G Pantaleo, C Graziosi, AS Fauci: The Immunopathogenesis of Human Immunodeficiency Virus Infection. N Engl J Med 328:327, 1993. Copyright © 1993 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)

TABLE 208-11 NIH/CDC/IDSA 2024 Guidelines for the Prevention of Opportunistic Infections in Persons Infected with HIV

PATHOGEN	INDICATIONS	FIRST CHOICE(S)	ALTERNATIVES
<i>Pneumocystis jirovecii</i>	CD4+ T cell count <100–200/μL if HIV RNA level above detection limits or CD4+ T cell count <100/μL regardless of HIV RNA level OR Oropharyngeal candidiasis or Prior bout of PCP May stop prophylaxis if CD4+ T cell count >200/μL for ≥3 months	Trimethoprim-sulfamethoxazole (TMP-SMX), 1 DS tablet qd PO or TMP-SMX, 1 SS tablet qd PO	Dapsone 50 mg bid PO or 100 mg/d PO or Dapsone 50 mg/d PO + Pyrimethamine 50 mg/week PO + Leucovorin 25 mg/week PO or (Dapsone 200 mg PO + Pyrimethamine 75 mg PO + Leucovorin 25 mg weekly PO) or Aerosolized pentamidine, 300 mg via Respigard II nebulizer every month or Atovaquone 1500 mg/d PO or TMP-SMX 1 DS tablet 37/week PO
<i>Mycobacterium tuberculosis</i>		Important to double check drug–drug interactions with ART regimen	
Isoniazid sensitive	Skin test >5 mm or Positive IFN-γ release assay or Prior positive test without treatment or Close contact with case of active pulmonary TB Same with high probability of exposure to drug-resistant TB	Isoniazid 15 mg/kg PO once weekly (900 mg maximum) plus rifapentine (600 mg for 25.1–32 kg; 750 mg for 32.1–49.9 kg; 900 mg ≥ 50.0 kg) Pyridoxine 50 mg PO weekly for 12 weeks or Isoniazid 300 mg PO daily + rifampin 600 mg PO daily + pyridoxine 25–50 mg PO daily for 3 months	Isoniazid 300 mg PO daily + pyridoxine 25–50 mg PO daily for 6–9 months Rifampin 600 mg PO daily for 4 months Isoniazid 300 mg PO daily + rifapentine (<35 kg: 300 mg; 35–45 kg: 450 mg; >45 kg: 600 mg) PO daily + pyridoxine 25–50 mg PO daily for 4 weeks (only for persons on an efavirenz-based ART)
Drug resistant	Consult local public health authorities		
<i>Mycobacterium avium</i> complex	CD4+ T cell count <50/μL unless ART immediately initiated—or if not able to achieve viral suppression or Prior documented disseminated disease May stop prophylaxis once fully suppressive ART in place	Azithromycin 1200 mg weekly PO or 600 mg twice weekly PO or Clarithromycin 500 mg twice daily PO	Rifabutin (dose adjusted based on cART regimen)
<i>Toxoplasma gondii</i>	<i>T. gondii</i> IgG antibody positive and CD4+ T cell count <100/μL or Prior toxoplasmic encephalitis and CD4+ T cell count <200/μL May stop prophylaxis if CD4+ T cell count >200/μL and HIV RNA suppressed for ≥3 months	TMP-SMX 1 DS tablet PO daily or Sulfadiazine 2000–4000 mg in 2–4 divided doses daily PO + Pyrimethamine 25–50 mg/d PO + Leucovorin 10–25 mg/d PO	TMP-SMX 1 DS 3 times weekly PO or TMP-SMX, 1 SS PO daily or Dapsone 50 mg/d PO + Pyrimethamine 50 mg weekly PO + Leucovorin 25 mg weekly PO or (Dapsone 200 mg PO + Pyrimethamine 75 mg PO + Leucovorin 25 mg PO) weekly or Atovaquone 1500 mg PO daily ± (Pyrimethamine 75 mg PO + Leucovorin 10 mg PO) daily or (Clindamycin 600 mg q8h PO + Pyrimethamine 25 mg/d PO + Leucovorin 10mg/d PO) daily
Varicella zoster virus	Significant exposure to chickenpox or shingles in a patient with no history of immunization or prior exposure to either	Varicella zoster immune globulin, IM, within 10 d of exposure (800-843-7477)	Acyclovir 800 mg PO 5 times a day for 5–7 days or Valacyclovir 1 g PO tid for 5–7 days

(Continued)

TABLE 208-11 NIH/CDC/IDSA 2024 Guidelines for the Prevention of Opportunistic Infections in Persons Infected with HIV (Continued)

PATHOGEN	INDICATIONS	FIRST CHOICE(S)	ALTERNATIVES
<i>Cryptococcus neoformans</i>	Prior documented disease—primary prophylaxis not recommended in United States May stop secondary prophylaxis 1 year after initiation of therapy if CD4+ T cell count >100/μL, no evidence of active fungal infection, and HIV RNA levels suppressed for >3 months	Fluconazole 200 mg/d PO	Itraconazole 200 mg/d PO
<i>Histoplasma capsulatum</i>	Prior documented disease or CD4+ T cell count <150 μL and high risk (endemic area or occupational exposure) May stop prophylaxis after 1 year if CD4+ T cell count >150/μL and patient on effective ART for ≥6 months	Itraconazole 200 mg bid PO	Fluconazole 400 mg/d PO
<i>Coccidioides immitis</i>	Prior documented disease or positive serology and CD4+ T cell count <250/μL if from a disease endemic area. (For this indication prophylaxis can be stopped if CD4+ T cell count ≥250 for 6 months.)	Fluconazole 400 mg PO daily	
<i>Talaromyces</i> (formerly <i>Penicillium</i>) <i>marneffei</i>	Prior documented disease Patients with CD4+ T cell counts <100 who live or stay in northern Thailand, Southern China, or Vietnam May stop secondary prophylaxis in patients on ART with CD4+ T cell count >100/μL for ≥6 months	Itraconazole 200 mg/d PO	Fluconazole 400 mg PO once weekly
<i>Salmonella</i> species	Prior recurrent bacteremia—primary prophylaxis not generally recommended	Based on sensitivity of initial infection	
<i>Bartonella</i>	Prior infection May stop after 3–4 months of therapy and with a CD4+ T cell count >200/μL for >3 months and effective ART for at least 6 months	Doxycycline 200 mg/d PO or Azithromycin 1200 mg weekly PO or Clarithromycin 500 mg bid PO	
Cytomegalovirus	Prior end-organ disease—primary prophylaxis not recommended May stop secondary prophylaxis if CD4+ T cell count >100/μL for 6 months and no evidence of active CMV disease Restart if CD4+ T cells <100/μL	Valganciclovir 900 mg once daily PO	
Immunizations Generally Recommended			
Hepatitis B virus	All susceptible (anti-HBc– and anti-HBs–negative) patients	Hepatitis B vaccine: 3 doses	
Hepatitis A virus	All susceptible (anti-HAV–negative) patients	Hepatitis A vaccine: 2 doses	
Influenza virus	All patients annually	Inactivated trivalent influenza virus vaccine 1 dose yearly	
COVID-19	Annually or per updated CDC recommendations (www.cdc.gov/covid/vaccines/immunocompromised-people.html)		
<i>Streptococcus pneumoniae</i>	All patients, preferably before CD4+ T cell count ≤200/μL Reimmunize persons initially immunized at a CD4+ T cell count <100/μL whose CD4+ T cell count then increases to >200/μL	Pneumococcal conjugated vaccine (15 or 20) 0.5 mL IM; if 15-valent is used, follow in 8 weeks or more by pneumococcal polysaccharide vaccine (23)	
Human papillomavirus	All patients 13–26 years of age	HPV vaccine; 3 doses	
Mpox	All patients with HIV who have potential for mpox exposure	Mpox vaccine (MVA-BN); 2 doses	
Meningococcal disease	All patients with HIV infection >18 years	Two doses of quadrivalent meningococcal conjugate vaccine	
Zoster	Age <18 years	Two-dose series of recombinant zoster vaccine (Shingrix) IM 2–6 months apart	

(Continued)

TABLE 208-11 NIH/CDC/IDSA 2024 Guidelines for the Prevention of Opportunistic Infections in Persons Infected with HIV (Continued)			
PATHOGEN	INDICATIONS	FIRST CHOICE(S)	ALTERNATIVES
Recommended for Prevention of Severe or Frequent Recurrences			
Herpes simplex	Frequent/severe recurrences	Valacyclovir 500 mg twice daily PO or Acyclovir 400 mg twice daily PO or Famciclovir 500 mg twice daily PO	
<i>Candida</i> (mucocutaneous)	Frequent/severe recurrences—primary prophylaxis not recommended	Fluconazole 100–200 mg daily PO	Posaconazole oral suspension 400 mg daily PO or Itraconazole 200 mg oral suspension daily or Posaconazole tablet 300 mg PO daily

Abbreviations: ART, antiretroviral therapy; bid, twice daily; cART, combination antiretroviral therapy; DS, double-strength; IFN- γ , interferon γ ; IM, intramuscular; PCP, *Pneumocystis jirovecii* pneumonia; PO, by mouth; qd, daily; SS, single-strength; TB, tuberculosis; tid, three times a day.

secondary infections has decreased dramatically (Fig. 208-34). Overall, the clinical spectrum of HIV disease is constantly changing as patients live longer and new and better approaches to treatment and prophylaxis are developed. In addition to the classic, original AIDS-defining illnesses, patients with HIV infection also have an increase in several serious non-AIDS illnesses, including non-AIDS-related cancers and cardiovascular, renal, and hepatic disease. Non-AIDS events (Table 208-4) now dominate the disease burden for patients

with HIV infection successfully treated with ART. In developed countries, AIDS-related illnesses are responsible for only ~25% of deaths in patients with HIV infection. A similar percentage of deaths are due to non-AIDS-defining malignancies. Cardiovascular disease and liver disease each account for approximately 15% of deaths with 3% of deaths due to suicide. The physician providing care to a patient with HIV infection must be well versed in general internal medicine as well as HIV-related opportunistic diseases. In general, it should be stressed that a key element of treatment of symptomatic complications of HIV disease, whether they are primary or secondary, is achieving good control of HIV replication with the use of ART and instituting primary and secondary prophylaxis for opportunistic infections as indicated.

Diseases of the Respiratory System Acute bronchitis and sinusitis are prevalent during all stages of HIV infection. The most severe cases tend to occur in patients with lower CD4+ T cell counts. Sinusitis presents as fever, nasal congestion, and headache. The diagnosis is made by CT or MRI. The maxillary sinuses are most commonly involved; however, disease is also frequently seen in the ethmoid, sphenoid, and frontal sinuses. While some patients may improve without antibiotic therapy, radiographic improvement is quicker and more pronounced in patients who have received antimicrobial therapy. It is postulated that this high incidence of sinusitis results from an increased frequency of infection with encapsulated organisms such as *H. influenzae* and *S. pneumoniae*. In patients with low CD4+ T cell counts one may see mucormycosis infections of the sinuses. In contrast to the course of this infection in other patient populations, mucormycosis of the sinuses in patients with HIV infection may progress more slowly. In this setting aggressive, frequent local debridement in addition to local and systemic amphotericin B may result in effective treatment.

Pulmonary disease is one of the most frequent complications of HIV infection. The most common manifestation of pulmonary disease is pneumonia. Three of the 10 most common AIDS-defining illnesses are recurrent bacterial pneumonia, tuberculosis, and pneumonia due to the unicellular fungus *P. jirovecii*. Other major causes of pulmonary infiltrates include other mycobacterial infections, other fungal infections, nonspecific interstitial pneumonitis, KS, and lymphoma.

Bacterial pneumonia is seen with an increased frequency in patients with HIV infection, with 0.8–2.0 cases per 100 person-years. Patients with HIV infection are particularly prone to infections with encapsulated organisms. *S. pneumoniae* (Chap. 153) and *H. influenzae* (Chap. 162) are responsible for most cases of bacterial pneumonia in patients with AIDS. This may be a consequence of altered B cell function and/or defects in neutrophil function that may be secondary to HIV disease (see above). Pneumonias due to *S. aureus* (Chap. 152) and *P. aeruginosa* (Chap. 170) also are reported to occur with an increased frequency in patients with HIV infection. *S. pneumoniae* (pneumococcal) infection may be the earliest serious infection to occur in patients with

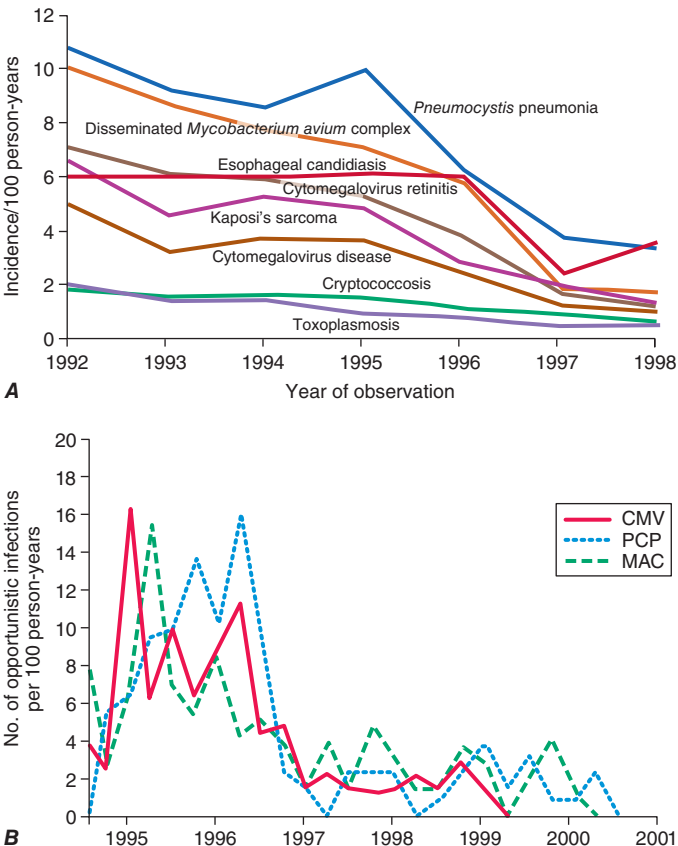


FIGURE 208-34 A. Decrease in the incidence of opportunistic infections and Kaposi's sarcoma in people with HIV with CD4+ T cell counts <100/ μ L from 1992 through 1998. (Reproduced with permission from JE Kaplan et al: Epidemiology of human immunodeficiency virus-associated opportunistic infections in the United States in the era of highly active antiretroviral therapy. Clin Infect Dis 30:S5-14, 2000.) B. Quarterly incidence rates of cytomegalovirus (CMV), *Pneumocystis jirovecii* pneumonia (PCP), and *Mycobacterium avium* complex (MAC) from 1995 to 2001. (Reproduced with permission from Palella FJ Jr et al; HIV Outpatient Study Investigators. Durability and predictors of success of highly active antiretroviral therapy for ambulatory patients with HIV. AIDS 16:1617, 2002.)

HIV disease. This can present as pneumonia, sinusitis, and/or bacteremia. Patients with untreated HIV infection have a 6-fold increase in the incidence of pneumococcal pneumonia and a 100-fold increase in the incidence of pneumococcal bacteremia. Pneumococcal disease may be seen in patients with relatively intact immune systems. In one study, the baseline CD4+ T cell count at the time of a first episode of pneumococcal pneumonia was ~300/ μ L. Of interest is the fact that the inflammatory response to pneumococcal infection appears proportional to the CD4+ T cell count. Due to this high risk of pneumococcal disease, immunization with the conjugated pneumococcal vaccine followed by booster immunization with the 23-valent pneumococcal polysaccharide vaccine is one of the generally recommended prophylactic measures for patients with HIV infection. This is likely most effective if given while the CD4+ T cell count is $>200/\mu$ L and, if given to patients with lower CD4+ T cell counts, should be repeated once the count has been above 200 for 6 months. Although clear guidelines do not exist, it also makes sense to repeat immunization every 5 years. The incidence of bacterial pneumonia is cut in half when patients quit smoking.

Pneumocystis pneumonia (PCP; **Chap. 227**) is caused by the fungus *P. jirovecii* and was once the hallmark of AIDS. It has dramatically declined in incidence following the development of effective prophylactic regimens and the widespread use of ART. It is, however, still the single most common cause of pneumonia in patients with HIV infection in the United States and can be identified as a likely etiologic agent in 25% of cases of pneumonia in patients with HIV infection, with an incidence of about 1 case per 100 person-years. Approximately 30% of cases of HIV-associated PCP occur in patients who are unaware of their HIV status. The risk of PCP is greatest among those who have experienced a previous bout of PCP and those who have CD4+ T cell counts of $<200/\mu$ L. Overall, 79% of patients with PCP have CD4+ T cell counts $<100/\mu$ L and 95% of patients have CD4+ T cell counts $<200/\mu$ L. Recurrent fever, night sweats, thrush, and unexplained weight loss also are associated with an increased incidence of PCP. For these reasons, it is strongly recommended that all patients with CD4+ T cell counts $<200/\mu$ L (or a CD4 percentage <15) receive some form of PCP prophylaxis. The incidence of PCP is approaching zero in patients with known HIV infection receiving appropriate ART and prophylaxis. In the United States, primary PCP is now occurring at a median CD4+ T cell count of 36/ μ L, while secondary PCP is occurring at a median CD4+ T cell count of 10/ μ L.

Patients with PCP generally present with fever and a cough that is usually nonproductive or productive of only scant amounts of white sputum. They may complain of a characteristic retrosternal chest pain that is worse on inspiration and is described as sharp or burning. HIV-associated PCP may have an indolent course characterized by weeks of vague symptoms and should be included in the differential diagnosis of fever, pulmonary complaints, or unexplained weight loss in any patient with HIV infection and <200 CD4+ T cells/ μ L. The most common finding on chest x-ray is either a normal film, if the disease is suspected early, or a faint bilateral interstitial infiltrate. The classic finding of a dense perihilar infiltrate is unusual in patients with AIDS. In patients with PCP who have been receiving aerosolized pentamidine for prophylaxis, one may see an x-ray picture of upper lobe cavitary disease, reminiscent of TB. Other less common findings on chest x-ray include lobar infiltrates and pleural effusions. Thin-section CT may demonstrate a patchy ground-glass appearance. Routine laboratory evaluation is usually of little help in the differential diagnosis of PCP. A mild leukocytosis is common, although this may not be obvious in patients with prior neutropenia. Elevation of lactate dehydrogenase is common. Arterial blood-gases may indicate hypoxemia with a decline in P_{aO_2} and an increase in the arterial-alveolar (a-A) gradient. Arterial blood-gas measurements not only aid in making the diagnosis of PCP but also provide important information for staging the severity of the disease and directing treatment (see below). A definitive diagnosis of PCP requires demonstration of the organism in samples obtained from induced sputum, bronchoalveolar lavage, transbronchial biopsy, or open-lung biopsy. PCR has been used to detect specific DNA sequences for *P. jirovecii* in clinical specimens where histologic examinations have failed to make a diagnosis.

In addition to pneumonia, other clinical problems have been reported in people with HIV as a result of infection with *P. jirovecii*. Otic involvement may be seen as a primary infection, presenting as a polypoid mass involving the external auditory canal. In patients receiving aerosolized pentamidine for prophylaxis against PCP, one may see a variety of extrapulmonary manifestations of *P. jirovecii*. These include ophthalmic lesions of the choroid, a necrotizing vasculitis that resembles Buerger disease, bone marrow hypoplasia, and intestinal obstruction. Other organs that have been involved include lymph nodes, spleen, liver, kidney, pancreas, pericardium, heart, thyroid, and adrenals. Organ infection may be associated with cystic lesions that may appear calcified on CT or ultrasound.

The standard treatment for PCP or disseminated pneumocystosis is trimethoprim-sulfamethoxazole (TMP-SMX). A high (20–85%) incidence of side effects, particularly skin rash and bone marrow suppression, is seen with TMP-SMX in patients with HIV infection. Alternative treatments for mild to moderate PCP include dapsone/trimethoprim, clindamycin/primaquine, and atovaquone. IV pentamidine is the treatment of choice for severe disease in the patient unable to tolerate TMP-SMX. For patients with a $P_{aO_2} <70$ mmHg or with an a-A gradient >35 mmHg, adjunct glucocorticoid therapy should be used in addition to specific antimicrobials. Overall, treatment should be continued for 21 days and followed by secondary prophylaxis. Prophylaxis for PCP is indicated for any individual with HIV who has experienced a prior bout of PCP, any patient with a CD4+ T cell count of $<200/\mu$ L or a CD4 percentage <15 , any patient with unexplained fever for >2 weeks, and any patient with a recent history of oropharyngeal candidiasis. The preferred regimen for prophylaxis is TMP-SMX, one double-strength tablet daily. This regimen also provides protection against toxoplasmosis and some bacterial respiratory pathogens. For patients who cannot tolerate TMP-SMX, alternatives for prophylaxis include dapsone plus pyrimethamine plus leucovorin, aerosolized pentamidine administered by the Respirgard II nebulizer, and atovaquone. Primary or secondary prophylaxis for PCP can be discontinued in those patients treated with ART who maintain good suppression of HIV (<50 copies/mL) and CD4+ T cell counts $>200/\mu$ L for at least 3 months.

M. tuberculosis, once thought to be on its way to extinction in the United States, experienced a resurgence associated with the HIV epidemic (**Chap. 183**). Worldwide, approximately one-third of all AIDS-related deaths are associated with TB, and TB is the primary cause of death for 10–15% of patients with HIV infection. In the United States ~5% of untreated AIDS patients have active TB. Patients with HIV infection are more likely to have active TB by a factor of 100 when compared with an HIV-negative population. For an asymptomatic HIV-negative person with a positive purified protein derivative (PPD) skin test, the risk of reactivation TB is around 1% per year. For the patient with untreated HIV infection, a positive PPD skin test, and no signs or symptoms of TB, the rate of reactivation TB is 7–10% per year. Untreated TB can accelerate the course of HIV infection. Levels of plasma HIV RNA increase in the setting of active TB and decline in the setting of successful TB treatment. Active TB is most common in patients 25–44 years of age, in African Americans and Hispanics, in patients in New York City and Miami, and in patients in developing countries. In these demographic groups, 20–70% of the new cases of active TB are in patients with HIV infection. The epidemic of TB embedded in the epidemic of HIV infection probably represents the greatest health risk to the general public and the health care profession associated with the HIV epidemic. In contrast to infection with atypical mycobacteria such as MAC, active TB often develops relatively early in the course of HIV infection and may be an early clinical sign of HIV disease. In one study, the median CD4+ T cell count at presentation of TB was 326/ μ L.

The clinical manifestations of TB in people with HIV are quite varied and generally show different patterns as a function of the CD4+ T cell count. In patients with relatively high CD4+ T cell counts, the typical pattern of pulmonary reactivation occurs: patients present with fever, cough, dyspnea on exertion, weight loss, night sweats, and a chest x-ray revealing cavitary apical disease of the upper lobes. In patients with lower CD4+ T cell counts, disseminated disease is more common.

In these patients the chest x-ray may reveal diffuse or lower-lobe bilateral reticulonodular infiltrates consistent with miliary spread, pleural effusions, and hilar and/or mediastinal adenopathy. Infection may be present in bone, brain, meninges, GI tract, lymph nodes (particularly cervical lymph nodes), and viscera. Some patients with advanced HIV infection and active TB may have no symptoms of illness, and thus screening for TB should be part of the initial evaluation of every patient with HIV infection. Approximately 60–80% of people with HIV and TB have pulmonary disease, and 30–40% have extrapulmonary disease. Respiratory isolation and a negative-pressure room should be used for patients in whom a diagnosis of pulmonary TB is being considered. This approach is critical to limit nosocomial and community spread of infection. Culture of the organism from an involved site provides a definitive diagnosis. Blood cultures are positive in 15% of patients. This figure is higher in patients with lower CD4+ T cell counts. In the setting of fulminant disease, one cannot rely on the accuracy of a negative PPD skin test to rule out a diagnosis of TB. In addition, IFN- γ release assays may be difficult to interpret due to high backgrounds as a consequence of HIV-associated immune activation. TB is one of the conditions associated with HIV infection for which cure is possible with appropriate therapy. Therapy for TB is generally the same in the patient with HIV as in the HIV-negative patient (Chap. 183). Due to the possibility of multidrug-resistant or extensively drug-resistant TB, drug susceptibility testing should be performed to guide therapy. Due to pharmacokinetic interactions, adjusted doses of rifabutin and/or changes in ART are required when treating TB in the setting of HIV infection. Treatment is most effective in programs that involve directly observed therapy. Initiation of ART and/or anti-TB therapy may be associated with clinical deterioration due to immune reconstitution inflammatory syndrome (IRIS) reactions. These are most common in patients initiating both treatments at the same time, may occur as early as 1 week after initiation of ART therapy, and are seen more frequently in patients with advanced HIV disease. For these reasons it is recommended that initiation of ART be delayed in antiretroviral-naïve patients with CD4 counts ≥ 50 cells/ μ L until 2–8 weeks following the initiation of treatment for TB. For patients with CD4 counts < 50 cells/ μ L the benefits of more immediate ART outweigh the risks of IRIS, and ART should be started as soon as possible in those patients. Effective prevention of active TB can be a reality if the health care professional is aggressive in looking for evidence of latent or active TB by making sure that all patients with HIV infection receive a PPD skin test or evaluation with an IFN- γ release assay. Anergy testing is not of value in this setting. Since these tests rely on the host mounting an immune response to *M. tuberculosis*, patients with CD4+ T cell counts < 200 cells/ μ L should be retested if their CD4+ T cell counts rise to persistently above 200. Patients at risk of continued exposure to TB should be tested annually. People with HIV with a skin-test reaction of > 5 mm, those with a positive IFN- γ release assay, or those who are close household contacts of persons with active TB should receive treatment with 12 weeks of once-weekly isoniazid and rifapentin.

Atypical mycobacterial infections also are seen with an increased frequency in patients with HIV infection. Infections with at least 12 different mycobacteria have been reported, including *M. bovis* and representatives of all four Runyon groups. The most common atypical mycobacterial infection is with *M. avium* or *M. intracellulare* species—the *Mycobacterium avium* complex (MAC). Infections with MAC are seen mainly in patients in the United States and are rare in Africa. It has been suggested that prior infection with *M. tuberculosis* decreases the risk of MAC infection. MAC infections probably arise from organisms that are ubiquitous in the environment, including both soil and water. There is little evidence for person-to-person transmission of MAC infection. The presumed portals of entry are the respiratory and GI tracts. MAC infection is a late complication of HIV infection, occurring predominantly in patients with CD4+ T cell counts of < 50 / μ L. The average CD4+ T cell count at the time of diagnosis is 10/ μ L. The most common presentation is disseminated disease with fever, weight loss, and night sweats. At least 85% of patients with MAC infection are mycobacteremic, and large numbers of organisms can often be demonstrated on bone marrow biopsy. The chest x-ray is abnormal

in ~25% of patients, with the most common pattern being that of a bilateral, lower-lobe infiltrate suggestive of miliary spread. Alveolar or nodular infiltrates and hilar and/or mediastinal adenopathy also can occur. Other clinical findings include endobronchial lesions, abdominal pain, diarrhea, and lymphadenopathy. Anemia and elevated liver alkaline phosphatase are common. The diagnosis is made by the culture of blood or involved tissue. The finding of two consecutive sputum samples positive for MAC is highly suggestive of pulmonary infection. Cultures may take 2 weeks to turn positive. Therapy consists of a macrolide, usually clarithromycin, with ethambutol. Some physicians elect to add a third drug from among rifabutin, ciprofloxacin, or amikacin in patients with extensive disease. Therapy is continued until resolution of clinical signs and symptoms, negative cultures, and CD4+ T cell counts > 100 / μ L for 3–6 months in the setting of ART. Primary prophylaxis for MAC is indicated in patients with HIV infection and CD4+ T cell counts < 50 / μ L not immediately starting ART (Table 208-11). This may be discontinued in patients in whom ART induces a sustained suppression of viral replication regardless of the change in CD4+ T cell count.

Rhodococcus equi is a gram-positive, pleomorphic, acid-fast, non-spore-forming bacillus that can cause pulmonary and/or disseminated infection in patients with advanced HIV infection. Fever and cough are the most common presenting signs. Radiographically one may see cavitary lesions and consolidation. Blood cultures are often positive. Treatment is based on antimicrobial sensitivity testing.

Fungal infections of the lung, in addition to PCP, can be seen in patients with AIDS. Patients with pulmonary cryptococcal disease present with fever, cough, dyspnea, and, in some cases, hemoptysis. A focal or diffuse interstitial infiltrate is seen on chest x-ray in $> 90\%$ of patients. In addition, one may see lobar disease, cavitary disease, pleural effusions, and hilar or mediastinal adenopathy. More than half of patients are fungemic, and 90% of patients have concomitant CNS infection. *Coccidioides immitis* is a mold that is endemic in the southwest United States. It can cause a reactivation pulmonary syndrome in patients with HIV infection. Most patients with this condition will have CD4+ T cell counts < 250 / μ L. Patients present with fever, weight loss, cough, and extensive, diffuse reticulonodular infiltrates on chest x-ray. One may also see nodules, cavities, pleural effusions, and hilar adenopathy. While serologic testing is of value in the immunocompetent host, serologies are negative in 25% of patients with HIV and coccidioidal infection. Invasive aspergillosis is not an AIDS-defining illness and is generally not seen in patients with AIDS in the absence of neutropenia or administration of glucocorticoids. When it does occur, *Aspergillus* infection may have an unusual presentation in the respiratory tract of patients with AIDS, where it gives the appearance of a pseudomembranous tracheobronchitis. Primary pulmonary infection of the lung may be seen with *histoplasmosis*. The most common pulmonary manifestation of histoplasmosis, however, is in the setting of disseminated disease, presumably due to reactivation. In this setting respiratory symptoms are usually minimal, with cough and dyspnea occurring in 10–30% of patients. The chest x-ray is abnormal in ~50% of patients, showing either a diffuse interstitial infiltrate or diffuse small nodules, and the urine will often be positive for *Histoplasma* antigen.

Two forms of *idiopathic interstitial pneumonia* have been identified in patients with HIV infection: lymphoid interstitial pneumonitis (LIP) and nonspecific interstitial pneumonitis (NIP). LIP, a common finding in children, is seen in about 1% of adult patients with untreated HIV infection. This disorder is characterized by a benign infiltrate of the lung and is thought to be part of the polyclonal activation of lymphocytes seen in the context of HIV and EBV infections. Transbronchial biopsy is diagnostic in 50% of the cases, with an open-lung biopsy required for diagnosis in the remainder of cases. This condition is generally self-limited, and no specific treatment is necessary. Severe cases have been managed with brief courses of glucocorticoids. Although rarely a clinical problem since the use of ART, evidence of NIP may be seen in up to half of all patients with untreated HIV infection. Histologically, interstitial infiltrates of lymphocytes and plasma cells in a perivascular and peribronchial distribution are present. When symptomatic, patients present with fever and nonproductive cough occasionally accompanied by mild chest discomfort. Chest x-ray is

usually normal or may reveal a faint interstitial pattern. Like LIP, NIP is a self-limited process for which no therapy is indicated other than appropriate management of the underlying HIV infection. HIV-related pulmonary arterial hypertension (HIV-PAH) is seen in ~0.5% of people with HIV. Patients may present with an array of symptoms including shortness of breath, fatigue, syncope, chest pain, and signs of right-sided heart failure. Chest x-ray reveals dilated pulmonary vessels and right-sided cardiomegaly with right ventricular hypertrophy seen on electrocardiogram. ART does not appear to be of clear benefit, and the prognosis is quite poor with a median survival in the range of 2 years.

Neoplastic diseases of the lung including KS and lymphoma are discussed below in the section on neoplastic diseases.

Diseases of the Cardiovascular System Heart disease is a relatively common postmortem finding in people with HIV (25–75% in autopsy series). The most common form of heart disease is coronary heart disease. In one large series the overall rate of myocardial infarction (MI) was 3.5/1000 patient-years, 28% of these events were fatal, and MI was responsible for 7% of all deaths in the cohort. In patients with HIV infection, cardiovascular disease may be associated with classic risk factors such as smoking, a direct consequence of HIV infection and the accompanying immune activation, or a complication of ART. In general, patients with HIV infection have higher levels of triglycerides, lower levels of high-density lipoprotein cholesterol, and a higher prevalence of smoking than cohorts of individuals without HIV infection. The finding that the rate of cardiovascular disease events was lower in patients on antiretroviral therapy than in those randomized to undergo a treatment interruption identified a clear association between HIV replication and risk of cardiovascular disease. In one study, a baseline CD4+ T cell count of <500/ μ L was found to be an independent risk factor for cardiovascular disease comparable in magnitude to that attributable to smoking. While the precise pathogenesis of this association remains unclear, it is likely related to the immune activation and increased propensity for coagulation seen because of HIV replication. Exposure to HIV protease inhibitors and certain reverse transcriptase inhibitors has been associated with increases in total cholesterol and/or risk of MI. Any increases in the risk of death from MI resulting from the use of certain antiretrovirals must be balanced against the marked increases in overall survival brought about by these drugs.

Another form of heart disease associated with HIV infection is a dilated cardiomyopathy associated with congestive heart failure (CHF) referred to as *HIV-associated cardiomyopathy*. This generally occurs as a late complication of HIV infection and, histologically, displays elements of myocarditis. For this reason, some have advocated it be treated with IV immunoglobulin (IVIg). HIV can be directly demonstrated in cardiac tissue in this setting, and there is debate over whether HIV plays a direct role in this condition. Patients present with typical findings of CHF including edema and shortness of breath. Patients with HIV infection may also develop cardiomyopathy as side effects of IFN- α or nucleoside analogue therapy. These are reversible once therapy is stopped. KS, cryptococcosis, Chagas' disease, and toxoplasmosis can involve the myocardium, leading to cardiomyopathy. In one series, most patients with HIV infection and a treatable myocarditis were found to have myocarditis associated with toxoplasmosis. Most of these patients also had evidence of CNS toxoplasmosis. Thus, MRI or double-dose contrast CT scan of the brain should be included in the workup of any patient with advanced HIV infection and cardiomyopathy.

A variety of other cardiovascular problems are found in patients with HIV infection. Pericardial effusions may be seen in the setting of advanced HIV infection. Predisposing factors include TB, CHF, mycobacterial infection, cryptococcal infection, pulmonary infection, lymphoma, and KS. While pericarditis is quite rare, in one series 5% of patients with HIV disease had pericardial effusions considered to be moderate or severe. Tamponade and death have occurred in association with pericardial KS, presumably owing to acute hemorrhage. Nonbacterial thrombotic endocarditis has been reported and should be considered in patients with unexplained embolic phenomena. IV

pentamidine, when given rapidly, can result in hypotension as a consequence of cardiovascular collapse.

Diseases of the Oropharynx and Gastrointestinal System

Oropharyngeal and GI diseases are common features of HIV infection. They are most frequently due to secondary infections. In addition, oral and GI lesions may occur with KS and lymphoma.

Oral lesions, including *thrush*, *hairy leukoplakia*, and *aphthous ulcers* (Fig. 208-35), are particularly common in patients with untreated HIV infection. Thrush, due to *Candida* infection, and oral hairy leukoplakia, associated with EBV, are usually indicative of fairly advanced immunologic decline; they generally occur in patients with CD4+ T cell counts of <300/ μ L. In one study, 59% of patients with oral candidiasis went on to develop AIDS in the next year. Thrush appears as a white, cheesy exudate, often on an erythematous mucosa in the posterior oropharynx. While most commonly seen on the soft palate, early lesions are often found along the gingival vestibule. The diagnosis is made by direct examination of a scraping for pseudohyphal elements. Culturing is of no diagnostic value, as patients with HIV infection may have a positive throat culture for *Candida* in the absence of thrush. Oral hairy leukoplakia presents as white, frondlike lesions, generally along the lateral borders of the tongue and sometimes on the adjacent buccal mucosa (Fig. 208-35). Despite its name, oral hairy leukoplakia is not considered a premalignant condition. Lesions are associated with florid replication of EBV. While usually more disconcerting as a sign of HIV-associated immunodeficiency than a clinical problem in need of treatment, severe cases of oral hairy leukoplakia have been reported to respond to topical podophyllin or systemic therapy with anti-herpesvirus agents. Aphthous ulcers of the posterior oropharynx also are seen with regularity in patients with untreated HIV infection (Fig. 208-35). These lesions are of unknown etiology and can be quite painful and interfere with swallowing. Topical anesthetics provide immediate symptomatic relief of short duration. The fact that thalidomide is an effective treatment for this condition suggests that the pathogenesis may involve the action of tissue-destructive cytokines. Palatal, glossal, or gingival ulcers may also result from cryptococcal disease or histoplasmosis.

Esophagitis (Fig. 208-36) may present with odynophagia and retrosternal pain. Upper endoscopy is generally required to make an accurate diagnosis. Esophagitis may be due to *Candida*, CMV, or HSV. While CMV tends to be associated with a single large ulcer, HSV infection is more often associated with multiple small ulcers. The esophagus may also be the site of KS and lymphoma. Like the oral mucosa, the esophageal mucosa may have large, painful ulcers of unclear etiology that may respond to thalidomide. While achlorhydria is a common problem in patients with HIV infection, other gastric problems are generally rare. Among the neoplastic conditions involving the stomach are KS and lymphoma.

Infections of the small and large intestine leading to diarrhea, abdominal pain, and occasionally fever are among the most significant GI problems in people with HIV. They include infections with bacteria, protozoa, and viruses.

Bacteria may be responsible for infections of the GI tract in patients with HIV infection. Infections with enteric pathogens such as *Salmonella*, *Shigella*, and *Campylobacter* are more common in men who have sex with men and are often more severe and more apt to relapse in patients with HIV infection. Patients with untreated HIV have approximately a 20-fold increased risk of infection with *S. typhimurium*. They may present with a variety of nonspecific symptoms including fever, anorexia, fatigue, and malaise of several weeks' duration. Diarrhea is common but may be absent. Diagnosis is made by culture of blood and stool. Long-term therapy with ciprofloxacin is the recommended treatment. HIV-infected patients also have an increased incidence of *S. typhi* infection in areas of the world where typhoid is a problem. *Shigella* spp., particularly *S. flexneri*, can cause severe intestinal disease in people with HIV. Up to 50% of patients with GI disease will develop bacteremia. *Campylobacter* infections occur with an increased frequency in patients with HIV infection. While *C. jejuni* is the strain most frequently isolated, infections with many other strains have been



FIGURE 208-35 Various oral lesions in people with HIV. **A.** Thrush. **B.** Hairy leukoplakia. **C.** Aphthous ulcer. **D.** Kaposi's sarcoma.



FIGURE 208-36 Barium swallow of a patient with *Candida* esophagitis. The flow of barium along the mucosal surface is grossly irregular.

reported. Patients usually present with crampy abdominal pain, fever, and bloody diarrhea. Infection may also present as proctitis. Stool examination reveals the presence of fecal leukocytes. Systemic infection can occur, with up to 10% of infected patients exhibiting bacteremia. Most strains are sensitive to erythromycin. Abdominal pain and diarrhea may be seen with MAC infection, and patients with HIV infection may have persistent diarrhea due to enteroaggregative *E. coli*.

Fungal infections including histoplasmosis, coccidioidomycosis, and penicilliosis have all been identified as a cause of fever and diarrhea in patients with HIV infection. Peritonitis has been seen with *C. immitis*.

Cryptosporidia, microsporidia, and *Isospora belli* (Chap. 236) are the most common opportunistic protozoa that infect the GI tract and cause diarrhea in patients with HIV. Cryptosporidial infection may present in a variety of ways, ranging from a self-limited or intermittent diarrheal illness in patients in the early stages of HIV infection to a severe, life-threatening diarrhea in severely immunodeficient individuals. In patients with untreated HIV infection and CD4⁺ T cell counts of <300/ μ L, the incidence of cryptosporidiosis is ~1% per year. In 75% of cases the diarrhea is accompanied by crampy abdominal pain, and 25% of patients have nausea and/or vomiting. Cryptosporidia may also cause biliary tract disease in the patient with HIV, leading to cholecystitis with or without accompanying cholangitis and pancreatitis secondary to papillary stenosis. The diagnosis of cryptosporidial diarrhea is made by stool examination or biopsy of the small intestine. The

diarrhea is noninflammatory, and the characteristic finding is the presence of oocysts that stain with acid-fast dyes. Therapy is predominantly supportive and marked improvements have been reported in the setting of effective ART. Treatment with up to 2000 mg/d of nitazoxanide (NTZ) is associated with improvement in symptoms or a decrease in shedding of organisms in about half of patients. Its overall role in the management of this condition remains unclear. Patients can minimize their risk of developing cryptosporidiosis by avoiding contact with human and animal feces, by not drinking untreated water from lakes or rivers, and by not eating raw shellfish.

Microsporidia are small, unicellular, obligate intracellular parasites that reside in the cytoplasm of enteric cells (Chap. 236). The main species causing disease in humans is *Enterocytozoon bienersi*. The clinical manifestations are similar to those described for cryptosporidiosis and include abdominal pain, malabsorption, diarrhea, and cholangitis. The small size of the organism may make it difficult to detect; however, with the use of chromotrope-based stains, organisms can be identified in stool samples by light microscopy. Definitive diagnosis generally depends on electron-microscopic examination of a stool specimen, intestinal aspirate, or intestinal biopsy specimen. In contrast to cryptosporidiosis, microsporidia have been noted in a variety of extraintestinal locations, including the eye, brain, sinuses, muscle, and liver, and they have been associated with conjunctivitis and hepatitis. The most effective way to deal with microsporidia in a patient with HIV infection is to restore the immune system by treating the HIV infection with ART. Albendazole, 400 mg bid, has been reported to be of benefit in some patients.

I. belli is a coccidian parasite (Chap. 236) most commonly found as a cause of diarrhea in patients from tropical and subtropical regions. Its cysts appear in the stool as large, acid-fast structures that can be differentiated from those of cryptosporidiosis based on size, shape, and number of sporocysts. The clinical syndromes of *Isospora* infection are identical to those caused by cryptosporidiosis. The important distinction is that infection with *Isospora* is generally easy to treat with TMP-SMX. While relapses are common, a thrice-weekly regimen of TMP-SMX appears adequate to prevent recurrence.

CMV colitis (Chap. 200) was once seen as a consequence of advanced immunodeficiency in 5–10% of patients with AIDS. It is much less common with the advent of ART. CMV colitis presents as diarrhea, abdominal pain, weight loss, and anorexia. The diarrhea is usually nonbloody, and the diagnosis is achieved through endoscopy and biopsy. Multiple mucosal ulcerations are seen at endoscopy, and biopsies reveal characteristic intranuclear and cytoplasmic inclusion bodies. Secondary bacteremias may result as a consequence of thinning of the bowel wall. Treatment is with either valganciclovir/ganciclovir or foscarnet for 3–6 weeks. Relapses are common, and maintenance therapy is typically necessary in patients whose HIV infection is poorly controlled. Patients with CMV disease of the GI tract should be carefully monitored for evidence of CMV retinitis.

In addition to disease caused by specific secondary infections, patients with HIV infection may also experience a chronic diarrheal syndrome for which no etiologic agent other than HIV can be identified. This entity is referred to as *AIDS enteropathy* or *HIV enteropathy*. It is most likely a direct result of HIV infection in the GI tract and improves with ART. Histologic examination of the small bowel in these patients reveals low-grade mucosal atrophy with a decrease in mitotic figures, suggesting a hyporegenerative state. Patients often have decreased or absent small-bowel lactase and malabsorption with accompanying weight loss.

The initial evaluation of a patient with HIV infection and diarrhea should include a set of stool examinations, including culture, examination for ova and parasites, and examination for *Clostridium difficile* toxin. Approximately 50% of the time this workup will demonstrate infection with pathogenic bacteria, mycobacteria, or protozoa. If the initial stool examinations are negative, additional evaluation, including upper and/or lower endoscopy with biopsy, will yield a diagnosis of microsporidial or mycobacterial infection of the small intestine ~30% of the time. In patients for whom this diagnostic evaluation is nonrevealing, a presumptive diagnosis of HIV enteropathy can be made if the

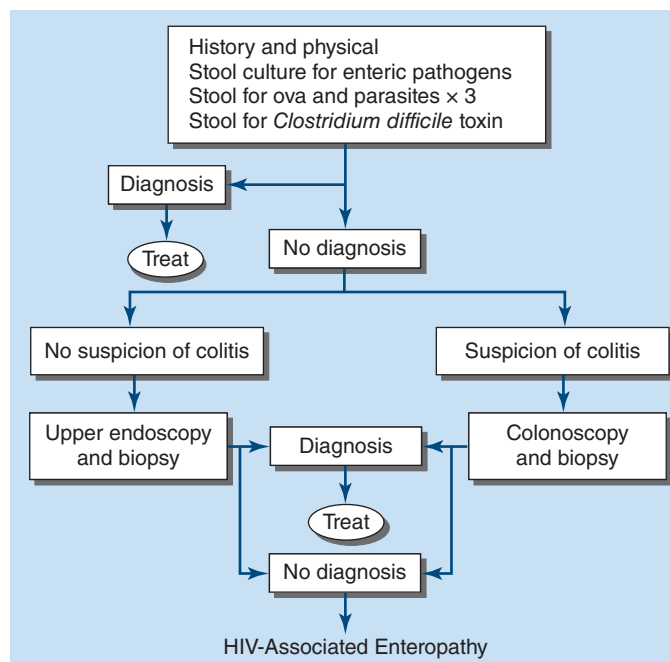


FIGURE 208-37 Algorithm for the evaluation of diarrhea in a patient with HIV infection. HIV-associated enteropathy is a diagnosis of exclusion and can be made only after other, generally treatable, forms of diarrheal illness have been ruled out.

diarrhea has persisted for >1 month. An algorithm for the evaluation of diarrhea in patients with HIV infection is given in Fig. 208-37.

Rectal lesions are common in patients with HIV, particularly the perirectal ulcers and erosions due to the reactivation of HSV (Fig. 208-38). These lesions may appear quite atypical, as denuded skin without vesicles. They typically respond well to treatment with valacyclovir, famciclovir, or foscarnet. Other rectal lesions encountered in patients with HIV infection include condylomata acuminata, KS, and intraepithelial neoplasia (see below).

Hepatobiliary Diseases Diseases of the hepatobiliary system are a major problem in patients with HIV infection. It has been estimated that approximately 15% of the deaths of patients with HIV infection



FIGURE 208-38 Severe, erosive perirectal herpes simplex in a patient with AIDS.

are related to liver disease. While this is predominantly a reflection of the problems encountered in the setting of co-infection with hepatitis B or C, it is also a reflection of the hepatic injury, ranging from hepatic steatosis to hypersensitivity reactions to immune reconstitution, that can be seen in the context of ART.

The prevalence of co-infection with HIV and hepatitis viruses varies by geographic region. In the United States, ~90% of people with HIV have evidence of infection with HBV; 6–14% have chronic HBV infection; 5–50% of patients are co-infected with HCV; and co-infections with hepatitis D, E, and/or G viruses are common. Among IV drug users with HIV infection, rates of HCV infection range from 70 to 95%. HIV infection has a significant impact on the course of hepatitis virus infection. It is associated with approximately a 3-fold increase in the development of persistent hepatitis B surface antigenemia. Patients infected with both HBV and HIV have decreased evidence of inflammatory liver disease. The presumption that this is due to the immunosuppressive effects of HIV infection is supported by the observations that this situation can be reversed, and one may see the development of more severe hepatitis following the initiation of effective ART. In studies of the impact of HIV on HBV infection, 4- to 10-fold increases in liver-related mortality rates have been noted in patients with HIV and active HBV infection compared to rates in patients with either infection alone. There is, however, only a slight increase in overall mortality rate in people with HIV who are also hepatitis B surface antigen (HBsAg)-positive. IFN- α is less successful as treatment for HBV in patients with HIV co-infection. Lamivudine, emtricitabine, adefovir/tenofovir/entecavir, and telbivudine alone or in combination are useful in the treatment of hepatitis B in patients with HIV infection. It is important to remember that all the above-mentioned drugs also have activity against HIV and should not be used alone in patients with HIV infection, to avoid the emergence of quasiespecies of HIV resistant to these drugs. For this reason, the treatment of hepatitis B infection in a patient with HIV infection should always be done in the setting of ART, and alterations in ART need to take into account that the current regimen is also treating HBV. HCV infection is more severe in the patient with HIV infection; it does not appear to affect overall mortality rates in people with HIV when other variables such as age, baseline CD4+ T cell count, and use of ART are taken into account. In the setting of HIV and HCV co-infection, levels of HCV are approximately 10-fold higher than in the HIV-negative patient with HCV infection. There is a 50% higher overall mortality rate with a five-fold increased risk of death due to liver disease in patients chronically infected with both HCV and HIV. Use of directly acting agents for the treatment of HCV leads to cure rates approaching 100%, even in patients with HIV co-infection. Successful treatment of HCV in patients with HIV decreases mortality. Hepatitis A virus infection is not seen with an increased frequency in patients with HIV infection. It is recommended that all patients with HIV infection who have not experienced natural infection be immunized with hepatitis A and/or hepatitis B vaccines. Infection with hepatitis G virus, also known as GB virus C, is seen in ~50% of patients with HIV infection. For reasons that are currently unclear, there are data to suggest that patients with HIV infection co-infected with this virus have a decreased rate of progression to AIDS.

A variety of other infections also may involve the liver. Granulomatous hepatitis may be seen as a consequence of mycobacterial or fungal infections, particularly MAC infection. Hepatic masses may be seen in the context of TB, peliosis hepatis, or fungal infection. Among the fungal opportunistic infections, *C. immitis* and *Histoplasma capsulatum* (Chap. 218) are those most likely to involve the liver. Biliary tract disease in the form of papillary stenosis or sclerosing cholangitis has been reported in the context of cryptosporidiosis, CMV infection, and KS. When no diagnosis can be made, the term *AIDS cholangiopathy* is used. Hemophagocytic lymphohistiocytosis of the liver has been seen in the setting of Hodgkin's disease and may occur prior to diagnosis of the underlying neoplasm.

Many of the drugs used to treat HIV infection are metabolized by the liver and can cause liver injury. Fatal hepatic reactions have been reported with a wide array of antiretrovirals including nucleoside analogues, nonnucleoside analogues, and protease inhibitors. Nucleoside

analogues work by inhibiting DNA synthesis. This can result in toxicity to mitochondria, which can lead to disturbances in oxidative metabolism. This may manifest as hepatic steatosis and, in severe cases, lactic acidosis and fulminant liver failure. It is important to be aware of this condition and to watch for it in patients with HIV infection receiving nucleoside analogues. It is reversible if diagnosed early and the offending agent(s) discontinued. Nevirapine has been associated with at times fatal fulminant and cholestatic hepatitis, hepatic necrosis, and hepatic failure. Indinavir may cause mild to moderate elevations in serum bilirubin in 10–15% of patients in a syndrome similar to Gilbert's syndrome. A similar pattern of hepatic injury may be seen with atazanavir. In the patient receiving ART with an unexplained increase in hepatic transaminases, strong consideration should be given to drug toxicity.

Pancreatic injury is most commonly a consequence of drug toxicity, notably that secondary to pentamidine or dideoxynucleosides. While up to half of patients in some series have biochemical evidence of pancreatic injury, <5% of patients show any clinical evidence of pancreatitis that is not linked to a drug toxicity.

Diseases of the Kidney and Genitourinary Tract Diseases of the kidney or genitourinary tract may be a direct consequence of HIV infection, due to an opportunistic infection or neoplasm, or related to drug toxicity. Overall, microalbuminuria is seen in ~20% of untreated patients with HIV; significant proteinuria is seen in closer to 2%. The presence of microalbuminuria has been associated with an increase in all-cause mortality. *HIV-associated nephropathy* (HIVAN) was first described in IDUs and was initially thought to be IDU nephropathy in patients with HIV infection; it is now recognized as a true direct complication of HIV infection. Although most patients with this condition have CD4+ T cell counts <200/ μ L, HIV-associated nephropathy can be an early manifestation of HIV infection and is also seen in children. More than 90% of reported cases have been in African-American or Hispanic individuals; the disease is not only more prevalent in these populations but also more severe and at one point was the third leading cause of end-stage renal failure among African Americans age 20–64 years in the United States. Proteinuria is the hallmark of this disorder. Edema and hypertension are rare. Ultrasound examination reveals enlarged, hyperechogenic kidneys. A definitive diagnosis is obtained through renal biopsy. Histologically, focal segmental glomerulosclerosis is present in 80%, and mesangial proliferation in 10–15% of cases. Prior to effective antiretroviral therapy, this disease was characterized by relatively rapid progression to end-stage renal disease. Patients with HIV-associated nephropathy should be treated for their HIV infection. Treatment with angiotensin-converting enzyme (ACE) inhibitors and/or prednisone, 60 mg/d, have been reported to be of benefit in some cases. The incidence of this disease in patients receiving adequate ART has not been well defined; however, the impression is that it has decreased in frequency and severity. It is the leading cause of end-stage renal disease in patients with HIV infection.

Among the drugs commonly associated with renal damage in patients with HIV disease are pentamidine, amphotericin, adefovir, didanosine, tenofovir, and foscarnet. Switching from tenofovir disoproxil fumarate (TDF) to tenofovir alafenamide (TAF) may lead to a decrease in renal injury from TDF. TMP-SMX may compete for tubular secretion with creatinine and cause an increase in the serum creatinine level. The pharmacokinetic booster cobicistat, a component of several fixed-drug ART formulations, inhibits renal tubular secretion of creatinine leading to increased serum creatinine levels without a true decline in glomerular filtration rate. Sulfadiazine may crystallize in the kidney and result in an easily reversible form of renal shutdown, while indinavir or atazanavir may form renal calculi. Adequate hydration is the mainstay of treatment and prevention for these latter two conditions.

Genitourinary tract infections are seen with a high frequency in patients with HIV infection; they present with skin lesions, dysuria, hematuria, and/or pyuria and are managed in the same fashion as in patients without HIV infection. Infections with HSV are covered below ("Dermatologic Diseases"). Infections with *T. pallidum*, the etiologic agent of syphilis, play an important role in the HIV epidemic. In HIV-negative individuals, genital syphilitic ulcers as well as the ulcers

of chancroid are major predisposing factors for heterosexual transmission of HIV infection. While most people with HIV and syphilis have a typical presentation of the latter, a variety of formerly rare clinical problems may be encountered in the setting of dual infection. Among them are *lues maligna*, an ulcerating lesion of the skin due to a necrotizing vasculitis and associated with unexplained fever; nephrotic syndrome; and neurosyphilis. The most common presentation of syphilis in the patient with HIV is that of *condylomata lata*, a form of secondary syphilis. Neurosyphilis may be asymptomatic or may present as acute meningitis, neuroretinitis, deafness, or stroke. The rate of neurosyphilis may be as high as 1% in patients with HIV infection, and one should consider a lumbar puncture to look for neurosyphilis in all patients with HIV infection and secondary syphilis. As a consequence of the immunologic abnormalities seen in the setting of HIV infection, diagnosis of syphilis through standard serologic testing may be challenging. On the one hand, a significant number of patients have false-positive Venereal Disease Research Laboratory (VDRL) tests due to polyclonal B cell activation. On the other hand, the development of a new positive VDRL may be delayed in patients with new infections, and the anti-fluorescent treponemal antibody (anti-FTA) test may be negative in the setting of immunodeficiency. Thus, dark-field examination of appropriate specimens should be performed in any patient in whom syphilis is suspected, even if the patient has a negative VDRL. Similarly, any patient with a positive serum VDRL test, neurologic findings, and an abnormal spinal fluid examination should be considered to have neurosyphilis and treated accordingly, regardless of the CSF VDRL result. In any setting, patients treated for syphilis need to be carefully monitored to ensure adequate therapy. Approximately one-third of patients with HIV infection will experience a Jarisch-Herxheimer reaction upon initiation of therapy for syphilis.

Vulvovaginal candidiasis is a common problem in women with HIV infection. Symptoms include pruritus, discomfort, dyspareunia, and dysuria. Vulvar infection may present as a morbilliform rash that may extend to the thighs. Vaginal infection is usually associated with a white discharge, and plaques may be seen along an erythematous vaginal wall. Diagnosis is made by microscopic examination of the discharge for pseudohyphal elements in a 10% potassium hydroxide solution. Mild disease can be treated with topical therapy. More serious disease can be treated with fluconazole. Other causes of vaginitis include *Trichomonas* and mixed bacteria.

Diseases of the Endocrine System and Metabolic Disorders

A variety of endocrine and metabolic disorders are seen in the context of HIV infection. These may be a direct consequence of HIV infection, secondary to opportunistic infections or neoplasms, or related to medication side effects. Between 33% and 75% of patients with HIV infection receiving thymidine analogues or protease inhibitors as a component of ART develop a syndrome often referred to as *lipodystrophy*, consisting of elevations in plasma triglycerides, total cholesterol, and apolipoprotein B, as well as hyperinsulinemia and hyperglycemia. Many of the patients have been noted to have a characteristic set of body habitus changes associated with fat redistribution, consisting of truncal obesity coupled with peripheral wasting (Fig. 208-39). Truncal obesity is apparent as an increase in abdominal girth related to increases in mesenteric fat, a dorsocervical fat pad ("buffalo hump") reminiscent of patients with Cushing's syndrome, and enlargement of the breasts. The peripheral wasting, or lipoatrophy, is particularly

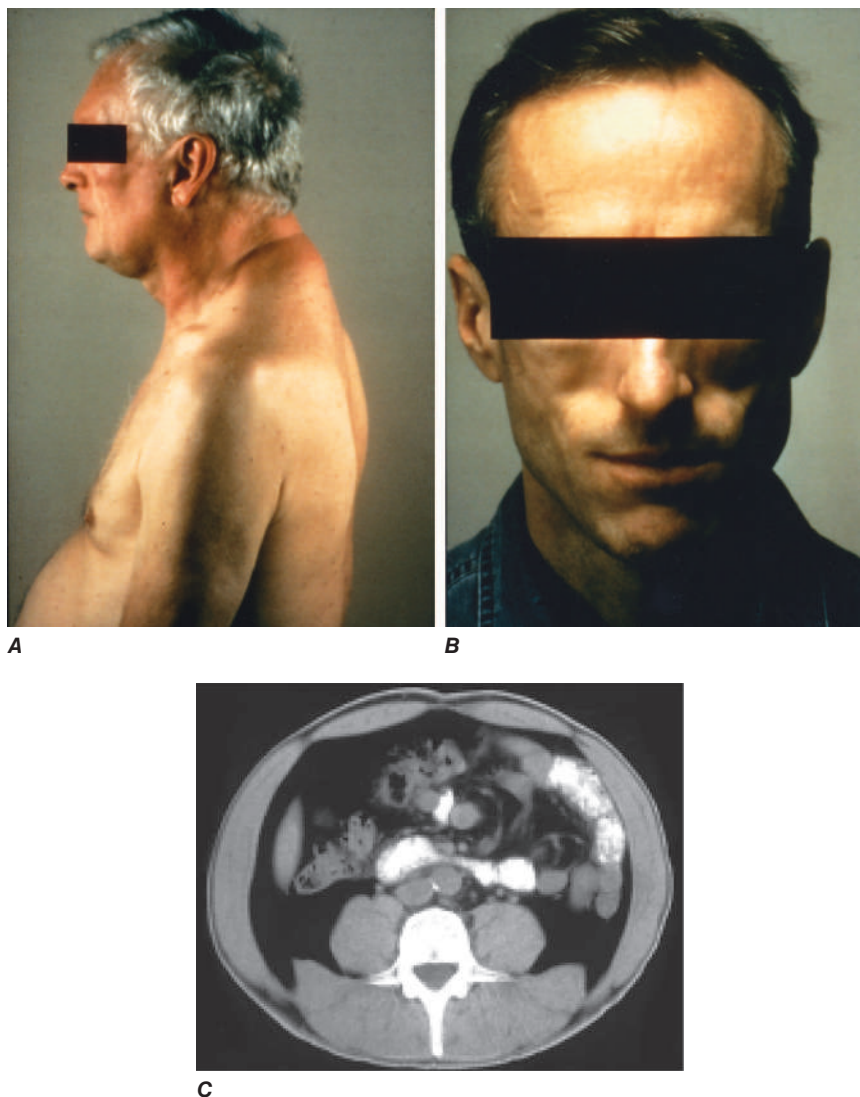


FIGURE 208-39 Characteristics of lipodystrophy. A. Truncal obesity and buffalo hump. **B.** Facial wasting. **C.** Accumulation of intraabdominal fat on CT scan.

noticeable in the face and buttocks and by the prominence of the veins in the legs. These changes may develop at any time ranging from ~6 weeks to several years following the initiation of ART. Approximately 20% of the patients with HIV-associated lipodystrophy meet the criteria for the *metabolic syndrome* as defined by the International Diabetes Federation or the U.S. National Cholesterol Education Program (NCEP) Adult Treatment Panel III. The lipodystrophy syndrome has been reported in association with regimens containing a variety of different drugs, and while initially reported in the setting of protease inhibitor therapy, it appears that similar changes can also be induced by protease-sparing regimens. It has been suggested that the lipoatrophy changes are particularly severe in patients receiving the thymidine analogues stavudine and zidovudine. Current treatment guidelines avoid these drugs and recommend drugs with fewer of these side effects. NCEP guidelines should be followed in the management of these lipid abnormalities (Chap. 419), with a further recommendation that moderate-intensity statin therapy be given to all individuals with HIV infection between the ages of 40 and 75 years unless their 10-year risk of cardiovascular disease is <5%. In all patients, consideration should be given to changing the components of ART with avoidance of thymidine analogues (azidothymidine and stavudine) and offending protease inhibitors. Due to concerns regarding drug interactions, the most utilized lipid-lowering agents in this setting are gemfibrozil and atorvastatin. Lactic acidosis is associated with certain ART medications. This is most often seen with the nucleoside analogue reverse transcriptase inhibitors and can be fatal.

Patients with advanced HIV disease may develop hyponatremia due to the syndrome of inappropriate antidiuretic hormone (vasopressin) secretion (SIADH) because of increased free-water intake and decreased free-water excretion. SIADH is usually seen in conjunction with pulmonary or CNS disease. Low serum sodium may also be due to adrenal insufficiency; a concomitant high serum potassium should alert one to this possibility. Hyperkalemia may be secondary to adrenal insufficiency, HIV nephropathy; or medications, particularly trimethoprim and pentamidine. Hypokalemia may be seen in the setting of tenofovir or amphotericin therapy. Adrenal gland disease may be due to mycobacterial infections, CMV disease, cryptococcal disease, histoplasmosis, or ketoconazole toxicity. Iatrogenic Cushing's syndrome with suppression of the hypothalamic-pituitary-adrenal axis may be seen with the use of local glucocorticoids (injected or inhaled) in patients receiving ritonavir or cobicistat. This is due to inhibition of the hepatic enzyme CYP3A4 by ritonavir leading to prolongation of the glucocorticoid half-life.

Thyroid function may be altered in 10–15% of patients with HIV infection. Both hypo- and hyperthyroidism may be seen. The predominant abnormality is subclinical hypothyroidism. In the setting of ART, up to 10% of patients have been noted to have elevated thyroid-stimulating hormone levels, suggesting that this may be a manifestation of immune reconstitution. Immune-reconstitution Graves' disease may occur as a late (9–48 months) complication of ART. In advanced HIV disease, infection of the thyroid gland may occur with opportunistic pathogens, including *P. jirovecii*, CMV, mycobacteria, *Toxoplasma gondii*, and *Cryptococcus neoformans*. These infections are generally associated with a nontender, diffuse enlargement of the thyroid gland. Thyroid function is usually normal. Diagnosis is made by fine-needle aspirate or open biopsy.

Depending on the severity of disease, HIV infection is associated with *hypogonadism* in 20–50% of men and is lowest in the setting of ART. While this is generally a complication of underlying illness, testicular dysfunction may also be a side effect of ganciclovir therapy. In some surveys, up to two-thirds of patients report decreased libido and one-third complain of erectile dysfunction. Androgen-replacement therapy should be considered in patients with symptomatic hypogonadism. HIV infection does not seem to have a significant effect on the menstrual cycle outside the setting of advanced disease.

Immunologic and Rheumatologic Diseases Immunologic and rheumatologic disorders are common in patients with HIV infection and range from excessive immediate-type hypersensitivity reactions (Chap. 363) to an increase in the incidence of reactive arthritis (Chap. 386) to conditions characterized by a diffuse infiltrative lymphocytosis. The occurrence of these phenomena is an apparent paradox in the setting of the profound immunodeficiency and immunosuppression that characterizes HIV infection and reflects the complex nature of the immune system and its regulatory mechanisms.

Drug allergies are the most significant allergic reactions occurring in patients with HIV and appear to become more common as the disease progresses. They occur in up to 65% of patients who receive therapy with TMP-SMX for PCP. In general, these drug reactions are characterized by erythematous, morbilliform eruptions that are pruritic, tend to coalesce, and are often associated with fever. Nonetheless, ~33% of patients can be maintained on the offending therapy, and thus these reactions are not an immediate indication to stop the drug. Anaphylaxis is extremely rare in patients with HIV infection, and patients who have a cutaneous reaction during a single course of therapy can still be considered candidates for future treatment or prophylaxis with the same agent. The one exception to this is the nucleoside analogue abacavir, where fatal hypersensitivity reactions have been reported with rechallenge. This hypersensitivity is strongly associated with the HLA-B5701 haplotype, and a hypersensitivity reaction to abacavir is an absolute contraindication to future therapy. For other agents, including TMP-SMX, desensitization regimens are moderately successful. While the mechanisms underlying these allergic-type reactions remain unknown, patients with HIV infection have been noted to have elevated IgE levels that increase as the CD4+ T cell count declines. The

numerous examples of patients with multiple drug reactions suggest that a common pathway is involved.

HIV infection shares many similarities with a variety of autoimmune diseases, including a substantial polyclonal B cell activation that is associated with a high incidence of antiphospholipid antibodies, such as anticardiolipin antibodies, VDRL antibodies, and lupus-like anticoagulants. In addition, people with HIV have an increased incidence of antinuclear antibodies. Despite these serologic findings, there is no evidence that people with HIV have an increase in two of the more common autoimmune diseases, i.e., systemic lupus erythematosus and rheumatoid arthritis. In fact, it has been observed that these diseases may be somewhat ameliorated by the concomitant presence of HIV infection, suggesting that an intact CD4+ T cell limb of the immune response plays an integral role in the pathogenesis of these conditions. Similarly, there are anecdotal reports of patients with common variable immunodeficiency (Chap. 362), characterized by hypogammaglobulinemia, who have had a normalization of Ig levels following the development of HIV infection, suggesting a possible role for overactive CD4+ T cell immunity in certain forms of that syndrome. The one autoimmune disease that may occur with an increased frequency in patients with HIV infection is a variant of primary Sjögren's syndrome (Chap. 373) in which patients with HIV infection develop a syndrome consisting of parotid gland enlargement, dry eyes, and dry mouth. This condition is associated with lymphocytic infiltrates of the salivary gland and lung. One also can see peripheral neuropathy, polymyositis, renal tubular acidosis, and hepatitis. In contrast to Sjögren's syndrome, in which the lymphocytic infiltrates are composed predominantly of CD4+ T cells, in patients with HIV infection the infiltrates are composed predominantly of CD8+ T cells. In addition, while patients with Sjögren's syndrome are mainly women who have autoantibodies to Ro and La and who frequently have HLA-DR3 or B8 MHC haplotypes, people with HIV who have this syndrome are usually African-American men who do not have anti-Ro or anti-La and who most often have the HLA-DR5 haplotype. This syndrome appears to be less common with the increased use of effective ART. The term *diffuse infiltrative lymphocytosis syndrome* (DILS) is used to describe this entity and to distinguish it from Sjögren's syndrome.

Approximately one-third of people with HIV experience arthralgias; furthermore, 5–10% are diagnosed as having some form of reactive arthritis, such as Reiter's syndrome or psoriatic arthritis as well as undifferentiated spondyloarthropathy (Chap. 374). These syndromes occur with increasing frequency as the competency of the immune system declines. This association may be related to an increase in the number of infections with organisms that may trigger a reactive arthritis with progressive immunodeficiency or to a loss of important regulatory T cells. Reactive arthritides in people with HIV generally respond well to standard treatment; however, therapy with methotrexate has been associated with an increase in the incidence of opportunistic infections and should be used with caution and only in severe cases.

People with HIV also experience a variety of joint problems without obvious cause that are referred to generically as *HIV- or AIDS-associated arthropathy*. This syndrome is characterized by subacute oligoarticular arthritis developing over a period of 1–6 weeks and lasting 6 weeks to 6 months. It generally involves the large joints, predominantly the knees and ankles, and is nonerosive with only a mild inflammatory response. X-rays are nonrevealing. Nonsteroidal anti-inflammatory drugs are only marginally helpful; however, relief has been noted with the use of intraarticular glucocorticoids. A second form of arthritis also thought to be secondary to HIV infection is called *painful articular syndrome*. This condition, reported as occurring in up to 10% of AIDS patients, presents as an acute, severe, sharp pain in the affected joint. It affects primarily the knees, elbows, and shoulders; lasts 2–24 h; and may be severe enough to require narcotic analgesics. The cause of this arthropathy is unclear; however, it is thought to result from a direct effect of HIV on the joint. This condition is reminiscent of the fact that other lentiviruses, in particular the caprine arthritis-encephalitis virus, are capable of directly causing arthritis.

A variety of other immunologic or rheumatologic diseases have been reported in people with HIV, either *de novo* or in association

1604 with opportunistic infections or drugs. Using the criteria of widespread musculoskeletal pain of at least 3 months' duration and the presence of at least 11 of 18 possible tender points by digital palpation, 11% of a cohort of people with HIV containing 55% IDUs were diagnosed as having *fibromyalgia* (Chap. 385). While the incidence of frank arthritis was less in this population than in other studied populations that consisted predominantly of men who have sex with men, these data support the concept that there are musculoskeletal problems that occur as a direct result of HIV infection. CNS angiitis and polymyositis also have been reported in people with HIV. Septic arthritis is surprisingly rare, especially given the increased incidence of staphylococcal bacteremias seen in this population. When septic arthritis has been reported, it has usually been due to *Staphylococcus aureus*, systemic fungal infection with *C. neoformans*, *Sporothrix schenckii*, or *H. capsulatum* or to systemic mycobacterial infection with *M. tuberculosis*, *M. haemophilum*, *M. avium*, or *M. kansasii*.

Patients with HIV infection treated with ART have been found to have an increased incidence of osteonecrosis or avascular necrosis of the hip and shoulders. In a study of asymptomatic patients, 4.4% were found to have evidence of osteonecrosis on MRI. While precise cause-and-effect relationships have been difficult to establish, this complication has been associated with the use of lipid-lowering agents, systemic glucocorticoids, and testosterone; bodybuilding exercise; alcohol consumption; and the presence of anticardiolipin antibodies. Osteoporosis has been reported in 7% of women with HIV infection, with 41% of women demonstrating some degree of osteopenia. Several studies have documented decreases in bone mineral density of 2–6% in the first 2 years following the initiation of ART. This may be particularly apparent with tenofovir-containing regimens.

Immune Reconstitution Inflammatory Syndrome (IRIS)
Following the initiation of effective ART, a paradoxical worsening of pre-existing, untreated, or partially treated opportunistic infections may be noted. One may also see exacerbations of pre-existing autoimmune conditions or the development of new autoimmune conditions following the initiation of antiretrovirals (Table 208-12). IRIS related to a known pre-existing infection or neoplasm is referred to as *paradoxical IRIS*, while IRIS associated with a previously undiagnosed condition is referred to as *unmasking IRIS*. The term *immune reconstitution disease* (IRD) is sometimes used to distinguish IRIS manifestations related to opportunistic diseases from IRIS manifestations related to autoimmune diseases. IRD is particularly common in patients with underlying untreated mycobacterial or fungal infections. Some form of IRIS is seen in 10–30% of patients following the initiation of ART, depending on the clinical setting, and is most common in patients starting therapy with CD4+ T cell counts <50 cells/μL who have a precipitous drop in HIV RNA levels following the initiation of ART. Signs and symptoms may appear anywhere from 2 weeks to 2 years after the initiation of ART and can include localized lymphadenitis, prolonged fever, pulmonary infiltrates, hepatitis, increased intracranial pressure, uveitis, sarcoidosis, and Graves' disease. The clinical course can be protracted, and severe cases can be fatal. The underlying mechanism appears to be related to a phenomenon similar to type IV hypersensitivity reactions and reflects the immediate improvements in immune function that occur as levels of HIV RNA drop and the

TABLE 208-12 Characteristics of Immune Reconstitution Inflammatory Syndrome (IRIS)

Paradoxical worsening of an existing clinical condition or abrupt appearance of a new clinical finding (unmasking) is seen following the initiation of antiretroviral therapy
Occurs weeks to months following the initiation of antiretroviral therapy
Is most common in patients starting therapy with a CD4+ T cell count <50/μL who experience a precipitous drop in viral load
Is frequently seen in the setting of tuberculosis, particularly when ART is started soon after initiation of anti-TB therapy
Can be fatal

Abbreviations: ART, antiretroviral therapy; TB, tuberculosis.

TABLE 208-13 Causes of Bone Marrow Suppression in Patients with HIV Infection

DISEASES	MEDICATIONS
HIV infection	Zidovudine
Mycobacterial infections	Dapsone
Fungal infections	Trimethoprim-sulfamethoxazole
B19 parvovirus infection	Pyrimethamine
Lymphoma	5-flucytosine
	Ganciclovir
	Interferon α
	Trimetrexate
	Foscarnet

immunosuppressive effects of HIV infection are controlled. In severe cases, the use of immunosuppressive drugs such as glucocorticoids may be required to blunt the inflammatory component of these reactions while specific antimicrobial therapy takes effect.

Diseases of the Hematopoietic System Disorders of the hematopoietic system including lymphadenopathy, anemia, leukopenia, and/or thrombocytopenia are common throughout the course of HIV infection and may be the direct result of HIV, manifestations of secondary infections and neoplasms, or side effects of therapy (Table 208-13). Direct histologic examination and culture of lymph node or bone marrow tissue are often diagnostic. A significant percentage of bone marrow aspirates from patients with HIV infection have been reported to contain lymphoid aggregates, the precise significance of which is unknown. Initiation of ART will lead to reversal of most hematologic complications that are the direct result of HIV infection.

Some patients, otherwise asymptomatic, may develop *persistent generalized lymphadenopathy* as an early clinical manifestation of HIV infection. This condition is defined as the presence of enlarged lymph nodes (>1 cm) in two or more extrainguinal sites for >3 months without an obvious cause. The lymphadenopathy in this setting is due to marked follicular hyperplasia in the node in response to HIV infection. The nodes are generally discrete and freely movable. This feature of HIV disease may be seen at any point in the spectrum of immune dysfunction and is not associated with an increased likelihood of developing AIDS. Paradoxically, a loss in lymphadenopathy or a decrease in lymph node size outside the setting of ART may be a prognostic marker of disease progression. In patients with CD4+ T cell counts >200/μL, the differential diagnosis of lymphadenopathy includes TB, KS, Castleman's disease, and lymphoma. In patients with more advanced disease, lymphadenopathy may also be due to atypical mycobacterial infection, toxoplasmosis, systemic fungal infection, or bacillary angiomatosis. While indicated in patients with CD4+ T cell counts <200/μL, lymph node biopsy is not indicated in patients with early-stage disease unless there are signs and symptoms of systemic illness, such as fever and weight loss, or unless the nodes begin to enlarge, become fixed, or coalesce. Monoclonal gammopathy of unknown significance (MGUS) (Chap. 116), defined as the presence of a serum monoclonal IgG, IgA, or IgM in the absence of a clear cause, has been reported in 3% of patients with HIV infection. The overall clinical significance of this finding in patients with HIV infection is unclear, although it has been associated with other viral infections, non-Hodgkin's lymphoma, and plasma cell malignancy.

Anemia is the most common hematologic abnormality in patients with HIV and, in the absence of a specific treatable cause, is independently associated with a poor prognosis. While generally mild, anemia can be quite severe and require chronic blood transfusions. Among the specific reversible causes of anemia in the setting of HIV infection are drug toxicity, systemic fungal and mycobacterial infections, nutritional deficiencies, and parvovirus B19 infections. The antiretroviral zidovudine may block erythroid maturation prior to its effects on other marrow elements. A characteristic feature of zidovudine therapy is an elevated mean corpuscular volume (MCV). Another drug used in

patients with HIV infection that has a selective effect on the erythroid series is dapson. This drug can cause a serious hemolytic anemia in patients who are deficient in glucose-6-phosphate dehydrogenase and can create a functional anemia in others through induction of methemoglobinemia. Folate levels are usually normal in people with HIV; however, vitamin B₁₂ levels may be depressed as a consequence of achlorhydria or malabsorption. True autoimmune hemolytic anemia is rare, although ~20% of patients with HIV infection may have a positive direct antiglobulin test as a consequence of polyclonal B cell activation. Infection with parvovirus B19 also may cause anemia. It is important to recognize this possibility given the fact that it responds well to treatment with IVIg. Erythropoietin levels in patients with HIV infection and anemia are generally lower than expected given the degree of anemia. Treatment with erythropoietin may result in an increase in hemoglobin levels. An exception to this is a subset of patients with zidovudine-associated anemia in whom erythropoietin levels may be quite high.

During the course of HIV infection, neutropenia may be seen in approximately half of patients. In most instances it is mild; however, it can be severe and can put patients at risk of spontaneous bacterial infections. This is most frequently seen in patients with severely advanced HIV disease and in patients receiving potentially myelosuppressive therapies. In the setting of neutropenia, diseases not commonly seen in patients with HIV, such as aspergillosis or mucormycosis, may occur. Both granulocyte colony-stimulating factor (G-CSF) and GM-CSF increase neutrophil counts in patients with HIV infection regardless of the cause of the neutropenia. Earlier concerns about the potential of these agents to also increase levels of HIV were not confirmed in controlled clinical trials.

Thrombocytopenia may be an early consequence of HIV infection. Approximately 3% of patients with untreated HIV infection and CD4+ T cell counts $\geq 400/\mu\text{L}$ have platelet counts $< 150,000/\mu\text{L}$. For untreated patients with CD4+ T cell counts $< 400/\mu\text{L}$, this incidence increases to 10%. Thrombocytopenia is more common in patients with hepatitis C co-infection, cirrhosis, and/or ongoing high-level HIV replication. Thrombocytopenia is rarely a serious clinical problem in patients with HIV infection and generally responds well to successful ART. Clinically, it resembles the thrombocytopenia seen in patients with idiopathic thrombocytopenic purpura (Chap. 120). Immune complexes containing anti-gp120 antibodies and anti-anti-gp120 antibodies have been noted in the circulation and on the surface of platelets in patients with HIV infection. Patients with HIV infection have also been noted to have a platelet-specific antibody directed toward a 25-kDa component of the surface of the platelet. Other data suggest that the thrombocytopenia in patients with HIV infection may be due to a direct effect of HIV on megakaryocytes. Whatever the cause, it is very clear that the most effective medical approach to this problem has been the use of ART. For patients with platelet counts $< 20,000/\mu\text{L}$, a more aggressive approach combining IVIg or anti-Rh Ig for an immediate response and ART for a more lasting response is appropriate. Rituximab has been used with some success in otherwise refractory cases. Splenectomy is a rarely needed option and is reserved for patients refractory to medical management. Because of the risk of serious infection with encapsulated organisms, all patients with HIV infection about to undergo splenectomy should be immunized with vaccines to prevent disease from *S. pneumoniae*, *N. meningitidis*, and *H. influenzae* type b. It should be noted that, in addition to causing an increase in the platelet count, removal of the spleen will result in an increase in the peripheral blood lymphocyte count, making CD4+ T cell counts unreliable markers of immunocompetence. In this setting, the clinician should rely on the CD4+ T cell percentage for making diagnostic decisions with respect to the likelihood of opportunistic infections. A CD4+ T cell percentage of 15 is approximately equivalent to a CD4+ T cell count of $200/\mu\text{L}$. In patients with early HIV infection, thrombocytopenia has also been reported as a consequence of classic thrombotic thrombocytopenic purpura (Chap. 120). This clinical syndrome, consisting of fever, thrombocytopenia, hemolytic anemia, and neurologic and renal dysfunction, is a rare complication of early HIV infection. As in other settings, the appropriate management is the use of salicylates and plasma

exchange. Other causes of thrombocytopenia include lymphoma, mycobacterial infections, and fungal infections.

The incidence of venous thromboembolic disease such as deep-vein thrombosis or pulmonary embolus is approximately 1% per year in patients with HIV infection. This is approximately 10 times higher than that seen in an age-matched population. Factors associated with an increased risk of clinical thrombosis include age > 45 , history of an opportunistic infection, lower CD4 count, and estrogen use. Abnormalities of the coagulation cascade, including decreased protein S activity, increases in factor VIII, anticardiolipin antibodies, PAR-1 expression on T cells, or lupus-like anticoagulant, have been reported in more than 50% of patients with HIV infection. The clinical significance of this increased propensity toward thromboembolic disease is likely reflected in the observation that elevations in D-dimer are strongly associated with all-cause mortality in patients with HIV infection (Table 208-9).

Dermatologic Diseases Dermatologic problems occur in $> 90\%$ of patients with HIV infection. From the macular, roseola-like rash seen with the acute seroconversion syndrome to extensive end-stage KS, cutaneous manifestations of HIV disease can be seen throughout the course of HIV infection. Among the more common nonneoplastic problems are seborrheic dermatitis, folliculitis, and opportunistic infections. Extrapulmonary pneumocystosis may cause a necrotizing vasculitis. Neoplastic conditions are covered in a separate section below.

Seborrheic dermatitis occurs in 3% of the general population and in up to 50% of patients with HIV infection. Seborrheic dermatitis increases in prevalence and severity as the CD4+ T cell count declines. In patients with HIV, seborrheic dermatitis may be aggravated by concomitant infection with *Pityrosporum*, a yeastlike fungus; use of topical antifungal agents has been recommended in cases refractory to standard topical treatment.

Folliculitis is among the most prevalent dermatologic disorders in patients with HIV infection and is seen in ~20% of patients. It is more common in patients with CD4+ T cell counts < 200 cells/ μL . *Pruritic papular eruption* is one of the most common pruritic rashes in patients with HIV infection. It appears as multiple papules on the face, trunk, and extensor surfaces and may improve with ART. *Eosinophilic pustular folliculitis* is a rare form of folliculitis that is seen with increased frequency in patients with HIV infection. It presents as multiple, urticarial perifollicular papules that may coalesce into plaque-like lesions. Skin biopsy reveals an eosinophilic infiltrate of the hair follicle, which in certain cases has been associated with the presence of a mite. Patients typically have an elevated serum IgE level and may respond to treatment with topical anthelmintics. Pruritus is a common symptom in patients with HIV infection and can lead to *prurigo nodularis*. Patients with HIV infection have also been reported to develop a severe form of *Norwegian scabies* with hyperkeratotic psoriasiform lesions.

Both *psoriasis* and *ichthyosis*, although they are not reported to be increased in frequency, may be particularly severe when they occur in patients with HIV infection. Pre-existing psoriasis may become guttate in appearance and more refractory to treatment in the setting of HIV infection.

Reactivation herpes zoster (shingles) is seen in 10–20% of patients with HIV infection. This reactivation syndrome of varicella-zoster virus indicates a modest decline in immune function and may be the first indication of clinical immunodeficiency. In one series, patients who developed shingles did so an average of 5 years after HIV infection. In a cohort of patients with HIV infection and localized zoster, the subsequent rate of the development of AIDS was 1% per month. In that study, AIDS was more likely to develop if the outbreak of zoster was associated with severe pain, extensive skin involvement, or involvement of cranial or cervical dermatomes. The clinical manifestations of reactivation zoster in patients with HIV, although indicative of immunologic compromise, are not as severe as those seen in other immunodeficient conditions. Thus, while lesions may extend over several dermatomes, involve the spinal cord, and/or be associated with frank cutaneous dissemination, visceral involvement has not been reported. In contrast to patients without a known underlying immunodeficiency

state, patients with HIV infection tend to have recurrences of shingles with a relapse rate of ~20%. Valacyclovir, famciclovir, or acyclovir is the treatment of choice. IV acyclovir can be used in severe cases, and foscarnet may be of value in patients with acyclovir-resistant virus.

Infection with *herpes simplex virus* in people with HIV is associated with recurrent orolabial, genital, and perianal lesions as part of recurrent reactivation syndromes (Chap. 197). As HIV disease progresses and the CD4+ T cell count declines, these infections become more frequent and severe. Lesions often appear as beefy red, are exquisitely painful, and tend to occur high in the gluteal cleft (Fig. 208-38). Perirectal HSV may be associated with proctitis and anal fissures. HSV should be high in the differential diagnosis of any patient with HIV who has a poorly healing, painful perirectal lesion. In addition to recurrent mucosal ulcers, recurrent HSV infection in the form of *herpetic whitlow* can be a problem in patients with HIV infection, presenting with painful vesicles or extensive cutaneous erosion. Valacyclovir, famciclovir, or acyclovir is the treatment of choice in these settings. It is noteworthy that even subclinical reactivation of herpes simplex may be associated with increases in plasma HIV RNA levels.

Diffuse skin eruptions due to *Molluscum contagiosum* may be seen in patients with advanced HIV infection. These flesh-colored, umbilicated lesions resemble those of *Talaromyces* (formerly *Penicillium*) *marneffe* or *Cryptococcosis*. They tend to regress with effective ART and can also be treated with local therapy. Similarly, *condyloma acuminatum* lesions may be more severe and more widely distributed in patients with low CD4+ T cell counts. Imiquimod cream may be helpful in some cases. Atypical mycobacterial infections may present as erythematous cutaneous nodules, as may fungal infections, *Bartonella*, *Acanthamoeba*, and KS. Cutaneous infections with *Aspergillus* have been noted at the site of IV catheter placement.

The skin of patients with HIV infection is often a target organ for drug reactions (Chap. 63). Although most skin reactions are mild and not necessarily an indication to discontinue therapy, some patients may have particularly severe cutaneous reactions to drugs, including erythroderma, *Stevens-Johnson syndrome*, and toxic epidermal necrolysis. This is particularly true for sulfa drugs, nonnucleoside reverse transcriptase inhibitors, abacavir, amprenavir, darunavir, fosamprenavir, and tipranavir. Similarly, patients with HIV infection are often quite photosensitive and burn easily following exposure to sunlight or as a side effect of radiation therapy (Chap. 64).

HIV infection and its treatment may be accompanied by cosmetic changes of the skin that are not of great clinical importance but may be troubling to patients. Yellowing of the nails and straightening of the hair, particularly in African-American patients, have been reported as a consequence of HIV infection. Zidovudine therapy has been associated with elongation of the eyelashes and the development of a bluish discoloration to the nails, again more common in African-American patients. Therapy with clofazimine may cause a yellow-orange discoloration of the skin and urine.

Neurologic Diseases Clinical disease of the nervous system accounts for a significant degree of morbidity in a high percentage of patients with HIV infection (Table 208-14). The neurologic problems that occur in people with HIV may be either primary to the pathogenic processes of HIV infection or secondary to opportunistic infections or neoplasms. Among the more frequent opportunistic diseases that involve the CNS are toxoplasmosis, cryptococcosis, progressive multifocal leukoencephalopathy, and primary CNS lymphoma. Other less common problems include mycobacterial infections; syphilis; and infection with CMV, herpes zoster, HTLV-1, *Trypanosoma cruzi*, or *Acanthamoeba*. Overall, secondary diseases of the CNS have been reported to occur in approximately one-third of patients with AIDS. These data antedate the widespread use of ART, and this frequency is considerably lower in patients with suppressed viral replication. Primary processes related to HIV infection of the nervous system are reminiscent of those seen with other lentiviruses, such as the maedi-visna virus of sheep.

Neurologic problems directly attributable to HIV occur throughout the course of infection and may be inflammatory, demyelinating,

TABLE 208-14 Neurologic Diseases in Patients with HIV Infection	
HIV-1 infection	Opportunistic infections
Aseptic meningitis	Toxoplasmosis
HIV-associated neurocognitive disorders (HAND), including HIV encephalopathy/AIDS dementia complex	Cryptococcosis
Myelopathy	Progressive multifocal leukoencephalopathy
Vacuolar myelopathy	Cytomegalovirus
Pure sensory ataxia	Syphilis
Paresthesia/dysesthesia	<i>Mycobacterium tuberculosis</i>
Peripheral neuropathy	HTLV-1 infection
Acute inflammatory demyelinating polyneuropathy (Guillain-Barré syndrome)	Amebiasis
Chronic inflammatory demyelinating polyneuropathy (CIPD)	Neoplasms
Mononeuritis multiplex	Primary CNS lymphoma
Distal symmetric polyneuropathy	Kaposi's sarcoma
Myopathy	

or degenerative in nature. The term *HIV-associated neurocognitive disorders* (HAND) is used to describe a spectrum of disorders that range from asymptomatic neurocognitive impairment (ANI) to minor neurocognitive disorder (MND) to clinically severe dementia. The most severe form, *HIV-associated dementia* (HAD), also referred to as the *AIDS dementia complex*, or *HIV encephalopathy*, is considered an AIDS-defining illness. Many HIV-infected patients have some neurologic problem during the course of their disease. Even in the setting of suppressive ART, approximately 50% of people with HIV can be shown to have mild to moderate neurocognitive impairment using sensitive neuropsychiatric testing. As noted in the section on pathogenesis, damage to the CNS may be a direct result of viral infection of the CNS macrophages or glial cells or may be secondary to the release of neurotoxins and potentially toxic cytokines such as IL-1 β , TNF- α , IL-6, and TGF- β . It has been reported that people with HIV who have the E4 allele for apoE are at increased risk for AIDS encephalopathy and peripheral neuropathy. Virtually all patients with HIV infection have some degree of nervous system involvement with the virus. This is evidenced by the fact that CSF findings are abnormal in ~90% of untreated patients, even during the asymptomatic phase of HIV infection. CSF abnormalities include pleocytosis (50–65% of patients), detection of viral RNA (~75%), elevated CSF protein (35%), and evidence of intrathecal synthesis of anti-HIV antibodies (90%). It is important to point out that evidence of infection of the CNS with HIV does not imply impairment of cognitive function. The neurologic function of a person with HIV should be considered normal unless clinical signs and symptoms suggest otherwise.

Aseptic meningitis may occur at any time in the course of HIV infection; however, it is rare following the development of AIDS. This suggests that clinical aseptic meningitis in the context of HIV infection is an immune-mediated disease. In the setting of acute primary infection, patients may experience a syndrome of headache, photophobia, and meningismus. Rarely, an acute encephalopathy due to encephalitis may occur. Cranial nerve involvement may be seen, predominantly cranial nerve VII but occasionally V and/or VIII. CSF findings include a lymphocytic pleocytosis, elevated protein level, and normal glucose level. This syndrome, which cannot be clinically differentiated from other viral meningitides (Chap. 143), usually resolves spontaneously within 2–4 weeks; however, in some patients, signs and symptoms may become chronic.

Fungal meningitis is the leading infectious cause of meningitis in patients with AIDS (Chaps. 143 and 144). While the vast majority of these are due to *C. neoformans*, up to 12% may be due to *C. gattii*. Cryptococcal meningitis is the initial AIDS-defining illness in ~2% of patients and generally occurs in patients with CD4+ T cell counts <100/ μ L. Cryptococcal meningitis is particularly common in untreated patients with AIDS in Africa, occurring in ~5% of patients. Most patients present with a picture of subacute meningoencephalitis with

fever, nausea, vomiting, altered mental status, headache, and meningeal signs. The incidence of seizures and focal neurologic deficits is low. The CSF profile may be normal or may show only modest elevations in WBC or protein levels and decreases in glucose. The opening pressure in the CSF is usually elevated. In addition to meningitis, patients may develop cryptococcomas and cranial nerve involvement. Approximately one-third of patients also have pulmonary disease. Uncommon manifestations of cryptococcal infection include skin lesions that resemble *molluscum contagiosum*, lymphadenopathy, palatal and glossal ulcers, arthritis, gastroenteritis, myocarditis, and prostatitis. The prostate gland may serve as a reservoir for smoldering cryptococcal infection. The diagnosis of cryptococcal meningitis is made by identification of organisms in spinal fluid with India ink examination or by the detection of cryptococcal antigen. Blood cultures for fungus are often positive. A biopsy may be needed to make a diagnosis of CNS cryptococcoma and to distinguish inadequately treated infection from immune reconstitution syndrome. Initial treatment is with IV amphotericin B 0.7 mg/kg daily, or liposomal amphotericin 3–4 mg/kg daily, with flucytosine 25 mg/kg qid for at least 2 weeks if possible. Decreases in renal function in association with amphotericin can lead to increases in flucytosine levels and subsequent bone marrow suppression. Therapy continues with amphotericin alone until the CSF culture turns negative followed by fluconazole 800 mg/d PO for 8 weeks, and then fluconazole 200 mg/d until the CD4+ T cell count has increased to >200 cells/ μ L for 6 months in response to ART. Repeated lumbar puncture may be required to manage increased intracranial pressure. Symptoms may recur with initiation of ART as an immune reconstitution syndrome (see above). For this reason, it is recommended that patients receive 4–6 weeks of antifungal therapy prior to initiation of ART. Other fungi that may cause meningitis in patients with HIV infection are *C. immitis* and *H. capsulatum*. Meningoencephalitis has also been reported due to *Acanthamoeba* or *Naegleria*.

HIV-associated dementia consists of a constellation of signs and symptoms of CNS disease. While this is generally a late complication of HIV infection that progresses slowly over months, it can be seen in patients with CD4+ T cell counts >350 cells/ μ L. A major feature of this entity is the development of dementia, defined as a decline in cognitive ability from a previous level. It may present as impaired ability to concentrate, increased forgetfulness, difficulty reading, or increased difficulty performing complex tasks. Initially these symptoms may be indistinguishable from findings of situational depression or fatigue. In contrast to “cortical” dementia (such as Alzheimer’s disease), aphasia, apraxia, and agnosia are uncommon, leading some investigators to classify HIV encephalopathy as a “subcortical dementia” characterized by defects in short-term memory and executive function (see below). In addition to dementia, patients with HIV encephalopathy may also have motor and behavioral abnormalities. Among the motor problems are unsteady gait, poor balance, tremor, and difficulty with rapid alternating movements. Increased tone and deep tendon reflexes may be found in patients with spinal cord involvement. Late stages may be complicated by bowel and/or bladder incontinence. Behavioral problems include apathy, irritability, and lack of initiative, with progression to a vegetative state in some instances. Some patients develop a state of agitation or mild mania. These changes usually occur without significant changes in level of alertness. This contrasts with the finding of somnolence in patients with dementia due to toxic/metabolic encephalopathies.

HIV-associated dementia is the initial AIDS-defining illness in ~3% of patients with HIV infection and thus only rarely precedes clinical evidence of immunodeficiency. Clinically significant encephalopathy eventually develops in ~25% of untreated patients with AIDS. As immunologic function declines, the risk and severity of HIV-associated dementia increases. Autopsy series suggest that 80–90% of patients with HIV infection have histologic evidence of CNS involvement. Several classification schemes have been developed for grading HIV encephalopathy; a commonly used clinical staging system is outlined in Table 208-15.

The precise cause of HIV-associated dementia remains unclear, although the condition is thought to be a result of a combination of

TABLE 208-15 Clinical Staging of HAND According to Frascati Criteria

STAGE	NEUROCOGNITIVE STATUS ^a	FUNCTIONAL STATUS ^b
Asymptomatic	1 SD below mean in 2 cognitive domains	No impairments in activities of daily living
Mild neurocognitive disorder	1 SD below mean in 2 cognitive domains	Impairments in activities of daily living
HIV-associated dementia	2 SD below mean in 2 cognitive domains	Notable impairments in activities of daily living

^aNeurocognitive testing should include assessment of at least 5 domains, including attention/information processing, language, abstraction/executive function, complex perceptual motor skills, memory (including learning and recall), simple motor skills, or sensory perceptual skills. Appropriate norms must be available to establish the number of domains in which performance is below 1 SD. ^bFunctional status is typically assessed by self-reporting but might be corroborated by a collateral source. No agreed measures exist for HIV-associated neurocognitive disorder criteria. Note that, for diagnosis of HIV-associated neurocognitive disorder, other causes of dementia must be ruled out and potential confounding effects of substance use or psychiatric illness should be considered.

Source: Adapted from A Antinori et al: Neurology 69:1789, 2007.

direct effects of HIV on the CNS and associated immune activation. HIV has been found in the brains of patients with HIV encephalopathy by Southern blot, in situ hybridization, PCR, and electron microscopy. Multinucleated giant cells, macrophages, and microglial cells appear to be the main cell types harboring virus in the CNS. Histologically, the major changes are seen in the subcortical areas of the brain and include pallor and gliosis, multinucleated giant cell encephalitis, and vacuolar myelopathy. Less commonly, diffuse or focal spongiform changes occur in the white matter. Areas of the brain involved in motor function, language, and judgment are most severely affected.

There are no specific criteria for a diagnosis of HIV-associated dementia, and this syndrome must be differentiated from other diseases that affect the CNS of patients with HIV (Table 208-14). The diagnosis of dementia depends on demonstrating a decline in cognitive function. This can be accomplished objectively with the use of a Mini-Mental State Examination (MMSE) in patients for whom prior scores are available. For this reason, it is advisable for all patients with a diagnosis of HIV infection to have a baseline MMSE. However, changes in MMSE scores may be absent in patients with mild HIV encephalopathy. Imaging studies of the CNS, by either MRI or CT, often demonstrate evidence of cerebral atrophy (Fig. 208-40). MRI may also reveal small areas of increased density on T2-weighted images. Lumbar puncture is an important element of the evaluation of patients with



FIGURE 208-40 AIDS dementia complex. Postcontrast CT scan through the lateral ventricles of a 47-year-old man with AIDS, altered mental status, and dementia. The lateral and third ventricles and the cerebral sulci are abnormally prominent. Mild white matter hypodensity is seen adjacent to the frontal horns of the lateral ventricles.

HIV infection and neurologic abnormalities. It is generally most helpful in ruling out or making a diagnosis of opportunistic infections. In HIV encephalopathy, patients may have the nonspecific findings of an increase in CSF cells and protein level. While HIV RNA can often be detected in the spinal fluid and HIV can be cultured from the CSF, this finding is not specific for HIV encephalopathy. There appears to be no correlation between the presence of HIV in the CSF and the presence of HIV encephalopathy. Elevated levels of monocyte chemoattractant protein-1 (MCP-1), β_2 -microglobulin, neopterin, and quinolinic acid (a metabolite of tryptophan reported to cause CNS injury) have been noted in the CSF of patients with HIV encephalopathy. These findings suggest that these factors as well as inflammatory cytokines may be involved in the pathogenesis of this syndrome.

Combination antiretroviral therapy is of benefit in patients with HIV-associated dementia. Improvement in neuropsychiatric test scores has been noted for both adult and pediatric patients treated with antiretrovirals. The rapid improvement in cognitive function noted with the initiation of ART suggests that at least some component of this problem is quickly reversible, again supporting at least a partial role of soluble mediators in the pathogenesis. It should also be noted that these patients have an increased sensitivity to the side effects of neuroleptic drugs. The use of these drugs for symptomatic treatment is associated with an increased risk of extrapyramidal side effects; therefore, patients with HIV encephalopathy who receive these agents must be monitored carefully. It is felt by many physicians that the decrease in the prevalence of severe cases of HAND brought about by ART has resulted in an increase in the prevalence of milder forms of this disorder.

Seizures may be a consequence of opportunistic infections, neoplasms, or HIV encephalopathy (Table 208-16). The seizure threshold is often lower than normal in patients with advanced HIV infection due in part to the frequent presence of electrolyte abnormalities. Seizures are seen in 15–40% of patients with cerebral toxoplasmosis, 15–35% of patients with primary CNS lymphoma, 8% of patients with cryptococcal meningitis, and 7–50% of patients with HIV encephalopathy. Seizures may also be seen in patients with CNS tuberculosis, aseptic meningitis, and progressive multifocal leukoencephalopathy. Seizures may be the presenting clinical symptom of HIV disease. In one study of 100 patients with HIV infection presenting with a first seizure, cerebral mass lesions were the most common cause, responsible for 32 of the 100 new-onset seizures. Of these 32 cases, 28 were due to toxoplasmosis and 4 to lymphoma. HIV encephalopathy accounted for an additional 24 new-onset seizures. Cryptococcal meningitis was the third most common diagnosis, responsible for 13 of the 100 seizures. In 23 cases, no cause could be found, and it is possible that these cases represent a subcategory of HIV encephalopathy. Of these 23 cases, 16 (70%) had 2 or more seizures, suggesting that anticonvulsant therapy is indicated in all patients with HIV infection and seizures unless a rapidly correctable cause is found. Due to a variety of drug–drug interactions between antiseizure medications and antiretrovirals, drug levels need to be monitored carefully.

Patients with HIV infection may present with focal neurologic deficits from a variety of causes. The most common causes are toxoplasmosis, progressive multifocal leukoencephalopathy, and CNS lymphoma. Other causes include cryptococcal infections (discussed above; also Chap. 221), stroke, and reactivation of Chagas’ disease.

TABLE 208-16 Causes of Seizures in Patients with HIV Infection		
DISEASE	OVERALL CONTRIBUTION TO FIRST SEIZURE, %	FRACTION OF PATIENTS WHO HAVE SEIZURES, %
HIV encephalopathy	24–47	7–50
Cerebral toxoplasmosis	28	15–40
Cryptococcal meningitis	13	8
Primary central nervous system lymphoma	4	15–30
Progressive multifocal leukoencephalopathy	1	20

Source: From DM Holtzman et al: Am J Med 87:173, 1989.

Toxoplasmosis (Chap. 235) has been one of the most common causes of secondary CNS infections in patients with AIDS, but its incidence is decreasing in the era of ART. It is most common in patients from the Caribbean and from France, where the seroprevalence of *T. gondii* is around 50%. This figure is closer to 15% in the United States. Toxoplasmosis is generally a late complication of HIV infection and usually occurs in patients with CD4+ T cell counts <200/ μ L. Cerebral toxoplasmosis is thought to represent a reactivation of latent tissue cysts. It is 10 times more common in patients with antibodies to the organism than in patients who are seronegative. Patients diagnosed with HIV infection should be screened for IgG antibodies to *T. gondii* during the time of their initial workup. Those who are seronegative should be counseled about ways to minimize the risk of primary infection including avoiding the consumption of undercooked meat and careful hand washing after contact with soil or changing the cat litter box. The most common clinical presentation of cerebral toxoplasmosis in patients with HIV infection is fever, headache, and focal neurologic deficits. Patients may present with seizure, hemiparesis, or aphasia as a manifestation of these focal deficits or with a picture more influenced by the accompanying cerebral edema and characterized by confusion, dementia, and lethargy, which can progress to coma. The diagnosis is usually suspected on the basis of MRI findings of multiple lesions in multiple locations, although in some cases only a single lesion is seen. Pathologically, these lesions generally exhibit inflammation and central necrosis and, as a result, demonstrate ring enhancement on contrast MRI (Fig. 208-41) or, if MRI is unavailable or contraindicated, on double-dose contrast CT. There is usually evidence of surrounding edema. In addition to toxoplasmosis, the differential diagnosis of single or multiple enhancing mass lesions in the patient with HIV includes primary CNS lymphoma and, less commonly, TB or fungal or bacterial abscesses. The definitive diagnostic procedure is brain biopsy. However, given the morbidity rate that can accompany this procedure, it is usually reserved for the patient who has failed 2–4 weeks of empiric therapy for toxoplasmosis. If the patient is seronegative for *T. gondii*, the likelihood that a mass lesion is due to toxoplasmosis is <10%. In that setting, one may choose to be more aggressive and perform a brain biopsy sooner. Standard treatment is sulfadiazine and pyrimethamine with leucovorin as needed for a minimum of 4–6 weeks. Alternative therapeutic regimens include clindamycin in combination with pyrimethamine; atovaquone plus pyrimethamine; and azithromycin plus pyrimethamine plus rifabutin. Relapses are common, and it is recommended that patients with a history of prior toxoplasmic encephalitis receive maintenance therapy with sulfadiazine, pyrimethamine, and leucovorin as long as their CD4+ T cell counts remain <200 cells/ μ L. Patients with CD4+ T cell counts <100/ μ L and IgG antibody to *Toxoplasma* should receive primary prophylaxis for toxoplasmosis.

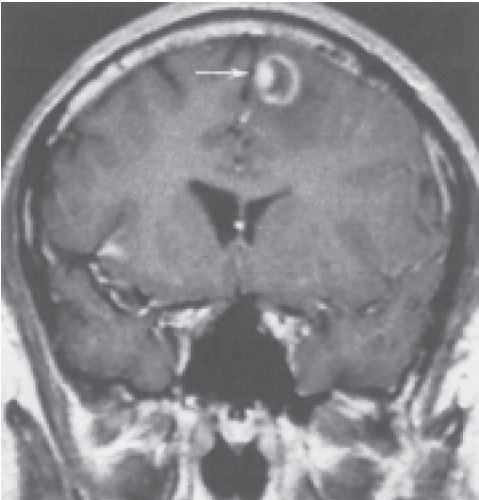


FIGURE 208-41 Central nervous system toxoplasmosis. A coronal postcontrast T1-weighted MRI scan demonstrates a peripheral enhancing lesion in the left frontal lobe, associated with an eccentric nodular area of enhancement (arrow); this so-called eccentric target sign is typical of toxoplasmosis.

Fortunately, the same daily regimen of a single double-strength tablet of TMP-SMX used for *P. jirovecii* prophylaxis provides adequate primary protection against toxoplasmosis. Secondary prophylaxis/maintenance therapy for toxoplasmosis may be discontinued in the setting of effective ART and increases in CD4+ T cell counts to >200/ μ L for 6 months.

JC virus, a human polyomavirus that is the etiologic agent of *progressive multifocal leukoencephalopathy* (PML), is an important opportunistic pathogen in patients with AIDS (**Chap. 142**). While ~80% of the general adult population has antibodies to JC virus, indicative of prior infection, <10% of healthy adults show any evidence of ongoing viral replication. PML is the only known clinical manifestation of JC virus infection. It is a late manifestation of AIDS and is seen in ~1–4% of patients with AIDS. The lesions of PML begin as small foci of demyelination in subcortical white matter that eventually coalesce. The cerebral hemispheres, cerebellum, and brainstem may all be involved. Patients typically have a protracted course with multifocal neurologic deficits, with or without changes in mental status. Approximately 20% of patients experience seizures. Ataxia, hemiparesis, visual field defects, aphasia, and sensory defects may occur. Headache, fever, nausea, and vomiting are rarely seen. Their presence should suggest another diagnosis. MRI typically reveals multiple, nonenhancing white matter lesions that may coalesce and have a predilection for the occipital and parietal lobes. The lesions show signal hyperintensity on T2-weighted images and diminished signal on T1-weighted images. The measurement of JC virus DNA levels in CSF has a diagnostic sensitivity of 76% and a specificity of close to 100%. Prior to the availability of ART, most patients with PML died within 3–6 months of the onset of symptoms. Paradoxical worsening of PML has been seen with initiation of ART as an immune reconstitution syndrome. There is no specific treatment for PML; however, a median survival of 2 years and survival of >15 years have been reported in patients with PML treated with ART for their HIV disease. Despite having a significant impact on survival, only ~50% of patients with HIV infection and PML show neurologic improvement with ART. Studies with other antiviral agents such as cidofovir have failed to show clear benefit. Factors influencing a favorable prognosis for PML in the setting of HIV infection include a CD4+ T cell count >100/ μ L at baseline and the ability to maintain an HIV viral load of <500 copies/mL. Baseline HIV-1 viral load does not have independent predictive value of survival. PML is one of the few opportunistic infections that continues to occur with some frequency despite the widespread use of ART.

Reactivation American trypanosomiasis may present as acute meningoencephalitis with focal neurologic signs, fever, headache, vomiting, and seizures. Accompanying cardiac disease in the form of arrhythmias or heart failure should increase the index of suspicion. The presence of antibodies to *T. cruzi* supports the diagnosis. In South America, reactivation of *Chagas' disease* is considered to be an AIDS-defining condition and may be the initial AIDS-defining condition. Most cases occur in patients with CD4+ T cell counts <200 cells/ μ L. Lesions appear radiographically as single or multiple hypodense areas, typically with ring enhancement and edema. They are found predominantly in the subcortical areas, a feature that differentiates them from the deeper lesions of toxoplasmosis. *T. cruzi* amastigotes, or trypanosomes, can be identified from biopsy specimens or CSF. Other CSF findings include elevated protein and a mild (<100 cells/ μ L) lymphocytic pleocytosis. Organisms can also be identified by direct examination of the blood. Treatment consists of benzimidazole (2.5 mg/kg bid) or nifurtimox (2 mg/kg qid) for at least 60 days, followed by maintenance therapy for the duration of immunodeficiency with either drug at a dose of 5 mg/kg three times a week. As is the case with cerebral toxoplasmosis, successful therapy with antiretrovirals may allow discontinuation of maintenance therapy for Chagas' disease.

Stroke may occur in patients with HIV infection. In contrast to the other causes of focal neurologic deficits in patients with HIV infection, the symptoms of a stroke are sudden in onset. Patients with HIV infection have an increased prevalence of many classic risk factors associated with stroke, including smoking and diabetes. It has been reported that HIV infection itself can lead to an increase in carotid

artery stiffness. The relative increase in risk for stroke as a consequence of HIV infection is more pronounced in women and in individuals between the ages of 18 and 29. Among the secondary infectious diseases in patients with HIV infection that may be associated with stroke are vasculitis due to cerebral varicella zoster or neurosyphilis and septic embolism in association with fungal infection. Other elements of the differential diagnosis of stroke in the patient with HIV infection include atherosclerotic cerebral vascular disease, thrombotic thrombocytopenic purpura, and cocaine or amphetamine use.

Primary CNS lymphoma is discussed below in the section on neoplastic diseases.

Spinal cord disease, or myelopathy, is present in ~20% of patients with AIDS, often as part of HIV-associated neurocognitive disorder. In fact, 90% of the patients with HIV-associated myelopathy have some evidence of dementia, suggesting that similar pathologic processes may be responsible for both conditions. Three main types of spinal cord disease are seen in patients with AIDS. The first of these is a vacuolar myelopathy, as mentioned above. This condition is pathologically similar to subacute combined degeneration of the cord, such as that occurring with pernicious anemia. Although vitamin B₁₂ deficiency can be seen in patients with AIDS as a primary complication of HIV infection, it does not appear to be responsible for most cases of myelopathy seen in patients with HIV infection. However, it should be included in the differential diagnosis. Vacuolar myelopathy is characterized by a subacute onset and often presents with gait disturbances, predominantly ataxia and spasticity; it may progress to include bladder and bowel dysfunction. Physical findings include evidence of increased deep tendon reflexes and extensor plantar responses. The second form of spinal cord disease involves the dorsal columns and presents as a pure sensory ataxia. The third form is also sensory in nature and presents with paresthesias and dysesthesias of the lower extremities. In contrast to the cognitive problems seen in patients with HIV encephalopathy, these spinal cord syndromes do not respond well to antiretroviral drugs, and therapy is mainly supportive.

One important disease of the spinal cord that also involves the peripheral nerves is a *myelopathy* and *polyradiculopathy* seen in association with CMV infection. This entity is generally seen late in the course of HIV infection and is fulminant in onset, with lower extremity and sacral paresthesias, difficulty in walking, areflexia, ascending sensory loss, and urinary retention. The clinical course is rapidly progressive over a period of weeks. CSF examination reveals a predominantly neutrophilic pleocytosis, and CMV DNA can be detected by CSF PCR. Intravenous therapy with ganciclovir or foscarnet can lead to rapid improvement, and prompt initiation of therapy is important in minimizing the degree of permanent neurologic damage. Combination therapy with both drugs should be considered in patients who have been previously treated for CMV disease. Other diseases involving the spinal cord in patients with HIV infection include HTLV-1-associated myelopathy (HAM) (**Chap. 207**), neurosyphilis (**Chap. 187**), infection with herpes simplex (**Chap. 197**) or varicella-zoster (**Chap. 198**), TB (**Chap. 183**), and lymphoma (**Chap. 113**).

Peripheral neuropathies are common in patients with HIV infection. They occur at all stages of illness and take a variety of forms. Early in the course of HIV infection, an acute inflammatory demyelinating polyneuropathy resembling Guillain-Barré syndrome may occur (**Chap. 458**). In other patients, a progressive or relapsing-remitting inflammatory neuropathy resembling chronic inflammatory demyelinating polyneuropathy (CIDP) has been noted. Patients commonly present with progressive weakness, areflexia, and minimal sensory changes. CSF examination often reveals a mononuclear pleocytosis, and peripheral nerve biopsy demonstrates a perivascular infiltrate suggesting an autoimmune etiology. Plasma exchange or IVIg has been tried with variable success. Because of the immunosuppressive effects of glucocorticoids, they should be reserved for severe cases of CIDP refractory to other measures. Another autoimmune peripheral neuropathy seen in patients with AIDS is mononeuritis multiplex (**Chap. 457**) due to a necrotizing arteritis of peripheral nerves. The most common peripheral neuropathy in patients with HIV infection is a *distal sensory polyneuropathy* (DSPN) also referred to as painful sensory neuropathy

(HIV-SN), predominantly sensory neuropathy, or distal symmetric peripheral neuropathy. This condition may be a direct consequence of HIV infection or a side effect of ART with dideoxynucleosides. It is more common in taller individuals, older individuals, and those with lower CD4 counts. Two-thirds of patients with AIDS may be shown by electrophysiologic studies to have some evidence of peripheral nerve disease. Presenting symptoms are usually painful burning sensations in the feet and lower extremities. Findings on examination include a stocking-type sensory loss to pinprick, temperature, and touch sensation and a loss of ankle reflexes. Motor changes are mild and are usually limited to weakness of the intrinsic foot muscles. Response of this condition to antiretrovirals has been variable, perhaps because antiretrovirals are responsible for the problem in some instances. When due to dideoxynucleoside therapy, patients with lower extremity peripheral neuropathy may complain of a sensation that they are walking on ice. Other entities in the differential diagnosis of peripheral neuropathy include diabetes mellitus, vitamin B₁₂ deficiency, and side effects from metronidazole or dapsone. For distal symmetric polyneuropathy that fails to resolve following the discontinuation of dideoxynucleosides, therapy is symptomatic; gabapentin, carbamazepine, tricyclics, or analgesics may be effective for dysesthesias. Treatment-naïve patients may respond to ART.

Myopathy may complicate the course of HIV infection; causes include HIV infection itself, zidovudine, and the generalized wasting syndrome (discussed below). HIV-associated myopathy may range in severity from an asymptomatic elevation in creatine kinase levels to a subacute syndrome characterized by proximal muscle weakness and myalgias. Quite pronounced elevations in creatine kinase may occur in asymptomatic patients, particularly after exercise. The clinical significance of this as an isolated laboratory finding is unclear. A variety of both inflammatory and noninflammatory pathologic processes have been noted in patients with more severe myopathy, including myofiber necrosis with inflammatory cells, nemaline rod bodies, cytoplasmic bodies, and mitochondrial abnormalities. Profound muscle wasting, often with muscle pain, may be seen after prolonged zidovudine therapy. This toxic side effect of the drug is dose-dependent and is related to its ability to interfere with the function of mitochondrial polymerases. It is reversible following discontinuation of the drug. Red ragged fibers are a histologic hallmark of zidovudine-induced myopathy.

Ophthalmologic Diseases Ophthalmologic problems occur in ~50% of patients with advanced HIV infection. The most common abnormal findings on fundoscopic examination are cotton-wool spots. These are hard white spots that appear on the surface of the retina and often have an irregular edge. They represent areas of retinal ischemia secondary to microvascular disease. At times they are associated with small areas of hemorrhage and thus can be difficult to distinguish from CMV retinitis. In contrast to CMV retinitis, however, these lesions are not associated with visual loss and tend to remain stable or improve over time.

One of the most devastating consequences of HIV infection is CMV retinitis. Patients at high risk of CMV retinitis (CD4+ T cell count <100/ μ L) should undergo an ophthalmologic examination every 3–6 months. The majority of cases of CMV retinitis occur in patients with a CD4+ T cell count <50/ μ L. Prior to the availability of ART, this CMV reactivation syndrome was seen in 25–30% of patients with AIDS. In the ART era this has dropped to close to 2%. CMV retinitis usually presents as a painless, progressive loss of vision. Patients may also complain of blurred vision, “floaters,” and scintillations. The disease is usually bilateral, although typically it affects one eye more than the other. The diagnosis is made on clinical grounds by an experienced ophthalmologist. The characteristic retinal appearance is that of perivascular hemorrhage and exudate. In situations where the diagnosis is in doubt due to an atypical presentation or an unexpected lack of response to therapy, vitreous or aqueous humor sampling with molecular diagnostic techniques may be of value. CMV infection of the retina results in a necrotic inflammatory process, and the visual loss that develops is irreversible. CMV retinitis may be complicated by rhegmatogenous retinal detachment as a consequence of retinal atrophy

in areas of prior inflammation. Therapy for CMV retinitis consists of oral valganciclovir, IV ganciclovir, or IV foscarnet, with cidofovir as an alternative. Combination therapy with ganciclovir and foscarnet has been shown to be slightly more effective than either ganciclovir or foscarnet alone in the patient with relapsed CMV retinitis. A 3-week induction course is followed by maintenance therapy with oral valganciclovir. If CMV disease is limited to the eye, intravitreal injections of ganciclovir or foscarnet may be considered. Intravitreal injections of cidofovir are generally avoided due to the increased risk of uveitis and hypotony. Maintenance therapy is continued until the CD4+ T cell count remains >100/ μ L for >6 months. The majority of patients with HIV infection and CMV disease develop some degree of uveitis with the initiation of ART. The etiology of this is unknown; however, it has been suggested that this may be due to the generation of an enhanced immune response to CMV as an IRIS (see above). In some instances, this has required the use of topical glucocorticoids.

Both HSV and varicella zoster virus can cause a rapidly progressing, bilateral, necrotizing retinitis referred to as the *acute retinal necrosis syndrome*, or *progressive outer retinal necrosis* (PORN). This syndrome, in contrast to CMV retinitis, is associated with pain, keratitis, and iritis. It is often associated with orolabial HSV or trigeminal zoster. Ophthalmologic examination reveals widespread pale gray peripheral lesions. This condition is often complicated by retinal detachment. It is important to recognize and treat this condition with IV ganciclovir or IV acyclovir (if definitely due to HSV) as quickly as possible to minimize the loss of vision.

Several other secondary infections may cause ocular problems in patients with HIV. *P. jirovecii* can cause a lesion of the choroid that may be detected as an incidental finding on ophthalmologic examination. These lesions are typically bilateral, are from half to twice the disc diameter in size, and appear as slightly elevated yellow-white plaques. They are usually asymptomatic and may be confused with cotton-wool spots. Chorioretinitis due to toxoplasmosis can be seen alone or, more commonly, in association with CNS toxoplasmosis. KS may involve the eyelid or conjunctiva, while lymphoma may involve the retina. Syphilis may lead to a uveitis that is highly associated with the presence of neurosyphilis.

Additional Disseminated Infections and Wasting Syndrome Infections with species of the small, gram-negative, *Rickettsia*-like organism *Bartonella* (**Chap. 177**) are seen with increased frequency in patients with HIV infection. While it is not considered an AIDS-defining illness by the CDC, many experts view infection with *Bartonella* as indicative of a severe defect in cell-mediated immunity. It is usually seen in patients with CD4+ T cell counts <100/ μ L and is a significant cause of unexplained fever in patients with advanced HIV infection. Among the clinical manifestations of *Bartonella* infection are bacillary angiomatosis, cat-scratch disease, and trench fever. *Bacillary angiomatosis* is usually due to infection with *B. henselae* and is linked to exposure to flea-infested cats. It is characterized by a vascular proliferation that leads to a variety of skin lesions that have been confused with the skin lesions of KS. In contrast to the lesions of KS, the lesions of bacillary angiomatosis generally blanch, are painful, and typically occur in the setting of systemic symptoms. Infection can extend to the lymph nodes, liver (peliosis hepatis), spleen, bone, heart, CNS, respiratory tract, and GI tract. *Cat-scratch disease* is also due to infection with *B. henselae* and generally begins with a papule at the site of inoculation. This is followed several weeks later by the development of regional adenopathy and malaise. Infection with *B. quintana* is transmitted by lice and has been associated with case reports of trench fever, endocarditis, adenopathy, and bacillary angiomatosis. The organism is quite difficult to culture, and diagnosis often relies on identifying the organism in biopsy specimens using the Warthin-Starry or similar stains, PCR, and/or seroconversion. Treatment is with either doxycycline or erythromycin for at least 3 months.

Histoplasmosis is an opportunistic infection that is seen most frequently in patients in the Mississippi and Ohio River valleys, Puerto Rico, the Dominican Republic, and South America. These are all areas in which infection with *H. capsulatum* is endemic (**Chap. 218**).

Because of this limited geographic distribution, histoplasmosis is only seen in approximately 0.5% of AIDS cases in the United States. Histoplasmosis is generally a late manifestation of HIV infection; however, it may be the initial AIDS-defining condition. In one study, the median CD4+ T cell count for patients with histoplasmosis and AIDS was 33/ μ L. While disease due to *H. capsulatum* may present as a primary infection of the lung, disseminated disease, presumably due to reactivation, is the most common presentation in patients with HIV. Patients usually present with a 4- to 8-week history of fever and weight loss. Hepatosplenomegaly and lymphadenopathy are each seen in about 25% of patients. CNS disease, either meningitis or a mass lesion, is seen in 15% of patients. Bone marrow involvement is common, with thrombocytopenia, neutropenia, and anemia occurring in 33% of patients. Approximately 7% of patients have mucocutaneous lesions consisting of a maculopapular rash and skin or oral ulcers. Respiratory symptoms are usually mild, with chest x-ray showing a diffuse infiltrate or diffuse small nodules in ~50% of cases. The gastrointestinal tract may be involved. Diagnosis is made by silver staining of tissue, by culturing the organisms from blood, bone marrow, or tissue, or by detecting antigen in blood or urine. Treatment is typically with liposomal amphotericin B followed by maintenance therapy with oral itraconazole until the serum *Histoplasma* antigen is <2 units, the patient has been on antiretrovirals for at least 6 months, and the CD4 count is >150 cells/ μ L. In the setting of mild infection, it may be appropriate to initiate therapy with itraconazole alone.

Following the spread of HIV infection to southeast Asia, disseminated infection with the fungus *Talaromyces* (formerly *Penicillium*) *marneffei* was recognized as a complication of HIV infection and is considered an AIDS-defining condition in those parts of the world where it occurs. *T. marneffei* is the third most common AIDS-defining illness in Thailand, following TB and cryptococcosis. It is more frequently diagnosed in the rainy than the dry season. Clinical features include fever, generalized lymphadenopathy, hepatosplenomegaly, anemia, thrombocytopenia, and papular skin lesions with central umbilication resembling the lesions of *Molluscum contagiosum*. Treatment is with amphotericin B followed by itraconazole until the CD4+ T cell count is >100 cells/ μ L for at least 6 months.

Visceral leishmaniasis (Chap. 233) is recognized with increasing frequency in patients with HIV infection who live in or travel to areas endemic for this protozoal infection transmitted by sandflies. The clinical presentation is one of hepatosplenomegaly, fever, and hematologic abnormalities. Lymphadenopathy and other constitutional symptoms may be present. A chronic, relapsing course is seen in two-thirds of co-infected patients. Organisms can be detected by PCR and, with special techniques, isolated from cultures of bone marrow aspirates. Histologic stains are often diagnostic but may be negative. Antibody titers are of little help. Patients with HIV infection usually respond well initially to standard therapy with amphotericin B or pentavalent antimony compounds. Eradication of the organism is difficult, however, and relapses are common.

Patients with HIV infection are at a slightly increased risk of infection with malaria and of clinical malaria. This is particularly true for patients from nonendemic areas who are at risk for primary infection and in patients with lower CD4+ T cell counts. HIV-positive

individuals with CD4+ T cell counts <300 cells/ μ L have a poorer response to malaria treatment than others. Co-infection with malaria is associated with a modest increase in HIV viral load. The risk of malaria may be decreased with TMP-SMX prophylaxis.

Generalized wasting is an AIDS-defining condition; it is defined as involuntary weight loss of >10% associated with intermittent or constant fever and chronic diarrhea or fatigue lasting >30 days in the absence of a defined cause other than HIV infection. Prior to the widespread use of ART it was the initial AIDS-defining condition in ~10% of patients with AIDS in the United States. Generalized wasting is rarely seen today with the earlier initiation of antiretrovirals. A constant feature of this syndrome is severe muscle wasting with scattered myofiber degeneration and occasional evidence of myositis. Glucocorticoids may be of some benefit; however, this approach must be carefully weighed against the risk of compounding the immunodeficiency of HIV infection. Androgenic steroids, growth hormone, and total parenteral nutrition have been used as therapeutic interventions with variable success.

Neoplastic Diseases The neoplastic diseases considered to be AIDS-defining conditions are Kaposi's sarcoma, non-Hodgkin's lymphoma, and invasive cervical carcinoma. In addition, there is also an increase in the incidence of a variety of non-AIDS-defining malignancies including Hodgkin's disease; multiple myeloma; leukemia; melanoma; and cervical, brain, testicular, oral, lung, gastric, liver, renal, and anal cancers. Since the introduction of potent ART, there has been a marked reduction in the incidence of KS (Fig. 208-34). The non-AIDS-defining malignancies now account for more morbidity and mortality in patients with HIV infection than the AIDS-defining malignancies and are responsible for approximately 10% of the deaths in patients with HIV infection. Rates of non-Hodgkin's lymphoma have declined; however, this decline has not been as dramatic as the decline in rates of KS. In contrast, ART has had little effect on human papillomavirus (HPV)-associated malignancies. As patients with HIV infection live longer, a wider array of cancers is seen in this population. While some may only reflect known risk factors (e.g., smoking, alcohol consumption, co-infection with other viruses such as hepatitis B) that are increased in patients with HIV infection, some may be a direct consequence of HIV and are clearly increased in patients with lower CD4+ T cell counts.

Kaposi's sarcoma is a multicentric neoplasm presenting as multiple vascular nodules in the skin, mucous membranes, and viscera. The clinical course of KS ranges from indolent, with only minor skin or lymph node involvement, to fulminant, with extensive cutaneous and visceral involvement. In the initial period of the AIDS epidemic, KS was a prominent clinical feature of the first cases of AIDS, occurring in 79% of the patients diagnosed in 1981. By 1989 it was seen in only 25% of cases, by 1992 the number had decreased to 9%, and by 1997 the number was <1%. HHV-8 (KSHV) has been strongly implicated as a viral cofactor in the pathogenesis of KS.

Clinically, KS has varied presentations and may be seen at any stage of HIV infection, even in the presence of a normal CD4+ T cell count. The initial lesion may be a small, raised, reddish-purple nodule on the skin (Fig. 208-42), a discoloration on the oral mucosa (Fig. 208-35D),



FIGURE 208-42 Kaposi's sarcoma in three patients with AIDS demonstrating (A) periorbital edema and bruising; (B) classic truncal distribution of lesions; and (C) upper extremity lesions.

or a swollen lymph node. Lesions often appear in sun-exposed areas, particularly the tip of the nose, and have a propensity to occur in areas of trauma (Koebner phenomenon). Because of the vascular nature of the tumors and the presence of extravasated red blood cells in the lesions, their colors range from reddish to purple to brown and often take the appearance of a bruise, with yellowish discoloration and tattooing. Lesions range in size from a few millimeters to several centimeters in diameter and may be either discrete or confluent. KS lesions most commonly appear as raised macules; however, they can also be papular, particularly in patients with higher CD4⁺ T cell counts. Confluent lesions may give rise to surrounding lymphedema and may be disfiguring when they involve the face and disabling when they involve the lower extremities or the surfaces of joints. Apart from skin, the lymph nodes, GI tract, and lung are the organ systems most commonly affected by KS. Lesions have been reported in virtually every organ, including the heart and the CNS. In contrast to most malignancies, in which lymph node involvement implies metastatic spread and a poor prognosis, lymph node involvement may be seen very early in KS and is of no special clinical significance. In fact, some patients may present with disease limited to the lymph nodes. These are generally patients with relatively intact immune function and thus the patients with the best prognosis. Pulmonary involvement with KS generally presents with shortness of breath. Some 80% of patients with pulmonary KS also have cutaneous lesions. The chest x-ray characteristically shows bilateral lower lobe infiltrates that obscure the margins of the mediastinum and diaphragm (Fig. 208-43). Pleural effusions are seen in 70% of cases of pulmonary KS, a fact that is often helpful in the differential diagnosis. GI involvement is seen in 50% of patients with KS and usually takes one of two forms: (1) mucosal involvement, which may lead to bleeding that can be severe; these patients sometimes also develop symptoms of GI obstruction if lesions become large; and (2) biliary tract involvement. KS lesions may infiltrate the gallbladder and biliary tree, leading to a clinical picture of obstructive jaundice similar to that seen with sclerosing cholangitis. Several staging systems have been proposed for KS. One in common use was developed by the National Institute of Allergy and Infectious Diseases AIDS Clinical Trials Group; it distinguishes patients on the basis of tumor extent, immunologic function, and presence or absence of systemic symptoms (Table 208-17).

A diagnosis of KS is based on biopsy of a suspicious lesion. Histologically one sees a proliferation of spindle cells and endothelial cells, extravasation of red blood cells, hemosiderin-laden macrophages, and, in early cases, an inflammatory cell infiltrate. Included in the differential diagnosis are lymphoma (particularly for oral lesions), bacillary angiomatosis, and cutaneous mycobacterial infections.



FIGURE 208-43 Chest x-ray of a patient with AIDS and pulmonary Kaposi's sarcoma. The characteristic findings include dense bilateral lower lobe infiltrates obscuring the heart borders and pleural effusions.

TABLE 208-17 National Institute of Allergy and Infectious Diseases AIDS Clinical Trials Group TIS Staging System for Kaposi's Sarcoma

PARAMETER	GOOD RISK (STAGE 0): ALL OF THE FOLLOWING	POOR RISK (STAGE 1): ANY OF THE FOLLOWING
Tumor (T)	Confined to skin and/or lymph nodes and/or minimal oral disease	Tumor-associated edema or ulceration Extensive oral lesions GI lesions Nonnodal visceral lesions
Immune system (I)	CD4 ⁺ T cell count $\geq 200/\mu\text{L}$	CD4 ⁺ T cell count $< 200/\mu\text{L}$
Systemic illness (S)	No B symptoms ^a	B symptoms ^a present
	Karnofsky performance status ≥ 70	Karnofsky performance status < 70
	No history of opportunistic infection, neurologic disease, lymphoma, or thrush	History of opportunistic infection, neurologic disease, lymphoma, or thrush

^aDefined as unexplained fever, night sweats, $>10\%$ involuntary weight loss, or diarrhea persisting for more than 2 weeks.

Management of KS (Table 208-18) should be conducted in consultation with an expert since definitive treatment guidelines do not exist. In the majority of cases, effective ART will go a long way in achieving control. ART has been associated with the spontaneous regression of KS lesions. Paradoxically, it has also been associated with the initial appearance of KS as a form of IRIS. For patients in whom tumor persists or is compromising vital functions or in whom control of HIV replication is not possible, a variety of options exist. In some cases, lesions remain quite indolent, and many of these patients can be managed with no specific treatment. Fewer than 10% of AIDS patients with KS die as a consequence of their malignancy, and death from secondary infections is considerably more common. Thus, whenever possible one should avoid treatment regimens that may further suppress the immune system and increase susceptibility to opportunistic infections. Treatment is indicated under two main circumstances. The first is when a single lesion or a limited number of easily accessible lesions are causing significant discomfort or cosmetic problems, such as with prominent facial lesions, lesions overlying a joint, or lesions in the oropharynx that interfere with swallowing or breathing. Under these circumstances, treatment with localized radiation, intralesional vinblastine, topical 9-*cis*-retinoic acid, or cryotherapy may be helpful. It should be noted that patients with HIV infection are particularly sensitive to the side effects of radiation therapy. This is especially true with respect to the development of radiation-induced mucositis; doses of radiation directed at mucosal surfaces, particularly in the head and

TABLE 208-18 Management of AIDS-Associated Kaposi's Sarcoma

Observation and optimization of antiretroviral therapy
Single or limited number of lesions
Radiation
Intralesional vinblastine
Cryotherapy
Extensive disease; inadequate response to ART
Initial therapy
Interferon α (if CD4 ⁺ T cells $>150/\mu\text{L}$)
Liposomal daunorubicin
Subsequent therapy
Liposomal doxorubicin
Paclitaxel
Pomalidomide
Combination chemotherapy with low-dose doxorubicin, bleomycin, and vinblastine (ABV)
Targeted radiation

neck region, should be adjusted accordingly. The second indication for KS-directed treatment is for patients with a large number of lesions or in patients with visceral involvement. In these patients, systemic therapy, either IFN- α or chemotherapy, should be considered. The single most important determinant of response appears to be the CD4+ T cell count. This relationship between response rate and baseline CD4+ T cell count is particularly true for IFN- α . The response rate to IFN- α for patients with CD4+ T cell counts $>600/\mu\text{L}$ is $\sim 80\%$, while the response rate for patients with counts $<150/\mu\text{L}$ is $<10\%$. In contrast to the other systemic therapies, IFN- α provides an added advantage of having antiretroviral activity; thus, it may be the appropriate first choice for single-agent systemic therapy for early patients with disseminated disease. A variety of chemotherapeutic agents also have been shown to have activity against KS. Five of them—liposomal daunorubicin, liposomal doxorubicin, vinblastine, paclitaxel, and the thalidomide analogue pomalidomide—have been approved by the FDA for this indication. Liposomal daunorubicin and pomalidomide are approved as first-line therapy for patients with advanced KS despite ART. They have fewer side effects than conventional chemotherapy. In contrast, liposomal doxorubicin and paclitaxel are approved only for KS patients who have failed standard chemotherapy. Response rates vary from 23% to 88%, appear to be comparable to what had been achieved earlier with combination chemotherapy regimens, and are greatly influenced by CD4+ T cell count. Vinblastine is most commonly used as an intralésional injection or as part of a combination regimen.

Lymphomas (Chaps. 113 and 114) occur with an increased frequency in patients with congenital or acquired T cell immunodeficiencies. AIDS is no exception; at least 6% of all patients with AIDS develop lymphoma at some time during the course of their illness. This is a 10- to 20-fold increase in incidence compared with the general population. In contrast to the situation with KS, primary CNS lymphoma, and most opportunistic infections, the incidence of AIDS-associated systemic lymphomas has not experienced a dramatic decrease as a consequence of the widespread use of effective ART. Lymphoma occurs in all risk groups, with the highest incidence in patients with hemophilia and the lowest incidence in patients from the Caribbean or Africa with heterosexually acquired infection. Lymphoma is a late manifestation of HIV infection, generally occurring in patients with CD4+ T cell counts $<200/\mu\text{L}$. As HIV disease progresses, the risk of lymphoma increases. The attack rate for lymphoma increases exponentially with increasing duration of HIV infection and decreasing level of immunologic function. At 3 years following a diagnosis of HIV infection, the risk of lymphoma is 0.8% per year; by 8 years after infection, it is 2.6% per year. As individuals with HIV infection live longer as a consequence of improved ART and better treatment and prophylaxis of opportunistic infections, it is anticipated that the incidence of lymphomas may increase.

Three main categories of lymphoma are seen in patients with HIV infection: grade III or IV immunoblastic lymphoma, Burkitt's lymphoma, and primary CNS lymphoma. Approximately 90% of these lymphomas are B cell in phenotype; more than half contain EBV DNA. Some are associated with KSHV. These tumors may be either monoclonal or oligoclonal in nature and are probably in some way related to the pronounced polyclonal B cell activation seen in patients with HIV infection.

Immunoblastic lymphomas account for $\sim 60\%$ of the cases of lymphoma in patients with AIDS. The majority of these are diffuse large B cell lymphomas (DLBCL). They are generally high grade and would have been classified as diffuse histiocytic lymphomas in earlier classification schemes. This tumor is more common in older patients, increasing in incidence from 0% in people with HIV <1 year old to $>3\%$ in those >50 years of age. Two variants of immunoblastic lymphoma that are seen primarily in patients with HIV are primary effusion lymphoma (PEL) and its solid variant, plasmacytic lymphoma of the oral cavity. PEL, also referred to as body cavity lymphoma, presents with lymphomatous pleural, pericardial, and/or peritoneal effusions in the absence of discrete nodal or extranodal masses. The tumor cells do not express surface markers for B cells or T cells and are felt to represent a plasmacytic stage of differentiation. While both KSHV and EBV

DNA sequences have been found in the genomes of the malignant cells from patients with body cavity lymphoma, KSHV is felt to be the driving force behind the oncogenesis (see above).

Small noncleaved cell lymphoma (Burkitt's lymphoma) accounts for $\sim 20\%$ of the cases of lymphoma in patients with AIDS. It is most frequent in patients 10–19 years old and usually demonstrates characteristic *c-myc* translocations from chromosome 8 to chromosome 14 or 22. Burkitt's lymphoma is not commonly seen in the setting of immunodeficiency other than HIV-associated immunodeficiency, and the incidence of this particular tumor is more than 1000-fold higher in the setting of HIV infection than in the general population. In contrast to African Burkitt's lymphoma, where 97% of the cases contain EBV genome, only 50% of HIV-associated Burkitt's lymphomas are EBV-positive.

Primary CNS lymphoma accounts for $\sim 20\%$ of the cases of lymphoma in patients with HIV infection. In contrast to HIV-associated Burkitt's lymphoma, primary CNS lymphomas are usually positive for EBV. In one study, the incidence of Epstein-Barr positivity was 100%. This malignancy does not have a predilection for any particular age group. The median CD4+ T cell count at the time of diagnosis is $\sim 50/\mu\text{L}$. Thus, CNS lymphoma generally presents at a later stage of HIV infection than does systemic lymphoma. This may explain, at least in part, the poorer prognosis for this subset of patients.

The clinical presentation of lymphoma in patients with HIV infection is quite varied, ranging from focal seizures to rapidly growing mass lesions in the oral mucosa (Fig. 208-44) to persistent unexplained fever. At least 80% of patients present with extranodal disease, and a similar percentage have B-type symptoms of fever, night sweats, and/or weight loss. Virtually any site in the body may be involved. The most common extranodal site is the CNS, involved in approximately one-third of all patients with lymphoma. Approximately 60% of these cases are primary CNS lymphoma. Primary CNS lymphoma generally presents with focal neurologic deficits, including cranial nerve findings, headaches, and/or seizures. MRI or CT generally reveals a limited number (one to three) of 3- to 5-cm lesions (Fig. 208-45). The lesions often show ring enhancement on contrast administration and may occur in any location. Contrast enhancement is usually less pronounced than that seen with toxoplasmosis. Lesions of CNS lymphoma are most commonly seen deep in the white matter. The main diseases in the differential diagnosis are cerebral toxoplasmosis and cerebral Chagas' disease. In addition to the 20% of lymphomas in people with HIV that are primary CNS lymphomas, CNS disease is also seen in patients with HIV and systemic lymphoma. Approximately 20% of patients with systemic lymphoma have CNS disease in the form of leptomeningeal involvement. This fact underscores the importance of lumbar puncture in the staging evaluation of patients with systemic lymphoma.

Systemic lymphoma is seen at earlier stages of HIV infection than primary CNS lymphoma. In one series the mean CD4+ T cell count was $226/\mu\text{L}$. In addition to lymph node involvement, systemic



FIGURE 208-44 Immunoblastic lymphoma involving the hard palate of a patient with AIDS.



FIGURE 208-45 Central nervous system lymphoma. Postcontrast T1-weighted MRI scan in a patient with AIDS, altered mental status, and hemiparesis. Multiple enhancing lesions, some ring-enhancing, are present. The left sylvian lesion shows gyral and subcortical enhancement, and the lesions in the caudate and splenium (arrowheads) show enhancement of adjacent ependymal surfaces.

lymphoma may commonly involve the GI tract, bone marrow, liver, and lung. GI tract involvement is seen in ~25% of patients. Any site in the GI tract may be involved, and patients may complain of difficulty swallowing or abdominal pain. The diagnosis is usually suspected on the basis of CT or MRI of the abdomen. Bone marrow involvement is seen in ~20% of patients and may lead to pancytopenia. Liver and lung involvement are each seen in ~10% of patients. Pulmonary disease may present as a mass lesion, multiple nodules, or an interstitial infiltrate.

Both conventional and unconventional approaches have been employed in an attempt to treat HIV-related lymphomas. Systemic lymphoma is generally treated by the oncologist with combination chemotherapy. Earlier disappointing figures are being replaced with more optimistic results for the treatment of systemic lymphoma following the availability of more effective ART and the use of rituximab in CD20+ tumors. While there is some controversy regarding the use of antiretrovirals during chemotherapy, there is no question that their use overall in patients with HIV lymphoma has improved survival. Concerns regarding synergistic bone marrow toxicities with chemotherapy and ART are mitigated with the use of ART regimens that avoid myelosuppressive antiretrovirals. As in most situations in patients with HIV disease, those with higher CD4+ T cell counts tend to fare better although not as well as patients with lymphoma without HIV. Response rates as high as 72% with a median survival of 33 months and disease-free intervals up to 9 years have been reported. Treatment of primary CNS lymphoma remains a significant challenge. Treatment is complicated by the fact that this illness usually occurs in patients with advanced HIV disease. Palliative measures such as radiation therapy provide some relief. The prognosis remains poor in this group, with a 2-year survival of 20–30%.

Multicentric Castleman's disease (MCD) is a KSHV-associated lymphoproliferative disorder that is seen with an increased frequency in patients with HIV infection. While the incidence of Kaposi's sarcoma has decreased, the incidence of MCD has increased in the setting of ART. While not a true malignancy, MCD shares many features with lymphoma including generalized lymphadenopathy, hepatosplenomegaly, and systemic symptoms of fever, fatigue, and weight loss. Pulmonary symptoms may be seen in ~50% of patients. KS is present in 75–82% of cases. Lymph node biopsies reveal a predominance of interfollicular plasma cells and/or germinal centers with vascularization and an "onion skin" (hyaline vascular) appearance. Prior to the availability of ART, patients with HIV and multicentric Castleman's disease had a 15-fold increased risk of developing non-Hodgkin's lymphoma compared with patients with HIV in general. Treatment typically involves chemotherapy. Rituximab may be of benefit, but it has been associated

with worsening of coexisting KS. Similarly, the use of corticosteroids may increase the risk of developing KS. The median survival of patients with treated multicentric Castleman's disease pre-ART was initially reported as 14 months. This has increased to a 2-year survival of more than 90% in the era of ART.

Evidence of infection with *human papillomavirus* (HPV), associated with *intraepithelial dysplasia of the cervix or anus*, is approximately twice as common in people with HIV as in the general population and can lead to intraepithelial neoplasia and eventually invasive cancer. In a series of studies, men with HIV were examined for evidence of anal dysplasia, and Papanicolaou (Pap) smears were found to be abnormal in 20–80%. These changes tend to persist and are generally not affected by ART, raising the possibility of a subsequent transition to a more malignant condition. While the incidence of an abnormal Pap smear of the cervix is ~5% in otherwise healthy women, the incidence of abnormal cervical smears in women with HIV infection is 30–60%, and *invasive cervical cancer* is included as an AIDS-defining condition. While only small increases in the absolute numbers of cervical or anal cancers have been seen as a consequence of HIV infection, the relative risk of these conditions when one compares men and women with HIV to men and women without HIV infection is on the order of 10- to 100-fold and decreased with ART. Given the high rates of dysplasia and relative risks for cervical and anal cancer, a comprehensive gynecologic and rectal examination, including Pap smear, is indicated at the initial evaluation and 6 months later for all patients with HIV infection. If these examinations are negative at both time points, the patient should be followed with yearly evaluations. If an initial or repeat Pap smear shows evidence of severe inflammation with reactive squamous changes, the next Pap smear should be performed at 3 months. If, at any time, a Pap smear shows evidence of squamous intraepithelial lesions, colposcopic examination with biopsies as indicated should be performed. The 2-year survival rate for women with HIV and invasive cervical cancer is 64% compared with 79% for women without HIV. In addition to rectal and cervical lesions, HPV can also lead to head and neck cancers. In one study of men who have sex with men, 25% were found to have oral HPV; high-risk HPV genotypes were three times more common in men with HIV. The most common HPV genotypes in the general population and the genotypes upon which current HPV vaccines are based are 6, 11, 16, and 18. In the population of people with HIV, other genotypes such as 58 and 53 also are prominent. This raises a concern about the level of effectiveness of the current HPV vaccines for patients with HIV. Despite this, it is recommended that patients with HIV infection be vaccinated against HPV.

IDIOPATHIC CD4+ T LYMPHOCYTOPENIA

A syndrome was recognized in 1992 characterized by an absolute CD4+ T cell count of <300/ μ L or <20% of total T cells on a minimum of two occasions at least 6 weeks apart; no evidence of HIV-1, HIV-2, HTLV-1, or HTLV-2 on testing; and the absence of any defined immunodeficiency or therapy associated with decreased levels of CD4+ T cells. By mid-1993, ~100 patients had been described. After extensive multicenter investigations, a series of reports were published in early 1993, which together allowed a number of conclusions. Idiopathic CD4+ lymphocytopenia (ICL) is a very rare syndrome, as determined by studies of blood donors and cohorts of HIV-seronegative men who have sex with men. Cases were clearly identified as early as 1983. The definition of ICL based on CD4+ T cell counts coincided with the ready availability of testing for CD4+ T cells in patients suspected of being immunodeficient. However, as a result of immune deficiency, certain patients with ICL develop some of the opportunistic diseases (particularly cryptococcosis, nontuberculous mycobacterial infections, and HPV disease) seen in patients with HIV. In one study, HPV-associated disease was seen in 29% of patients, cryptococcosis in 24%, molluscum contagiosum in 95%, and nontuberculous mycobacterial disease in 5%. Approximately 10–30% of patients may exhibit an autoimmune disease; this is more common in those with higher CD4 counts. The syndrome is demographically, clinically, and immunologically unlike HIV infection and AIDS. Fewer than half of the reported ICL patients had risk factors for HIV infection, and there were wide geographic

and age distributions. The fact that a significant proportion of initially diagnosed patients did have risk factors probably reflects a selection bias, in that physicians who take care of patients with HIV were more likely to monitor CD4+ T cells. Approximately half of the patients are women, compared with approximately one-fifth among people with HIV in the United States. Many patients with ICL remained clinically stable, and their condition may not deteriorate progressively as is common with seriously immunodeficient patients with HIV. Approximately 15% of patients with ICL experience spontaneous reversal of the CD4+ T lymphocytopenia. Immunologic abnormalities in ICL are somewhat different from those of HIV infection. ICL patients often have increases in CD4+ T cell activation with decreases in CD8+ T cells and B cells. Furthermore, immunoglobulin levels are either normal or, more commonly, decreased in patients with ICL, compared with the usual hypergammaglobulinemia of people with HIV. Virologic studies of these patients have revealed no evidence of HIV-1, HIV-2, HTLV-1, or HTLV-2 or of any other mononuclear cell-tropic virus. Furthermore, there has been no epidemiologic evidence to suggest that a transmissible microbe was involved. The cases of ICL have been widely dispersed, with no clustering. Close contacts and sexual partners who were studied were clinically well and were serologically, immunologically, and virologically negative for HIV. ICL is a heterogeneous syndrome, and it is highly likely that there is no common cause; however, there may be common causes among subgroups of patients that are currently unrecognized.

Patients who present with laboratory data consistent with ICL should be worked up for underlying diseases that could be responsible for the immune deficiency. If no underlying cause is detected, no specific therapy should be initiated. However, if opportunistic diseases occur, they should be treated appropriately (see above). Depending on the level of the CD4+ T cell count, patients should receive prophylaxis for the commonly encountered opportunistic infections.

TREATMENT

AIDS and Related Disorders

GENERAL PRINCIPLES OF PATIENT MANAGEMENT

The CDC guidelines call for the testing for HIV infection to be a part of routine medical care. It is recommended that the patient be informed of the intention to test, as is the case with other routine laboratory determinations, and be given the opportunity to “opt out.” Such an approach is critical to the goal of identifying as many infected individuals as possible since 13% of the 1.2 million individuals in the United States with HIV are not aware of their status. In the setting of routine testing, although it is difficult, pretest counseling is an important part of the process. No matter how well prepared a patient is for adversity, the discovery of a diagnosis of HIV infection is a devastating event. Thus, physicians should be sensitive to this fact and, where possible, utilize pretest counseling to at least partially prepare the patient should the results demonstrate the presence of HIV infection. Following a diagnosis of HIV infection, the health care provider should be prepared to immediately activate support systems for the newly diagnosed patient and initiate ART. These supports should include individuals who can spend time talking to the newly diagnosed person and ensuring that he or she is emotionally stable and ready to begin therapy. Most communities have HIV support centers that can be of great help in these difficult situations.

The treatment of patients with HIV infection requires not only a comprehensive knowledge of the possible disease processes that may occur and up-to-date knowledge of and experience with ART, but also the ability to deal with the problems of a chronic, potentially life-threatening illness. A comprehensive knowledge of internal medicine is required to deal with the changing spectrum of illnesses associated with HIV infection, many of which are similar to a state of accelerated aging. The appropriate use of potent ART and other treatment and prophylactic interventions is of critical importance in providing each patient with the best opportunity to

live a long and healthy life with HIV infection. In contrast to the earlier days of this epidemic, a diagnosis of HIV infection needs no longer be equated with having an inevitably fatal disease. In addition to medical interventions, the health care provider has a responsibility to provide each patient with appropriate counseling and education concerning their disease as part of a comprehensive care plan. Patients must be educated about the potential transmissibility of their infection and about the fact that while health care providers may refer to levels of the virus as “undetectable,” this is only a reflection of the sensitivity of the assay being used to measure the virus, rather than a comment on the presence or absence of the virus. It is important for patients to be aware that the virus is still present in virtually all patients who have ever been diagnosed with HIV infection and capable of being transmitted in the absence of effective ART. Thus, there must be frank discussions concerning sexual practices and the sharing of syringes and other paraphernalia used in illicit drug use. The treating physician not only must be aware of the latest medications available for patients with HIV infection but also must educate patients concerning the natural history of their illness, listen to their concerns, and be sensitive to their fears. As with other diseases, therapeutic decisions should be made in consultation with the patient, when possible, and with the patient’s proxy if the patient is incapable of making decisions. In this regard, it is recommended that all patients with HIV infection, and in particular those with CD4+ T cell counts <200/ μ L, designate a trusted individual with durable power of attorney to make medical decisions on their behalf, if necessary.

Following a diagnosis of HIV infection, several examinations and laboratory studies should be performed to help determine the extent of disease and provide baseline standards for future reference (Table 208-19). In addition to routine chemistry, fasting lipid profile, aspartate aminotransferase, alanine aminotransferase, total and direct bilirubin, fasting glucose and hematology screening panels, Pap smear, urinalysis, and chest x-ray, one should also obtain a CD4+ T cell count, a plasma HIV RNA level, an HIV resistance test, a rapid plasma reagin (RPR) or Venereal Disease Research Laboratory (VDRL) test, an anti-*Toxoplasma* antibody titer, and serologies for hepatitis A, B, and C. A PPD test or IFN- γ release assay should be done and an MMSE performed and recorded. A pregnancy test should be done for any women for whom the drug efavirenz is being considered, and HLA-B5701 testing should be done in all patients in whom the drug abacavir is being considered.

TABLE 208-19 Initial Evaluation of the Patient with HIV Infection

History and physical examination
Routine chemistry and hematology
AST, ALT, alkaline phosphatase, direct and indirect bilirubin
Lipid profile and fasting glucose
CD4+ T lymphocyte count
Plasma HIV RNA level
HIV resistance testing
HLA-B5701 screening
RPR or VDRL test
Anti- <i>Toxoplasma</i> antibody titer
Urinalysis
PPD skin test or IFN- γ release assay
Mini-Mental Status Examination
Serologies for hepatitis A, hepatitis B, and hepatitis C
Immunization with pneumococcal polysaccharide; influenza; HPV as indicated
Immunization with hepatitis A and hepatitis B if seronegative
Counseling regarding natural history and transmission
Help contacting others who might be infected

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; PPD, purified protein derivative; RPR, rapid plasma reagin; VDRL, Venereal Disease Research Laboratory.

TABLE 208-20 HIV Resources Available on the World Wide Web

CLINICALINFO.HIV.GOV	US-approved HIV treatment guidelines
CLINICALTRIALS.GOV	Information on clinical trials
HIV.GOV	Epidemiological data and prevention information

Patients should be immunized with pneumococcal polysaccharide, with annual influenza and COVID-19 shots, and, if seronegative for these viruses, with HPV, hepatitis A, and hepatitis B vaccines. The status of hepatitis C infection should be determined. In addition, patients should be counseled with regard to sexual practices and needle sharing, and counseling should be offered to people whom the patient knows, or suspects, also may be infected. Once these baseline activities are performed, short- and long-term medical management strategies should be developed based on the most recent information available and modified as new information becomes available. The field of HIV medicine is changing rapidly, and it is difficult to remain fully up to date. Fortunately, there are a series of excellent sites on the Internet that are frequently updated, and they provide the most recent information on a variety of topics, including consensus panel reports on treatment (Table 208-20).

ANTIRETROVIRAL THERAPY

Combination antiretroviral therapy (ART), also referred to as highly active antiretroviral therapy (HAART), is the cornerstone of management of patients with HIV infection and should be initiated as soon as possible following a diagnosis of HIV infection. One exception to immediate initiation of ART is in the setting of cryptococcal meningitis or TB where several weeks of specific antimicrobial therapy prior to initiation of ART may decrease the risk of severe IRIS. Following the initiation of widespread use of ART in the United States in 1995–1996, marked declines were noted in the incidence of most AIDS-defining conditions (Fig. 208-34). Suppression of HIV replication is an important component in prolonging and improving the quality of life for the patient as well as minimizing the risk of transmission of HIV to others. Adequate suppression of HIV replication requires strict adherence to prescribed regimens of antiretroviral drugs. This has been facilitated by the coformulations of antiretrovirals and the development of once-daily, monthly regimens, and every-six-month regimens with the expectation that even longer-lasting drugs are on the horizon. Among the decisions that need to be made in the context of prescribing ART are selection of the best initial regimen, determining when a given regimen should be changed, and deciding what regimen should be selected when a change is made. The care provider and patient must come to a mutually agreeable plan based on the best available data. In an effort to facilitate this process, the U.S. Department of Health and Human Services makes available on the Internet (clinicalinfo.hiv.gov/en/guidelines) a series of periodically updated guidelines, including “Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents” and “Guidelines for the Prevention and Treatment of Opportunistic Infections in Persons Infected with Human Immunodeficiency Virus.” At present, an extensive clinical trials network, involving both clinical investigators and patient advocates, is in place attempting to develop improved approaches to therapy. Consortia comprising representatives of academia, industry, independent foundations, and the federal government engage in drug development, including a wide-ranging series of clinical trials. As a result, new therapies and new therapeutic strategies are continually emerging. New drugs are often available through expanded-access programs prior to official licensure. Given the complexity of this field, decisions regarding ART are best made in consultation with experts.

Currently available drugs for the treatment of HIV infection as part of a combination regimen fall into five categories: those that inhibit the viral reverse transcriptase enzyme (nucleoside and nucleotide reverse transcriptase inhibitors; nonnucleoside reverse transcriptase inhibitors), those that inhibit the viral protease enzyme (protease inhibitors), those that inhibit the viral integrase enzyme (integrase strand transfer inhibitors), those that interfere with viral

entry (fusion inhibitors; CCR5 antagonists; CD4 antagonists) and those that interfere with the viral capsid (Table 208-21; Fig. 208-46). A typical regimen will include two nucleoside/nucleotide reverse transcriptase inhibitors (usually a tenofovir-based drug or abacavir + 3TC or FTC) plus a nonnucleoside reverse transcriptase inhibitor, an integrase inhibitor, or a protease inhibitor boosted with a pharmacokinetic enhancer (ritonavir or cobicistat). More recent studies have also supported the two-drug regimen of dolutegravir plus 3TC for initial therapy in hepatitis B–negative patients with baseline HIV RNA levels <500,000 copies/mL. Numerous fixed-drug formulations combining two or more of these antiretroviral drugs have been licensed (Table 208-22). Prior to initiation of therapy and at any time a change in therapy due to treatment failure is being considered, drug resistance testing should be performed to help guide the selection of drugs to be used in combination. A summary of known resistance mutations for antiretroviral drugs is shown in Fig. 208-47.

While most patients with HIV infection will be infected with HIV-1, some patients, especially those with an epidemiologic link to West Africa, may be infected with HIV-2. While the principles of treatment are the same as those for persons infected with HIV-1, it is important to note that the nonnucleoside reverse transcriptase inhibitors, enfuvirtide, and fostemsavir are not active against HIV-2 and should not be used as part of ART regimens in HIV-2–infected individuals.

The FDA-approved reverse transcriptase inhibitors include the *nucleoside analogues* zidovudine, didanosine, zalcitabine, stavudine, lamivudine, abacavir, and emtricitabine; the *nucleotide analogues* tenofovir disoproxil and tenofovir alafenamide; and the *nonnucleoside reverse transcriptase inhibitors* nevirapine, delavirdine, efavirenz, etravirine, rilpivirine, long-acting rilpivirine, and doravirine (Table 208-21). These represent the first class of drugs licensed for the treatment of HIV infection. They are indicated for this use as part of combination regimens. It should be stressed that none of these drugs should be used as monotherapy for HIV infection due to the relative ease with which drug resistance may develop under such circumstances. Thus, when lamivudine, emtricitabine, or tenofovir is used to treat hepatitis B infection in the setting of HIV infection, one should ensure that the patient is also on additional antiretroviral medication. Similarly, when any of these three medications are discontinued, one needs to be vigilant for a flare of hepatitis B in coinfecting patients. The reverse transcriptase inhibitors block the HIV replication cycle at the point of RNA-dependent DNA synthesis, the reverse transcription step. While the nonnucleoside reverse transcriptase inhibitors are quite selective for the HIV-1 reverse transcriptase, the nucleoside and nucleotide analogues inhibit a variety of DNA polymerases in addition to those of the HIV-1 reverse transcriptase. For this reason, serious side effects are more varied with the nucleoside analogues and include mitochondrial damage that can lead to hepatic steatosis and lactic acidosis as well as peripheral neuropathy and pancreatitis. The use of either of the thymidine analogues zidovudine and stavudine has been associated with a syndrome of hyperlipidemia, glucose intolerance/insulin resistance, and fat redistribution often referred to as *lipodystrophy syndrome* (discussed in “Diseases of the Endocrine System and Metabolic Disorders,” above). For these reasons, the older drugs in this class, zidovudine, didanosine, zalcitabine, and stavudine, are no longer recommend for use in the United States due to their side-effect profiles. The nucleoside and nucleotide reverse transcriptase inhibitors preferred for use in combination regimens according to the DHHS Panel on the use of antiretroviral drugs are lamivudine, emtricitabine, abacavir, tenofovir disoproxil, and tenofovir alafenamide. Given its renal toxicity, tenofovir disoproxil should be limited to use in patients with creatinine clearance (CrCl) >70 while tenofovir alafenamide should generally be limited to use in patients with CrCl >30. The preferred nonnucleoside reverse transcriptase inhibitors are efavirenz, rilpivirine, and doravirine. Of note, rilpivirine is approved for treatment only in ART-naïve patients with HIV RNA levels <100,000 copies/mL and is contraindicated in patients taking proton pump inhibitors.

TABLE 208-21 Antiretroviral Drugs Licensed in the United States for the Treatment of HIV Infection

DRUG ^a	INDICATION	DOSE IN COMBINATION	SUPPORTING DATA	TOXICITY
Nucleoside or Nucleotide Reverse Transcriptase Inhibitors				
Lamivudine (Epivir, 2'-3'-dideoxy-3'-thiacytidine, 3TC)	In combination with other antiretroviral agents for the treatment of HIV infection	150 mg bid 300 mg qd	In combination with zidovudine superior to zidovudine alone with respect to changes in CD4+ T cell counts in 495 patients who were zidovudine-naïve and 477 patients who were zidovudine-experienced; overall CD4+ T cell counts for the zidovudine group were at baseline by 24 weeks, while in the group treated with zidovudine plus lamivudine they were 10–50 cells/μL above baseline; 54% decrease in progression to AIDS/death compared with zidovudine alone	Flare of hepatitis in HBV-co-infected patients who discontinue drug
Emtricitabine (FTC, Emtriva)	In combination with other antiretroviral agents for the treatment of HIV infection	200 mg qd	Comparable to lamivudine in combination with stavudine and nevirapine/efavirenz	Hepatotoxicity in HBV-co-infected patients who discontinue drug, skin discoloration
Abacavir (Ziagen)	For treatment of HIV infection in combination with other antiretroviral agents	300 mg bid	Abacavir + zidovudine + 3TC equivalent to indinavir + zidovudine + 3TC with regard to viral load suppression (~60% in each group with <400 HIV RNA copies/mL plasma) and CD4+ T cell increase (~100/μL in each group) at 24 weeks	Hypersensitivity reaction In HLA-B5701+ individuals (can be fatal); fever, rash, nausea, vomiting, malaise or fatigue, and loss of appetite
Tenofovir disoproxil fumarate (Viread)	For use in combination with other antiretroviral agents when treatment is indicated	300 mg qd	Reduction of ~0.6 log in HIV-1 RNA levels when added to background regimen in treatment-experienced patients	Renal, osteomalacia, flare of hepatitis in HBV-co-infected patients who discontinue drug
Tenofovir alafenamide (Vemlidy)	In combination with emtricitabine and other antiretroviral agents for treatment of HIV-1 infection	25 mg qd	92% of patients treated in combination with emtricitabine, elvitegravir, and cobicistat had HIV-1 RNA levels <50 copies/mL	Nausea, less renal toxicity than tenofovir disoproxil fumarate
Nonnucleoside Reverse Transcriptase Inhibitors				
Nevirapine (Viramune)	In combination with other antiretroviral agents for treatment of progressive HIV infection	200 mg/d × 14 days then 200 mg bid or 400 mg extended release qd	Increase in CD4+ T cell count, decrease in HIV RNA when used in combination with nucleosides	Skin rash, hepatotoxicity
Efavirenz (Sustiva)	For treatment of HIV infection in combination with other antiretroviral agents	600 mg qhs	Efavirenz + zidovudine + 3TC comparable to indinavir + zidovudine + 3TC with regard to viral load suppression (a higher percentage of the efavirenz group achieved viral load <50 copies/mL, but the discontinuation rate in the indinavir group was unexpectedly high, accounting for most treatment “failures”); CD4 cell increase (~140/μL in each group) at 24 weeks	Rash, dysphoria, elevated liver function tests, drowsiness, abnormal dreams, depression, lipid abnormalities, potentially teratogenic
Etravirine (Intelence)	In combination with other antiretroviral agents in treatment-experienced patients whose HIV is resistant to nonnucleoside reverse transcriptase inhibitors and other antiretroviral medications	200 mg bid	Higher rates of HIV RNA suppression to <50 copies/mL (56% vs 39%); greater increases in CD4+ T cell count (89 vs 64 cells) compared to placebo when given in combination with an optimized background regimen	Rash, nausea, hypersensitivity reactions
Rilpivirine (Edurant)	In combination with other drugs in previously untreated patients when treatment is indicated	25 mg qd	Noninferior to efavirenz with respect to suppression at week 48 in 1368 treatment-naïve individuals, except in patients with pretherapy HIV RNA levels >100,000 where it was inferior	Nausea, dizziness, somnolence, vertigo, less CNS toxicity and rash than efavirenz
Protease Inhibitors				
Ritonavir (Norvir)	In combination with other antiretroviral agents for treatment of HIV infection when treatment is warranted	600 mg bid (also used in lower doses as pharmacokinetic booster)	Reduction in the cumulative incidence of clinical progression or death from 34% to 17% in patients with CD4+ T cell count <100/μL treated for a median of 6 months	Nausea, abdominal pain, hyperglycemia, fat redistribution, lipid abnormalities, may alter levels of many other drugs, paresthesias, hepatitis
Atazanavir (Reyataz)	For treatment of HIV infection in combination with other antiretroviral agents	400 mg qd or 300 mg qd + ritonavir 100 mg qd when given with efavirenz	Comparable to efavirenz when given in combination with zidovudine + 3TC in a study of 810 treatment-naïve patients; comparable to nelfinavir when given in combination with stavudine + 3TC in a study of 467 treatment-naïve patients	Hyperbilirubinemia, PR prolongation, nausea, vomiting, hyperglycemia, fat maldistribution, rash transaminase elevations, renal stones

(Continued)

TABLE 208-21 Antiretroviral Drugs Licensed in the United States for the Treatment of HIV Infection (Continued)

DRUG ^a	INDICATION	DOSE IN COMBINATION	SUPPORTING DATA	TOXICITY
Darunavir (Prezista)	In combination with 100 mg ritonavir for combination therapy in treatment-experienced adults	600 mg + 100 mg ritonavir bid with food	At 24 weeks, patients with prior extensive exposure to antiretrovirals treated with a new combination including darunavir showed a –1.89-log change in HIV RNA levels and a 92-cell increase in CD4+ T cells compared with –0.48 log and 17 cells in the control arm	Diarrhea, nausea, headache, skin rash, hepatotoxicity, hyperlipidemia, hyperglycemia
Entry Inhibitors				
Enfuvirtide (Fuzeon)	In combination with other agents in treatment-experienced patients with evidence of HIV-1 replication despite ongoing antiretroviral therapy	90 mg SC bid	In treatment of experienced patients, superior to placebo when added to new optimized background (37% vs 16% with <400 HIV RNA copies/mL at 24 weeks; + 71 vs + 35 CD4+ T cells at 24 weeks)	Local injection reactions, hypersensitivity reactions, increased rate of bacterial pneumonia
Maraviroc (Selzentry)	In combination with other antiretroviral agents in adults infected with only CCR5-tropic HIV-1	150–600 mg bid depending on concomitant medications (see text)	At 24 weeks, among 635 patients with CCR5-tropic virus and HIV-1 RNA >5000 copies/mL despite at least 6 months of prior therapy with at least 1 agent from 3 of the 4 antiretroviral drug classes, 61% of patients randomized to maraviroc achieved HIV RNA levels <400 copies/mL compared with 28% of patients randomized to placebo	Hepatotoxicity, nasopharyngitis, fever, cough, rash, abdominal pain, dizziness, musculoskeletal symptoms
Ibalizumab (Trogarzo)	In combination with other antiretroviral agents in patients with multidrug-resistant HIV-1	Single loading dose of 2000 mg followed by a maintenance dose of 800 mg every 2 weeks	At 25 weeks, 50% of patients with multidrug-resistant HIV-1 with HIV-1 RNA >1000 copies/mL treated with an optimized background of 1 active drug and ibalizumab achieved HIV RNA levels <200 copies/mL	Rash, diarrhea, nausea
Fostemsavir (Rukobia)	In combination with other antiretroviral agents in patients with multi-drug resistant HIV-1	600 mg bid	At 96 weeks, 64% of patients with multi-drug resistant HIV-1 treated with an optimized background and fostemsavir achieved an HIV-1 RNA level <200 copies/mL	QTc prolongation at higher doses, elevation in liver enzymes in those with hepatitis B or C
Integrase Inhibitors				
Raltegravir (Isentress)	In combination with other antiretroviral agents	400 mg bid	At 24 weeks, among 436 patients with 3-class drug resistance, 76% of patients randomized to receive raltegravir achieved HIV RNA levels <400 copies/mL compared with 41% of patients randomized to receive placebo	Nausea, headache, diarrhea, CPK elevation, muscle weakness, rhabdomyolysis
Elvitegravir (Available only in combination with cobicistat, tenofovir, and emtricitabine [Stribild])	Fixed-dose combination	1 tablet daily	Noninferior to raltegravir or atazanavir/ritonavir in treatment-experienced patients	Diarrhea, nausea, upper respiratory infections, headache
Dolutegravir (Tivicay)	In combination with other antiretroviral agents	50 mg daily for treatment-naïve patients 50 mg twice daily for treatment-experienced patients or those also receiving efavirenz or rifampin	Noninferior to raltegravir, superior to efavirenz or darunavir/ritonavir	Insomnia, headache, hypersensitivity reactions, hepatotoxicity
Bictegravir (Available only in combination with tenofovir alafenamide and emtricitabine [Biktarvy])	For treatment of HIV infection in adults	50 mg bictegravir/25 mg tenofovir alafenamide/200 mg emtricitabine qd	Noninferior to dolutegravir/tenofovir/emtricitabine and noninferior to dolutegravir/abacavir/lamivudine	Nausea, diarrhea, headache
Cabotegravir (Vocabria)	In combination with rilpivirine for treatment of HIV infection in adults	Oral lead-in of 30 mg + 25 mg rilpivirine for 1 month; followed by an initial injection of 600 mg (3 mL) IM + 900 mg (3 mL) rilpivirine IM; followed by monthly injections of 400 mg (2 mL) IM + 600 mg (2 mL) rilpivirine IM	Noninferior to abacavir/dolutegravir/lamivudine or dolutegravir + 2 nucleoside/tide reverse transcriptase inhibitors Noninferior to nonnucleoside reverse transcriptase inhibitor + 2 nucleoside/tide reverse transcriptase inhibitors or a protease inhibitor + 2 nucleoside/tide reverse transcriptase inhibitors or an integrase inhibitor and 2 nucleoside/tide reverse transcriptase inhibitors	Injection-site reactions

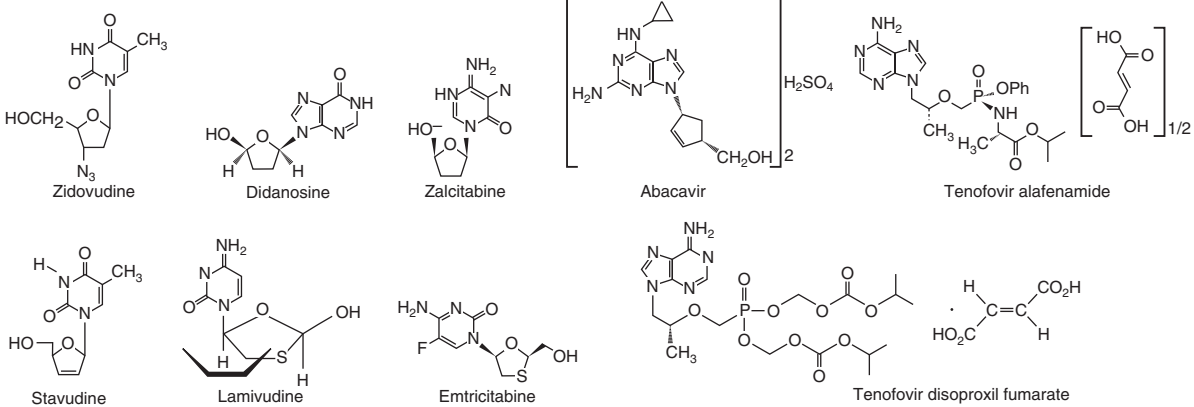
(Continued)

TABLE 208-21 Antiretroviral Drugs Licensed in the United States for the Treatment of HIV Infection (Continued)				
DRUG ^a	INDICATION	DOSE IN COMBINATION	SUPPORTING DATA	TOXICITY
Lenacapavir (Sunlenca)	In combination with other agents for treatment of multidrug-resistant HIV infection in adults	Induction with either (927 mg SC + 600 mg orally day 1; 600 mg orally day 2) or (600 mg orally days 1 and 2; 300 mg orally day 8; 927 mg SC day 15) followed by 927 mg SC every 6 months	At 52 weeks, 83% of patients with multi-drug resistant HIV-1 treated with an optimized background and lenacapavir achieved an HIV-1 level <50 copies/mL	Injection site reactions, nausea

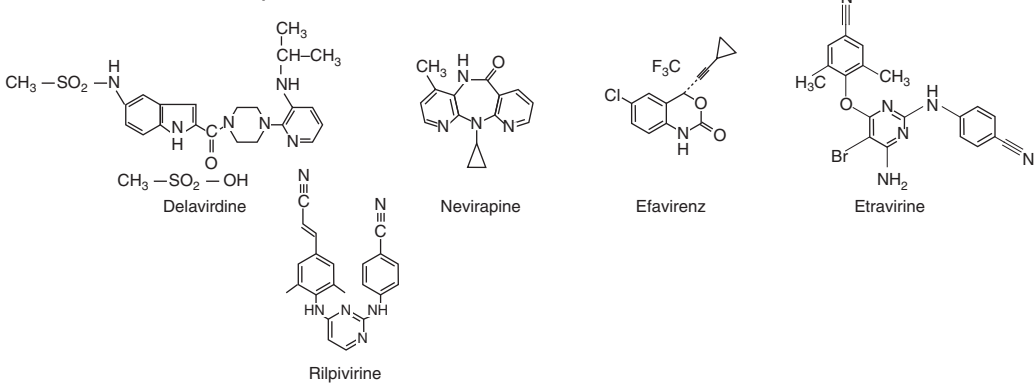
^aInitial trade names are provided. Generic forms may be available.

Abbreviations: ARC, AIDS-related complex; bid, twice daily; CPK, creatine phosphokinase; HBV, hepatitis B virus; IM, intramuscular; qd, once daily; qhs, once daily at bedtime; SC, subcutaneous.

Nucleoside or Nucleotide Reverse Transcriptase Inhibitors



Nonnucleoside Reverse Transcriptase Inhibitors



Protease Inhibitors

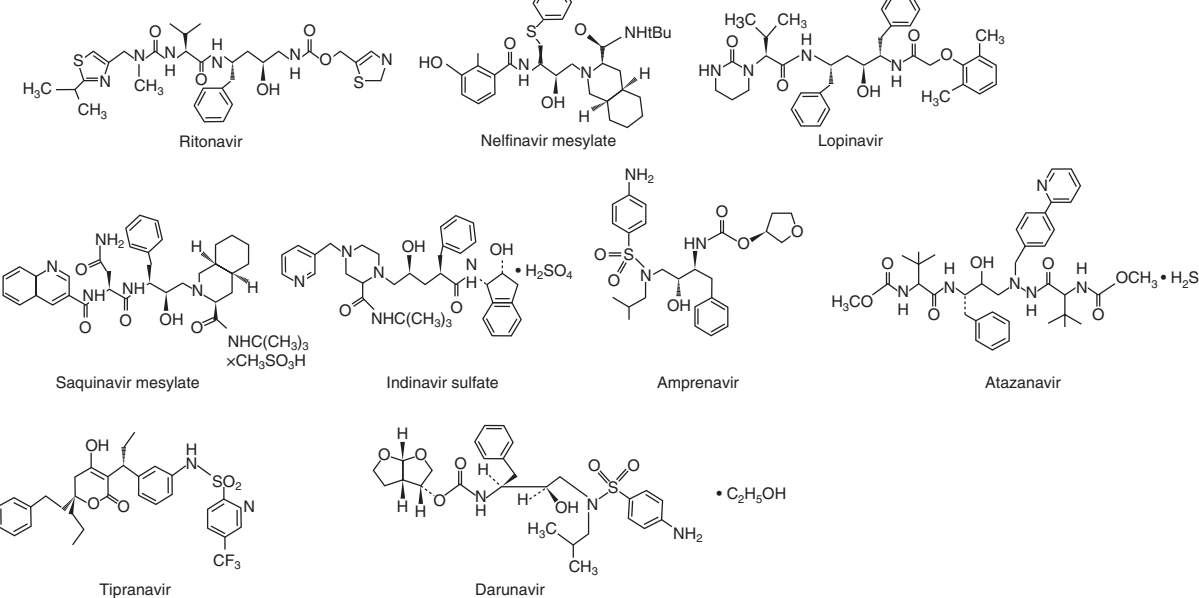
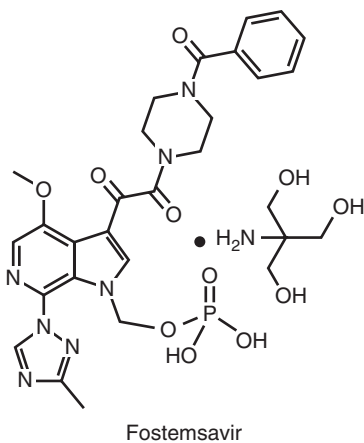
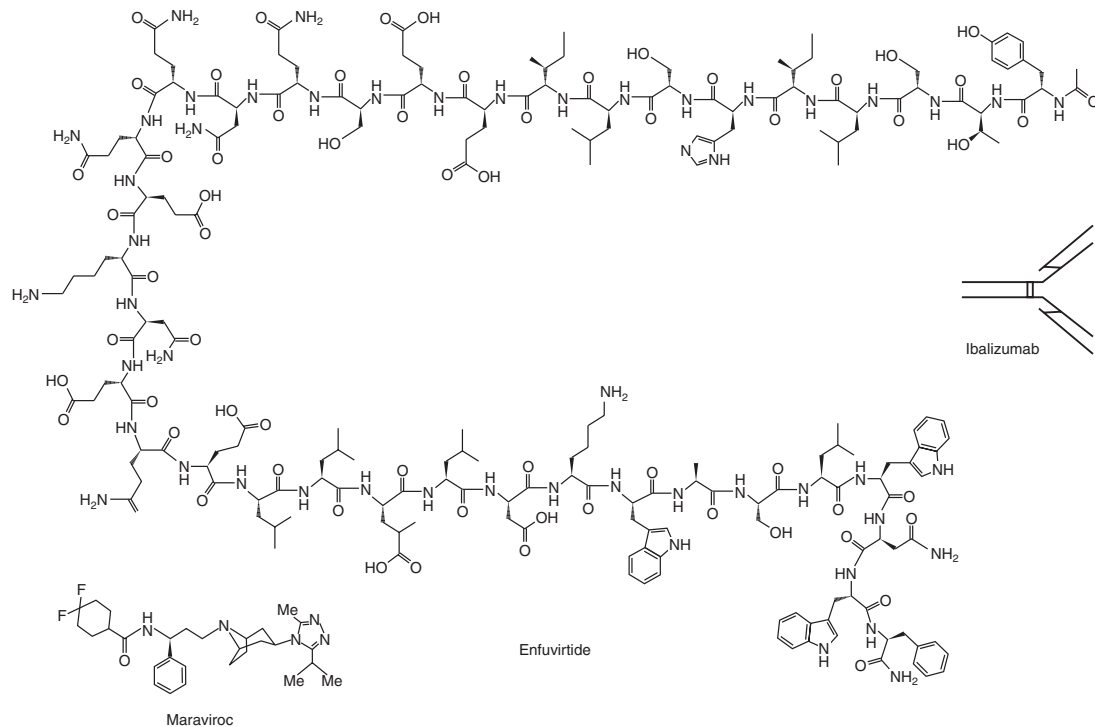
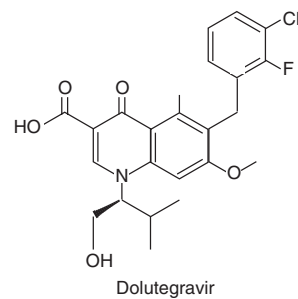
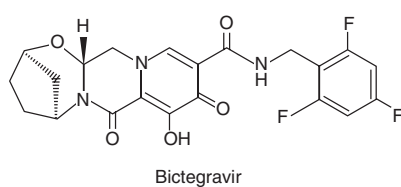
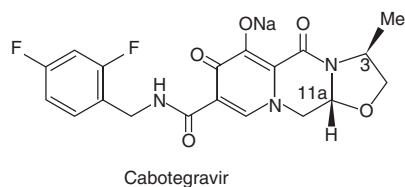
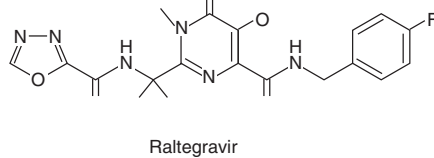
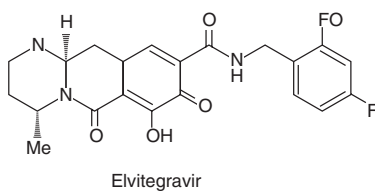


FIGURE 208-46 Molecular structures of antiretroviral agents.



Integrase Inhibitors



Capsid Inhibitor

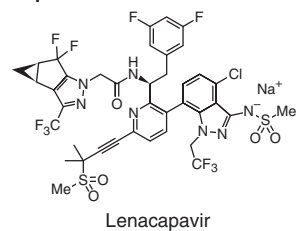


FIGURE 208-46 (Continued)

TABLE 208-22 Combination Formulations of Antiretroviral Drugs

NAME	COMBINATION
ABC/3TC (generic)	Abacavir 600 mg/lamivudine 300 mg
EFV/TDF/FTC (generic) ^a	Efavirenz 600 mg/tenofovir disoproxil fumarate 300 mg/emtricitabine 200 mg
Biktarvy ^a	Bictegravir 50 mg/tenofovir alafenamide 25 mg/emtricitabine 200 mg
Cabenuva ^a	Cabotegravir + rilpivirine (long-acting injection)
Cimduo	Tenofovir disoproxil fumarate 300 mg/lamivudine 300 mg
Complera ^a	Rilpivirine 25 mg/tenofovir disoproxil fumarate 300 mg/emtricitabine 200 mg
Delstrigo ^a	Doravirine 100 mg/tenofovir disoproxil fumarate 300 mg/lamivudine 300 mg
Descovy	Tenofovir alafenamide 25 mg/emtricitabine 200 mg
Dovato ^a	Dolutegravir 50 mg/lamivudine 300 mg
Genvoya ^a	Elvitegravir 150 mg/cobicistat 150 mg/tenofovir alafenamide 10 mg/emtricitabine 200 mg
Juluca ^a	Dolutegravir 50 mg/rilpivirine 25 mg
Odefsey ^a	Rilpivirine 25 mg/tenofovir alafenamide 25 mg/emtricitabine 200 mg
Stribild ^a	Elvitegravir 150 mg/cobicistat 150 mg/tenofovir disoproxil fumarate 300 mg/emtricitabine 200 mg
Symfi ^a	Efavirenz 600 mg/tenofovir disoproxil fumarate 300 mg/lamivudine 300 mg
Symfi Lo ^a	Efavirenz 400 mg/tenofovir disoproxil fumarate 300 mg/lamivudine 300 mg
Symtuza ^a	Darunavir 800 mg/cobicistat 150 mg/tenofovir alafenamide 10 mg/emtricitabine 200 mg
Triumeq ^a	Dolutegravir 50 mg/abacavir 600 mg/lamivudine 300 mg
Truvada	Tenofovir disoproxil fumarate 300 mg/emtricitabine 200 mg

^aComplete, once-daily, single-tablet regimens.

Source: Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents, USPHS.

The HIV-1 *protease inhibitors* (saquinavir, indinavir, ritonavir, nelfinavir, amprenavir, fosamprenavir, lopinavir/ritonavir, atazanavir, atazanavir/cobicistat, tipranavir, darunavir, and darunavir/cobicistat) are an important part of the therapeutic armamentarium of antiretrovirals. While possessing antiviral properties of its own, ritonavir is typically used as a pharmacokinetic enhancer due to its high affinity for several isoforms of cytochrome P450 (3A4, 2D6) leading to large increases in the plasma concentrations of co-administered drugs metabolized by these pathways. As in the case of reverse transcriptase inhibitors, resistance to protease inhibitors can develop rapidly in the setting of monotherapy, and thus these agents should be used only as part of combination therapeutic regimens. Based on superior efficacy and side-effect profile, ritonavir-boosted darunavir in combination with emtricitabine and tenofovir (disoproxil or alafenamide) is the preferred protease inhibitor strategy according to the DHHS Panel on the use of antiretroviral drugs.

Integrase strand transfer inhibitors act by blocking the action of the HIV integrase enzyme and thus preventing integration of the HIV provirus into the host cell genome. They are among the most potent and safest of the antiretroviral drugs and frequently part of initial combination regimens. The five licensed integrase inhibitors are raltegravir, cabotegravir, elvitegravir, dolutegravir, and bictegravir. Cabotegravir is an integrase inhibitor that is given in combination with rilpivirine as a monthly injection. Prior to initiation of the monthly injections, patients should initially be treated with oral preparations of the two drugs to be sure they are well tolerated. Elvitegravir is always given in combination with cobicistat, which acts to boost the concentrations of elvitegravir. Cobicistat also inhibits tubular secretion of creatinine, resulting in increases in serum creatinine, and is not recommended for patients with estimated creatinine clearances <70 mL/min. Bictegravir is available only in combination with tenofovir alafenamide and emtricitabine. When used as part of initial ART, integrase inhibitor-containing regimens have been associated with greater weight gain

than nonnucleoside reverse transcriptase inhibitor- or protease inhibitor-containing regimens.

Entry inhibitors act by interfering with the binding of HIV to its receptor or co-receptor or by interfering with the process of fusion (see above). The first drug in this class to be licensed was the fusion inhibitor enfuvirtide, or T-20, followed by the CCR5 antagonist maraviroc. The anti-CD4 monoclonal antibody ibalizumab was licensed in 2018, and the small molecule fostemsavir in 2020. Given that maraviroc is effective only against CCR5-tropic viruses, a co-receptor tropism assay should be performed when use of this agent is being considered.

The *capsid inhibitor* lenacapavir acts by interfering with viral replication by binding to the interface between p24 capsid subunits in hexamers. This then interferes with multiple steps in the viral life-cycle including nuclear uptake of proviral DNA, virus assembly and release, and capsid core formation. It is licensed for treatment of individuals who are heavily treatment-experienced with multidrug-resistant virus that is not suppressed on the current regimen. It is given by subcutaneous injection every 6 months after an oral or oral plus subcutaneous loading regimen.

PRINCIPLES OF THERAPY

The principles of therapy for HIV infection have been articulated by a panel sponsored by the U.S. Department of Health and Human Services as a working group of the NIH Office of AIDS Research Advisory Council. These principles are summarized in [Table 208-23](#). As noted in these guidelines, ART of HIV infection does not lead to eradication or cure of HIV. The possible exceptions are a limited number of individuals with HIV infection and cancer who received allogeneic stem cell transplants from donors who were homozygous for the CCR5Δ32 mutation (see above) and thus resistant to HIV infection.

Treatment decisions must consider the fact that one is dealing with a chronic infection that requires life-long therapy. Patients initiating antiretroviral therapy must be willing to commit to ongoing treatment and understand the importance of adherence to their prescribed regimen. The importance of adherence is illustrated by the observation that treatment interruption is associated with rapid increases in HIV RNA levels, rapid declines in CD4+ T cell counts, and an increased risk of clinical progression. While it seems reasonable to assume that the complications associated with ART could be minimized by intermittent treatment regimens designed to minimize exposure to the drugs in question, all efforts to do so have paradoxically been associated with an increase in serious adverse events in the patients randomized to intermittent therapy, demonstrating that some “non-AIDS-defining” serious adverse events such as heart attack and stroke are linked to HIV replication. Thus, unless contraindicated for reasons of toxicity, patients started on ART should remain on ART.

At present, the U.S. Department of Health and Human Services Guidelines panel recommends that everyone with HIV infection be treated with ART and that therapy be initiated as soon as possible after diagnosis with some exceptions noted below. ART has been associated with a decrease in disease progression in patients at all stages of HIV infection and leads to a decrease in the risk of transmission of infection. In addition, one may wish to administer a 6-week course of therapy to uninfected individuals immediately following a high-risk exposure to HIV. The combination of tenofovir and emtricitabine is also licensed for pre-exposure prophylaxis in individuals at high risk of HIV infection, as is an injectable, long-acting formulation of cabotegravir that may be even more effective. For patients diagnosed with an opportunistic infection and HIV infection at the same time and a CD4+ count ≥50 cells/μL, one may consider a 2- to 4-week delay in the initiation of antiretroviral therapy during which time treatment is focused on the opportunistic infection. This delay may decrease the severity of any subsequent immune reconstitution inflammatory syndrome by lowering the antigenic burden of the opportunistic infection. This is particularly true for patients with TB or cryptococcal meningitis. For patients with advanced HIV infection (CD4+ <50 cells/μL), however, ART should be initiated as soon as possible.

Nucleoside and Nucleotide Analogue Reverse Transcriptase Inhibitors (nRTIs)

69 Insertion Complex (affects all nRTIs currently approved by the US Food and Drug Administration)

Multi-nRTI Resistance	M L	A 62 V	▼ 69 Insert	K 70 R	L 210 W	T 215 Y	K 219 Q E
-----------------------	--------	--------------	-------------------	--------------	---------------	---------------	--------------------

151 Complex (affects all nRTIs currently approved by the US FDA except tenofovir)

Multi-nRTI Resistance	A 62 V	V 75 I	F 77 L	F 116 Y	Q 151 M
-----------------------	--------------	--------------	--------------	---------------	---------------

Thymidine Analogue-Associated Mutations (TAMs; affect all nRTIs currently approved by the US FDA other than emtricitabine and lamivudine)

Multi-nRTI Resistance	M L	K 70 R	L 210 W	T 215 Y	K 219 Q E
-----------------------	--------	--------------	---------------	---------------	--------------------

Abacavir	K 65 R E N	L 74 V	Y 115 F	M 184 V			
Emtricitabine/ Lamivudine	K 65 R E N			M 184 V I			
Tenofovir	K 65 R E N	K 70 E					
Zidovudine	M 41 L	D 67 N	K 70 R	L 210 W	T 215 Y	K 219 Q E	
<i>Didanosine</i>	K 65 R E N	L 74 V					
<i>Stavudine</i>	M 41 L	K 65 R E N	D 67 N	K 70 R	L 210 W	T 215 Y	K 219 Q E

Nonnucleoside Analogue Reverse Transcriptase Inhibitors (NNRTIs)

Doravirine	A 98 G	V 106 A I M T	Y 188 L	G 190 E	P 225 H	F 227 C	M 230 L	L 234 I	Y 318 F	
Efavirenz	L 100 I	K 101 P	K 103 N	V 106 M	V 108 I	Y 181 C I	Y 188 L	G 190 S A	P 225 H	M 230 L
Etravirine	V 90 I	A 98 G	L 100 I	K 101 E H P	V 106 I	E 138 A G K Q	V 179 D F T	Y 181 C I V	G 190 S A	M 230 L
Nevirapine	L 100 I	K 101 P	K 103 N	V 106 A S	V 108 I M	Y 181 C I	Y 188 L	G 190 S A H	M 230 L	
Rilpivirine	L 100 I	K 101 E P	E 138 A G K Q R	V 179 L	Y 181 C I V	Y 188 L	H 221 Y	F 227 C	M 230 I L	

FIGURE 208-47 Amino acid substitutions conferring resistance to antiretroviral drugs. For each amino acid residue, the letter above the bar indicates the amino acid associated with wild-type virus and the letter(s) below indicate the substitution(s) that confer viral resistance. The number shows the position of the mutation in the protein. Mutations selected by protease inhibitors in Gag cleavage sites are not listed. HR1, first heptad repeat; NAMs, nRTI-associated mutations; NNRTI, nonnucleoside reverse transcriptase inhibitor; nRTI, nucleoside reverse transcriptase inhibitor; PI, protease inhibitor. Amino acid abbreviations: A, alanine; C, cysteine; D, aspartate; E, glutamic acid; F, phenylalanine; G, glycine; H, histidine; I, isoleucine; K, lysine; L, leucine; M, methionine; N, asparagine; P, proline; Q, glutamine; R, arginine; S, serine; T, threonine; V, valine; W, tryptophan; Y, tyrosine. (Reproduced with permission from AM Wensing et al: 2022 Update of the drug resistance mutations in HIV-1. 13, 2022. Reproduced with permission from IAS–USA.)

Lenacapavir	L 56 I	M Q 66 67 I H	K N 70 74 H D N S R	A T 105 107 T N
-------------	--------------	---------------------	---------------------------------	-----------------------

MUTATIONS IN THE PROTEASE GENE ASSOCIATED WITH RESISTANCE TO PROTEASE INHIBITORS (PIs)

Atazanavir +/- ritonavir	L 10 F	K L 20 24 T I	V L 32 33 I F	M 46 I L	G 48 V	I F I 50 53 54 L L V L Y M L T A L S	G 73 C S T A	V 82 A T F L M S	I I N 84 85 88 V V S	L 90 M		
Darunavir/ ritonavir	V 11 I		V L 32 33 I F	I 47 V	I 50 V	I 54 M L	T L 74 76 P V	I 84 V	L 89 V			
Lopinavir/ ritonavir	L 10 F I R V	K L 20 24 M I	V L 32 33 I F	M I 46 47 I V L A	I F I 50 53 54 V L V L A M T S	A G 71 73 V T S	L 76 V	V 82 A F T S	I 84 V	L 90 M		
Tipranavir/ ritonavir	L 10 V		L 33 F	M 36 I L V	K 43 T	M I 46 47 L V	I 54 A M V	Q 58 E	H 69 K R	T 74 P	V N I 82 83 84 L D V	L 89 I M V
Fosamprenavir/ ritonavir	L 10 F I R V		V 32 I	M I 46 47 L V	I 50 V	I 54 L V M	G L 73 76 S V	V 82 A F S T	I 84 V	L 90 M		
Indinavir/ ritonavir	L 10 I R V	K L 20 24 M I	V 32 I	M 36 I	M 46 I L	I 54 V	A G 71 73 V T A	L V 76 77 V I	V 82 A F T	I 84 V	L 90 M	
Nelfinavir	L 10 F I	D 30 N	M 36 I	M 46 I L			A 71 V T	V 77 I	V 82 A F T S	I 84 V	N 88 D S	L 90 M
Saquinavir/ ritonavir	L 10 I R V	L 24 I			G 48 V	I 54 V L	I 62 V	A G 71 73 V T S	V 77 I	V 82 A F T S	I 84 V	L 90 M

MUTATIONS IN THE ENVELOPE GENE ASSOCIATED WITH RESISTANCE TO ENTRY INHIBITORS

Fostemsavir	
Ibalizumab	
Maraviroc	
Enfuvirtide	G I V Q Q N N 36 37 38 39 40 42 43 D V A R H T D S M E

MUTATIONS IN THE INTEGRASE GENE ASSOCIATED WITH RESISTANCE TO INTEGRASE STRAND TRANSFER INHIBITORS (INSTIs)

Bictegravir					G 118 R	E 138 A K T	G 140 A C R S	Q 148 H K R	S 153 F Y		R 263 K
Cabotegravir	T 66 K	L 74 I*		T 97 A	G 118 R	E 138 A K T	G 140 A C R S	Q 148 H K R	S 153 F Y	N 155 H	R 263 K
	</										

*Relevant to subtype A6 only

FIGURE 208-47 (Continued)

TABLE 208-23 Principles of Therapy of HIV Infection

1. Ongoing HIV replication leads to immune system damage, progression to AIDS, and systemic immune activation.
2. Plasma HIV RNA levels indicate the magnitude of HIV replication and the rate of CD4⁺ T cell destruction. CD4⁺ T cell counts indicate the current level of competence of the immune system.
3. Maximal suppression of viral replication is a goal of therapy; the greater the suppression the less likely the appearance of drug-resistant quasiespecies.
4. The most effective therapeutic strategies involve the simultaneous initiation of combinations of effective anti-HIV drugs with which the patient has not been previously treated and that are not cross-resistant with antiretroviral agents that the patient has already received.
5. The antiretroviral drugs used in combination regimens should be used according to optimum schedules and dosages.
6. The number of available drugs is limited. Any decisions on antiretroviral therapy have a long-term impact on future options for the patient.
7. Women should receive optimal antiretroviral therapy regardless of pregnancy status.
8. The same principles apply to children and adults, but the treatment of HIV-infected children involves unique pharmacologic, virologic, and immunologic considerations.
9. Compliance is an important part of ensuring maximal effect from a given regimen. The simpler the regimen, the easier it is for the patient to be compliant.

Source: Modified from *Principles of Therapy of HIV Infection*, USPHS, and the Henry J. Kaiser Family Foundation.

Once the decision has been made to initiate therapy, the health care provider must decide which drugs to use as the first regimen. The decision regarding choice of drugs not only will affect the immediate response to therapy but also will have implications regarding options for future therapeutic regimens. The initial regimen is usually the most effective insofar as the virus has yet to be under any selective pressure to develop significant drug resistance. HIV is capable of rapidly developing resistance to any single agent, and therapy must be given as a multidrug combination. Given that patients can be infected with viruses that harbor drug resistance mutations, it is recommended that a viral genotype be done at the time of initiation of therapy to ensure that appropriate antiretroviral agents have been selected. The combination regimens currently recommended for initial therapy in most treatment-naïve patients are listed in [Table 208-24](#). It is currently debated whether treatment-naïve individuals with <50 copies/mL of HIV RNA benefit from ART. While these individuals are at low risk of disease progression in the short term, they do have evidence of persistent immune activation that may have long-term consequences.

Following the initiation of therapy, one should expect a rapid, at least 1-log (10-fold) reduction in plasma HIV RNA levels within 1–2 months and then a slower decline in plasma HIV RNA levels to <50 copies/mL within 6 months. During this same time there should be a rise in the CD4⁺ T cell count of 100–150/cells μ L that is also

TABLE 208-24 Initial Combination Regimens Recommended for Most Treatment-Naïve Patients Regardless of HIV RNA Level or CD4 Count

For people who do not have a history of using Carbotegravir-LA as PrEP:

Bictegravir + tenofovir alafenamide + emtricitabine (Biktarvy)

Dolutegravir + tenofovir^a + emtricitabine^b

Dolutegravir + lamivudine (Dovato) only for those with HIV RNA $\leq$ 500,000 copies/mL

For people who have a history of Carbotegravir-LA as PrEP, INSTI genotype resistance testing should be performed; if starting prior to resistance testing results:

Darunavir/cobicistat or darunavir/ritonavir + tenofovir^a + emtricitabine^b

^aTenofovir alafenamide and tenofovir disoproxil fumarate are two forms of tenofovir approved by FDA. Tenofovir alafenamide has fewer bone and renal toxicities, while tenofovir disoproxil fumarate is associated with lower lipid levels. ^bLamivudine may substitute for emtricitabine and vice versa.

Source: Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents, USPHS.

TABLE 208-25 Indications for Changing Antiretroviral Therapy in Patients with HIV Infection^a

Less than a 1-log drop in plasma HIV RNA by 4 weeks following the initiation of therapy

A reproducible significant increase (defined as threefold or greater) from the nadir of plasma HIV RNA level not attributable to intercurrent infection, vaccination, or test methodology

Persistently declining CD4⁺ T cell numbers

Clinical deterioration

Side effects

^aGenerally speaking, a change should involve the initiation of at least two drugs felt to be effective in the given patient. The exception to this is when a change is being made to manage toxicity, in which case a single substitution is reasonable.

Source: Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents, USPHS.

particularly brisk during the first month of therapy. Subsequently, one should anticipate a CD4⁺ T cell count increase of 50–100 cells/year until numbers approach normal. Many clinicians feel that failure to achieve these endpoints is an indication for a change in therapy. Other reasons for a change in therapy include a persistently declining CD4⁺ T cell count, a consistent increase in HIV RNA levels to >200 copies/mL, clinical deterioration, or drug toxicity ([Table 208-25](#)). As in the case of initiating therapy, changing therapy may have a lasting impact on future therapeutic options. When changing therapy because of treatment failure (clinical progression or worsening laboratory parameters), it is important to attempt to provide a regimen with at least two new active drugs. This decision can be guided by resistance testing (see below). In the patient in whom a change is made for reasons of drug toxicity, a simple replacement of one drug is reasonable. It should be stressed that in attempting to sort out a drug toxicity it may be advisable to hold all therapy for a period of time to distinguish between drug toxicity and disease progression. Drug toxicity will usually begin to show signs of reversal within 1–2 weeks. Prior to changing a treatment regimen because of drug failure, it is important to ensure that the patient has been adherent to the prescribed regimen. As in the case of initial therapy, the simpler the new therapeutic regimen, the easier it is for the patient to be compliant. Plasma HIV RNA levels should be monitored within 2–4 weeks after initiation of ART or following a change in regimen, every 4–8 weeks until HIV RNA levels are suppressed to <200 copies/mL, and then every 3–6 months during therapy.

In order to determine an optimal therapeutic regimen for initial therapy or for a patient on a failing regimen, one may attempt to measure antiretroviral drug susceptibility through genotyping or phenotyping of HIV quasiespecies and to determine adequacy of dosing through measurement of drug levels. Genotyping may be done through cDNA sequencing. Phenotypic assays typically measure the enzymatic activity of viral enzymes in the presence or absence of different concentrations of different drugs and have also been used to determine co-receptor tropism. These assays will generally detect quasiespecies present at a frequency of $\geq$ 10%. Next-generation sequencing may allow detection of quasiespecies at frequencies down to 1%. It is generally recommended that resistance testing be used in confirming initial therapy choices in settings where the risk of transmission of resistant virus is high (such as the United States and Europe) and in determining new regimens for patients experiencing virologic failure while on therapy. Resistance testing may be of particular value in distinguishing drug-resistant virus from poor patient compliance. Due to the rapid rate at which drug-resistant viruses revert to wild-type, it is recommended that resistance testing performed in the setting of drug failure be conducted while the patient is still on the failing regimen. Measurement of plasma drug levels can also be used to tailor individual treatment. The inhibitory quotient, defined as the trough blood level/IC₅₀ of the patient's virus, is used by some to determine the adequacy of dosing of a given treatment regimen. Despite the best of efforts there will still be patients with ongoing high levels of HIV

replication while receiving the best available therapy. These patients will receive benefit from remaining on antiretroviral therapy even though it is not fully suppressive.

In addition to the licensed medications discussed above, a large number of experimental agents are being evaluated as possible therapies for HIV infection. Therapeutic strategies are being developed to interfere with virtually every step of the replication cycle of the virus (Fig. 208-3) and in an attempt to eliminate the reservoir of infected cells to “cure” HIV infection. In addition to directly acting antiviral drugs, other strategies, generically referred to as “immune-based therapies,” are being developed as a complement to antiviral therapy. Among the antiviral agents in early clinical trials are additional nucleoside and nucleotide analogues, protease inhibitors, fusion inhibitors, receptor and co-receptor antagonists, and integrase inhibitors—as well as new antiviral strategies including long-acting injectables, antisense nucleic acids, and maturation inhibitors. Among the immune-based therapies being evaluated are monoclonal antibodies, bone marrow transplantation, adoptive transfer of lymphocytes genetically modified to resist infection or enhance HIV-specific immunity, active immunotherapy with inactivated HIV or its components, IL-7, and IL-15.

HIV AND THE HEALTH CARE WORKER

Health care workers, especially those who deal with large numbers of patients with HIV, have a small but definite risk of becoming infected with HIV as a result of professional activities (see “Occupational Transmission of HIV: Health Care Workers, Laboratory Workers, and the Health Care Setting,” above).

In the United States, 58 health care workers for whom case investigations have been completed have had documented seroconversions to HIV following occupational exposures. Only one of these has occurred since 1999. Approximately 85% of the exposures resulting in infection have been due to percutaneous (puncture/cut injury) exposures to HIV-infected blood. In addition, at least 150 possible cases of occupationally acquired HIV infection have been reported among health care personnel in the United States. The number of these workers who actually acquired their infection through occupational exposures is not known. Taken together, data from several large studies suggest that the risk of HIV infection following a percutaneous exposure to HIV-contaminated blood in an individual who does not receive postexposure prophylaxis (PEP) is ~0.23%, and after a mucous membrane exposure, ~0.09%. Although episodes of HIV transmission after nonintact skin exposure have been documented, the average risk for transmission by this route has not been precisely quantified but is estimated to be less than the risk for mucous membrane exposures. The risk for transmission after exposure to body fluids or tissues other than HIV-infected blood also has not been quantified but is probably considerably lower than for blood exposures. A seroprevalence survey of 3420 orthopedic surgeons, 75% of whom practiced in an area with a relatively high prevalence of HIV infection and 39% of whom reported percutaneous exposure to patient blood, usually through an accident involving a suture needle, failed to reveal any cases of possible occupational infection, suggesting that the risk of infection with a suture needle may be considerably less than that with a blood-drawing (hollow-bore) needle.

Most cases of health care worker seroconversion have occurred as a result of needle-stick injuries. When one considers the circumstances that result in needle-stick injuries, it is immediately obvious that adhering to the standard guidelines for dealing with sharp objects would result in a significant decrease in this type of accident. In one study, 27% of needle-stick injuries resulted from improper disposal of the needle (more than half of these were due to recapping the needle), 23% occurred during attempts to start an IV line, 22% occurred during blood drawing, 16% were associated with an IM or SC injection, and 12% were associated with giving an IV infusion.

Occupational exposures to HIV should be considered as a medical emergency to ensure timely postexposure management and administration of PEP. A delay of even sending 72 h in the initiation of PEP may be the difference preventing and not preventing infection.

Recommendations regarding PEP must take into account that a variety of circumstances determine the risk of transmission of HIV following occupational exposure. In this regard, several factors have been associated with an increased risk for occupational transmission of HIV infection, including deep injury, the presence of visible blood on the instrument causing the exposure, injury with a device that had been placed in the vein or artery of the source patient, and advanced HIV disease in the source patient. Other important considerations when considering PEP in the health care worker include known or suspected pregnancy or breast-feeding, the possibility of exposure to drug-resistant virus, and the toxicities of different PEP regimens. Regardless of the decision to use PEP, the wound should be cleansed immediately and antiseptic applied. If a decision is made to offer PEP, U.S. Public Health Service guidelines recommend that PEP regimens contain three (or more) antiretroviral drugs administered for a 4-week duration for all occupational exposures to HIV. Detailed guidelines are available from the *Updated U.S. Public Health Service Guidelines for the Management of Occupational Exposures to HIV and Recommendations for Postexposure Prophylaxis* (CDC, 2018). The report emphasizes the importance of adherence to PEP when it is indicated, and close follow-up of exposed workers should be provided including counseling, baseline and follow-up HIV testing, and monitoring for drug toxicity. Follow-up appointments should begin within 72 h of an HIV exposure and may be concluded 4 months after exposure. For consultation on the treatment of occupational exposures to HIV and other bloodborne pathogens, the clinician managing the exposed patient can call the National Clinicians’ Post-Exposure Prophylaxis Hotline (PEPline) at 888-448-4911. This service is available 24 hours a day at no charge. (Additional information is available at www.ncccf.edu.) PEPline support may be especially useful in challenging situations, such as when drug-resistant HIV strains are suspected or if the health care worker is pregnant.

Health care workers can minimize their risk of occupational HIV infection by following the CDC guidelines of June 2015, which include adherence to universal precautions and assuming that blood and other body fluids from all patients are potentially infectious. Therefore, the following infection control precautions should be adhered to at all times: (1) routinely use barriers (such as gloves and/or goggles) when anticipating contact with blood or body fluids; (2) immediately wash hands and other skin surfaces after contact with blood or body fluids; and (3) carefully handle and dispose of sharp instruments during and after use. For further information contact the CDC at 800-CDC-INFO (232-4636) or see www.cdc.gov/cdc-info/. The risk of HBV infection following a needle-stick injury from a hepatitis antigen-positive patient is much higher than the risk of HIV infection (see “Transmission,” above). There are multiple examples of needle-stick injuries where the patient was positive for both HBV and HIV and the health care worker became infected only with HBV. For these reasons, it is advisable, given the high prevalence of HBV infection in people with HIV, that all health care workers dealing with patients with HIV be immunized with the HBV vaccine.

TB is another infection common to patients with HIV that can be transmitted to the health care worker. For this reason, all health care workers should know their PPD status, have it checked yearly, and, where appropriate, receive 6 months of isoniazid treatment if their skin test converts to positive. In addition, all patients in whom a diagnosis of active pulmonary TB is being entertained should be placed immediately in respiratory isolation, pending results of the diagnostic evaluation. The emergence of drug-resistant organisms, including extensively drug-resistant TB strains, has made TB an increasingly important problem for health care workers. This is particularly true for the health care worker with pre-existing HIV infection.

HIV PREVENTION

Many proven interventions, usually applied in combination, have a role in preventing the transmission of HIV (Fig. 208-48). Education, counseling, and behavior modification are the cornerstones of any HIV prevention strategy. A major problem in the United States and elsewhere is that many infections are passed on by those who do not know that they are infected. Of the ~1.2 million persons in the United States

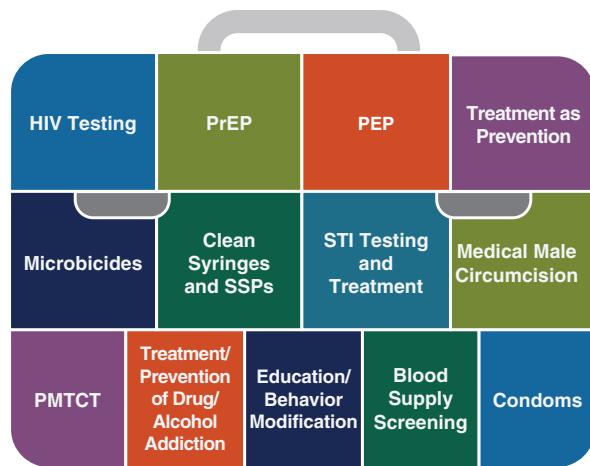


FIGURE 208-48 The HIV prevention “toolkit.” See text for detailed description. PMTCT, prevention of mother-to-child transmission of HIV; PEP, postexposure prophylaxis with antiretroviral drugs; PrEP, pre-exposure prophylaxis with antiretroviral drugs; SSPs, syringe services programs. (From: *The White House. 2021. National HIV/AIDS Strategy for the United States 2022–2025. Washington, DC.*)

with HIV, it is estimated that ~13% do not know their HIV status and that a substantial proportion of all new infections are transmitted by those people. In this regard, the CDC has recommended HIV testing as part of routine medical care and that all individuals between the ages of 13 and 64 years be tested at least once. These individuals should be informed of the testing and be tested without the need for written informed consent. Each individual can “opt out” of testing; however, testing should otherwise be routinely administered. Individuals who are practicing high-risk behavior should be tested more often and should use pre-exposure prophylaxis (PrEP) (see below). Partners engaged in monogamous sexual relationships who wish to be assured of safety should both be tested for HIV antibody. If both are negative, it must be understood that any divergence from monogamy puts both partners at risk; open discussion of the importance of honesty in such relationships should be encouraged.

When the HIV status of either partner is not known, or when one partner is positive, there are a number of options. *Use of condoms* can markedly decrease the chance of HIV transmission. It should be remembered that condoms are not 100% effective in preventing transmission of HIV infection, and there is a ~10% failure rate of condoms used for contraceptive purposes. Most condom failures result from breakage or improper usage, such as not wearing the condom for the entire period of intercourse. Latex condoms are preferable since viruses have been shown to leak through natural skin condoms. Petroleum-based gels should never be used for lubrication of the condom, since they increase the likelihood of condom rupture.

Microbicides composed of gels or rings containing antiretroviral drugs have been shown to be variably efficacious in preventing acquisition of HIV infection in women engaging in vaginal intercourse. The considerable degree of variability in efficacy relates to the generally poor adherence of participants to the use of the intervention. One product, a vaginal ring that releases the antiretroviral drug dapivirine from the ring into the vagina slowly over 28 days, has been recommended by WHO as an additional prevention choice for women at substantial risk of HIV infection as part of combination prevention approaches.

Large, prospective clinical trials have clearly demonstrated that ART for people with HIV has an important role in HIV prevention. The initial results of the HPTN 052 clinical trial published in 2011 demonstrated a 96% reduction in HIV transmission risk among heterosexual HIV-discordant couples where the partner with HIV started ART immediately versus delayed ART initiation. The final results of HPTN 052, published in 2016, reported no HIV transmissions within these couples when the partner with HIV had a suppressed viral load (defined as having a viral load of <400 copies of HIV RNA per milliliter). Three subsequent studies reported similar results, with no

genetically linked infections while the partner with HIV was virally suppressed even though couples were engaging in sex without a condom and not using PrEP. These three studies included >500 HIV-discordant heterosexual couples and >1100 HIV-discordant couples of men who have sex with men. Combined, these couples engaged in >125,000 sex acts without a condom or PrEP over more than 2600 couple-years of observation. Collectively, the studies demonstrated that if the viral load of the infected partner is decreased to below detectable levels by antiretroviral therapy, sexual transmission to the uninfected partner does not occur. This is true for heterosexuals and men who have sex with men, leading, as noted above, to the commonly used phrase “undetectable equals untransmittable” or U = U.

Pre-exposure prophylaxis (PrEP) with antiretroviral medication also is highly effective in preventing HIV acquisition by at-risk uninfected men who have sex with men and heterosexual men and women. Accumulated data indicate that high adherence to a PrEP regimen of emtricitabine + tenofovir disoproxil fumarate, taken as 1 pill per day or on demand (immediately before and following a sexual encounter), is 99% effective in preventing HIV acquisition if subjects adhere strictly to the regimen. Subsequent studies indicated similar, if not better, efficacy with cabotegravir injections given every 2 months as a maintenance regimen. A recent phase 3 clinical trial conducted in South Africa and Uganda indicated that the twice-yearly injectable HIV capsid inhibitor lenacapavir demonstrated even greater efficacy in preventing HIV infection in cisgender women. Additional phase 3 studies of lenacapavir for PrEP are being conducted in cisgender men, transgender women, transgender men, and gender-nonbinary individuals. More limited data demonstrate the utility of PrEP for people who inject drugs. CDC estimates that approximately 1.2 million people in the United States are at “substantial” risk for HIV infection and should be counseled about PrEP.

Adult male circumcision, which has been shown to result in a 50–65% reduction in HIV acquisition in the circumcised subject, is currently being pursued, particularly in developing nations, as a component of HIV prevention (see above). The most effective way to prevent transmission of HIV infection among IDUs is to stop the use of injectable drugs. Unfortunately, that is extremely difficult to accomplish unless the individual enters a treatment program. For those who will not or cannot participate in a drug treatment program and who will continue to inject drugs, the avoidance of sharing of needles and other paraphernalia (“works”) is the next best way to avoid transmission of infection. However, the cultural and social factors that contribute to the sharing of paraphernalia are complex and difficult to overcome. Under these circumstances, paraphernalia should be cleaned after each usage with a virucidal solution, such as sodium hypochlorite (undiluted household bleach). *Needle exchange programs* have been highly successful in decreasing HIV transmission among injection drug users without increasing the use of injection drugs. As noted, above, oral PrEP also is effective in preventing acquisition of HIV infections among IDUs. It is important for IDUs to be tested for HIV infection and counseled to avoid transmission to their sexual partners. Prevention of transmission through blood or blood products and prevention of mother-to-child transmission are discussed in “Transmission,” above.

■ HIV VACCINES

There is currently no safe and effective vaccine approved for the prevention of HIV infection. Successful vaccines for other diseases are predicated on the assumptions that the body can mount an adequate immune response to the microbe or virus in question during natural infection and that the vaccine will mimic the natural response to infection. Even with serious diseases, such as smallpox, poliomyelitis, measles, and influenza among others, the body usually clears the infectious agent and provides protection, which is usually lifelong against future exposure to the same pathogen. Unfortunately, this is not the case with HIV infection since the natural immune response to HIV infection is unable to clear the virus from the body and cases of superinfection are not rare.

Some of the factors that contribute to the problematic nature of developing a preventive HIV vaccine are (1) the high mutability of the

virus; (2) the fact that the infection can be transmitted by cell-free or cell-associated virus; (3) the fact that the HIV provirus integrates itself into the genome of the target cell and may remain in a latent form unexposed to the immune system; (4) the likely need for the development of effective mucosal immunity; and, importantly, (5) the difficulty that the immune system has in readily mounting broadly neutralizing antibodies in response to natural infection with HIV (see below).

Early attempts to develop a vaccine with the envelope protein gp120 aimed at inducing neutralizing antibodies in humans were unsuccessful; the elicited antisera failed to neutralize primary isolates of HIV. In this regard, two phase 3 trials were undertaken in the United States and Thailand using soluble gp120, and the vaccines failed to protect human volunteers from HIV infection. In addition, two separate vaccine trials aimed at eliciting CD8⁺ T cell responses to prevent infection and, if unsuccessful in preventing infection, to control postinfection viremia, also failed at both goals. In 2009, a vaccine using a poxvirus vector prime expressing various viral proteins followed by an envelope protein boost was assessed in a 16,000-person clinical trial (RV144) conducted in Thailand among predominantly low-HIV-prevalence heterosexuals. The vaccine provided the first positive, albeit very modest, signal ever reported in an HIV vaccine trial, showing 31% protection against acquisition of infection. Such a result is certainly not sufficient justification for clinical use of the vaccine; however, it served as an important first step in the direction of the development of a safe and effective vaccine against HIV infection.

Follow-up studies of RV144 indicate that nonneutralizing or weakly neutralizing antibody responses against certain constant epitopes in the otherwise highly variable V1–V2 region of the HIV envelope may be associated with the modest degree of protection observed in that clinical trial. Three additional similar studies were undertaken in high-HIV-prevalence countries in sub-Saharan Africa as well as in the Americas and certain European countries in attempts to improve on the results of RV144 by a variety of approaches, including increasing the number of vaccine boosts with envelope protein, the use of mosaic antigens, and the addition of adjuvant. Unfortunately, all three of these phase 3 studies of candidate vaccines failed to show efficacy. Another study was terminated early due to lack of efficacy.

An area of HIV vaccine research that is currently being actively pursued is the attempt to induce broadly neutralizing antibodies by developing as immunogens for vaccination certain epitopes on the HIV envelope that are the targets of naturally occurring broadly neutralizing antibodies during HIV infection (Fig. 208-30). It is curious that only about 20% of people with HIV develop broadly neutralizing antibodies in response to natural infection and they do so only after 2–3 years of ongoing infection. By the time these antibodies appear, they can neutralize a broad range of primary HIV isolates, but they appear to be ineffective against the autologous virus in the infected subject. Upon close examination, these broadly neutralizing antibodies manifest a high degree of somatic mutations that accumulated over time and are responsible for their affinity maturation and broadly neutralizing capacity. The goal of current efforts is to develop the conformationally correct HIV envelope epitopes that, when used as immunogens, would direct the immune response of an uninfected individual to the production of broadly neutralizing antibodies over a reasonable time frame by sequential immunizations. It remains to be seen whether this approach will be feasible.

FURTHER READING

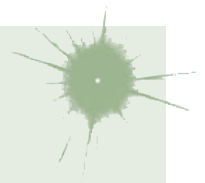
- BEKKER LG et al: HIV infection. *Nat Rev Dis Primers* 9:42, 2023.
BEYRER C et al: Is HIV epidemic control by 2030 realistic? *Lancet HIV* 7:e489, 2024.
CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC): Clinical Guidance for PrEP. Available at www.cdc.gov/hiv/nexus/hcp/prep/.
CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC): Clinical Guidance for PEP. Available at www.cdc.gov/hiv/nexus/hcp/pep/.
CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC): Clinical Care of HIV. Available at www.cdc.gov/hiv/nexus/hcp/clinical-care/.
COHN LB et al: Biology of the HIV-1 latent reservoir and implications for cure strategies. *Cell Host Microbe* 27:519, 2020.

- COLLINS DR et al: CD8⁺ T cells in HIV control, cure and prevention. *Nat Rev Immunol* 20:471, 2020.
EISINGER RW et al: HIV viral load and transmissibility of HIV infection: Undetectable equals untransmittable. *JAMA* 321:451, 2019.
FAUCI AS, LANE HC: Four decades of HIV/AIDS—much accomplished, much to do. *N Engl J Med* 383:1, 2020.
GROSSO TM et al: HIV and aging, biological mechanisms, and therapies: What do we know? *AIDS Rev* 25:79, 2022.
HAYNES BF et al: Strategies for HIV-1 vaccines that induce broadly neutralizing antibodies. *Nat Rev Immunol* 23:142, 2023.
HORBERG M et al: Primary care guidance for persons with human immunodeficiency virus: 2024 update by the HIV Medicine Association of the Infectious Diseases Society of America. *Clin Infect Dis*, 2024. [Epub ahead of print.]
KAZER SW: Evolution and diversity of immune responses during acute HIV infection. *Immunity* 53:908, 2020.
LANDOVITZ RJ et al: Prevention, treatment and cure of HIV infection. *Nat Rev Microbiol* 21:657, 2023.
LISCO A et al: Reappraisal of idiopathic CD4 lymphocytopenia at 30 years. *N Engl J Med* 388:1681, 2023.
MOIR S, FAUCI AS: B-cell responses to HIV infection. *Immunol Rev* 275:33, 2017.
NKOLOLA JP, BAROUCH DH: Prophylactic HIV-1 vaccine trials: Past, present, and future. *Lancet HIV* 11:e117, 2024.
PANEL ON OPPORTUNISTIC INFECTIONS IN ADULTS AND ADOLESCENTS WITH HIV: Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents with HIV. Available at clinicalinfo.hiv.gov/en/guidelines/hiv-clinical-guidelines-adult-and-adolescent-opportunistic-infections/whats-new.
SAEZ-CIRION A, SERETI I: Immunometabolism and HIV-1 pathogenesis: Food for thought. *Nat Rev Immunol* 21:5, 2021.
UN JOINT PROGRAMME ON HIV/AIDS (UNAIDS): 2024 global AIDS report — The Urgency of Now: AIDS at a Crossroads. Available at www.unaids.org/en/resources/documents/2024/global-aids-update-2024.
U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES PANEL ON ANTIRETROVIRAL GUIDELINES FOR ADULTS AND ADOLESCENTS: Guidelines for the use of antiretroviral agents in adults and adolescents living with HIV. Available at clinicalinfo.hiv.gov/en/guidelines/hiv-clinical-guidelines-adult-and-adolescent-arv/whats-new.
VENTER WDF et al: The long wait for long-acting HIV prevention and treatment formulations. *Lancet HIV* 11:e711, 2024.

Section 15 Infections Due to RNA Viruses

209 Viral Gastroenteritis

Umesh D. Parashar, Roger I. Glass



Acute infectious gastroenteritis is a common illness that affects persons of all ages worldwide. It is a leading cause of death among children in developing countries, accounting for an estimated 0.5 million deaths each year, and is responsible for up to 6–8% of all hospitalizations among children in industrialized countries, including the United States. Elderly persons, especially those with debilitating health conditions, also are at risk of severe complications and death from acute gastroenteritis. Among healthy young adults, acute gastroenteritis is rarely fatal but incurs substantial medical and social costs, including those of time lost from work.

Several enteric viruses have been recognized as important etiologic agents of acute infectious gastroenteritis (Table 209-1, Fig. 209-1).

TABLE 209-1 Viral Causes of Gastroenteritis among Humans

VIRUS	FAMILY	GENOME	PRIMARY AGE GROUP AT RISK	CLINICAL SEVERITY	DETECTION ASSAYS
Group A rotavirus	Reoviridae	Double-strand segmented RNA	Children <5 years	+++	EIA (commercial), RT-PCR, EM, PAGE
Norovirus	Caliciviridae	Positive-sense single-strand RNA	All ages	++	RT-PCR, EM, EIA (commercial)
Sapovirus	Caliciviridae	Positive-sense single-strand RNA	Children <5 years	+	RT-PCR, EM
Astrovirus	Astroviridae	Positive-sense single-strand RNA	Children <5 years	+	EIA, RT-PCR, EM
Adenovirus (mainly types 40 and 41)	Adenoviridae	Double-strand DNA	Children <5 years	+/++	EIA (commercial), PCR, EM

Abbreviations: EIA, enzyme immunoassay; EM, electron microscopy; PAGE, polyacrylamide gel electrophoresis; PCR, polymerase chain reaction; RT-PCR, reverse-transcription PCR.

Although most viral gastroenteritis is caused by RNA viruses, the DNA viruses that are occasionally involved (e.g., adenovirus types 40 and 41) are included in this chapter. Illness caused by these viruses is characterized by the acute onset of vomiting and/or diarrhea, which may be accompanied by fever, nausea, abdominal cramps, anorexia, and malaise. As shown in Table 209-2, several features can help distinguish gastroenteritis caused by viruses from that caused by bacterial agents. However, the distinction based on clinical and epidemiologic parameters alone is often difficult, and laboratory tests are required to confirm the diagnosis.

HUMAN CALICIVIRUSES

Etiologic Agent The Norwalk virus is the prototype strain of a group of small (27–40 nm), nonenveloped, round, icosahedral viruses with relatively amorphous surface features on visualization by electron microscopy. Molecular cloning and characterization have demonstrated that the viruses have a single, positive-strand RNA genome ~7.5 kb in length and possess a single virion-associated protein—similar to that of typical caliciviruses—with a molecular mass of 60 kDa. On the basis of these molecular characteristics, these viruses are presently classified into two genera belonging to the family Caliciviridae, the *noroviruses* and the *sapoviruses*, which are further classified into genogroups and genotypes. Of the 10 recognized norovirus genogroups in humans and animals, 35 different genotypes belonging to 5 genogroups (GI, GII, GIV, GVIII, and GIX) are known to infect humans.

Epidemiology Infections with human caliciviruses are common worldwide, and most adults have antibodies to these viruses. Antibody is acquired at an earlier age in developing countries—a pattern consistent with the presumed fecal–oral mode of transmission. Infections occur year-round, although, in temperate climates, a distinct increase has been noted in cold-weather months. Noroviruses may be the most common infectious agents of mild gastroenteritis in the community

and affect all age groups, whereas sapoviruses primarily cause gastroenteritis in children. Noroviruses also cause traveler’s diarrhea, and outbreaks have occurred among military personnel deployed to various parts of the world. The limited data available indicate that norovirus may be the second most common viral agent (after rotavirus) among young children and the most common agent among older children and adults. In the United States and some other developed countries, with the decline in severe rotavirus disease following implementation of a rotavirus vaccination program, norovirus has become the leading cause of medically attended gastroenteritis in young children. Noroviruses are also recognized as the major cause of epidemics of gastroenteritis worldwide. In the United States, ~50% of all reported outbreaks of gastroenteritis are caused by noroviruses.

Norovirus is transmitted predominantly by the fecal–oral route but is also present in vomitus. Because an inoculum with very few viruses can be infectious, transmission can occur by aerosolization, by contact with contaminated fomites, and by person-to-person contact. Viral shedding and infectivity are greatest during the acute illness, but challenge studies with Norwalk virus in volunteers indicate that viral antigen may be shed by asymptotically infected persons and also by symptomatic persons before the onset of symptoms and for several weeks after the resolution of illness. Viral shedding can be prolonged in immunocompromised individuals.

Pathogenesis The exact sites and cellular receptors for attachment of viral particles have not been determined. Data suggest that carbohydrates that are similar to human histo-blood group antigens (HBGA) and are present on the gastroduodenal epithelium of individuals with the secretor phenotype may serve as ligands for the attachment of norovirus. Additional studies must more fully elucidate norovirus–carbohydrate interactions, including strain-specific variations. After the infection of volunteers, reversible lesions are noted in the upper jejunum, with broadening and blunting of the villi, shortening of the microvilli, vacuolization of the lining epithelium, crypt hyperplasia,

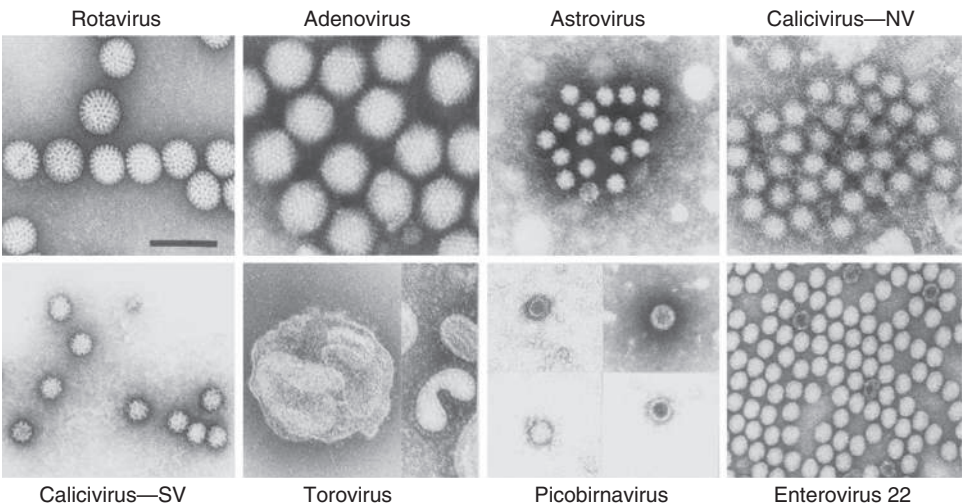


FIGURE 209-1 Viral agents of gastroenteritis. NV, norovirus; SV, sapovirus.

TABLE 209-2 Characteristics of Gastroenteritis Caused by Viral and Bacterial Agents

FEATURE	VIRAL GASTROENTERITIS	BACTERIAL GASTROENTERITIS
Setting	Incidence similar in developing and developed countries	More common in settings with poor hygiene and sanitation
Infectious dose	Low (10–100 viral particles) for most agents	High (>10 ⁵ bacteria) for <i>Escherichia coli</i> , <i>Salmonella</i> , <i>Vibrio</i> ; medium (10 ² –10 ⁵ bacteria) for <i>Campylobacter jejuni</i> ; low (10–100 bacteria) for <i>Shigella</i>
Seasonality	In temperate climates, winter seasonality for most agents; year-round occurrence in tropical areas	More common in summer or rainy months, particularly in developing countries with a high disease burden
Incubation period	1–3 days for most agents; can be shorter for norovirus	1–7 days for common agents (e.g., <i>Campylobacter</i> , <i>E. coli</i> , <i>Shigella</i> , <i>Salmonella</i>); a few hours for bacteria producing preformed toxins (e.g., <i>Staphylococcus aureus</i> , <i>Bacillus cereus</i>)
Reservoir	Primarily humans	Depending on bacterial species, human (e.g., <i>Shigella</i> , <i>Salmonella</i>), animal (e.g., <i>Campylobacter</i> , <i>Salmonella</i> , <i>E. coli</i>), and water (e.g., <i>Vibrio</i>) reservoirs exist
Fever	Common with rotavirus and norovirus; uncommon with other agents	Common with agents causing inflammatory diarrhea (e.g., <i>Salmonella</i> , <i>Shigella</i>)
Vomiting	Prominent and can be the only presenting feature, especially in children	Common with bacteria producing preformed toxins; less prominent in diarrhea due to other agents
Diarrhea	Common; nonbloody in almost all cases	Prominent and occasionally bloody with agents causing inflammatory diarrhea
Duration	1–3 days for norovirus and sapovirus; 2–8 days for other viruses	1–2 days for bacteria producing preformed toxins; 2–8 days for most other bacteria
Diagnosis	This is often a diagnosis of exclusion in clinical practice. Commercial enzyme immunoassays are available for detection of rotavirus and adenovirus, but identification of other agents is limited to research and public health laboratories.	Fecal examination for leukocytes and blood is helpful in differential diagnosis. Culture of stool specimens, sometimes on special media, can identify several pathogens. Molecular techniques are useful epidemiologic tools but are not routinely used in most laboratories.
Treatment	Supportive therapy to maintain adequate hydration and nutrition should be given. Antibiotics and antimotility agents are contraindicated.	Supportive hydration therapy is adequate for most patients. Antibiotics are recommended for patients with dysentery caused by <i>Shigella</i> or diarrhea caused by <i>Vibrio cholerae</i> and for some patients with <i>Clostridium difficile</i> colitis.

and infiltration of the lamina propria by polymorphonuclear neutrophils and lymphocytes. The lesions persist for at least 4 days after the resolution of symptoms and are associated with malabsorption of carbohydrates and fats and a decreased level of brush-border enzymes. Adenylate cyclase activity is not altered. No histopathologic changes are seen in the stomach or colon, but gastric motor function is delayed, and this alteration is believed to contribute to the nausea and vomiting that are typical of this illness.

Clinical Manifestations Gastroenteritis caused by human caliciviruses has a sudden onset following an average incubation period of 24 h (range, 12–72 h). The illness generally lasts 12–60 h and is characterized by one or more of the following symptoms: nausea, vomiting, abdominal cramps, and diarrhea. Vomiting is more prevalent among children, whereas a greater proportion of adults develop diarrhea. Constitutional symptoms are common, including headache, fever, chills, and myalgias. The stools are characteristically loose and watery, without blood, mucus, or leukocytes. White cell counts are generally normal; rarely, leukocytosis with relative lymphopenia may be observed. Death is a rare outcome and usually results from severe dehydration in vulnerable persons (e.g., elderly patients with debilitating health conditions).

Immunity Approximately 50% of persons challenged with Norwalk virus become ill and acquire short-term immunity against the infecting strain. In early human volunteer studies, immunity to Norwalk virus appeared to correlate inversely with level of antibody; i.e., persons with higher levels of preexisting antibody to Norwalk virus were more susceptible to illness on rechallenge. This paradoxical observation was later explained by data indicating that some individuals have a genetic predisposition to illness, with specific HBGA phenotypes influencing susceptibility to norovirus infection. Contemporary data show that functional antibodies that block norovirus binding to HBGA correlate with protective immunity in human volunteer challenge and vaccination studies. Furthermore, initial studies have demonstrated that norovirus grown in vitro in the newly developed human intestinal enteroid (HIE) cell-based system can be neutralized by sera containing blocking antibodies.

Diagnosis Cloning and sequencing of the genomes of Norwalk and several other human caliciviruses have allowed the development

of assays based on polymerase chain reaction (PCR) for detection of virus in stool and vomitus. PCR-based detection assays for norovirus combined with those for multiple enteric pathogens are commercially available and are increasingly used in clinical settings. Virus-like particles (VLPs) produced by expression of capsid proteins in a recombinant baculovirus vector have been used to develop enzyme immunoassays (EIAs) for detection of virus in stool or a serologic response to a specific viral antigen. Commercial EIA kits for detection in stool have limited sensitivity and usefulness in clinical practice and are of greatest utility in outbreaks, in which many specimens are tested and only a few need be positive to identify norovirus as the cause.

TREATMENT

Infections with Norwalk and Related Human Caliciviruses

The disease is self-limited, and oral rehydration therapy is generally adequate. If severe dehydration develops, IV fluid therapy is indicated. No specific antiviral therapy is available.

Prevention Epidemic prevention relies on situation-specific measures, such as control of contamination of food and water, exclusion of ill food handlers, and reduction of person-to-person spread through good personal hygiene and disinfection of contaminated fomites. The role of immunoprophylaxis is not clear, given the lack of long-term immunity from natural disease, but efforts to develop norovirus vaccines are ongoing. Vaccines based on VLPs are being tested in human volunteers. In a proof-of-concept trial, the efficacy of a monovalent GI.1 VLP vaccine was 47% among volunteers who received the vaccine intranasally and were then challenged with a homologous strain. In a second trial, norovirus disease severity was reduced in volunteers who received a bivalent GI.1/GII.4 VLP vaccine intramuscularly (with the GII.4 component including a consensus sequence from three different GII.4 strains) and were subsequently challenged with a GII.4 norovirus strain. Data from the first field efficacy study of this bivalent vaccine conducted in ~4700 healthy U.S. Navy recruits given 1 intramuscular injection of the bivalent vaccine were recently reported. While the primary endpoint of

1630 protection against homotypic infection could not be evaluated because only 6 total moderate/severe cases due to GI.1 or GII.4 norovirus strains occurred during the trial, the vaccine efficacy was 61.8% (95.01% confidence interval, 20.8–81.6%) for moderate/severe norovirus acute gastroenteritis due to any type. These initial data are encouraging; however, key issues to be further studied include the duration of protection and the level of heterotypic protection against antigenically distinct strains, particularly given the continuing and rapid natural evolution leading to the emergence of novel norovirus strains.

■ ROTAVIRUS

Etiologic Agent Rotaviruses are members of the family Reoviridae. The viral genome consists of 11 segments of double-strand RNA enclosed in a triple-layered, nonenveloped, icosahedral capsid 75 nm in diameter. Viral protein 6 (VP6), the major structural protein, is the target of commercial immunoassays and determines the group specificity of rotaviruses. Seven major groups of rotavirus (A through G) exist; human illness is caused primarily by group A and, to a much lesser extent, by groups B and C. Two outer-capsid proteins, VP7 (G-protein) and VP4 (P-protein), determine serotype specificity, induce neutralizing antibodies, and form the basis for binary classification of rotaviruses (G and P types). The segmented genome of rotavirus allows genetic reassortment (i.e., exchange of genome segments between viruses) during co-infection—a property that plays a role in viral evolution and that has been utilized in the development of reassortant animal/human rotavirus-based vaccines.

Epidemiology Worldwide, nearly all children are infected with rotavirus by 3–5 years of age. Neonatal infections are common but are often asymptomatic or mild, presumably because of protection by maternal antibody or breast milk. Compared with rotavirus disease in industrialized countries, disease in developing countries occurs at a younger age, is less seasonal, is more frequently caused by uncommon or multiple rotavirus strains, and is more often fatal. Moreover, because of suboptimal access to hydration therapy, rotavirus is a leading cause of diarrheal death among children in the developing world, with the

highest mortality rates among children in sub-Saharan Africa and southern Asia (Fig. 209-2).

First infections after 3 months of age are likely to be symptomatic, and the incidence of disease peaks among children 4–23 months of age. Reinfections are common, but the severity of disease decreases with each repeat infection. Therefore, severe rotavirus infections are less common among older children and adults than among younger individuals. Nevertheless, rotavirus can cause illness in parents and caretakers of children with rotavirus diarrhea, immunocompromised persons, travelers, and elderly individuals and should be considered in the differential diagnosis of gastroenteritis among adults.

In tropical settings, rotavirus disease occurs year-round, with less pronounced seasonal peaks than in temperate settings, where rotavirus disease occurs predominantly during the cooler fall and winter months. Before the introduction of rotavirus vaccine in the United States, the rotavirus season each year began in the Southwest during the autumn and early winter (October through December) and migrated across the continent, peaking in the Northeast during late winter and spring (March through May). The reasons for this characteristic pattern are not clear but may be correlated with state-specific differences in birth rates, which could influence the rate of accumulation of susceptible infants after each rotavirus season. After the implementation of routine vaccination of U.S. infants against rotavirus in 2006, the characteristic prevaccine geotemporal pattern of U.S. rotavirus was dramatically altered, and these changes were accompanied by substantial declines in rotavirus detections by a national network of sentinel laboratories.

During episodes of rotavirus-associated diarrhea, virus is shed in large quantities in stool (10^7 – 10^{12} /g). Viral shedding detectable by EIA usually subsides within 1 week but may persist for >30 days in immunocompromised individuals; it may be detected for longer periods by sensitive molecular assays, such as PCR. The virus is transmitted predominantly through the fecal–oral route. Spread through respiratory secretions, person-to-person contact, or contaminated environmental surfaces has been postulated to explain the rapid acquisition of antibody in the first 3 years of life, regardless of sanitary conditions.

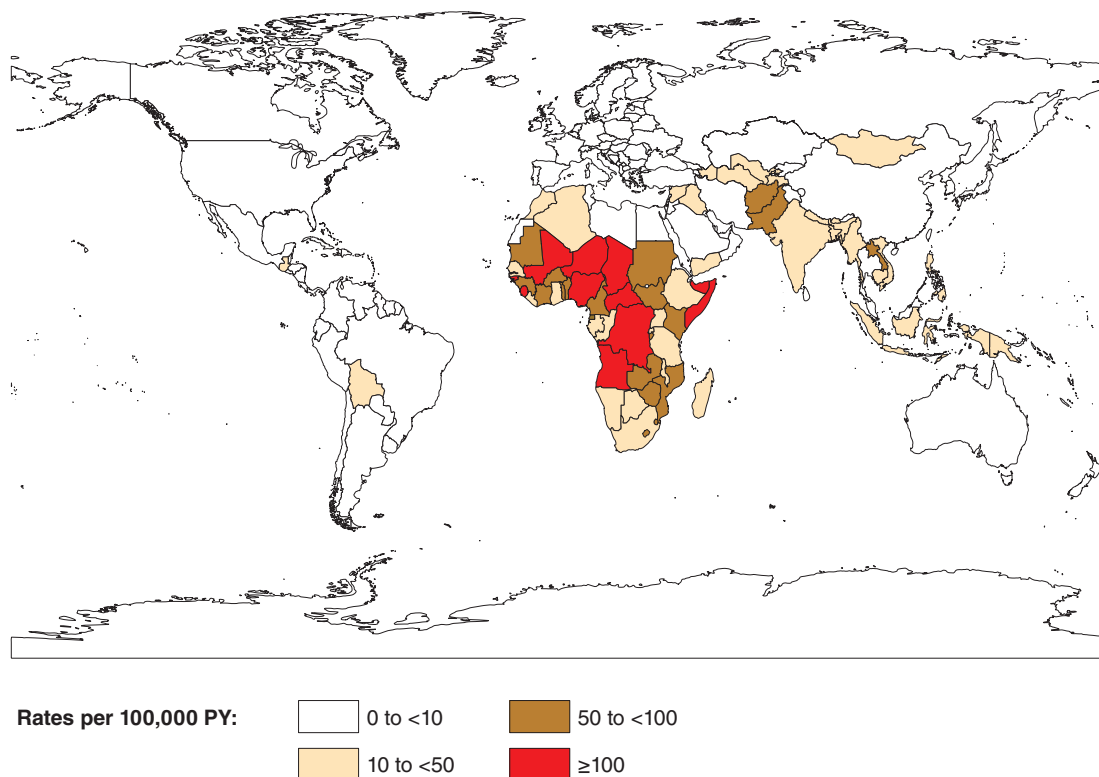


FIGURE 209-2 Rotavirus mortality rates by country, per 100,000 children <5 years of age. (From JE Tate et al: *Global, regional, and national estimates of rotavirus mortality in children <5 years of age, 2000–2013. Clin Infect Dis* 62:S96, 2016.)

At least 10 different G serotypes of group A rotavirus have been identified in humans, but only 5 types (G1 through G4 and G9) are common. While human rotavirus strains that possess a high degree of genetic homology with animal strains have been identified, animal-to-human transmission appears to be uncommon.

Group B rotaviruses have been associated with several large epidemics of severe gastroenteritis among adults in China since 1982 and have also been identified in India. Group C rotaviruses have been associated with a small proportion of pediatric gastroenteritis cases in several countries worldwide.

Pathogenesis Rotaviruses infect and ultimately destroy mature enterocytes in the villous epithelium of the proximal small intestine. The loss of absorptive villous epithelium, coupled with the proliferation of secretory crypt cells, results in secretory diarrhea. Brush-border enzymes characteristic of differentiated cells are reduced, and this change leads to the accumulation of unmetabolized disaccharides and consequent osmotic diarrhea. Studies in mice indicate that a non-structural rotavirus protein, NSP4, functions as an enterotoxin and contributes to secretory diarrhea by altering epithelial cell function and permeability. In addition, rotavirus may evoke fluid secretion through activation of the enteric nervous system in the intestinal wall. Rotavirus antigenemia and viremia are common among children with acute rotavirus infection, although the antigen and RNA levels in serum are substantially lower than those in stool.

Clinical Manifestations The clinical spectrum of rotavirus infection ranges from subclinical infection to severe gastroenteritis leading to life-threatening dehydration. After an incubation period of 1–3 days, the illness has an abrupt onset, with vomiting frequently preceding the onset of diarrhea. Up to one-third of patients may have a temperature of $>39^{\circ}\text{C}$. The stools are characteristically loose and watery and only infrequently contain red or white cells. Gastrointestinal symptoms generally resolve in 3–7 days.

Respiratory and neurologic features in children with rotavirus infection have been reported, but causal associations have not been proven. Moreover, rotavirus infection has been associated with a variety of other clinical conditions (e.g., sudden infant death syndrome, necrotizing enterocolitis, intussusception, Kawasaki disease, and type 1 diabetes), but no causal relationship has been confirmed with any of these syndromes.

Rotavirus does not appear to be a major opportunistic pathogen in children with HIV infection. In severely immunodeficient children, rotavirus can cause protracted diarrhea with prolonged viral excretion and, in rare instances, can disseminate systemically. Persons who are immunosuppressed for bone marrow transplantation also are at risk for severe or even fatal rotavirus disease.

Immunity Protection against rotavirus disease is correlated with the presence of virus-specific secretory IgA antibodies in the intestine and, to some extent, the serum. Because virus-specific IgA production at the intestinal surface is short-lived, complete protection against disease is only temporary. However, each infection and subsequent reinfection confers progressively greater immunity; thus, severe disease is most common among young children with first or second infections. Immunologic memory is believed to be important in the attenuation of disease severity upon reinfection.

Diagnosis Illness caused by rotavirus is difficult to distinguish clinically from that caused by other enteric viruses. Because large quantities of virus are shed in feces, the diagnosis can usually be confirmed by a wide variety of commercially available EIAs or by techniques for detecting viral RNA, such as gel electrophoresis, probe hybridization, or PCR.

TREATMENT

Rotavirus Infections

Rotavirus gastroenteritis can lead to severe dehydration; appropriate treatment should be instituted early. Standard oral rehydration therapy is successful for most children who can take fluids

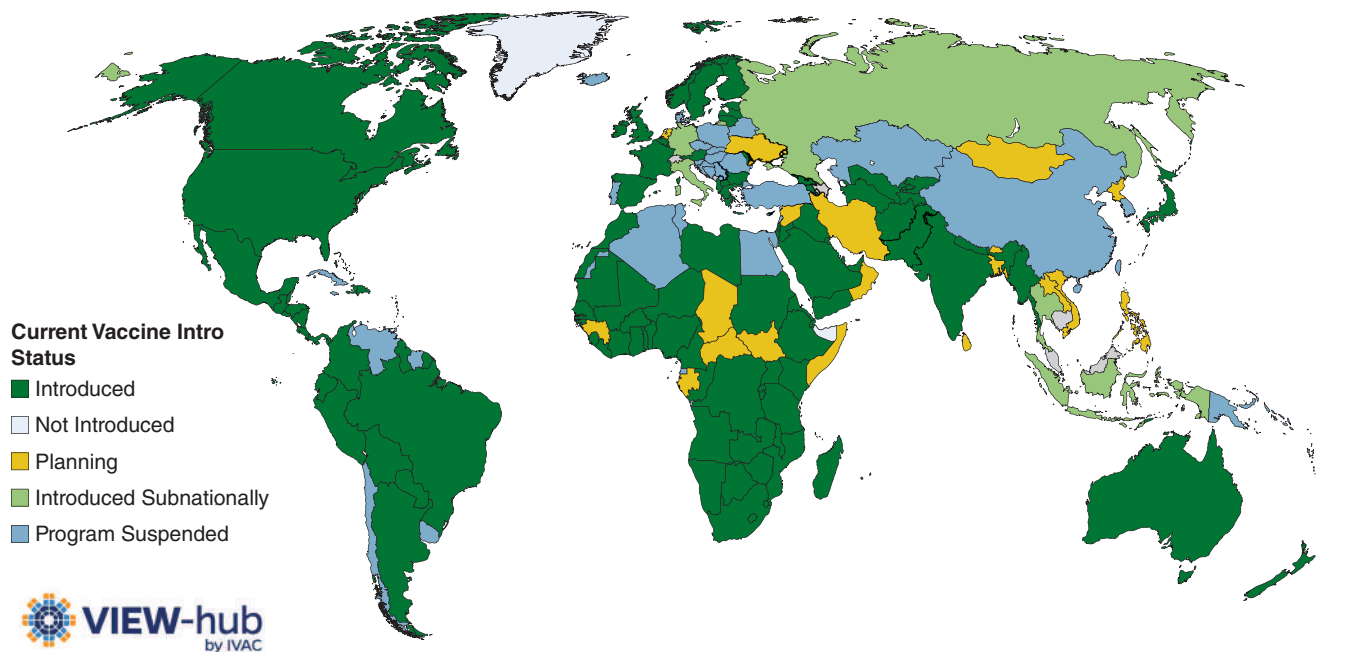
by mouth, but IV fluid replacement may be required for patients who are severely dehydrated or are unable to tolerate oral therapy because of frequent vomiting. The therapeutic roles of probiotics, bismuth subsalicylate, enkephalinase inhibitors, and nitazoxanide have been evaluated in clinical studies but are not clearly defined. Antibiotics and antimotility agents should be avoided. In immunocompromised children with chronic symptomatic rotavirus disease, orally administered immunoglobulins or colostrum may result in the resolution of symptoms, but the best choices regarding agents and their doses have not been well studied, and treatment decisions are often empirical.

Prevention Efforts to develop rotavirus vaccines were pursued because it was apparent—given the similar rates in less developed and industrialized nations—that improvements in hygiene and sanitation were unlikely to reduce disease incidence. The first rotavirus vaccine licensed in the United States in 1998 was withdrawn from the market within 1 year because it was linked with a low incidence of intussusception, a form of bowel obstruction.

In 2006, promising safety and efficacy (85–98% against severe rotavirus disease) data for two new rotavirus vaccines—RotaTaq (Merck, United States) and Rotarix (GlaxoSmithKline, Belgium)—were reported from large clinical trials conducted in North America, Europe, and Latin America. Both vaccines are now recommended for routine immunization of all U.S. infants, and their use has led to a >70 –80% decline in rotavirus hospitalizations and emergency department visits at hospitals across the United States. Somewhat unexpectedly, rotavirus vaccination of young infants has also resulted in the added benefit of declines in rotavirus disease among children who miss vaccination and even among older children and adults who are not eligible for vaccination in some settings. The reason is likely to be a reduction in community transmission of rotavirus because of vaccination—i.e., herd protection. In April 2009, the World Health Organization (WHO) recommended the use of rotavirus vaccines in all countries worldwide. As of December 2003, over 120 countries, including several low-income countries in Africa and Asia, have incorporated rotavirus vaccine into their national childhood immunization programs (Fig. 209-3). Large declines in severe morbidity and mortality from childhood diarrhea have been documented in many countries. Postmarketing surveillance has identified a low risk of intussusception in some high- and middle-income countries; however, the benefits of vaccination exceed the risks, and no changes in vaccine administration policy have been implemented. An intussusception risk has not been identified in several postmarketing evaluations in developing countries to date.

The different epidemiology of rotavirus disease and the greater prevalence of co-infection with other enteric pathogens, of comorbidities, and of malnutrition in developing countries may adversely affect the performance of oral rotavirus vaccines, as is the case with oral vaccines against poliomyelitis, cholera, and typhoid in these regions. Therefore, evaluation of the efficacy of rotavirus vaccines in resource-poor settings of Africa and Asia was specifically recommended, and these trials have now been completed. As anticipated, the efficacy of rotavirus vaccines was moderate (50–65%) in these settings when compared with that in industrialized countries. Despite modest efficacy, routine use of rotavirus vaccines in low-income African countries with a heavy disease burden has yielded substantial public health benefits.

Several manufacturers in emerging markets, including India, China, Vietnam, Indonesia, and Brazil, are developing candidate rotavirus vaccines. Beginning in 2016, two Indian-made rotavirus vaccines—Rotavac (Bharat Biotech, India) and Rotasiil (Serum Institute, India)—were implemented in India's routine childhood immunization program, which has since expanded to all Indian states with a birth cohort of >25 million. In trials conducted in low-income countries, the efficacy of Rotavac and Rotasiil ranged from 36 to 66%, similar to the efficacy of multinational vaccines in these settings. In 2018, these two vaccines were prequalified by WHO, allowing their procurement with funding support from Gavi, the Vaccine Alliance, in low-income countries outside India.



January 8, 2024 © The International Vaccine Access Center (IVAC)

FIGURE 209-3 Countries that have implemented national rotavirus vaccination programs, January 8, 2024. (Source: View-Hub, <http://www.view-hub.org/viz/>. © The International Vaccine Access Center [IVAC].)

■ OTHER VIRAL AGENTS OF GASTROENTERITIS

Enteric *adenoviruses* of serotypes 40 and 41 belonging to subgroup F are 70- to 80-nm viruses with double-strand DNA that cause ~2–12% of all diarrhea episodes in young children. Unlike adenoviruses that cause respiratory illness, enteric adenoviruses are difficult to cultivate in cell lines, but they can be detected with commercially available EIAs. Adenovirus types 31 and 42–49 have been linked to diarrhea in HIV-infected and other immunocompromised persons.

Astroviruses are 28- to 30-nm viruses with a characteristic icosahedral structure and a positive-sense, single-strand RNA. At least seven serotypes have been identified, of which serotype 1 is most common. Astroviruses are primarily pediatric pathogens, causing ~2–10% of cases of mild to moderate gastroenteritis in children. The availability of simple immunoassays to detect virus in fecal specimens and of molecular methods to confirm and characterize strains will permit more comprehensive assessment of the etiologic role of these agents.

Toroviruses are 100- to 140-nm, enveloped, positive-strand RNA viruses that are recognized as causes of gastroenteritis in horses (Berne virus) and cattle (Breda virus). Their role as a cause of diarrhea in humans is still unclear, but studies from Canada have demonstrated associations between torovirus excretion and both nosocomial gastroenteritis and necrotizing enterocolitis in neonates. These associations require further evaluation.

Picobirnaviruses are small, bisegmented, double-strand RNA viruses that cause gastroenteritis in a variety of animals. Their role as primary causes of gastroenteritis in humans remains unclear, but several studies have found an association between picobirnaviruses and gastroenteritis in HIV-infected adults.

Several other viruses (e.g., enteroviruses, reoviruses, pestiviruses, Aichivirus, and parvovirus B) have been identified in the feces of patients with diarrhea, but their etiologic role in gastroenteritis has not been proven. Diarrhea has also been noted as a manifestation of infection with recently recognized viruses that primarily cause severe respiratory illness: the SARS-CoV viruses, the influenza A/H5N1 virus, and the current pandemic strain of influenza A/H1N1 virus.

■ FURTHER READING

ARMAH G et al: Vaccine value profile for norovirus. *Vaccine* 41 Suppl 2:S134, 2023.

CLARK A et al: Estimating the global impact of rotavirus vaccines on child mortality. *Int J Infect Dis* 137:90, 2023.

COHEN A et al: Aetiology and incidence of diarrhoea requiring hospitalisation in children under 5 years of age in 28 low-income and middle-income countries: findings from the Global Pediatric Diarrhea Surveillance. *BMJ Global Health* 7:e009548, 2022.

FLYNN T et al: Viral gastroenteritis. *Lancet* 403:862, 2024.

TATE J et al: Global, regional, and national estimates of rotavirus mortality in children <5 years of age, 2000–2013. *Clin Infect Dis* 62:S96, 2016.

210 Enterovirus, Parechovirus, and Reovirus Infections

Jeffrey I. Cohen

ENTEROVIRUSES

■ CLASSIFICATION AND CHARACTERIZATION

Enteroviruses, members of the family Picornaviridae, are so designated because of their ability to multiply in the gastrointestinal tract. Despite their name, these viruses are not a prominent cause of gastroenteritis. Enteroviruses encompass more than 115 human serotypes: 3 serotypes of poliovirus, 23 serotypes of coxsackievirus A, 6 serotypes of coxsackievirus B, 29 serotypes of echovirus, enteroviruses 68–71, and multiple new enteroviruses (beginning with enterovirus 73) that have been identified by molecular techniques. Human enteroviruses have been reclassified into four species designated A–D. Echoviruses 22 and 23 have been reclassified as parechoviruses 1 and 2 on the basis of low nucleotide homology and differences in viral proteins. Enterovirus and parechovirus surveillance conducted in the United States by the Centers for Disease Control and Prevention (CDC) in 2022 showed that the most common enteroviruses and parechoviruses were enterovirus D68 (41.1% of cases) and human parechovirus 2 (20.7%), followed by

coxsackievirus A6, A9, B3, and B5, each of which accounted for 4–5% of all isolates.

Human enteroviruses contain a single-stranded RNA genome surrounded by an icosahedral capsid comprising four viral proteins. These viruses have no lipid envelope and are stable in acidic environments, including the stomach. They are susceptible to chlorine-containing cleansers but resistant to inactivation by standard disinfectants (e.g., alcohol, detergents) and can persist for days at room temperature.

■ PATHOGENESIS AND IMMUNITY

Much of what is known about the pathogenesis of enteroviruses has been derived from studies of poliovirus infection. After ingestion, poliovirus is thought to infect epithelial cells in the mucosa of the gastrointestinal tract and then to spread to and replicate in the submucosal lymphoid tissue of the tonsils and Peyer's patches. The virus next spreads to the regional lymph nodes, a viremic phase ensues, and the virus replicates in organs of the reticuloendothelial system. In some cases, a second episode of viremia occurs and the virus replicates further in various tissues, sometimes causing symptomatic disease.

It is uncertain whether poliovirus reaches the central nervous system (CNS) during viremia or whether it also spreads via peripheral nerves. Since viremia precedes the onset of neurologic disease in humans, it has been assumed that the virus enters the CNS via the bloodstream. The poliovirus receptor is a member of the immunoglobulin superfamily. Poliovirus infection is limited to primates, largely because their cells express the viral receptor. Studies demonstrating the poliovirus receptor in the end-plate region of muscle at the neuromuscular junction suggest that, if the virus enters the muscle during viremia, it could travel across the neuromuscular junction up the axon to the anterior horn cells. Studies of monkeys and of transgenic mice expressing the poliovirus receptor show that, after IM injection, poliovirus does not reach the spinal cord if the sciatic nerve is cut. Taken together, these findings suggest that poliovirus can spread directly from muscle to the CNS by neural pathways.

Poliovirus can usually be cultured from the blood 3–5 days after infection, before the development of neutralizing antibodies. While viral replication at secondary sites begins to slow 1 week after infection, it continues in the gastrointestinal tract. Poliovirus is shed from the oropharynx for up to 3 weeks after infection and from the gastrointestinal tract for as long as 12 weeks; hypogammaglobulinemic patients can shed poliovirus for >20 years. During replication in the gastrointestinal tract, attenuated oral poliovirus can mutate, reverting to a more neurovirulent phenotype within a few days; however, additional mutations are probably required for full neurovirulence. One patient with hypogammaglobulinemia who had been infected 12 years earlier and was receiving IV immune globulin suddenly developed quadriplegia and respiratory muscle paralysis and died; analysis showed that the virus had reverted to a more wild-type sequence.

Humoral and secretory immunity in the gastrointestinal tract is important for the control of enterovirus infections. Enteroviruses induce specific IgM, which usually persists for <6 months, and specific IgG, which persists for life. Capsid protein VP1 is the predominant target of neutralizing antibody, which generally confers lifelong protection against subsequent disease caused by the same serotype but does not prevent infection or virus shedding. Enteroviruses also induce cellular immunity of uncertain significance. Patients with impaired cellular immunity are not known to develop unusually severe disease when infected with enteroviruses. In contrast, the severe infections in patients with agammaglobulinemia emphasize the importance of humoral immunity in controlling enterovirus infections. Disseminated enterovirus infections have occurred in hematopoietic cell transplant recipients. IgA antibodies are instrumental in reducing poliovirus replication in and shedding from the gastrointestinal tract. Breast milk contains IgA specific for enteroviruses and can protect humans from infection.

■ EPIDEMIOLOGY

Enteroviruses have a worldwide distribution. More than 50% of nonpoliovirus enterovirus infections and >90% of poliovirus infections are subclinical. When symptoms do develop, they are usually

nonspecific and occur in conjunction with fever; only a minority of infections are associated with specific clinical syndromes. The incubation period for most enterovirus infections ranges from 2 to 14 days but usually is <1 week.

Enterovirus infection is more common in socioeconomically disadvantaged areas, especially in those where conditions are crowded and in tropical areas where hygiene is poor. Infection is most common among infants and young children; serious illness develops most often during the first few days of life and in older children and adults. In developing countries, where children are infected at an early age, poliovirus infection has less often been associated with paralysis; in countries with better hygiene, older children and adults are more likely to be seronegative, become infected, and develop paralysis. Passively acquired maternal antibody reduces the risk of symptomatic infection in neonates. Young children are the most frequent shedders of enteroviruses and are usually the index cases in family outbreaks. In temperate climates, enterovirus infections occur most often in the summer and fall; no seasonal pattern is apparent in the tropics.

Most enteroviruses are transmitted primarily by the fecal–oral or oral–oral route. Patients are most infectious shortly before and after the onset of symptomatic disease, when virus is present in the stool and throat. The ingestion of virus-contaminated food or water also can cause disease. Certain enteroviruses (such as enterovirus 70, which causes acute hemorrhagic conjunctivitis) can be transmitted by direct inoculation from the fingers to the eye. Airborne transmission is important for some viruses that cause respiratory tract disease, such as coxsackievirus A21. Enteroviruses can be transmitted across the placenta from mother to fetus, causing severe disease in the newborn. The transmission of enteroviruses through blood transfusions or insect bites has not been documented. Nosocomial spread of coxsackievirus and echovirus has taken place in hospital nurseries. Outbreaks of enteroviruses correlate with levels of preexisting immunity to specific serotypes and birth rates.

■ CLINICAL FEATURES

Poliovirus Infection Most infections with poliovirus are asymptomatic. After an incubation period of 3–6 days, ~5% of patients present with a minor illness (abortive poliomyelitis) manifested by fever, malaise, sore throat, anorexia, myalgias, and headache. This condition usually resolves in 3 days. About 1% of patients present with aseptic meningitis (nonparalytic poliomyelitis). Examination of cerebrospinal fluid (CSF) reveals lymphocytic pleocytosis, a normal glucose level, and a normal or slightly elevated protein level; CSF polymorphonuclear leukocytes may be present early. In some patients, especially children, malaise and fever precede the onset of aseptic meningitis.

PARALYTIC POLIOMYELITIS The least common presentation is that of paralytic disease. After one or several days, signs of aseptic meningitis are followed by severe back, neck, and muscle pain and by the rapid or gradual development of motor weakness. In some cases, the disease appears to be biphasic, with aseptic meningitis followed first by apparent recovery but then (1–2 days later) by the return of fever and the development of paralysis; this form is more common among children than among adults. Weakness is generally asymmetric, is proximal more than distal, and may involve the legs (most commonly); the arms; or the abdominal, thoracic, or bulbar muscles. Paralysis develops during the febrile phase of the illness and usually does not progress after defervescence. Urinary retention also may occur. Examination reveals weakness, fasciculations, decreased muscle tone, and reduced or absent reflexes in affected areas. Transient hyperreflexia sometimes precedes the loss of reflexes. Patients frequently report sensory symptoms, but objective sensory testing usually yields normal results. Bulbar paralysis may lead to dysphagia, difficulty in handling secretions, or dysphonia. Respiratory insufficiency due to aspiration, involvement of the respiratory center in the medulla, or paralysis of the phrenic or intercostal nerves may develop, and severe medullary involvement may lead to circulatory collapse. Most patients with paralysis recover some function weeks to months after infection. About two-thirds of patients have residual neurologic sequelae.

Paralytic disease is more common among older individuals, pregnant women, and persons exercising strenuously or undergoing trauma at the time of CNS symptoms. Tonsillectomy predisposes to bulbar poliomyelitis, and IM injections increase the risk of paralysis in the involved limb(s).

VACCINE-ASSOCIATED POLIOMYELITIS The risk of developing poliomyelitis after oral vaccination is estimated at 1 case per 2.5 million doses. The risk is ~2000 times higher among immunodeficient persons, especially persons with hypo- or agammaglobulinemia. Before 1997, an average of eight cases of vaccine-associated poliomyelitis occurred—in both vaccinees and their contacts—in the United States each year. With the change in recommendations first to a sequential regimen of inactivated poliovirus vaccine (IPV) and oral poliovirus vaccine (OPV) in 1997 and then to an all-IPV regimen in 2000, the number of cases of vaccine-associated polio declined. From 1997 to 1999, six such cases were reported in the United States followed by one case in 2022.

POSTPOLIO SYNDROME The *postpolio syndrome* presents as new onset of weakness, fatigue, fasciculations, and pain with additional atrophy of the muscle group involved during the initial paralytic disease 20–40 years earlier. The syndrome is more common among women and with increasing time after acute disease. The onset is usually insidious, and weakness occasionally extends to muscles that were not involved during the initial illness. The prognosis is generally good; progression to further weakness is usually slow, with plateau periods of 1–10 years. The postpolio syndrome is thought to be due to progressive dysfunction and loss of motor neurons that compensated for the neurons lost during the original infection and not to persistent or reactivated poliovirus infection.

Other Enteroviruses An estimated 5–10 million cases of symptomatic disease due to enteroviruses other than poliovirus occur in the United States each year. Among neonates, enteroviruses are the most common cause of aseptic meningitis and nonspecific febrile illnesses. Certain clinical syndromes are more likely to be caused by certain serotypes (Table 210-1).

NONSPECIFIC FEBRILE ILLNESS (SUMMER GRIPPE) The most common clinical manifestation of enterovirus infection is a nonspecific febrile illness. After an incubation period of 3–6 days, patients present

with an acute onset of fever, malaise, and headache. Occasional cases are associated with upper respiratory symptoms, and some cases include nausea and vomiting. Symptoms often last for 3–4 days, and most cases resolve in a week. While infections with other respiratory viruses occur more often from late fall to early spring, febrile illness due to enteroviruses frequently occurs in the summer and early fall.

GENERALIZED DISEASE OF THE NEWBORN Most serious enterovirus infections in infants develop during the first week of life, although severe disease can occur up to 3 months of age. Neonates often present with an illness resembling bacterial sepsis, with fever, irritability, and lethargy. Laboratory abnormalities include leukocytosis with a left shift, thrombocytopenia, elevated values in liver function tests, and CSF pleocytosis. The illness can be complicated by myocarditis and hypotension, fulminant hepatitis and disseminated intravascular coagulation, meningitis or meningoencephalitis, or pneumonia. It may be difficult to distinguish neonatal enterovirus infection from bacterial sepsis, although a history of a recent virus-like illness in the mother provides a clue.

ASEPTIC MENINGITIS AND ENCEPHALITIS In children and young adults, enteroviruses are the cause of up to 90% of cases of aseptic meningitis in which an etiologic agent can be identified. Patients with aseptic meningitis typically present with an acute onset of fever, chills, headache, photophobia, and pain on eye movement. Nausea and vomiting also are common. Examination reveals meningismus without localizing neurologic signs; drowsiness or irritability also may be apparent. In some cases, a febrile illness may remit and return several days later in conjunction with signs of meningitis. Other systemic manifestations may provide clues to an enteroviral cause, including diarrhea, myalgias, rash, pleurodynia, myocarditis, and herpangina. Examination of the CSF invariably reveals pleocytosis; the CSF cell count shows a shift from neutrophil to lymphocyte predominance within 1 day of presentation, and the total cell count does not exceed 1000/μL. The CSF glucose level is usually normal (in contrast to the low CSF glucose level in mumps), with a normal or slightly elevated protein concentration. Partially treated bacterial meningitis may be particularly difficult to exclude in some instances. Enteroviral meningitis is more common in summer and fall in temperate climates, while viral meningitis of other etiologies is more common in winter and spring. Symptoms ordinarily resolve within a week, although CSF abnormalities can persist for several weeks. Enteroviral meningitis is often more severe in adults than in children. Neurologic sequelae are rare, and most patients have an excellent prognosis.

Enteroviral encephalitis is much less common than enteroviral aseptic meningitis. Occasional highly inflammatory cases of enteroviral meningitis may be complicated by a mild form of encephalitis that is recognized on the basis of progressive lethargy, disorientation, and sometimes seizures. Less commonly, severe primary encephalitis may develop. An estimated 10–35% of cases of viral encephalitis are due to enteroviruses. Immunocompetent patients generally have a good prognosis.

Patients with hypogammaglobulinemia, agammaglobulinemia, or severe combined immunodeficiency may develop chronic meningitis or encephalitis; about half of these patients have a dermatomyositis-like syndrome, with peripheral edema, rash, and myositis. They may also have chronic hepatitis. Patients may develop neurologic disease while receiving immunoglobulin replacement therapy. Echoviruses (especially echovirus 11) are the most common pathogens in this situation.

Paralytic disease due to enteroviruses other than poliovirus occurs sporadically and is usually less severe than poliomyelitis. Most cases are due to enterovirus 70 or 71 or to coxsackievirus A7 or A9. Guillain-Barré syndrome is also associated with enterovirus infection. While earlier studies suggested a link between enteroviruses and chronic fatigue syndrome, most recent studies have not demonstrated such an association.

ACUTE FLACCID MYELITIS Patients with acute flaccid myelitis present with fever or respiratory symptoms and progress within hours to a few days to flaccid paralysis in one or more limbs. The disease is much more frequent in children. Less commonly, the disease can affect

TABLE 210-1 Manifestations Commonly Associated with Enterovirus Serotypes		
MANIFESTATION	SEROTYPE(S) OF INDICATED VIRUS	
	COXSACKIEVIRUS	ECHOVIRUS (E) AND ENTEROVIRUS (Ent)
Acute hemorrhagic conjunctivitis	A24	E70
Aseptic meningitis	A2, 4, 7, 9, 10; B1–5	E4, 6, 7, 9, 11, 13, 16, 18, 19, 30, 33; Ent70, 71
Encephalitis	A9; B1–5	E3, 4, 6, 7, 9, 11, 18, 25, 30; Ent71
Exanthem	A4, 5, 6, 9, 10, 16; B1, 3–5	E4–7, 9, 11, 16–19, 25, 30; Ent71
Generalized disease of the newborn	B1–5	E4–7, 9, 11, 14, 16, 18, 19
Hand-foot-and-mouth disease	A5–7, 9, 10, 16; B1, 2, 5	Ent71
Herpangina	A1–10, 16, 22; B1–5	E6, 9, 11, 16, 17, 25, 30; Ent71
Myocarditis, pericarditis	A4, 9, 16; B1–5	E6, 9, 11, 22
Paralysis	A4, 7, 9; B1–5	E2–4, 6, 7, 9, 11, 18, 30; EntD68, 70, 71
Pleurodynia	A1, 2, 4, 6, 9, 10, 16; B1–6	E1–3, 6, 7, 9, 11, 12, 14, 16, 19, 24, 25, 30
Pneumonia	A9, 16; B1–5	E6, 7, 9, 11, 12, 19, 20, 30; EntD68, 71

cranial nerves and respiratory or bulbar muscles. Like polio and some other enteroviruses, the disease affects the anterior horn cells in the spinal cord; gray matter changes can be seen on MRI of the spinal cord. The CSF shows a lymphocytic pleocytosis and often a mildly elevated protein. Cases of acute flaccid myelitis have occurred in late summer or early fall since 2012. Several studies have shown antibodies to enteroviruses in the CSF; antibodies to enterovirus D68 are most frequently detected. While enterovirus D68 has been detected in respiratory, stool, and nasopharyngeal samples from patients with acute flaccid myelitis, the virus has been rarely detected in the CSF. Treatment is supportive, and most patients have persistent neurologic deficits.

PLEURODYNIA (BORNHOLM DISEASE) Patients with pleurodynia present with an acute onset of fever and spasms of pleuritic chest or upper abdominal pain. Chest pain is more common in adults, and abdominal pain is more common in children. Paroxysms of severe, knifelike pain usually last 15–30 min and are associated with diaphoresis and tachypnea. Fever peaks within an hour after the onset of paroxysms and subsides when pain resolves. The involved muscles are tender to palpation, and a pleural rub may be detected. The white blood cell count and chest x-ray results are usually normal. Most cases are due to coxsackievirus B and occur during epidemics. Symptoms resolve in a few days, and recurrences are rare. Treatment includes the administration of nonsteroidal anti-inflammatory agents or the application of heat to the affected muscles.

MYOCARDITIS AND PERICARDITIS Enteroviruses are estimated to cause up to one-third of cases of acute myocarditis. Coxsackievirus B and its RNA have been detected in pericardial fluid and myocardial tissue in some cases of acute myocarditis and pericarditis. Most cases of enteroviral myocarditis or pericarditis occur in newborns, adolescents, or young adults. More than two-thirds of patients are male. Patients often present with an upper respiratory tract infection that is followed by fever, chest pain, dyspnea, arrhythmias, and occasionally heart failure. A pericardial friction rub is documented in half of cases, and the electrocardiogram shows ST-segment elevations or ST- and T-wave abnormalities. Serum levels of myocardial enzymes are often elevated. Neonates commonly have severe disease, while older children and adults recover completely. Up to 10% of cases progress to chronic dilated cardiomyopathy. Chronic constrictive pericarditis also may be a sequela.

EXANTHEMS Enterovirus infection is the leading cause of exanthems in children in the summer and fall. While exanthems are associated with many enteroviruses, certain types have been linked to specific syndromes. Echoviruses 9 and 16 have frequently been associated with exanthem and fever. Rashes may be discrete or confluent, beginning on the face and spreading to the trunk and extremities. Echovirus 9 is the most common cause of a rubelliform (discrete) rash. Unlike the rash of rubella, the enteroviral rash occurs in the summer and is not associated with lymphadenopathy. Roseola-like rashes develop after defervescence, with macules and papules on the face and trunk. The Boston exanthem, caused by echovirus 16, is a roseola-like rash. A variety of other rashes have been associated with enteroviruses, including erythema multiforme (see Fig. A1-24A) and vesicular, urticarial, petechial, bullous, or purpuric lesions. Enanthems also occur, including lesions that resemble the Koplik's spots seen with measles (see Fig. A1-2).

HAND-FOOT-AND-MOUTH DISEASE (FIG. 210-1) After an incubation period of 4–6 days, patients with hand-foot-and-mouth disease present with fever, anorexia, and malaise; these manifestations are followed by the development of sore throat and vesicles (see Fig. A1-22) on the buccal mucosa and often on the tongue and then by the appearance of tender vesicular lesions on the dorsum of the hands, sometimes with involvement of the palms. The vesicles may form bullae and quickly ulcerate. About one-third of patients also have lesions on the palate, uvula, or tonsillar pillars, and one-third have a rash on the feet (including the soles) or on the buttocks. Generalized rashes also have been reported. The disease is highly infectious, with attack rates of close to 100% among young children. The lesions usually resolve in 1 week. Most cases are due to coxsackievirus A16 or enterovirus 71.

An epidemic of enterovirus 71 infection in Taiwan in 1998 resulted in thousands of cases of hand-foot-and-mouth disease or herpangina (see below). Severe complications included CNS disease, myocarditis, and pulmonary hemorrhage. About 90% of those who died were children ≤ 5 years old, and death was associated with pulmonary edema or pulmonary hemorrhage. CNS disease included aseptic meningitis, flaccid paralysis (similar to that seen in poliomyelitis), and rhombencephalitis with myoclonus and tremor or ataxia. The mean age of patients with CNS complications was 2.5 years, and MRI in cases with encephalitis usually showed brainstem lesions. Follow-up of children at 6 months showed persistent dysphagia, cranial nerve palsies, hypoventilation, limb weakness, and atrophy; at 3 years, persistent neurologic sequelae were documented, with delayed development and impaired cognitive function.

Yearly epidemics of enterovirus 71 infection have occurred in China since 2008, with thousands of cases and hundreds of deaths each year. Infections have been associated with fever, rash, brainstem encephalitis with myoclonic jerks, and limb trembling; some cases have progressed to seizures and coma. Lung findings include pulmonary edema and hemorrhage. While the level of creatine kinase MB is sometimes elevated, myocardial necrosis generally is not found.

Cyclic epidemics occur every 2–3 years in other Asian countries. However, the virus circulates at lower rates in the United States, Europe, and Africa. In the United States, hand-foot-and-mouth disease is most commonly associated with coxsackievirus A16. Between November 2011 and February 2012, outbreaks of hand-foot-and-mouth disease



FIGURE 210-1 Vesicular eruptions of the hand (A), knee (B), and mouth (C) of a 6-year-old boy with coxsackievirus A6 infection. Several of his fingernails were shed 2 months later (D). (Images reprinted courtesy of Centers for Disease Control and Prevention/Emerging Infectious Diseases.)

HERPANGINA Herpangina is usually caused by coxsackievirus A and presents as acute-onset fever, sore throat, odynophagia, and grayish-white papulovesicular lesions on an erythematous base that ulcerate. The lesions can persist for weeks; are present on the soft palate, anterior pillars of the tonsils, and uvula; and are concentrated in the posterior portion of the mouth. In contrast to herpes stomatitis, enteroviral herpangina is not associated with gingivitis. Acute lymphonodular pharyngitis associated with coxsackievirus A10 presents as white or yellow nodules surrounded by erythema in the posterior oropharynx. The lesions do not ulcerate.

ACUTE HEMORRHAGIC CONJUNCTIVITIS Patients with acute hemorrhagic conjunctivitis present with an acute onset of severe eye pain, blurred vision, photophobia, and watery discharge from the eye. Examination reveals edema, chemosis, and subconjunctival hemorrhage and often shows punctate keratitis and conjunctival follicles as well (Fig. 210-2). Preauricular adenopathy is often found. Epidemics and nosocomial spread have been associated with enterovirus 70 and coxsackievirus A24. Outbreaks have been due to coxsackievirus A24 in China and India (2010), Japan (2011), and Thailand (2014). Systemic symptoms, including headache and fever, develop in 20% of cases, and recovery is usually complete in 10 days. The sudden onset and short duration of the illness help to distinguish acute hemorrhagic conjunctivitis from other ocular infections, such as those due to adenovirus and *Chlamydia trachomatis*. Paralysis has been associated with some cases of acute hemorrhagic conjunctivitis due to enterovirus 70 during epidemics.

OTHER MANIFESTATIONS Enteroviruses are an infrequent cause of childhood pneumonia and the common cold. From mid-August 2014 to January 2015, enterovirus D68 infection was confirmed in more than 1000 persons with mild to severe respiratory illnesses in 49 U.S. states. Nearly all reported cases were in children, many of whom had asthma. Severe respiratory illness due to enterovirus D68 continues to occur; case numbers were lower in 2020, likely due to mitigation efforts for COVID-19, but rose in 2022. A prospective study of >300 children showed that prolonged shedding of enteroviruses in the stool was associated with development of islet cell autoantibodies and type 1 diabetes. Coxsackievirus B has been isolated at autopsy from the pancreas of a few children presenting with type 1 diabetes mellitus; however, most attempts to isolate the virus have been unsuccessful. Other diseases that have been associated with enterovirus infection include parotitis, bronchitis, bronchiolitis, croup, infectious lymphocytosis, polymyositis, acute arthritis, and acute nephritis.

■ DIAGNOSIS

Isolation of enterovirus in cell culture had been the traditional diagnostic procedure; PCR is used now more often. Cultures of stool,

nasopharyngeal, or throat samples from patients with enterovirus diseases do not prove that the virus is directly associated with disease because these sites are frequently colonized for weeks in patients with subclinical infections. Isolation of virus from the throat is more likely to be associated with disease than is isolation from the stool since virus is shed for shorter periods from the throat. Cultures of CSF, serum, fluid from body cavities, or tissues are positive less frequently, but a positive result is indicative of disease caused by enterovirus. Cultures are more likely to be positive earlier than later in the course of infection. Cultures may be negative because of the presence of neutralizing antibody, lack of susceptibility of the cells used, or inappropriate handling of the specimen. Coxsackievirus A may require inoculation into special cell-culture lines or into suckling mice.

Identification of the enterovirus serotype is useful primarily for epidemiologic studies and, with a few exceptions, has little clinical utility. It is important to identify serious infections with enterovirus during epidemics and to distinguish the vaccine strain of poliovirus from the other enteroviruses in the throat or in the feces. Stool and throat samples for culture as well as acute- and convalescent-phase serum specimens should be obtained from all patients with suspected poliomyelitis. In the absence of a positive CSF culture, a positive culture of stool obtained within the first 2 weeks after the onset of symptoms is most often used to confirm the diagnosis of poliomyelitis. If poliovirus infection is suspected, two or more fecal and throat swab samples should be obtained at least 1 day apart and cultured for enterovirus as soon as possible. If poliovirus is isolated, it should be sent to the CDC for identification as either wild-type or vaccine virus.

Reverse-transcription polymerase chain reaction (PCR) has been used to amplify viral nucleic acid from CSF, serum, urine, stool, conjunctiva, throat swabs, and tissues. A pan-enterovirus PCR assay can detect all human enteroviruses. With the proper controls, PCR of the CSF is highly sensitive (70–100%) and specific (>80%) and is more rapid than culture. PCR of the CSF is less likely to be positive when patients present ≥ 3 days after the onset of meningitis or with enterovirus 71 infection; in these cases, PCR of throat or rectal swabs—although less specific than PCR of CSF—should be considered.

PCR of serum is also highly sensitive and specific in the diagnosis of disseminated disease. PCR may be particularly helpful for the diagnosis and follow-up of enterovirus disease in immunodeficient patients receiving immunoglobulin therapy, whose viral cultures may be negative. Antigen detection is less sensitive than PCR.

Serologic diagnosis of enterovirus infection is limited by the large number of serotypes and the lack of a common antigen. Demonstration of seroconversion may be useful in rare cases for confirmation of culture results, but serologic testing is usually limited to epidemiologic studies. Serum should be collected and frozen soon after the onset of disease and again ~4 weeks later. Measurement of neutralizing titers is the most accurate method for antibody determination; measurement of complement-fixation titers is usually less sensitive. Titers of virus-specific IgM are elevated in both acute and chronic infection.

TREATMENT

Enterovirus Infections

Most enterovirus infections are mild and resolve spontaneously; however, intensive supportive care may be needed for cardiac, hepatic, or CNS disease. IV, intrathecal, or intraventricular immunoglobulin has been used with apparent success in some cases for the treatment of chronic enterovirus meningoencephalitis and dermatomyositis in patients with hypogammaglobulinemia or agammaglobulinemia. The disease may stabilize or resolve during therapy; however, some patients decline inexorably despite therapy. IV immunoglobulin often prevents severe enterovirus disease in these patients. IV administration of immunoglobulin with high titers of antibody to the infecting virus has been used in some cases of life-threatening infection in neonates, who may not have maternally acquired antibody. In one trial involving neonates with enterovirus infections, immunoglobulin containing very high titers

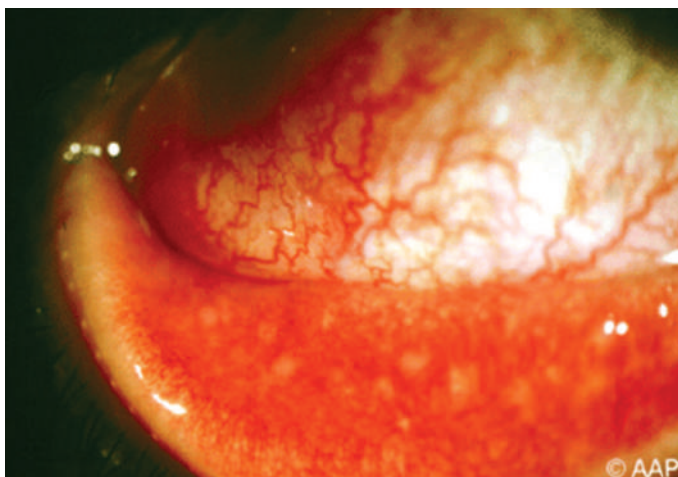


FIGURE 210-2 Acute hemorrhagic conjunctivitis due to enterovirus 70. (Image reprinted courtesy of Jerri Ann Jenista, MD.)

of antibody to the infecting virus reduced rates of viremia; however, the study was too small to show a substantial clinical benefit. The level of enteroviral antibodies varies with the immunoglobulin preparation. While a phase 2 trial of pleconaril for neonatal enterovirus sepsis showed that the time to serum PCR negativity was reduced and the survival rate increased in newborns who had confirmed enterovirus infections and were treated with the drug, the differences did not reach significance and the drug is not available on a compassionate-use basis. Pocapavir and vapendavir also are being tested for enterovirus infections; resistance developed rapidly to OPV in a clinical trial of pocapavir. Glucocorticoids are contraindicated.

Good hand-washing practices and the use of gowns and gloves are important in limiting nosocomial transmission of enteroviruses during epidemics. Enteric precautions are indicated for 7 days after the onset of enterovirus infections. Inactivated enterovirus 71 vaccines have been licensed in China.

PREVENTION AND ERADICATION OF POLIOVIRUS

(See also Chap. 129) After a peak of 57,879 cases of poliomyelitis in the United States in 1952, the introduction of IPV in 1955 and of OPV in 1961 ultimately eradicated disease due to wild-type poliovirus in the Western Hemisphere. Such disease has not been documented in the United States since 1979, when cases occurred among religious groups who had declined immunization. In the Western Hemisphere, paralysis due to wild-type poliovirus was last documented in 1991. Paralysis due to vaccine-derived poliovirus (VDPV) was reported in 2022 in New York (see below).

In 1988, when ~350,000 cases of polio occurred in 125 countries, the World Health Organization adopted a resolution to eradicate poliomyelitis by the year 2000. Wild-type poliovirus type 2 and wild-type poliovirus type 3 were declared eradicated in 2015 and 2019, respectively. The Americas were certified free of indigenous wild-type poliovirus transmission in 1994, the Western Pacific Region in 2000, the European Region in 2002, and Southeast Asia in 2014. After a nadir of 496 cases in 2001, 21 countries that had previously been free of polio reported cases imported from 6 polio-endemic countries in 2002–2005. By 2006, polio transmission had been reduced in most of these 21 countries. In 2017, there were 22 cases of wild-type polio, the lowest ever reported for 1 year—all of these cases were from Pakistan and Afghanistan. In 2021 wild-type polio reemerged in Africa. After another peak of 176 cases in 2019, the number of cases of wild-type polio had fallen to 12 in 2023, all from 2 countries and all due to polio type 1 (Table 210-2). Polio is a source of concern for unimmunized or partially immunized travelers. Clearly, global eradication of polio is necessary to eliminate the risk of importation of wild-type virus.

Outbreaks are thought to have been facilitated by suboptimal rates of vaccination, isolated pockets of unvaccinated children, poor sanitation and crowding, improper vaccine-storage conditions, and a reduced level of response to one of the serotypes in the vaccine. While the global eradication campaign has markedly reduced the number of cases of endemic polio, doubts have been raised as to whether eradication is a realistic goal, given the large number of asymptomatic infections and the political instability in developing countries.

Use of OPV, especially in areas with low vaccination rates, has been associated with vaccine-derived polio due to mutations that result in restoration of viral fitness and neurovirulence during prolonged replication in individuals or person-to-person transmission. Vaccine-derived polio was recognized in Egypt in 1983–1993, and hundreds of cases have been reported in many countries, including 385 cases in Nigeria in 2005–2012. Epidemics have been rapidly terminated after intensive vaccination with OPV. In 2005, a case of vaccine-derived polio occurred in an unvaccinated U.S. woman returning from a visit to Central and South America. In the same year, an unvaccinated immunocompromised infant in Minnesota was found to be shedding VDPV; further investigation identified 4 of 22 infants in the same community who were shedding the virus. All 5 infants were asymptomatic. These outbreaks emphasize the need for maintaining high levels of vaccine coverage and continued surveillance for circulating virus. In 2016, only 5 cases were reported; however, this number increased with a peak of 1113 cases in 2020 and subsequently declined to 526 cases of vaccine-derived polio in 2023 from 24 countries; 96% of these cases were from Africa and 3% were from the Eastern Mediterranean Region (Table 210-2). This decline in VDPV type 2 is associated with the use of a safer, novel type 2 OPV (see below). A case of vaccine-derived polio type 2 occurred in an unvaccinated adult in New York in 2022 and was genetically linked to wastewater collected in the area about 1 month before and after the case; the patient had not traveled internationally during the incubation period. VDPV due to OPV2 has been detected in wastewater from other countries in 2022, including Canada, Israel, and the United Kingdom, where virus transmission has been eliminated. From 2018 to March 2020, 92% of cases of vaccine-derived polio were due to type 2 virus. Cessation of vaccination with type 2 OPV is believed to be responsible for this increase in polio type 2. IPV is used in most industrialized countries and OPV in most developing countries, including those in which polio still is or recently was endemic. While IM injections of other vaccines (live or attenuated) can be given concurrently with OPV, unnecessary IM injections should be avoided during the first month after OPV vaccination because they increase the risk of vaccine-associated paralysis. Since 1988, an enhanced-potency inactivated poliovirus vaccine has been available in the United States.

After several doses of OPV alone, the seropositivity rate for individual poliovirus serotypes may still be suboptimal for children in developing countries; one or more supplemental doses of IPV can increase the rate of seropositivity for these serotypes. Against a given serotype, monovalent OPV containing only that serotype is more immunogenic than trivalent vaccine because of a lack of interference from other serotypes. Given the eradication of wild-type poliovirus type 2 and the establishment of OPV type 2 as the primary cause of vaccine-derived polio, bivalent OPV (types 1 and 3), which had been shown to be superior to trivalent OPV in inducing antibodies to types 1 and 3, replaced trivalent OPV vaccine in April 2016. However, outbreaks of vaccine-derived polio due to polio type 2 have required vaccination with monovalent OPV type 2. A novel type 2 oral poliovirus vaccine (nOPV2) was engineered to be impaired in reversion to neurovirulence. After ~700 million doses of nOPV2 were administered in response to circulating VDPV type 2, from 2021 to 2023, 61 cases of paralysis were associated with nOPV2, with all the cases in Africa. The rate of emergence of paralysis with nOPV2 is about 10-fold lower than with monovalent OPV2. Another approach to reduce VDPV is to vaccinate with IPV followed by OPV. Addition of at least one dose of trivalent IPV after immunization with bivalent OPV will also reduce the risk of vaccine-derived polio associated with type 2 virus and enhance immunity to poliovirus types 1 and 3. Accordingly, in 2016, ~90% of countries included trivalent IPV in their immunization schedules.

TABLE 210-2 Laboratory-Confirmed Cases of Poliomyelitis in 2023

COUNTRY	WILD-TYPE POLIO	VACCINE-DERIVED POLIO
Pakistan	6	0
Afghanistan	6	0
Democratic Republic of the Congo	0	223
Nigeria	0	87
Chad	0	55
Guinea	0	47
Madagascar	0	24
Mali	0	15
Central African Republic	0	14
Others	0	61 ^a
Total	12	526

^aOthers with <10 cases; Kenya, Somalia, Yemen 8 cases; Cote d'Ivoire, Indonesia 6 cases; Mozambique 5 cases; Benin, Burkina Fossa, Niger, South Sudan 3 cases; United Republic of Tanzania 2 cases; Burundi, Ethiopia, Israel, Mauritania, Zambia, Zimbabwe 1 case each.

TABLE 210-3 Recommendations for Poliovirus Vaccination of Adults

1. Most adults in the United States have little risk for exposure to polioviruses, and most are immune as a result of vaccination during childhood. Vaccination with IPV is recommended for those at greater risk for exposure to polioviruses than the general population:
 - a. travelers to areas or countries where polio is epidemic or endemic;
 - b. laboratory workers who handle specimens that might contain polioviruses;
 - c. health care workers or other caregivers who have close contact with patients who might be excreting wild-type polioviruses; and
 - d. adults who are identified by public health authorities as being part of a group or population at increased risk because of an outbreak.
2. Adults who are unvaccinated or whose vaccination status is unknown and who are at increased risk should receive three doses of IPV. Two doses of IPV should be administered at intervals of 4–8 weeks; a third dose should be administered 6–12 months after the second.
3. Adults who have had a primary series of polio vaccine and who are at increased risk should receive another dose of IPV. Currently, data do not indicate a need for more than a single lifetime booster dose with IPV for adults.

Abbreviation: IPV, inactivated poliovirus vaccine.

Source: From <https://www.cdc.gov/vaccines/vpd/polio/hcp/recommendations.html>.

As the frequency of wild-type polio declines and reports of polio associated with circulating VDPV increase, the World Health Organization is investigating whether IPV can be produced from OPV strains that require less biocontainment, ultimately replacing OPV.

OPV and IPV induce antibodies that persist for at least 5 years. Both vaccines induce IgG and IgA antibodies. Compared with recipients of IPV, recipients of OPV shed less virus and less frequently develop reinfection with wild-type virus after exposure to poliovirus. Although IPV is safe and efficacious, OPV offers the advantages of ease of administration, lower cost, and induction of intestinal immunity resulting in a reduction in the risk of community transmission of wild-type virus. Because of progress toward global eradication of polio and the continued occurrence of cases of vaccine-associated polio, an all-IPV regimen was recommended in 2000 for childhood poliovirus vaccination in the United States, with vaccine administration at 2, 4, and 6–18 months and 4–6 years of age. The risk of vaccine-associated polio should be discussed before OPV is administered. Recommendations for vaccination of adults are listed in Table 210-3.

There are concerns about discontinuing vaccination in the event that endemic spread of poliovirus is eliminated. Among the reasons for these concerns are that poliovirus is shed from some immunocompromised persons for >25 years, that vaccine-derived poliovirus can circulate and cause disease, and that wild-type poliovirus is present in research laboratories and vaccine manufacturing facilities. Antivirals and monoclonal antibodies are in development to reduce or terminate shedding of poliovirus by long-term virus excretors. Pocapavir was shown to reduce shedding of OPV type 1 in a clinical trial, but rapid development of resistance with virus transmission, despite reduced shedding, indicates that combination therapy with other antivirals and/or monoclonal antibodies will be needed.

PARECHOVIRUSES

Human parechoviruses (HPeVs), like enteroviruses, are members of the family Picornaviridae. The 19 serotypes of HPeV commonly cause infections in early childhood. Infections with HPeV type 1 (HPeV-1) occur throughout the year, while other parechovirus infections occur more commonly in summer and fall. Infections with HPeVs present similarly to those due to enteroviruses and may cause generalized disease of the newborn, aseptic meningitis, encephalitis, seizures, paralysis, exanthems, respiratory tract disease, rash, hepatitis, and gastroenteritis. While HPeV-1 is the most common serotype and generally causes mild disease, deaths of infants in the United States have been associated with HPeV-1, HPeV-3, and HPeV-6. HPeVs can be isolated from the same sites as enteroviruses, including the nasopharynx, stool, and respiratory tract secretions. PCR using pan-enterovirus primers does not detect HPeVs, and while PCR assays are performed by the

CDC and research laboratories, many commercial laboratories do not perform the test.

REOVIRUSES

Reoviruses are double-stranded RNA viruses encompassing three serotypes. Serologic studies indicate that most humans are infected with reoviruses during childhood. Most infections either are asymptomatic or cause mild upper respiratory tract symptoms. Reovirus is considered a rare cause of mild gastroenteritis or meningitis in infants and children. Speculation regarding an association of reovirus type 3 with idiopathic neonatal hepatitis and extrahepatic biliary atresia is based on an elevated prevalence of antibody to reovirus in some affected patients and the detection of viral RNA by PCR in hepatobiliary tissues in some studies. Orthoreoviruses have been associated with human disease—e.g., Melaka and Kampar viruses with fever and acute respiratory disease in Malaysia and Nelson Bay virus with acute respiratory disease in a traveler from Bali.

FURTHER READING

LEE SE et al: Progress toward poliomyelitis eradication – Worldwide, January 2021–March 2023. *MMWR Morb Mortal Wkly Rep* 72:517, 2023.

LINK-GELLES R et al: Public health response to a case of paralytic poliomyelitis in an unvaccinated person and detection of poliovirus in wastewater - New York, June–August 2022. *Morb Mortal Wkly Rep* 19:1065, 2022.

MA KC et al: Increase in acute respiratory illnesses among children and adolescents associated with rhinoviruses and enteroviruses, including enterovirus D68—United States, July–September 2022. *MMWR Morb Mortal Wkly Rep* 71:1265, 2022.

PALLANSCH MA: Circulating poliovirus in New York—new instance of an old problem. *N Engl J Med* 387:1725, 2022.

TOMATIS SOUVERBIELE C et al: Update on nonpolio enterovirus and parechovirus infections in neonates and young infants. *Curr Opin Pediatr* 35:380, 2023.

211 Measles (Rubeola)

Alex C. Kong, William J. Moss

DEFINITION

Measles is a highly contagious viral disease characterized by a prodromal illness of fever, cough, coryza, and conjunctivitis followed by the appearance of a generalized maculopapular rash. Before the widespread use of measles vaccines, it was estimated that measles caused >2 million deaths worldwide each year.

GLOBAL CONSIDERATIONS

Remarkable progress has been made in reducing global measles incidence and mortality rates through measles vaccination. In the Americas, intensive vaccination and surveillance efforts—based in part on the successful Pan American Health Organization strategy of periodic nationwide measles vaccination campaigns (supplementary immunization activities [SIAs])—and high levels of routine measles vaccine coverage interrupted endemic transmission of measles virus. The World Health Organization's (WHO's) Region of the Americas was declared to have eliminated measles in September 2016—the first region in the world to do so. However, endemic measles virus transmission was reestablished, and the region lost its measles elimination status. As such, no WHO region has achieved and sustained measles elimination status, highlighting the importance of maintaining high measles vaccination coverage.

In the United States, high-level coverage with two doses of measles vaccine eliminated endemic measles virus transmission in 2000. However, imported cases and low measles vaccine coverage in some communities threaten this goal. The 1274 measles cases reported in the United States in 2019 represent the highest count since 1992. Historically low levels of measles cases were reported early after the COVID-19 pandemic began but were not sustained, with resurgence of measles globally and in the United States.

Progress also has been made in reducing measles incidence and mortality rates in sub-Saharan Africa and Asia because of increasing routine measles vaccine coverage and provision of a second dose of measles vaccine through mass measles vaccination campaigns and routine childhood immunization programs. From 2000 to 2022, the estimated annual number of global measles deaths per year decreased 82%, from 772,854 (95% confidence interval [CI]: 580,969–1,064,580) to 136,216 (95% CI: 97,058–190,234). Measles vaccination prevented an estimated 57.2 million deaths over this period. Despite this progress, the fact that >100,000 children die each year from a preventable disease such as measles attests to the need for greater resources and efforts to identify and reach unvaccinated children.

The COVID-19 pandemic caused severe disruptions to immunization activities, further threatening progress toward measles control and elimination. Almost 40 million children were estimated to have missed a dose of measles vaccine in 2021, including 25 million children who missed their first dose. Large-scale measles outbreaks occurred in 22 countries in 2021 and 37 countries in 2022. Over the same period, the estimated number of measles cases and deaths increased by 18% and 43%, respectively.

The Measles and Rubella Partnership (MRP)—formerly the Measles and Rubella Initiative (MRI)—is working to improve immunization coverage and address setbacks caused by the pandemic. Since its inception in 2001, MRI has played an important role in reducing global measles incidence and mortality rates, providing governments and communities in 88 countries with technical and financial support for routine immunization activities, mass vaccination campaigns, and disease surveillance systems. In 2023, MRI was rebranded as MRP, and the partnership that had historically been led by the American Red Cross, the United Nations Foundation, UNICEF, and the U.S. Centers for Disease Control and Prevention (CDC) expanded to formally include longtime partners Gavi, the Vaccine Alliance and the Bill and Melinda Gates Foundation as core partners.

■ ETIOLOGY

Measles virus is a spherical, nonsegmented, single-stranded, negative-sense RNA virus and a member of the *Morbillivirus* genus in the family *Paramyxoviridae*. Measles was originally a zoonotic infection, arising from animal-to-human transmission of an ancestral morbillivirus thousands of years ago, when human populations attained sufficient size to sustain virus transmission. Although RNA viruses typically have high mutation rates, measles virus is an antigenically monotypic virus, i.e., the surface proteins responsible for inducing protective immunity retained their antigenic structure across time and distance because of their key role in binding cellular receptors. The public health significance of this stability is that measles vaccines developed decades ago from a single strain of measles virus remain protective worldwide. Both wild-type and attenuated measles viruses are inactivated by ultraviolet light and heat, necessitating a cold chain for vaccine transport and storage.

■ EPIDEMIOLOGY

Measles virus is one of the most highly contagious directly transmitted pathogens. Outbreaks can occur in populations in which <10% of persons are susceptible. Chains of transmission are common among household contacts, school-age children, and health care workers. There are no latent or persistent measles virus infections that result in a prolonged infectious period, nor are there animal reservoirs for the virus. Thus, measles virus can be maintained in human populations only by an unbroken chain of acute infections, which requires a continuous supply of susceptible individuals. Newborns become susceptible

to measles virus infection when passively acquired maternal antibodies are lost, generally before 6–9 months of age. When not immunized, these infants account for the bulk of new susceptible individuals that sustain measles virus transmission.

Endemic measles has a typical temporal pattern characterized by yearly seasonal epidemics superimposed on longer epidemic cycles of 2–5 years or more. In temperate climates, annual measles outbreaks typically occur in the late winter and early spring. These annual outbreaks are probably attributable to social networks facilitating transmission (e.g., congregation of children at school) and environmental factors favoring the viability and transmission of measles virus. Measles cases continue to occur during interepidemic periods in large populations but at low incidence. The longer epidemic cycles occurring every several years result from the accumulation of susceptible persons over successive birth cohorts and the subsequent decline in the number of susceptibles following an outbreak.

Secondary attack rates among susceptible household and institutional contacts generally exceed 90%. The average age at which measles occurs depends on rates of contact with infected persons, protective maternal antibody decline, and vaccine coverage. In densely populated urban settings with low-level vaccination coverage, measles is a disease of infants and young children. The cumulative incidence can reach 50% by 1 year of age, with a significant proportion of children acquiring measles before 9 months—the age at which the first of two routine vaccine doses are administered in many countries, in line with the schedule recommended by the WHO's Expanded Programme on Immunization. As measles vaccine coverage increases or population density decreases, the age distribution shifts toward older children. In such situations, measles cases predominate in school-age children. Infants and young children, although susceptible if not protected by maternal antibodies or vaccination, are not exposed to measles virus at a rate sufficient to cause a heavy disease burden in this age group. As vaccination coverage increases further, the age distribution of cases may be shifted into adolescence and adulthood. This distribution is seen in measles outbreaks in the United States and necessitates targeted measles vaccination programs for these older age groups. Some countries have a bimodal distribution, with measles cases predominantly in young infants and adults.

Persons with measles are infectious for several days before and after the onset of rash, when levels of measles virus in blood and body fluids are highest and when cough, coryza, and sneezing that facilitate virus spread are most severe. The contagiousness of measles before the onset of recognizable disease hinders the effectiveness of isolation measures.

Medical settings are well-recognized sites of measles virus transmission. Children may present to health care facilities during the prodrome, when the diagnosis is not obvious, although the child is infectious and is likely to infect susceptible contacts. Susceptible health care workers can acquire measles from infected children and transmit measles virus to others. Nosocomial transmission can be reduced by maintenance of a high index of clinical suspicion particularly during outbreaks, use of appropriate isolation precautions when measles is suspected, administration of measles vaccine to susceptible children and health care workers, and documentation of health care workers' immunity to measles (i.e., proof of receipt of two doses of measles vaccine or detection of IgG antibodies to measles virus).

As efforts at measles control are increasingly successful, public perceptions of the risk of measles diminish and may be replaced by concerns about possible adverse events associated with measles vaccine. Consequently, measles outbreaks have occurred because of opposition to vaccination on religious or philosophical grounds or unfounded fears of serious adverse events (see “Active Immunization,” below, and [Chap. 3](#)).

■ PATHOGENESIS

Measles virus is transmitted primarily by respiratory droplets over short distances and, less commonly, by small-particle aerosols that remain suspended in the air for long periods. Airborne transmission appears to be important in certain settings, including schools, physicians' offices, hospitals, and enclosed public places. The virus can

1640 be transmitted by direct contact with infected secretions but does not survive for long on fomites.

The incubation period for measles is ~10 days to fever onset and 14 days to rash onset. This period may be shorter in infants and longer (up to 3 weeks) in adults. Infection is initiated when measles virus is deposited in the respiratory tract, oropharynx, or conjunctivae (Fig. 211-1A). During the first 2–4 days after infection, measles virus proliferates locally in the respiratory mucosa, primarily in dendritic cells and lymphocytes, and spreads to draining lymph nodes. Virus then enters the bloodstream by budding from infected lymphocytes, producing the viremia that disseminates infection throughout the body. Replication of measles virus in the target organs, together with the host's immune response, are responsible for the signs and symptoms of measles that occur 8–12 days after infection and mark the end of the incubation period (Fig. 211-1B).

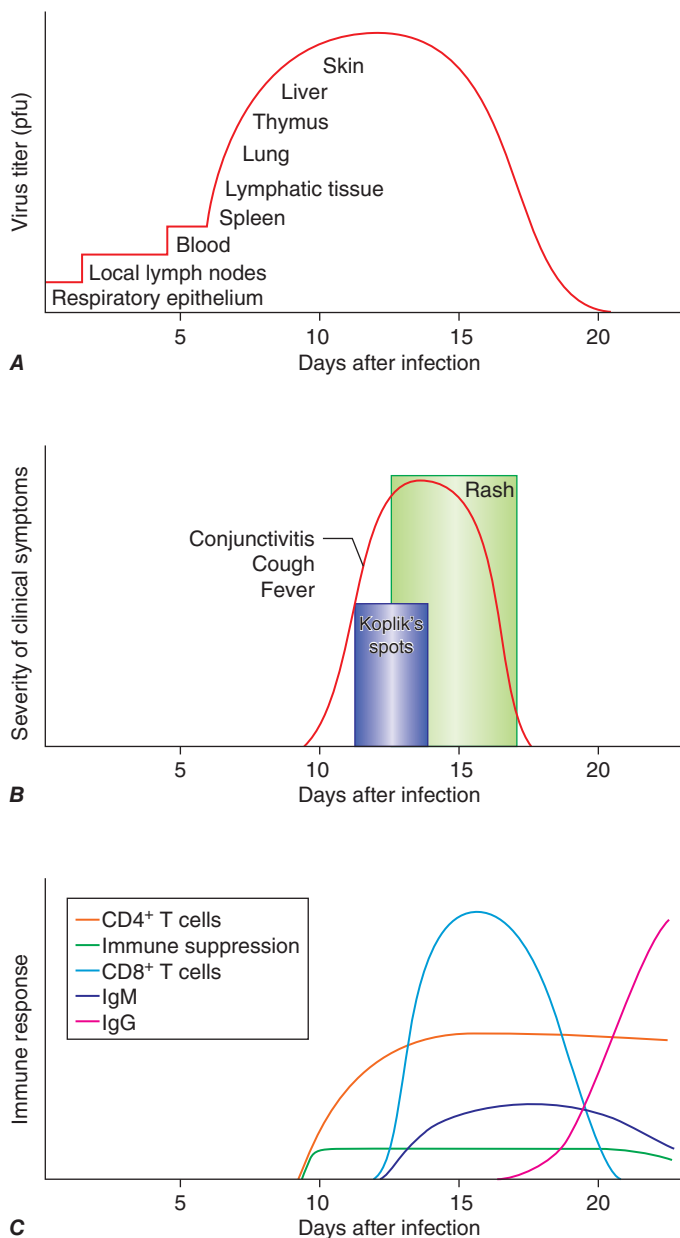


FIGURE 211-1 Measles virus infection: pathogenesis, clinical features, and immune responses. **A.** Spread of measles virus, from initial infection of the respiratory tract through dissemination to the skin. **B.** Appearance of clinical signs and symptoms, including Koplik's spots and rash. **C.** Antibody and T-cell responses to measles virus. The signs and symptoms of measles arise coincident with the host immune response. (Reproduced with permission from WJ Moss and DE Griffin: *Global measles elimination. Nat Rev Microbiology* 4:900, 2006.)

IMMUNE RESPONSES

Host immune responses to measles virus are essential for viral clearance, clinical recovery, and the establishment of long-term protective immunity (Fig. 211-1C). Early nonspecific (innate) immune responses during the prodromal phase include activation of natural killer cells and increased production of antiviral proteins. The adaptive immune responses consist of measles virus-specific antibody and cellular responses. The protective efficacy of antibodies to measles virus is illustrated by the immunity conferred to infants from passively acquired maternal antibodies and the protection of exposed, susceptible individuals after administration of anti-measles virus immunoglobulin. The first measles virus-specific antibodies produced after infection are of the IgM subtype, with a subsequent switch to predominantly IgG1 and IgG3 isotypes. The IgM antibody response is typically absent following reexposure or revaccination and serves as a marker of primary infection.

The importance of cellular immunity to measles virus is demonstrated by the ability of children with agammaglobulinemia (congenital inability to produce antibodies) to recover fully from measles and the contrasting picture for children with severe defects in T lymphocyte function who often develop severe or fatal disease (Chap. 362). The initial predominant T_H1 response (characterized by interferon- γ) is essential for viral clearance, and the later T_H2 response (characterized by interleukin-4) promotes the development of measles virus-specific antibodies that are critical for protection against reinfection.

The duration of protective immunity following wild-type measles virus infection is generally thought to be lifelong. Immunologic memory to measles virus includes both continued production of measles virus-specific antibodies by long-lived plasma cells and memory B cells as well as circulation of measles virus-specific CD4⁺ and CD8⁺ T lymphocytes.

However, the intense immune responses induced by measles virus infection are paradoxically associated with depressed responses to unrelated (non-measles virus) antigens. This state of immunosuppression persists for at least several weeks to months beyond resolution of the acute illness, enhances susceptibility to secondary infections with bacteria and viruses that cause pneumonia and diarrhea, and is thus responsible for a substantial proportion of measles-related morbidity and deaths. Delayed-type hypersensitivity responses to recall antigens, such as tuberculin, are suppressed, and cellular and humoral responses to new antigens are impaired. Reactivation of latent tuberculosis and remission of autoimmune diseases after measles have been described and are attributed to this period of immune suppression. Importantly, measles results in reductions in the magnitude and diversity of antibodies against previously encountered viral and bacterial antigens, impairing immunologic memory. This mechanism may explain why child morbidity and mortality due to other infectious diseases may be increased for >2 years after measles.

APPROACH TO THE PATIENT

Measles

Clinicians should consider measles in persons presenting with fever and generalized erythematous rash, particularly when measles virus is known to be circulating or the patient has a history of travel to endemic areas. Appropriate precautions must be taken to prevent nosocomial transmission. The diagnosis requires laboratory confirmation except during large outbreaks in which an epidemiologic link to a confirmed case can be established. Care is largely supportive and consists of the administration of vitamin A and antibiotics for secondary bacterial infections (see "Treatment," below). Complications of measles, including bacterial infections and encephalitis, may occur after acute illness and require careful monitoring, particularly in immunocompromised persons.

CLINICAL MANIFESTATIONS

In most persons, the signs and symptoms of measles are highly characteristic (Fig. 211-1B). Fever and malaise beginning ~10 days after exposure are followed by cough, coryza, and conjunctivitis. These

signs and symptoms increase in severity over 4 days. Koplik's spots (see Fig. A1-2) develop on the buccal mucosa ~2 days before the rash appears. Koplik's spots are pathognomonic of measles and consist of bluish white dots ~1 mm in diameter surrounded by erythema. The lesions appear first on the buccal mucosa opposite the lower molars but rapidly increase in number and may involve the entire buccal mucosa. They fade with the onset of rash.

The characteristic rash of measles (see Fig. A1-3) begins 2 weeks after infection, when the clinical manifestations are most severe, and signal the host's immune response to the replicating virus. Headache, abdominal pain, vomiting, diarrhea, and myalgia may be present. The rash of measles begins as erythematous macules behind the ears and on the neck and hairline. The rash progresses to involve the face, trunk, and arms, with involvement of the legs and feet by the end of the second day. Areas of confluent rash appear on the trunk and extremities, and petechiae may be present. The rash fades slowly in the same order of progression as it appeared, usually beginning on the third or fourth day after onset. Resolution of the rash may be followed by desquamation, particularly in undernourished children.

Because the characteristic rash of measles is a consequence of the cellular immune response, it may not develop in persons with impaired cellular immunity. These persons have a high case-fatality rate and frequently develop giant cell pneumonitis caused by measles virus.

■ DIFFERENTIAL DIAGNOSIS

The differential diagnosis of measles includes other causes of fever, morbilliform rash, and conjunctivitis, including rubella, Kawasaki disease, infectious mononucleosis, roseola, scarlet fever, Rocky Mountain spotted fever, enterovirus or adenovirus infection, and drug sensitivity. Rubella is a milder illness without cough and with distinctive posterior auricular or suboccipital lymphadenopathy. The rash of roseola (exanthem subitum) (see Fig. A1-5) appears after fever has subsided. The atypical lymphocytosis in infectious mononucleosis contrasts with the leukopenia commonly observed in children with measles.

■ DIAGNOSIS

Measles is readily diagnosed on clinical grounds by clinicians familiar with the disease, particularly during outbreaks. Koplik's spots are especially helpful because they appear early and are pathognomonic. Clinical diagnosis is more difficult (1) during the prodromal illness; (2) when the rash is attenuated by passively acquired antibodies or prior immunization; (3) when the rash is absent or delayed in immunocompromised children or severely undernourished children with impaired cellular immunity; and (4) in regions where the incidence of measles is low and other pathogens are responsible for most illnesses with fever and rash. The CDC case definition for measles requires (1) a generalized maculopapular rash of at least 3 days' duration; (2) fever of at least 38.3°C (101°F); and (3) cough, coryza, or conjunctivitis.

Serology is the most common method of laboratory diagnosis. The detection of measles virus-specific IgM in a single specimen of serum or oral fluid is considered diagnostic of acute infection, as is a four-fold or greater increase in measles virus-specific IgG antibody levels between acute- and convalescent-phase serum specimens. Primary infection in the immunocompetent host results in antibodies that are often detectable within 1–3 days of rash onset and reach peak levels in 2–4 weeks. However, measles virus-specific IgM antibodies may not be detectable until 4–5 days or more after rash onset, resulting in false-negative test results if the specimen is obtained too early, and usually fall to undetectable levels within 4–8 weeks of rash onset.

Several methods for measurement of antibodies to measles virus are available. Neutralization tests are sensitive and specific, and the results are highly correlated with protective immunity. However, these tests require propagation of measles virus in cell culture and thus are expensive and laborious. Commercially available measles IgM enzyme immunoassays are most frequently used. Measles can also be diagnosed by isolation of the virus in cell culture from respiratory secretions, nasopharyngeal or conjunctival swabs, blood, or urine. Direct detection of giant cells in respiratory secretions, urine, or tissue obtained by biopsy provides another method of diagnosis.

For detection of measles virus RNA by reverse-transcription polymerase chain reaction, primers targeted to highly conserved regions of measles virus genes are used. Extremely sensitive and specific, this assay may also permit identification and characterization of measles virus genotypes for molecular epidemiologic studies and can distinguish wild-type from vaccine virus strains.

TREATMENT

Measles

There is no specific antiviral therapy for measles. Treatment consists of general supportive measures such as hydration and administration of antipyretic agents. Because secondary bacterial infections are a major cause of morbidity and death attributable to measles, effective case management involves prompt antibiotic treatment for patients who have clinical evidence of bacterial infection, including pneumonia and otitis media.

Vitamin A (available in oral and parenteral formulations) is effective for the treatment of measles and can markedly reduce rates of morbidity and mortality. The WHO recommends administration of once-daily oral doses of 200,000 IU of vitamin A for 2 consecutive days to all children with measles who are ≥12 months of age. Lower doses are recommended for younger children: 100,000 IU per day for children 6–11 months of age and 50,000 IU per day for children <6 months old. A third dose is recommended 2–6 weeks later for children with evidence of vitamin A deficiency. While such deficiency is not a widely recognized problem in the United States, many American children with measles do, in fact, have low serum levels of vitamin A, and these children experience increased measles-associated morbidity.

Anecdotal reports have described the recovery of previously healthy pregnant and immunocompromised patients with measles pneumonia and of immunocompromised patients with measles encephalitis after treatment with aerosolized and IV ribavirin. However, the clinical benefits of ribavirin in measles have not been conclusively demonstrated in clinical trials.

■ COMPLICATIONS

Most complications of measles involve the respiratory tract and include the effects of measles virus itself and secondary bacterial infections. Giant cell pneumonitis due to replication of measles virus in the lungs can develop in immunocompromised persons. Acute laryngotracheobronchitis (croup) can occur during measles and may result in airway obstruction, particularly in young children. Many children with measles develop diarrhea, which contributes to and can exacerbate existing undernutrition.

Most complications of measles result from secondary bacterial infections of the respiratory tract that are attributable to a state of immune suppression after acute measles. Otitis media and bronchopneumonia are most common. Recurrence of fever or failure of fever to subside with the rash suggests secondary bacterial infection.

Severe complications of measles involve the central nervous system (CNS). Post-measles encephalomyelitis complicates ~1 in 1000 cases, affecting mainly older children and adults. Encephalomyelitis occurs within 2 weeks of rash onset and is characterized by fever, seizures, and a variety of neurologic abnormalities. The finding of periventricular demyelination, the induction of immune responses to myelin basic protein, and the absence of measles virus in the brain suggest that post-measles encephalomyelitis is an autoimmune disorder triggered by measles virus infection. Rarer CNS complications that occur months to years after acute infection are measles inclusion body encephalitis (MIBE) and subacute sclerosing panencephalitis (SSPE). In contrast to post-measles encephalomyelitis, MIBE and SSPE are caused by persistent measles virus infection. MIBE is a rare but fatal complication that affects individuals with defective cellular immunity and typically occurs months after infection. SSPE is a slowly progressive disease characterized by seizures and progressive deterioration of cognitive and motor functions, with death occurring 5–15 years after measles

■ PROGNOSIS

Most persons with measles recover and develop long-term protective immunity to reinfection. Measles case–fatality proportions vary with the average age of infection, the nutritional and immunologic status of the population, measles vaccine coverage, and access to health care. Among previously vaccinated persons who do become infected, disease is less severe and mortality rates are significantly lower. In most developed countries, the case–fatality rate is 0.01–0.1%, but in endemic areas of sub-Saharan Africa, the measles case–fatality rate may be 5–10% or even higher. Measles is a major cause of childhood deaths in refugee camps and in internally displaced populations, where case–fatality rates have been as high as 20–30%.

■ PREVENTION

Passive Immunization Human immunoglobulin given shortly after exposure can attenuate the clinical course of measles. In immunocompetent persons, administration of immunoglobulin within 72 h of exposure usually prevents measles virus infection and almost always prevents clinical measles. Administered up to 6 days after exposure, immunoglobulin will still prevent or modify the disease. Prophylaxis with immunoglobulin is recommended for susceptible household and nosocomial contacts who are at risk of developing severe measles, particularly children <1 year of age, immunocompromised persons (including immunocompromised persons living with HIV who were previously immunized with live attenuated measles vaccine), and pregnant women. Except for premature infants, children <6 months of age usually will be partially or completely protected by passively acquired maternal antibody. Infants born to women with vaccine-induced measles immunity become susceptible to measles at a younger age than infants born to women with acquired immunity from natural infection. If measles is diagnosed in a household member, all unimmunized children in the household should receive immunoglobulin. The recommended dose is 0.5 mL/kg given intramuscularly with a maximum total dose of 15 mL. Immunocompromised and pregnant persons should receive 400 mg/kg intravenously. IV immunoglobulin contains antibodies to measles virus, and the usual dose of 100–400 mg/kg generally provides adequate prophylaxis for measles exposures occurring as long as 3 weeks or more after IV immunoglobulin administration.

Active Immunization The first live attenuated measles vaccine was developed by passage of the Edmonston strain in chick embryo fibroblasts to produce the Edmonston B virus, which was licensed in 1963 in the United States but was reactogenic. Further passage of Edmonston B virus produced the more attenuated and less reactogenic Schwarz vaccine. The Moraten (“more attenuated Enders”) strain, which was licensed in 1968 and is used in the United States, is genetically identical to the Schwarz strain. The Edmonston–Zagreb vaccine, also derived from the Edmonston B strain, is widely used in many countries and was passaged in human diploid cells.

Lyophilized measles vaccines are relatively stable, but reconstituted vaccine rapidly loses potency. Live attenuated measles vaccines are inactivated by light and heat and lose about half their potency at 20°C and almost all their potency at 37°C within 1 h after reconstitution. Therefore, a cold chain must be maintained before and after reconstitution. Antibodies first appear 12–15 days after vaccination, and titers peak at 1–3 months. Measles vaccines are often combined with other live attenuated virus vaccines, such as those for mumps and rubella (MMR) and for mumps, rubella, and varicella (MMRV).

The recommended age of first vaccination varies from 6 to 15 months and represents a balance between the optimal age for seroconversion and the probability of acquiring measles before that age. The proportions of children who develop protective levels of antibody after the first measles vaccination approximate 85% at 9 months of age and 95% at 12 months. Common childhood illnesses concomitant with vaccination may reduce the level of immune response, but such illnesses are not valid reasons to withhold vaccination. Measles vaccines have

been well tolerated and immunogenic in children and adults living with HIV, although antibody levels may wane more rapidly. Because of the potential severity of wild-type measles virus infection in children living with HIV, routine measles vaccination is recommended except for those who are severely immunocompromised. Measles vaccination is contraindicated in individuals with other severe deficiencies of cellular immunity because of the possibility of disease due to progressive pulmonary or CNS infection with the vaccine virus.

The duration of vaccine-induced immunity is at least several decades, if not longer. Rates of secondary vaccine failure 10–15 years after immunization have been estimated at ~5% but are likely lower when vaccination takes place after 12 months of age. Decreasing antibody concentrations do not necessarily imply a complete loss of protective immunity as a secondary immune response usually develops after reexposure to measles virus, with a rapid rise in antibody titers in the absence of overt clinical disease.

Standard doses of currently licensed measles vaccines are safe for immunocompetent children and adults. Fever up to 39.4°C (103°F) occurs in ~5–15% of seronegative vaccine recipients, and ~5% of vaccine recipients develop a transient rash. Mild transient thrombocytopenia has been reported, with an incidence of 1 case per ~40,000 MMR recipients.

Since the publication of a report in 1998 falsely hypothesizing that MMR vaccine may cause a syndrome of autism and intestinal inflammation, much public attention has focused on this purported association. The events that followed publication of this report led to diminished vaccine coverage in the United Kingdom and provide important lessons in the misinterpretation of epidemiologic evidence and the communication of scientific results to the public. The publication that incited the concern was a case series describing 12 children with a regressive developmental disorder and chronic enterocolitis; 9 of these children had autism. In 8 of the 12 cases, the parents associated onset of the developmental delay with MMR vaccination. This simple temporal association was misinterpreted and misrepresented as a possible causal relationship, first by the lead author of the study and then by elements of the media and the public. Subsequently, many comprehensive reviews and additional epidemiologic studies refuted evidence of a causal relationship between MMR vaccination and autism, and the offending publication was retracted.

■ PROSPECTS FOR MEASLES ERADICATION

Progress in global measles control has renewed discussion of measles eradication. In contrast to poliovirus eradication, the eradication of measles virus will not entail challenges posed by prolonged shedding of potentially virulent vaccine viruses and asymptomatic reservoirs. However, in comparison with smallpox eradication, higher levels of population immunity will be necessary to interrupt measles virus transmission, more highly skilled health care workers will be required to administer measles vaccines, and containment through case detection and ring vaccination will be more difficult for measles virus because of infectivity before rash onset.

New tools, such as microarray patches (MAPs) to deliver measles vaccine, could facilitate mass vaccination campaigns and vaccination of hard-to-reach children such as those residing in remote rural areas. In May 2023, Micron Biomedical reported positive results in a phase 1/2 clinical trial in The Gambia comparing the results of immunization by measles- and rubella-containing (MR) vaccine administered by MAP versus subcutaneous injection. This trial was the first of its kind to use microarray technology in children and found similar rates of seroconversion in MR vaccine-naïve children (92.9–100% for MAP and 89.7–100% for subcutaneous) and seroprotection in all age groups (93.2–100% for MAP and 89.8–100% for subcutaneous). Further studies and the overcoming of manufacturing and regulatory hurdles will be needed before MAPs become available.

Despite enormous progress, measles remains a leading vaccine-preventable cause of childhood mortality worldwide and continues to cause outbreaks in communities with low vaccination coverage rates. As the world looks to rebuild immunization services disrupted by the COVID-19 pandemic and improve these services through the ambitious MRP agenda, measles outbreaks will continue to remind us of the challenges to be overcome.

ACKNOWLEDGMENT

The authors acknowledge the substantial contributions of Kaitlin Rainwater-Lovett, who coauthored prior editions of this chapter.

FURTHER READING

- DE SWART RL, MOSS WJ: The immunological basis for immunization series: Module 7: Measles. Update 2020. Geneva: World Health Organization, 2020.
- GRIFFIN DE: Measles immunity and immunosuppression. *Curr Opin Virol* 46:9, 2020.
- HÜBSCHEN JM et al: Measles. *Lancet* 399:678, 2022.
- MINA MJ et al: Measles virus infection diminishes preexisting antibodies that offer protection from other pathogens. *Science* 366:599, 2019.
- MINTA AA et al: Progress toward measles elimination—Worldwide, 2000–2022. *MMWR Morb Mortal Wkly Rep* 72:1262, 2023.
- MOSS WJ: Measles. *Lancet* 380:2490, 2017.
- MOSS WJ et al: Feasibility assessment of measles and rubella eradication. *Vaccine* 39:3544, 2021.
- PHADKE VK et al: Vaccine refusal and measles outbreaks in the US. *JAMA* 324:1344, 2020.
- STREBEL PM, ORENSTEIN WA: Measles. *N Engl J Med* 381:349, 2019.
- WORLD HEALTH ORGANIZATION: Measles vaccines: WHO position paper—April 2017. *Wkly Epidemiol Rec* 92:205, 2017.

212 Rubella (German Measles)

Alan C. Ou, Ludmila M. Perelygina,
Laura A. Zimmerman, Susan E. Reef

Rubella was historically viewed as a variant of measles or scarlet fever. After an epidemic of rubella in Australia in the early 1940s, the ophthalmologist Norman Gregg noticed the occurrence of congenital cataracts among infants whose mothers had reported rubella during early pregnancy, and congenital rubella syndrome (CRS; see “Clinical Manifestations,” below) was first described. Not until 1962 was a separate viral agent for rubella isolated.

ETIOLOGY

Rubella virus is a member of the Matonaviridae family the genus *Rubivirus*. This single-strand RNA enveloped virus measures 40–80 nm in diameter. Its nucleocapsid consists of ~10-kb positive-sense RNA genome surrounded by a protein shell composed of a core protein and a single-layer lipoprotein envelope with spike-like projections containing two glycoproteins, E1 and E2. There is only one antigenic type of rubella virus, and humans are its only known reservoir.

PATHOGENESIS AND PATHOLOGY

Although the pathogenesis of postnatal (acquired) rubella has been well documented, data on pathology are limited because of the mildness of the disease. Rubella virus is spread from person to person via respiratory droplets. Primary implantation and replication in the nasopharynx are followed by spread to the lymph nodes. Subsequent viremia occurs, which in pregnant women often results in infection of the placenta. Placental virus replication may lead to infection of fetal organs. The pathology of CRS in the infected fetus is well defined, with almost all organs found to be infected; however, the pathogenesis of CRS is only poorly delineated. In tissue, infections with rubella virus have diverse effects, ranging from no obvious impact to cell destruction. The hallmark of fetal infection is a chronic infection with persistence throughout fetal development in utero and for up to 1 year after birth.

Individuals with acquired rubella may shed virus from 7 days before rash onset to ~5–7 days thereafter. Both clinical and subclinical

infections are considered contagious. Infants with CRS may shed large quantities of virus from bodily secretions, particularly from the throat and in the urine, up to 1 year of age. Outbreaks of rubella, including some in nosocomial settings, have originated with index cases of CRS. Thus, only individuals immune to rubella virus should have contact with infants who have CRS or who are congenitally infected with rubella virus but are not showing signs of CRS.

EPIDEMIOLOGY

The largest recent rubella epidemic in the United States took place in 1964–1965, when an estimated 12.5 million cases occurred, resulting in ~20,000 cases of CRS. Since the introduction of the routine rubella vaccination program in the United States in 1969, the number of rubella cases reported each year has dropped by >99%; the rate of vaccination coverage with rubella-containing vaccine (RCV) has been >90% among children 19–35 months old since 1996. In the United States, a goal for the elimination of rubella and CRS by 2000 was set in 1989. Interruption of endemic transmission of rubella virus was achieved by 2001. In 2004, a panel of experts agreed unanimously that rubella was no longer an endemic disease in the United States. The criteria used to document lack of endemic transmission included low disease incidence, high nationwide rubella antibody seroprevalence, outbreaks that were few and contained (i.e., small numbers of cases), and lack of endemic virus transmission (as assessed by genetic sequencing). Although interruption of endemic transmission has been sustained since 2001, rubella virus importations continue to occur, and cases continue to develop among susceptible persons. During 2010–2022, 66 cases of rubella were reported; 71% of these cases were in persons 20–49 years old—an age group that includes women of childbearing age. During this period, 13 cases of CRS were reported, all from foreign-born mothers. Therefore, health care providers should remain vigilant, considering the possibility of rubella virus infection in adults (especially those emigrating or returning from countries without rubella control programs) and recognizing the potential for CRS among their infants.

The Global Measles and Rubella Strategic Framework 2021–2030 envisions a “world free from measles and rubella” with the goal to “achieve and sustain the regional measles and rubella elimination goals.” By 2023, five of the six World Health Organization (WHO) regions had rubella elimination goals (see “Prevention” below). Although rubella and CRS are no longer endemic in the WHO Region of the Americas, they remain important public health problems globally. The number of rubella cases reported worldwide in 2000 was ~700,000; this figure declined to 17,407 in 2022. However, the number of rubella cases may be underestimated because cases are often mild, patients may not seek care, cases may not be recognized or may not be reported, and, in some countries, cases are identified through measles surveillance systems that are not specific for rubella. Despite a continued increase in the number of countries with rubella vaccination programs, 25% of the world’s children remained unvaccinated against rubella in 2022. In 2010, it was estimated that 105,000 cases of CRS occurred annually globally.

CLINICAL FEATURES

Acquired Rubella Acquired rubella commonly presents with a generalized maculopapular rash that usually lasts up to 3 days (Fig. 212-1), although as many as 50% of cases may be subclinical or without rash. When the rash occurs, it is usually mild and may be difficult to detect in persons with darker skin. In younger children, rash is usually the first sign of illness. However, in older children and adults, a 1- to 5-day prodrome often precedes the rash and may include low-grade fever, malaise, and upper respiratory symptoms. The incubation period is 14 days (range, 12–23 days).

Lymphadenopathy, particularly occipital and postauricular, may be noted during the second week after exposure. Although acquired rubella is usually thought of as a benign disease, arthralgia and arthritis are common in infected adults, particularly women. Thrombocytopenia and encephalitis are less common complications.

Congenital Rubella Syndrome The most serious consequence of rubella virus infection can develop when a woman becomes infected



FIGURE 212-1 Mild maculopapular rash of rubella in a child.

during pregnancy, particularly during the first trimester. The resulting complications may include miscarriage, fetal death, premature delivery, or live birth with congenital defects. Infants infected with rubella virus in utero may have myriad physical defects (Table 212-1), which most commonly relate to the eyes, ears, and heart. This constellation of severe birth defects is known as CRS. In addition to permanent manifestations, there are a host of transient physical manifestations, including thrombocytopenia with purpura/petechiae (e.g., dermal erythropoiesis, “blueberry muffin syndrome”). Some infants may be born with congenital rubella virus infection but have no apparent signs or symptoms of CRS and are referred to as “infants with congenital rubella virus infection only.”

■ DIAGNOSIS

Acquired Rubella Clinical diagnosis of acquired rubella is difficult because of the mimicry of many illnesses with rashes, the varied clinical presentations, and the high rates of subclinical and mild disease. Illnesses that may be similar to rubella in presentation include scarlet fever, roseola, toxoplasmosis, fifth disease, measles, Zika, and illnesses with suboccipital and postauricular lymphadenopathy. Thus, laboratory documentation of rubella virus infection is considered the only reliable way to confirm acute disease.

Laboratory assessment of rubella virus infection is conducted by serologic and virus detection methods. For acquired rubella, serologic diagnosis is most common and depends on the demonstration of IgM antibodies in an acute-phase serum specimen or a fourfold rise in IgG antibody titer between acute- and convalescent-phase specimens. To

detect a rise in IgG antibody titer indicative of acute disease, the acute-phase serum specimen should be collected within 7–10 days after onset of illness and the convalescent-phase specimen ~14–21 days after the first specimen. The enzyme-linked immunosorbent assay IgM capture technique is considered most accurate for serologic diagnosis, but the indirect IgM assays also are acceptable. After rubella virus infection, IgM antibody may be detectable for up to 6 weeks. In case of a negative result for IgM in specimens taken earlier than day 5 after rash onset, serologic testing should be repeated.

Although uncommon, reinfection with rubella virus is possible, and IgM antibodies may be present. In this instance, IgG avidity testing is used in conjunction with IgG testing to distinguish primary rubella infection from reinfection. The detection of low-avidity antibodies in a patient’s serum indicates recent infection. The presence of mature (high-avidity) IgG antibodies most likely indicates an infection occurring at least 2 months previously. Avidity testing may be particularly useful in diagnosing rubella in pregnant women and assessing the risk of CRS.

Rubella virus is typically detected in the nasopharynx during the prodromal period and for as long as 2 weeks after rash onset. However, viral specimens (nasopharyngeal swabs are preferred but throat swabs or urine are also acceptable) should be collected as soon after symptom onset as possible, preferably 1–3 days after onset, but no later than 7 days after onset. Rubella is usually diagnosed by viral RNA detection in a reverse-transcriptase polymerase chain reaction (RT-PCR) assay; rubella virus isolation can also be used to diagnose rubella.

Congenital Rubella Syndrome The classic triad of CRS—clinical manifestations of cataracts, hearing impairment, and heart defects—is seen in ~10% of infants with CRS. Infants may present with different combinations of defects depending on when infection occurs during gestation. Hearing impairment is the most common single defect of CRS. However, as with acquired rubella, laboratory diagnosis of congenital infection is highly recommended, particularly because most features of the clinical presentation are nonspecific and may be associated with other intrauterine infections. Early diagnosis of CRS allows the prompt implementation of infection control measures and facilitates appropriate medical intervention for specific disabilities.

Diagnostic tests used to confirm CRS include serologic assays and virus detection. In an infant with congenital infection, serum IgM antibodies are normally present for up to 6 months but may be detectable for up to 1 year after birth. In some instances, IgM may not be detectable until 1 month of age; thus, infants who have symptoms consistent with CRS but who test negative shortly after birth should be retested at 1 month. A rubella serum IgG titer persisting beyond the time expected after passive transfer of maternal IgG antibody (i.e., a rubella titer that does not decline at the expected rate of a twofold dilution per month) is another serologic criterion used to confirm CRS.

In congenital infection, rubella virus is detected most commonly from nasopharyngeal and throat swabs and urine. Infants with congenital rubella may excrete virus for up to 1 year, but specimens for RT-PCR are most likely to be positive if obtained within the first 6 months after birth. Rubella virus in infants with CRS can also be detected by virus culture.

Rubella Diagnosis in Pregnant Women In the United States, screening for rubella IgG antibodies is recommended as part of routine prenatal care. Pregnant women with a positive IgG antibody serologic test are considered immune. Susceptible pregnant women should be vaccinated postpartum.

A susceptible pregnant woman exposed to rubella virus should be tested for IgM antibodies and, if positive, confirmed by testing for low-avidity IgG antibodies to determine whether she was infected during pregnancy. Pregnant women with evidence of acute infection must be clinically monitored, and gestational age at the time of maternal infection must be determined to assess the possibility of risk to the fetus. Among women infected with rubella virus during the first 10 weeks of gestation, the risk of delivering an infant with CRS is 90%. The risk of birth defects declines with infection later in gestation, and fetal defects are rarely associated with maternal

TABLE 212-1 Common Transient and Permanent Manifestations in Infants with Congenital Rubella Syndrome	
TRANSIENT MANIFESTATIONS	PERMANENT MANIFESTATIONS
Hepatosplenomegaly	Hearing impairment/deafness
Interstitial pneumonitis	Congenital heart defects (patent ductus arteriosus, pulmonary arterial stenosis)
Thrombocytopenia with purpura/petechiae (e.g., dermal erythropoiesis or “blueberry muffin syndrome”)	Eye defects (cataracts, cloudy cornea, microphthalmos, pigmentary retinopathy, congenital glaucoma)
Hemolytic anemia	Microcephaly
Bony radiolucencies	Central nervous system sequelae (mental and motor delay, autism)
Intrauterine growth retardation	
Adenopathy	
Meningoencephalitis	

rubella after the 16th week of gestation, although sensorineural hearing deficit may occur with infection as late as 20 weeks. Because of the potential for false-positive results, rubella IgM antibody testing is not recommended for pregnant women with no history of illness or contact with a rubella-like illness.

TREATMENT

Rubella

No specific therapy is available for rubella virus infection. Symptom-based treatment for various manifestations, such as fever and arthralgia, is appropriate. Immunoglobulin does not prevent rubella virus infection after exposure and therefore is not recommended as routine postexposure prophylaxis. Although immunoglobulin may modify or suppress symptoms, it can create an unwarranted sense of security: infants with congenital rubella have been born to women who received immunoglobulin shortly after exposure. Administration of immunoglobulin should be considered only if a pregnant woman who has been exposed to a person with rubella will not consider termination of the pregnancy under any circumstances. In such cases, IM administration of 20 mL of immunoglobulin within 72 h of rubella exposure may reduce—but does not eliminate—the risk of rubella.

PREVENTION

After the isolation of rubella virus in the early 1960s and the occurrence of a devastating rubella pandemic in 1964–1965, a vaccine for rubella was developed and licensed in 1969. The majority of rubella-containing vaccines (RCVs) used worldwide are combined measles and rubella (MR) or measles, mumps, and rubella (MMR) formulations. A tetravalent measles, mumps, rubella, and varicella (MMRV) vaccine is available but is not widely used. Available rubella-containing vaccines are live attenuated vaccine virus.

The public health burden of rubella virus infection is measured primarily through the occurrence of CRS cases among women who were infected during pregnancy. The 1964–1965 rubella epidemic in

the United States resulted in >30,000 infections during pregnancy. CRS occurred in ~20,000 infants born alive, including >11,000 infants who were deaf, >3500 infants who were blind, and almost 2000 infants with intellectual disability. The medical cost of this epidemic exceeded \$1.5 billion. It has been estimated that the lifetime medical costs for children with CRS range from \$11,255 in low-income countries to \$934,000 in high-income countries.

In most countries, there are few data to document the epidemiology of CRS, but clusters of CRS cases have been reported in developing countries. Before the introduction of routine immunization against rubella in the United States, the incidence of CRS was 0.1–0.2 case per 1000 live births during endemic periods and 1–4 cases per 1000 live births during epidemic periods. Where rubella virus is circulating and women of childbearing age are susceptible, CRS cases will continue to occur.

The most effective method of preventing acquired rubella and CRS is through vaccination with an RCV. One dose induces seroconversion in ≥95% of persons ≥1 year of age. Immunity is considered long-term and is probably lifelong. The most commonly used vaccine globally is the RA27/3 virus strain. The recommendation for routine rubella vaccination schedules in the United States is a first dose of MMR vaccine at 12–15 months of age and a second dose at 4–6 years. Other persons recommended to receive a dose of a rubella-containing vaccine include adolescents and adults without documented evidence of immunity, individuals in congregate settings (e.g., college students, military personnel, childcare and health care workers), international travelers, and susceptible women before and after pregnancy.

Because of the theoretical risk of transmission of live attenuated rubella vaccine virus to the developing fetus, women known to be pregnant should not receive RCV. In addition, pregnancy should be avoided for 28 days after receipt of RCV. In follow-up studies of ~3000 unknowingly pregnant women who received rubella vaccine, no infant was born with CRS. Receipt of RCV during pregnancy is not ordinarily a reason to consider termination of the pregnancy.

In 2022, 175 (90%) of the 194 member countries of the WHO recommended inclusion of RCV in the routine childhood vaccination schedule (Fig. 212-2). Goals for the elimination of rubella and CRS

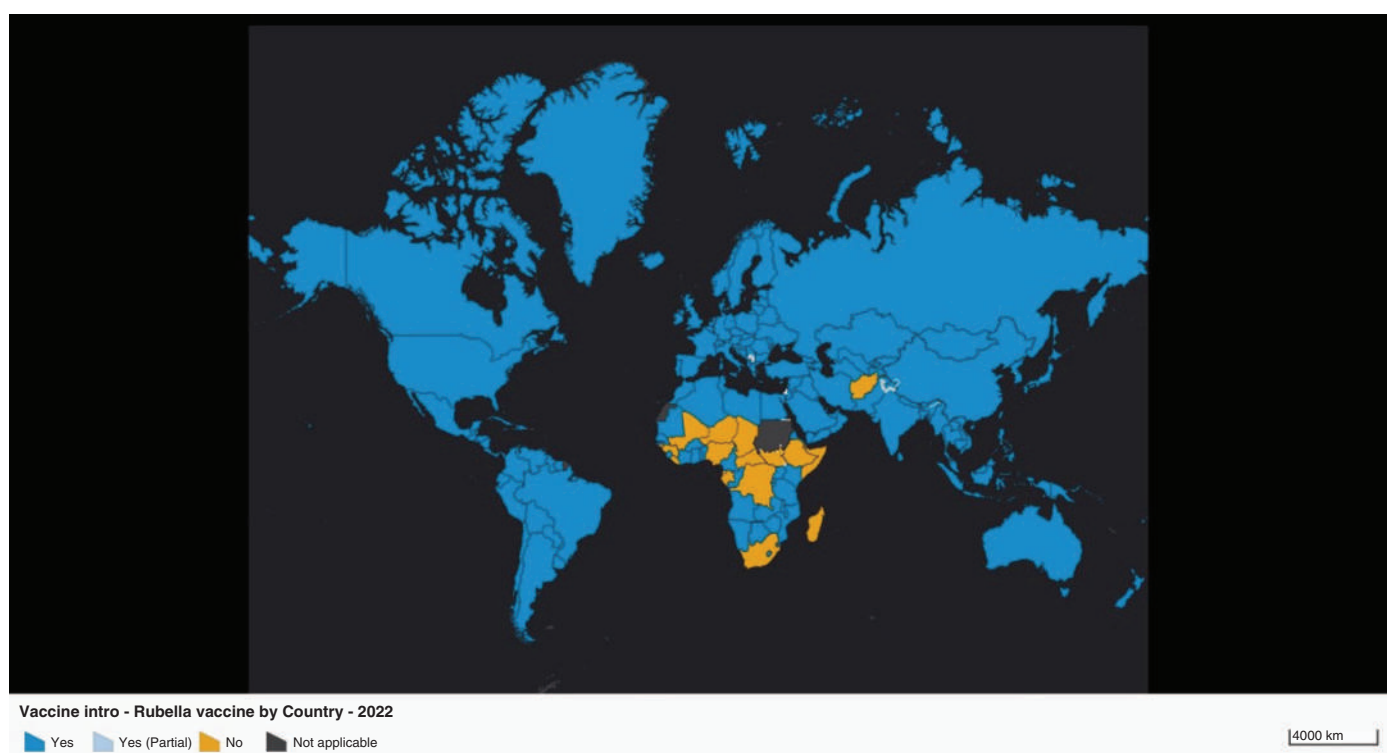


FIGURE 212-2 Countries using rubella vaccine in national childhood immunization schedules, 2022. Disclaimer—The boundaries and names shown and the designations used on this map do not imply the expression or any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city, or area nor of its authorities, or concerning the delimitation of its frontiers or boundaries. *Dotted or dashed lines* on maps represent approximate border lines for which there may not be full agreement. (Reproduced with permission from the World Health Organization; 2024.)

have been established in the WHO American, European, Southeast Asian, African, and Western Pacific regions. The Eastern Mediterranean region has not yet set such goals. Unfortunately, the COVID-19 pandemic, which began in late 2019, led to extensive disruptions of routine vaccination services in many countries. Thus, it is essential that all children are up to date with rubella vaccination, especially those who missed vaccination during the pandemic. To protect against rubella vaccine throughout the life course, additional strategies to immunize adolescents and adults will be needed to ensure adults of childbearing age are protected from the risk of having an infant with CRS.

FURTHER READING

CENTERS FOR DISEASE CONTROL AND PREVENTION: Control and prevention of rubella: Evaluation and management of suspected outbreaks, rubella in pregnant women, and surveillance for congenital rubella syndrome. *MMWR Morb Mortal Wkly Rep* 50:1, 2001.

CENTERS FOR DISEASE CONTROL AND PREVENTION: Notice to readers: Revised ACIP recommendation for avoiding pregnancy after receiving a rubella-containing vaccine. *MMWR Morb Mortal Wkly Rep* 50:1117, 2001.

CENTERS FOR DISEASE CONTROL AND PREVENTION: *Manual for the Surveillance of Vaccine-Preventable Diseases*, SW Roush et al (eds). Atlanta, Centers for Disease Control and Prevention, Last Reviewed: March 6, 2020, Chapters 14–15 (Rubella & Congenital Rubella Syndrome). Available at <https://www.cdc.gov/vaccines/pubs/surv-manual/chapters.html>. Accessed March 5, 2024.

CENTERS FOR DISEASE CONTROL AND PREVENTION: Rubella, in *Epidemiology and Prevention of Vaccine Preventable Diseases*, 14th ed. E Hall et al (eds). Washington, DC, Public Health Foundation, August 2021, Chapter 20. Available at <https://www.cdc.gov/vaccines/pubs/pinkbook/front-matter.html>. Accessed March 5, 2024.

OU AC et al: Progress toward rubella and congenital rubella syndrome control and elimination—Worldwide, 2012–2022. *MMWR Morb Mortal Wkly Rep* 73:162, 2024.

REEF SE, PLOTKIN SA: Rubella vaccine, in *Vaccines*, SA Plotkin, WA Orenstein (eds). Philadelphia, Saunders, 2024, pp 1025–1056.

THOMPSON K, ODAHOWSKI C: The costs and valuation of health impacts of measles and rubella risk management policies. *Risk Anal* 36:1357, 2016.

VYNNYCKY E et al: Estimates of the global burden of congenital rubella syndrome, 1996–2019. *Int J Infect Dis* 137:149, 2023.

WORLD HEALTH ORGANIZATION: Rubella, module 11, in *The Immunological Basis for Immunization Series*. Geneva, WHO, 2008. Available at <https://www.who.int/publications/i/item/9789241596848>. Accessed August 10, 2023.

WORLD HEALTH ORGANIZATION: Rubella vaccines: WHO position paper. *Wkly Epidemiol Rec* 95:301, 2020. Available at <https://www.who.int/publications/i/item/WHO-WER9527>. Accessed August 10, 2023.

WORLD HEALTH ORGANIZATION: Global Vaccine Action Plan 2011–2020. Geneva, WHO, 2013. Available at http://www.who.int/immunization/global_vaccine_action_plan/GVAP_doc_2011_2020/en/. Accessed August 10, 2023.

WORLD HEALTH ORGANIZATION: Measles and rubella strategic framework 2021–2030. Geneva: WHO; 2020. License: CC BY-NC-SA 3.0 IGO.

WORLD HEALTH ORGANIZATION: Immunization Agenda 2030: A global strategy to leave no one behind. Available at <https://www.who.int/teams/immunization-vaccines-and-biologicals/strategies/ia2030>. Accessed March 5, 2024.

213

Mumps

Jessica Leung, Nina B. Masters



Mumps is an acute, self-limited, systemic viral illness typically characterized by parotitis or other salivary gland swelling. Although mumps was once considered a universal childhood disease in the United States, routine mumps vaccination—with a one-dose policy implemented in 1977 and a two-dose policy in 1989—led to a >99% reduction in cases by the early 2000s. However, since 2006, there has been an increase in mumps cases in the United States, the majority among fully vaccinated persons. Mumps should be suspected in all patients with parotitis or mumps complications (see “Clinical Manifestations”), regardless of age, vaccination status, or travel history.

ETIOLOGIC AGENT

Mumps is an acute viral illness caused by a paramyxovirus from the *Rubulavirus* genus in the *Paramyxoviridae* family. This single-stranded, negative-sense, enveloped RNA virus is ~15.3 kb in size and encodes several minor proteins and seven major proteins. There is only one mumps virus serotype. One of the seven major encoded proteins, the small hydrophobic (SH) protein exhibits hypervariability among strains; thus, the SH gene nucleotide sequence (316 nucleotides) is used to genotype the virus for molecular epidemiologic purposes. Mumps virus is rapidly inactivated by formalin, ether, chloroform, heat, and ultraviolet light.

The 12 known genotypes of mumps virus are designated by the letters A to N (except E and M). In the United States, >98% of mumps virus specimens genotyped from 2015 through 2017 were genotype G. Most mumps vaccines licensed globally are composed of virus strains from genotype A, B, or N. The mumps virus strain (Jeryl Lynn) used in vaccines in the United States is genotype A.

EPIDEMIOLOGY

Mumps occurs worldwide and is endemic in many countries. In the absence of routine vaccination, the annual incidence of mumps is 100–1000 cases per 100,000 population, with epidemic peaks every 2–5 years. From 1999 to 2018, on average, >500,000 mumps cases were reported to the World Health Organization annually, with fewer reported cases from 2019 to 2021 (between 160,000 and 270,000). However, the global mumps incidence is challenging to estimate, as few countries routinely collect the pertinent data. As of 2021, mumps vaccine was introduced in 123 WHO member states. Mumps incidence has been reduced by 97–99% in countries with a routine two-dose measles, mumps, and rubella (MMR) vaccination schedule and by 87–88% in those with a one-dose vaccination program. However, since the mid-2000s, large mumps outbreaks have been reported among populations with high two-dose MMR coverage in countries with routine mumps immunization programs. Most outbreaks have occurred in settings with intense or frequent close contact, such as universities, close-knit communities, and correctional facilities, and most of these cases have occurred in fully vaccinated persons. Despite these outbreaks, mumps incidence is still much higher in countries that do not have routine mumps vaccination.

In the United States, prior to licensure of a vaccine for mumps in 1967, >100,000 mumps cases occurred annually. After the implementation of a one-dose mumps vaccination policy in 1977 and a subsequent two-dose policy in 1989, reported mumps cases declined to an annual average of ~300 by the early 2000s. However, since 2006, there has been an increase in mumps cases reported in the United States, with several peak years (Fig. 213-1). During the highest peak in recent cases, from January 2016 through June 2017, 150 mumps outbreaks and 9200 outbreak-associated cases were reported in a range of settings and groups, including schools, universities, athletic teams and facilities, church groups, workplaces, and large parties and events. While a majority of cases occurred in fully vaccinated young adults in association with large university outbreaks, about one-third of cases have

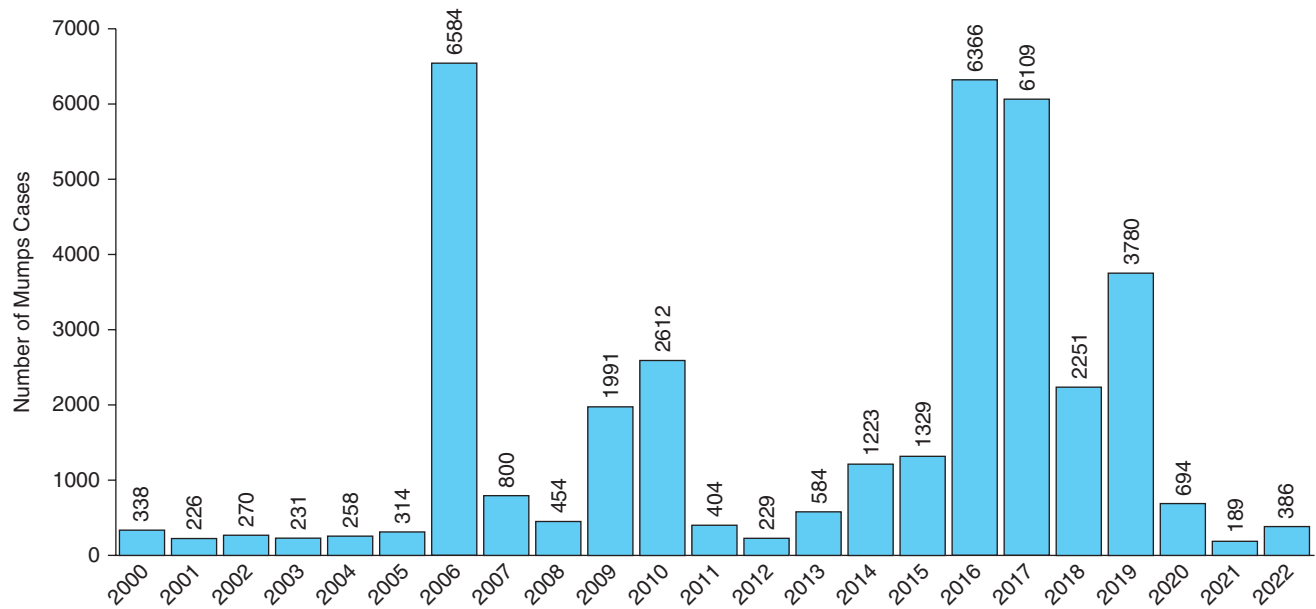


FIGURE 213-1 Reported mumps cases: United States, 2000–2022. (Source: National Notifiable Diseases Surveillance System (NNDS), Notifiable Infectious Disease Data Tables. Atlanta, GA, CDC Division of Health Informatics and Surveillance, 2022. Available at <https://www.cdc.gov/nndds/data-statistics/infectious-tables/index.html>.)

affected children or adolescents, most of whom were vaccinated. Since 2020, there has been a large reduction in reported mumps cases likely due to social distancing and other COVID-19 prevention measures, with 150–700 reported annually. As of 2023, mumps is endemic in the United States, and there are no elimination goals for the disease.

Multiple factors are likely involved in being at risk for mumps infection among vaccinated persons. Following vaccination, these factors include (1) failure to develop an immune response, (2) the development of a low-level immune response that is insufficient for protection, (3) a decrease in immunity over time (waning immunity) after initial development of a vaccine-induced immune response, (4) lower levels of vaccine-induced antibodies to the circulating wild-type virus strains than to the vaccine virus strain, and (5) a lower frequency of subclinical immunologic boosting due to lack of exposure to wild-type virus during periods of low disease incidence.

■ PATHOGENESIS

Humans are the only known natural reservoir for mumps virus, which is transmitted through direct contact with respiratory droplets or saliva of an infected person. The average incubation period is 16–18 days, with a range of 12–25 days. A person is most infectious from 2 days before until 5 days after onset of parotitis or other salivary gland swelling. However, mumps virus has been detected in saliva as early as 7 days before onset and as late as 9 days after onset of these manifestations. Mumps virus has been isolated from urine and seminal fluid up to 14 days after onset of parotitis, although no studies have assessed transmissibility of the virus through these fluids.

Primary mumps virus replication likely occurs in the nasal mucosa or upper respiratory mucosal epithelium. Given the range of symptoms, it is assumed that, after infection of the respiratory mucosa, the virus spreads to regional lymph nodes. Mononuclear cells and cells within regional lymph nodes can become infected; such infection facilitates the development of viremia, which usually lasts 3–5 days. Viremia can result in a range of acute inflammatory reactions, most commonly in the salivary glands (resulting in parotitis) and the testes (resulting in orchitis). Other sites of virus dissemination include the kidneys (reflected in the frequency of viruria), the central nervous system (CNS), the pancreas, the heart, the ovaries, the mammary glands, the perilymphatic fluid within the cochlea, and (during pregnancy) the fetus.

Little is known about the pathology of mumps since the disease is rarely fatal. Affected salivary glands contain perivascular and interstitial mononuclear-cell infiltrates and exhibit hemorrhage with prominent edema. Serum and urine amylase levels may be elevated as a result

of inflammation and tissue damage in the parotid gland. Necrosis of acinar and epithelial duct cells is evident in the salivary glands and in the germinal epithelium of the seminiferous tubules of the testes. The virus probably enters cerebrospinal fluid (CSF) through the choroid plexus or via transiting mononuclear cells during plasma viremia. Although relevant data are limited, in many cases, mumps encephalitis appears to be a para- or postinfectious process (as suggested by perivenous demyelination and perivascular mononuclear-cell inflammation) rather than the result of a direct cytotoxic effect caused by viral invasion of the CNS. However, although rare, primary mumps encephalitis does occur, as shown by mumps virus isolation from brain tissue. Infection of the perilymphatic fluid likely develops via retrograde penetration by the virus from the cervical lymph nodes following viremia, but infection could also occur via the CSF in cases of mumps CNS infection, given that the perilymph communicates with the CSF. Virus in the perilymph can result in infection of the cochlea and damage to the organ of Corti and the tectorial membrane, leading to transient or permanent deafness. Evidence of placental and intrauterine spread has been found in both early and late gestation. Virus frequently disseminates to the kidneys, but kidney involvement in mumps is almost always benign.

■ CLINICAL MANIFESTATIONS

While typically presenting with parotitis or other salivary gland swelling, mumps infection can be asymptomatic or present as nonspecific respiratory symptoms, though serious complications such as sensorineural hearing loss can occur. Fully vaccinated persons can contract mumps, but vaccinated persons are at a lower risk for mumps and mumps complications. Mumps infection is asymptomatic in ~20% of unvaccinated patients; the proportion asymptomatic among vaccinated persons is unknown.

Parotitis can be preceded by several days by a prodrome of low-grade fever, malaise, myalgia, headache, and anorexia. Parotitis typically lasts for 5 days (range, 3–7 days); most cases resolve within 10 days. Parotitis is generally bilateral and may not occur synchronously on both sides; unilateral involvement occurs in about one-third of cases. Swelling of the parotid gland is accompanied by tenderness and obliteration of the space between the earlobe and the angle of the mandible (**Figs. 213-2 and 213-3**). The patient frequently reports an earache and jaw pain and finds it difficult to eat, swallow, or talk. The orifice of the parotid duct is commonly red and swollen. The submaxillary and sublingual glands are involved less often than the parotid gland and are rarely involved alone. In ~6% of mumps cases, obstruction of lymphatic drainage secondary to bilateral salivary gland swelling may lead to presternal

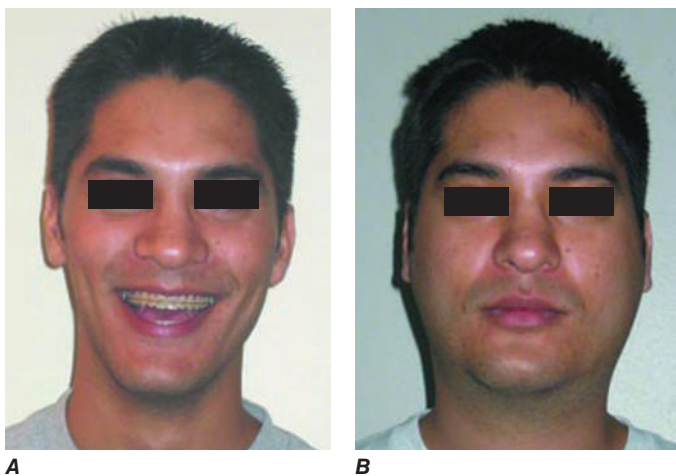


FIGURE 213-2 The same person before mumps acquisition (A) and on day 3 of acute bilateral parotitis (B). (Courtesy of patient C.M. From JD Shanley: *The resurgence of mumps in young adults and adolescents*. *Cleve Clin J Med* 74:42, 2007. Reprinted with permission. Copyright © 2007 Cleveland Clinic Foundation. All rights reserved.)

pitting edema, associated often with submandibular adenitis and rarely with the more life-threatening supraglottic edema.

The most frequent complications of mumps include orchitis, oophoritis, mastitis, pancreatitis, hearing loss, meningitis, and encephalitis. Complications can occur in the absence of parotitis and are more common among adults than among children and among males than among females, likely due to rates of orchitis.

Orchitis (testicular inflammation), usually accompanied by fever, is the most common complication, developing in up to 30% of unvaccinated and 6% of vaccinated postpubertal males. This complication is rare in children. Orchitis typically occurs during the first week of parotitis but can develop up to 6 weeks after parotitis. Both testes are involved in ~10–30% of cases. The testis is painful and tender and can be enlarged to several times its normal size. Pain and swelling may last for 1 week, while tenderness may last for several weeks. Testicular atrophy develops in ~30–50% of affected testicles. The development of anti-sperm antibodies, reduced testosterone production, and impaired sperm mobility through oligospermia, azospermia, or asthenospermia may lead to temporary sterility or subfertility. However, no studies have assessed the risk of permanent infertility in men with mumps orchitis.

Approximately 7% of unvaccinated and ≤1% of vaccinated postpubertal women develop oophoritis, which may be associated with lower abdominal pain and vomiting. The rate of mastitis in mumps has been estimated to be as high as 30% among unvaccinated postpubertal women and as low as ≤1% among vaccinated postpubertal women. Pancreatitis occurs in ~4% of unvaccinated and <1% of vaccinated mumps patients. Mumps pancreatitis, which may present as abdominal

pain, is difficult to diagnose because an elevated serum amylase level can be associated with either parotitis or pancreatitis. However, serum lipase is elevated in pancreatitis and the presence of both elevated serum amylase and lipase can help determine if pancreatitis is present in addition to parotitis. Hearing loss associated with mumps infection can occur in up to 4% of unvaccinated and <1% of vaccinated mumps patients. Mumps-related hearing loss is usually sudden in onset, unilateral, and transient and may be associated with vestibular symptoms. Bilateral and permanent hearing loss are rare.

Mumps virus is highly neurotropic, with subclinical CNS involvement occurring in up to 55% of patients as manifested by CSF pleocytosis. However, symptomatic CNS infection is less common. Aseptic meningitis occurs in ≤1% of vaccinated patients and up to 10% of unvaccinated patients and is a self-limited manifestation without significant risk of death or long-term sequelae. Symptoms of aseptic meningitis, including stiff neck, headache, and drowsiness, typically appear ~5 days after parotitis. Encephalitis develops in ≤1% of patients, who present with high fever, marked changes in the level of consciousness, seizures, and focal neurologic symptoms. Electroencephalographic abnormalities may be seen. Permanent sequelae are sometimes identified in survivors, and adult infections more commonly have poor outcomes than pediatric infections. The mortality rate associated with mumps encephalitis is ~1.5%. Other CNS problems occasionally associated with mumps include cerebellar ataxia, facial palsy, transverse myelitis, hydrocephalus, Guillain-Barré syndrome, flaccid paralysis, and behavioral changes.

Although rare and self-limited, myocarditis and endocardial fibroelastosis may represent severe complications of mumps infection; however, mumps-associated electrocardiographic abnormalities have been reported in up to 15% of cases. Other unusual complications include thyroiditis, nephritis, arthritis, hepatic disease, keratouveitis, and thrombocytopenic purpura. Abnormal renal function is common, but severe, life-threatening nephritis is rare.

Mumps infection in pregnant women is generally benign and is not more severe than in women who are not pregnant. Evidence suggesting an association between maternal mumps infection and an increased rate of spontaneous abortion or intrauterine fetal death is inconclusive.

Both mumps reinfection after natural infection and recurrent infection (in which parotid gland swelling resolves and then, weeks to months later, develops on the same or the other side) can occur. In the past, mumps reinfection was thought to be rare, but more recent data have suggested that it may be more common than previously thought.

Death due to mumps is exceedingly rare.

■ DIFFERENTIAL DIAGNOSIS

Mumps is the only cause of epidemic parotitis, although an increase in parotitis cases may also result from increased influenza activity—specifically, infection with influenza A virus subtype H3N2. Other infectious causes of parotitis include parainfluenza virus types 1–3, Epstein-Barr virus, human herpesviruses 6A and 6B, herpes simplex viruses types 1 and 2, coxsackievirus A, adenovirus, parvovirus B19, echovirus, lymphocytic choriomeningitis virus, and HIV. Laboratory testing for sporadic parotitis cases caused by these infectious pathogens can help rule out mumps.

Parotitis can also develop in the setting of sarcoidosis, Sjögren's syndrome, Mikulicz's syndrome, Parinaud's oculoglandular syndrome, uremia, diabetes mellitus, laundry starch ingestion, malnutrition, cirrhosis, and some drug treatments. Unilateral parotitis can be caused by ductal obstruction, cysts, and tumors. In the absence of parotitis or other salivary gland enlargement, symptoms of other visceral-organ and/or CNS involvement may predominate, and a laboratory diagnosis is required. Other entities should be considered when manifestations consistent with mumps appear in organs other than the parotid. For example, testicular torsion may produce a painful scrotal mass resembling that seen in mumps orchitis. Orchitis can also be caused by bacterial infections in the prostate and urinary tract, sexually transmitted diseases such as chlamydia and gonorrhea, and other viral infections such as those with coxsackievirus, varicella, echovirus, and cytomegalovirus. Oophoritis can also be caused by sexually transmitted diseases

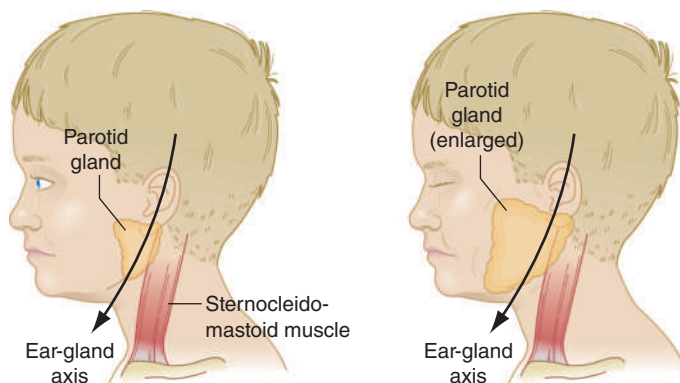


FIGURE 213-3 Schematic drawings of a normal parotid gland (left) and a parotid gland infected with mumps virus (right). An enlarged cervical lymph node is usually posterior to the imaginary line. (Reproduced with permission from A Gershon et al: *Krugman's Infectious Diseases of Children*, 11th ed. Philadelphia, Elsevier, 2004.)

such as chlamydia and gonorrhea. A number of viruses (e.g., enteroviruses) can cause aseptic meningitis that is clinically indistinguishable from that due to mumps virus.

LABORATORY DIAGNOSIS

If mumps is suspected, infection is confirmed by virologic methods, but serologic testing can aid in diagnosis. Especially in vaccinated patients, a negative virologic or serologic result in a person with clinical signs of mumps does not rule out mumps infection.

Virologic methods for confirming mumps include reverse transcription polymerase chain reaction (RT-PCR) and viral culture. RT-PCR is preferred because of its sensitivity, specificity, and timeliness. Mumps virus and viral RNA can be detected in blood, saliva, urine, and CSF. Buccal swabs provide the best specimens for virus detection. The parotid gland should be massaged for 30 s prior to collection of the buccal swab sample. As maximal viral shedding occurs within 5 days after symptom onset, specimens for mumps virologic testing ideally should be collected as close to parotitis onset as possible. The diagnostic yield of urine specimens increases over time up to 10 days after parotitis onset, but buccal specimens are more likely than urine specimens to result in virus detection at any time point.

Serologic methods that can aid in the diagnosis of mumps include detection of mumps-specific IgM antibodies or a fourfold rise between acute- and convalescent-phase IgG antibodies. In unvaccinated persons, IgM antibody is usually detectable within 5 days after onset, reaches a maximal level a week after onset, and remains elevated for weeks or months. Failure to detect mumps IgM in vaccinated patients is very common, as the IgM response is often undetectable, transient, or delayed in these individuals. Collection of specimens >3 days after onset may improve IgM detection. Additionally, IgM can yield false-positive results due to serologic cross-reactions. Use of IgG testing is generally not recommended, as IgG titers in vaccinated or previously infected patients may already be elevated at the time of acute-phase specimen collection, such that a fourfold rise is not detected in the convalescent-phase specimen.

TREATMENT

Mumps

Mumps is generally a benign, self-resolving illness. Therapy for parotitis and other clinical manifestations is symptom based and supportive. The administration of analgesics and the application of warm or cold compresses to the parotid area may be helpful. Testicular pain may be minimized by the local application of cold compresses and gentle support for the scrotum. Anesthetic blocks also may be used. Neither the administration of glucocorticoids nor incision of the tunica albuginea is of proven value in severe orchitis. Mumps immune globulin is not recommended for postexposure prophylaxis or treatment.

PREVENTION

Vaccination is the best prevention measure against mumps. Mumps vaccine is commonly included as part of the combination measles-mumps-rubella (MMR) vaccine or the combination measles-mumps-rubella-varicella (MMRV) vaccine. All mumps vaccines currently on the market are live attenuated virus vaccines. Strains used in mumps vaccines have included Jeryl Lynn, RIT 4385, Urabe Am9, Rubini, Leningrad-3 and Leningrad-Zagreb. Urabe- and Rubini-containing vaccines are no longer available. The Jeryl Lynn and RIT 4385 strains are the only strains used in mumps vaccines in the United States since 1967.

In the United States, children are recommended to receive the first MMR dose at 12–15 months of age and the second dose at 4–6 years. MMR vaccine is licensed for use in persons age ≥12 months; MMRV vaccine is licensed for use in persons age 12 months through 12 years. Due to the potential increased risk of febrile seizures, it is recommended that MMR vaccine and varicella vaccine be administered for the first dose in children 12–15 months of age. Adequate vaccination

against mumps is defined as two doses of MMR for school-aged children (i.e., grades K–12) and for adults at high risk (i.e., health care workers, international travelers, and students at post-high school educational institutions) and one dose for preschool-aged children and adults not at high risk. During an outbreak, a second dose should be considered for children age 1–4 years and adults who have received one dose. In 2017, after an increase in cases among persons with two MMR doses and a study demonstrating added benefit of a third MMR vaccine dose for individual protection, a third dose was recommended for use during outbreaks, specifically for groups whom public health authorities identify as at increased risk of acquiring mumps; public health authorities will inform providers of these groups at increased risk. As the duration of protection provided by a third dose of MMR vaccine is unknown and may be short term (<1 year), there is no current recommendation for a routine third dose.

The effectiveness of Jeryl Lynn containing MMR vaccine in preventing mumps was 72% after one dose (RR 0.24, 95% CI 0.08 to 0.76; 6 cohort studies; 9915 children; moderate certainty evidence), 86% after two doses (RR 0.12, 95% CI 0.04 to 0.35; 5 cohort studies; 7792 children; moderate certainty evidence). The effectiveness of the mumps component is lower than that of the measles component (two-dose effectiveness of 97%) and the rubella component (one-dose effectiveness of 97%). Incremental vaccine effectiveness of a third MMR dose—compared with two doses—during outbreaks is estimated at 78% (range, 61–88%).

In general, most recipients of mumps vaccine will seroconvert after vaccination and will have detectable antibodies to mumps virus; however, antibody levels start to decline soon after vaccination. Vaccine-induced neutralizing antibodies to wild-type strains may be lower in titer and may decline more rapidly than antibodies to the vaccine strain (Jeryl Lynn). However, most young adults given two vaccine doses in childhood appear to retain memory B cells.

Mumps vaccines are generally very safe. Urabe and Leningrad-Zagreb mumps strain vaccines have been associated with a slightly increased risk of aseptic meningitis, but there is no evidence of this risk for Jeryl Lynn and RIT 4385 mumps strain vaccines. There is a twofold greater risk of febrile seizures among children 12–23 months of age after receipt of the first dose of MMRV vaccine than after the first dose of MMR vaccine, with or without simultaneous varicella vaccination; this risk has not been found among vaccinated children 4–6 years of age.

There is no known immune correlate of protection for mumps; a positive IgG titer indicates only that a person has been exposed to mumps virus through either vaccination or natural infection and does not predict protection against infection. Therefore, all close contacts of a mumps patient should be advised to self-monitor for mumps symptoms for 25 days after their last exposure. Further, IgG titers should not be used to infer immunity in close contacts as it may indicate acute infection rather than immunity. MMR vaccine has not been shown to prevent illness or alter clinical severity in persons already infected with mumps virus and is not recommended as postexposure prophylaxis for immediate close contacts of mumps patients.

ACKNOWLEDGMENT

The authors acknowledge and thank Drs. Mariel Marlow and Stephen Rubin, authors of prior editions of this chapter.

FURTHER READING

- DI PIETRANTONJ C et al: Vaccines for measles, mumps, rubella, and varicella in children. *Cochrane Database Syst Rev* 4:CD004407, 2020.
- KROW-LUCAL E et al: Measles, mumps, rubella vaccine (PRIORIX): Recommendations of the Advisory Committee on Immunization Practices—United States, 2022. *MMWR Morb Mortal Wkly Rep* 71:1465, 2022.
- MARIN M et al: Recommendation of the Advisory Committee on Immunization Practices for use of a third dose of mumps virus-containing vaccine in persons at increased risk for mumps during an outbreak. *MMWR Morb Mortal Wkly Rep* 67:33, 2018.
- MASARANI M et al: Mumps orchitis. *J R Soc Med* 99:573, 2006.

- 1650 MCCLEAN HQ et al: Prevention of measles, rubella, congenital rubella syndrome, and mumps, 2013: Summary recommendations of the Advisory Committee on Immunization Practices (ACIP). *MMWR Recomm Rep* 62:1, 2013.
- RASHEED MAU et al: Decreased humoral immunity to mumps in young adults immunized with MMR vaccine in childhood. *Proc Natl Acad Sci USA* 116:19071, 2019.
- ROTA JS et al: Comparison of the sensitivity of laboratory diagnostic methods from a well-characterized outbreak of mumps in New York City in 2009. *Clin Vaccine Immunol* 20:391, 2013.
- RUBIN S et al: Molecular biology, pathogenesis and pathology of mumps virus. *J Pathol* 235:242, 2015.
- SHEPERSKY L et al: Mumps in vaccinated children and adolescents: 2007–2019. *Pediatrics* 148:e2021051873, 2021.
- TAPPE J et al: Characteristics of reported mumps cases in the United States: Vaccine 2018–2023. Vaccine 2024. Available at <https://pubmed.ncbi.nlm.nih.gov/39019661/>.
- WORLD HEALTH ORGANIZATION: WHO immunological basis for immunization series. Module 16: Mumps update 2020. Available at <https://apps.who.int/iris/bitstream/handle/10665/338004/9789240017504-eng.pdf>.

human disease. Rabies virus is a lyssavirus that infects a broad range of mammals and causes serious neurologic disease when transmitted to humans. This single-strand RNA virus has a nonsegmented, negative-sense (antisense) genome that consists of 11,932 nucleotides and encodes five proteins: nucleocapsid protein, phosphoprotein, matrix protein, glycoprotein, and a large polymerase protein. Rabies virus variants, which can be characterized by distinctive nucleotide sequences, are associated with specific animal reservoirs. Six other non-rabies virus species in the *Lyssavirus* genus have been reported to cause a clinical picture similar to rabies. Vesicular stomatitis virus, a vesiculovirus, causes vesiculation and ulceration in cattle, horses, and other animals and causes a self-limited, mild, systemic illness in humans (see “Other Rhabdoviruses,” below).

■ EPIDEMIOLOGY

Rabies is a zoonotic infection that occurs in a variety of mammals throughout the world except in Antarctica and on some islands. Rabies virus is usually transmitted to humans by the bite of an infected animal. Worldwide, endemic canine rabies is estimated to cause 59,000 human deaths annually. Most of these deaths occur in Asia and Africa, with rural populations and children disproportionately affected. Thus, in many resource-poor and resource-limited countries, canine rabies continues to be a threat to humans. However, in Latin America, rabies control efforts in dogs have been quite successful in recent years. Endemic canine rabies has been eliminated from the United States and most other resource-rich countries. Rabies is endemic in wildlife species, and a variety of animal reservoirs have been identified in different countries of the world (Fig. 214-1). Surveillance data from 2021 identified 3579 confirmed animal cases of rabies in the United States and Puerto Rico. Only 9.6% of these cases were in domestic animals, including 222 cases in cats, 50 in dogs, and 42 in cattle. In North American wildlife reservoirs, including bats, raccoons, skunks, and foxes, the infection is endemic, with involvement of one or more rabies virus variants in each reservoir species (Fig. 214-2). “Spillover” of rabies to other wildlife species and to domestic animals occurs. Bat rabies virus variants are present in every state except Hawaii and are responsible for most indigenously acquired human rabies cases in the United States. Raccoon rabies is endemic along the entire eastern coast of the United States. Skunk rabies is present in the midwestern states, with another focus in California. Rabies in foxes occurs in New Mexico, Arizona, and Alaska. In the United States, there were five human rabies deaths in 2021 and none in 2019, 2020, and 2022.

In Canada and Europe, epizootics of rabies in red foxes have been well controlled with the use of baits containing rabies vaccine. A similar approach, along with additional measures, is used in Canada to control incursions of raccoon rabies from the United States.

214 Rabies and Other Rhabdovirus Infections

Alan C. Jackson

RABIES

Rabies is a rapidly progressive, acute infectious disease of the central nervous system (CNS) in humans and animals that is caused by infection with rabies virus. The infection is normally transmitted from animal vectors via a bite exposure. Rabies has encephalitic and paralytic forms that progress to death.

■ ETIOLOGIC AGENT

Rabies virus is a member of the family *Rhabdoviridae*. Two genera in this family, *Lyssavirus* and *Vesiculovirus*, contain species that cause

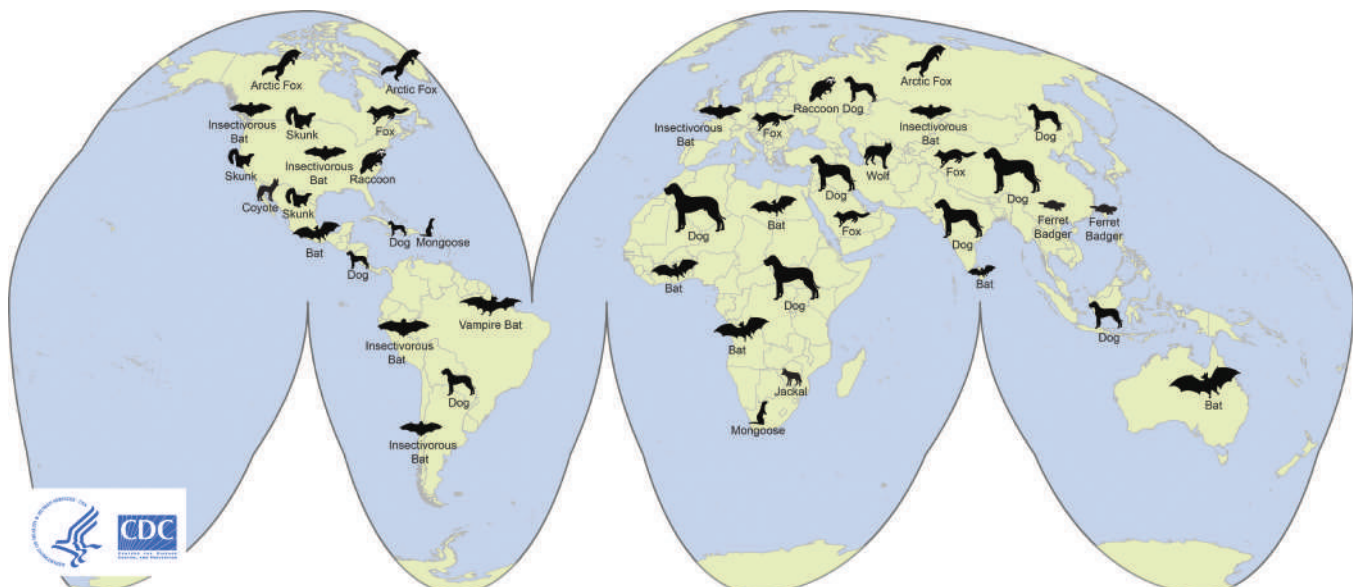


FIGURE 214-1 Distribution of global rabies vectors. (Courtesy of the Centers for Disease Control and Prevention.)

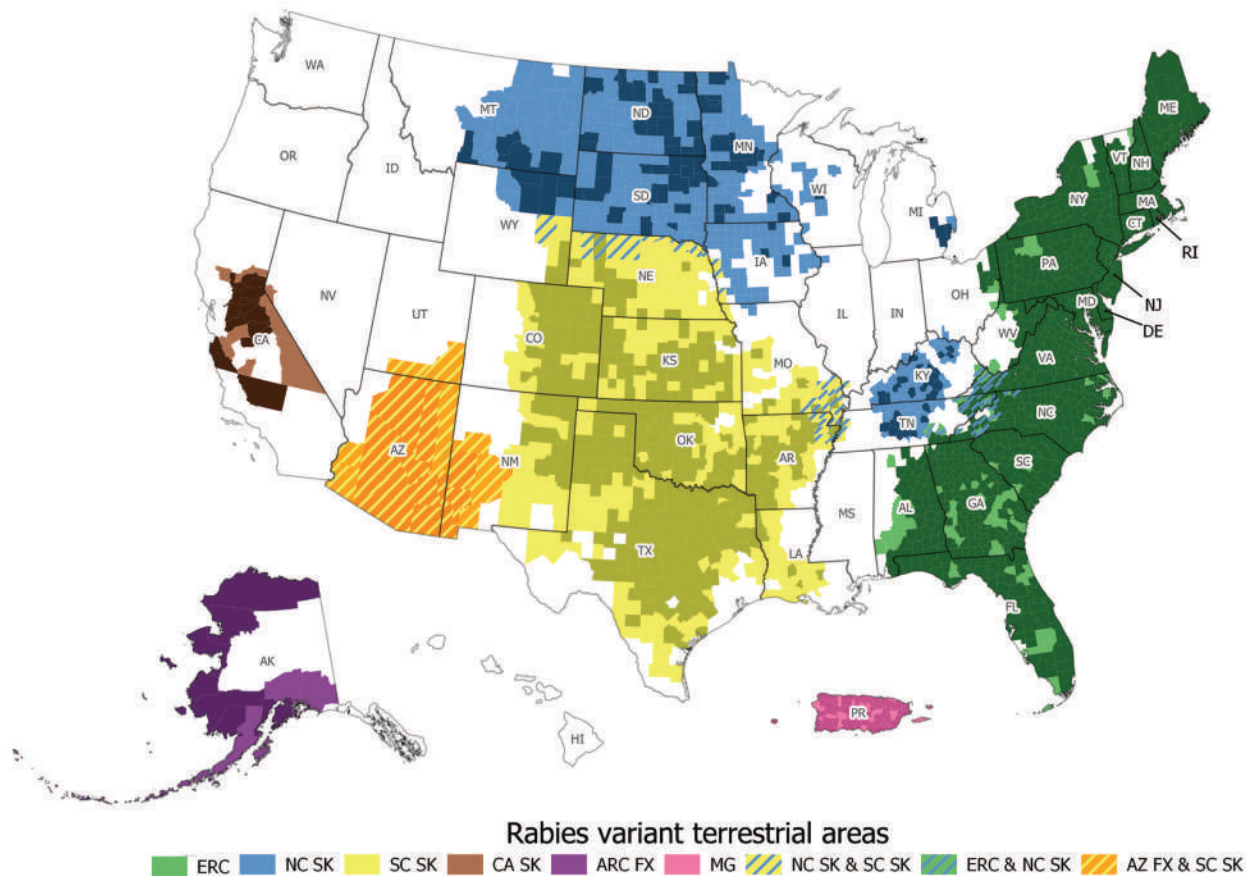


FIGURE 214-2 Distribution of the major rabies virus variants among wild terrestrial reservoirs in the United States and Puerto Rico, 2017–2021. Darker shading indicates counties with confirmed animal rabies cases in the past 5 years; lighter shading represents counties bordering enzootic counties without animal rabies cases that did not satisfy criteria for adequate surveillance. Small nonenzootic areas with no rabies cases reported in the past 15 years are shaded if they are in the vicinity of known enzootic counties and do not satisfy criteria for adequate surveillance. ARC FX rabies virus variant, Arctic fox RVV; AZ FX, Arizona fox RVV; CA SK, California skunk RVV; ERC, Eastern raccoon RVV; MG, Dog-mongoose RVV; NC SK, North central skunk RVV; SC SK, South central skunk RVV. (Reproduced with permission from X Ma et al: Rabies surveillance in the United States during 2021. *J Am Vet Med Assoc* 261:1045, 2023.)

Rabies virus variants isolated from humans or other mammalian species can be identified by reverse-transcription polymerase chain reaction (RT-PCR) amplification and sequencing or by characterization with monoclonal antibodies. These techniques are helpful in human cases with no known history of exposure. Worldwide, most human rabies (~99%) is transmitted from dogs in countries with endemic canine rabies and dog-to-dog transmission, and human cases can be imported by travelers returning from these regions. In North America, indigenously acquired human disease is usually associated with transmission from bats; there may be no known history of bat bite or other bat exposure in these cases. Most human cases are due to a bat rabies virus variant associated with silver-haired and tricolored bats and, less commonly, from Brazilian (Mexican) free-tail bats. These are small bats whose bite may not be recognized, and the virus has adapted for replication at skin temperature and in cell types that are present in the skin.

Transmission from nonbite exposures is relatively uncommon. Aerosols generated in the laboratory or in caves containing millions of Brazilian free-tail bats have rarely caused human rabies. Transmission has resulted from corneal transplantation and also from solid-organ transplantation and a vascular conduit (for a liver transplant) from undiagnosed donors with rabies in Texas, Florida, Germany, Kuwait, Syria, and China. Human-to-human transmission is extremely rare, although hypothetical concern about transmission to health care workers has prompted the implementation of barrier techniques to prevent exposures from patients with rabies.

■ PATHOGENESIS

The incubation period of rabies (defined as the interval between exposure and the onset of clinical disease) is usually 20–90 days, but in rare

cases is either as short as a few days or >1 year. During most of the incubation period, rabies virus is thought to be present at or close to the site of inoculation (Fig. 214-3). In muscles, the virus is known to bind to nicotinic acetylcholine receptors on postsynaptic membranes at neuromuscular junctions, but the exact details of viral entry into the skin and subcutaneous tissues have not yet been clarified. Rabies virus spreads centripetally along peripheral nerves toward the spinal cord or brainstem via retrograde fast axonal transport (rate, up to ~250 mm/d), with delays at intervals of ~12 h at each synapse. Once the virus enters the CNS, it rapidly disseminates to other regions of the CNS via fast axonal transport along neuroanatomic connections. Neurons are prominently infected in rabies; infection of astrocytes is unusual. After CNS infection becomes established, there is centrifugal spread along sensory and autonomic nerves to other tissues, including the salivary glands, heart, adrenal glands, and skin. Rabies virus replicates in acinar cells of the salivary glands and is secreted in the saliva of rabid animals that serve as vectors of the disease. There is no well-documented evidence for hematogenous spread of rabies virus.

Pathologic studies show mild inflammatory changes in the CNS in rabies, with mononuclear inflammatory infiltration in the leptomeninges, perivascular regions, and parenchyma, including microglial nodules called *Babes nodules*. Degenerative neuronal changes usually are not prominent, and there is little evidence of neuronal death; neuronophagia is observed occasionally. The pathologic changes are surprisingly mild in light of the clinical severity and fatal outcome of the disease. The most characteristic pathologic finding in rabies is the *Negri body* (Fig. 214-4). Negri bodies are eosinophilic cytoplasmic inclusions in brain neurons that are composed of rabies virus proteins and viral RNA. These inclusions occur in a minority of infected neurons, are commonly observed in Purkinje cells of the cerebellum

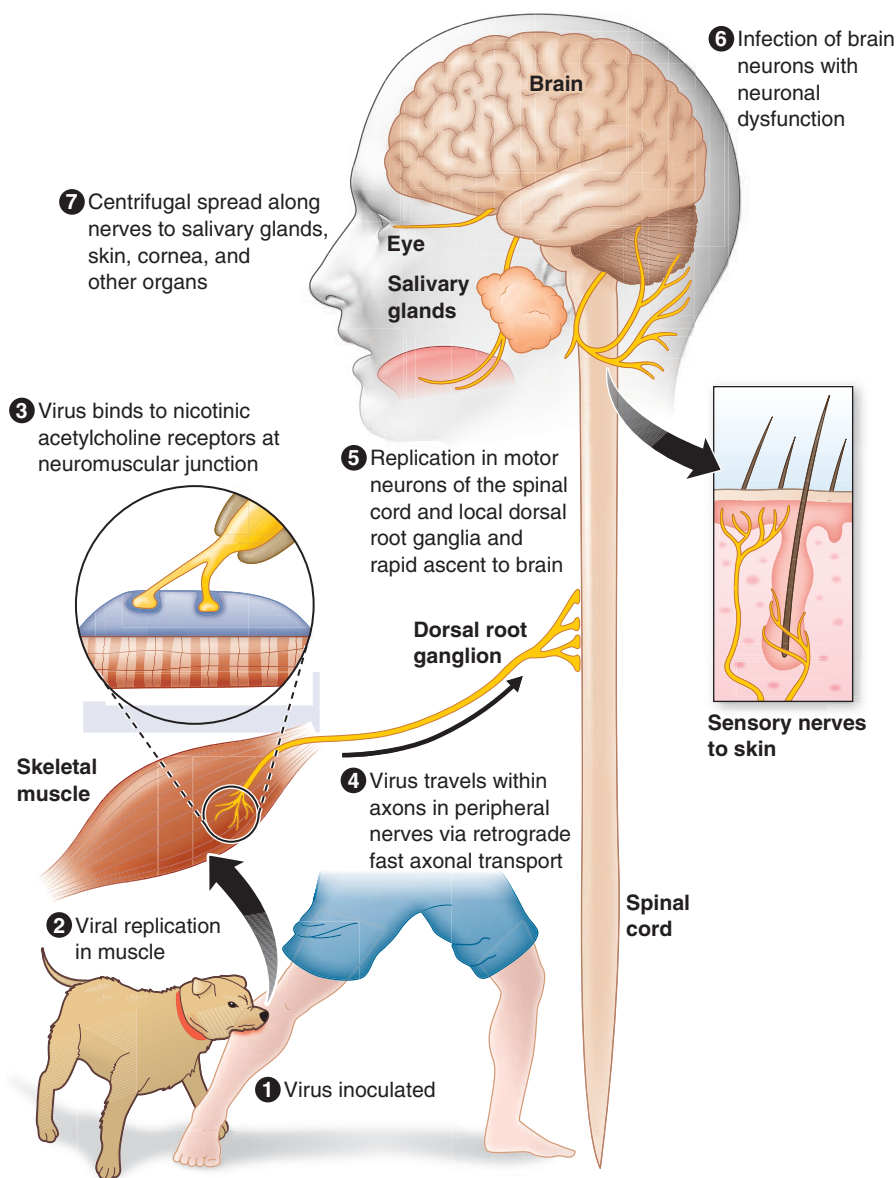


FIGURE 214-3 Schematic representation of events in rabies pathogenesis following peripheral inoculation of rabies virus by an animal bite. (Reproduced with permission from AC Jackson: *Pathogenesis, in Rabies: Scientific Basis of the Disease and Its Management*, 3rd ed. Oxford, UK, Elsevier Academic Press, 2013.)

and in pyramidal neurons of the hippocampus, and are less frequently seen in cortical and brainstem neurons. Negri bodies are not observed in all cases of rabies. The lack of prominent degenerative neuronal changes has led to the concept that neuronal dysfunction—rather than neuronal death—is responsible for clinical disease in rabies. Experimental studies indicate that oxidative stress due to mitochondrial dysfunction plays an important role. The basis for behavioral changes, including the aggressive behavior of rabid animals, is not well understood but may be related to infection of serotonergic neurons in the brainstem.

CLINICAL MANIFESTATIONS

For rabies prevention, the emphasis must be on postexposure prophylaxis (PEP) initiated after a recognized exposure and before any symptoms or signs develop. Rabies should usually be suspected on the basis of the clinical presentation with or without a history of an exposure. The disease generally presents as atypical encephalitis with relative preservation of consciousness. Rabies may be difficult to recognize late in the clinical course when progression to coma has occurred. A minority of patients (~20%) present with acute flaccid paralysis. There are prodromal, acute neurologic, and comatose phases that usually progress to death despite aggressive therapy ([Table 214-1](#)).

Prodromal Features The clinical features of rabies begin with nonspecific prodromal manifestations, including fever, malaise, headache, nausea, and vomiting. Anxiety or agitation also may occur. The earliest specific neurologic symptoms of rabies include paresthesias, pain, or pruritus near the site of the exposure, one or more of which occur in 50–80% of patients and strongly suggest rabies. The wound has usually healed by this point, and these symptoms probably reflect infection with associated inflammatory changes in local dorsal root or cranial sensory ganglia.

Encephalitic Rabies Two acute neurologic forms of rabies are seen in humans: the encephalitic (furious) form in 80% and the paralytic form in 20%. Some of the manifestations of encephalitic rabies, including fever, confusion, hallucinations, combativeness, and seizures, may be seen in other viral encephalitides as well. Autonomic dysfunction is common in rabies and may result in hypersalivation, gooseflesh, cardiac arrhythmia, and priapism. In encephalitic rabies, episodes of hyperexcitability are typically followed by periods of complete lucidity that become shorter as the disease progresses. Rabies encephalitis is distinguished by early brainstem involvement, which results in the classic features of *hydrophobia* (involuntary, painful contraction of the diaphragm and accessory respiratory, laryngeal, and pharyngeal muscles in response to swallowing liquids) ([Fig. 214-5](#)) and *aerophobia* (the same features caused by stimulation from a draft of air). These symptoms are probably due to dysfunction of infected brainstem neurons that normally inhibit inspiratory neurons near the nucleus ambiguus, resulting in exaggerated defense reflexes that protect the respiratory tract. The combination of hypersalivation and pharyngeal dysfunction is responsible for the classic appearance of “foaming at the mouth.” Brainstem dysfunction progresses rapidly, and coma—followed within days by death—is the rule unless the course is prolonged by supportive measures. With such measures, late complications can include cardiac

and/or respiratory failure, disturbances of water balance (syndrome of inappropriate antidiuretic hormone secretion or diabetes insipidus), noncardiogenic pulmonary edema, and gastrointestinal hemorrhage. Cardiac arrhythmias may be due to dysfunction affecting vital centers in the brainstem or autonomic pathways or to myocarditis. Multiple-organ failure is common in patients treated aggressively in critical care units.

Paralytic Rabies About 20% of patients have paralytic rabies in which muscle weakness predominates and cardinal features of encephalitic rabies (hyperexcitability, hydrophobia, and aerophobia) are lacking. There is early and prominent flaccid muscle weakness, often beginning in the bitten extremity and spreading to produce quadriplegia and facial weakness. Sphincter involvement is common, sensory involvement is usually mild, and these cases are commonly misdiagnosed as Guillain-Barré syndrome. Patients with paralytic rabies generally survive a few days longer than those with encephalitic rabies, but multiple-organ failure nevertheless ensues.

LABORATORY INVESTIGATIONS

Most routine laboratory tests in rabies yield normal results or show nonspecific abnormalities. Complete blood counts are usually normal. Examination of cerebrospinal fluid (CSF) often reveals mild

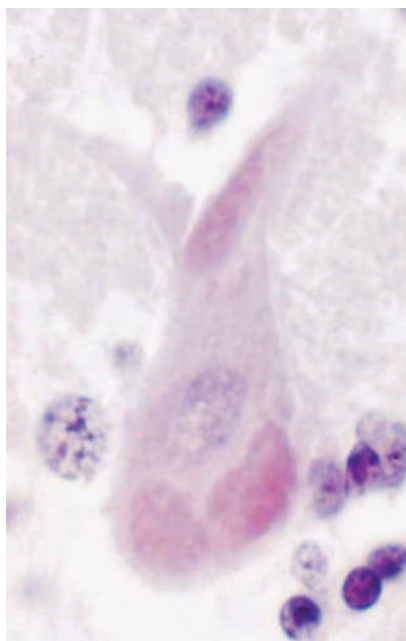


FIGURE 214-4 Three large Negri bodies in the cytoplasm of a cerebellar Purkinje cell from an 8-year-old boy who died of rabies after being bitten by a rabid dog in Mexico. (Reproduced with permission from AC Jackson, E Lopez-Corella. *N Engl J Med* 335:568, 1996; © Massachusetts Medical Society.)

mononuclear-cell pleocytosis with a mildly elevated protein level. Severe pleocytosis (>1000 white cells/ μL) is unusual and should prompt a search for an alternative diagnosis. Imaging is usually performed to exclude other diagnostic possibilities. Computed tomography (CT) head scans are usually normal in rabies. Magnetic resonance imaging (MRI) brain scans may show signal abnormalities in the brainstem or other gray-matter areas, but these findings are variable and nonspecific. Electroencephalograms typically show only nonspecific abnormalities. Of course, important tests in suspected cases of rabies include those that may identify an alternative, potentially treatable diagnosis (see “Differential Diagnosis,” below).

DIAGNOSIS

In North America, a diagnosis of rabies often is not considered until relatively late in the clinical course, even with a typical clinical presentation. This diagnosis should be considered in patients presenting with acute atypical encephalitis or acute flaccid paralysis, including those in whom Guillain-Barré syndrome is suspected. The absence of an animal-bite history is common in North America, particularly due to unrecognized bat exposures. The lack of hydrophobia is not unusual in rabies. Once rabies is suspected, rabies-specific laboratory tests should be performed to confirm the diagnosis. Diagnostically

TABLE 214-1 Clinical Stages of Rabies		
STAGE	TYPICAL DURATION	SYMPTOMS AND SIGNS
Incubation period	20–90 days	None
Prodrome	2–10 days	Fever, malaise, anorexia, nausea, vomiting; paresthesias, pain, or pruritus at the wound site
Acute Neurologic Disease		
Encephalitic (80%)	2–7 days	Anxiety, agitation, hyperactivity, bizarre behavior, hallucinations, autonomic dysfunction, hydrophobia
Paralytic (20%)	2–10 days	Flaccid paralysis in limb(s) progressing to quadriplegia with facial paralysis
Coma, death ^a	0–14 days	

^aRecovery is rare.

Source: Adapted from MAW Hattwick: Rabies virus, in *Principles and Practice of Infectious Diseases*, GL Mandell et al (eds). New York, Wiley, 1979.



FIGURE 214-5 Hydrophobic spasm of inspiratory muscles associated with terror in a patient with encephalitic (furious) rabies who is attempting to swallow water. (Copyright DA Warrell, Oxford, UK; with permission.)

useful specimens include serum, CSF, fresh saliva, skin biopsy samples from the neck, and brain tissue (rarely obtained before death). Because skin biopsy relies on the demonstration of rabies virus antigen and/or RNA in cutaneous nerves at the base of hair follicles, samples are usually taken from hairy skin at the nape of the neck. Corneal impression smears are of low diagnostic yield and are generally not performed. Negative antemortem rabies-specific laboratory tests never exclude a diagnosis of rabies, and tests may need to be repeated after an interval for diagnostic confirmation.

Rabies Virus–Specific Antibodies In a previously unimmunized patient, serum neutralizing antibodies to rabies virus are diagnostic. However, because rabies virus infects immunologically privileged neuronal tissues, serum antibodies may not develop until late in the disease. Antibodies may be detected within a few days after the onset of symptoms, but some patients die without detectable antibodies. The presence of rabies virus–specific neutralizing antibodies in the CSF suggests rabies encephalitis, regardless of immunization status. A diagnosis of rabies is questionable in patients who recover from their illness without developing serum neutralizing antibodies to rabies virus.

RT-PCR Amplification Detection of rabies virus RNA by RT-PCR is highly sensitive and specific. This technique can detect virus in fresh saliva samples, skin biopsy specimens, CSF (less sensitive), and brain tissues. In addition, RT-PCR with genetic sequencing can distinguish among rabies virus variants, permitting identification of the probable source of an infection.

Direct Fluorescent Antibody Testing Direct fluorescent antibody (DFA) testing with rabies virus antibodies conjugated to fluorescent dyes is highly sensitive and specific for the detection of rabies virus antigen in tissues; the test can be performed quickly and applied to skin biopsy and brain tissue samples. In skin biopsy samples, rabies virus antigen may be detected in cutaneous nerves at the base of hair follicles.

DIFFERENTIAL DIAGNOSIS

The diagnosis of rabies may be difficult without a history of animal exposure, and no exposure to an animal (e.g., a bat) may be recalled. The presentation of rabies is usually quite different from that of acute

viral encephalitis due to most other causes, including herpes simplex encephalitis and arboviral (e.g., West Nile) encephalitis. Early neurologic symptoms may occur at the site of the bite, and there may be early features of brainstem involvement with preservation of consciousness. Anti-N-methyl-D-aspartate receptor (anti-NMDA) encephalitis occurs in young patients (especially females) and is characterized by behavioral changes, autonomic instability, hypoventilation, and seizures. Many other antibodies are also associated with autoimmune encephalitis. Postinfectious (immune-mediated) encephalomyelitis may follow influenza, measles, mumps, and other infections; it may also occur as a sequela of immunization with rabies vaccines derived from neural tissues, which are now infrequently used and only in resource-limited and resource-poor countries. Rabies may present with unusual neuropsychiatric symptoms and may be misdiagnosed as a psychiatric disorder. Rabies hysteria (now classified as a somatic symptom disorder) may occur as a psychological response to the fear of rabies and is often characterized by a shorter incubation period than rabies, aggressive behavior, inability to communicate, and a long course with recovery.

As previously mentioned, paralytic rabies may mimic Guillain-Barré syndrome. In these cases, fever, bladder dysfunction, a normal sensory examination, and CSF pleocytosis favor a diagnosis of rabies. Conversely, Guillain-Barré syndrome may occur as a complication of rabies vaccination with a neural tissue-derived product (e.g., suckling mouse brain vaccine) and may be mistaken for paralytic rabies (i.e., vaccine failure).

TREATMENT

Rabies

There is no established treatment for rabies. Aggressive management with supportive care in critical care units has resulted in the survival of at least 33 patients with rabies. Many of these survivors have recently been reported from India. There have been many recent treatment failures (>60) with the combination of therapeutic (induced) coma, antiviral drugs, and sometimes ketamine—measures that were used in a healthy survivor in whom neutralizing antibodies to rabies virus were detected at presentation. No antiviral therapy has demonstrated any efficacy in the treatment of rabies. Expert opinion is recommended before a course of experimental therapy is embarked upon. A palliative approach may be appropriate for many patients who are not considered candidates for aggressive management.

PROGNOSIS

Rabies is an almost uniformly fatal disease but is nearly always preventable after recognized exposures with appropriate postexposure therapy during the early incubation period (see below). All but 2 of 33 documented survivors of rabies received one or more doses of rabies vaccine before disease onset. The survivors who had not received vaccine had neutralizing antibodies to rabies virus in CSF at clinical presentation. Most patients with rabies die within several days of the onset of illness, despite aggressive care in a critical care unit.

PREVENTION

Postexposure Prophylaxis Since there is no effective therapy for rabies, it is extremely important to prevent the disease after an animal exposure. **Figure 214-6** shows the steps involved in making decisions about PEP. Based on the exposure history and local epidemiologic information, the physician must decide whether initiation of PEP is warranted. Healthy dogs, cats, or ferrets may be confined and observed for 10 days. PEP is not necessary if the animal remains healthy. If the animal develops signs of rabies during the observation period, it should be euthanized immediately; the head should be transported to the laboratory under refrigeration, rabies virus should be sought by DFA testing, and viral isolation should be attempted by cell culture and/or mouse inoculation. Any animal other than a dog, cat, or ferret should be euthanized immediately and the head submitted for laboratory

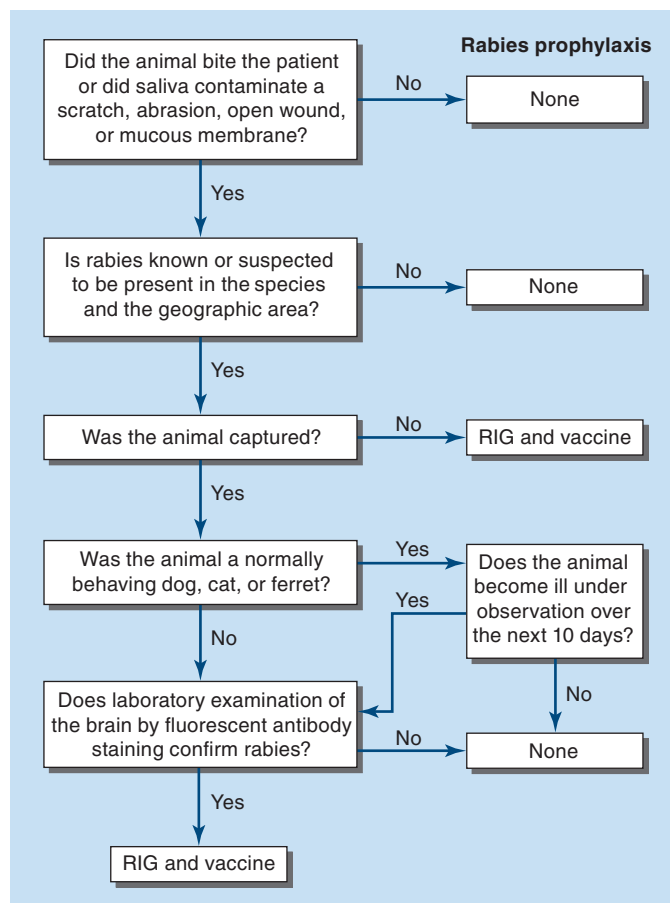


FIGURE 214-6 Algorithm for rabies postexposure prophylaxis. RIG, rabies immune globulin. (From L. Corey, in *Harrison's Principles of Internal Medicine*, 15th ed. E Braunwald et al [eds]: New York, McGraw-Hill, 2001.)

examination. In high-risk exposures and in areas where canine rabies is endemic, rabies prophylaxis should be initiated without waiting for laboratory results. If the laboratory results prove to be negative, it may safely be concluded that the animal's saliva did not contain rabies virus, and immunization should be discontinued. If an animal escapes after an exposure, it must be considered rabid, and PEP must be initiated unless information from public health officials indicates otherwise (i.e., there is no endemic rabies in the area). Although controversial, the use of PEP may be warranted when a person (e.g., a small child or a sleeping adult) has been present in the same space as a bat and an unrecognized bite cannot be reliably excluded.

PEP includes local wound care and both active and passive immunization. It is important that current recommendations are followed very closely because minor deviations can lead to failure of prophylactic measures. Local wound care is essential and may greatly decrease the risk of rabies virus infection. Wound care should not be delayed, even if the initiation of immunization is postponed pending the results of the 10-day observation period. All bite wounds and scratches should be washed thoroughly with soap and water. Devitalized tissues should be debrided, tetanus prophylaxis given, and antibiotic treatment initiated whenever indicated.

Previously unvaccinated persons (but not those who have previously been immunized) should be passively immunized with rabies immune globulin (RIG). If RIG is not immediately available, it should be administered no later than 7 days after the first vaccine dose. After day 7, endogenous antibodies are being produced, and passive immunization may actually be counterproductive. If anatomically feasible, the entire dose of RIG (20 IU/kg) should be infiltrated at the site of the bite, and any RIG remaining after infiltration of the bite site should be administered IM at a distant site. Recent recommendations by the World Health Organization indicate that under certain circumstances the remainder of the dose does not need to be administered after local

infiltration of the wound(s). With multiple or large wounds, the RIG preparation may need to be diluted in order to obtain a sufficient volume for adequate infiltration of all wound sites. If the exposure involves a mucous membrane, the entire dose should be administered IM. Rabies vaccine and RIG should never be administered at the same site or with the same syringe. Commercially available RIG in the United States is purified from the serum of hyperimmunized human donors. These human RIG preparations are much better tolerated than are the equine-derived preparations still in use in some countries (see below). Serious adverse effects of human RIG are uncommon. Local pain and low-grade fever may occur.

Two purified inactivated rabies vaccines are available for rabies PEP in the United States. They are highly immunogenic and remarkably safe compared with earlier vaccines. Four 1-mL doses of rabies vaccine should be given IM in the deltoid area. (The anterolateral aspect of the thigh also is acceptable in children.) Gluteal injections, which may not always reach muscle, should not be given and have been associated with rare vaccine failures. Ideally, the first dose should be given as soon as possible after exposure; failing that, it should be given without further delay. The three additional doses should be given on days 3, 7, and 14; a fifth dose on day 28 is no longer recommended except in immunocompromised patients. Pregnancy is not a contraindication for immunization. Glucocorticoids and other immunosuppressive medications may interfere with the development of active immunity and should not be administered during PEP unless they are essential. Routine measurement of serum neutralizing antibody titers is not required, but titers should be measured 2–4 weeks after immunization in immunocompromised persons. Local reactions (pain, erythema, edema, and pruritus) and mild systemic reactions (fever, myalgias, headache, and nausea) are common; anti-inflammatory and antipyretic medications may be used, but immunization should not be discontinued. Systemic allergic reactions are uncommon, but anaphylaxis does occur rarely and can be treated with epinephrine and antihistamines. The risk of rabies development should be carefully considered before the decision is made to discontinue vaccination because of an adverse reaction.

Most of the burden of rabies PEP is borne by persons with the fewest resources. In addition to the rabies vaccines discussed above, vaccines grown in either primary cell lines (hamster or dog kidney) or continuous cell lines (Vero cells) are satisfactory and are available in many countries outside the United States. Less expensive vaccines derived from neural tissues are still used in a diminishing number of developing countries; however, these vaccines are associated with serious neuroparalytic complications, including postinfectious encephalomyelitis and Guillain-Barré syndrome. The use of these vaccines should be discontinued as soon as possible, and progress has been made in this regard. Worldwide, more than 10 million individuals receive postexposure rabies vaccine each year.

If human RIG is unavailable, purified equine RIG can be used in the same manner at a dose of 40 IU/kg. The incidence of anaphylactic reactions and serum sickness has been low with recent equine RIG products.

Preexposure Rabies Vaccination Preexposure rabies prophylaxis should be considered for people with an occupational or recreational risk of rabies exposures and also for certain travelers to rabies-endemic areas. The primary schedule consists of two doses of rabies vaccine given on days 0 and 7. Serum neutralizing antibody tests help determine the need for subsequent booster doses. When a previously immunized individual is exposed to rabies, two booster doses of vaccine should be administered on days 0 and 3. Wound care remains essential. As stated above, RIG should not be administered to previously vaccinated persons.

OTHER RHABDOVIRUSES

■ OTHER LYSSAVIRUSES

A growing number of lyssaviruses other than rabies virus have been discovered to infect bat populations in Europe, Africa, Asia, and Australia. Six of these viruses have produced a very small number of cases of a human disease indistinguishable from rabies, including European bat lyssaviruses 1 and 2, Australian bat lyssavirus, Irkut virus,

and Duvenhage virus. Mokola virus, a lyssavirus that has been isolated from shrews with an unknown reservoir species in Africa, may also produce human disease indistinguishable from rabies.

■ VESICULAR STOMATITIS VIRUS

Vesicular stomatitis is a viral disease of cattle, horses, pigs, and some wild mammals. Vesicular stomatitis virus is a member of the genus *Vesiculovirus* in the family *Rhabdoviridae*. Outbreaks of vesicular stomatitis in horses and cattle occur sporadically in the southwestern United States. The animal infection is associated with severe vesiculation and ulceration of oral tissues, teats, and feet and may be clinically indistinguishable from the more dangerous foot-and-mouth disease. Epidemics are usually seasonal, typically beginning in the late spring, and are probably due to arthropod vectors. Direct animal-to-animal spread can also occur, although the virus cannot penetrate intact skin. Transmission to humans usually results from direct contact with infected animals (particularly cattle) and occasionally follows laboratory exposure. In human disease, early conjunctivitis is followed by an acute influenza-like illness with fever, chills, nausea, vomiting, headache, retrobulbar pain, myalgias, substernal pain, malaise, pharyngitis, and lymphadenitis. Small vesicular lesions may be present on the buccal mucosa or on the fingers. Encephalitis is very rare. The illness usually lasts 3–6 days, with complete recovery. Subclinical infections are common. A serologic diagnosis can be made on the basis of a rise in titer of complement-fixing or neutralizing antibodies. Therapy is symptom-based.

■ FURTHER READING

- FOOKS AR et al: Current status of rabies and prospects for elimination. *Lancet* 384:1389, 2014.
- FOOKS AR, JACKSON AC (eds): *Rabies: Scientific Basis of the Disease and Its Management*, 4th ed. London, Elsevier Academic Press, 2020.
- JACKSON AC: Treatment of rabies. In: Post TW, ed. *UpToDate*. Waltham, Massachusetts: Wolters Kluwer, 2023. www.uptodate.com.
- LETCHWORTH GJ et al: Vesicular stomatitis. *Vet J* 157:239, 1999.
- MANNING SE et al: Human rabies prevention—United States, 2008: Recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep* 57:1, 2008.
- WORLD HEALTH ORGANIZATION: *WHO Expert Consultation on Rabies: Third Report* (WHO Technical Report Series No. 1012). Geneva, World Health Organization, 2018. Available at <https://iris.who.int/bitstream/handle/10665/272364/9789241210218-eng.pdf>. Accessed September 20, 2024.

215

Arthropod-Borne and Rodent-Borne Virus Infections

Jens H. Kuhn, Ian Crozier



This chapter summarizes the major features of selected arthropod-borne and rodent-borne viruses and associated infections and/or disease. Numerous viruses of this category are spread in nature among animals without ever infecting humans. Other viruses incidentally infect humans, with few causing disease. Some viruses are regularly introduced into the human population by arthropods (specifically, insects and ticks) or by chronically infected rodents.

These zoonotic viruses are taxonomically diverse and therefore differ fundamentally in virion morphology, replication strategy, genomic organization, and genome sequence. Although a virus's classification in a taxon is enlightening regarding natural maintenance strategies, sensitivity to antiviral agents, and aspects of pathogenesis, it does not

necessarily predict which clinical symptoms and signs (if any) the viral infection will cause in humans. Zoonotic viruses are evolving, and “new” zoonotic viruses are regularly discovered. The epizootiology and epidemiology of zoonotic viruses continue to change because of environmental alterations affecting vectors, reservoirs, and hosts, including wildlife, livestock, and humans. Zoonotic viruses are most numerous in the tropics but are also found in temperate and frigid climates. The distribution and seasonal activity of zoonotic viruses may vary in a manner likely to depend largely on ecologic conditions (e.g., rainfall and temperature), which can affect the density of virus vectors and reservoirs, likelihood of exposure, and development of infection.

Arthropod-borne viruses (arboviruses) infect their vectors after ingestion of blood meals from viremic, usually nonhuman vertebrates; some infections may be the result of saliva-activated transmission. Arthropod vectors develop chronic systemic infection as the viruses penetrate their guts and spread throughout their bodies to their salivary glands; such virus dissemination, referred to as *extrinsic incubation*, typically lasts 1–3 weeks in mosquitoes. If the salivary glands become involved, the arthropod vectors are competent to continue the chain of transmission by infecting vertebrates during subsequent blood meals. Alternatively, virus maintenance in arthropod vectors may be achieved through *transovarial transmission* to progeny. Generally, the arthropods are unharmed by the infections and natural vertebrate partners have only transient viremia without overt disease.

Rodent-borne viruses are maintained in nature by transmission among rodents, which become chronically infected. A high degree of rodent–virus specificity may be observed, and overt disease in the reservoir hosts is rare.

ETIOLOGY

Arthropod-borne and rodent-borne zoonotic viruses belong mostly to the classes *Alsuviricetes* (family *Togaviridae*), *Bunyaviricetes* (families *Arenaviridae*, *Hantaviridae*, *Nairoviridae*, *Peribunyaviridae*, and *Phenuiviridae*), *Flasuviricetes* (family *Flaviviridae*), *Insthoviricetes* (family *Orthomyxoviridae*), *Monjiviricetes* (families *Bornaviridae* and *Rhabdoviridae*), and *Resentoviricetes* (families *Sedoreoviridae* and *Spinareoviridae*) (Table 215-1). An exception is Syr-Darya Valley fever virus, an ixodid tick-borne cardiavirus (*Pisoniviricetes: Picornaviridae*) that causes febrile disease and has been found in Central Asia.

■ ALSUVIRICETES: TOGAVIRIDAE

Members of the family *Togaviridae* have linear, positive-sense RNA genomes (≈9.7–11.8 kb) and form enveloped icosahedral virions (≈60–70 nm in diameter) that bud from the plasma membrane of the infected cell. The togavirids discussed in this chapter are all members of the genus *Alphavirus* and are transmitted to vertebrates by mosquitoes.

■ BUNYAVIRICETES: ARENAVIRIDAE

Members of the family *Arenaviridae* that infect humans are all assigned to the genus *Mammarenavirus*. Mammarenaviruses form spherical, oval, or pleomorphic enveloped and spiked virions (≈50–300 nm in diameter) that bud from the plasma membrane of the infected cell. The particles contain two genomic single-stranded RNAs (small [S], ≈3.5 kb, and large [L], ≈7.5 kb), encoding structural proteins in an ambisense orientation. Most mammarenaviruses persist by chronically infecting rodents. Human mammarenaviruses are maintained by muroid rodents that are often continuously viremic and commonly transmit viruses vertically and horizontally. One mammarenavirus associated with human infections is maintained by shrews. Strikingly, each mammarenavirus is predominantly adapted to one particular type of rodent. Humans usually become infected through inhalation of or direct contact with infected rodent excreta or secretions (e.g., from aerosolized whole rodents in harvesting machines or aerosolized dried urine or feces when sweeping floors in barns or houses) and direct contact with rodents in traps. Person-to-person transmission of mammarenaviruses occurs but is uncommon.

■ BUNYAVIRICETES: HANTAVIRIDAE, NAIROVIRIDAE, PERIBUNYAVIRIDAE, AND PHENUIVIRIDAE

Members of these families that infect humans form spherical to pleomorphic enveloped virions containing three genomic RNAs

(S, ≈1–2 kb; medium [M], 3.6–5.3 kb; and L, 6.4–12.3 kb) of negative-sense (hantavirids, nairovirids, and peribunyavirids) or negative-sense or ambisense polarities (phenuivirids). These bunyaviricetes mature into particles ≈80–120 nm in diameter in the Golgi complexes of infected cells and exit these cells by exocytosis.

Hantavirids that infect humans are classified in the genus *Orthohantavirus* and are maintained by muroid rodents that chronically shed virions. As with mammarenaviruses, individual orthohantaviruses are usually specifically adapted to a particular type of rodent. However, orthohantaviruses do not cause chronic viremia in their rodent hosts and are transmitted only horizontally from rodent to rodent. Also like mammarenaviruses, orthohantaviruses infect humans primarily through inhalation of or direct contact with rodent excreta or secretions, and person-to-person transmission occurs but is not common (with the notable exception of Andes virus).

Nairovirids that infect humans are classified in the genus *Orthonairovirus*. Orthonairoviruses are predominantly maintained by ixodid ticks, which transmit these viruses vertically (transovarially and transstadially) to progeny and horizontally through contact with viremic vertebrate hosts and subsequent transmission to healthy vertebrates. Humans are usually infected via a tick bite or during handling of infected vertebrates.

Peribunyavirids of one genus (*Orthobunyavirus*) infect humans. Orthobunyaviruses are primarily mosquito-borne (and, in rare circumstances, transmitted by midges or sandflies) and have viremic vertebrate intermediate hosts. Many orthobunyaviruses are transmitted vertically (transovarially) to their mosquito hosts. Numerous orthobunyaviruses have been associated with human infection and disease.

Phenuivirids are transmitted vertically (transovarially) in their arthropod vectors and horizontally through viremic vertebrate hosts. Human phenuivirids are found in four genera: *Bandavirus*, *Phlebovirus*, *Tanzavirus*, and *Uukuvirus*. Bandaviruses and uukuviruses are transmitted by ticks, whereas human phleboviruses are transmitted by sandflies (the ecology of tanzaviruses is unclear).

■ FLASUVIRICETES: FLAVIVIRIDAE

The family *Flaviviridae* currently includes only one genus (*Orthoflavivirus*) that comprises arthropod-borne human viruses. Orthoflaviviruses have single-stranded positive-sense RNA genomes (≈11 kb) and form spherical enveloped particles or virions (40–60 nm in diameter). The orthoflaviviruses discussed in this chapter belong to two phylogenetically and antigenically distinct groups that are transmitted among vertebrates by mosquitoes (generally, *Aedes* and *Culex* spp.) and, typically, ixodid ticks, respectively. Vectors are usually infected when they feed on viremic hosts. As in the case of most other viruses discussed in this chapter, humans are accidental hosts, typically infected by arthropod bites. Arthropods maintain orthoflavivirus infections horizontally, although vertical (transovarial) transmission has been documented. Under some circumstances, orthoflaviviruses can be transmitted by aerosol or via contaminated food products; in particular, raw milk can transmit tick-borne encephalitis virus.

■ INSTHOVIRICETES: ORTHOMYXOVIRIDAE

The family *Orthomyxoviridae* includes two genera of medically relevant arthropod-borne viruses: *Quarantavirus* and *Thogotovirus*. Quarantaviruses are transmitted among birds by argasid ticks, whereas thogotoviruses have a predilection for mammalian host reservoirs and can be transmitted by argasid and ixodid ticks and mosquitoes.

■ MONJIVIRICETES: BORNAVIRIDAE AND RHABDOVIRIDAE

Bornavirids and rhabdovirids that infect humans have linear non-segmented, negative-sense RNA genomes (bornavirids, ≈9 kb; rhabdovirids, ≈11–15 kb) and form spherical (bornavirids, 90–130 nm in diameter) and bullet-shaped to pleomorphic enveloped particles (rhabdovirids, 100–430 nm long and 45–100 nm wide). The only known rodent-borne bornavirid that infects humans is a member of genus *Orthobornavirus* and is transmitted by squirrels. The rhabdovirid genus *Tibrovirus* contains several uncharacterized human pathogens

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
<i>Alphavirus</i> (family <i>Togaviridae</i>)	Barmah Forest virus (BFV)	Cattle, horses, marsupials?	Biting midges (<i>Culicoides marksii</i>), mosquitoes (<i>Aedes camptorhynchus</i> , <i>Ae. normanensis</i> , <i>Ae. notoscriptus</i> , <i>Ae. vigilax</i> , <i>Culex annulirostris</i>)	A&R
	Chikungunya virus (CHIKV)	Bats, nonhuman primates	Mosquitoes (predominantly <i>Aedes aegypti</i> and <i>Ae. Albopictus</i>)	A&R ^c
	Eastern equine encephalitis virus (EEEV)	Freshwater swamp passeriform birds, but also opportunistic amphibians, other birds (emu, gallinaceous poultry, pheasants), reptiles, and mammals (goats, horses, pigs, rodents)	Mosquitoes (<i>Aedes</i> , <i>Coquillettidia</i> , <i>Culex</i> spp., <i>Anopheles quadrimaculatus</i> , <i>Culiseta melanura</i> , <i>Mansonia perturbans</i> , <i>Psorophora</i> spp.)	E
	Everglades virus (EVEV)	Cotton deermice (<i>Peromyscus gossypinus</i>), hispid cotton rats (<i>Sigmodon hispidus</i>)	Mosquitoes (<i>Culex cedecei</i>)	F&M, E
	Madariaga virus (MADV)	Likely bats, marsupials, reptiles, rodents	Mosquitoes (<i>Culex</i> , <i>Culiseta</i> spp.)	F&M, E
	Mayaro virus (MAYV)	Nonhuman primates, possums, rodents; possibly caimans, horses, sheep	Mosquitoes (predominantly <i>Haemagogus</i> spp., but also <i>Aedes</i> , <i>Culex</i> , <i>Mansonia</i> , <i>Psorophora</i> , <i>Sabethes</i> spp.)	A&R
	Mucambo virus (MUCV)	Nonhuman primates, rodents	Mosquitoes (<i>Culex</i> , <i>Ochlerotatus</i> spp.)	F&M, E
	O'nyong-nyong virus ^d (ONNV)	Unknown	Mosquitoes (in particular <i>Anopheles gambiae</i> , <i>A. funestus</i> , <i>Mansonia</i> spp.)	A&R
	Ross River virus (RRV)	Marsupials, rodents	Mosquitoes (<i>Aedes normanensis</i> , <i>Ae. vigilax</i> , <i>Culex annulirostris</i>)	A&R
	Semliki Forest virus (SFV)	Birds, rodents	Mosquitoes (<i>Aedes</i> , <i>Culex</i> spp.)	A&R
	Sindbis virus ^e (SINV)	Typically birds, but also frogs and rats	Typically mosquitoes (<i>Culex</i> , <i>Culiseta</i> spp.), but tick isolation has been reported	A&R
	Tonate virus (TONV)	Birds, Suriname crested oropendolas (<i>Psarocolius decumanus</i>)	Mosquitoes (<i>Anopheles</i> , <i>Coquillettidia</i> , <i>Culex</i> , <i>Mansonia</i> , <i>Uranotaenia</i> , <i>Wyeomyia</i> spp.), sandflies (<i>Lutzomyia</i> spp.)	F&M, E
	Una virus (UNAV)	Birds, horses, nonhuman primates, rodents	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Coquillettidia</i> , <i>Culex</i> , <i>Ochlerotatus</i> , <i>Psorophora</i> spp.)	F&M
	Venezuelan equine encephalitis virus (VEEV)	Equids, rodents	Mosquitoes (<i>Aedes</i> , <i>Culex</i> spp., <i>Psorophora confinnis</i>)	F&M, E
	Western equine encephalitis virus (WEEV)	Equids, lagomorphs, passeriform birds, pheasants	Mosquitoes (<i>Aedes</i> spp., <i>Culex tarsalis</i> , <i>Culiseta</i> spp.)	E
<i>Bandavirus</i> (family <i>Phenuiviridae</i>)	Bhanja virus ^f (BHAV)	Cattle, four-toed hedgehog (<i>Atelerix albiventris</i>), goats, sheep, striped ground squirrels (<i>Xerus erythropus</i>)	Ixodid ticks (<i>Amblyomma</i> , <i>Dermacentor</i> , <i>Haemaphysalis</i> , <i>Hyalomma</i> , <i>Rhipicephalus</i> spp.)	F&M, E
	Heartland virus (HRTV)	Cattle, deer, elk, goats, raccoons, sheep?	Ixodid ticks (<i>Amblyomma americanum</i>)	F&M
	Severe fever with thrombocytopenia syndrome virus ^g (SFTSV)	Camels, cats, cattle, chickens, dogs, goats, hedgehogs, pigs, rodents, sheep?	Ixodid ticks (predominantly <i>Haemaphysalis longicornis</i> , but also <i>Amblyomma testudinarium</i> , <i>H. concinna</i> , <i>H. flava</i> , <i>Ixodes nipponensis</i> , <i>Rhipicephalus microplus</i>)	F&M, VHF
<i>Coltivirus</i> (family <i>Spinareoviridae</i>)	Colorado tick fever virus (CTFV)	Predominantly golden-mantled ground squirrels (<i>Spermophilus lateralis</i>), but also bushy-tailed woodrats (<i>Neotoma cinerea</i>), Columbian ground squirrels (<i>Spermophilus columbianus</i>), eastern deermice (<i>Peromyscus maniculatus</i>), least chipmunks (<i>Tamias minimus</i>), North American porcupines (<i>Erethizon dorsata</i>), Richardson ground squirrel (<i>Spermophilus richardsonii</i>), Uinta chipmunks (<i>Tamias umbrinus</i>), yellow-pine chipmunks (<i>Tamias amoenus</i>) among lesser important mammal hosts	Ixodid ticks (<i>Dermacentor andersoni</i> , possibly <i>D. occidentalis</i>)	F&M, E
	Eyach virus (EYAV)	Lagomorphs, rodents	Ixodid ticks (<i>Ixodes ricinus</i> , <i>I. ventralis</i>)	F&M, E
	Salmon River virus (SRV)	Unknown	Ixodid ticks (<i>Ixodes</i> spp.)	F&M, E
	Chapare virus (CHAPV)	Unidentified cricetid rodents	None	VHF
<i>Mammarenavirus</i> (family <i>Arenaviridae</i>)	Flexal virus (FLEV)	Unidentified cricetid rodents	None	(F&M)
	Guanarito virus (GTOV)	Predominantly short-tailed zygodonts (<i>Zygodontomys brevicauda</i>) but also Alston's cotton rats (<i>Sigmodon alstoni</i>)	None	VHF

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
	Junín virus (JUNV)	Predominantly drylands lauchas (<i>Calomys musculus</i>) but also Azara's akodonts (<i>Akodon azarae</i>) and little lauchas (<i>Calomys laucha</i>)	None	VHF
	Lassa virus (LASV)	Predominantly Natal mastomys (<i>Mastomys natalensis</i>), but also reddish-white mastomys (<i>M. erythroleucus</i>), African hylomyscus (<i>Hylomyscus pamfi</i>), and Baoule's mice (<i>Mus baoulei</i>)	None	F&M, VHF
	Lymphocytic choriomeningitis virus (LCMV)	Predominantly house mice (<i>Mus musculus</i>), but also murid long-tailed field mice (<i>Apodemus sylvaticus</i>), soft-furred mice (<i>Praomys</i> spp.), and golden hamsters (<i>Mesocricetus auratus</i>)	None	F&M, E, (VHF)
	Lujo virus (LUJV)	Unknown	None	VHF
	Machupo virus (MACV)	Big lauchas (<i>Calomys callosus</i>)	None	VHF
	Sabiá virus (SBAV)	Unidentified cricetid rodents	None	VHF
	Whitewater Arroyo virus (WWAV)	White-throated woodrats (<i>Neotoma albigula</i>)	None	(E)
<i>Orbivirus</i> (family <i>Sedoreoviridae</i>)	Kemerovo virus (KEMV)	Birds, rodents	Ixodid ticks (<i>Ixodes persulcatus</i>)	F&M, E
	Lebombo virus (LEBV)	Unknown	Mosquitoes (<i>Aedes</i> , <i>Mansonia</i> spp.)	F&M
	Orungo virus (ORUV)	Camels, cattle, goats, nonhuman primates, sheep	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> spp.)	F&M, E
	Tribeč virus (TRBV) ^m	Bank voles (<i>Myodes glareolus</i>), birds, common pine voles (<i>Microtus subterraneus</i>), goats, hares	Ixodid ticks (<i>Ixodes persulcatus</i> , <i>I. ricinus</i>)	F&M
<i>Orthobornavirus</i> (family <i>Bornaviridae</i>)	Variegated squirrel bornavirus 1 (VSBV1)	Finlayson's squirrel (<i>Callosciurus finlaysonii</i>), Prevost's squirrels (<i>Callosciurus prevostii</i>), Swinhoe's striped squirrel (<i>Tamias swinhoei</i>), variegated squirrels (<i>Sciurus variegatoides</i>)	None	E
<i>Orthobunyavirus</i> (family <i>Peribunyaviridae</i>)	Apeú virus (APEUV)	Bare-tailed woolly opossums (<i>Caluromys philander</i>) and other opossums; rodents; black howlers (<i>Alouatta caraya</i>), tufted capuchins (<i>Cebus apella</i>)	Mosquitoes (<i>Aedes</i> , <i>Culex</i> spp.)	F&M
	Bangui virus (BGIV)	Unknown	Unknown	F&M
	Batai virus (BATV) ⁿ	Birds, camels, cattle, goats, rodents, sheep	Mosquitoes (<i>Aedes abnormalis</i> , <i>A. curtipes</i> , <i>Anopheles barbirostris</i> , <i>A. maculipennis</i> , <i>A. phillippinensis</i> , <i>Culex gelidus</i> , other spp.)	F&M
	Bunyamwera virus (BUNV)	Birds, cows, goats, horses, sheep	Mosquitoes (<i>Aedes</i> spp.)	F&M
	Bwamba virus (BWAV)	Unknown	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Mansonia</i> spp.)	F&M
	Cache Valley virus (CVV)	Cattle, deer, dogs, foxes, groundhogs, horses, nonhuman primates, pigs, rabbits, raccoons, rodents	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Coquillettia</i> , <i>Culiseta</i> , <i>Psorophora</i> spp.)	F&M, E
	California encephalitis virus (CEV)	Lagomorphs, rodents	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Culiseta</i> , <i>Psorophora</i> spp.)	F&M, E
	Caraparú virus (CARV)	Rodents, tufted capuchins (<i>C. apella</i>)	Mosquitoes (<i>Culex</i> spp.)	F&M
	Catú virus (CATUV)	Bats, capuchins (<i>Cebus</i> spp.), opossums, rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Cristoli virus (/)	Unknown	Mosquitoes?	E
	Fort Sherman virus (FSV)	Cattle, goats, horses, sheep?	Mosquitoes?	F&M
	Gan Gan virus (GGV)	Unknown	Mosquitoes (<i>Aedes</i> , <i>Culex</i> spp.)	A&R
	Germiston virus (GERV)	Rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Guamá virus (GMAV)	Bats, capuchins (<i>Cebus</i> spp.), howlers (<i>Alouatta</i> spp.), marsupials, rodents	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Limatus</i> , <i>Mansonia</i> , <i>Psorophora</i> , <i>Trichoprosopon</i> spp.)	F&M
	Guaroa virus (GROV)	Unknown	Mosquitoes (<i>Anopheles</i> spp.)	F&M
	Ilesha virus (ILEV)	Unknown	Mosquitoes (<i>Anopheles gambiae</i>)	F&M, (VHF)
	Inkoo virus (INKV)	Cattle, foxes, hares, moose, reindeer, rodents	Mosquitoes (<i>Aedes</i> spp.)	F&M, E

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
	Iquitos virus (IQTV)	Unknown	Unknown	F&M
	Itaqui virus (ITQV)	Capuchins (<i>Cebus</i> spp.), opossums, rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Jamestown Canyon virus (JCV)	Predominantly white-tailed deer (<i>Odocoileus virginianus</i>), but also bison, elk, and moose	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Coquillettia</i> , <i>Culiseta</i> , <i>Ochlerotatus</i> spp.)	F&M, E
	Keystone virus (KEYV)	Rabbits, squirrels	Mosquitoes (<i>Ochlerotatus atlanticus</i>)	E
	La Crosse virus (LACV)	Chipmunks, squirrels	Mosquitoes (<i>Ochlerotatus triseriatus</i>)	F&M, E
	Lumbo virus (LUMV)	Unknown	Mosquitoes (<i>Aedes pembraensis</i>)	F&M, E
	Madrid virus (MADV)	Capuchins (<i>Cebus</i> spp.), opossums, rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Maguari virus (MAGV)	Birds, cattle, horses, sheep, water buffalo	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> , <i>Psorophora</i> , <i>Wyeomyia</i> spp.)	F&M
	Marituba virus (MTBV)	Capuchins (<i>Cebus</i> spp.), opossums, rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Murutucú virus (MURV)	Capuchins (<i>Cebus</i> spp.), opossums, pale-throated sloths (<i>Bradypus tridactylus</i>), rodents	Mosquitoes (<i>Coquillettia</i> , <i>Culex</i> spp.)	F&M
	Nepuyo virus (NEPV)	Bats (<i>Artibeus</i> spp.), rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Ngari virus (NRIV)	Cattle, goats, sheep	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> spp.)	F&M, VHF
	Nyando virus (NDV)	Unknown	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> spp.), sandflies (<i>Lutzomyia</i> spp.)	F&M
	Oriboca virus (ORIV)	Capuchins (<i>Cebus</i> spp.), opossums, rodents	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Mansonia</i> , <i>Psorophora</i> spp.)	F&M
	Oropouche virus (OROV)	Marmosets (<i>Callithrix</i> spp.), pale-throated sloths (<i>B. tridactylus</i>)	Biting midges (<i>Culicoides paraensis</i>), mosquitoes (<i>Coquillettia venezuelensis</i> , <i>Culex quinquefasciatus</i> , <i>Mansonia</i> spp., <i>Ochlerotatus serratus</i>)	F&M, E
	Ossa virus (OSSAV)	Rodents	Mosquitoes (<i>Culex</i> spp.)	F&M
	Pongola virus (PGAV)	Cattle, donkeys, goats, sheep	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Mansonia</i> spp.)	F&M
	Restan virus (RESV)	Unknown	Mosquitoes (<i>Culex</i> spp.)	F&M
	Shokwe virus (SHOV)	Rodents	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Mansonia</i> spp.)	F&M
	Shuni virus (SHUV)	Horses, livestock	Mosquitoes (<i>Culex theileri</i> , <i>Culicoides</i> spp.)	E
	Snowshoe hare virus (SSHV)	Collared lemmings, snowshoe hares, squirrels, voles	Mosquitoes (<i>Aedes</i> , <i>Culiseta</i> , <i>Ochlerotatus</i> , <i>Simulium</i> spp.)	F&M, E
	Tacaiuma virus (TCMV)	Nonhuman primates	Mosquitoes (<i>Anopheles</i> , <i>Haemagogus</i> spp.)	F&M
	Tahyña virus (TAHV)	Boars, cattle, deer, dogs, eulipotyphla, foxes, hares, horses, pigs, rodents	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> , <i>Culiseta</i> spp.)	F&M, E
	Tataguine virus (TATV)	Unknown	Mosquitoes (<i>Anopheles</i> spp.)	F&M
	Trubanaman virus (TRUV)	Unknown	Mosquitoes (<i>Anopheles</i> , <i>Culex</i> spp.)	(A&R)
	Wyeomyia virus (WYOV)	Unknown	Mosquitoes (<i>Wyeomyia</i> spp.)	F&M
	Xingu virus (XINV)	Unknown	Unknown	F&M
	Zungarococha virus (ZUNV)	Unknown	Unknown	F&M
<i>Orthoflavivirus</i> (family <i>Flaviviridae</i>)	Alkhurma hemorrhagic fever virus (AHFV) ^j	Livestock?	Argasid ticks (<i>Ornithodoros savignyi</i>), ixodid ticks (<i>Hyalomma dromedarii</i> , <i>H. rufipes</i>)	VHF
	Apoi virus (APOIV)	Field mice, voles	Unknown	E
	Banzi virus (BANV)	Rodents?	Mosquitoes (<i>Culex rubinotus</i>)	F&M
	Bussuquara virus (BSQV)	Cotton rats, monkeys, spiny-rats	Mosquitoes (<i>Culex</i> spp.)	F&M
	Cacipacoré virus (CPCV)	Birds, horses, nonhuman primates, water buffalo	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> spp.)	F&M
	Dakar bat virus (DBV)	Bats	Unknown	F&M
	Dengue viruses 1–4 (DENV 1–4)	Nonhuman primates	Mosquitoes (predominantly <i>Aedes aegypti</i> , <i>A. albopictus</i>)	F&M, VHF
	Edge Hill virus (EHV)	Bandicoots, dogs, wallabies	Mosquitoes (<i>Aedes vigilax</i> , <i>Culex annulirostris</i>)	F&M
	Ilhéus virus (ILHV)	Birds, coatis, nonhuman primates, rodents, reptiles, sloths, water buffalo	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Coquillettia</i> , <i>Haemagogus</i> , <i>Ochlerotatus</i> , <i>Psorophora</i> , <i>Sabethes</i> , <i>Trichoprosopon</i> spp.)	E

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
	Japanese encephalitis virus (JEV)	Ardeid wading birds (in particular herons), horses, pigs	Mosquitoes (<i>Culex</i> spp., in particular <i>C. tritaeniorhynchus</i>)	E
	Karshi virus (KSIV)	Great gerbils (<i>Rhombomys opimus</i>)	Argasid ticks (<i>Ornithodoros capensis</i>), ixodid ticks (<i>Hyalomma asiaticum</i>)	F&M, E
	Kédougou virus (KEDV)	Unknown	Mosquitoes (<i>Culex</i> spp.)	F&M
	Kokohera virus (KOKV)	Macropods, horses	Mosquitoes (<i>Culex</i> spp.)	A&R
	Koutango virus (KOUV)	Gerbils	Mosquitoes (<i>Aedes</i> spp.)	F&M
	Kyasanur Forest disease virus (KFDV) ^k	Indomalayan vandeureias (<i>Vandeleuria oleracea</i>), roof rats (<i>Rattus rattus</i>)	Ixodid ticks (predominantly <i>Haemaphysalis spinigera</i>)	VHF
	Modoc virus (MODV)	North American deermice (<i>Peromyscus maniculatus</i>)	Unknown	E
	Murray Valley encephalitis virus ^h (MVEV)	Birds (egrets, herons)	Mosquitoes (predominantly <i>C. annulirostris</i>)	E
	Ntaya virus (NTAV)	Birds, domestic animals	Mosquitoes (<i>Culex</i> spp.)	F&M
	Omsk hemorrhagic fever virus (OHFV)	Migratory birds, rodents	Ixodid ticks (predominantly <i>Dermacentor</i> spp.)	VHF
	Powassan virus (POWV)	Red squirrels (<i>Tamiasciurus hudsonicus</i>), white-footed deermice (<i>Peromyscus leucopus</i>), woodchucks (<i>Marmota monax</i>), other small mammals	Ixodid ticks (in particular <i>Ixodes cookei</i> , other <i>Ixodes</i> spp., <i>Dermacentor</i> spp.)	E
	Rio Bravo virus (RBV)	Bats	Unknown	F&M
	Rocio virus (ROCV)	Rufous-collared sparrows (<i>Zonotrichia capensis</i>), various migratory birds, horses, water buffalo (<i>Bubalus bubalis</i>)	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Psorophora</i> spp.)	E
	Sepik virus (SEPV)	Unknown	Mosquitoes (<i>Mansonia septempunctata</i>)	F&M
	Spondweni virus (SPOV)	Cattle, sheep	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Mansonia</i> spp.)	F&M
	St. Louis encephalitis virus (SLEV)	Columbiform and passeriform birds (finches, sparrows)	Mosquitoes (predominantly <i>Culex</i> spp., in particular <i>C. nigripalpus</i> , <i>C. pipiens</i> , <i>C. quinquefasciatus</i> , <i>C. tarsalis</i>)	E
	Tick-borne encephalitis virus (TBEV)	Passeriform birds, deer, eulipotyphla, goats, grouse, small mammals, rodents, sheep	Ixodid ticks (<i>Ixodes gibbosus</i> , <i>I. persulcatus</i> , <i>I. ricinus</i> ; sporadically <i>Dermacentor</i> , <i>Haemaphysalis</i> , <i>Hyalomma</i> spp.)	F&M, E, (VHF)
	Usutu virus (USUV)	Accipitriform, columbiform, passeriform, and strigiform birds	Mosquitoes (<i>Culex</i> spp., in particular <i>C. pipiens</i> , but also <i>Aedes</i> spp, <i>Anopheles</i> spp., <i>Ochlerotatus</i> spp.)	(E)
	Wesselsbron virus (WESSV)	Cattle, goats, rodents, sheep	Mosquitoes (<i>Aedes</i> spp.)	F&M
	West Nile virus (WNV) ⁱ	Passeriform birds (blackbirds, crows, finches, sparrows), small mammals, horses	Mosquitoes (<i>Culex</i> spp., in particular <i>C. pipiens</i> , <i>C. quinquefasciatus</i> , <i>C. restuans</i> , <i>C. tarsalis</i>)	E
	Yellow fever virus (YFV)	Nonhuman primates (<i>Alouatta</i> , <i>Ateles</i> , <i>Cebus</i> , <i>Cercopithecus</i> , <i>Colobus</i> spp.)	Mosquitoes (<i>Aedes</i> spp., in particular <i>Ae. aegypti</i>)	VHF
	Zika virus (ZIKV)	Nonhuman primates (<i>Macaca</i> , <i>Pongo</i> spp.)	Mosquitoes (<i>Aedes</i> spp.)	F&M, A&R
Orthohantavirus (family Hantaviridae)	Amur virus (AMRV)	Korean field mice (<i>Apodemus peninsulae</i>)	None	VHF
	Anajatuba virus (ANJV)	Fornes' colilargos (<i>Oligoryzomys fornesi</i>)	None	P
	Andes virus (ANDV)	Long-tailed colilargos (<i>Oligoryzomys longicaudatus</i>)	None	P
	Araraquara virus (ARAV)	Hairy-tailed akodonts (<i>Necomys lasiurus</i>)	None	P
	Araucária virus (ARAUV)	Black-footed colilargos (<i>Oligoryzomys nigripes</i>)	None	P
	Bayou virus (BAYV)	Marsh rice rats (<i>Oryzomys palustris</i>)	None	P
	Bermejo virus (BMJV)	Chacoan colilargos (<i>Oligoryzomys chacoensis</i>)	None	P
	Black Creek Canal virus (BCCV)	Hispid cotton rats (<i>Sigmodon hispidus</i>)	None	P
	Blue River virus (BRV)	White-footed deermice (<i>Peromyscus leucopus</i>)	None	P
	Caño Delgadito virus (CADV)	Alston's cotton rats (<i>Sigmodon alstoni</i>)	None	P

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
	Castelo dos Sonhos virus (CASV)	Brazilian colilargos (<i>Oligoryzomys elii</i>)	None	P
	Catacamas virus (CATV)	Coues' oryzomys (<i>Oryzomys couesi</i>)	None	P
	Choclo virus (CHOV)	Fulvous colilargos (<i>Oligoryzomys fulvescens</i>)	None	F&M, P
	Dobrava virus (DOBV)	Caucasus field mice (<i>Apodemus ponticus</i>), striped field mice (<i>A. agrarius</i>), yellow-necked field mice (<i>A. flavicollis</i>)	None	VHF
	El Moro Canyon virus (ELMCV)	Western harvest mice (<i>Reithrodontomys megalotis</i>)	None	VHF
	Göu virus (GOUV)	Brown rats (<i>Rattus norvegicus</i>), roof rats (<i>R. rattus</i>), Oriental house rats (<i>R. tanezum</i>)	None	VHF
	Hantaan virus (HTNV)	Striped field mice (<i>Apodemus agrarius</i>)	None	VHF
	Juquitiba virus (JUQV)	Black-footed colilargos (<i>Oligoryzomys nigripes</i>)	None	P
	Kurkino virus (KURV)	Striped field mice (<i>Apodemus agrarius</i>)	None	VHF
	Laguna Negra virus (LANV)	Little lauchas (<i>Calomys laucha</i>)	None	P
	Lechiguanas virus (LECV)	Flavescent colilargos (<i>Oligoryzomys flavescens</i>)	None	P
	Maciel virus (MCLV)	Dark-furred akodonts (<i>Necomys obscurus</i>)	None	P
	Maripa virus (MARV)	Unknown	None	P
	Monongahela virus (MGLV)	Eastern deermice (<i>Peromyscus maniculatus</i>)	None	P
	Muju virus (MUJV)	Korean red-backed voles (<i>Myodes regulus</i>)	None	VHF
	New York virus (NYV)	White-footed deermice (<i>Peromyscus leucopus</i>)	None	P
	Orán virus (ORNV)	Long-tailed colilargos (<i>Oligoryzomys longicaudatus</i>)	None	P
	Paranoá virus (PARV)	Unknown	None	P
	Pergamino virus (PRGV)	Azara's akodonts (<i>Akodon azarae</i>)	None	P
	Puumala virus (PUUV)	Bank voles (<i>Myodes glareolus</i>)	None	P, VHF
	Rio Mamoré virus (RIOMV)	Common bristly mice (<i>Neacomys spinosus</i>)	None	P
	Saaremaa virus (SAAV)	Striped field mice (<i>Apodemus agrarius</i>)	None	VHF
	Seoul virus (SEOV)	Brown rats (<i>Rattus norvegicus</i>), roof rats (<i>R. rattus</i>)	None	VHF
	Sin Nombre virus (SNV)	Western deermice (<i>Peromyscus sonoriensis</i>)	None	P
	Sochi virus (SOCV)	Caucasus field mice (<i>Apodemus ponticus</i>)	None	VHF
	Tula virus (TULV)	Common voles (<i>Microtus arvalis</i>), East European voles (<i>M. levis</i>), field voles (<i>M. agrestis</i>)	None	(P), VHF
	Tunari virus (TUNV)	Unknown	None	P
<i>Orthonairovirus</i> (family <i>Nairoviridae</i>)	Aigai virus (AIGV)	Cattle, goats, tortoises	Ixodid ticks (<i>Hyalomma</i> spp., <i>Rhipicephalus</i> spp.)	VHF
	Avalon virus (AVAV)	European herring gulls (<i>Larus argentatus</i>)	Ixodid ticks (<i>Ixodes uriae</i>)	(Polyradiculoneuritis?)
	Crimean-Congo hemorrhagic fever virus (CCHFV)	Cattle, dogs, goats, hares, hedgehogs, mice, ostriches, sheep	Predominantly ixodid ticks (<i>Hyalomma</i> spp.)	VHF
	Dugbe virus (DUGV)	Northern giant pouched rats (<i>Cricetomys gambianus</i>), Zébu cattle (<i>Bos primigenius</i>)	Biting midges (<i>Culicoides</i> spp.), ixodid ticks (<i>Amblyomma</i> , <i>Hyalomma</i> , <i>Rhipicephalus</i> spp.)	F&M
	Erve virus (ERVEV)	Greater white-toothed shrews (<i>Crocidura russula</i>)	Unknown	(Thunderclap headache?)
	Issyk-kul virus (ISKV)	Bats, birds	Biting midges (<i>Culicoides schultzei</i>), horseflies (<i>Tabanus agrestis</i>), mosquitoes (<i>Aedes caspius</i> , <i>Anopheles hyrcanus</i>), argasid ticks (<i>Argas vespertilionis</i> , <i>A. pusillus</i>), ixodid ticks (<i>Ixodes vespertilionis</i>)	F&M

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
	Nairobi sheep disease virus ^c (NSDV)	Sheep	Ixodid ticks (<i>Amblyomma</i> , <i>Haemaphysalis</i> , <i>Rhipicephalus</i> spp.), mosquitoes (<i>Culex</i> spp.)	F&M
	Söngling virus (SGLV)	Great gerbils (<i>Rhombomys opimus</i>)	Ixodid ticks (<i>Ixodes crenulatus</i> , <i>Ixodes persulcatus</i> , <i>Haemaphysalis concinna</i> , and <i>Haemaphysalis longicornis</i>)	F&M
	Táchéng tick virus 1 (TcTV1)	Sheep, tortoises	Ixodid ticks (<i>Dermacentor marginatus</i> , <i>Hyalomma aegyptium</i>)	F&M
	Tamdy virus (TAMV)	Gerbils, other mammals (including Bactrian camels), birds	Ixodid ticks (<i>Dermacentor</i> , <i>Hyalomma</i> spp.)	F&M
	Wetland virus (WELV)	Horses, pigs, sheep, North China zokors	Ixodid ticks (<i>Dermacentor</i> , <i>Ixodes</i> , <i>Haemaphysalis</i> spp.)	F&M
	Yezo virus (YEZV)	Unknown	Ixodid ticks (<i>Ixodes persulcatus</i>)	F&M
<i>Phlebovirus</i> (family <i>Phenuiviridae</i>)	Adria virus (ADRV)	Unknown	Sandflies	E
	Alenquer virus (ALEV)	Unknown	Unknown	F&M
	Candirú virus (CDUV)	Unknown	Unknown	F&M
	Chagres virus (CHGV)	Unknown	Sandflies (<i>Lutzomyia</i> spp.)	F&M
	Chios virus (/)	Unknown	Unknown	E
	Coclé virus (CCLV)	Unknown	Sandflies	F&M
	Echarate virus (ECHV)	Unknown	Unknown	F&M
	Granada virus (GRV = GRAV)	Unknown	Sandflies	F&M
	Maldonado virus (MLOV)	Unknown	Unknown	F&M
	Morumbi virus (MRBV = MRMBV)	Unknown	Unknown	F&M
	Punta Toro virus (PTV)	Unknown	Sandflies (<i>Lutzomyia</i> spp.)	F&M
	Rift Valley fever virus (RVFV)	Cattle, sheep	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Coquillettidia</i> , <i>Culex</i> , <i>Eretmapodites</i> , <i>Mansonia</i> spp.)	F&M, E, VHF
	Sandfly fever Cyprus virus (SFCV)	Unknown	Unknown	F&M
	Sandfly fever Ethiopia virus (SFEV)	Unknown	Sandflies	F&M
	Sandfly fever Naples virus (SFNV)	Unknown	Sandflies (<i>Phlebotomus papatasi</i> , <i>P. perfliewi</i> , <i>P. perniciosus</i>)	F&M
	Sandfly fever Sicilian virus (SFSV)	Eulipotyphla, least weasels (<i>Mustela nivalis</i>), rodents	Sandflies (predominantly <i>Phlebotomus papatasi</i>)	F&M
	Sandfly fever Turkey virus (SFTV)	Unknown	Sandflies (<i>Phlebotomus</i> spp.)	F&M
	Serra Norte virus (SRNV)	Unknown	Unknown	F&M
	Toscana virus (TOSV)	Migratory birds?	Sandflies (<i>Phlebotomus papatasi</i> , <i>P. perfliewi</i> , <i>Sergentomyia</i> spp.)	F&M, E
<i>Quarantavirus</i> (family <i>Orthomyxoviridae</i>)	Quaranfil virus (QRFV)	Birds	Argasid ticks (<i>Argas arboreus</i>)	F&M
<i>Seadornavirus</i> (family <i>Sedoreoviridae</i>)	Banna virus (BAV)	Cattle, pigs	Biting midges (<i>Culicoides</i> spp.), mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culiseta</i> spp.), ticks	E
<i>Tanzavirus</i> (family <i>Phenuiviridae</i>)	Dar es Salaam virus (DeSV)	Unknown	Unknown	F&M
<i>Tibrovirus</i> (family <i>Rhabdoviridae</i>)	Bas-Congo virus (BASV)	Artiodactyls?	Biting midges?	(VHF)
	Ekpoma virus 1 (EKV1)	Artiodactyls?	Biting midges?	F&M
	Ekpoma virus 2 (EKV2)	Artiodactyls?	Biting midges?	F&M
	Mundri virus (MUNV)	Artiodactyls?	Biting midges?	E (New onset nodding syndrome)
<i>Thogotovirus</i> (family <i>Orthomyxoviridae</i>)	Bourbon virus (BRBV)	Raccoons (<i>Procyon lotor</i>), white-tailed deer (<i>Odocoileus virginianus</i>)?	Ixodid ticks (<i>Amblyomma americanum</i>)	F&M
	Dhori virus (DHOV) ^p	Bats, camels, horses	Mosquitoes (<i>Aedes</i> , <i>Anopheles</i> , <i>Culex</i> spp.), argasid ticks (<i>Ornithodoros</i> spp.), ixodid ticks (<i>Dermacentor</i> , <i>Hyalomma</i> spp.)	F&M, E
	Thogoto virus (THOV)	Camels, cattle	Ixodid ticks (<i>Amblyomma</i> , <i>Hyalomma</i> , <i>Rhipicephalus</i> spp.)	F&M, E

(Continued)

TABLE 215-1 Zoonotic Arthropod- and Rodent-Borne Viruses That Infect Humans (Continued)

VIRUS TAXON	VIRUS (ABBREVIATION)	MAJOR NONHUMAN HOST(S) ^a	VECTOR(S)	SYNDROME ^b
<i>Uukuvirus</i> (family <i>Phenuiviridae</i>)	Tácheng tick virus 2 (TcTV2)	Unknown	Ixodid ticks (<i>Dermacentor marginatus</i> , <i>Dermacentor nuttalli</i> , <i>Dermacentor silvarum</i> , <i>Hyalomma asiaticum</i>)	(E)
	Uukuniemi virus (UUKV)	Birds, cattle, rodents	Ixodid ticks (<i>Ixodes</i> spp.)	F&M
<i>Vesiculovirus</i> (family <i>Rhabdoviridae</i>)	Chandipura virus (CHPV)	Hedgehogs	Mosquitoes (<i>Aedes aegypti</i>), sandflies (<i>Phlebotomus</i> , <i>Sergentomyia</i> spp.)	F&M, E
	Isfahan virus (ISFV)	Great gerbils (<i>Rhombomys opimus</i>)	Sandflies (<i>Phlebotomus papatasi</i>)	F&M
	Piry virus (PIRYV)	Gray four-eyed opossums (<i>Philander opossum</i>)	Mosquitoes (<i>Aedes</i> , <i>Culex</i> , <i>Toxorhynchites</i> spp.)	F&M
	Vesicular stomatitis Indiana virus (VSIV)	Cattle, horses, pigs	Sandflies (<i>Lutzomyia</i> spp.)	F&M
	Vesicular stomatitis New Jersey virus (VSNJV)	Cattle, horses, pigs	Biting midges (<i>Culicoides</i> spp.), chloropid flies, mosquitoes (<i>Culex</i> , <i>Mansonia</i> spp.), muscoid flies (<i>Musca</i> spp.), simuliid flies	F&M

^aMammalian names as listed in Wilson & Reeder's *Mammal Species of the World*, 3rd edition (<https://www.departments.bucknell.edu/biology/resources/msw3/>).

^bAbbreviations refer to the syndromes most associated with the viruses: A&R, arthritis and rash; E, encephalitis; F&M, fever and myalgia; P, pulmonary; VHF, viral hemorrhagic fever. Abbreviations are placed in parentheses when cases are controversial. ^cIn the older literature, chikungunya virus often is also listed as a causative agent of VHF. However, later studies revealed that, in most cases, people with "chikungunya hemorrhagic fever" were co-infected with one or more dengue viruses, an observation suggesting that the VHF was severe dengue. ^dAlso known as Igbo-Ora virus. ^eAlso known as Ockelbo virus (OCKV), Pogosta virus, and Karelian fever virus (KFV). ^fAlso known as Palma virus (PALV). ^gAlternatives used in the literature are Huáiyángshān virus (HYSV) and Hénán fever virus (HNFV). ^hAlso known as Alfuy virus (ALFV). ⁱAlso includes Kunjin virus (KUNV). ^jAlso spelled Alkhurma hemorrhagic fever virus (AHFV) and known as Alkhurma/Alkhurma virus (ALKV). ^kAlso known as Nánjiányí(n) virus. ^lWhitewater Arroyo virus is often listed as a causative agent of VHF in the literature, but convincing data associating this virus with VHF have not been published. ^mAlso known as Brezová virus, Cvilin virus, Kharagsh virus, Koliba virus, or Lipovnik virus. ⁿAlso known as Čalovo virus (CVOV) or Chittoor virus (CHITV). ^oAlso known as Ganjam virus (GV). ^pAlso known as Astra virus and Batken virus (BKNV).

that cause fever and myalgia (F&M) syndrome and possibly viral hemorrhagic fever, whereas the genus *Vesiculovirus* includes confirmed human arthropod-borne viruses, all of which are transmitted by insects (biting midges, mosquitoes, and sandflies). **The general properties of rhabdovirids are discussed in more detail in Chap. 214.**

■ RESENTOVIRICETES: SEDOREOVIRIDAE AND SPINAREOVIRIDAE

The families *Sedoreoviridae* and *Spinareoviridae* are established for viruses with linear, multisegmented, double-stranded RNA genomes (≈16–29 kb in total). These viruses produce particles that have icosahedral symmetry and are 60–80 nm in diameter. In contrast to all other virions discussed in this chapter, resectovirions are not enveloped and thus are insensitive to detergent inactivation. Human arthropod-borne viruses are found within the genera *Coltivirus* (family *Spinareoviridae*), *Orbivirus*, and *Seadornavirus* (both in family *Sedoreoviridae*). Arthropod-borne coltiviruses possess 12 genome segments and are transmitted by numerous tick types transstadially but not transovarially. Therefore, maintenance of the transmission cycle involves viremic mammalian hosts infected by tick bites. Arthropod-borne orbiviruses have 10 genome segments and are transmitted by mosquitoes or ixodid ticks, whereas relevant seadornaviruses have 12 genome segments and are transmitted exclusively by mosquitoes.

EPIDEMIOLOGY

Arthropod-borne and rodent-borne viruses are generally confined to the areas inhabited by their reservoir hosts and/or vectors. Consequently, a patient's geographic origin or travel history can provide important clues in the differential diagnosis. **Table 215-2** lists the approximate geographic distribution of most arthropod-borne and rodent-borne infections. Many of these diseases can be acquired in rural or urban settings. However, diseases commonly associated with urban outbreaks include yellow fever (YF), dengue without/with warning signs (previously called dengue fever), severe dengue (previously called dengue hemorrhagic fever and dengue shock syndrome), chikungunya virus disease, hemorrhagic fever with renal syndrome (HFRS) caused by Seoul virus, sandfly fever caused by sandfly fever Naples and Sicilian viruses, and Oropouche virus disease.

DIAGNOSIS

In patients with suspected viral infection, a history of mosquito bite(s) has little diagnostic significance, but a history of tick bite(s) is more useful. Patients infected with mammarenaviruses or orthohantaviruses

sometimes report exposure to rodents. Although clinical signs and epidemiologic clues may enable presumptive etiologic diagnosis in epidemic settings, laboratory diagnosis is required in all cases. For most arthropod-borne and rodent-borne virus diseases, acute-phase serum samples (collected within 3 or 4 days of onset) have yielded isolates. Paired serum samples have been used to show rising antibody titers. Intensive efforts to develop rapid tests for viral hemorrhagic fevers (VHFs) have resulted in reliable antigen-detection enzyme-linked immunosorbent assays (ELISAs), IgM-capture ELISAs, and multiplex polymerase chain reaction (PCR) assays. These tests can provide a diagnosis based on a single serum sample within a few hours and are particularly useful for assessment of patients with severe disease. More sensitive reverse transcriptase PCR (RT-PCR) assays may yield diagnoses based on samples without detectable antigen and may also provide useful truly quantifiable genetic information about the etiologic agent.

At diagnosis, patients with encephalitis generally are no longer viremic or antigenemic and usually do not have virions in cerebrospinal fluid (CSF). In this situation, either serologic detection of IgM or RT-PCR detection of viral nucleic acid may enable diagnosis. Increasingly, IgM-capture ELISAs are used for the simultaneous testing of serum and CSF. IgG ELISA or classic serology is useful in the evaluation of past exposure to viruses, many of which circulate in areas with minimal medical infrastructures and sometimes cause only mild or subclinical infections (and are not recognized).

CLINICAL DISEASE SYNDROMES

There is a wide spectrum of possible human responses to infection with arthropod-borne or rodent-borne viruses, and knowledge of the outcome of most of these infections is limited. People infected with these viruses may not develop or recognize symptoms or signs of illness. If viral disease is recognized, it can usually be grouped into one of five broad syndromic categories: F&M, arthritis and rash (A&R), encephalitis, pulmonary disease, or VHF (**Table 215-3**). Although a useful clinical heuristic, it should be acknowledged that these categories often overlap in the complex spectra of disease caused by arthropod-borne and rodent-borne viruses. Furthermore, illness caused by many of these viruses is often best known by the most severe disease phenotypes, which typically do not result in the most common disease manifestations. For example, infections with West Nile virus (WNV) and Venezuelan equine encephalitis virus (VEEV) are discussed in this chapter as encephalitis, but during epidemics, many patients present with much milder F&M. Similarly, Rift Valley fever virus (RVFV)

TABLE 215-2 Geographic Distribution of Zoonotic Arthropod-Borne or Rodent-Borne Viral Diseases

AREA ^a	TYPE OF DISEASE ^b					
	<i>BUNYAVIRICETES</i> (<i>Arenaviridae</i> , <i>Hantaviridae</i> , <i>Nairoviridae</i> , <i>Peribunyaviridae</i> , <i>Phenuiviridae</i>)	<i>FLASUVIRICETES</i> (<i>Flaviviridae</i>)	<i>INSTHOVIRICETES</i> (<i>Orthomyxoviridae</i>)	<i>RESENTOVIRICETES</i> (<i>Sedoreoviridae</i> , <i>Spinareoviridae</i>)	<i>MONJIVIRICETES</i> (<i>Bornaviridae</i> , <i>Rhabdoviridae</i>)	<i>ALSUVIRICETES</i> (<i>Togaviridae</i>)
Africa	Aigai, Bangui, Batai, Bhanja, Bunyamwera, and Bwamba virus infections, Crimean-Congo hemorrhagic fever, Dar es Salaam, Dugbe, Germiston, Ilesha virus infections, Lassa fever, Lujo virus, Nairobi sheep disease virus infection, Ngari, Nyando, and Pongola virus infections, Rift Valley fever, sandfly fever, Shokwe, Shuni, Tataguine virus infections	Alkhurma hemorrhagic fever, Banzi virus infection, Dakar bat, dengue without/with warning signs/severe dengue, Kédougou, Koutango, Ntaya, Spondweni, Usutu, Wesselsbron, West Nile virus infections, yellow fever, Zika virus disease	Dhori, Quarantil, Thogoto virus infections	Lebombo, Orungo, Tribeč virus infections	(Bas-Congo virus infection,) Ekpoma virus 1, Ekpoma virus 2, Mundri virus infections	Chikungunya virus disease, o'nyong-nyong fever, Semliki Forest, Sindbis virus infections
Central Asia	Bhanja, Issyk-kul virus infections, Crimean-Congo hemorrhagic fever, sandfly fever, Tshyňa, Tamdy virus infections	Far Eastern tick-borne encephalitis, Karshi, Powassan, West Nile virus infections	Dhori virus infections	—	Isfahan virus infection	Sindbis virus infection
Eastern Asia	Crimean-Congo hemorrhagic fever, hemorrhagic fever with renal syndrome, sandfly fever, severe fever with thrombocytopenia syndrome, Táchéng tick virus 1 (and 2) and Tshyňa, Tamdy, Songling, wetland, and Yezo virus infections	Apoi virus infection, dengue without/with warning signs/severe dengue, Far Eastern tick-borne encephalitis, Japanese encephalitis, Karshi virus infection, Kyasanur Forest disease	—	Banna virus infection	—	—
Southern Asia	Batai, Bhanja virus infections, Crimean-Congo hemorrhagic fever, hemorrhagic fever with renal syndrome, Nairobi sheep disease virus infection, sandfly fever	Dengue without/with warning signs/severe dengue, Japanese encephalitis, Kyasanur Forest disease, West Nile virus infection, Zika virus disease	Dhori, Quarantil, Thogoto virus infections	—	Chandipura, Isfahan virus infections, variegated squirrel bornavirus 1?	Chikungunya virus disease
South-Eastern Asia	Batai virus infection, hemorrhagic fever with renal syndrome	Dengue without/with warning signs/severe dengue, Japanese encephalitis, West Nile virus infection, Zika virus disease	—	—	—	Chikungunya virus disease
Western Asia	Aigai, Batai, Bhanja virus infections, Crimean-Congo hemorrhagic fever, hemorrhagic fever with renal syndrome, sandfly fever, Táchéng tick virus 1, Tamdy virus infections	Alkhurma hemorrhagic fever, Central European tick-borne encephalitis, dengue without/with warning signs/severe dengue, West Nile virus infection	Dhori, Quarantil virus infections	—	—	Chikungunya virus disease
Latin/Central America and the Caribbean	Alenquer, Apeú virus infections, Argentinian hemorrhagic fever, Bolivian hemorrhagic fever, "Brazilian hemorrhagic fever", Bunyamwera, Candirú, Caraparú, Catú, Chagres, Chapare, Coclé, Echarate, (Flexal,) Fort Sherman, Guamá, Guaroa virus infections, hantavirus pulmonary syndrome, Iquitos, Itaquí, Juquitiba virus infections, lymphocytic choriomeningitis, Madrid, Maguari, Maldonado, Marituba, Mayaro, Morumbi, Murutucú, Nepuyo, Oriboca virus infections, Oropouche virus disease, Ossa, Punta Toro, Restan, Serra Norte, Tacaiuma, Trinidad virus infections, Venezuelan hemorrhagic fever, Wyeomyia, Xingu, Zungarococha virus infections	Bussuquara, Cacipacoré virus infections, dengue without/with warning signs/severe dengue, Ilhéus, Rio Bravo virus infections, Rocio viral encephalitis, Spondweni, St Louis encephalitis, yellow fever, Zika virus disease	—	—	Piry fever, vesicular stomatitis fever	Chikungunya virus disease, Madariaga, Mayaro, Mucambo, Tonate, Una virus infections, Venezuelan equine encephalitis

(Continued)

TABLE 215-2 Geographic Distribution of Zoonotic Arthropod-Borne or Rodent-Borne Viral Diseases (Continued)

AREA ^a	TYPE OF DISEASE ^b					
	<i>BUNYAVIRICETES</i> (<i>Arenaviridae</i> , <i>Hantaviridae</i> , <i>Nairoviridae</i> , <i>Peribunyaviridae</i> , <i>Phenuiviridae</i>)	<i>FLASUVIRICETES</i> (<i>Flaviviridae</i>)	<i>INSTHOVIRICETES</i> (<i>Orthomyxoviridae</i>)	<i>RESENTOVIRICETES</i> (<i>Sedoreoviridae</i> , <i>Spinareoviridae</i>)	<i>MONJIVIRICETES</i> (<i>Bornaviridae</i> , <i>Rhabdoviridae</i>)	<i>ALSUVIRICETES</i> (<i>Togaviridae</i>)
Northern America	(Avalon,) Cache Valley virus infections, California encephalitis, hantavirus pulmonary syndrome, Heartland and Keystone virus infections, lymphocytic choriomeningitis, Nepuyo, snowshoe hare, (Whitewater Arroyo virus) infections	Dengue without/with warning signs/severe dengue, Modoc virus infection, Powassan virus disease, Rio Bravo virus infection, St Louis encephalitis, West Nile virus infection, Zika virus disease	Bourbon virus infection	Colorado tick fever, Salmon River virus infection	Vesicular stomatitis fever	Eastern equine encephalitis, Everglades virus infection, western equine encephalitis
Europe	Adria, Aigai, (Avalon,) Batai, Bhanja, Cristoli virus infections, California encephalitis, Crimean-Congo hemorrhagic fever, (Erve virus infection), hemorrhagic fever with renal syndrome, Inkoo, Issyk-kul virus infections, lymphocytic choriomeningitis, sandfly fever, snowshoe hare, Tahyňa, Uukuniemi virus infections	Central European tick-borne encephalitis, dengue without/with warning signs/severe dengue, Ntaya virus infection, Omsk hemorrhagic fever, Powassan, Usutu, West Nile virus infections	Dhori, Thogoto virus infections	Eyach, Kemerovo, Tribeč virus infections	—	Chikungunya virus disease, Sindbis virus infection
Oceania	Batai, Gan Gan, (Trubananman) virus infections	Dengue without/with warning signs/severe dengue, Edge Hill virus infection, Japanese encephalitis, Kokobera virus infection, Murray Valley encephalitis, Sepik virus, West Nile virus infections, Zika virus disease	—	—	—	Barmah Forest virus infection, Ross River disease, Sindbis virus infection

^aGeographic names here and throughout the chapter are as recommended by the UN geoscheme (<https://unstats.un.org/unsd/methodology/m49/>). ^bDisease names according to the World Health Organization's International Classification of Diseases 11th revision (ICD-11; <https://icd.who.int/browse11/l-m/en>). Quotation marks indicate common usage in the absence of ICD-11 recognition. Diseases not acknowledged by the ICD-11 are designated as "virus infection(s)."

is best known as a cause of VHF, but the prevalence of milder F&M syndrome is far higher after infection, with encephalitis and ocular disease also occurring occasionally. Lymphocytic choriomeningitis virus (LCMV) is classified in this chapter as a cause of F&M syndrome because it is the most common disease manifestation. Even when central nervous system (CNS) disease evolves during infection with this virus, neurologic manifestations are usually mild and preceded by F&M. However, this virus may also cause fetal microcephaly. Overlap between syndromic categories is further complicated by evolving nomenclature around their classification. For example, infection with any dengue virus (DENV-1–4) is considered as a cause of F&M syndrome because this disease presentation, historically called "dengue fever," is by far the most common manifestation. However, severe manifestations of DENV infection have a complicated pathogenesis: the historical classification of disease as "dengue hemorrhagic fever" included a subset of patients with "dengue shock syndrome," which is of tremendous consequence for pediatric populations in particular areas of the world. Further complicating this overlap, a relatively recent World Health Organization (WHO) revision of disease classification recommended a less descriptive but more pragmatic use of "dengue without warning signs," "dengue with warning signs," and "severe dengue" to describe the same spectrum and enhance clinical management and case reporting.

Unfortunately, most of the known arthropod-borne and rodent-borne viral diseases have not been studied in detail with modern medical approaches. Thus, available data may be incomplete, at relatively low resolution, or biased toward severe disease. Data on geographic distribution are often difficult to interpret. Frequently, the literature is not clear as to whether the data pertain to the distribution of a particular virus or to the areas where human disease has been observed. In

addition, the designations for viruses and viral diseases have changed multiple times over decades. In this chapter, virus and taxon names are in line with the latest reports of the International Committee on Taxonomy of Viruses, and disease names are in accordance with the WHO's International Classification of Diseases 11th revision (ICD-11). When needed for clarity or historical reference, alternative nomenclature is used. Considering this syndromic approach, it should be noted that the variable clinical manifestations of particular viruses may be captured over a number of sections.

■ FEVER AND MYALGIA (F&M)

F&M is by far the most common clinical syndrome and has the most favorable outcomes after arthropod-borne and rodent-borne infections. However, it is also a common prodrome en route to more invasive syndromes associated with these infections. In logical sequence, therefore, F&M will be discussed initially but should not be considered in isolation. Indeed, clinicians evaluating patients with F&M should pay careful attention for symptoms and signs that suggest the A&R, neuroinvasive, pulmonary, or VHF syndromes that are subsequently discussed in this chapter. Although many of the viruses listed in Table 215-1 probably cause F&M, only some of these viruses have prominent associations with the syndrome that are considered biomedically important.

F&M syndrome typically begins with the abrupt onset of fever, chills, intense myalgia, and malaise. Patients may also report joint pain, but true arthritis is not found. Anorexia is characteristic and may be accompanied by nausea or vomiting. Headache is common and may be severe, with photophobia and retroorbital pain. Physical findings are minimal and usually confined to conjunctival injection, pharyngeal erythema/exudate, muscle tenderness, abdominal tenderness,

TABLE 215-3 Clinical Syndromes Caused by Zoonotic Arthropod-Borne or Rodent-Borne Viruses

SYNDROME	VIRUS
Fever and myalgia (F&M)	<i>Arenaviridae</i>: (Flexal,) Lassa and lymphocytic choriomeningitis viruses <i>Flaviviridae</i>: Bussuquara, Banzi, Cacipacoré, Dakar bat, dengue 1–4, Edge Hill, Karshi, Kédougou, Koutango, Ntaya virus, Rio Bravo, Sepik, Spondweni, tick-borne encephalitis, Wesselsbron, and Zika viruses <i>Hantaviridae</i>: Choclo virus <i>Nairoviridae</i>: Dugbe, Issyk-kul, Nairobi sheep disease, Sönglǎng viruses, Táchéng tick virus 1, Tamdy, wetland, Yezo virus <i>Orthomyxoviridae</i>: Bourbon, Dhori, and Thogoto viruses <i>Peribunyaviridae</i>: Apeú, Bangui, Batai, Bunyamwera, Bwamba, Cache Valley, California encephalitis, Caraparú, Catú, Fort Sherman, Germiston, Guamá, Guaroa, Ilesha, Inkoo, Iquitos, Itaquí, Jamestown Canyon, La Crosse, Lumbo, Madrid, Maguari, Marituba, Nepuyo, Ngari, Nyando, Oriboca, Oropouche, Ossa, Pongola, Restan, Shokwe, snowshoe hare, Tacaiuma, Tahyña, Tataguine, Wyemyia, Xingu, and Zungarococha viruses <i>Phenuiviridae</i>: Alenquer, Bhanja, Candirú, Chagres, Dar es Salaam, Echarate, Heartland, Maldonado, Morumbi, Punta Toro, Rift Valley fever, sandfly fever Cyprus, sandfly fever Ethiopia, sandfly fever Naples, sandfly fever Sicilian, sandfly fever Turkey, Serra Norte, severe fever with thrombocytopenia syndrome, Toscana, and Uukuniemi viruses <i>Sedoreoviridae</i>: Kemerovo, Lebombo, Orungo, and Tribeč viruses <i>Spinareoviridae</i>: Colorado tick fever, Eyach, and Salmon River viruses <i>Rhabdoviridae</i>: Ekpoma virus 1, Ekpoma virus 2, Chandipura, Isfahan, Piry, vesicular stomatitis Indiana, and vesicular stomatitis New Jersey viruses <i>Togaviridae</i>: Everglades, Madariaga, Mucambo, Tonate, Una, and Venezuelan equine encephalitis viruses
Arthritis and rash (A&R)	<i>Flaviviridae</i>: Kokobera and Zika viruses <i>Peribunyaviridae</i>: Gan Gan virus, (Trubanaman virus) <i>Togaviridae</i>: Barmah Forest, chikungunya, Mayaro, o'nyong-nyong, Ross River, Semliki Forest, and Sindbis viruses
Encephalitis	<i>Arenaviridae</i>: lymphocytic choriomeningitis virus, (Whitewater Arroyo virus) <i>Flaviviridae</i>: Apoi, Ilhéus, Japanese encephalitis, Karshi, Modoc, Murray Valley encephalitis, Powassan, Rocio, St. Louis encephalitis, tick-borne encephalitis, Usutu, and West Nile viruses <i>Orthomyxoviridae</i>: Dhori and Thogoto viruses <i>Bornaviridae</i>: variegated squirrel bornavirus 1 <i>Peribunyaviridae</i>: Cache Valley, California encephalitis, Cristoli, Inkoo, Jamestown Canyon, Keystone, La Crosse, Lumbo, Oropouche, snowshoe hare, Shuni, and Tahyña viruses <i>Phenuiviridae</i>: Adria, Bhanja, Chios, Rift Valley fever, (Táchéng tick virus 2), and Toscana viruses <i>Sedoreoviridae</i>: Banna, Kemerovo, and Orungo viruses <i>Spinareoviridae</i>: Colorado tick fever, Eyach, and Salmon River viruses <i>Rhabdoviridae</i>: Chandipura, Mundri viruses <i>Togaviridae</i>: eastern equine encephalitis, Everglades, Madariaga, Mucambo, Tonate, Venezuelan equine encephalitis, and western equine encephalitis viruses
Pulmonary disease	<i>Hantaviridae</i>: Anajatuba, Andes, Araucária, bayou, Bermejo, Black Creek Canal, Blue River, Caño Delgadito, Castelo dos Sonhos, Catacamas, Choclo, Juquitiba, Laguna Negra, Lechiguana, Maciel, Monongahela, New York, Orán, Paranoá, Pergamino, Puumala, Rio Mamoré, Sin Nombre, (Tula,) and Tunari viruses
Viral hemorrhagic fever (VHF)	<i>Arenaviridae</i>: Chapare, Guanarito, Junín, Lassa, Lujo, (lymphocytic choriomeningitis,) Machupo, and Sabiá viruses <i>Hantaviridae</i>: Amur, Dobrava, El Moro Canyon, gou, Hantaan, Kurkino, Muju, Puumala, Saaremaa, Seoul, Sochi, and Tula viruses <i>Nairoviridae</i>: Crimean-Congo hemorrhagic fever virus <i>Peribunyaviridae</i>: (Ilesha virus,) Ngari viruses <i>Phenuiviridae</i>: Rift Valley fever and severe fever with thrombocytopenia syndrome viruses <i>Flaviviridae</i>: Alkhurma hemorrhagic fever, dengue 1–4, Kyasanur Forest disease, Omsk hemorrhagic fever, (tick-borne encephalitis,) and yellow fever viruses <i>Rhabdoviridae</i>: (Bas-Congo virus)

Viruses are placed in parentheses if cases were controversial.

and possibly the presence of a nonpruritic maculopapular rash that may have a petechial component. The spectrum of disease varies from subclinical to temporarily incapacitating. When present, the evolution and duration of symptoms/signs are quite variable (generally 2–5 days) and may be biphasic after some viral infections. Less-common findings include epistaxis (not necessarily indicating a bleeding diathesis) and signs of aseptic meningitis. Even in the presence of headache, photophobia, and meningismus, challenges in obtaining and examining the CSF in remote areas makes diagnosis difficult. Although upper and lower respiratory symptoms/signs and radiographic evidence of pulmonary infiltrates are noted in some patients, the agents causing this syndrome are not primary respiratory pathogens.

F&M syndrome is the most nonspecific of the disease syndromes caused by arboviral infections. The early stages of other syndromes discussed in this chapter begin similarly and are encompassed in a broad differential diagnosis that also includes community-acquired parasitic infections (e.g., malaria), bacterial infections (e.g., anicteric

leptospirosis and rickettsial diseases), and other viral infections. F&M syndrome is often described as “influenza-like,” but the usual absence of cough and coryza makes influenza an unlikely confounder except at the earliest stages. Treatment is supportive, but acetylsalicylic acid is generally avoided because of the potential for exacerbated bleeding or Reye syndrome. Complete recovery is the expected outcome for symptomatic patients, although prolonged asthenia and nonspecific symptoms have been described, particularly after infection with LCMV or DENV-1–4. Nonetheless, it must be reemphasized that this non-specific syndrome may be the prodrome for arthritic, neuroinvasive, pulmonary, or VHF syndromes with less favorable outcomes.

Efforts to prevent viral infections causing this syndrome are best targeted to vector control, which, however, may be expensive or impossible. Destruction of mosquito breeding sites is generally the most economically and environmentally sound approach. Emerging containment technologies include the release of genetically modified mosquitoes and the spread of *Wolbachia* bacteria to limit mosquito

multiplication rates. Depending on the vector and its habits, other possible strategies include the use of window screens on dwellings or other barriers (e.g., permethrin-impregnated bed nets), judicious application of arthropod repellents (e.g., *N,N*-diethyltoluamide [DEET]) to the skin, use of long-sleeved (ideally permethrin-impregnated) clothing, and avoidance of vector habitats, particularly at peak feeding times.

Bunyaviricetes F&M syndrome is caused by numerous bunyaviricetes, many of which cause isolated individual infections and usually do not cause epidemics. These viruses include arenavirids (mammarenaviruses), hantavirids (orthohantaviruses),airovirids (orthonairoviruses), peribunyavirids (orthobunyaviruses), and phenuivirids (bandaviruses and phleboviruses).

ARENAVIRIDS Infection with LCMV is the only human mammarenavirus infection resulting predominantly in F&M syndrome. LCMV is transmitted to humans from the common house mouse (*Mus musculus*) by aerosols of excreta or secretions. The virus is maintained in the mouse mainly by vertical transmission from infected dams. Infected mice remain viremic and shed virus for life, with high concentrations of virus in all tissues. Mouse-acquired infections of colonies of pet hamsters also can serve as a link for human acquisition. Patients may have a history of residence in rodent-infested housing or other exposure to rodents. An antibody prevalence of ~5–10% has been reported among urban adults from Argentina, Germany, and the United States. In addition, laboratory infections among scientists and animal caretakers can occur because the virus is widely used in immunology laboratories to study T-lymphocyte function and can silently infect cell cultures and passaged tumor lines. Transmission has occurred via organ transplantation.

Lymphocytic choriomeningitis (LCM) differs from the general F&M syndrome in a characteristic gradual onset of the otherwise typical illness. Orchitis, transient alopecia, arthritis, pharyngitis, cough, and maculopapular rash are associated with LCM. An estimated one fourth (or less) of patients experience an initial febrile phase of 3–6 days. After a brief remission, many develop recurrent fever accompanied by meningeal (severe headache, photophobia, nausea and vomiting, and neck stiffness) and/or less common encephalitic signs (altered mental status or level of consciousness, sensorimotor deficits) as part of a ~1-week CNS phase. Patients with neuroinvasive disease, including patients with clear-cut signs of encephalitis, almost always recover fully. Rarely, disease is complicated by transient hydrocephalus (that may require shunting) and myelitis. During the initial febrile phase, leukopenia and thrombocytopenia are common, and virus can usually be isolated from blood. During the CNS phase, the virus may be found in the CSF, and antibodies are detected in the blood. The pathogenesis of LCM is thought to resemble manifestations resulting from direct intracranial inoculation of the virus into adult mice. The onset of the immune response leads to T-cell-mediated immunopathologic meningitis. During the meningeal phase, CSF monocyte counts range from the hundreds to the low thousands per microliter, and, unusually for a viral meningitis, hypoglycorrhachia is found in one-third of patients.

IgM-capture ELISA, immunochemistry, and RT-PCR are used in the diagnosis of lymphocytic choriomeningitis. IgM-capture ELISA of serum and CSF usually yields positive results; RT-PCR assays have been developed for CSF detection. Particular diagnostic challenges arise in immunosuppressed patients with fulminant infections transmitted by recent organ transplantation: in the absence of typical immune responses, molecular diagnosis via RT-PCR or immunohistochemistry may be required. Infection should be suspected in acutely ill febrile patients with marked leukopenia and thrombocytopenia and meningitis syndromes. In patients with aseptic meningitis, a diagnosis of LCM is suggested by the following: a well-marked febrile prodrome, adult age, occurrence in the autumn, low CSF glucose levels, or CSF monocyte counts of >1,000/μL. Throughout a pregnancy, maternal-to-fetal transmission may occur, leading to fetal death (in the first trimester) or to consequent congenital hydrocephalus, microcephaly, and/or chorioretinitis (in the second or third trimesters). Because a mild maternal infection may be unrecognized or unrecalled, specific

antibodies to the virus should be sought in both mother and fetus under suspicious circumstances, particularly in neonatal hydrocephalus testing negative for toxoplasmosis, rubella, cytomegalovirus, herpes simplex, and HIV-1/HIV-2 (TORCH) pathogens.

PERIBUNYAVIRIDS Apeú, Caraparú, Itaqui, Madrid, Marituba, Murutucú, Nepuyo, Oriboca, Ossa, Restan, and Zungarococha viruses are among the most common causes of arboviral infection in humans entering South American jungles. These viruses cause acute febrile disease and are transmitted by mosquitoes in neotropical forests. Oropouche virus is transmitted in urban settings in Central and South America by biting midges (*Culicoides paraensis*), which often breed to high density in cacao husks and other vegetable detritus found in towns and cities. Explosive epidemics involving thousands of patients with Oropouche virus disease have been reported in Brazil and Peru, with smaller outbreaks in Panama and French Guiana and most recent outbreaks in Cuba and Haiti. Direct human-to-human transmission does not occur. Sloths and nonhuman primates may play a role as vertebrate hosts during maintenance sylvatic cycles. After a 3- to 10-day incubation period, most patients develop typical F&M syndrome with a biphasic evolution (recurrence of symptoms in the second week of illness) and, uncommonly, hemorrhagic manifestations. However, a small percentage of patients develop severe illness with neurologic (meningoencephalitis, Guillain-Barré) or hemorrhagic syndromes that have been associated with fatal outcome. Recently, infection of pregnant women has been associated with spontaneous miscarriage, intrauterine fetal death, and congenital anomalies. Oropouche virus has been detected in umbilical cord blood and fetal tissues in these clinical settings. Serologic detection (IgM or a rise in antibody titer on paired samples) or PCR confirms the diagnosis of acute F&M syndrome; IgM and viral nucleic acid have been detected in the CSF in rare cases of neuroinvasive disease. Postacute sequelae have not been described, though some patients have prolonged asthenia after the acute syndrome. Specific therapeutics or vaccines are not available.

Iquitos virus, a recently discovered reassortant and close relative of Oropouche virus, causes disease that is easily mistaken for Oropouche virus disease; its overall epidemiologic significance remains to be determined.

PHENUIVIRIDS The genus *Phlebovirus* includes numerous viruses that may cause human infection. Sandfly fever Cyprus virus, sandfly fever Ethiopia virus, sandfly fever Sicilian virus, and sandfly fever Turkey virus (and the encephalitis-causing Chios virus) are genetically and antigenically very closely related. In contrast, sandfly fever Naples virus (SFNV) is only genetically and antigenically distantly related to these viruses. SFNV has not been detected in sandflies, humans, or nonhuman vertebrates since the 1980s and therefore may be extinct, but closely related viruses, such as Granada virus and Toscana virus, continue to circulate. Toscana virus is thus far the only *Phlebovirus* transmitted by sandflies that is known to cause central and peripheral nervous system disease, such as encephalitis, meningitis, or polymyeloradiculopathy. *Phlebotomus* sandflies transmit the virus, probably by biting small mammals and humans. Female sandflies may be infected orally during blood meals and may transmit the virus to progeny when they lay their eggs. This prominent transovarial transmission confounds virus control.

Sandfly fever is found in the circum-Mediterranean area, extending to the east through the Balkans into parts of China as well as into Western Asia. Sandflies are found in both rural and urban settings and are known for their short flight ranges and small sizes; the latter enables them to penetrate standard mosquito screens and netting. Epidemics have been described in the wake of natural disasters and wars. After World War II, extensive spraying in parts of Europe to control malaria greatly reduced sandfly populations and transmission of SFNV; the incidence of sandfly fever continues to be low.

A common pattern of disease in endemic areas consists of high attack rates among visitors (including military personnel) but little or no disease in the local population, having developed protective immunity after childhood infection. Toscana virus infection is common during the summer among rural residents and vacationers, particularly in

Italy, Spain, and Portugal; several cases have been identified in travelers returning to Germany and Scandinavia. Clinical presentation is typical for F&M syndrome, with onset 3–6 days after a sandfly bite (a papule at the bite site may be observed). Notably, outbreaks of aseptic meningitis (virus can be isolated from the CSF) or meningoencephalitis have been associated with Toscana virus infection.

Coclé and Punta Toro viruses are not directly related to sandfly fever viruses but, like them, are transmitted by sandflies and cause a sandfly-fever-like disease. These two viruses are maintained in Latin American and Caribbean tropical forests, respectively, where the vectors rest on tree buttresses. Epidemics have not been reported, but antibody prevalence among inhabitants of villages in endemic areas indicates a cumulative lifetime exposure rate of >50% in the case of Punta Toro virus.

Heartland virus is notable as an emerging virus in the eastern, mid-western, and southern United States (>60 cases since its first description in 2012), where it causes often-fatal infection among males >50 years of age with comorbidities.

Flavivirids The most clinically significant orthoflaviviruses that cause F&M syndrome are DENV-1–4 and Zika virus (ZIKV). In fact, dengue without or with warning signs (“dengue,” historically called “dengue fever”—to be distinguished from severe dengue) is probably the most prevalent arthropod-borne viral disease worldwide, with ≈400 million infections occurring per year, of which ≈100 million (25%) cause clinical illness. Dengue is endemic in >100 countries worldwide, including those in Africa, the Americas, the eastern Mediterranean, Southeastern Asia, and the western Pacific. More than half of the world’s population is considered at risk, although Asia bears 70% of the global burden, with alarming increases over the past decade including, for example, >400,000 cases in 2019 in the Philippines. Year-round transmission of DENV-1–4 occurs between latitudes 25°N and 25°S, but seasonal forays of the viruses into the United States and Europe have been documented. The principal vectors for all four viruses are YF mosquitoes (*Aedes aegypti*). Through increasing spread of mosquitoes throughout the tropics and subtropics and international travel by infected humans, large areas of the world have become vulnerable to the introduction of DENV-1–4. Thus, dengue and severe dengue (see “Viral Hemorrhagic Fever,”) are becoming increasingly common. For instance, conditions favorable to DENV-1–4 transmission via YF mosquitoes exist in Hawaii and the southern United States. The range of a lesser vector of DENV-1–4, the Asian tiger mosquito (*Aedes albopictus*), now extends from Asia to the continental United States, the Indian Ocean, parts of Europe, and Hawaii. Also anthropophilic, YF mosquitoes typically breed near human habitation, using relatively fresh water in locations such as jars, vases, discarded containers, coconut husks, and old tires. These mosquitoes usually bite during the day. Bursts of dengue and severe dengue cases are to be expected in the southern United States, particularly along the Mexican border, where containers of water may be infested with YF mosquitoes. Air-conditioned buildings with screened vents may inhibit transmission of many arboviruses, including DENV-1–4.

Most primary DENV infections are subclinical. After an incubation period of ≈4–7 days, symptomatic patients present with three evolving phases: febrile, critical, and recovery. Although most patients presenting with F&M syndrome do not go through a critical phase, early recognition of the critical phase consistent with severe dengue must be considered in all patients. In most patients, dengue begins with the typical sudden onset of high-grade fever, frontal headache, retroorbital pain, back pain, and severe myalgia. These symptoms and signs gave rise to the colloquial designation of dengue as “break-bone fever.” A transient macular rash is often present at illness onset, and conjunctival redness, pharyngeal erythema, lymphadenopathy, and hepatomegaly may be noted on physical examination. The illness may last a week and include additional symptoms and signs (anorexia, nausea or vomiting, and marked cutaneous hypersensitivity). Near the time of defervescence (days 3–5), a maculopapular rash begins on the trunk and spreads to the extremities and the face. Epistaxis and scattered petechiae are often noted in uncomplicated dengue (without/with warning signs), and preexisting gastrointestinal lesions may bleed

during the acute illness. A positive tourniquet test—i.e., the detection of 10 or more new petechiae in one square inch of the upper arm after a 5-min blood pressure cuff inflation to midway between systolic and diastolic pressure—may demonstrate microvascular damage, but this finding is more likely to be associated with severe dengue. After defervescence and a brief remission, a subset of patients has recurrence of fever and other signs in a “saddle-back” pattern. Regardless, most patients recover from acute illness by 7–10 days, though convalescence can be prolonged.

Laboratory findings of dengue (without/with warning signs) include leukopenia, thrombocytopenia, and, in many cases, modest elevations of serum aspartate aminotransferase (AST) activity without hepatic synthetic dysfunction. The diagnosis is made by antigen-detection ELISA or RT-PCR during the acute phase or by IgM ELISA or paired serology during recovery. Virus is readily isolated from blood in the acute phase if mosquito inoculation or mosquito cell culture is used. Further clinical management and vaccine prevention are discussed in reference to severe dengue.

ZIKV is an emerging pathogen that is transmitted to nonhuman primates and humans by *Aedes* mosquitoes. The virus was discovered 1947 in a sentinel rhesus monkey (*Macaca mulatta*) and *Aedes africanus* mosquitoes in the Zika Forest in what was then the British Protectorate of Uganda. Human ZIKV infection was first documented during a YF outbreak in 1954 in Nigeria. Later, ZIKV infections were recognized in Southeastern Asia and Southern Asia. Prior to 2007, only 14 clinically identified cases of Zika virus disease had been reported. In recent years, the number of reported ZIKV infections has increased steadily and rapidly, with large but generally mild disease outbreaks on Yap Island and Micronesia (2007) and in Cambodia (2010), the Philippines (2012), and French Polynesia (2013–2014). ZIKV disease in the Americas was first reported on Easter Island, Chile (2014), and in Brazil (2015). By the end of 2015, an estimated 440,000 to 1.3 million cases had occurred in Brazil. At the end of May 2017, ZIKV infections had been recorded on five continents in 85 countries, including Mexico and the United States. Beginning in 2018, the global activity of ZIKV declined rather rapidly for unknown reasons.

Phylogenetic analysis of all available African ZIKV isolates revealed two geographically overlapping clades (Eastern and Western Africa). A descendant Asian lineage, represented by viruses collected from mosquitoes trapped in homes in Malaysia, was first reported in 1969. All ZIKV isolates causing human cases outside of Africa trace back to this Asian lineage.

Human infections are usually asymptomatic or benign and self-resolving and are most commonly misdiagnosed as dengue without/with warning signs or influenza. Typically, ZIKV disease is characterized by low-grade fever, an itchy maculopapular rash, arthralgia/myalgia, nonpurulent conjunctivitis, headache, and malaise. Other clinical manifestations include involvement of the gastrointestinal (nausea, vomiting, abdominal pain, and/or diarrhea), genitourinary (hematospermia), and, less commonly, cardiac (myocarditis and/or pericarditis), ocular (uveitis), and auditory (hearing loss) systems. The most concerning complications of ZIKV infection are neurologic syndromes, such as, Guillain-Barré syndrome and congenital fetal microcephaly, which may be associated either with fetal death or serious developmental delay in newborn infants. Other neurologic complications include encephalitis, meningoencephalitis, transverse myelitis, demyelinating polyneuropathies, cerebrovascular ischemia, retinopathies, and neurologic birth defects. Although most human ZIKV infections are acquired after bites by infected female mosquitoes, transmission may also occur perinatally or via breast-feeding, sexual contact with an infected person, transfusion of blood products, or organ transplantation. Infectious ZIKV or ZIKV RNA has been documented in the semen of male patients for up to 69 and 188 days after illness onset, respectively; sexual transmission associated with persistence has been documented up to 41 days after illness onset. Antiviral treatments (curative or preventive) and licensed vaccines against ZIKV are not yet available. Of note, some evidence suggests that prior exposure to ZIKV infection may increase the risk of severe disease syndrome and poor outcomes from DENV infection.

Orthomyxovirids Bourbon virus is emerging as a rare cause of F&M in humans in at least 14 states of the United States (≈ 60 cases since discovery in 2015). Onset is typical, but severe disease progression occurs among older individuals with comorbidities, including multisystem organ (kidney, respiratory, and/or hemodynamic) failure that has been fatal.

Sedoreovirids Several orbiviruses (Lebombo, Kemerovo, Orungo, and Tribeč viruses) can cause F&M in humans. Lebombo and Orungo viruses are transmitted by mosquitoes, whereas Kemerovo and Tribeč viruses are transmitted by ticks.

Spinareovirids Several coltivirus (Colorado tick fever, Eyach, and Salmon River viruses) can cause F&M in humans. All are transmitted by ticks. The most significant spinareovirid arthropod-borne disease is Colorado tick fever. Several hundred patients with this disease are reported annually in the United States and Canada. The infection is acquired between March and November through the bite of Rocky Mountain wood ticks (*Dermacentor andersoni*) in mountainous western regions at altitudes of 1200–3000 m. Small mammals serve as amplifying hosts. After a tick exposure (reported by almost all patients) and a mean incubation period of 1–14 days, the most common presentation is F&M, often with headache; rash develops in a minority of patients. Meningoencephalitis is not uncommon, especially in children, and disseminated intravascular coagulation (DIC) with hemorrhagic manifestations, pericarditis, myocarditis, orchitis, and pulmonary disease have also been reported. Leukopenia and thrombocytopenia are noted. The disease usually lasts 7–10 days and is often biphasic. Historically, the most important differential diagnostic considerations have been Rocky Mountain spotted fever (although Colorado tick fever is much more common in Colorado) and tularemia. Notably, seropositivity may be delayed for 10–14 days, limiting the utility of typical serologic testing early in the disease course; molecular RT-PCR testing may be used in this setting. Also, Colorado tick fever virus replicates for several weeks in erythropoietic cells and can be found in erythrocytes. This feature, detected in erythroid smears stained by immunofluorescence, can be diagnostically helpful and is important during screening of blood donors.

■ ARTHRITIS AND RASH (A&R)

Arthritides are common clinical presentations (or manifestations) of several viral diseases, such as hepatitis B, hepatitis C, parvovirus B19 infection, and rubella, and occasionally accompany infection due to adenovirids, enteroviruses, herpesvirids, mumps virus, or HIV-1/HIV-2. Arthropod-borne alphaviruses are also common causes of arthritides—usually acute febrile syndromes with joint involvement often accompanied by a maculopapular rash. Rheumatic involvement includes arthralgia with or without typical inflammatory signs, including periarticular redness, swelling (less commonly, joint effusions), and immobility. Most alphavirus infections are less severe and have fewer articular manifestations in children than in adults. In temperate climates, these ailments are summer diseases. The most significant alphavirus causes of arthritides are chikungunya virus disease, Ross River disease, Barmah Forest virus infection, and Sindbis virus infection. ZIKV infections may also be associated with joint manifestations. Less significant but historically notable are viruses that caused isolated cases or epidemics. A large (>2 million cases), albeit isolated, epidemic was caused by o'nyong-nyong virus from 1959 to 1962 (o'nyong-nyong fever). Mayaro virus, Semliki Forest virus, and Una virus caused isolated cases or limited and infrequent outbreaks (30 to several hundred cases per year). Symptoms and signs of infections with these viruses often are similar to those observed with chikungunya virus disease. Two orthobunyaviruses—Gan Gan virus and Trubanaman virus—and the orthoflavivirus Kokobera virus have been associated with single cases of polyarthritic disease. Except for a vaccine to prevent chikungunya virus disease, no specific therapies or licensed vaccines exist.

Chikungunya Virus Disease Historically, chikungunya virus (CHIKV) was considered endemic in rural areas of Africa, with intermittent outbreaks occurring in towns and cities of both Africa and Asia.

In 2004, a large epidemic began in the Indian Ocean region (specifically on the islands of Réunion and Mauritius) and was most likely spread by travelers. YF mosquitoes and Asian tiger mosquitoes are the major CHIKV vectors. All regions with established populations of these two vectors have now documented local mosquito-borne transmission, and the virus has now been identified across Africa, Asia, Europe, and the Americas, including in the continental United States, where suitable vector mosquitoes are present in southern states and local transmission has been documented. From December 2013 to June 2023, >3.6 million cases were reported from 50 countries or territories in the Americas.

The disease is more common in adults. After an incubation period of 2–10 days, the abrupt onset of fever (often severe, with a saddle-back pattern) and severe arthralgia are accompanied by constitutional symptoms and signs, including chills, abdominal pain, anorexia, conjunctival injection, headache, nausea, and photophobia. Migratory polyarthritides mainly affects the small joints of the ankles, feet, hands, and wrists, but the larger joints may be involved. Rash may appear at the outset or several days into the illness; its development often coincides with defervescence around day 2 or 3 of clinical illness. The rash is most intense on the trunk and limbs and may desquamate. In hospitalized patients with severe acute disease, uncommon but reported manifestations include cardiorespiratory, neurologic, renal, neurologic, and ocular complications; deaths have been reported, typically in the context of large outbreaks. Young children develop less prominent symptoms and signs and are therefore less frequently hospitalized. Children also often develop a bullous rather than a maculopapular/petechial rash. Although pregnant women are not at risk of more severe disease, there is a high risk of maternal–fetal transmission, particularly during the intrapartum period, that, in some cases, has led to fetal death. Recovery may require weeks, and a significant portion of middle-aged to older patients develop chronic arthritis or arthralgia syndromes (typically involving the same joints) that may be disabling. The risk of chronic musculoskeletal syndromes is increased in patients with preexisting osteoarthritis or severe acute disease who test positive for the human leukocyte antigen B27 subtype (HLA-B27) and may be associated with cryoglobulins. Although petechiae and epistaxis are occasionally seen, CHIKV should not be considered as a cause of VHF. Laboratory abnormalities may include transient lymphopenia and mild thrombocytopenia, as well as elevated activities of AST and concentrations of C-reactive protein (CRP). Treatment of chikungunya virus disease relies on acetaminophen and nonsteroidal anti-inflammatory drugs (NSAIDs). For patients with refractory arthritis, glucocorticoids or disease-modifying antirheumatic drugs (DMARDs) may be considered. A live-attenuated vaccine to prevent chikungunya virus disease has been approved by the U.S. Food and Drug Administration (FDA) and is undergoing postmarketing evaluation. Other vaccine and therapeutic candidates in preclinical development may benefit from the designation of CHIKV as the representative pathogen for arthritic alphaviruses.

Ross River Disease and Barmah Forest Virus Infection Ross River virus (RRV) and Barmah Forest virus (BFV) cause diseases that are clinically indistinguishable (hence the previously common disease designation of “epidemic polyarthritides” for both infections). RRV has caused epidemics in Australia, Papua New Guinea, and the South Pacific since the beginning of the 20th century. In 1979 and 1980, the virus swept through the Pacific Islands, causing $>500,000$ infections. From 1991 to 2011, the virus caused 92,559 infections in rural and suburban areas of Australia. From 2014 to 2015, $>10,000$ cases were recorded in Australia. RRV is predominantly transmitted by *Aedes normanensis*, *Aedes vigilax*, and *Culex annulirostris* mosquitoes. Marsupials (e.g., kangaroos, wallabies, and koalas) and rodents are probably the main vertebrate hosts, but infection also occurs in horses and other livestock. BFV infections have been increasingly documented since the early 1990s. For instance, from 1991 to 2011, 21,815 cases were recorded in Australia, and new data indicate that the disease also occurs in Papua New Guinea. BFV is transmitted by both *Aedes* and *Culex* mosquitoes and has been isolated from biting midges. The vertebrate hosts remain to be determined, but serologic studies implicate horses and possums.

Of the human RRV and BFV infections surveyed, 55–75% were asymptomatic; however, clinical illness can be debilitating when it occurs. After a typical incubation period of 3–9 days, patients develop the sudden onset of symmetric joint pain predominantly involving the peripheral extremities. Generally, a nonitchy, diffuse, maculopapular rash (more common in BFV infection) develops coincidentally or follows shortly; but, in some patients, rash can precede joint pain by several days. Constitutional symptoms (e.g., low-grade fever, asthenia, headache, myalgia, and nausea) are not prominent or are absent in many cases. Most patients are incapacitated for considerable periods (6 months or more) by joint involvement that interferes with grasping, sleeping, and walking. Ankle, interphalangeal, knee, metacarpophalangeal, and wrist joints are most often involved, although elbows, shoulders, and toes may also be affected. Periarticular swelling and tenosynovitis are common, and one third of patients have true arthritis (more common in Ross River disease). Myalgia and nuchal stiffness may accompany joint pain. Only half of all patients with arthritis can resume normal activities within 4 weeks, and 10% continue to limit their activities after 3 months. Occasionally, patients are symptomatic for >1 year but without progressive arthropathy.

In the diagnosis of either infection, clinical laboratory values are normal or variable. Tests for rheumatoid factor and antinuclear antibodies are negative, and the erythrocyte sedimentation rate is acutely elevated. Joint fluid contains 1000–60,000 monocytes per microliter, and viral antigen can usually be detected in macrophages. Serologic demonstration of rising antibody titers is the cornerstone of diagnosis, though it should be noted that virus-specific IgM antibodies may persist for month to years. Isolation of the virus from blood after mosquito inoculation or growth of the virus in cell culture is possible early in the illness. Because of the great economic impact of annual epidemics in Australia, an inactivated RRV vaccine has been under advanced development; phase 3 trials were completed in 2015 with promising results, but the candidate vaccine has not yet been developed for the market. Analgesics or NSAIDs are used for symptomatic treatment.

Sindbis Virus Infection Sindbis virus is typically transmitted to birds primarily by infected *Culex* and *Culiseta* mosquitoes that also vector human transmission over a range that includes Africa, Asia, Europe, and Australia. Infections with northern European or southern African variants are particularly likely in rural environments. Although infections may be subclinical, rash and arthralgia typically develop after an incubation period of <1 week. Constitutional clinical signs are not marked, and fever is modest or absent. The rash, which lasts ≈1 week, begins on the trunk, spreads to the extremities, and evolves from macules to papules that often vesiculate. The arthritis is polyarticular, migratory, and incapacitating, with resolution of the acute phase in a few days. The ankles, elbows, knees, phalangeal joints, wrists, and—to a much lesser extent—proximal and axial joints are involved. Persistence of joint pain and occasionally arthritis may continue for months or even years despite lack of deformities.

■ ENCEPHALITIS

The major encephalitis viruses are found in the families *Flaviviridae*, *Peribunyaviridae*, *Rhabdoviridae*, and *Togaviridae*. However, individual agents of other families, including Dhori virus and Thogoto virus (*Orthomyxoviridae*) and Banna virus (*Sedoreoviridae*), have caused isolated cases of encephalitis. Arboviral encephalitides are seasonal diseases, commonly occurring in the warmer months. Their incidence varies markedly with time and place, depending on ecologic factors that determine vector activity and human exposure. The causative viruses differ substantially in terms of the ratio of clinical to subclinical infections, case-fatality rate (CFR), and postacute sequelae. Humans are not considered important amplifiers of these viruses.

All the viral encephalitides discussed in this section share a similar pathogenesis. An infected arthropod ingests blood from a human and, through this contact, initiates infection. Initial viremia, thought to originate from the lymphoid system, leads to multifocal entry into the CNS, presumably via infection of olfactory neuroepithelium and subsequent passage through the cribriform plate via infected macrophages

or infection of brain capillary endothelial cells. During the viremic phase, there may be little or no recognizable disease except in tick-borne orthoflavivirus encephalitides, which have clearly delineated phases of fever and systemic illness.

CNS dysfunction or damage and associated clinical symptoms/signs arise partly from direct neuronal infection and subsequent damage and partly from edema, inflammation, and other indirect effects of the host response. The usual pathologic features of arboviral encephalitides are focal necroses of neurons, inflammatory glial nodules, and perivascular lymphoid cuffing. Involved areas display the “luxury perfusion” phenomenon, with normal or increased total blood flow and low oxygen extraction. The typical patient presents with a prodrome of nonspecific constitutional symptoms and signs, including fever, abdominal pain, sore throat, and respiratory signs. Headache, meningeal signs, photophobia, and vomiting quickly follow. The severity of human infection widely varies from an absence of symptoms/signs to febrile headache, aseptic meningitis, and full-blown encephalitis. The proportions and severity of these manifestations vary with the infecting virus and with known (e.g., age) and mostly unknown host immunogenetic determinants. Encephalitic involvement of deeper brain structures may be signaled by lethargy, somnolence, and cognitive deficit detected by mental status examination. More severely affected patients are obviously disoriented and may become comatose. Tremors, loss of abdominal reflexes, cranial nerve palsies, hemiparesis, monoparesis, difficulty swallowing, limb-girdle syndrome, and frontal lobe signs are common. Spinal and motor neuron diseases have been documented after WNV and Japanese encephalitis virus infections. Seizures and focal signs may be evident early or may appear during the disease. Some patients present with an abrupt onset of fever, convulsions, and other signs of CNS involvement. Encephalitis usually lasts a few days to several weeks and may be fatal, or recovery may be slow (with weeks or months before the return of maximal recoverable function) or incomplete (with persisting long-term deficits). Difficulty concentrating, fatigability, tremors, and personality changes are common during recovery.

The diagnosis of arboviral encephalitides depends on careful history-taking (travel/exposures), physical examination (in particular, of a febrile patient with signs of CNS dysfunction), and virus-specific or serology-specific laboratory testing. Clinicians should (1) consider empirical acyclovir treatment for herpesvirus meningoencephalitis and antibiotic treatment for bacterial meningitis until test results are received; (2) exclude intoxication and metabolic or oncologic causes, including paraneoplastic syndromes, hyperammonemia, liver failure, and anti-*N*-methyl-D-aspartate (NMDA) receptor encephalitis; and (3) rule out a brain abscess or a cerebrovascular event. Leptospirosis, neurosyphilis, Lyme disease, cat-scratch disease, opportunistic infections in immunocompromised patients, and more recently described viral encephalitides (e.g., Nipah virus infection), among others, should be considered in the differential diagnosis if epidemiologically relevant. CSF examination usually shows a modest increase in white blood cell (WBC) counts—in the tens or hundreds or perhaps a few thousand. In early phases of disease, a significant proportion of CSF WBCs may be polymorphonuclear leukocytes (PMNs), but monocytes later predominate. CSF glucose concentrations are generally normal. There are exceptions to this pattern of findings. In eastern equine encephalitis, for example, PMNs may predominate during the first 72 h of disease, and hypoglycorrhachia may be detected. In lymphocytic choriomeningitis, lymphocyte counts may be in the thousands, and glucose concentrations may be diminished. A humoral immune response is usually detectable at or near the onset of disease. Both serum (sampled in the acute or convalescent phase) and CSF should be examined for IgM antibodies, and plaque-reduction neutralization testing and/or RT-PCR should be conducted to detect viruses. Viruses generally cannot be isolated from blood or CSF, although Japanese encephalitis virus has been recovered from CSF of patients with severe disease. RT-PCR analysis of CSF may yield positive results but cannot exclude diagnoses. Metagenomic/high-throughput sequencing of CSF may support diagnosis of arboviral encephalitides. Viral antigen is present in brain tissue, although its distribution may be focal. Electroencephalography usually shows diffuse abnormalities and thus is not directly helpful.

Experience with medical imaging is still evolving but is generally nonspecific, as most patients do not present with pathognomonic abnormalities; however, imaging may be useful to rule out other suspected causes of disease. Both computed tomography (CT) and magnetic resonance imaging (MRI) scans may show normal results except for evidence of preexisting conditions or occasional diffuse edema. Notably, images acquired early may be normal and imaging abnormalities may only be detected later as the disease evolves. Although not pathognomonic, characteristic imaging abnormalities for some arboviral encephalitides have been reported, such as with eastern equine encephalitis (focal abnormalities) and severe Japanese encephalitis (hemorrhagic bilateral thalamic abnormalities).

Supportive care for severely ill patients may require management of elevated intracranial pressure, the syndrome of inappropriate secretion of antidiuretic hormone, respiratory failure, or seizures. Specific therapies for these viral encephalitides are not available. The only practical preventive measures are vector management and personal protective measures against the arthropod transmitting the virus. Direct human-to-human transmission has not been confirmed. For Japanese or Central European/Far Eastern tick-borne encephalitides, vaccination should be considered in specific circumstances (see relevant sections that follow).

Flavivirids The most significant orthoflavivirus encephalitides are Central European/Far Eastern tick-borne encephalitis, Japanese encephalitis, St. Louis encephalitis, and WNV infection. Murray Valley encephalitis and Rocio virus infection resemble Japanese encephalitis but are documented only occasionally in Australia and Brazil. Powassan virus has caused ≈ 400 cases of often-severe disease (CFR $\approx 8\%$), frequently occurring among children in eastern Canada and the United States. Usutu virus causes sporadic human infections (e.g., 105 cases from 2012 to 2021 in Europe), but such infections may be underdiagnosed.

CENTRAL EUROPEAN/FAR EASTERN TICK-BORNE ENCEPHALITIDES Tick-borne encephalitis virus (TBEV) is currently subdivided into four groups: the western/European subtype (previously called Central European encephalitis virus), the (Ural-)Siberian subtype (previously called Russian spring-summer encephalitis virus), the Far Eastern subtype, and the louping ill subtype (previously called louping ill virus, or, in Japan, Negishi virus). Small mammals, grouse, deer, and sheep are the vertebrate amplifiers for these viruses, which are transmitted by ticks. The risk of infection varies by geographic area and can be highly localized. Human infections usually follow outdoor activities during which tick bites occur or consumption of raw (unpasteurized) milk or cheese from infected goats or, less commonly, infected cows or sheep. Milk seems to represent the main transmission route for louping ill subtype viruses, which cause disease very rarely. Several thousand infections with TBEV are recorded each year among people of all ages. Tick-borne encephalitis occurs between April and October, with a peak in June and July.

Western/European viruses classically caused bimodal disease. After an incubation period of 7–14 days, the illness begins with an influenza-like febrile prodrome (fever, arthralgia/myalgia, headaches, and nausea and/or vomiting) that lasts for 2–4 days and is thought to correlate with viremia. A subsequent remission for several days is followed by the recurrence of fever and the onset of neurologic signs. The neuro-invasive phase (7–10 days before onset of improvement) varies from mild aseptic meningitis (more common among younger patients) to severe (meningo)encephalitis with coma, seizures, tremors, and motor signs. Spinal and medullary involvement can lead to typical limb-girdle paralysis and respiratory paralysis. Most patients with western/European virus infections recover (CFR $\approx 1\text{--}2\%$), and only a minority of patients have significant residual deficits. However, patients with (Ural-)Siberian virus infections have worse outcomes (CFR $\approx 7\text{--}8\%$) and are more likely to experience prolonged infections and develop chronic disability.

Infections with Far Eastern viruses generally run a more abrupt and severe course. The encephalitic syndrome caused by these viruses

sometimes begins without a remission from the fever-myalgia phase and has more severe manifestations than the western/European syndrome. CFR is high ($\approx 20\text{--}40\%$), and major sequelae—most notably, lower motor neuron paralyzes of the proximal muscles of the extremities, trunk, and neck—are common, developing in approximately half of patients. Thrombocytopenia may occur during the initial febrile illness, resembling the early hemorrhagic phase of some other tick-borne orthoflavivirus infections, such as Kyasanur Forest disease. In the early stage of the illness, virus may be detected by PCR or isolated from the blood; however, after the onset of CNS manifestations, virus cannot typically be detected in or isolated from the CSF, and diagnosis requires detection of IgM antibodies in serum and/or CSF.

Diagnosis of Central European/Far Eastern tick-borne encephalitis primarily relies on serology and detection of viral genomes by RT-PCR. There is no specific therapy for infection. However, effective alum-adsorbed, formalin-inactivated virus vaccines (FSME-IMMUN and Encepur) are produced in Austria, Germany, and Russia in chicken embryo cells. Two doses of the Austrian vaccine separated by an interval of 1–3 months appear to be effective in the field, though antibody responses are similar when vaccine doses are given 2 weeks apart. Because rare cases of postvaccination Guillain-Barré syndrome have been reported, vaccination should be reserved for people likely to experience rural exposure in an endemic area during the season of transmission. Cross-neutralization for the western/European and Far Eastern variants has been established, but there are no published field studies on cross-protection among formalin-inactivated vaccines.

Because up to 4% of ticks in endemic areas may be infected, the use of immunoglobulin prophylaxis of Central European/Far Eastern tick-borne encephalitis has increased in some regions. Prompt administration of high-titer-specific immunoglobulin is routine in some areas (e.g., Russia) but has been discontinued in many European countries because of concerns for antibody-mediated enhancement of infections and/or disease.

JAPANESE ENCEPHALITIS Japanese encephalitis is the most significant viral encephalitis in Asia. Each year, $\approx 68,000$ cases and $\approx 13,600\text{--}20,400$ deaths are reported. Japanese encephalitis virus is found throughout Asia—including in the Russian Far East, Japan, China, India, Pakistan, and Southeastern Asia—and causes occasional epidemics on western Pacific Islands. The virus has previously been considered endemic in Australia's Torres Strait Islands; from 2021 to 2022, 45 cases causing seven deaths were identified across a broad region of eastern and southern Australia. The virus is particularly common in areas where irrigated rice fields attract the natural avian vertebrate hosts and provide abundant breeding sites for *Culex tritaeniorhynchus* mosquitoes, which transmit the virus to humans. Amplification by pigs, which subsequently abort pregnancies, and horses, which develop encephalitis, may be significant, as well. Vaccination of domestic pigs and horses may reduce the transmission of the virus. After an incubation period of 5–15 days, clinical signs of Japanese encephalitis range from nonspecific febrile illness (nausea, vomiting, diarrhea, and/or cough) to aseptic meningitis, meningoencephalitis, acute flaccid paralysis, and severe encephalitis. Common findings are cerebellar signs, cranial nerve palsies, and cognitive and speech impairments. A Parkinsonian presentation and seizures are typical in severe cases. Patients may present with neuropsychiatric manifestations, including abnormal behavior and acute psychosis. MRI detection of thalamic abnormalities is specific but insensitive. CFR in hospitalized patients is high ($\approx 20\text{--}30\%$) and long-term neurologic dysfunction and disability are common in survivors. However, most long-term residents of endemic areas eventually seroconvert after natural environmental exposures and develop protective immunity. Effective vaccines are available and indicated for naïve individuals who are relocating or traveling frequently or over extended periods to endemic rural areas. Usually, two intramuscular doses of the vaccine are given 28 days apart, with the second dose administered at least 1 week prior to travel.

ST. LOUIS ENCEPHALITIS St. Louis encephalitis virus is transmitted among birds by mosquitoes. This virus causes a low-level endemic infection among rural residents of the central and western United States,

where *Culex tarsalis* mosquitoes serve as vectors. Urban mosquitoes (*Culex pipiens* and *Culex quinquefasciatus*) have been responsible for epidemics, resulting in hundreds or even thousands of cases in cities of the central and eastern United States. Most cases occur in June through October in the United States, but sporadic cases of the disease have been noted throughout the year in Latin/Central America and the Caribbean. The urban mosquitoes breed in accumulations of stagnant water and sewage with high organic content. The elimination of open sewers and trash-filled drainage systems is expensive and may not be possible. However, installation of screens on windows and vents of houses and implementation of personal protective measures may be effective in preventing infection. Mosquitoes are most active outdoors at dusk; bites can be avoided by modifying activities and using repellents.

Most St. Louis encephalitis virus infections are subclinical; susceptibility to severe disease increases with age. Infections that result in aseptic meningitis or mild encephalitis are concentrated among children and young adults, whereas severe and/or fatal cases primarily affect the elderly. Infection rates are similar in all age groups; a pathophysiologic explanation for higher risk of severe disease in older individuals is unclear. After an incubation period of 4–21 days, patients typically present with a nonspecific prodrome (fever, malaise, myalgia, and/or headache), followed by rapid-onset CNS manifestations with neurologic abnormalities that commonly include nuchal rigidity, hypotonia, hyperreflexia, myoclonus, and tremors. Severe cases can include cranial nerve palsies, hemiparesis, and seizures. Of interest, during and after the prodrome, patients often report dysuria and may have viral antigenuria and pyuria. Overall CFR is $\approx 7\%$ but may be increased up to $\approx 20\%$ among patients >60 years of age. Recovery is slow, and emotional lability, difficulty concentrating, memory issues, asthenia, and tremors are commonly prolonged in older convalescent patients. The CSF of patients with St. Louis encephalitis usually contains tens to hundreds of WBCs, with a lymphocytic predominance and a left shift. The CSF glucose concentration is normal in these patients.

WEST NILE VIRUS INFECTION WNV is now the primary cause of arboviral encephalitis in the United States. From 1999 to 2022, 28,684 cases of neuroinvasive disease (e.g., meningitis, encephalitis, and acute flaccid paralysis), with 2641 deaths, and 26,891 cases of nonneuroinvasive infection, with 135 deaths, were reported. WNV was initially described as being transmitted among wild birds by *Culex* mosquitoes in Africa, Asia, and southern Europe. In addition, the virus has been implicated in severe and fatal hepatic necrosis in Africa. WNV was introduced into New York City via diseased birds in 1999 and subsequently spread to other areas of the northeastern United States, causing die-offs among crows, exotic zoo birds, and other birds. The virus has continued to spread and is now found in all states (of the United States) as well as in Canada, Mexico, South America, and the Caribbean islands. *C. pipiens* mosquitoes remain the major vectors in the northeastern United States, but mosquitoes of several other *Culex* species and Asian tiger mosquitoes are also involved. Jays compete with crows and other corvids as amplifiers elsewhere in the United States.

The majority of WNV infections are subclinical. After an incubation period of 3–14 days, 20% of those infected develop typical F&M syndrome (without CNS involvement); $<1\%$ develop aseptic meningitis and severe encephalitis, particularly among the elderly. The febrile syndrome associated with WNV infection is notable for the frequent—rather than occasional—appearance of a maculopapular rash concentrated on the trunk (especially in children) and the development of lymphadenopathy. Back pain, fatigue, headache, myalgia, retroorbital pain, sore throat, nausea and vomiting, and arthralgia (but not arthritis) are common symptoms. Most patients fully recover, though asthenia may persist for several weeks. Patients who develop neuroinvasive disease typically have meningitis, encephalitis, or acute flaccid paralysis syndromes, often with some overlap. The risk of encephalitis, neurologic sequelae, and death is increased in male, elderly, diabetic, and hypertensive patients and in patients with previous CNS disease, but the pathophysiologic determinants of such a wide spectrum of disease phenotype and severity have otherwise been unexplained.

Host genetic risk factors for neuroinvasive disease and death may include chemokine receptor CCR5 deficiency. Recent identification of a high prevalence ($\approx 35\%$) of preexisting autoantibodies against type I interferons in patients with encephalitis (vs patients with subclinical WNV infection, in whom the prevalence approximated the general population) suggests this predisposition may underlie 40% of cases of WNV encephalitis. In addition to the more severe motor and cognitive sequelae, milder neurologic findings may include cranial nerve palsies, tremor, slight abnormalities in motor skills, and loss of executive functions. Given the burden of disease, intense clinical interest, and widely available diagnostics, more unusual features are increasingly described (including chorioretinitis, myocarditis, myositis and rhabdomyolysis, orchitis, flaccid paralysis with histologic lesions resembling poliomyelitis, and initial presentation with fever and focal neurologic deficits in the absence of diffuse encephalitis). Immunosuppressed patients may have fulminant courses or may develop persistent CNS infection. Postacute neurologic sequelae are common, especially in patients with severe neuroinvasive disease syndromes, and thus far have persisted in some patients for >5 years. Virus transmission through both transplantation and blood transfusion has necessitated screening of blood and organ donors by nucleic acid–based tests. A low but non-zero risk of maternal-to-fetal/neonatal transmission has been reported. Diagnosis rests upon detection of IgM antibodies in serum or CSF. In patients with potential prior exposures to sero-cross-reactive flaviviruses, plaque-reduction neutralization testing for specific neutralizing antibodies or molecular testing (PCR) may aid in specific diagnosis. Treatment is supportive only, and ventilatory support may be required for severe neuroinvasive disease. Although an equine vaccine is available, prevention of WNV infection in humans relies on avoidance of mosquito bites, vector control, and safe handling of potentially infected carcasses.

Peribunyavirids • CALIFORNIA ENCEPHALITIS California encephalitis virus has been implicated in only a very few cases of encephalitis (California encephalitis *sensu stricto*), whereas its close relative, La Crosse virus (LACV), is the major cause of this disease (historically ≈ 80 – 100 cases per year in the United States, although prevalence has been lower in recent years). La Crosse encephalitis is most reported in the upper midwestern United States but is also found in other areas of the central and southeastern parts of the country, such as West Virginia, Tennessee, North Carolina, and Georgia. Inkoo, Jamestown Canyon, Lumbo, snowshoe hare, and Tahyňa viruses are close relatives of LACV that also cause human disease (California encephalitis *sensu lato*, including La Crosse encephalitis). Transovarial infection is a strong component of transmission of these viruses in *Aedes* and *Ochlerotatus* mosquitoes. The vector of LACV is the *Ochlerotatus triseriatus* mosquito. These mosquitoes are infected by transovarial transmission, venereal transmission, and feeding on viremic chipmunks and other mammals. *O. triseriatus* mosquitoes breed in standing water in locations such as tree holes and abandoned tires; they bite during daylight hours. Risk factors for human cases include recreation in forested areas, residence at a forest's edge, and nearby standing water. Intensive environmental modification based on these findings has reduced the incidence of disease in a highly endemic area in the midwestern United States.

Most humans are infected from July through September. Asian tiger mosquitoes efficiently transmit LACV to mice and have transmitted it transovarially in the laboratory. This aggressive anthropophilic mosquito has the capacity to urbanize, and its possible transmission of virus to humans is of concern. The prevalence of antibody to LACV in humans is 20% or higher in endemic areas, indicating that infection is common but most often asymptomatic or subclinical.

Neuroinvasive disease after LACV infection varies from aseptic meningitis accompanied by confusion to severe and occasionally fatal encephalitis (CFR $<0.5\%$). CNS disease has been recognized primarily in children <15 years of age. After an incubation period of ≈ 3 – 7 days, a usual nonspecific febrile prodrome is followed by the sudden onset of CNS manifestations, including headache and lethargy, often with nausea, vomiting, convulsions (in half of patients), and coma (in

one-third of patients). Focal seizures, hemiparesis, tremor, aphasia, chorea, Babinski signs, and other evidence of significant neurologic dysfunction are common during acute neuroinvasive disease. Most patients recover completely, but $\approx 10\%$ develop postacute sequelae, including recurrent seizures, focal neurologic deficits, and cognitive and behavioral problems that may affect learning abilities.

The WBC count is commonly elevated and left-shifted in patients with LACV infection, sometimes reaching 20,000/ μL . CSF leukocyte counts are typically 30–500/ μL , usually with a monocyte predominance (although 25–90% of cells are PMNs in some patients). The blood protein concentration is normal or slightly increased, and the glucose concentration is normal. Specific virologic diagnosis based on IgM-capture assays of serum and CSF is efficient. The only human anatomic site from which virus has been isolated is the brain.

Treatment is supportive over a 1- to 2-week acute phase and focused on the prevention and management of status epilepticus, cerebral edema, and the syndrome of inappropriate secretion of antidiuretic hormone. A phase 2B clinical trial of IV ribavirin in children with LACV infection was discontinued during dose escalation because of adverse effects.

Jamestown Canyon virus has been implicated in several cases of encephalitis in adults (≈ 10 –40 cases per year since 2011), usually with a significant respiratory illness at onset. Human infection with this virus has been documented in Northern America (in the United States in at least 10 states and in Canada), where the vector mosquito (*Aedes stimulans*) feeds on its main host, the white-tailed deer (*Odocoileus virginianus*). Tahyna virus can be found in Africa, China, Central Europe, and Russia. The virus is a prominent cause of undifferentiated febrile disease but can also cause pharyngitis, pulmonary syndromes, aseptic meningitis, or meningoencephalitis.

Rhabdovirids • CHANDIPURA VIRUS INFECTION Chandipura virus is an emerging and increasingly significant human virus in India, where it is transmitted among hedgehogs by mosquitoes and sandflies. In humans, the disease begins as a rapid-onset influenza-like illness, with fever, headache, abdominal pain, nausea, and vomiting. These manifestations are followed by infection-related or autoimmune-mediated encephalitis that may cause altered mental status and seizures. Notably, neither the virus nor abnormal cells are detected in CSF; the brain has not been examined histopathologically, but limited cerebral imaging data query whether CNS disease is more likely related to cerebrovascular events (vasospasm/vasculitis) than to infectious encephalitis. Several hundred cases of Chandipura virus infection are recorded in India every year in children, with a CFR ≈ 45 –80%. Infections with other arthropod-borne rhabdovirids (Isfahan, Piry, vesicular stomatitis Indiana, and vesicular stomatitis New Jersey viruses) may imitate the early febrile stage of Chandipura virus infection.

Togavirids • EASTERN EQUINE ENCEPHALITIS This disease is encountered primarily in swampy foci along the east coast and, more recently and specifically, the southeastern coast of the United States, with a few inland foci as far removed from the coastlines as Michigan, Wisconsin, Arkansas, and Montana (≈ 4 –40 cases/year). Humans are mostly infected from June through October. During this period, *Culiseta* mosquitoes infect birds and virus spills over into other vectors, such as *Aedes sollicitans* or *Aedes vexans* mosquitoes, which are more likely to feed on mammals. Especially with spread to the southeastern coast, there is concern over the amplifying role of introduced Asian tiger mosquitoes, which have been found to be infected with eastern equine encephalitis virus and are an effective experimental vector in the laboratory. Horses are a common target for the virus. Contact with unvaccinated horses may be associated with human disease, but horses probably do not play a significant role in amplification of the virus.

Most infections are subclinical. After an incubation period of ≈ 5 –10 days, symptomatic individuals develop a nonspecific febrile prodrome, after which a small proportion (2% of adults, 6% of children) develop sudden and rapidly progressive neuroinvasive disease with (meningo)encephalitis syndromes associated with focal neurologic deficits, seizures, profoundly altered mental status, and coma.

Neuroinvasive disease is highly fatal (CFR ≥ 30 –50%) and $>50\%$ survivors have residual postacute sequelae. Acute PMN CSF pleocytosis, often occurring during the first 1–3 days of disease, is another indication of severity, and it parallels peripheral leukocytosis with a left shift. Extensive necrotic lesions and PMN infiltrates are found at post-mortem examination of the brain. Specific treatment is not available, although the therapeutic role of IV immunoglobulin (IVIG) remains under query. A formalin-inactivated vaccine has been used to protect laboratory workers but is not generally available or applicable.

VENEZUELAN EQUINE ENCEPHALITIS VEEV is separated into epizootic viruses (subtypes IA/B and IC) and enzootic viruses (subtypes ID, IE, and IF). Closely related enzootic viruses are Everglades virus, Mucambo virus, and Tonate virus. Enzootic viruses are found primarily in humid tropical-forest habitats and are maintained between culicoid mosquitoes and rodents. These viruses cause acute febrile syndromes in humans but are not pathogenic for horses and do not cause epizootics. Everglades virus has caused sporadic cases of encephalitis in humans in Florida. Extrapolation from the rate of genetic change suggests that Everglades virus may have been introduced into Florida <200 years ago. Everglades virus is most closely related to the ID-subtype viruses that appear to have given evolutionary rise to the epizootic variants active in South America.

Epizootic viruses have an unknown natural cycle but periodically cause extensive epizootics/epidemics in equids and humans in the Americas. These epizootics/epidemics are the result of high-level viremia in horses and mules, which transmit the infection to several types of mosquitoes, which in turn infect humans. Humans also have high-level viremia, but their role in virus transmission is unclear. Relatively restricted epizootics of Venezuelan equine encephalitis (VEE) occurred repeatedly in South America in <10 -year intervals from the 1930s until 1969, when a massive epizootic, including tens of thousands of equine and human infections, spread throughout Central America and Mexico, reaching southern Texas in 1971. Genetic sequencing suggested that the virus from that outbreak originated from residual “un-inactivated” IA/B-subtype virus in veterinary vaccines. The outbreak was terminated in Texas with a live-attenuated vaccine (TC-83), originally developed for human use by the U.S. Army; the epizootic virus was then used for further production of inactivated veterinary vaccines. No further major outbreaks occurred until 1995 and 1996, when large epizootics of VEE occurred in Colombia/Venezuela and Mexico, respectively. Of the $>85,000$ clinical cases, 4% (more children than adults) had neurologic symptoms/signs, resulting in 300 deaths. The viruses involved in these epizootics, as well as previously epizootic IC viruses, are close phylogenetic relatives of known enzootic ID viruses. This finding suggests that active virus evolution and selection of epizootic viruses are underway in South America.

During epizootics, extensive human infection is typical, with clinical disease occurring in 10–60% of infected individuals. Most infections result in notable acute febrile syndromes, whereas relatively few (5–15%) result in neurologic disease. A low rate of CNS invasion is supported by the absence of encephalitis after the many infections (from exposure to aerosols) that have occurred in the laboratory setting. Small-molecule and monoclonal antibody-based antiviral therapeutics are in preclinical development but not yet approved. These (and vaccine; see below) efforts may benefit from the designation of VEEV as a representative pathogen for the encephalitic alphaviruses.

The prevention of epizootic VEE depends on vaccination of horses with the attenuated TC-83 vaccine or with an inactivated vaccine prepared from that variant. Enzootic viruses are genetically and antigenically different from epizootic viruses, and protection against the former with vaccines prepared from the latter is relatively ineffective. Humans can be protected by immunization with similar vaccines prepared from Everglades virus, Mucambo virus, and VEEV, but the use of the vaccines is restricted to laboratory personnel because of reactogenicity, possible fetal pathogenicity, and limited availability.

WESTERN EQUINE ENCEPHALITIS The primary maintenance cycle of western equine encephalitis virus in the western United States and Canada involves *Aedes*, *C. tarsalis*, and *Culiseta* mosquitoes and

birds (principally sparrows and finches). Equids and humans become infected, and both suffer encephalitis without amplifying the virus. St. Louis encephalitis virus is transmitted via a similar cycle in the same regions harboring western equine encephalitis virus; disease caused by the former occurs about a month earlier than that caused by the latter (July through October). Large epidemics of western equine encephalitis occurred in the western and central United States and Canada from the 1930s through the 1950s but have been uncommon since then. From 1964 through 2010, only 640 cases were reported in the United States. This decline in incidence may reflect, in part, the integrated approach to mosquito management employed in irrigation projects and, in part, the increasing use of agricultural pesticides. The decreased incidence of western equine encephalitis almost certainly reflects the increased tendency for humans to be indoors behind closed or screened windows at dusk—the peak biting period of the major vector.

Most infections are subclinical or mild. After an incubation period of ≈ 5 –10 days, most symptomatic patients develop a nonspecific febrile prodrome with spontaneous recovery. The small minority of patients who develop neuroinvasive disease present with a typical viral meningoencephalitis syndrome (headache, vomiting, neck and back stiffness, and/or altered mental status). The frequency and morbidity of severe neurologic disease are increased among infants and very young children; in addition, CFR is high in this and in very elderly populations (3–7% overall). One-third of individuals who have convulsions during the acute illness have subsequent seizure activity. Infants <1 year of age—particularly those in the first months of life—are at serious risk of persistent motor and cognitive deficits. Of those 5–9 years of age, twice as many males as females develop clinical encephalitis; the contribution of biological versus behavioral factors (increased outdoor exposure to the vector) to incidence differences is uncertain. Specific treatment is not available. A formalin-inactivated vaccine has been used to protect laboratory workers but is not generally available.

■ PULMONARY DISEASE

Hantavirus pulmonary syndrome (HPS) was first described in 1993, but retrospective identification of cases by immunohistochemistry (1978) and serology (1959) supports the idea that HPS is a recently discovered disease rather than a truly new one. The causative agents are orthohantaviruses of a distinct phylogenetic lineage that is associated with the cricetid rodent subfamily Sigmodontinae. Sin Nombre virus, which chronically infects western deer mice (*Peromyscus sonoriensis*), is the most important agent of HPS in the United States. Several other related viruses (Anajatuba, Andes, Araraquara, Araucária, bayou, Bermejo, Black Creek Canal, Blue River, Caño Delgadito, Castelo dos Sonhos, Catacamas, Choclo, Juquitiba, Laguna Negra, Lechiguana, Maciel, Maripa, Monongahela, New York, Orán, Paranoá, Pergamino, Rio Mamoré, and Tunari viruses) cause the disease in Northern and South America. Andes virus is unusual in that it has been implicated in human-to-human transmission. HPS particularly affects rural residents living in dwellings permeable to rodent entry or workers in occupations that risk rodent exposure. Each type of rodent has particular habits; in the case of deer mice, these behaviors include living in and around human habitation.

After a typical incubation period of 2–3 weeks, HPS begins with a nonspecific prodrome of ≈ 3 –4 days (range 1–11 days) comprising fever, malaise, myalgia, and—in many cases—gastrointestinal disturbances (e.g., abdominal pain, nausea, and vomiting). Dizziness is common, and vertigo is occasional. Severe prodromal symptoms/signs may bring some patients to medical attention, but most cases are recognized as the pulmonary phase begins, with typical signs of slightly lowered blood pressure, tachycardia, tachypnea, mild hypoxemia, thrombocytopenia, and early radiographic signs of pulmonary edema. Physical findings in the chest are often surprisingly scant. The conjunctival and cutaneous signs of vascular involvement seen in orthohantavirus VHFs (see “Viral Hemorrhagic Fever,” below) are uncommon but may be more frequent with HPS outside Northern America. Early recognition and triage of patients at this stage are critical because, shortly after onset, hypotension and noncardiogenic edema progress rapidly to hypoxic respiratory failure.

The differential diagnosis of HPS includes undifferentiated sepsis syndromes and severe acute respiratory infections and should be informed by local epidemiology. Considerations include severe atypical pneumonia (*Legionella*, *Mycoplasma*, and Q fever), influenza, rickettsial disease, meningococcemia, septicemic plague, tularemia, leptospirosis, dengue, and YF. Fever with severe abdominal pain and tenderness merits consideration of the causes of a “surgical” abdomen and pyelonephritis. A specific diagnosis is best made when antigen-specific IgM, present in most symptomatic patients, is detected in acute-phase serum. Sin Nombre virus antigen testing detects antibodies to the related HPS-causing orthohantaviruses. Occasionally, heterotypic viruses will cross-react (only to the IgG ELISA), but the very low seroprevalence of these viruses in normal populations queries this laboratory finding. Orthohantaviral RNA is cleared quickly, although RT-PCR may be useful to detect virus in tissue biopsies, thrombus, or at autopsy. RT-PCR and DNA sequencing may be needed in specifically identifying the infecting virus in areas outside the home range of deer mice, in atypical cases, and for molecular epidemiology.

During the prodrome, differentiating HPS from other causes is difficult, but diagnostically helpful features emerge at or soon after initial presentation. A new dry cough, usually absent at disease onset, may signal the pulmonary phase. Early in disease, interstitial edema may be evident radiographically; with progression, bilateral alveolar infiltrates in a central pattern and with a normal-sized heart signal the development of noncardiogenic edema. Pleural effusions are often seen. Coincident thrombocytopenia, circulating atypical lymphocytes (lymphoblasts), and a left shift (often with marked leukocytosis) are considered almost diagnostic in experienced centers. Hemoconcentration, hypoalbuminemia, and proteinuria also aid in diagnosis. Although thrombocytopenia and partial thromboplastin time (PTT) prolongation are usually found, clinical evidence of coagulopathy or laboratory indications of DIC are found in only a minority of severely ill patients. Patients with severe illness also have acidosis and elevated serum lactate concentrations. Oliguria and mildly increased values in serum creatinine concentrations are common, but patients with severe HPS often have markedly decreased glomerular filtration. Some American orthohantaviruses other than Sin Nombre virus (e.g., Andes virus) have been associated with more severe renal dysfunction, but few such cases have been studied.

After recognition, early triage-based monitoring and management of hemodynamic, respiratory, and renal dysfunction during the first few hours are critical to improved outcomes. During this period, hypotension and hemoconcentration suggest intravascular volume depletion; however, aggressive IV fluid therapy is not advised in the setting of widespread capillary leak, frequently depressed cardiac output, and oliguria. Mild hypoxemia may be managed by oxygen administration alone, but mechanical ventilatory support of respiratory failure and shock is often needed. Pulmonary artery catheterization enables pulmonary capillary wedge pressure-guided inotropic and vasopressor support and cautious intravascular volume replacement. The most critically ill patients with a low cardiac index or refractory hypoxemia despite this support may benefit from extracorporeal membrane oxygenation, ideally before the onset of shock. CFR is high (≈ 30 –40%), even with appropriate management, but most patients surviving the first 48 h of hospitalization are extubated and discharged within a few days with no apparent long-term sequelae. Although ribavirin inhibits orthohantaviruses in vitro and has been effectively used in patients with HFRS due to Hantaan virus, an underpowered randomized clinical trial of ribavirin for HPS failed to show efficacy. Preclinical development of novel small-molecule- and monoclonal antibody-based therapeutics for HPS is ongoing.

■ VIRAL HEMORRHAGIC FEVER

VHF syndrome encompasses a small number of acute viral infections that cause severe disease and multisystem organ dysfunction sharing a mutual pathophysiologic disruption to vascular function, stability, and integrity. In practice, viral infections classified as VHFs vary widely across a spectrum of clinical phenotypes and severity; indeed, hemorrhagic manifestations are typically not the most prominent nor severe

features of disease. Nonetheless, the VHF designation largely captures a useful common theme: hemorrhagic manifestations are a sign of widespread vascular dysfunction or damage that plays a central role in local, organ-specific, and systemic expression of disease. Direct or indirect damage to the microvasculature leads to increased permeability and (particularly when platelet function is decreased) to actual disruption and local hemorrhage most evident in the skin and mucous membranes. Cutaneous flushing and conjunctival suffusion are examples of common observable abnormalities due to dysfunction of local circulation. Similarly, affected microvasculature may be fragile to trauma and external forces, such as the application of a blood pressure cuff causing a positive tourniquet sign. Although overt hemorrhage occurs infrequently, subtle local hemorrhagic signs (e.g., bleeding of the gums or at IV catheter sites) are often present on careful examination. Circulatory dysfunction may also contribute to specific organ dysfunction or damage that may be particularly prominent in some VHF cases; for instance, the kidneys are primary targets in HFRS, and the liver is a primary target in YF. Regarding severe systemic manifestations, hypotension and shock are not usually a direct result of life-threatening blood loss but rather an indication of generalized circulatory dysfunction contributing to hemodynamic instability. Despite this common theme, the pathogenesis of VHF is poorly understood and varies among the viruses regularly implicated in the syndrome. In some viral infections, direct damage to the vascular system or even to parenchymal cells of target organs is an important factor; in other viral infections, soluble mediators are thought to play a major role in the development of hemorrhage or hemodynamic derangement.

Symptom initiation in the acute phase in most VHF syndromes is associated with ongoing virus replication and viremia, typically with nonspecific fever and arthralgia/myalgia, usually of abrupt onset (mammarenavirus infections are exceptions, as they often develop gradually). Within a few days, patients present for medical attention with severe fatigue and prostration that is often accompanied by anorexia, severe headache, sore throat, chest pain, gastrointestinal symptoms (abdominal pain, nausea, vomiting, and/or diarrhea), dizziness, hyperesthesia, and photophobia. Initial examination often reveals only an acutely ill febrile patient with asthenia, conjunctival suffusion, tenderness to palpation of muscles or abdomen, and borderline hypotension or postural hypotension (perhaps with tachycardia). Petechiae (often best visualized in the axillae), flushing of the head and thorax, periorbital edema, and proteinuria are common. AST activities are usually elevated at presentation or within a day or two thereafter. Hemoconcentration from vascular leakage is usually evident and is most marked in HFRS and in severe dengue. Seriously ill patients develop more severe clinical signs, including initial organ dysfunction typical of the causative virus; continued high viral loads coupled with ineffective and dysregulated immune responses lead to progressive multiorgan dysfunction characterized by shock, acute kidney injury, CNS injury (encephalopathy, coma, seizures), severe hemorrhagic manifestations, and death.

One of the major diagnostic clues is travel to an endemic or active outbreak area within the incubation period for a given syndrome. Except in infections with Seoul virus, DENV-1–4, and yellow fever virus (YFV), which have urban hosts/vectors, rural travel exposures are more commonly associated with a VHF. In addition, several travel-associated diseases—falciparum malaria, shigellosis, typhoid fever, leptospirosis, relapsing fever, and rickettsial diseases—should also be considered in the differential diagnosis, as they are treatable despite being potentially fatal.

Early recognition of VHF is crucial for timely initiation of appropriate supportive clinical care and virus-specific therapy (if available). Key components of clinical care include prompt isolation and hospitalization even in absence of clinical signs; rapid triage and ongoing monitoring of vital signs; the prevention and management of dehydration, hypovolemia, and hemodynamic instability (with closely monitored IV fluids and vasopressors); the prevention and management of acute kidney injury and electrolyte abnormalities; empiric or specific treatment of common concurrent or secondary bacterial or parasitic infections; replacement of blood products (packed red blood cells, clotting factors,

and platelets as indicated); and the usual precautionary measures used in the treatment of patients with hemorrhagic diatheses. DIC should be treated only with clear laboratory evidence and if laboratory monitoring of therapy is feasible; there is no proven benefit of such therapy. Of note, VHF patients are commonly hypovolemic from insensible fluid and gastrointestinal losses and require intravascular fluids to maintain blood pressure and glomerular filtration; however, VHF patients may also have decreased cardiac output, capillary leak syndromes, and oliguria that require judicious fluid management and inotropic/vasopressor support. Specific therapy is available for several of the VHFs. Strict barrier nursing and other precautions against infection of medical staff and visitors are indicated when VHFs are encountered, except when the illness is due to DENV-1–4, orthohantaviruses, RVFV, or YFV.

Novel VHF-causing agents are still being discovered. Some of these viruses have recently emerged and continue to cause disease (e.g., severe fever with thrombocytopenia syndrome virus [SFTSV]). Others have caused isolated VHF-like cases without recurrence, including viruses for whom Koch's postulates are yet to be fulfilled (e.g., Bas-Congo virus).

Bunyaviricetes The most significant VHF-causing bunyaviricetes are arenavirids (Junín, Lassa, and Machupo viruses), hantavirids, nairovirids (Crimean-Congo hemorrhagic fever virus), and phenuivirids (RVFV and SFTSV). Other bunyaviricetes—e.g., the Garissa variant of Ngari virus and Ilesha virus (both orthobunyaviruses) or Chapare, Guanarito, Lujo, and Sabiá viruses (all mammarenaviruses)—have caused sporadic VHF outbreaks.

ARGENTINIAN AND BOLIVIAN HEMORRHAGIC FEVERS These severe diseases (with CFR reaching >50% in individual outbreaks) are caused by Junín virus and Machupo virus, respectively. Clinical presentations of both diseases are similar, but epidemiology differs because of the distribution and behavior of the distinct rodent reservoirs. Argentinian hemorrhagic fever has been recorded only in rural areas of Argentina, whereas Bolivian hemorrhagic fever seems to be confined to rural Bolivia. Unusually for arboviruses, infection with the causative agents almost always results in disease, and all ages and both sexes are affected. Person-to-person or nosocomial transmission is rare but has occurred. The transmission of Junín virus from male survivors to their partners suggests the need to avoid intimate contact in early convalescence (and counseling in this regard for all patients with mammarenavirus hemorrhagic fevers). In contrast to LF (see below), thrombocytopenia—often marked—is the rule, mucosal hemorrhage is common, and CNS dysfunction (e.g., marked confusion, tremors of the upper extremities and tongue, and cerebellar signs) is much more common in disease caused by Junín virus and Machupo virus (20–30%). Some cases follow a predominantly neurologic course, with a poor prognosis. Convalescence is protracted and frequently associated with sequelae such as hearing loss.

Typical clinical laboratory findings (thrombocytopenia, leukopenia, and proteinuria) support the diagnosis. Argentinian and Bolivian hemorrhagic fevers are readily treated with convalescent plasma given within the first 8 days of illness; however, the use of convalescent plasma may be associated with neurologic sequelae. In the absence of passive antibody therapy, (off-label) IV ribavirin in the dose recommended for LF is likely to be effective in all South American VHFs caused by mammarenaviruses. A safe, effective, live-attenuated vaccine exists for Argentinian hemorrhagic fever. After vaccination of >250,000 high-risk people in the endemic area, the incidence of this VHF decreased markedly. In experimental animals, this vaccine is cross-protective against Bolivian hemorrhagic fever. Other mammarenaviruses have caused isolated or sporadic VHF in South America, including Guanarito virus (Venezuelan hemorrhagic fever) and Chapare and Sabiá viruses. Multivalent vaccine approaches against all five VHF-causing mammarenaviruses in this region are in preclinical development.

LASSA FEVER Lassa virus (LASV) is known to cause endemic and epidemic disease in Nigeria, Sierra Leone, Guinea, and Liberia, although it is probably more widely distributed in Western Africa. In Western Africa alone, probably tens to hundreds of thousands of LASV

infections occur annually, and $\approx 20\%$ of these develop LF, which can be a prominent cause of febrile disease. For example, in one hospital in Sierra Leone, laboratory-confirmed LF is consistently responsible for one-fifth of admissions to the medical wards. LASV can be transmitted by close person-to-person contact. Nosocomial spread has occurred but is uncommon if appropriate infection prevention and control practices are followed. All ages and both sexes are affected; the incidence of disease is highest in the dry season, but transmission occurs year-round.

The majority of LASV infections are subclinical or mild. After a highly variable incubation period of 1–3 weeks, symptomatic patients have the gradual onset of a nonspecific febrile prodrome (among the VHF agents, only mammarenaviruses are typically associated with a gradual onset of illness). Progression in the second week of illness includes sore throat, cough, retrosternal chest and back pain, and gastrointestinal symptoms (abdominal pain, vomiting, and/or diarrhea). A maculopapular rash may be noted in light-skinned patients. Further disease progression includes conjunctival infection, head and neck edema, and respiratory (pneumonia and/or pleural effusions) and cardiac (pericarditis and/or pericardial effusions) manifestations. In late stages, multisystem organ dysfunction occurs with acute kidney injury, CNS manifestations (encephalopathy, coma, and/or seizures), shock, and hemorrhage (seen in only $\approx 15\text{--}30\%$ of LF patients overall). Laboratory abnormalities may include leukocytosis, thrombocytopenia, elevated serum creatinine levels, hyperkalemia, hypoalbuminemia, elevated AST and ALT activities, and proteinuria. Viral load, age, pregnancy status, clinical signs (CNS features, face and neck swelling, hypotension, hemorrhage, and/or jaundice), and laboratory abnormalities (high serum creatinine levels, AST activity, leukocytosis, and/or thrombocytopenia) have been associated with fatal outcome. Pregnancy is associated with higher case fatality, especially during the last trimester, when fetal death is 90%. Interruption of pregnancy via medical abortion may increase survival rates, but data on LF and pregnancy outcomes remain sparse. In survivors, a high rate ($\approx 30\%$) of sensorineural deafness (onset during the acute illness or in convalescence, usually bilateral) occurs; other postacute sequelae of LF are poorly defined but include neurologic and mental health issues. The virus is detected in the urine during convalescence and has been detected in seminal fluid early in recovery. Reinfection may occur but has not been associated with severe disease.

Aggressive supportive and critical care, including renal replacement therapy when available, is required to improve outcomes. Regarding LASV-specific therapy, observational studies in the 1980s informed the current practice of treating patients with (off-label) ribavirin (IV route preferred). Historically, ribavirin appeared to be partially effective in reducing fatality from that documented among retrospective controls. However, possible side effects, such as reversible anemia (which usually does not require transfusion), dependent hemolytic anemia, and bone marrow suppression, need to be kept in mind. When used, ribavirin should be given by slow IV infusion in a dose of 32 mg/kg; this dose should be followed by 16 mg/kg every 6 h for 4 days and then by 8 mg/kg every 8 h for 6 days. Small-molecule and monoclonal antibody therapeutics are in advanced preclinical development, and research networks in Western Africa are preparing for randomized clinical trials that will be critical to inform best specific therapeutic approaches. Inactivated LASV vaccines failed in preclinical studies, but several promising vaccine platforms are under experimental evaluation.

HEMORRHAGIC FEVER WITH RENAL SYNDROME HFRS is the most significant VHF today, with $>100,000$ cases of severe disease in Asia annually and thousands of mild infections in Europe. The disease is widely distributed in Eurasia. The major causative viruses are Puumala virus (Europe), Dobrava virus (the Balkans), and Hantaan virus (Eastern Asia). Amur, göu, Kurkino, Muju, Saaremaa, Sochi, and Tula viruses also cause HFRS but much less frequently and in more geographically confined areas that are determined by the distribution of reservoir hosts. Seoul virus is an exception; because it is associated with brown rats (*Rattus norvegicus*), which migrate on ships, the virus has a worldwide distribution. Despite the wide distribution of Seoul virus,

only mild or moderate HFRS occurs in Asia; human disease has been difficult to identify in many areas of the world, although an outbreak in North America has been described. Most cases of HFRS occur in rural residents or vacationers; again, the exception is Seoul virus infection, which may be acquired in an urban, rural, or laboratory setting. Classic Hantaan virus infection in Korea and in rural China is most common in the spring and fall and is related to rodent density and agricultural practices. Human infection is acquired primarily through aerosols of rodent urine, although virus is also present in rodent saliva and feces. Patients with HFRS are not infectious.

Most patients with hantavirus infections do not develop severe disease. Classic features of HFRS include fever, hypotension, hemorrhage, and acute kidney injury, with severe disease expressed in four identifiable stages after a typical 1- to 2-week incubation period:

1. The *febrile* stage lasts 3 or 4 days and is identified by the abrupt onset of fever, headache, severe myalgia, thirst, anorexia, and often nausea and vomiting. Photophobia, retroorbital pain, and pain on ocular movement are common, and the vision may become blurred with ciliary body inflammation. Flushing over the face, the anterior neck, and back is characteristic, as are pharyngeal injection, periorbital edema, and conjunctival suffusion. Petechiae often develop in areas of pressure, the conjunctivae, and the axillae. Back pain and tenderness to percussion at the costovertebral angle reflect massive retroperitoneal edema. Thrombocytopenia and mild to moderate DIC is present, and hemorrhagic manifestations may occur. Other early laboratory findings include proteinuria and an active urinary sediment.
2. During the *hypotensive* stage (lasting from a few hours to 48 h), the blood pressure falls (sometimes with shock) with a compensatory tachycardia (vs the relative bradycardia typical of the febrile phase). Kinin activation and capillary leak are marked, leading to hemoconcentration. Leukocytosis with a left shift develops, and thrombocytopenia persists. Atypical lymphocytes—which, in fact, are activated CD8⁺ and, to a lesser extent, CD4⁺ T cells—circulate. Renal perfusion is compromised from local and systemic circulatory changes resulting in necrosis of tubules, particularly at the corticomedullary junction, and oliguria. Consequent inability to concentrate the urine develops, and the urine's specific gravity falls to 1.010. Proteinuria is marked. Diffuse hemorrhage may appear, including hematemesis, hemoptysis, and melena.
3. During the *oliguric* stage, hemorrhagic tendencies continue, probably in large part because of uremic bleeding defects. Oliguria persists for 3–10 days before the return of renal function marks the onset of the polyuric stage.
4. During the *polyuric* stage (diuresis with hyposthenuria), the inability to concentrate the urine may lead to volume depletion and electrolyte abnormalities.

Mild cases of HFRS may be much less stereotypical, presenting only with fever, gastrointestinal abnormalities, and transient oliguria followed by hyposthenuria. Infections with Puumala virus, the most common cause of HFRS in Europe (“nephropathia epidemica”), result in a similar general but much-attenuated presentation. Bleeding manifestations are found in only 10% of patients, hypotension rather than shock is usually documented, and oliguria is present in only about half of patients. The dominant features may be fever, abdominal pain, proteinuria, mild oliguria, and sometimes blurred vision or glaucoma, followed by polyuria and hyposthenuria in recovery. CFR is $<1\%$.

HFRS should be suspected in patients with exposure in an endemic (typically rural) area. Prompt recognition of the disease permits rapid hospitalization and expectant management of shock and renal failure. Clinical laboratory parameters include leukocytosis (which may be leukemoid and with a left shift), thrombocytopenia, elevated serum creatinine level, proteinuria, and hematuria. HFRS is readily diagnosed by an IgM-capture ELISA that is positive at admission or within 24–48 h thereafter. The isolation of orthohantaviruses is difficult, but RT-PCR of renal biopsy, thrombus (collected early in the clinical course), or postmortem tissues may yield molecular diagnosis and should be used if definitive identification of the infecting virus is required.

Mainstays of therapy are management of shock with vasopressor support and modest crystalloid infusion, IV human serum albumin administration, timely renal replacement therapy to prevent volume overload that may result in pulmonary edema, and control of hypertension to prevent intracranial hemorrhage. Use of IV ribavirin has reduced CFR and morbidity in severe cases, provided treatment is begun within the first 4 days of illness. CFR approaches 15% but should be lower than 5% with adequate supportive and critical care. Generally, surviving patients make a full recovery, even of glomerular function, and postacute sequelae have not been identified.

CRIMEAN-CONGO HEMORRHAGIC FEVER (CCHF) CCHFV causes severe VHF over a wide endemic area (Africa, Asia, and Europe), determined primarily by the distribution of ixodid *Hyalomma* ticks. Incidence has been increasing in known endemic areas (particularly in Western Asia), and recent autochthonous infections in nonendemic areas, such as Spain, suggest that the at-risk geography is expanding. Because of the propensity of CCHFV-transmitting ticks to feed on domestic livestock and some wild mammals, veterinary serosurveys are the most effective mechanism for the monitoring of virus circulation. Human infections are acquired via tick bites, during the crushing of infected ticks, and contact with blood or body fluids of infected animals or humans (including nosocomial and vertical transmission). Domestic animals do not become ill but do develop viremia. Thus, risk of acquiring CCHFV occurs during sheep shearing, animal slaughter, or contact with infected hides or carcasses from recently slaughtered infected animals. Nosocomial outbreaks are common and are usually related to extensive blood exposure or needlesticks.

The majority of infections are subclinical. After an incubation period that varies by exposure to a tick bite (1–3 days) or blood or body fluids (3–7 days, perhaps shorter after nosocomial exposures to highly viremic patients), patients develop a sudden-onset nonspecific febrile prodrome typical to VHF. Severe illness is also typical, with the possible exceptions of a profound and dysregulated inflammatory component, more severe hemorrhagic manifestations (including extended ecchymoses and/or retinal and pulmonary hemorrhage), and more severe liver injury, resulting in jaundice in some patients. Recovery (for most patients; sequelae are not identified) or rapid progression to fatal disease occurs in the second week of illness with multiorgan (renal and respiratory) dysfunction, shock, and profound coagulopathy leading to massive bleeding. Clinical laboratory findings indicate coagulopathy as well as markers of liver (elevations in activities of AST/ALT and bilirubin concentrations) and muscle (elevations in creatine phosphokinase levels) injury. Predictors of fatal outcome include clinical features (hematemesis, melena, and/or altered mental status), laboratory biomarkers (viral load, thrombocytopenia, elevated AST/ALT levels, and/or obvious DIC), and delayed CCHFV-specific IgM and IgG responses. The mainstay of treatment is supportive and critical care, including extracorporeal support of organ dysfunction. The pathophysiology argues for combined virus-specific and disease-modifying immunomodulatory approaches. The benefit of IV ribavirin, suggested in observational studies that were likely confounded, has not been borne out in randomized trials. Small-molecule antivirals (favipiravir and molnupiravir), monoclonal antibodies, and the use of convalescent plasma are in preclinical evaluation, but clinical studies are needed. The routine use of steroids and/or IVIG is not currently recommended. No human or veterinary vaccines are recommended.

RIFT VALLEY FEVER The natural range of RVFV was previously confined to sub-Saharan Africa, with circulation of the virus markedly enhanced by substantial rainfall. The El Niño Southern Oscillation phenomenon of 1997 facilitated subsequent spread of Rift Valley fever (RVF) to the Arabian Peninsula, with epidemic disease in 2000. The virus has also been found in Madagascar and Egypt, where it caused major epidemics in 1977–1979, 1993, and thereafter. RVFV is maintained by transovarial transmission in floodwater *Aedes* mosquitoes and presumably also via a vertebrate amplifier. Increased transmission during particularly heavy rains leads to epizootics characterized by high-level viremia in cattle, goats, or sheep. Numerous

types of mosquitoes feed on these animals and become infected, thereby increasing the possibility of human infections. Remote sensing via satellite can detect the ecologic changes associated with high rainfall that predict the likelihood of RVFV transmission. High-resolution satellites can also detect the special depressions in floodwaters from which the mosquitoes emerge. The virus can be transmitted to humans by mosquitoes but most commonly by contact with blood or aerosols from domestic animals. Therefore, transmission risk is high during birthing, and both abortuses and placentas need to be handled with caution. Risk is also high during animal slaughter but decreases thereafter as anaerobic glycolysis in postmortem tissues results in an acidic environment that rapidly inactivates bunyaviricetes in carcasses. Neither person-to-person nor nosocomial transmission of RVFV has been documented.

RVFV is unusual in that it causes several clinical syndromes. Most infections are either subclinical or produce a typical F&M syndrome after an incubation period of 2–6 days. A much smaller proportion (<1%) of patients develop typical VHF notable for prominent liver involvement (jaundice) and an extremely rapid course with high case fatality (50%, within 3–6 days), often with renal failure and DIC/hemorrhagic manifestations. Further, <1% of patients develop a late onset (1–4 weeks) meningoencephalitis syndrome that is not usually fatal but does lead to severe neurologic sequelae. Perhaps 10% of otherwise mild infections develop a late-onset (1–3 weeks) retinal vasculitis causing blurred or loss of vision. Fundoscopic examination reveals edema, hemorrhages, and infarction of the retina as well as optic nerve degeneration. In a small proportion of patients (<1 in 200), retinal vasculitis is followed by viral encephalitis. Of interest, the late onset of both retinal disease and meningoencephalitis—after serum neutralizing antibody has developed—suggests (so far, undefined) immunopathology.

No proven therapy exists for RVF. Epidemic disease is best prevented by vaccination of livestock. The ability of this virus to propagate after introduction into Egypt suggests that other potentially receptive areas, including the United States, should develop response plans. RVF, like VEE, is likely to be controlled only with adequate stocks of an effective live-attenuated vaccine, but global stocks are unavailable. A formalin-inactivated vaccine confers immunity in humans; however, quantities are limited, and three injections are required. This vaccine is recommended for potentially exposed laboratory workers and for veterinarians working in sub-Saharan Africa. A new live-attenuated vaccine, MP-12, is being tested in humans (phase 2 trials have been completed). The vaccine is safe and licensed for use in sheep and cattle. A modified replication-incompetent chimpanzee adenovirus (ChAdOx1) vectored vaccine has recently been shown to be safe and immunogenic in phase 1 trials. In addition, several vaccines are being developed specifically for use in animals.

SEVERE FEVER WITH THROMBOCYTOPENIA SYNDROME This tick-borne disease is caused by SFTSV. The primary tick vector, the Asian long-horned tick (*Haemaphysalis longicornis*), is endemic in areas of southeast Asia and Oceania, and geographic distribution has been increasing, likely via migratory birds (in Asia) or by importation of pets or livestock (in the United States, where the vector is now widespread and considered invasive). Numerous human infections have been reported during the past few years from China, and cases have also been detected in Japan, the Republic of Korea, Vietnam, and Thailand. Seroprevalence in endemic areas of China approaches 5%. The clinical presentation ranges from mild nonspecific febrile syndromes to severe VHF with high CFR (>12%) particularly in older individuals. After a 1- to 2-week incubation period, patients develop a nonspecific febrile prodrome that may include abdominal pain, vomiting, and diarrhea. Common laboratory abnormalities include prominent thrombocytopenia, leukopenia (often with atypical lymphocytes), markers of hepatocyte injury, and elevated ferritin levels. Most patients recover spontaneously; however, in the second week of illness, some patients develop severe disease with multiorgan dysfunction that can include acute renal injury, myocarditis, meningoencephalitis, DIC with hemorrhage, and hemophagocytic lymphohistiocytosis. Diagnosis requires RT-PCR, as serologic tests are not available. No vaccines or approved

Flasuviricetes The most significant orthoflaviviruses that cause VHF are the mosquito-borne DENV-1–4 and YFV. These viruses are widely distributed and cause tens to hundreds of thousands of infections each year. Alkhurma hemorrhagic fever virus (>600 cases since discovery in 1995), Kyasanur Forest disease virus (≈10,000 cases over 60 years), and Omsk hemorrhagic fever virus (isolated infections every year with intermittent larger outbreaks) are geographically very restricted but prevalent tick-borne orthoflaviviruses that cause VHF, sometimes with subsequent viral encephalitis. Tick-borne encephalitis virus has caused VHF in a few patients. There is currently no therapy for infection with these VHFs, but an inactivated vaccine has been used in India to prevent Kyasanur Forest disease (and there is a vaccine to prevent tick-borne encephalitis; see above).

SEVERE DENGUE Although most individuals infected with DENV-1–4 have either subclinical infection or F&M syndrome (the febrile phase), some of these patients enter a critical (or plasma leakage) phase—often as fever declines—and develop the criteria for severe dengue: a definite or presumptive diagnosis of dengue infection plus either (1) plasma leakage severe enough to cause shock or respiratory distress, or (2) severe bleeding, or (3) severe organ dysfunction. Early recognition and action long before this critical period are crucial to initiate appropriate supportive care. Indeed, all patients with a presumptive diagnosis of dengue (of any severity) should be initially assessed for designated warning signs (dengue with warning signs, directing inpatient management) as well as for the criteria for severe dengue (i.e., a provisional diagnosis of severe dengue should be made clinically).

The complex determinants of risk for this progression to severe dengue include host and viral factors but center most notably around the potential for immune-mediated enhancement of disease. Several weeks after convalescence from infection with DENV-1–4, the transient protection conferred by that infection against reinfection with a heterotypic DENV usually wanes. Heterotypic reinfection may result in classic dengue without/with warning signs or, less commonly, in severe dengue. In the past 20 years, YF mosquitoes have progressively invaded Latin America and other areas, and frequent travel by infected individuals has introduced multiple variants of DENV-1–4 from many geographic areas. Thus, the pattern of hyperendemic transmission of multiple DENV serotypes established in the Americas and the Caribbean has led to the emergence of severe dengue as a major problem. Among the millions of DENV-1–4 infections, ≈500,000 cases of severe dengue occur annually, with a CFR of ≈2.5%. The induction of vascular permeability and shock depends on multiple factors, such as the presence or absence of enhancing and nonneutralizing antibodies, age (susceptibility to severe dengue drops considerably after 12 years of age), sex (females are more often affected than males), race (white people are more often affected than black people), nutritional status, and the timing and sequence of infections (e.g., DENV-1 infection followed by DENV-2 infection seems to be more severe than DENV-4 infection followed by DENV-2 infection). The presence of neutralizing antibodies is associated with decreased viremia on subsequent infection. In addition, considerable heterogeneity exists among variants in each DENV population. For instance, Southeastern Asian DENV-2 variants have more potential to cause severe dengue than do other variants. Recent evidence points to a key role for the DENV NS1 protein in the vascular leak phenomenon associated with severe dengue.

In milder cases of severe dengue, restlessness, lethargy, thrombocytopenia (<100,000/ μ L), and hemoconcentration are detected 2–5 days after the onset of typical dengue, usually at the time of defervescence. The maculopapular rash that often develops in dengue (without/with warning signs) may also appear in severe dengue. However, severe dengue is most notoriously identified as the consequence of a vascular leak syndrome leading to intravascular volume depletion, hypoalbuminemia, serosal effusions (pleural, ascitic), and, in severe cases, circulatory collapse (i.e., shock, typically lasting 2–3 days), often with an accompanying narrowed pulse pressure, hepatomegaly, and cyanosis. Bleeding tendencies (evidenced by a positive tourniquet test

and petechiae) or overt bleeding in the absence of underlying causes (e.g., preexisting gastrointestinal lesions) may be detected but are less common in children. Organ involvement may include mild hepatic injury, CNS abnormalities (e.g., altered mental status, seizures), cardiac abnormalities (e.g., arrhythmias), renal disturbances (e.g., acute kidney injury), and ocular dysfunction.

A virologic diagnosis of severe dengue can be made by the usual means (nucleic acid amplification or antigen detection) in the first 5 days of infection, after which diagnosis relies on serologic testing. Combination testing—point-of-care rapid tests for NS1 antigen and IgM antibody assays—is increasingly used in the clinical setting. The early detection of IgG (starting at 4 days after illness onset, rapidly rising) suggests a secondary rather than primary infection, in which IgG is detected later and at much lower titers. However, serologic cross-reactivity after infection with (or vaccination for) antigenically similar orthoflaviviruses (e.g., YFV, ZIKV, and Japanese encephalitis virus) confounds diagnosis.

As above, patients with severe dengue should be hospitalized and rapidly triaged (and then closely monitored) for hemodynamic instability and bleeding. In general, appropriate IV fluids should be administered as early as possible, preferably before shock develops. Based on clinical parameters (vital signs, urine output, hematocrit, bleeding, and signs of volume overload), algorithmic approaches have been developed to standardize appropriate fluid management (crystalloid vs colloid vs blood rate) and monitoring of patients with and without shock and/or bleeding. CFR varies greatly depending on early recognition and quality of treatment. However, most patients with severe dengue respond well to supportive therapy, and the overall CFR at experienced clinical centers in the tropics is probably as low as 1%. When patients enter the convalescent phase, significant fluid reabsorption (of extravascular fluid and effusions) occurs as the plasma leak resolves, blood pressure improves, and diuresis occurs. Notably, a patchy erythematous pruritic rash may develop and desquamate during recovery. Significant postacute sequelae include prolonged fatigue and depression.

The key to control of both dengue (without/with warning signs) and severe dengue is the control of YF mosquitoes, which also reduces the risk of urban YF and CHIKV circulation. Control efforts have been handicapped by the presence of nondegradable tires and long-lived plastic containers in trash repositories—creating perfect mosquito breeding grounds upon filling with water during rainfall—and by insecticide resistance. Urban poverty and an inability of the public health community to mobilize the human population to respond to the need to eliminate mosquito breeding sites are also factors in lack of mosquito control. New approaches that may be considered in the future of vector control include the release of *Aedes* mosquitoes infected with *Wolbachia* or carrying dominant lethal genetic mutations that will be passed on to offspring. A tetravalent live-attenuated dengue vaccine based on the attenuated YFV 17D platform (CYD-TDV, or Dengvaxia) was licensed in 2015 and registered in 20 countries for individuals 9–45 years of age. However, retrospective analysis of phase 3 trials in Latin America and Asia suggested protection from severe dengue only in previously seropositive individuals; indeed, the risk of severe dengue was actually increased in seronegative vaccine recipients over that in nonvaccinated seronegative individuals, suggesting that a “first serologic hit” from the vaccine predisposes naïve recipients to more severe primary dengue infection. Strategic revision to avoid vaccine-enhanced disease pivoted to include prevaccination serologic screening aimed to avoid vaccination of seropositive individuals; however, the requirement for prevaccine testing and decision-making is challenging. Recently, live-attenuated candidate vaccines based on modified recombinant DENVs have been or are being evaluated, with these learnings in mind. Qdenga (TAK-3), given in two doses 3 month apart, has shown overall efficacy against infection and hospitalization in seropositive and seronegative children, although not against DENV-3 (in fact, a very small signal of potentially increased risk of hospitalization after DENV-3 infection was observed), and the trial did not include DENV-4 infections. This vaccine has been approved for use in several European countries. Recommendations globally are currently for introduction in settings with high transmission intensity for maximal

public health impact while reducing potential risk in seronegative individuals. Further postmarket evaluation is ongoing. More recently, a phase 3 trial of Butantan-DV, a single-dose tetravalent vaccine, was shown to prevent symptomatic DENV-1 and DENV-2 infection; in the absence of DENV-3 and DENV-4 infections during the trial, efficacy against these serotypes is unknown. TVOO3, a single-dose vaccine designed to balance DENV serotype-specific components, is currently undergoing phase 3 clinical evaluation; similar concerns about safety are being addressed.

YELLOW FEVER YFV had caused major epidemics in Africa and Europe before its transmission by YF mosquitoes was discovered in 1900. Urban YF became established in the Americas because of colonization with YF mosquitoes—originally an African mosquito. Subsequently, different types of mosquitoes and nonhuman primates were found to maintain YFV in Africa and in Central and South American rain forests. Transmission to humans is incidental, occurring via bites from mosquitoes that have fed on viremic monkeys. After the identification of *Ae. aegypti* as the vector of YF, containment strategies were aimed at increased mosquito control. Today, urban YFV transmission occurs only in some African cities, but the threat exists in the cities of South America, where reinfestation by YF mosquitoes has taken place, and DENV-1–4 transmission by these mosquitoes is common. Despite the existence of a highly effective and safe vaccine, several hundred jungle cases of YF occur annually in South America, and 84,000–170,000 severe jungle and urban cases (resulting in 29,000–60,000 fatal outcomes) occurred in Africa in 2013 alone. In 2016, a large urban outbreak (Luanda, Angola) spilled over to generate local transmission in large cities in neighboring countries (e.g., Kinshasa, Democratic Republic of the Congo) as well as travel-related cases in China; the signal of a global threat that included exportation to Asia stimulated ongoing efforts to identify and vaccinate highest-risk populations in 40 targeted countries in Africa and South America, to reactively vaccinate people in outbreak settings, and to increase measures to prevent exportation.

YF is a typical VHF notable for prominent hepatic necrosis. After an incubation period of 3–6 days, patients present with a nonspecific febrile illness (fatigue, myalgia, backache, headaches, photophobia, anorexia, nausea or vomiting) associated with viremia typically lasting 3–4 days. After defervescence, 10–15% of patients develop recrudescent fever and “intoxication” characterized by severe dysfunction of the liver and other organs. Hepatic failure leads to the characteristic jaundice, bleeding (gastrointestinal tract, nasopharyngeal mucosa), abdominal pain with nausea and vomiting, and hyperammonemic encephalopathy; acute kidney injury leads to oliguria, azotemia, and marked albuminuria; and myocardial injury and encephalitis have been described. Abnormalities in liver function tests range from modest elevations of hepatic aminotransferase activities in mild cases to severe liver injury, hyperbilirubinemia, and the synthetic dysfunction of acute hepatic failure. After early leukopenia, leukocytosis occurs as disease progresses, and coagulation abnormalities are common. Treatment is supportive only. Although most infections are subclinical, 50% of patients who enter the toxic phase die in the next 7–10 days.

Urban YF can be prevented by the control of YF mosquitoes. The continuing sylvatic cycles require vaccination of all visitors to areas of potential transmission with live-attenuated variant 17D vaccine virus, which cannot be transmitted by mosquitoes. With few exceptions, reactions to the vaccine are minimal; immunity is provided within 10 days and lasts for at least 25–35 years. Recent meta-analytic review of 39 clinical studies suggests seroprotection is high (>90%) and perhaps lifelong after one vaccine dose in adults from nonendemic areas, raising questions about the utility of further boosting in this population. An egg allergy mandates caution in vaccine administration. Although fetal harm has not been documented, pregnant women should be immunized only if they are at risk of exposure to YFV. Because vaccination has been associated with several cases of encephalitis in children <6 months of age, it is contraindicated in this age group and not recommended for infants 6–8 months of age unless the risk of exposure is very high. Rare, serious, multisystemic adverse reactions (occasionally fatal), including vaccine-associated

“viscerotropic” YF, have been reported, particularly affecting the elderly and those with congenital (mutations affecting interferon α [IFN α] signaling) or acquired (autoantibodies to IFN α , thymoma) impairment affecting viral control. The risk-to-benefit ratio should be weighed before vaccine administration to individuals ≥ 60 years of age. Nevertheless, the number of deaths of unvaccinated travelers with YF exceeds the number of deaths from vaccination, and a liberal vaccination policy for travelers to involved areas should be pursued. Timely information on changes in YF distribution and YF vaccine requirements can be obtained from the U.S. Centers for Disease Control and Prevention (<https://wwwnc.cdc.gov/travel/yellowbook/2024/preparing/yellow-fever-vaccine-malaria-prevention-by-country>).

ACKNOWLEDGMENT

The authors gratefully acknowledge the major contributions of Clarence J. Peters and additional contributions by R  mie N. Charrel to this chapter in previous editions and thank Anya Crane (IRF-Frederick) for editing the manuscript.

FURTHER READING

CENTERS FOR DISEASE CONTROL AND PREVENTION: Arbovirus catalog. Available at <https://wwwnc.cdc.gov/arbocat/>. Accessed February 4, 2024.

HOWLEY PM, KNIPE DM (eds): *Fields Virology. Volume 1: Emerging Viruses*, 7th ed. Philadelphia, Wolters Kluwer/Lippincott Williams & Wilkins, 2020.

INTERNATIONAL COMMITTEE ON TAXONOMY OF VIRUSES (ICTV): Virus taxonomy: The ICTV report on virus classification and taxon nomenclature. Available at <https://ictv.global/report>. Accessed February 4, 2024.

LVOV DK et al: *Zoonotic Viruses of Northern Eurasia: Taxonomy and Ecology*. London, Elsevier/Academic Press, 2015.

SINGH SK, RUZEK D (eds): *Viral Hemorrhagic Fevers*. Boca Raton, FL, CRC Press, 2013.

VASILAKIS N, GUBLER DJ (eds): *Arboviruses: Molecular Biology, Evolution and Control*. Haverhill, UK, Caister Academic Press, 2016.

WEBSITE

INTERNATIONAL COMMITTEE ON TAXONOMY OF VIRUSES (ICTV). <https://ictv.global/>. Accessed February 4, 2024.

216 Filovirids: Orthoebolavirus and Orthomarburgvirus Infections

Jens H. Kuhn, Ian Crozier

Several viruses in the family *Filoviridae* cause severe infections in humans that are often fatal. Introduction of filovirids into human populations is an extremely rare event that most likely occurs by direct or indirect contact with reservoir hosts (known and unknown) or by contact with sick or deceased filovirid-infected mammals. Filovirids are highly infectious but not exceptionally contagious. Human-to-human transmission occurs through direct person-to-person contact or exposure to infected bodily fluids or tissues; there is no evidence of aerosol or respiratory droplet transmission in natural outbreak settings. Infections manifest initially with a nonspecific influenza-like febrile illness that rapidly progresses to commonly include gastrointestinal manifestations and, in severe illness, coagulopathy, multiple-organ dysfunction syndrome, shock, and death. Although the prevalence and source remain controversial, serologic footprints of subclinical

acute filovirid infections have been identified since the first descriptions of filovirid disease outbreaks. Filovirid disease survivors may be persistently infected in immune-privileged tissue compartments, commonly in the male reproductive tract, but also in the central nervous system (CNS), and in intraocular tissues and fluids; notably, filoviral persistence has been rarely associated with the re-ignition of outbreaks. Historically, the prevention of filovirid infections has consisted primarily of tried-and-true epidemiologic approaches (e.g., isolation of cases, contact tracing, effective infection prevention and control, and safe burial practices). Treatment of disease traditionally consisted only of limited supportive clinical care (often constrained by in-field capacity); indeed, filovirid-specific vaccines or therapeutic agents had not been rigorously evaluated in humans prior to the 2013–2016/2021 outbreaks of Ebola virus disease (EVD) that occurred in Western Africa. Building on the knowledge gained in Western Africa and during the 2018–2020 EVD outbreak in the Democratic Republic of the Congo, prevention and treatment strategies now include the widespread deployment of an effective vaccine specific to Ebola virus (EBOV); the use of effective therapeutics based on virus-specific monoclonal antibodies (mAbs), which were identified in a first-of-its-kind randomized controlled trial; and the delivery of improved supportive care. Although these advances have essentially become new standards for prevention and treatment of EVD, the same cannot yet be said for other filovirid diseases.

Filovirids are categorized as World Health Organization (WHO) Risk Group 4 pathogens. Consequently, all work with material potentially containing replicating filovirids should be conducted in maximum containment (biosafety level 4) laboratories, or the viruses should be inactivated prior to activities in biosafety level 2/3 laboratories. These viruses must be handled by experienced personnel wearing appropriate personal protective equipment (PPE; see “Control and Prevention,” below) and following rigorous standard operating procedures. In addition, when filovirid disease outbreaks are suspected or confirmed, the appropriate national authorities and WHO reference laboratories should be contacted immediately.

■ ETIOLOGY

The family *Filoviridae* includes nine official genera (Figs. 216-1 and 216-2). Human pathogens are found in two of these genera, *Orthoebolavirus* and *Orthomarburgvirus*. Collectively, these pathogens cause “filovirid” (“filovirus”) disease (FVD; International Classification of Diseases, Eleventh Revision [ICD-11], code 1D60). FVD is subdivided into “Ebola disease” (EBOD; ICD-11, code 1D60.0), caused by four of six classified orthoebolaviruses (Bundibugyo virus [BDBV], EBOV, Sudan virus [SUDV], and Taï Forest virus [TAFV]), and “Marburg disease” (MARD; ICD-11, code 1D60.1), caused by the two orthomarburgviruses, Marburg virus and Ravn virus. Accordingly, specific diseases are optimally named with the virus-specific cause such that, for example, “EVD” is distinguished from “Sudan virus disease (SVD).”

 Mammalian filovirids have linear, nonsegmented, negative-sense RNA genomes that are ≈19 kb in length. These genomes contain seven genes that encode seven structural proteins: a nucleoprotein (NP), a polymerase cofactor (VP35), a matrix protein (VP40), a glycoprotein (GP_{1,2}), a transcriptional activator (VP30), a ribonucleoprotein complex-associated protein (VP24), and a large (L) protein that contains an RNA-directed RNA polymerase domain. Orthoebolaviruses, but not orthomarburgviruses, additionally encode three nonstructural proteins of unknown functions (sGP, ssGP, and Δ-peptide). Filovirions are unique among human virus particles in that they are predominantly pleomorphic filaments but also assume torus-like or 6-like shapes (width ≈91–98 nm; average length <1 μm). These enveloped virions contain helical ribonucleoprotein capsids and are covered with GP_{1,2} spikes (Fig. 216-3).

■ EPIDEMIOLOGY

The majority of recorded FVD outbreaks, including the 2013–2016 EVD outbreak, can be traced back to single index patients who transmitted the infection to others. Although small outbreaks may not have been recorded historically, the epidemiology of EVD transmission chains suggests that only ≈50 natural host-to-human spillover events

Realm Riboviria

Kingdom Orthornavirae

Phylum Negarnaviricota

Subphylum Haploviricotina

Class Monjiviricetes

Order Mononegavirales

Family Filoviridae

Genus Cuevavirus

Species *Cuevavirus lloviense*

Virus: Lloviu virus (LLOV)

Genus Dianlovirus

Species “*Dianlovirus dehongense*” (proposed)

Virus: Déhóng virus (DEHV)

Species *Dianlovirus menglaense*

Virus: Měnglà virus (MLAV)

Genus Loebevirus

Species *Loebivirus percae*

Virus: Lötschberg virus (LTBV)

Genus Oblavirus

Species *Oblavirus percae*

Virus: Oberland virus (OBLV)

Genus Orthoebolavirus

Species *Orthoebolavirus bombaliense*

Virus: Bombali virus (BOMV)

Species *Orthoebolavirus bundibugyoense*

Virus: Bundibugyo virus (BDBV)

Species *Orthoebolavirus restonense*

Virus: Reston virus (RESTV)

Species *Orthoebolavirus sudanense*

Virus: Sudan virus (SUDV)

Species *Orthoebolavirus taiense*

Virus: Taï Forest virus (TAFV)

Species *Orthoebolavirus zairense*

Virus: Ebola virus (EBOV)

Genus Orthomarburgvirus

Species *Orthomarburgvirus marburgense*

Virus 1: Marburg virus (MARV)

Virus 2: Ravn virus (RAVV)

Genus Striavirus

Species *Striavirus antennarii*

Virus: Xilāng virus (XILV)

Genus Tapjovirus

Species *Tapjovirus bothropis*

Virus: Tapajós virus (TAPV)

Genus Thamnovirus

Species *Thamnovirus kanderense*

Virus: Kander virus (KNDV)

Species *Thamnovirus percae*

Virus: Fiwi virus (FIWIV)

Species *Thamnovirus thamnaconi*

Virus: Huángjiāo virus (HUJV)

Filovirids that are known to infect humans are depicted in the same nonblack colors as in figures.

FIGURE 216-1 Filovirid taxonomy (2024).

have occurred since the discovery of filovirids in 1967. However, outbreaks associated with reignition of human-to-human transmission from a persistently infected survivor of a previous outbreak have more recently been described. Outbreak frequency, size, and overall

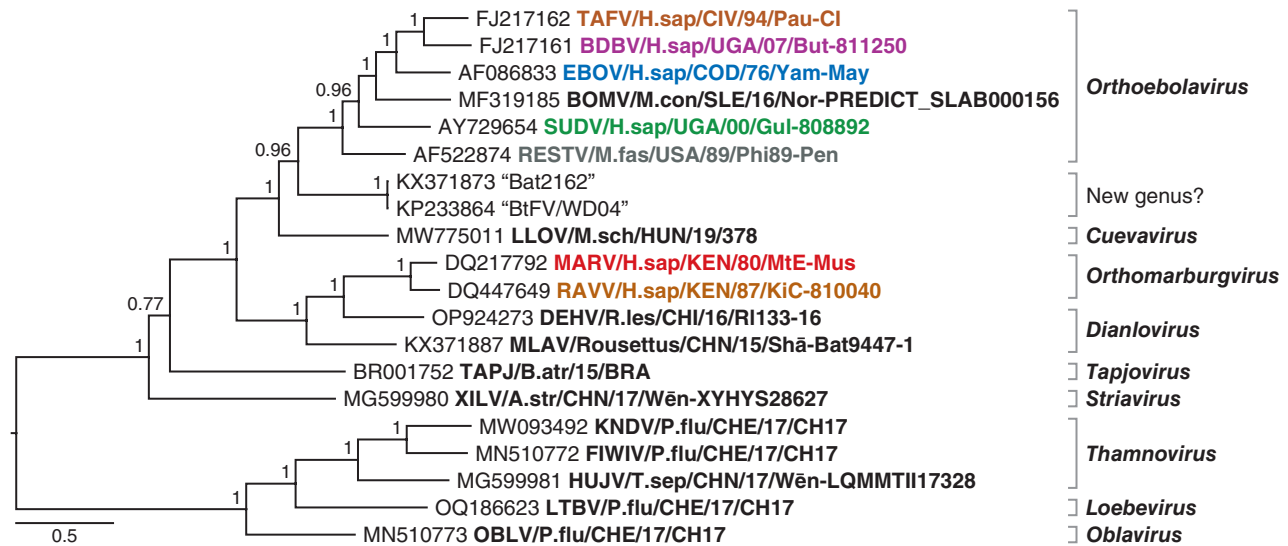


FIGURE 216-2 Filovirid phylogeny/evolution. Midpoint-rooted maximum-likelihood tree inferred by using filovirid large gene (*L*) sequences. Bootstrap values are shown at each node. The scale bar indicates nucleotide substitutions per site. Tips of branches are labeled with GenBank accession numbers followed by filovirid isolate designation. Filovirids that are known to infect humans are depicted in the same nonblack colors as in other figures. BDBV, Bundibugyo virus; BOMV, Bombali virus; DEHV, Dêhông virus; EBOV, Ebola virus; FIWIV, Fiwi virus; HUJV, Huángjiāo virus; KNDV, Kander virus; LLOV, Lloviu virus; LTBV, Löttschberg virus; MARV, Marburg virus; MLAV, Mênglà virus; OBLV, Oberland virus; RAVV, Ravn virus; RESTV, Reston virus; SUDV, Sudan virus; TAFV, Tai Forest virus; TAPV, Tapajós virus; XILV, Xiláng virus. (Adapted and expanded from JH Kuhn et al: *Filoviridae*, in *Fields Virology*, Vol 1, 7th ed, PM Howley et al (eds). Philadelphia, Wolters Kluwer/Lippincott Williams & Wilkins, 2020, pp 449–503. Analysis courtesy of Nicholas Di Paola, PhD, USAMRIID, Fort Detrick, MD, USA. Figure courtesy of Jiro Wada, NIH/NIAID/DCR/IRF-Frederick, Fort Detrick, MD, USA.)

case-fatality rate are likely the result of complex interactions of the specific filovirid, the reservoir hosts (known and unknown), the susceptible human population (e.g., varying with age, unknown genetic determinants of susceptibility and disease severity, and risk behavior), and the geographic setting (e.g., local public health capacity, socioeconomic conditions, and cultural practices).

As of November 13, 2024, 35,608 cases of human filovirid disease and 15,886 deaths had been recorded (Fig. 216-4). These numbers emphasize both the high case-fatality rate (number of deaths per number of sick people; 44.6%) and the overall low mortality (reflecting the impact on the healthy population) related to filovirid diseases. Of these totals, 28,652 cases and 11,325 deaths occurred during the 2013–2016 EVD outbreak in Western Africa; this was the largest of all recorded FVD outbreaks. Natural FVD outbreaks had not been considered a global threat until regional and then global spread during this outbreak challenged that tenet. Filovirids that are pathogenic in humans appear to be exclusively endemic to equatorial (Western, Middle, and Eastern)

Africa (Fig. 216-5), although this distribution may change if natural or artificial environmental alterations lead to filovirid host migration and increased contacts between nonhuman hosts and humans.

Outbreaks have been contained when high-risk activities (e.g., ritual washing as part of burial practices) have been limited or been made safer with appropriate infection prevention and control. Of particular importance is accessibility to health care centers with staff trained and equipped (e.g., with PPE) for adequate infection prevention and control, to limit both community spread and nosocomial amplification. The incidence of FVD may have increased over the past two decades (Fig. 216-4 and Fig. 216-5), but debate continues as to whether this increase is due to increased filovirid activity, more frequent human interaction with filovirid hosts, or improvement in surveillance capabilities.

FVD outbreaks are associated with distinct meteorologic and geographic conditions and are probably associated with distinct hosts or reservoirs. The four orthoebolaviruses that cause disease in humans appear to be endemic in humid rainforests. EVD outbreaks in particular have often been associated with hunting in forests or contact with bushmeat (i.e., meat from apes, other nonhuman primates, duikers, or bush pigs). Ecologic studies suggest that EBOV may play a role in extensive and often-fatal epizootics among wild ape populations. However, only one orthoebolavirus, TAFV, has been isolated from nonhuman primates in the wild. Orthomarburburgviruses, on the other hand, seem to infect hosts inhabiting arid woodlands, and associated outbreaks have almost always been epidemiologically linked to individuals visiting or working in natural or engineered caves or mines. The cave-dwelling Egyptian rousette (*Roussettus aegyptiacus* (Geoffroy, 1810)), a pteropodid (fruit) bat, serves as a natural and subclinically infected reservoir for both Marburg virus and Ravn virus. Although bats are suspected hosts for orthoebolaviruses as

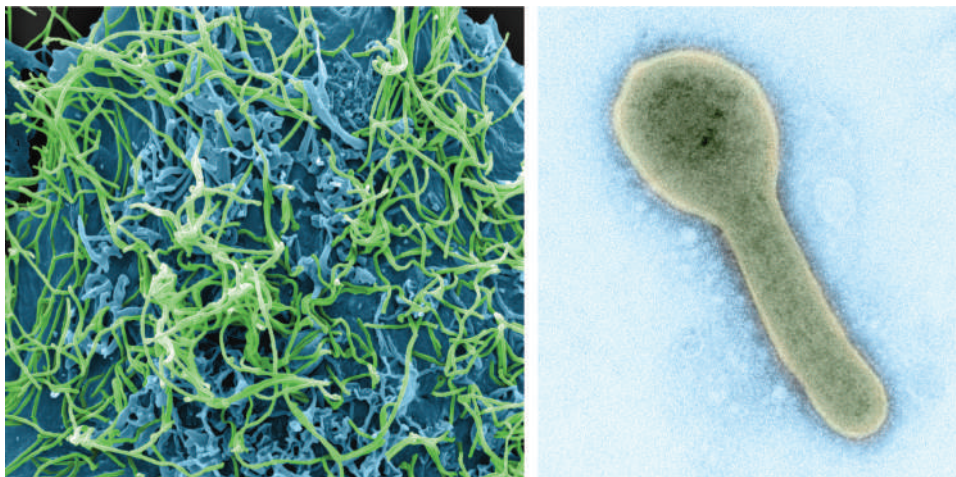


FIGURE 216-3 Ultrastructure of filovirions. Left: Colorized scanning electron micrograph of Ebola virus particles (green) attached to the surface of an infected grivet (*Chlorocebus aethiops* (Linnaeus, 1758)) Vero E6 producer cell (blue). Right: Colorized transmission electron micrograph of a Marburg virus particle collected from purified Vero E6 producer cell supernatant. (Figure courtesy of John G. Bernbaum and Jiro Wada, NIH/NIAID/DCR/IRF-Frederick, Fort Detrick, MD, USA.)

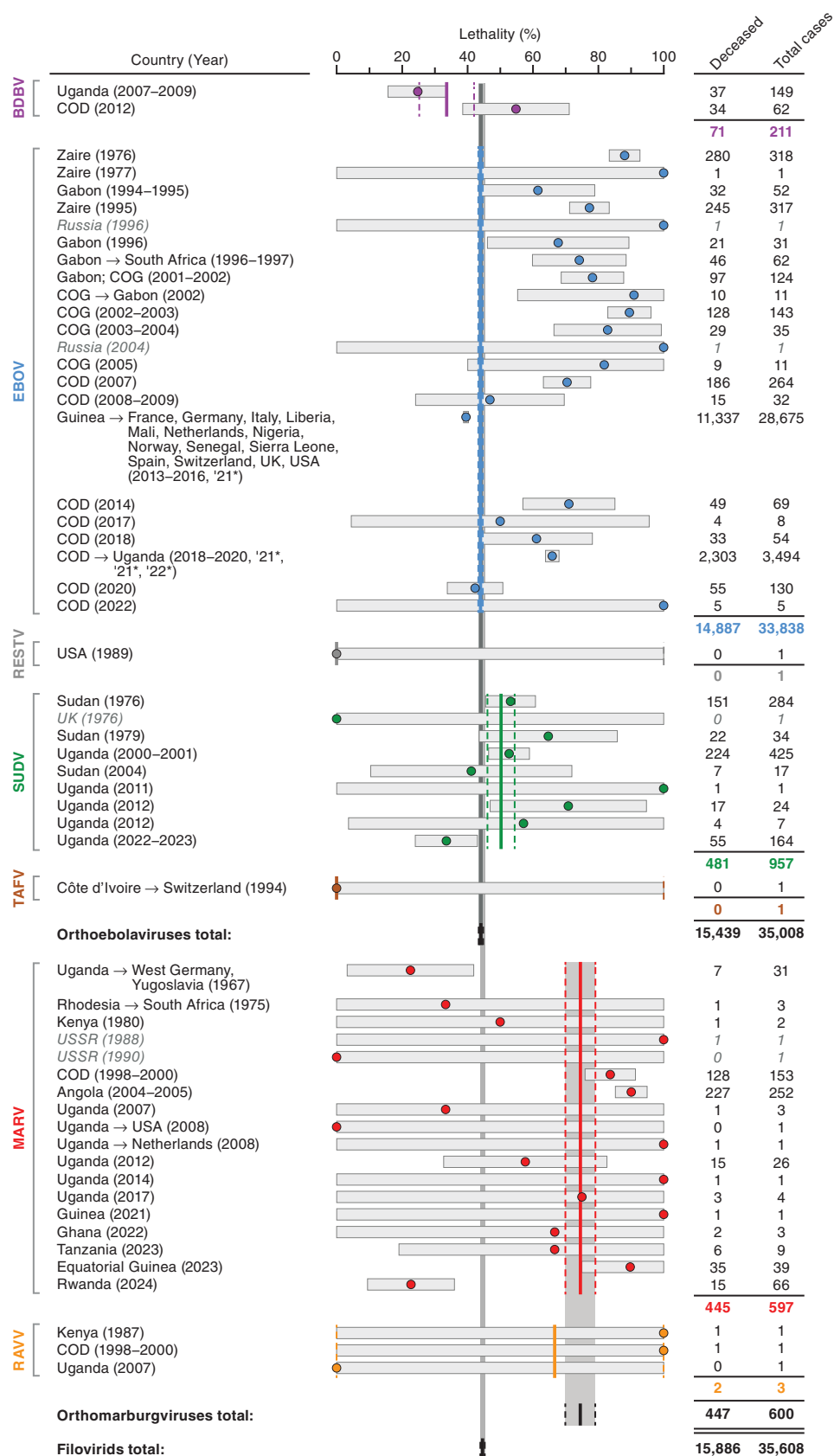


FIGURE 216-4 Characteristics of outbreaks of human filovirid disease. Seven of 17 classified filovirids have caused disease in humans. **Left column:** Outbreaks considered to be caused by independent zoonotic spillover are listed by virus in chronological order in the left column. Outbreaks considered to be related to reignition from a persistently infected survivor of a prior outbreak are included with the initial outbreak (denoted by asterisk). Laboratory infections are in *gray italicized text*. International case exportations are indicated with *arrows*. **Right column:** Numbers of fatal cases and total cases are summarized. **Middle column:** The case-fatality rate (*colored dots*) for each outbreak is plotted on a 0–100% scale along with 99% confidence intervals (*gray horizontal bars*). The overall case-fatality rate for disease caused by a particular virus is delineated by *vertical colored lines*, with *vertical colored dashed lines* indicating the corresponding 99% confidence intervals. The overall case-fatality rates for all *Orthoebolavirus*, all *Orthomarbuvirus*, and all filovirid disease outbreaks are shown by (underlaid) *vertical gray bars*. BDBV, Bundibugyo virus; COD, Democratic Republic of the Congo (formerly Zaire); COG, Republic of the Congo; EBOV, Ebola virus; MARV, Marburg virus; RAVV, Ravn virus; RESTV, Reston virus; SUDV, Sudan virus; TAFV, Tai Forest virus; UK, United Kingdom; USSR, Union of Soviet Socialist Republics (today Russia). (Adapted and expanded from JH Kuhn et al: *Evaluation of perceived threat differences posed by filovirus variants*. *Biosecure Bioterror* 9:361, 2011. Figure courtesy of Jiro Wada, NIH/NIAID/DCR/IRF-Frederick, Fort Detrick, MD, USA.)

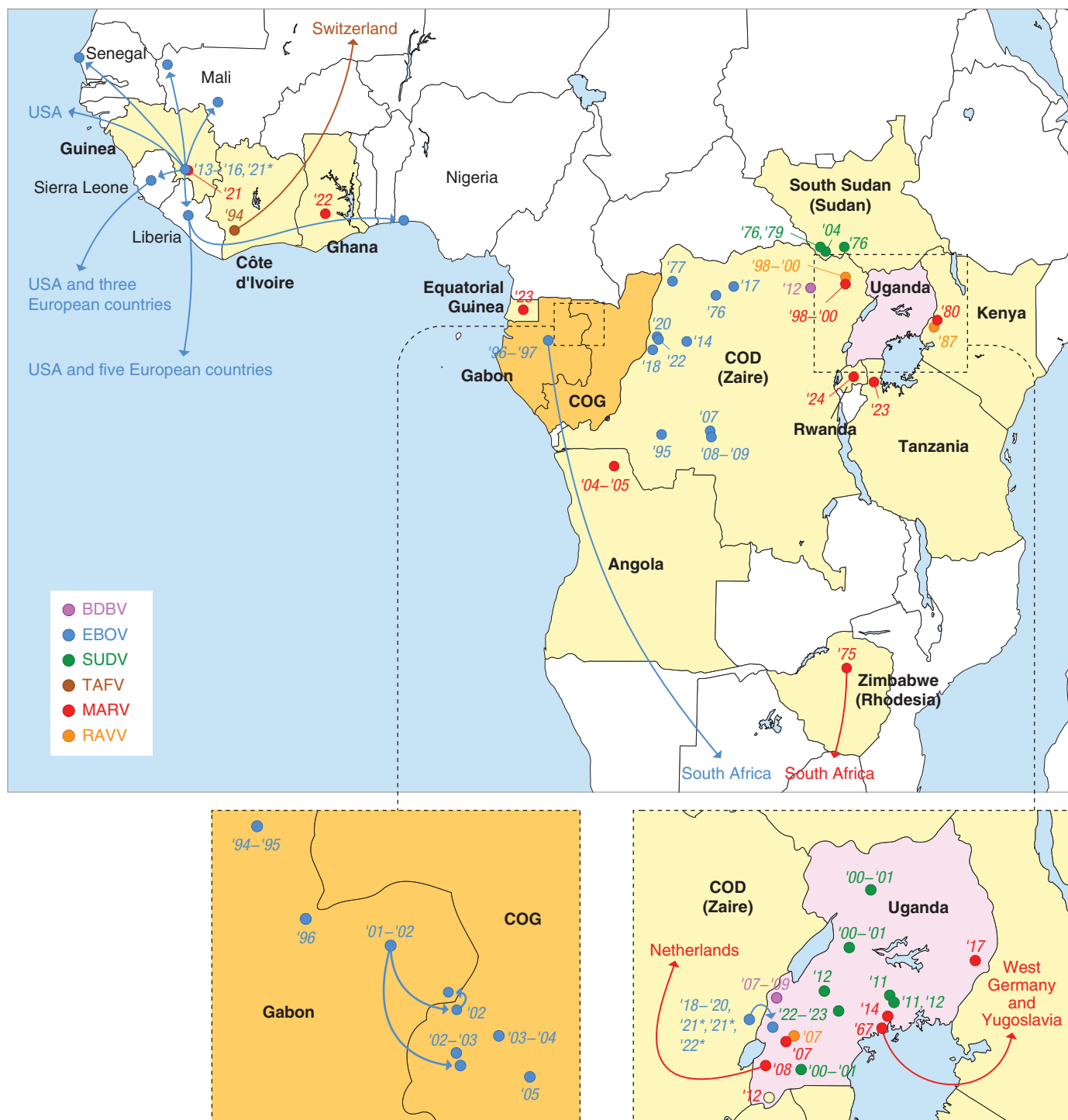


FIGURE 216-5 Geographic distribution of human filovirid disease outbreaks and years of occurrence. Arrows indicate international case exportations. BDBV, Bundibugyo virus; COD, Democratic Republic of the Congo (formerly Zaire); COG, Republic of the Congo; EBOV, Ebola virus; MARV, Marburg virus; RAVV, Ravn virus; SUDV, Sudan virus; TAFV, Taï Forest virus. (Figure courtesy of Jiro Wada, NIH/NIAID/DCR/IRF-Frederick, Fort Detrick, MD, USA.)

well, definitive proof is lacking. To date, the nonpathogenic Bombali virus is the only orthoebolavirus for which coding-complete genome nucleic acid has been detected in bats. EBOV and Reston virus have been loosely connected to frugivorous and insectivorous bats by means of antibody or genome fragment detection, whereas the hosts of BDBV, SUDV, and TAFV are enigmatic.

■ PATHOGENESIS

Human infections typically occur through direct exposure of skin lesions or mucosal surfaces to contaminated body fluids or material or by parenteral inoculation (e.g., via accidental needlesticks or reuse of needles in poorly equipped hospitals). Numerous studies, both in vitro and in vivo (in several animal models of human disease), have

illuminated aspects of FVD pathogenesis (Fig. 216-6). The GP_{1,2} spikes on the surface of filovirions determine their cell and tissue tropism by engaging cell-surface molecules and the intracellular filovirid receptor NPC intracellular cholesterol transporter 1 (NPC1).

One of the hallmarks of filovirid pathogenesis is a pronounced modulation and dysregulation of immune responses. The first targets of filovirions are local macrophages, monocytes, and dendritic cells. Several structural proteins of filovirions (i.e., VP35, VP40, and/or VP24) then suppress intrinsic and innate immune responses by, for example, inhibiting the type I interferon antiviral pathways. This immunomodulation ultimately enables a productive filovirid infection, resulting in very high viral titers (>10⁶ PFU/mL of serum in humans) with dissemination to most tissues. In tissues, filovirions infect additional

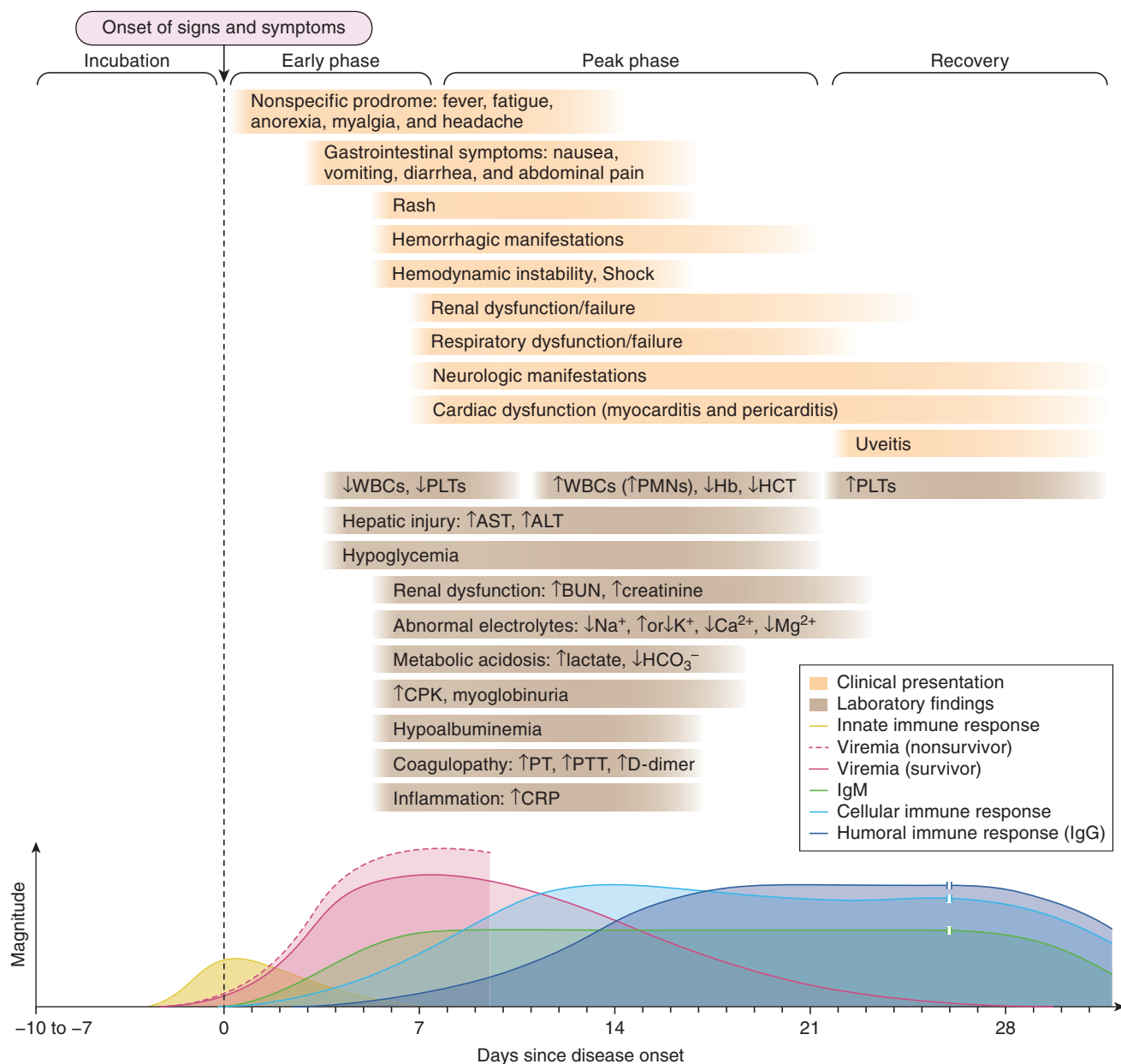


FIGURE 216-6 Ebola virus disease course. ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; CPK, creatine phosphokinase; CRP, C-reactive protein; Hb, hemoglobin; HCT, hematocrit; PLTs, platelets; PMNs, polymorphonuclear leukocytes; PT, prothrombin time; PTT, partial thromboplastin time; WBCs, white blood cells. (Adapted and expanded from JH Kuhn et al: *Filoviridae*, in *Fields Virology*, 7th ed, Vol. 1. Howley PM et al (eds). Philadelphia, Wolters Kluwer/Lippincott Williams & Wilkins, 2020, pp 449–503. Figure courtesy of Jiro Wada, NIH/NIAID/DCR/IRF-Frederick, Fort Detrick, MD, USA.)

phagocytic cells, including other macrophages (alveolar, peritoneal, and pleural macrophages; Kupffer cells in the liver; and microglia in the CNS), epithelial cells (e.g., adrenal cortical cells and hepatocytes), stromal cells (fibroblasts), and endothelial cells. Infection is cytolytic in some—but not all—infected cells; e.g., hepatocyte necrosis likely contributes to elevated aminotransferase activities, and hepatic synthetic dysfunction contributes to coagulopathy. Infection leads to the secretion of soluble signaling molecules (varying with the cell type) that likely contribute to forward dysregulation of immune responses and ultimately to multiorgan dysfunction syndromes. For instance, infected macrophages secrete proinflammatory cytokines, leading to further recruitment of monocyte-derived macrophages to the site of infection. In contrast, infected dendritic cells are not activated to secrete cytokines, and the expression of major histocompatibility class II antigens is partially suppressed, with consequently deficient antigen presentation. Immunosuppression also occurs in part by massive lymphoid depletion in lymph nodes, spleen, and thymus in the absence of effective humoral and cell-mediated immune responses, especially in severe or fatal disease. Results from animal studies suggest that depletion is

a direct consequence of considerable lymphocyte death; this explanation would also account for the severe lymphopenia that develops in patients. In addition to potential florid filovirid dissemination, another consequence of lymphocyte depletion may be susceptibility to secondary bacterial and fungal infections.

Other pathogenic hallmarks of filovirid infections include coagulopathy and endothelial dysfunction. Along with hepatic synthetic dysfunction, disseminated intravascular coagulation may contribute though this remains controversial. Thrombocytopenia, increased concentrations of tissue factor, consumption of clotting factors, increased concentrations of fibrin degradation products (D-dimers), and declining concentrations of protein C are typical features of severe disease. Consequently, fibrin deposition and microthrombotic small-vessel occlusion and necrotic/hypoxic infarction may occur in some tissues, particularly in the gonads and, less often, in the kidneys and spleen. In addition, petechiae, ecchymoses, extensive visceral effusions, and other hemorrhagic signs are observed in internal organs, mucous membranes, and skin. However, actual severe blood loss is a rare event (although it frequently occurs during or after childbirth). Most likely,

aberrance in cytokines or other factors, such as nitric oxide, and direct infection and activation of endothelial cells are responsible for upregulated permeability of blood-vessel endothelia. This upregulation leads to fluid redistribution and interstitial tissue edema and hypovolemic or septic shock are common features of severe disease.

Despite this long list of pathogenetic hallmarks, increasing evidence from clinical settings suggests that effective filovirid-specific adaptive immune responses do develop, coinciding with control and clearance of viremia and subsequent clinical improvement in surviving patients. However, depending on the severity of illness (including organ dysfunction and late complications), clinical illness may be protracted and recovery incomplete.

CLINICAL MANIFESTATIONS

Diseases caused by orthobolaviruses and orthomarburburgviruses present very similar clinical phenotypes that cannot be distinguished at the bedside and, for all practical purposes, may be considered the same disease; this approach may change as higher-resolution characterization of human FVD accrues. The prevalence and character of clinical signs do not differ significantly in disease caused by disparate filovirids (except for the possibly apathogenic Reston virus), although, with the exception of the 2013–2016 EVD outbreak, the numbers of deeply characterized clinical observations are very low. The incubation period is 2–25 days (most commonly 7–10 days), after which infected people develop a nonspecific influenza-like syndrome characterized by sudden onset of fever and chills, severe headaches, cough, myalgia, pharyngitis, arthralgia of the larger joints, development of a maculopapular rash, and other symptoms/signs. A subsequent second phase ($\approx$ 5–7 days after disease onset and thereafter) involves the gastrointestinal (nausea and vomiting and/or diarrhea, sometimes with abdominal pain), respiratory (chest pain, cough, and/or dyspnea), cardiovascular (hypotension and/or edema), and central nervous (confusion, headache, and/or coma) systems. Common hemorrhagic manifestations include subconjunctival injection, petechial rash, gingival bleeding, and bleeding at injection sites; epistaxis, hematemesis, hematuria, and melena occur but are less common. Patients usually succumb to acute disease 4–14 days after infection, often with severe multiorgan dysfunction that commonly includes shock and acute renal or respiratory failure.

Typical laboratory findings are leukopenia (with cell counts as low as 1000/ μ L) with a left shift prior to leukocytosis, thrombocytopenia (with counts as low as 50,000/ μ L), increased activities of liver enzymes (aspartate aminotransferase > alanine aminotransferase, γ -glutamyltransferase), increased creatinine and urea concentrations with proteinuria, electrolyte derangement (hypokalemia or hyperkalemia, hyponatremia, and/or hypocalcemia), hypoglycemia, hypoalbuminemia, prolonged prothrombin and partial thromboplastin times, and elevated creatine phosphokinase activities. Nonspecific markers of systemic inflammation (e.g., C-reactive protein concentrations) may be markedly elevated in severely ill patients.

DIAGNOSIS

Filovirid disease cannot be diagnosed based on clinical presentation alone. Numerous diseases common in equatorial Africa need to be considered in the differential diagnosis of a febrile patient. Almost all of these occur at a much higher incidence than FVD and are much more likely diagnoses in non-outbreak settings; however, during outbreaks and in peri-outbreak periods, timely and accurate laboratory diagnosis to rule in or rule out filovirid infection is critical. The most important infectious disease mimics of FVD are falciparum malaria and typhoid fever; also important are enterohemorrhagic *Escherichia coli* enteritis, gram-negative septicemia (including shigellosis), meningococcal septicemia, rickettsial infections, fulminant viral hepatitis, leptospirosis, measles, and other high-consequence viral infections (in particular, Lassa fever and yellow fever). Rarer noninfectious possibilities, including venomous snakebites, warfarin intoxication, and the many causes of acquired or inherited

coagulopathy, also must be considered in the bleeding patient. An exposure history—including exposure to caves or mines; direct contact with bats, nonhuman primates, or “bushmeat”; direct contact with severely ill local residents; or admission to rural hospitals with patient-to-patient or patient-to-health-care-worker clusters of illness—should raise the index of suspicion.

If FVD is suspected based on epidemiologic and/or clinical manifestations, infectious disease specialists and the proper public health authorities should be notified immediately. Laboratory diagnosis of FVD is relatively straightforward but ideally requires the capacity for maximum containment, which usually is not available in filovirid-endemic countries. Increasingly, laboratory diagnosis is performed with patient samples inactivated in mobile field “glove boxes” by on-site personnel trained in the safe use of diagnostic assays adapted for field use in lower-containment settings. Consequently, diagnostic samples should be collected and processed with great caution and with the use of appropriate PPE and strict barrier techniques. With adherence to established biosafety precautionary measures, samples should be sent in suitable transport media to national or international reference laboratories. Acute-phase blood/serum is the preferred diagnostic specimen because it usually contains high titers of filovirions.

The current assay of choice for the diagnosis of filovirid infection is reverse-transcription polymerase chain reaction (RT-PCR) targeting one or more filovirid genes; a typical detection limit is 1000–5000 PFU/mL of serum, depending on the assay. When available, safe, rapid-turnaround, and standardized in-field PCR-based approaches (e.g., the Cepheid GeneXpert platform) are now standardly deployed during outbreaks. Antigen-capture enzyme-linked immunosorbent assays (ELISAs) or immunochromatographic assays for the detection of filovirion proteins are in development and may be useful as point-of-care rapid diagnostic tests (RDTs) in the future; sensitivity limitations of EBOV-specific antigen-based RDTs restrict current use to diagnosis of postmortem samples only. Direct immunoglobulin M (IgM) capture, direct IgG capture, or IgM-capture ELISAs (including multiplex approaches) are used for the detection of filovirion-targeting antibodies from patients in the later stages of disease (i.e., those who have been able to mount a detectable antibody response), including survivors. All of these assays can be conducted on samples treated with guanidinium isothiocyanate (for RT-PCR), cobalt-60 irradiation (for ELISA), or other effective measures (heat or irradiation treatment) that render filovirids noninfectious. Virus isolation in cell culture and plaque assays for quantification or diagnostic confirmation are relatively simple but must be performed in maximum-containment laboratories. If available, electron microscopic examination of inactivated samples or cultures for the unique filamentous structures of filovirions can further support diagnosis (Fig. 216-3). Formalin-fixed skin biopsy samples and possibly skin swabs can be useful for safe postmortem diagnoses. In-field (or near in-field) rapid genome sequencing was first deployed to inform classic epidemiology in Western Africa during the 2013–2016 EVD outbreak and has become a mainstay of outbreak control and response, even in endemic settings.

TREATMENT

Filovirid Disease

Clinical management of patients with suspected or confirmed filovirid disease should be conducted by health care workers who have been well-trained in the complex care of FVD patients and who are using appropriate PPE in care environments with appropriate infection prevention and control measures (see “Control and Prevention,” below). Historically, management of FVD had been entirely supportive (and even that limited by resource constraints) because the efficacy and safety of specific antiviral countermeasures had not been rigorously studied outside of animal models of disease. The 2013–2016 EVD outbreak in Western Africa highlighted the need to conduct rigorous, feasible, and ethical clinical research during outbreaks. Building on challenges and lessons learned in that setting, the first-of-its-kind Pamoja Tulinde

Maisha (PALM) randomized clinical trial was conducted during the 2018–2020/2021 EVD outbreak in the Democratic Republic of the Congo and identified two therapeutics based on mAbs specific to EBOV to improve survival rates. mAb114 (ansuvimab-zykl) and REG-EB3 (atoltivimab, maftivimab, and odesivimab-ebgn) were subsequently approved for the treatment of EVD in PCR-confirmed adults (including pregnant women) and children. Both are administered via carefully monitored single-dose infusions initiated as soon as possible after diagnosis. In addition, efforts during this outbreak demonstrated the will and capacity to deliver more advanced supportive and critical care accompanying specific therapeutics in the African outbreak setting. Although evidence is lacking, consensus treatment strategies include those generally recommended for severe septicemia/sepsis/shock ([Chap. 315](#)) and should be applied with an emphasis on standard approaches—i.e., monitoring and response to respiratory dysfunction (e.g., oxygen), circulatory dysfunction (e.g., intravascular fluid repletion and vasopressor support), and CNS dysfunction (e.g., ruling out of reversible causes, notably hypoglycemia)—as well as the detection and management of acute kidney injury, hemorrhage, electrolyte derangements, and nutritional status and the prevention and treatment of secondary or co-infections. Pain management and administration of anti-pyretics, antiemetics, and antidiarrheal agents may be considered. Crucial strategies to improve outcomes in the most severely ill FVD patients include preventing organ dysfunction and providing safe and effective temporary organ support (e.g., mechanical ventilation and renal replacement) to expand the window for administration of medical countermeasures and development of effective endogenous immune responses.

■ COMPLICATIONS

Even in patients who have initial virologic or clinical improvement, complications in the second or third week of illness may include secondary infections; persistent renal dysfunction and electrolyte abnormalities; respiratory insufficiency; neurologic compromise (e.g., EBOV-related meningoencephalitis, cerebrovascular events, and seizures); cardiac dysfunction (e.g., myocarditis and pericarditis); and venous thrombosis. Pregnancy and labor are historically associated with severe complications and poor outcomes in filovirid disease due to clotting factor consumption, fetal loss, and/or severe blood loss during birth; in the era of EBOV-specific therapeutics and optimized supportive care, maternal and fetal outcomes are improved.

A number of sequelae have been self-reported or historically described in survivors of FVD, including prolonged and sometimes incapacitating arthralgia and myalgia, asthenia, alopecia, visual problems (including uveitis), hearing loss, memory loss and neurocognitive dysfunction, mental health conditions (anxiety, depression, and post-traumatic stress disorder), and reproductive problems. Well-controlled observational studies of EVD survivors were first conducted in the aftermath of the 2013–2016 Western Africa EVD outbreak. Compared with their close-contact controls, Liberian EVD survivors had an increased incidence of headache, arthralgia and myalgia, memory loss, fatigue, and urinary frequency as well as abnormal results in abdominal, chest, neurologic, musculoskeletal, and ocular exams. Of individual and public health significance is the potential for persistence of filovirids in immune-privileged tissue compartments (and their associated fluids) in FVD survivors, most commonly in the semen of male survivors (with the rare but documented potential for sexual transmission) and rarely in the CNS (causing recrudescent meningoencephalitis), the eye (causing recrudescent uveitis), and the placenta (causing transmission or placental insufficiency). True relapses that resemble the course of primary FVD are extremely rare but have been described. The risk of the remote reignition of outbreaks from persistently infected FVD survivors is apparently rare but obviously consequential. Notably, these events have only been detected after the two largest EVD outbreaks, suggesting a likely limitation, with the caveat that available samples and genomic tools to determine outbreak relatedness have historically not been available. EVD outbreaks in the

Democratic Republic of the Congo in 2021 were genomically linked to the previous 2018–2020 outbreak. After a 5- to 7-year interim, an EVD outbreak in Guinea in 2021 was genomically linked to the 2013–2016 Western African outbreak; this event remains enigmatic as corroborating epidemiologic links have not been found.

■ PROGNOSIS

Among the most severe of acute viral diseases in humans, FVD generally has a poor prognosis, although with much greater heterogeneity than was historically assumed; i.e., the 70–90% case-fatality rate ascribed for many decades to EVD has required revision. With an incomplete evidence base, the outcome probably depends on factors that include the particular filovirid causing the infection ([Fig. 216-4](#)), host factors (age, immune status, coinfections, and unknown host genetic factors), virus exposure route and dose, viral load, the presence and severity of organ dysfunction, and—critically—the availability of filovirid-specific countermeasures and requisite supportive care. Despite the receipt of effective EBOV-specific therapeutics, outcomes in severe EVD patients with high viral loads and complex organ dysfunction remain suboptimal and will require future research to evaluate and advance virus-targeted (e.g., combination therapeutic), disease-targeted (e.g., immunomodulatory), and optimized supportive care strategies. After resolution of acute disease, the long-term health and survival outcomes for FVD survivors are unknown. It is uncertain how increased access to filovirid-specific therapeutics and life-saving support will impact filovirid persistence in survivors; short- and long-term surveillance will be necessary to avoid individual consequences (e.g., relapse and/or recrudescent inflammatory syndromes) and public health consequences (e.g., reignition of outbreak transmission chains in the peri-outbreak or post-outbreak period).

Although remarkable progress in understanding and improving outcomes in EVD has occurred in recent years, the same cannot yet be said for disease caused by filovirids other than EBOV, either in acute disease or in convalescence.

■ CONTROL AND PREVENTION

Prevention of viral exposure in nature is challenged by an incomplete understanding of filovirid ecology. To prevent orthomarburburgvirus infection, people entering or living in areas where Egyptian rousettes can be found should avoid direct or indirect contact with these animals. Prevention in nature is more difficult in the case of pathogenic orthoebolaviruses, largely because ecologic reservoirs have not yet definitively been identified. EVD outbreaks have been associated with hunting or consumption of nonhuman primates more than exposure to bats. Indeed, the mechanism of introduction of orthoebolaviruses into nonhuman primate populations, if it occurs at all, is unclear. (Only one pathogenic orthoebolavirus, TAFV, has unequivocally been detected in wild nonhuman primates.) Therefore, to prevent orthoebolavirus infection, the only evidenced advice that can be offered to travelers and locals is to avoid contact with “bushmeat,” nonhuman primates, and bats. In any setting, the local involvement of medical anthropologists and careful community engagement is strongly advised to ensure preventive risk communication is not received as threatening or patronizing.

Biomedical prevention strategies have historically been limited to tried-and-true pillars of outbreak control, centering on the identification and isolation of cases, contact tracing, ensuring that health care workers and other response personnel have appropriate training and capacity in infection prevention and control, and preventing high-risk transmission events. Measures aimed at preventing and controlling infection, including relatively simple barrier nursing techniques, vigilant use of appropriate PPE, quarantine, and contact tracing, usually effectively terminate or at least contain FVD outbreaks. Isolation of infected people and their contacts and avoidance of direct person-to-person contact without appropriate PPE usually prevent further spread, as the virions are not transmitted through droplets or aerosols under natural conditions. Typical protective gear sufficient to prevent filovirid infections consists of disposable gloves, gowns, and shoe

covers and a face shield and/or goggles. If available, N-95 or N-100 respirators may be used to further limit infection risk. Positive-air-pressure respirators should be considered for high-risk medical procedures, such as intubation or suctioning. Medical equipment used in the care of an infected patient, such as gloves or syringes, should never be reused. Because filovirions are enveloped, disinfecting with detergents (e.g., 1% sodium deoxycholate, diethyl ether, or phenolic compounds) is relatively straightforward. Bleach solutions are recommended at 1:100 for surface disinfection and 1:10 for application to excreta or corpses. Whenever possible, potentially contaminated materials should be autoclaved, irradiated, or destroyed.

Emerging from research conducted during the 2013–2016 EVD outbreak in Western Africa, a vaccine based on a recombinant vesicular stomatitis Indiana virus expressing EBOV GP_{1,2} (rVSV-ZEBOV/Ervebo) was the first filovirid vaccine approved for use in the United States and the European Union (EU). It is now widely deployed in both a reactive-ring vaccination strategy, targeting close contacts and their contacts in EVD outbreak settings, and for the preexposure vaccination of health care workers in at-risk regions. More recently, a heterologous dose vaccine candidate incorporating EBOV GP_{1,2} into an adenovirus vector (Ad26.ZEBOV-GP/Zabdeno) followed by a vaccinia virus vector incorporating multiple filovirid antigens (MVA-BN-filo/Mvabea) has been shown to be safe and immunogenic in humans. Though evaluation of efficacy in a clinical trial has not been possible, immunobridging data gained during nonhuman primate experimentation led to regulatory authorization in the EU under “exceptional circumstances”; the two-dose requirement likely limits current use to proactive preexposure prevention in “peri-outbreak” settings rather than reactive “in-outbreak” reactive strategies. Development and evaluation of this and other vaccine candidates continue toward complementary preventive approaches for non-outbreak or peri-outbreak settings, with emphasis on the durability of immune responses and increases in preventive breadth toward other filovirids.

Even in the absence of high-level evidence, expert consensus informs the targeted use of EBOV-specific vaccine or postexposure prophylaxis (PEP) to prevent infection or disease in health care workers considered to have had a high-risk EBOV exposure (e.g., after needlestick injury). Evidence is needed to inform the use of PEP in high-risk contacts in the field outbreak setting. For male survivors, abstinence from sexual activity with a partner for at least 12 months after disappearance of clinical signs is recommended, unless testing proves semen to be free of filovirid RNA. (The use of condoms is generally recommended for all sexual activities.) Reproductive tract and CNS tissues, including ocular tissues and fluids from survivors, should be handled with appropriate precautions until demonstrated to be filovirid-RNA free. The role of filovirid-specific therapeutics in the prevention or treatment of filoviral persistence is unclear.

FURTHER READING

- CNOPS L et al: Essentials of filoviral load quantification. *Lancet Infect Dis* 16:e134, 2016.
- CROZIER I et al: The evolution of medical countermeasures for Ebola virus disease: Lessons learned and next steps. *Vaccines* (Basel) 10:1213, 2022.
- DUDAS G et al: Virus genomes reveal factors that spread and sustained the Ebola epidemic. *Nature* 544:309, 2017.
- HOENEN T et al: Therapeutic strategies to target the Ebola virus life cycle. *Nat Rev Microbiol* 17:593, 2019.
- JACOB ST et al: Ebola virus disease. *Nat Rev Dis Primers* 6:13, 2020.
- KUHN JH et al: *Filoviridae*, in *Fields Virology*, Vol 1, 7th ed, PM Howley et al (eds). Philadelphia, Wolters Kluwer/Lippincott Williams & Wilkins, 2020, pp 449–503.
- MATZ KM et al: Ebola vaccine trials: Progress in vaccine safety and immunogenicity. *Expert Rev Vaccines* 18:1229, 2019.
- MULANGU S et al: A randomized, controlled trial of Ebola virus disease therapeutics. *N Engl J Med* 381:2293, 2019.
- REGULES JA et al: A recombinant vesicular stomatitis virus Ebola vaccine. *N Engl J Med* 376:330, 2017.

217 Pathogenesis, Diagnosis, and Treatment of Fungal Infections

Michail S. Lionakis, John E. Edwards, Jr.



DEFINITION AND ETIOLOGY

In recent decades, human fungal infections have dramatically increased worldwide as a result of the AIDS pandemic, the widespread use of antibacterial agents, and the introduction of cytotoxic agents and precision medicine biologics for the treatment of autoimmune and neoplastic diseases and for use in patients undergoing solid organ transplantation or hematopoietic stem cell transplantation. Moreover, of great concern has been the recent rise in fungal infections caused by drug-resistant species, such as azole- and/or echinocandin-resistant *Candida glabrata* and *Candida auris* and azole-resistant *Aspergillus fumigatus*. Among the ~5 million fungal species, only a few cause human infections (Table 217-1).

Fungal infections are classified as *mucocutaneous* and *deep organ* infections on the basis of anatomic location and as *endemic* and *opportunistic* infections on the basis of epidemiology. Mucocutaneous infections can cause serious morbidity but are rarely fatal. Deep organ infections cause severe illness and often carry a high mortality rate. The endemic mycoses are caused by fungi that are not part of the normal human microbiota but are environmentally acquired. The opportunistic mycoses are caused by fungi (*Candida*, *Aspergillus*) that often are components of the human microbiota and whose ubiquity in nature renders them easily acquired by immunosuppressed hosts (Table 217-1). Opportunistic fungi cause serious infections when impaired host immune responses allow the organisms to transition from commensals to invasive pathogens. Endemic fungi typically cause self-limited disease in immunocompetent hosts but severe illness in immunosuppressed patients.

Fungi are morphologically classified as *yeast*, *mold*, and *dimorphic*. Yeasts are seen as round single cells or budding organisms. Molds grow as filamentous forms called *hyphae* both at room temperature and in tissue. *Aspergillus*, *Mucorales*, and dermatophytes that infect skin and nails are mold fungi. Variations exist within this classification system. For instance, when *Candida* infects tissue, both yeasts and filamentous forms (pseudohyphae) may be present (except in the cases of *C. glabrata* and *C. auris*, which form only yeasts in tissue); in contrast, *Cryptococcus* exists only in yeast form. *Dimorphic* is the term used to describe fungi that have two forms; they grow as yeasts or large spherical structures in tissue but as filamentous forms at room temperature in the environment (Table 217-1).

Patients acquire deep organ infection by molds and endemic dimorphic fungi via inhalation. Skin dermatophytes are primarily environmentally acquired, but human-to-human transmission may also occur. The commensal *Candida* invades deep tissues from sites of mucocutaneous colonization, usually in the gastrointestinal tract or the skin in the case of *C. auris*.

In this chapter, we outline general principles of immunology, diagnosis, and treatment related to the most common human fungal infections.

PATHOGENESIS

In the past decade, our understanding of fungal recognition pathways and of tissue-specific innate and adaptive antifungal host defense mechanisms has markedly expanded. A major breakthrough has been the discovery and functional characterization of the C-type lectin receptor/spleen tyrosine kinase/caspase recruitment domain-containing protein 9 (CLR/SYK/CARD9) signaling pathway, which mediates fungal polysaccharide recognition and orchestrates proinflammatory mediator production, leukocyte recruitment, inflammasome

TABLE 217-1 Major Fungal Infections, Associated At-Risk Patient Populations, and Diagnostic Tests

INFECTION (MOST COMMON FUNGAL GENERA AND SPECIES)	CLINICAL SYNDROME(S)	RISK FACTOR(S)	DIAGNOSTIC TEST(S)
Mold (Filamentous) Fungi			
Aspergillosis (<i>Aspergillus fumigatus</i> , <i>A. terreus</i> , <i>A. flavus</i> , <i>A. niger</i> , <i>A. nidulans</i> ^a)	Pneumonia or disseminated infection ABPA Keratitis	Neutropenia, glucocorticoids, HSCT, post-influenza or COVID-19, BTK inhibition Atopic individuals Direct inoculation	<i>Culture</i> of BAL fluid: low sensitivity, nonspecific (colonization, contamination) <i>Histologic examination</i> of tissue ^b : acute-angle septate hyphae <i>Biomarkers</i> : GM (BAL > serum); serum BDG (nonspecific)
Mucormycosis (<i>Rhizopus</i> , <i>Rhizomucor</i> , <i>Mucor</i> , <i>Cunninghamella</i> , and <i>Lichtheimia</i> spp.)	Sinopulmonary infection Rhinocerebral infection Necrotizing skin infection	Neutropenia, HSCT Diabetic ketoacidosis Direct inoculation (e.g., tornado victims)	<i>Culture</i> of BAL fluid or sinus tissue: very low sensitivity <i>Histologic examination</i> of tissue: ribbon-like aseptate hyphae <i>Biomarkers</i> : Negative
Fusariosis (<i>Fusarium solani</i> , <i>F. oxysporum</i>)	Pneumonia or disseminated infection Keratitis	Neutropenia Direct inoculation	<i>Culture</i> of tissue or blood: one of the few molds recovered from blood <i>Histologic examination</i> of tissue: acute-angle septate hyphae <i>Biomarkers</i> : GM can be positive; BDG (nonspecific)
Scedosporiosis (<i>Scedosporium apiospermum</i>)	Pneumonia or disseminated infection	Neutropenia, glucocorticoids, HSCT	<i>Culture</i> of BAL: low sensitivity, nonspecific (colonization, contamination) <i>Histologic examination</i> of tissue: acute-angle septate hyphae <i>Biomarkers</i> : BDG can be positive
Phaeohyphomycosis (<i>Cladophialophora</i> , <i>Alternaria</i> , <i>Phialophora</i> , <i>Rhinocladiella</i> , <i>Exophiala</i> , and <i>Exserohilum</i> spp.)	Sinopulmonary, CNS, or disseminated infection Skin infection Allergic sinusitis	HSCT, neutropenia, glucocorticoids, healthy individuals (for CNS), TNF- α inhibition Direct inoculation Atopic individuals	<i>Culture</i> of ordinarily sterile site <i>Histologic examination</i> of tissue: cell walls may appear dark brown or golden on H&E; Fontana-Masson may stain fungal melanin
Dermatophytosis (<i>Trichophyton</i> , <i>Microsporum</i> , and <i>Epidermophyton</i> spp.)	Skin and nail infections	Healthy individuals	<i>Culture</i> or <i>microscopic examination</i> of scrapings or clippings: chains of arthrospores (diagnostic)
Eumycetoma (<i>Madurella mycetomatis</i>)	Skin and subcutaneous infections	Healthy individuals	<i>Culture</i> and <i>macroscopic and histologic examination</i> of grains harvested from biopsy or aspiration
Yeast Fungi			
Mucosal candidiasis ^c (<i>Candida albicans</i> , <i>C. glabrata</i>)	Oropharyngeal or esophageal candidiasis Vulvovaginal candidiasis	AIDS, glucocorticoids Antibiotic use	<i>Culture</i> of mucosal surfaces <i>Histologic examination</i> of esophageal tissue or wet preparation (10% KOH) of vaginal discharge: yeast and/or pseudohyphae
Invasive candidiasis ^c (<i>C. albicans</i> , <i>C. glabrata</i> , <i>C. parapsilosis</i> , <i>C. tropicalis</i> , <i>C. auris</i>)	Candidemia Disseminated infection (spleen, liver, kidney, eye, heart, CNS)	Critical illness (ICU) Neutropenia, glucocorticoids	<i>Culture</i> of blood: low sensitivity <i>Histologic examination</i> of tissue: yeast and/or pseudohyphae <i>Biomarkers/other tests</i> : BDG (nonspecific); T2 magnetic resonance in whole blood
Cryptococcosis (<i>Cryptococcus neoformans</i> , <i>C. gattii</i>)	Pneumonia Osteomyelitis Meningoencephalitis	AIDS, glucocorticoids Sarcoidosis AIDS, AAbs to IFN- γ or GM-CSF, BTK or JAK inhibition	<i>Culture</i> of CSF, BAL fluid, blood <i>Microscopic examination</i> of tissue or CSF: encapsulated yeast (GMS, India ink, mucicarmine stain) <i>Biomarkers</i> : <i>Cryptococcus</i> Ag (serum, CSF) is sensitive and specific
Trichosporonosis ^d (<i>Trichosporon asahii</i> , <i>T. mucoides</i> , <i>T. asteroides</i>)	Superficial skin infection (white piedra) Disseminated infection (skin, eye)	Healthy individuals Neutropenia, glucocorticoids, HSCT, SOT	<i>Culture</i> of tissue or blood <i>Histologic examination</i> of tissue: yeasts, hyphae, and arthroconidia <i>Biomarkers</i> : BDG can be positive
Endemic Dimorphic Fungi			
Histoplasmosis (<i>Histoplasma capsulatum</i> , <i>H. duboisii</i> [in Africa])	Self-limited pneumonia Disseminated infection (liver, bone, bone marrow) Fibrosing mediastinitis	Healthy individuals AIDS, SOT, glucocorticoids, AAbs to IFN- γ , JAK or TNF- α inhibition	<i>Culture</i> of blood or tissue: low sensitivity; weeks needed for growth <i>Histologic examination</i> of tissue: yeast with narrow-based budding <i>Other tests</i> : <i>Histoplasma</i> Ag (urine > serum > BAL); BDG can be positive; serology (CF) can be useful in non-AIDS patients
Blastomycosis (<i>Blastomyces dermatitidis</i> , <i>B. gilchristii</i>)	Pneumonia Disseminated infection (skin, bone, mucosal surfaces, genitourinary tract)	Healthy individuals AIDS, glucocorticoids, TNF- α inhibition	<i>Culture</i> of BAL or tissue: low sensitivity; weeks needed for growth <i>Histologic examination</i> of tissue: yeast with broad-based budding <i>Other tests</i> : serology (CF, ID) has low sensitivity; <i>Blastomyces</i> Ag test cross-reacts with other endemic fungi; GM can be positive
Coccidioidomycosis (<i>Coccidioides immitis</i> , <i>C. posadasii</i>)	Self-limited pneumonia Disseminated infection (CNS, bone)	Healthy individuals AIDS, glucocorticoids, TNF- α inhibition	<i>Culture</i> is diagnostic ^e <i>Histologic examination</i> : spherules <i>Other tests</i> : serology (CF, ID); <i>Coccidioides</i> Ag test can be useful in CNS infection; BDG can be positive
Paracoccidioidomycosis (<i>Paracoccidioides brasiliensis</i> , <i>P. lutzii</i>)	Pneumonia Disseminated infection (skin, bone, mucosal surfaces)	Healthy individuals AIDS, glucocorticoids	<i>Culture</i> of tissue: active disease; several weeks needed for growth <i>Histologic examination</i> of KOH preparations or tissue: yeast with budding in steering-wheel pattern <i>Other tests</i> : serology (ID, CF); <i>Paracoccidioides</i> Ag test

(Continued)

TABLE 217-1 Major Fungal Infections, Associated At-Risk Patient Populations, and Diagnostic Tests (Continued)

INFECTION (MOST COMMON FUNGAL GENERA AND SPECIES)	CLINICAL SYNDROME(S)	RISK FACTOR(S)	DIAGNOSTIC TEST(S)
Sporotrichosis (<i>Sporothrix schenckii</i>)	Lymphocutaneous infection (ascending lymphangitis) Disseminated infection	Direct inoculation AIDS, glucocorticoids	Culture of tissue (diagnostic) Histologic examination: cigar-shaped yeast, often with surrounding asteroid body
Talaromycosis (<i>Talaromyces marneffe</i>)	Pneumonia Disseminated infection (skin, bone, mucosal surfaces)	Healthy individuals AIDS, glucocorticoids, AAbs to IFN- γ	Culture of tissue (diagnostic) Histologic examination of tissue: yeasts with transverse septa Biomarkers: GM is often positive
Adiaspiromycosis (<i>Emmonsia crescens</i> , <i>E. parva</i>)	Pneumonia	Occupational dust exposure	Culture: nonculturable Histologic examination: thick-walled adiaspore within granuloma
Emergomycosis (<i>Emergomycetes africanus</i> , <i>E. pasteurianus</i>)	Disseminated infection (lungs, skin)	AIDS, SOT	Culture of infected tissue Histologic examination of tissue: yeast with narrow-based budding Biomarkers: Histoplasma Ag can be positive
Chromoblastomycosis (<i>Fonsecaea pedrosoi</i> , <i>F. monophora</i>)	Skin and subcutaneous tissue infections	Healthy individuals	Culture of infected tissue Histologic examination of scrapings (KOH) or tissue (GMS): sclerotic bodies (pathognomonic)
Other Fungi			
Pneumocystosis ^f (<i>Pneumocystis jirovecii</i>)	Pneumonia Disseminated infection (eye, CNS, skin, gastrointestinal tract)	AIDS, glucocorticoids, BTK inhibition AIDS	Culture: nonculturable Histologic examination (gold standard): special (GMS, Diff-Quik) or immunofluorescence stains Biomarkers/other tests: BDG (nonspecific); BAL fluid PCR (sensitive; can be positive in colonized individuals)

^a*A. nidulans* is seen almost exclusively in chronic granulomatous disease. ^bGMS or PAS stains. ^cSome *Candida* species form pseudohyphae. ^d*Trichosporon* species are yeast-like fungi that also generate septate hyphae and arthroconidia. ^e*Coccidioides* is a laboratory hazard. It is important to notify the microbiology laboratory if this infection is suspected. ^f*Pneumocystis* is present in cyst and trophozoite forms.

Abbreviations: AAbs, autoantibodies; ABPA, allergic bronchopulmonary aspergillosis; Ag, antigen; BAL, bronchoalveolar lavage; BDG, β -D-glucan; BTK, Bruton's tyrosine kinase; CF, complement fixation; CNS, central nervous system; CSF, cerebrospinal fluid; GM, galactomannan; GM-CSF, granulocyte-macrophage colony-stimulating factor; GMS, Gomori methenamine silver; H&E, hematoxylin and eosin; HSCT, hematopoietic stem cell transplantation; ICU, intensive care unit; ID, immunodiffusion; IFN- γ , interferon γ ; JAK, Janus kinase; KOH, potassium hydroxide; PAS, periodic acid-Schiff; PCR, polymerase chain reaction; SOT, solid organ transplantation; TNF- α , tumor necrosis factor α .

activation, and Th17 cell differentiation upon fungal invasion. Human inherited CARD9 deficiency causes severe mucocutaneous and invasive fungal disease and is the only known primary immunodeficiency to feature fungus-specific infection susceptibility without a predisposition to other infections, autoimmunity, allergy, or cancer. Notably, CARD9-deficient patients develop infections by certain fungi in certain tissues, including (1) chronic mucocutaneous candidiasis linked to defective interleukin (IL) 17 responses; (2) infections of the central nervous system (CNS) caused by *Candida* (but also by *Aspergillus* and phaeohyphomycetes) linked to impaired microglial-neutrophilic responses; and (3) deep dermatophytosis. Thus, the clinical use of SYK inhibitors for autoimmunity and cancer may cause opportunistic fungal disease. Human inherited deficiency of Toll-like receptor (TLR) signaling does not lead to spontaneous fungal disease, yet polymorphisms in TLR pathway molecules may increase the risk of fungal disease in critically ill or immunosuppressed persons, and TLR stimulation may boost protective CLR immunity, as has been shown with the TLR7 agonist imiquimod in chromoblastomycosis.

The development of clinically relevant animal models of mycoses and the phenotypic characterization of fungal infections that develop in patients with primary immunodeficiencies and in recipients of immune pathway-targeting biologics have led to the delineation of fungus-, cell-, and tissue-specific requirements for antifungal host defense (Fig. 217-1).

At the mucosal interface, IL-17-producing lymphoid cells play a critical role in protection by driving epithelial cell production of antimicrobial peptides that restrict mucosal *Candida* invasion. Indeed, AIDS patients are at risk for mucosal—but not invasive—candidiasis. Concordantly, inherited deficiency of IL-17 signaling caused by mutations in *IL17F*, *IL17RA*, *IL17RC*, or *TRAF3IP2* (encoding the IL-17 receptor adaptor ACT1) or pharmacologic inhibition of IL-17 signaling by biologics that target IL-12p40, IL-23p19, IL-17A, IL-17A/IL-17F, or IL-17RA cause mucosal—but not invasive—candidiasis. Other conditions that underlie a predisposition to chronic mucocutaneous

candidiasis include primary immunodeficiencies due to mutations in *STAT3*, *STAT1*, *DOCK8*, *JNK1*, *IRF8*, *RORC*, and *CARD9*, all of which impair Th17 cells, as well as thymoma and autoimmune polyendocrinopathy–candidiasis–ectodermal dystrophy (APECED), which feature autoantibodies to IL-17A, IL-17F, and IL-22. In APECED, exacerbated T cell–derived interferon γ (IFN- γ)/STAT1 responses disrupt the integrity of the oral epithelial barrier, thereby promoting mucosal fungal invasion and infection; remission of candidiasis can be achieved with JAK inhibition in APECED and STAT1 gain of function (GOF). Of note, vaginal candidiasis (unlike oropharyngeal and esophageal candidiasis) develops in the setting of antibiotic treatment, not AIDS or IL-17–targeted biologics; this observation underscores the role of the microbiota in fungal control at the vaginal—but not the oral—mucosa.

On the other hand, neutrophils—but not lymphocytes—are critical for control of invasive infections caused by *Aspergillus* (and other inhaled molds) and *Candida* (Fig. 217-1). Indeed, patients with chemotherapy-induced neutropenia and patients undergoing allogeneic hematopoietic stem cell transplantation are at risk for invasive aspergillosis and candidiasis. Both oxidative and nonoxidative burst-dependent effector mechanisms are operational within neutrophils for fungal killing. Inherited deficiency in neutrophil superoxide generation due to mutations in the six subunits of the nicotinamide adenine dinucleotide phosphate (NADPH) oxidase complex causes chronic granulomatous disease, a prototypic primary immunodeficiency that carries a lifetime risk for invasive aspergillosis of ~40%; infrequently (i.e., in <5% of cases, mostly infants), chronic granulomatous disease predisposes to invasive candidiasis. The unexpected development of invasive mold infections in recipients of Bruton's tyrosine kinase (BTK) inhibitors has recently uncovered the critical role of BTK in promoting myeloid phagocyte-dependent antifungal effector functions.

Host defenses against fungi that reside within macrophages, such as *Cryptococcus*, *Pneumocystis*, and endemic dimorphic fungi, depend on the interplay of IFN- γ -producing lymphoid cells and IL-12–producing macrophages that enable intramacrophagic fungal killing (Fig. 217-1).

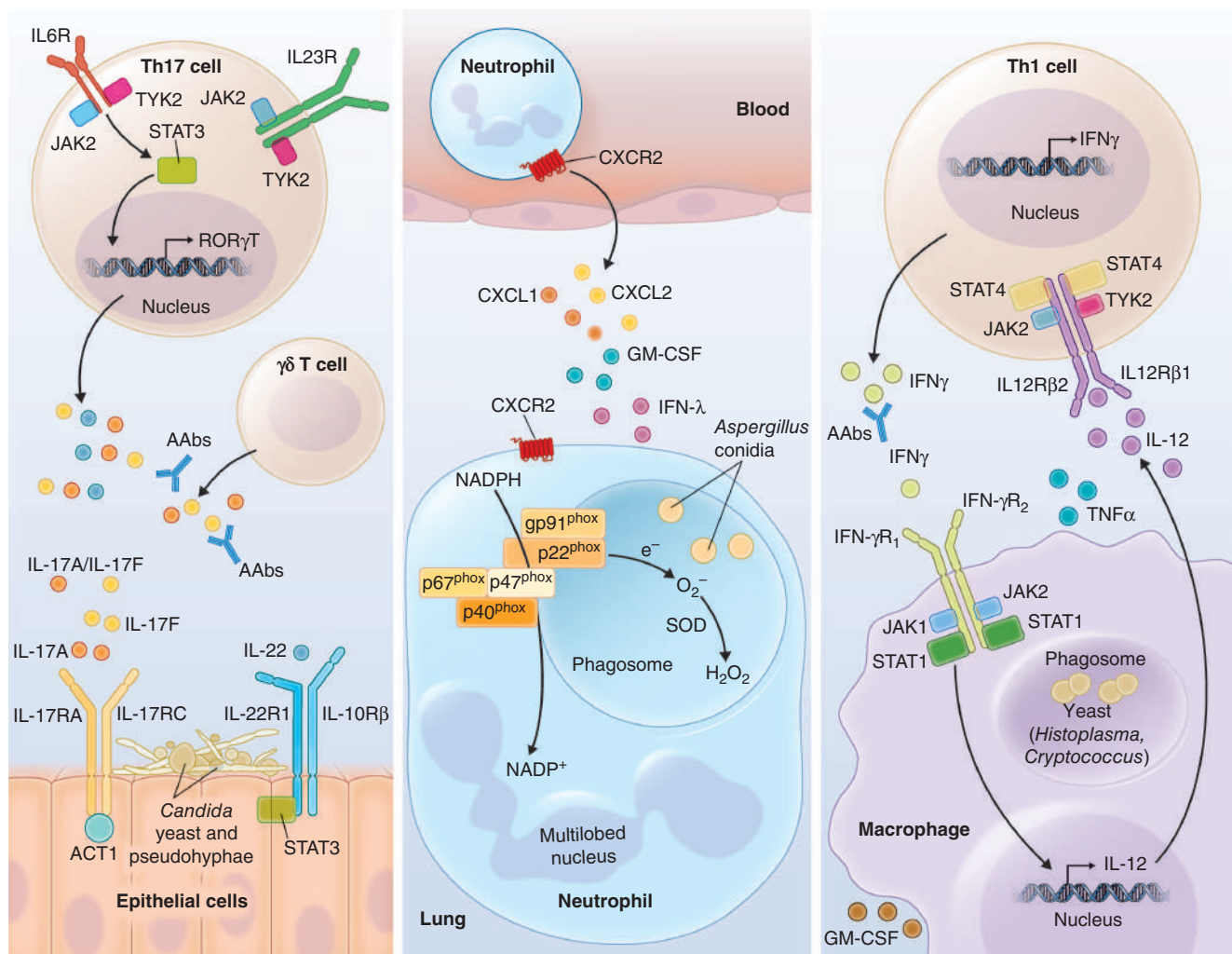


FIGURE 217-1 Host defense against fungi. **Left:** Production of IL-17A, IL-17F, and IL-22 by Th17 cells, γδ T cells, and innate lymphoid cells confers protection from mucosal *Candida* invasion. STAT3 promotes Th17 differentiation via RORγt induction. IL-17A and IL-17F bind to IL-17RA and IL-17RC on epithelial cells and signal via ACT1 to produce antimicrobial peptides that inhibit fungal growth. IL-22 binds to its receptor on epithelial cells and activates STAT3 to mediate epithelial proliferation and repair. **Middle:** Activation of CXCR2⁺ neutrophils recruited from blood in the *Aspergillus*-infected lung enables assembly of the six subunits of NADPH oxidase and superoxide generation that promotes fungal killing. Production of reactive oxygen species by neutrophils is facilitated by recruited monocyte-derived and plasmacytoid dendritic cells via type I and type III IFNs and GM-CSF. **Right:** The interaction of Th1 cells with macrophages is protective against intramacrophagic endemic dimorphic fungi, *Pneumocystis*, and *Cryptococcus*. Upon fungal uptake, macrophages produce IL-12 that binds to its receptor on T cells and activates STAT4, with consequent release of IFN-γ. IFN-γ binds to its receptor on macrophages and activates STAT1, thereby enabling fungal killing. TNF-α and GM-CSF are also critical for macrophage activation. AAbs, autoantibodies; IL, interleukin; IFN, interferon; JAK, Janus kinase; GM-CSF, granulocyte-macrophage colony-stimulating factor; NADPH, nicotinamide adenine dinucleotide phosphate; RORγt, RAR-related orphan receptor γ; SOD, superoxide dismutase; STAT, signal transducer and activator of transcription; TNF, tumor necrosis factor; TYK2, tyrosine kinase 2.

Indeed, AIDS patients and those receiving glucocorticoids, which affect lymphocytes and macrophages both quantitatively and qualitatively, are at risk for severe infections by these fungi. Accordingly, inherited impairment of the IL-12/IFN-γ signaling axis caused by mutations in *IL12RB1*, *IFNGR1*, *IFNGR2*, *STAT1*, *IRF8*, or *GATA2* underlies susceptibility to severe infection by intramacrophagic fungi (and other intramacrophagic pathogens, such as mycobacteria and salmonellae). In addition, the IFN-γ-targeting monoclonal antibody emapalumab, JAK inhibitors that block IFN-γ-dependent cellular responses, and autoantibodies to IFN-γ predispose to infection with intramacrophagic fungi, as do biologics targeting tumor necrosis factor α (TNF-α) and autoantibodies to granulocyte-macrophage colony-stimulating factor (GM-CSF). The latter predisposing factors reveal the central role of these two Th1-associated cytokines—TNF-α and GM-CSF—in macrophage activation.

Taken together, these observations show that the cellular and molecular factors that drive protective antifungal immune responses vary greatly with the anatomic site of the infection, the offending fungus, and the patient population (Table 217-1). The growing body of data on human immunologic responses to fungi promises to inform precision medicine strategies for risk assessment, prophylaxis, immunotherapy, and vaccination of vulnerable patients.

■ DIAGNOSIS

The diagnostic modalities used for various fungal infections are outlined in Table 217-1 and are detailed in the chapters on specific mycoses that follow in this section. Definitive diagnosis of a fungal infection requires histopathologic identification of the fungus invading tissue with parallel culture of the fungus from the specimen. Certain fungi have distinctive morphologic features that facilitate diagnosis (Table 217-1). The stains most often used to identify fungi are periodic acid–Schiff (PAS) and Gomori methenamine silver (GMS). *Candida*, unlike other fungi, is visible on Gram-stained tissue smears. Hematoxylin and eosin stains define accompanying histologic features of fungal disease (granuloma formation, angioinvasion, necrosis) but are insufficient to reliably identify fungi in tissue. A positive India ink stain of cerebrospinal fluid (CSF) is diagnostic for cryptococcosis. Most laboratories use calcofluor white staining coupled with fluorescence microscopy to identify fungi in fluid specimens. A positive fungal culture of blood or tissue may signify either patient colonization or lab contamination instead of true infection, with the most likely scenario depending on the fungus and the anatomic site. In blood, *Candida* can be detected with any of the widely used automated blood culture systems, but the lysis-centrifugation technique increases the sensitivity of blood cultures for

both *Candida* and other less common fungi (e.g., *Histoplasma*). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) is now used extensively for detection and speciation of fungi recovered from culture.

The several available fungal-antigen and serologic tests vary in sensitivity and specificity. The most reliable of these tests are the antibody to *Coccidioides*, *Histoplasma* antigen, and cryptococcal polysaccharide antigen. Serologic tests are also available for other endemic dimorphic fungi (Table 217-1). The galactomannan test—especially in the bronchoalveolar lavage fluid—is useful for the diagnosis of aspergillosis; however, false-negative results are common, particularly in patients receiving antifungal prophylaxis, and false-positive results may occur with other fungal infections. The β -glucan test has a high negative predictive value for invasive candidiasis but lacks specificity. T2 magnetic resonance is now approved by the U.S. Food and Drug Administration (FDA) for detection of *Candida* in blood. Several polymerase chain reaction and nucleic acid hybridization assays exist for fungal detection but are not standardized and are not widely used in the clinic.

■ ANTIFUNGAL DRUGS

This section provides a brief overview of available agents for the treatment of fungal infections. Drug regimens and schedules are detailed in the chapters on specific mycoses that follow in this section. Since fungal organisms, like human cells, are eukaryotic, the identification of drugs that selectively kill or inhibit fungi but that are not toxic to human cells poses challenges. Indeed, far fewer antifungal than antibacterial agents have been introduced into clinical medicine.

Early initiation of appropriate antifungal therapy is a critical determinant of favorable outcome, as has been shown for candidemia, aspergillosis, and mucormycosis. In addition, source control of the infection is important—e.g., with removal of the central venous catheter in candidemia, drainage of abdominal abscesses in intraabdominal candidiasis, and surgical debridement of sinus tissue in mucormycosis. Moreover, an essential factor in a favorable prognosis in patients with opportunistic mycoses is the achievement of immune reconstitution—e.g., with neutrophil recovery, tapering of glucocorticoids or other immunosuppressive drugs, or initiation of combination antiretroviral therapy in AIDS.

■ AMPHOTERICIN B

The advent of amphotericin B (AmB) in the 1950s revolutionized the treatment of deep-seated mycoses. Before the availability of AmB, cryptococcal meningitis and other disseminated fungal infections were nearly always fatal. AmB remains the broadest-spectrum antifungal agent. Its fungicidal mechanism of action involves forming extramembranous sponge-like aggregates that extract fungal ergosterol from lipid bilayers. AmB remains the preferred antifungal agent for the treatment of mucormycosis and fusariosis and for induction therapy for cryptococcal meningitis and disseminated infections caused by endemic dimorphic fungi. However, AmB has several limitations, including lack of a licensed oral formulation and significant toxicity from the intravenous preparations, primarily renal and infusion-related (fever, chills, thrombosis). The introduction of lipid AmB formulations has ameliorated these toxicities, and the lipid formulations have largely replaced the original deoxycholate formulation in resource-rich settings. In developing countries, AmB deoxycholate is still widely used because of the high cost of the lipid formulations. The two lipid formulations commonly used in the clinic are liposomal AmB and AmB lipid complex, which exhibit comparable efficacy, toxicity, and tissue penetration profiles.

■ AZOLES

Azoles offer important advantages over AmB, such as the availability of oral and IV formulations and a lack of renal toxicity. The mechanism of action of azoles involves inhibition of lanosterol 14 α -demethylase and ergosterol synthesis in the fungal cell membrane, with a consequent accumulation of toxic sterol intermediates and growth arrest. Unlike AmB, azoles are considered fungistatic.

Fluconazole Fluconazole plays an important role in the treatment of several fungal infections. Its major advantages are the availability of oral and IV formulations, a long half-life, penetration into most body

fluids (ocular fluid, CSF, urine), and minimal toxicity. This drug rarely causes liver toxicity; high doses may result in alopecia, dry mouth, and a metallic taste. Notably, the administration of even low doses of fluconazole to pregnant women for the treatment of vaginal candidiasis was recently linked to miscarriage and stillbirth. Fluconazole has no activity against molds and most endemic dimorphic fungi and is less active than the newer azoles against *C. glabrata* and *C. krusei*.

Fluconazole is the preferred agent for the treatment of coccidioid meningitis, although relapses may occur despite therapy. Fluconazole is also used as consolidation and maintenance therapy for cryptococcal meningitis and for the treatment of mucosal candidiasis. It is used for treating candidemia in patients who are not critically ill or immunosuppressed; in these patients, fluconazole was found to be as efficacious as AmB. Because of increasing rates of azole-resistant *Candida* strains, echinocandin treatment is preferred, which is then replaced by fluconazole once a susceptible *Candida* species is recovered. Fluconazole is effective as prophylaxis in recipients of high-risk liver and allogeneic bone marrow transplants, although many centers now use posaconazole in neutropenic patients, given its added activity against molds. Fluconazole prophylaxis in leukemic patients, in AIDS patients with low CD4+ T-cell counts, and in patients on surgical intensive care units is controversial.

Itraconazole Itraconazole is available in oral (capsule, suspension) and IV formulations and has broader antifungal activity—i.e., against molds and endemic dimorphic fungi—than fluconazole. Itraconazole is the drug of choice for mild to moderate histoplasmosis and blastomycosis and has also been used to treat chronic coccidioidomycosis, phaeohyphomycosis, sporotrichosis, and mucocutaneous mycoses such as oropharyngeal candidiasis, tinea versicolor, tinea capitis, and onychomycosis. Although it is approved by the FDA for use in febrile neutropenic patients, most centers now use newer azoles in such patients. Disadvantages of itraconazole include its poor CSF penetration, the use of cyclodextrin in its oral suspension and IV formulation, and its variable level of absorption in the capsule form, which requires monitoring of blood levels in patients receiving capsules for disseminated mycoses. A new formulation of itraconazole, called SUBA-itraconazole (for “super-bioavailability”), which results in improved absorption and less variable plasma levels, has recently been approved by the FDA. Itraconazole is a potent CYP3A4 inhibitor; this characteristic leads to significant drug interactions. The drug causes hepatotoxicity and cardiac toxicity that may manifest as congestive heart failure.

Voriconazole Voriconazole is also available in oral and IV formulations, has far broader antifungal activity than fluconazole (including *C. glabrata*, *C. krusei*, *Aspergillus*, *Scedosporium*, and endemic dimorphic fungi—but not *Mucorales*), and penetrates into most body fluids (ocular fluid, CSF). It is the preferred agent for the treatment of aspergillosis and also has been used to treat scedosporiosis and as step-down (but not primary) therapy for coccidioidomycosis, blastomycosis, and histoplasmosis. Voriconazole is considerably more expensive than fluconazole, and as with itraconazole, its use is associated with numerous interactions with drugs typically used in patients at risk for fungal infections. Hepatotoxicity, visual disturbances, and skin rashes (including photosensitivity) are relatively common, and long-term use requires skin cancer surveillance. A unique toxicity of voriconazole among azoles is fluorosis-associated periostitis. It is crucial to monitor drug levels because (1) voriconazole is metabolized in the liver by CYP2C9, CYP3A4, and CYP2C19; and (2) human genetic variation in CYP2C19 activity exists and can lead to significant interpatient variability in drug levels. Dosages should be reduced in patients with hepatic, but not renal, failure; however, because the IV formulation is prepared in cyclodextrin, it should be given with caution to patients with severe renal failure.

Posaconazole Posaconazole has broader activity than voriconazole, including activity against *Mucorales*. Both oral (suspension, tablet) and IV formulations are available. Posaconazole is approved by the FDA for antifungal prophylaxis in neutropenic leukemic patients and allogeneic hematopoietic stem cell transplant recipients as well as for treatment of oropharyngeal candidiasis, including infections refractory

to fluconazole or itraconazole. Posaconazole has been reported to be effective salvage therapy for aspergillosis, mucormycosis, fusariosis, cryptococcosis, histoplasmosis, and coccidioidomycosis, although controlled clinical trials are lacking. The tablet formulation is not hampered by the suboptimal absorption that occurs with the suspension; the tablet also results in higher and more reliable blood levels of the drug. Posaconazole is less hepatotoxic than voriconazole and does not cause the skin, visual, or bone toxicity that occurs with voriconazole. However, the use of posaconazole is linked to significant P450-related drug interactions and can lead to pseudohyperaldosteronism, which may manifest with hypokalemia, edema, and/or hypertension.

Isavuconazole Isavuconazole is available in oral and IV formulations and has broad antifungal activity similar to that of posaconazole. Isavuconazole is approved by the FDA for treatment of aspergillosis (on the basis of a randomized controlled trial that found it noninferior to voriconazole) and mucormycosis (on the basis of an open-label, non-comparative trial of 37 patients). Isavuconazole appears to be less hepatotoxic than voriconazole; it does not cause skin, bone, or visual toxicity, and it causes fewer P450-associated drug interactions than voriconazole.

Oteseconazole The newest azole—a tetrazole—oteseconazole, is approved by the FDA for the treatment of recurrent vulvovaginal candidiasis. It has potent activity against *Candida* (including azole-resistant strains) species and endemic dimorphic fungi. Its long half-life allows for once-weekly dosing following induction. Oteseconazole does not appear to cause hepatotoxicity or P450-associated drug interactions, but it is contraindicated in women of reproductive potential as it may cause fetal ocular toxicity.

■ ECHINOCANDINS

The echinocandins include the FDA-approved drugs caspofungin, anidulafungin, micafungin, and rezafungin, which are available solely as an IV formulation and inhibit β -1,3-glucan synthase, an enzyme that is crucial for fungal cell-wall synthesis but is not a constituent of human cells. Rezafungin, a new generation echinocandin, has a prolonged half-life that allows for weekly dosing with front-loading. The different echinocandins have comparable efficacy, toxicity, and tissue penetration profiles; are fungicidal for *Candida* and fungistatic for *Aspergillus*; and have no activity against other molds, *Cryptococcus*, or endemic dimorphic fungi. Their most common use to date is in candidal infections. These drugs offer three major advantages: minimal toxicity, minimal drug interactions, and activity against all *Candida* species. The minimal inhibitory concentrations (MICs) of echinocandins are higher against *Candida parapsilosis* than against other *Candida* species, but the higher MICs do not translate into less clinical efficacy against this species.

In controlled trials, caspofungin was as efficacious as AmB against candidemia and invasive candidiasis and as efficacious as fluconazole against candidal esophagitis. Caspofungin has also been efficacious as salvage therapy for aspergillosis. Anidulafungin is approved by the FDA as therapy for candidemia in nonneutropenic patients and for *Candida* esophagitis, abdominal infection, and peritonitis. In controlled trials, anidulafungin was noninferior and possibly superior to fluconazole against candidemia and invasive candidiasis and was as efficacious as fluconazole against candidal esophagitis. Micafungin is approved by the FDA for the treatment of candidal esophagitis and candidemia and for antifungal prophylaxis in hematopoietic stem cell transplantation. Moreover, micafungin yielded favorable results when used for the treatment of invasive aspergillosis and candidiasis in open-label trials. Rezafungin was recently approved by the FDA for the treatment of candidemia and invasive candidiasis.

■ IBREXAFUNGERP

Ibrexafungerp is an oral antifungal agent that belongs to a new class of glucan synthase inhibitors called triterpenoids and has potent activity against *Candida* species. Ibrexafungerp has been approved by the FDA for the treatment of vulvovaginal candidiasis and recurrent vulvovaginal candidiasis as it maintains its antifungal efficacy at low pH that is often observed in the vaginal mucosa. Its use is contraindicated in pregnant and lactating women.

■ FLUCYTOSINE (5-FLUOROCYTOSINE)

Flucytosine use has diminished as newer antifungal drugs have been developed. Its mechanism of action involves intrafungal conversion to 5-fluorouracil, which inhibits fungal DNA synthesis. The use of flucytosine in combination with AmB as induction therapy for cryptococcal meningitis is based on the drugs' synergistic interaction and favorable flucytosine CSF penetration that promotes a rapid decline of the cryptococcal burden in the CSF. Flucytosine is also used in combination with AmB for the treatment of candidal meningitis and endocarditis, although comparative trials with AmB monotherapy are lacking. Flucytosine monotherapy is not recommended as it is associated with the development of resistance. Flucytosine can cause bone marrow suppression and liver toxicity, which are intensified when the drug is used with AmB.

■ TERBINAFINE

Terbinafine inhibits squalene epoxidase and ergosterol synthesis, is used for onychomycosis and ringworm infection, and is as effective as itraconazole in both conditions. Although active against other fungi, terbinafine penetrates poorly into tissues beyond the skin and nails and therefore is not preferred for systemic mycoses. Terbinafine carries a risk for hepatotoxicity.

■ TOPICAL ANTIFUNGAL AGENTS

A detailed discussion of topical agents for mucocutaneous mycoses is beyond the scope of this chapter; the reader is referred to [Chap. 225](#) and the dermatology literature. Azoles such as clotrimazole, miconazole, and ketoconazole are often used topically to treat common cutaneous mycoses as well as oropharyngeal and vaginal candidiasis. In vaginal candidiasis, oral fluconazole given once has the advantage of not requiring repeated intravaginal application. The polyenes nystatin and AmB have also been used topically for oropharyngeal and vaginal candidiasis. Agents from other classes that are used to treat these conditions include ciclopirox, haloprogin, terbinafine, naftifine, tolnaftate, and undecylenic acid.

■ FURTHER READING

BENNETT JE: Introduction to mycoses, in *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*, 9th ed, JE Bennett et al (eds). Philadelphia, Elsevier Saunders, 2020, pp 3082–3086.

LIONAKIS MS et al: Immune responses to fungal pathogens and therapeutic prospects. *Nat Rev Immunol* 23:433, 2023.

PAPPAS PG et al: Clinical mycology today: A synopsis of the mycoses study group education and research consortium (MSGERC) second biennial meeting, September 27–30, 2018, Big Sky, Montana, a proposed global research agenda. *Med Mycol* 58:569, 2020.

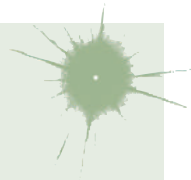
218 Histoplasmosis

Chadi A. Hage, L. Joseph Wheat

■ ETIOLOGY

Histoplasma capsulatum, a thermal dimorphic fungus, is the etiologic agent of histoplasmosis. In most endemic areas in North America, *H. capsulatum* var. *capsulatum* is the causative agent. In Central and South America, histoplasmosis is common and is caused by genetically different clades of *H. capsulatum* var. *capsulatum*. In Africa, *H. capsulatum* var. *duboisii* is also found. Yeasts of var. *duboisii* are larger than those of var. *capsulatum*.

Mycelia—the naturally infectious form of *Histoplasma*—have a characteristic appearance, with microconidial and macroconidial forms ([Fig. 218-1](#)). Microconidia are oval and are small enough



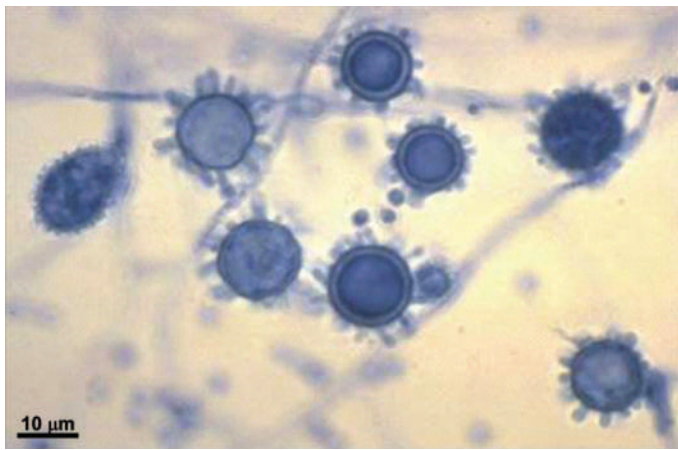


FIGURE 218-1 Spiked spherical conidia of *H. capsulatum* (lacto-phenol cotton blue stain) grown in the laboratory at room temperature.

(2–4 μm) to reach the terminal bronchioles and alveoli. Shortly after infecting the host, mycelia transform into the yeasts that are found inside macrophages and other phagocytes. The yeast forms are characteristically small (2–5 μm), with occasional narrow budding (Fig. 218-2). In the laboratory, mycelia are best grown at room temperature, whereas yeasts are grown at 37°C on enriched media.

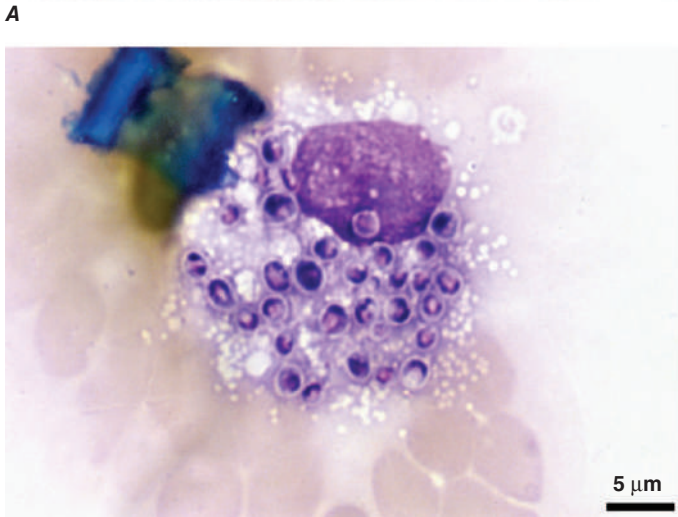
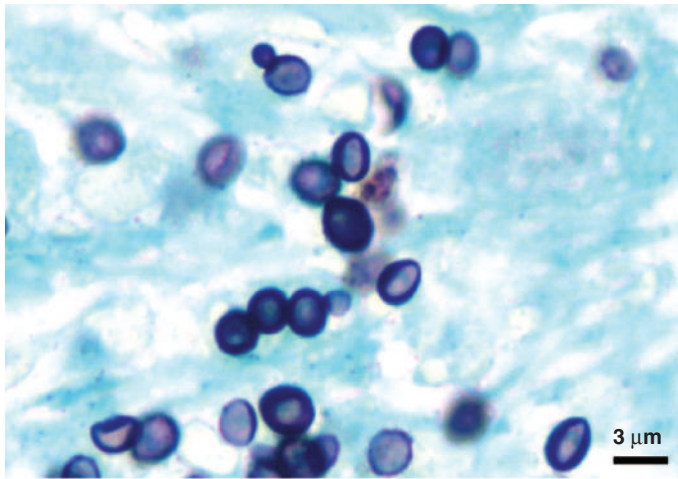


FIGURE 218-2 **A.** Small (2–5 μm) narrow budding yeasts of *H. capsulatum* from bronchoalveolar lavage fluid (Grocott's methenamine silver stain). **B.** Intracellular yeasts of *H. capsulatum* within an alveolar macrophage from a patient with AIDS and disseminated histoplasmosis (Giemsa stain).

■ EPIDEMIOLOGY

Histoplasmosis is the most prevalent endemic mycosis in North America. Although this fungal disease has been reported throughout the world, its endemicity is particularly notable in the Ohio and Mississippi river valleys of North America and in certain parts of Mexico, Central and South America (Brazil), Africa, and Asia. Histoplasmosis is increasingly reported outside of the traditionally known endemic areas. The geographic distribution of histoplasmosis is related to the humid and acidic nature of the soil in the endemic areas. Soil enriched with bird or bat droppings promotes the growth and sporulation of *Histoplasma*. Disruption of soil containing the organism leads to aerosolization of the microconidia and exposure of humans nearby. Activities associated with high-level exposure and high infection attack rate include spelunking, excavation, cleaning of chicken coops, demolition and remodeling of old buildings, and cutting of dead trees. Most cases seen outside of highly endemic areas represent imported disease, e.g., in Europe, histoplasmosis is diagnosed fairly often, mostly in emigrants from or travelers to endemic areas on other continents. The epidemiology of histoplasmosis is changing as a result of global climate changes and with the continued expansion of at-risk populations and the acceleration of intercontinental and international travel that brings this infection to areas of the world that are not known to be endemic. The population at risk for histoplasmosis continues to grow as a result of increasing numbers of patients receiving immunosuppressive therapies for autoimmune disorders, cancers, and organ transplants.

■ PATHOGENESIS AND PATHOLOGY

Infection follows inhalation of microconidia (Fig. 218-1). Once they reach the alveolar spaces, microconidia are rapidly recognized and engulfed by alveolar macrophages, where they transform into yeasts (Fig. 218-2), a process that is integral to the pathogenesis of histoplasmosis and is dependent on the availability of calcium and iron inside the phagocytes. The yeasts are capable of evading the immune system and multiplying inside resting macrophages. Neutrophils and then lymphocytes are attracted to the site of infection. Before the development of cellular immunity, yeasts use the phagosomes as a vehicle for translocation to local draining lymph nodes, whence they spread hematogenously throughout the reticuloendothelial system. Effective cellular immunity develops ~2 weeks after infection. T cells produce interferon-γ to assist the macrophages in killing the organism and controlling the progression of disease. Interleukin 12 and tumor necrosis factor α (TNF-α) play an essential role in cellular immunity to *H. capsulatum*. In the immunocompetent host, macrophages, lymphocytes, and epithelial cells eventually organize and form granulomas that contain the organisms. These granulomas typically fibrose and calcify; calcified lung nodules, mediastinal lymph nodes, and hepatosplenic calcifications are frequently found in healthy individuals from endemic areas. In immunocompetent hosts, infection with *H. capsulatum* confers protective immunity to reinfection. In patients with impaired cellular immunity, the infection is not properly contained and can disseminate throughout the reticuloendothelial system. Progressive disseminated histoplasmosis (PDH) can involve multiple organs, most commonly the lungs, bone marrow, spleen, liver (Fig. 218-3), adrenal glands, and mucocutaneous membranes. Unlike latent tuberculosis, inactive histoplasmosis does not reactivate. In patients with mildly impaired immune systems, active infection may smolder and eventually worsen with further decline in immunity.

Structural lung disease (e.g., emphysema) impairs the clearance of pulmonary histoplasmosis leading to the development of chronic pulmonary disease. This chronic process is characterized by progressive inflammation, tissue necrosis, and fibrosis mimicking cavitary tuberculosis.

■ CLINICAL MANIFESTATIONS

The clinical spectrum of histoplasmosis ranges from asymptomatic infection to life-threatening illness. The attack rate and the extent and severity of the disease depend on the intensity of exposure, the immune status of the exposed individual, and the underlying lung architecture of the host.

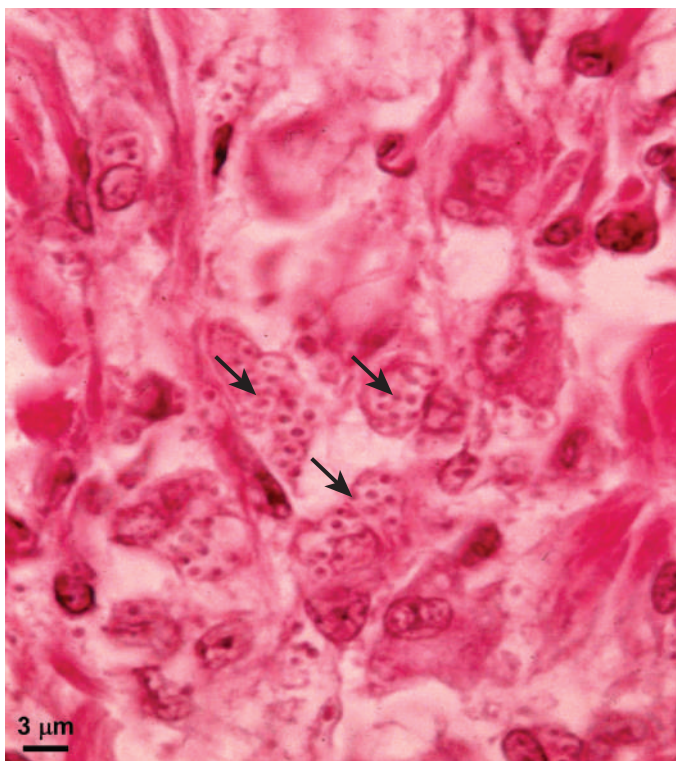


FIGURE 218-3 Intracellular yeasts (arrows) of *H. capsulatum* in a liver biopsy specimen (hematoxylin and eosin stain) from a patient who developed progressive disseminated histoplasmosis while receiving anti-tumor necrosis factor therapy for rheumatoid arthritis.

In immunocompetent individuals with low-level exposure, most *Histoplasma* infections are either asymptomatic or mild and self-limited. Of adults residing in endemic areas, up to 75% have immunologic and/or radiographic evidence of previous infection without clinical manifestations. Asymptomatic lung nodules representing controlled histoplasmosis are frequently found on chest computed tomography (CT) scans obtained during screening for lung cancer in smokers from endemic areas. When symptoms of acute histoplasmosis develop, they usually appear 1–4 weeks after exposure. Heavy exposure leads to a flulike illness with fever, chills, sweats, headache, myalgia, anorexia, dry cough, dyspnea, and chest pain. Chest radiographs usually show signs of pneumonitis with prominent hilar or mediastinal adenopathy. Pulmonary infiltrates may be focal with light exposure or diffuse with heavy exposure. Rheumatologic symptoms of arthralgia or arthritis, often associated with erythema nodosum, occur in 5–10% of patients with acute histoplasmosis. Pericarditis may also develop. These manifestations represent inflammatory responses to the acute pulmonary infection rather than extrapulmonary spread. Affected hilar or mediastinal lymph nodes may undergo necrosis and coalesce to form large mediastinal masses that can cause compression of great vessels, proximal airways, and the esophagus. These necrotic lymph nodes may also rupture and create fistulas between mediastinal structures (e.g., bronchoesophageal fistulae).

PDH is typically seen in immunocompromised individuals, who account for ~70% of cases. Common risk factors include AIDS (CD4+ T-cell count, <200/μL), extremes of age, the administration of immunosuppressive medications to prevent or treat rejection following transplantation (e.g., prednisone, mycophenolate, calcineurin inhibitors), and the use of methotrexate, anti-TNF-α agents, and other biologic response modifiers for autoimmune disorders. PDH may also occur in healthy individuals, some of whom may have rare undiagnosed genetic immunodeficiencies of the relevant immune pathways (IFN-γ and TNF-α). Workup for these conditions should be considered in healthy subjects with PDH.

The clinical spectrum of PDH ranges from an acute, rapidly fatal course—with diffuse interstitial or reticulonodular lung infiltrates

causing respiratory failure, shock, coagulopathy, and multiorgan failure—to a subacute or chronic course with a focal organ distribution. Common manifestations include fever, weight loss, hepatosplenomegaly, and thrombocytopenia. Other findings may include meningitis or focal brain lesions, ulcerations of the oral mucosa, gastrointestinal ulcerations and bleeding, and adrenal insufficiency. Prompt recognition of this devastating illness is of paramount importance in patients with severe manifestations or with underlying immunosuppression, especially those due to AIDS (**Chap. 208**) or anti-TNF therapy.

Chronic cavitary histoplasmosis is seen in smokers who have structural lung disease (e.g., bullous emphysema). This chronic illness is characterized by productive cough, dyspnea, low-grade fever, night sweats, and weight loss. Chest radiographs usually show upper-lobe infiltrates, cavitation, and pleural thickening—findings resembling those of tuberculosis. Without treatment, the course is slowly progressive.

Fibrosing mediastinitis is an uncommon but serious complication of histoplasmosis. In certain patients, acute infection is followed for unknown reasons by progressive fibrosis around the hilar and mediastinal lymph nodes, encasing mediastinal structures with potentially devastating consequences. Major manifestations include superior vena cava syndrome, obstruction of pulmonary vessels, and airway obstruction. Patients may experience recurrent pneumonia, hemoptysis, or respiratory failure. Fibrosing mediastinitis is fatal in up to one-third of cases.

In healed histoplasmosis, calcified mediastinal nodes or lung parenchymal nodules may erode through the walls of the airways and cause hemoptysis and expectoration of calcified material. This condition is called *broncholithiasis*.

The clinical features and management of histoplasmosis caused by the genetically different clades in Central and South America are similar to those of the disease in North America. African histoplasmosis caused by var. *duboisii* is clinically distinct and is characterized by frequent skin and bone involvement.

■ DIAGNOSIS

Recommendations for the diagnosis and treatment of histoplasmosis are summarized in **Table 218-1**. Once suspected, the diagnosis of histoplasmosis is usually straightforward as many diagnostic tools are now available in the United States. This is not the case in resource-limited endemic regions of Central America, South America, and Africa, where the diagnosis is often delayed, with consequently poor outcomes.

Fungal culture remains the gold standard diagnostic test for histoplasmosis. However, culture results may not be known for up to 1 month, and cultures are often negative in less severe cases. Cultures are positive in ~75% of patients with PDH and chronic pulmonary histoplasmosis. Cultures of bronchoalveolar lavage (BAL) fluid are positive in about half of patients with acute pulmonary histoplasmosis causing diffuse infiltrates and hypoxemia. In PDH, the culture yield is highest for BAL fluid, bone marrow aspirate, and blood. Cultures of sputum or bronchial washings are usually positive in chronic pulmonary histoplasmosis. Cultures are typically negative, however, in other forms of histoplasmosis.

Fungal stains of cytopathology or biopsy materials showing structures resembling *Histoplasma* yeasts are helpful in the diagnosis of PDH, yielding positive results in about half of cases. Yeasts can be seen in BAL fluid (Fig. 218-2) from patients with diffuse pulmonary infiltrates, in bone marrow biopsy samples, and in biopsy specimens of other involved organs (e.g., liver, adrenal glands). Occasionally, yeasts are seen within circulating phagocytes on blood smears from patients with severe PDH. However, staining artifacts and other fungal elements sometimes stain positively and may be misidentified as *Histoplasma* yeasts. Culture and pathology are no longer performed in most patients because diagnosis is more often established by antigen detection and/or serology, more rapidly and without subjecting the patient to invasive procedures.

The detection of *Histoplasma* antigen in body fluids is extremely useful in the diagnosis of PDH and acute diffuse pulmonary histoplasmosis.

TABLE 218-1 Recommendations for the Diagnosis and Treatment of Histoplasmosis

TYPE OF HISTOPLASMOSIS	DIAGNOSTIC TESTS	TREATMENT RECOMMENDATIONS	COMMENTS
Acute pulmonary, mild to moderate with no improvement by the time of diagnosis	<i>Histoplasma</i> antigen (BAL fluid, serum, urine) Cytopathology and fungal culture of BAL fluid <i>Histoplasma</i> serology (ID and CF), (EIA): IgG and IgM	Itraconazole (200 mg bid) for 6–12 weeks. Monitor renal and hepatic function.	Patients with mild cases usually recover without therapy, but itraconazole should be considered if the patient's condition is not already improving by the time the diagnosis is established.
Acute pulmonary, severe illness, respiratory failure (ARDS)	<i>Histoplasma</i> antigen (BAL fluid, serum, urine) Cytopathology and fungal culture of BAL fluid <i>Histoplasma</i> serology (ID and CF), (EIA): IgG and IgM	Lipid AmB (3–5 mg/kg per day) ± glucocorticoids for 1–2 weeks; then itraconazole (200 mg bid) for 6–12 weeks. Monitor renal and hepatic function.	Adjust itraconazole dose to achieve blood levels of 2–5 µg/mL.
Chronic/cavitary pulmonary	<i>Histoplasma</i> serology (ID and CF), (EIA): IgG and IgM Fungal culture of sputum or BAL fluid	Itraconazole (200 mg bid) Adjust dose to achieve blood levels of 2–5 µg/mL for at least 12 months. Monitor hepatic function.	Continue treatment until radiographic findings show no further improvement. Monitor for relapse after treatment is stopped.
Progressive disseminated	<i>Histoplasma</i> antigen (BAL fluid, serum, urine) <i>Histoplasma</i> serology (ID and CF), (EIA): IgG and IgM Fungal culture of blood or bone marrow aspirate Cytopathology on biopsy of affected organ	Lipid AmB (3–5 mg/kg per day) for 1–2 weeks; then itraconazole (200 mg bid); adjust dose to achieve blood levels of 2–5 µg/mL for at least 12 months. Monitor renal and hepatic function.	Liposomal AmB is preferred, but the AmB lipid complex may be used because of cost. Chronic antifungal maintenance therapy may be necessary if the degree of immunosuppression cannot be substantially reduced. Monitor antigen levels in urine and serum during treatment to ensure proper response.
Central nervous system	<i>Histoplasma</i> antigen CSF <i>Histoplasma</i> serology (ID and CF), (EIA): IgG and IgM Fungal culture of CSF	Liposomal AmB (5 mg/kg per day) for 4–6 weeks; then itraconazole (200 mg bid) Adjust dose to achieve blood levels of 2–5 µg/mL for at least 12 months. Monitor renal and hepatic function.	A longer course of lipid AmB is recommended because of the high risk of relapse. Itraconazole should be continued until CSF or MRI abnormalities clear.

Abbreviations: AmB, amphotericin B; BAL, bronchoalveolar lavage; CF, complement fixation; CSF, cerebrospinal fluid; EIA, enzyme immunoassay; ID, immunodiffusion; MRI, magnetic resonance imaging.

The sensitivity of this method is >95% in patients with PDH and >80% in patients with severe acute pulmonary histoplasmosis resulting from heavy exposure, if both urine and serum are tested. Antigen levels correlate with severity of illness in PDH and can be used to follow disease progression, as levels predictably decrease with effective therapy. Increasing antigen levels also predict relapse. *Histoplasma* antigen can be detected in cerebrospinal fluid from patients with *Histoplasma* meningitis and in BAL fluid from those with pulmonary histoplasmosis. Lateral flow antigen detection allows the diagnosis of histoplasmosis at the bedside as a point-of-care testing method, which might offer access to rapid testing in resource-limited areas of the world where traditional antigen testing might not be available. Cross-reactivity occurs with African histoplasmosis, blastomycosis, coccidioidomycosis, paracoccidioidomycosis, talaromycosis, and rarely aspergillosis.

Serologic tests, including immunodiffusion (ID), complement fixation (CF), and IgG and IgM enzyme immunoassay (EIA), are useful for the diagnosis of histoplasmosis, especially in immunocompetent patients. One month may be required for the detection of antibodies after the onset of infection by ID or CF, but antibodies may be detected earlier by more sensitive methods (EIA). IgM appears first then declines, and IgG appears later and increases during the infection. EIA for IgG and IgM antibodies provides a more accurate method for monitoring changes and antibody levels. Serologic tests are especially useful for the diagnosis of chronic pulmonary histoplasmosis. Limitations of ID and CF, however, include insensitivity early in the course of infection and reduced sensitivity in immunosuppressed patients, especially those receiving immunosuppression for organ transplantation. Also, antibodies may persist for several years after infection. Positive results from past infection may lead to a misdiagnosis of active histoplasmosis in a patient with another disease process.

TREATMENT

Histoplasmosis

Treatment is indicated for all patients with PDH or chronic pulmonary histoplasmosis as well as for most symptomatic patients with acute pulmonary histoplasmosis who have not improved by the time the diagnosis is established, especially in those with diffuse

infiltrates and difficulty breathing. In most other cases of pulmonary histoplasmosis, treatment is not recommended, especially if the immune system of the host is intact and the degree of exposure is not heavy. The symptoms usually are mild, subacute, and not progressive, and the illness resolves without therapy. Treatment should be considered if the symptoms are not improving within a month.

The preferred treatments for histoplasmosis (Table 218-1) include the lipid formulations of amphotericin B in severe cases and itraconazole in others. Liposomal amphotericin B is more effective and better tolerated than the deoxycholate formulation and is more effective in patients with AIDS and PDH. The deoxycholate formulation of amphotericin B is an alternative to a lipid formulation for patients at low risk for nephrotoxicity and if liposomal amphotericin B is not available. Posaconazole and isavuconazole are alternatives for patients who cannot take itraconazole. *Histoplasma* may develop resistance to fluconazole and voriconazole, and they are not the preferred alternative to itraconazole, especially in immunocompromised patients.

In severe cases requiring hospitalization, a lipid formulation of amphotericin B is used first for 2 weeks, followed by itraconazole. In patients with meningitis, a lipid formulation of amphotericin B should be given for 4–6 weeks before switching to itraconazole. In immunosuppressed patients, the degree of immunosuppression should be reduced if possible, although immune reconstitution inflammatory syndrome (IRIS) may ensue. Antiretroviral treatment improves the outcome of PDH in patients with AIDS and is recommended; however, whether antiretroviral treatment should be delayed to avoid IRIS is unknown.

Blood levels of itraconazole should be monitored to ensure adequate drug exposure, with target concentrations of the parent drug and its hydroxy metabolites measuring 2–5 µg/mL. Drug interactions should be carefully assessed; itraconazole not only is cleared by cytochrome P450 metabolism but also inhibits cytochrome P450. This profile causes interactions with many other medications used routinely in organ transplant recipients.

The duration of treatment for acute pulmonary histoplasmosis is 6–12 weeks, while that for PDH and chronic pulmonary histoplasmosis is at least 1 year. Antigen levels in urine and serum should be monitored during and for at least 1 year after therapy

for PDH. Stable or rising antigen levels suggest treatment failure or relapse and should raise concerns regarding proper intake of itraconazole (capsule formulation with food), adherence to treatment, drug absorption, itraconazole serum concentrations, and drug interactions.

Lifelong itraconazole maintenance therapy is recommended for patients with persistently suppressed immunity but not for those with immune recovery—e.g., patients with AIDS who complete at least 1 year of itraconazole and show no signs of active infection including *Histoplasma* antigen levels <2 ng/mL, CD4+ T-cell count recovery to at least 150/μL (preferably >250/μL), and HIV suppression with viral load <50 copies/mL on antiretroviral therapy. Similarly, maintenance therapy may not be necessary in other immunocompromised patients if the clinical findings have cleared, antigen levels are <2 ng/mL, and immunosuppression is substantially reduced.

Fibrosing mediastinitis, which represents a chronic fibrotic reaction to past mediastinal histoplasmosis rather than an active infection, does not respond to antifungal therapy. Often patients with mediastinal granuloma have chronic or progressive courses and receive treatment with itraconazole and corticosteroids to reduce disease progression.

FURTHER READING

- ABDALLAH W et al: Diagnosis of histoplasmosis using the MVista *Histoplasma* galactomannan antigen qualitative lateral flow-based immunoassay: A multicenter study. *Open Forum Infect Dis* 8:ofab454, 2021.
- AZAR MM et al: Clinical perspectives in the diagnosis and management of histoplasmosis. *Clin Chest Med* 38:403, 2017.
- AZAR MM et al: Current concepts in the epidemiology, diagnosis, and management of histoplasmosis. *Semin Respir Crit Care Med* 41:13, 2020.
- BAHR NC et al: Histoplasmosis infections worldwide: Thinking outside of the Ohio River valley. *Curr Trop Med Rep* 2:70, 2015.
- HAGE CA et al: A multicenter evaluation of tests for diagnosis of histoplasmosis. *Clin Infect Dis* 53:448, 2011.

areas, including in eastern Washington state and in northeastern Utah, have also been described, and a recent climate model suggests that the endemic range may expand further east and north over time. Outside the United States, coccidioidomycosis is endemic to northern Mexico as well as to localized regions of Central America. In South America, there are endemic foci in Colombia, Venezuela, northeastern Brazil, Paraguay, Bolivia, and north-central Argentina.

The fungus is known to inhabit the native soils within the endemic regions. Because of the difficulty in isolating *Coccidioides* from environmental sites, the precise characteristics of potentially infectious soil are not known, but alluvial alkaline deposits in regions of relative aridity with moderate temperature ranges are most closely associated with endemicity. In the United States, several outbreaks of coccidioidomycosis have been associated with soil from archeologic excavations of Amerindian sites both within and outside of the recognized endemic region. When isolated from the soil, *Coccidioides* is found no more than 2–20 cm below the surface, nor is it usually isolated from cultivated soil. In addition to its saprophytic soil phase, *Coccidioides* appears to have an endozoan life cycle, likely infecting burrowing rodents. This would suggest that the organisms are concentrated in these rodents and their burrows rather than extant in the soil.

In endemic areas, most cases of human coccidioidomycosis occur without obvious soil or dust exposure. Climatic factors may increase the risk of infection. Periods of aridity following rainy seasons are associated with increases in the number of symptomatic infections. The number of reported cases of coccidioidomycosis has been steadily increasing over the past two decades. In 2019, a total of 20,003 cases were reported. The majority of these were either from Arizona or California, in approximately equal numbers, with Nevada, New Mexico, and Utah reporting a small fraction of cases. The factors associated with these increases have not been elucidated but likely include an influx of older individuals without prior coccidioidal infection into endemic areas, construction activity, increased reporting, and changing climate.

PATHOGENESIS, PATHOLOGY, AND IMMUNE RESPONSE

The life cycle of *Coccidioides* is depicted in [Fig. 219-1](#). On agar media and in the environment, *Coccidioides* organisms exist as filamentous molds or mycelia. Individual filaments (*hyphae*) elongate and branch, some growing upward. Portions of these aerial hyphae thicken and septate, and alternating cells then degenerate by autolysis, leaving barrel-shaped viable spores called *arthroconidia*. Measuring approximately 2 μm by 5 μm, individual arthroconidia can easily dislodge from the hypha and become airborne. These may persist in the air for prolonged periods. When arthroconidia are inhaled by a susceptible host, their small size allows them to evade initial mechanical mucosal defenses and reach deep into the bronchial tree, where infection is initiated.

Once within a susceptible host, the individual arthroconidia enlarge, become rounded, and develop internal septations. The resulting structures, called *spherules*, may attain sizes up to 200 μm and are unique to *Coccidioides*. The septations encompass uninuclear elements called *endospores*. Spherules may rupture and release packets of endospores that can themselves develop into new spherules, thus propagating infection locally. If returned to artificial media or the soil, the fungus reverts to its mycelial stage.

Clinical observations and data from animal studies strongly support the critical role of a robust cellular immune response in the host's control of coccidioidomycosis. Necrotizing granulomas containing spherules are typically identified in patients with resolved pulmonary infection. In disseminated disease in which infection is not controlled, granulomas are generally poorly formed or do not develop at all, and a polymorphonuclear leukocyte response is frequently seen. In patients who are asymptomatic or in whom the initial pulmonary infection has resolved, delayed-type hypersensitivity to coccidioidal skin antigens has been routinely documented and T-cell activation has been observed in peripheral blood stimulated with coccidioidal antigens. In persons who have developed protective cellular immunity, spherules exist in the lungs for prolonged periods, possibly lifelong, in a latent

219 Coccidioidomycosis

Neil M. Ampel

DEFINITION AND ETIOLOGY

Coccidioidomycosis, commonly known as Valley fever (see “Epidemiology,” below), is caused by dimorphic soil-dwelling fungi of the genus *Coccidioides*. Genetic analysis has demonstrated the existence of at least two species, *C. immitis* and *C. posadasii*. These species are indistinguishable with regard to the clinical disease they cause and their appearance on routine laboratory media, although *C. posadasii* may grow more quickly at higher temperatures. Thus, the organisms will be referred to simply as *Coccidioides* for the remainder of this chapter.

EPIDEMIOLOGY

Coccidioidomycosis is confined to the Western Hemisphere between the latitudes of 40°N and 40°S. In the United States, areas of high endemicity include the southern San Joaquin Valley of California (hence the sobriquet “Valley fever”) and the south-central region of Arizona. However, infection may be acquired in other areas of the southwestern United States, including the other southern counties in California, southern Nevada, southwestern Utah, southern New Mexico, and western Texas. Cases acquired well outside the recognized endemic

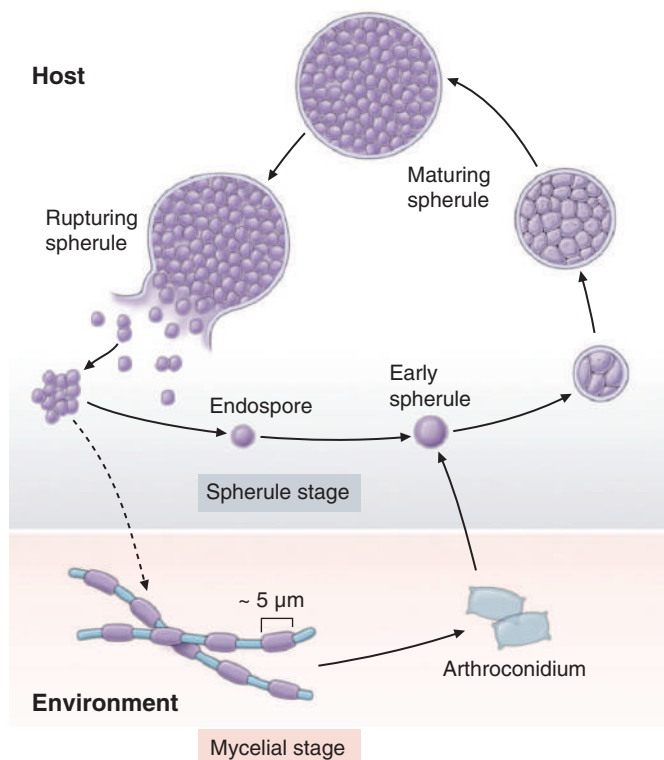


FIGURE 219-1 Life cycle of *Coccidioides*, including the mycelial phase in the environment and the spherule phase in the host.

state. This results in the persistence of protective immunity but also places those infected at risk of recurrent active disease if that immunity should wane.

CLINICAL AND LABORATORY MANIFESTATIONS

After infection, 60% of individuals are completely asymptomatic. The other 40% have symptoms that are related primarily to pulmonary infection, including fever, cough, and pleuritic chest pain. Symptoms generally occur from several days to up to 3 weeks after inhalation of arthroconidia. The risk of symptomatic illness increases with age.

In addition to these local symptoms indicative of infection, there are several manifestations of primary pulmonary coccidioidomycosis that are due to an immunologic response. Most prominent among these are cutaneous reactions. A diffuse, erythematous maculopapular rash, known as toxic erythema, has been noted in some cases. In addition, erythema nodosum (see Fig. A1-39)—typically over the lower extremities—and erythema multiforme (see Fig. A1-24)—usually in a necklace distribution—may occur. Lesions consistent with Sweet syndrome have also been reported (Chap. 21). Cutaneous manifestations are especially common in women. Symmetrical arthralgias (“desert rheumatism”), often involving the ankles, knees, and hips, may also occur with or without cutaneous manifestations.

Primary pulmonary coccidioidomycosis is often misdiagnosed as a community-acquired bacterial pneumonia. However, the diagnosis of primary pulmonary coccidioidomycosis is strongly suggested by the findings of rash or symmetrical arthralgias in a patient with an appropriate exposure history. The finding of any of the following is also strongly suggestive of coccidioidomycosis: a history of night sweats, marked fatigue, peripheral-blood eosinophilia, failure to improve with antibacterial therapy, and an upper lobe infiltrate or hilar or mediastinal lymphadenopathy on chest imaging.

In most patients, primary pulmonary coccidioidomycosis resolves without sequelae over several weeks. However, several pneumonic complications may arise. Pulmonary nodules are residua of the primary pneumonia. Generally single, frequently located in the upper lobes, and usually ≤ 4 cm in diameter, nodules are often discovered on routine chest radiograph in asymptomatic patients. Calcification is uncommon. Coccidioidal pulmonary nodules can be difficult

to distinguish radiographically from pulmonary malignancies. Like malignancies, coccidioidal nodules often enhance on positron emission tomography. However, unlike malignancies, routine computed tomography (CT) imaging often demonstrates multiple nodules and there may be microsatellite lesions, scattered smaller nodules surrounding the larger one. These findings are not specific, and biopsy may be required to distinguish between these two entities.

Pulmonary cavities occur when a nodule extrudes its contents into the bronchial tree, resulting in a thin-walled shell. Frequently asymptomatic, these cavities may on occasion be associated with persistent cough, hemoptysis, and pleuritic chest pain. They may also become secondarily infected with bacterial oral flora, environmental fungi such as *Aspergillus* species, or even with *Coccidioides* growing from the cavity wall. Rarely, a cavity may rupture into the pleural space, causing pyopneumothorax. In such cases, patients present with acute dyspnea, and the chest radiograph reveals a collapsed lung with a pleural air-fluid level. Chronic or persistent pulmonary coccidioidomycosis manifests with prolonged fever, cough, and weight loss and is radiographically associated with pulmonary scarring, fibrosis, and cavities. It is most common in patients who already have chronic lung disease due to other etiologies.

In some cases, primary pneumonia presents as a diffuse reticulo-nodular pulmonary process in association with dyspnea and fever. This manifestation of pulmonary coccidioidomycosis may occur in settings of intense environmental exposure or in those with profoundly suppressed cellular immunity, such as persons with untreated HIV-1 infection and markedly depressed peripheral blood CD4 cell count. In the latter case, this picture is associated with unrestrained fungal growth and is frequently associated with fungemia.

Clinical dissemination occurs in approximately 1% of infected individuals and is defined as finding the fungus outside the thoracic cavity. Dissemination is more likely to occur in male patients, particularly those of African and perhaps Filipino ancestry, and in persons with depressed cellular immunity, including patients with HIV-1 infection and peripheral blood CD4 counts of $<250/\mu\text{L}$, those receiving chronic glucocorticoid therapy, those with allogeneic solid-organ transplants, and those being treated with tumor necrosis factor- α antagonists or other biological response modifiers. Women who acquire new coccidioidal infection during the second or third trimester of pregnancy or postpartum also are at significant risk for disseminated disease. Common sites for dissemination include the skin, bones, joints, soft tissues, and meninges. Dissemination may follow symptomatic or asymptomatic pulmonary infection and may involve only one site or multiple anatomic foci. When it occurs, clinical dissemination is usually evident within the first 6 months after primary pulmonary infection and is generally associated with a lack of cellular immunity to *Coccidioides*.

Of the disseminated syndromes, coccidioidal meningitis is the most dire and is uniformly fatal if untreated. Patients usually present with a persistent dull headache, often accompanied by lethargy, and confusion. Nuchal rigidity, if present, is not severe. Examination of cerebrospinal fluid (CSF) demonstrates lymphocytic pleocytosis with profound hypoglycorrachia and elevated protein levels. CSF eosinophilia is occasionally observed. The diagnosis is usually established by finding coccidioidal complement-fixing antibody (see below) in the CSF in association with the CSF inflammatory pattern described above. The fungus is isolated from the CSF in fewer than one-third of cases. Magnetic resonance imaging with gadolinium frequently demonstrates enhancement in the basilar meninges. With or without appropriate therapy, patients may develop hydrocephalus, usually communicating, which presents clinically as a marked decline in mental status, often with gait disturbances. An elevated opening CSF pressure on lumbar puncture and dilated ventricles on brain imaging are hallmarks of this condition.

DIAGNOSIS

Serology is the most common method to establish the diagnosis of coccidioidomycosis. Several techniques are available, including the traditional tube-precipitin (TP) and complement-fixation (CF) assays, immunodiffusion TP and CF (IDTP and IDCF), and enzyme

immunoassay (EIA) to detect IgM and IgG antibodies. TP and IgM antibodies are found in serum soon after infection and persist for weeks to months. They are not useful for gauging severity of disease. The CF and IgG antibodies occur later in the course of the disease and persist longer than TP and IgM antibodies. Rising CF titers are a reflection of fungal growth and are associated with clinical progression, and the presence of CF antibody in CSF is indicative of coccidioidal meningitis. Antibodies disappear over time in persons whose clinical illness resolves and the presence of coccidioidal antibodies in the serum does not indicate immunity or control of coccidioidomycosis.

Because of its commercial availability, relatively rapid turnaround time, and higher sensitivity than other tests, EIAs for IgM and IgG are recommended for the initial diagnosis of coccidioidomycosis. Both tests, but particularly the IgM EIA, are occasionally falsely positive, and results should be interpreted cautiously when used for screening purposes in asymptomatic individuals. If either the IgM or IgG EIA serology is positive, immunodiffusion tests (IDTP and IDCF) or traditional TP and CF assays should be performed for confirmation. If the IDCF or traditional CF assays are qualitatively positive, a CF titer should be requested. The optical density obtained from the EIA IgG does not correlate with the serologic titer of either the IDCF or traditional CF assay and should not be used as a surrogate for a titer. The CF titer is an important prognostic tool. High titers, for example, those $\geq 1:32$, suggest unrestrained fungal growth, predict disease activity, and suggest the possibility of extrathoracic dissemination.

Coccidioides grows within 3–7 days at 37°C on a variety of artificial media, including blood agar. Therefore, it is always useful to obtain samples of sputum or other respiratory fluids and tissue for culture in suspected cases of coccidioidomycosis. The clinical laboratory should always be alerted to the possibility of this diagnosis, since *Coccidioides* poses a significant laboratory hazard if it is inadvertently inhaled. The organism can also be identified directly. While treatment of samples with potassium hydroxide is rarely fruitful, examination of sputum or other respiratory fluids after Papanicolaou, Gomori methenamine silver, or calcofluor white staining may reveal spherules in a significant proportion of patients with pulmonary coccidioidomycosis. For fixed tissues (e.g., those obtained from biopsy specimens), spherules with surrounding inflammation can be demonstrated with hematoxylin-eosin or Gomori methenamine silver staining.

A commercially available test for coccidioidal antigenuria and antigenemia has been developed and appears to be useful in immunosuppressed patients with severe or disseminated disease. It is also useful when the CSF is assayed in cases of suspected coccidioidal meningitis. False-positive results may occur in cases of histoplasmosis or blastomycosis. Some laboratories offer genomic detection by polymerase chain reaction; this assay does not appear to be more sensitive than culture but can be more rapid.

TREATMENT

Coccidioidomycosis

Currently, two classes of antifungal agents are useful for the treatment of coccidioidomycosis (Table 219-1). While once prescribed routinely, amphotericin B in all its formulations is now reserved for only the most severe cases of dissemination and for intrathecal or intraventricular administration in patients with coccidioidal meningitis in whom other antifungal therapy has failed. The original formulation of amphotericin B, which is dispersed with deoxycholate, is usually administered intravenously in doses of 0.7–1.0 mg/kg either daily or three times per week. The newer lipid-based formulations are associated with less renal toxicity, but there are no studies indicating whether they lead to better improvement than the deoxycholate formulation in coccidioidomycosis. The lipid dispersions are administered intravenously at doses of 3–5 mg/kg daily or three times per week.

Triazole antifungals are currently the principal drugs used to treat most cases of coccidioidomycosis. Clinical trials have demonstrated the usefulness of both fluconazole and itraconazole as initial

TABLE 219-1 Clinical Presentations of Coccidioidomycosis, Their Frequency, and Recommended Initial Therapy for the Immunocompetent Host

CLINICAL PRESENTATION	FREQUENCY, %	RECOMMENDED THERAPY
Asymptomatic infection	60	None
Primary pneumonia (focal)	40	In most cases, none ^a
Diffuse pneumonia	<1	Amphotericin B followed by prolonged oral triazole therapy
Pulmonary sequelae	5	
Nodule		None
Cavity		In most cases, none ^b
Chronic pneumonia		Prolonged triazole therapy
Disseminated disease	≤1	
Skin, bone, joint, soft tissue disease		Prolonged triazole therapy ^c
Meningitis		Lifelong triazole therapy ^d

^aTreatment is indicated for hosts with depressed cellular immunity as well as for those with prolonged symptoms and signs of increased severity, including night sweats for >3 weeks, weight loss of >10%, a complement-fixation titer of >1:16, and extensive pulmonary involvement on chest radiography. ^bTreatment (usually with the oral triazoles fluconazole and itraconazole) is recommended for persistent symptoms. ^cIn severe cases, some clinicians would use amphotericin B as initial therapy. ^dIntraventricular or intrathecal amphotericin B is recommended in cases of triazole failure. Hydrocephalus may occur, requiring a cerebrospinal fluid shunt.

Note: See text for dosages and durations.

agents. Fluconazole has been the triazole of choice for the treatment of coccidioidomycosis in the past because of availability, cost, predictable oral absorption, and perceived lack of severe adverse events. However, there are no studies demonstrating its superiority over other triazole antifungals, and evidence indicates that itraconazole is more effective against bone and joint disease and may be more effective for other forms of coccidioidomycosis. For both drugs, a minimal oral adult dosage of 400 mg/d should be used. The maximum dose of itraconazole is 200 mg three times daily, but higher doses of fluconazole may be given and 800 mg is frequently prescribed for coccidioidal meningitis. The newer triazole antifungals, voriconazole and posaconazole, are useful for all types of clinical disease, including meningitis, and should be considered in cases where fluconazole or itraconazole therapy has failed. To date, isavuconazole has been used in limited circumstances in coccidioidomycosis but also appears to be effective, including for meningitis. High-dose triazole therapy may be teratogenic during the first trimester of pregnancy, and triazole therapy should be avoided in pregnant women during this period.

There are several new antifungal agents that are promising for the treatment of coccidioidomycosis in the future. These include olorofim, oteseconazole, ibrexafungerp, fosmanogepix, and oral liquid nanocrystal amphotericin B. At this time, only olorofim has been studied in human coccidioidomycosis, and none are currently approved for use in coccidioidomycosis. However, olorofim has been studied in human coccidioidomycosis and has received U.S. Food and Drug Administration breakthrough therapy designation for difficult-to-treat *Coccidioides* meningitis.

Most patients with focal primary pulmonary coccidioidomycosis do not require antifungal therapy. Patients for whom antifungal therapy should be considered include those with underlying cellular immunodeficiencies and those with prolonged symptoms and signs of extensive disease. Specific criteria include symptoms persisting for ≥ 2 months, night sweats occurring for >3 weeks, weight loss of >10%, a serum CF antibody titer of >1:16, and extensive pulmonary involvement apparent on chest radiography. When antifungal therapy is used, either fluconazole or itraconazole at 400 mg daily anywhere from 6 weeks to 6 months is considered appropriate.

Diffuse pulmonary coccidioidomycosis represents a special situation. Because most patients with this form of disease are

profoundly hypoxemic and critically ill, many clinicians favor beginning therapy with an amphotericin B formulation combined with an oral triazole antifungal. The triazole antifungal therapy is continued alone once clinical improvement occurs and should be continued for 6 months to 1 year.

The nodules that may occur after primary pulmonary coccidioidomycosis do not require treatment. As noted above, these nodules are not easily distinguished from pulmonary malignancies by means of radiographic imaging. Close clinical follow-up and biopsy may be required to separate these two entities. Most pulmonary cavities do not require therapy. Antifungal treatment should be considered in patients with persistent cough, pleuritic chest pain, and hemoptysis. Occasionally, pulmonary coccidioid cavities become secondarily infected (see above). This development is often manifested by an air-fluid level within the cavity. Bacterial oral flora or *Aspergillus* species are commonly involved, and therapy directed at these organisms should be considered. Surgical removal of the cavity may be required in cases of persistent productive cough and hemoptysis or in those cases of a persistently growing cavity. In addition, cavities >4 cm in diameter are unlikely to resolve spontaneously, and surgical extirpation should be considered. Surgery is always required in cases of pyopneumothorax. For chronic pulmonary coccidioidomycosis, prolonged antifungal therapy—lasting for at least 1 year—is usually required, with monitoring of symptoms, radiographic changes, sputum cultures, and serologic titers.

Most cases of disseminated coccidioidomycosis require prolonged antifungal therapy. The duration of treatment is based on clinical improvement in conjunction with a significant decline in serum CF antibody titer. Such therapy routinely is continued for at least several years. Relapse occurs in 15–30% of individuals once therapy is discontinued and it is important to continue to monitor such patients on a regular basis (e.g., every 3–4 months) after antifungal therapy is discontinued.

Coccidioidal meningitis poses a special challenge. While most patients with this form of disease respond to treatment with oral triazoles, 80% experience relapse when therapy is stopped. Therefore, lifelong therapy is recommended. In cases of triazole failure, intrathecal or intraventricular amphotericin B may be used. Installation requires considerable expertise and should be undertaken only by an experienced health care provider. Shunting of CSF in addition to appropriate antifungal therapy is required in cases of meningitis complicated by hydrocephalus. It is prudent to obtain expert consultation in all cases of coccidioidal meningitis.

PREVENTION

There are no proven methods to reduce the risk of acquiring coccidioidomycosis among residents of an endemic region, but avoidance of inhalation of uncultivated soil or dust is a reasonable measure. For individuals with suppressed cellular immunity, the risk of developing symptomatic coccidioidomycosis is greater than that in the general population. Among those about to undergo allogeneic solid-organ transplantation, antifungal therapy is appropriate prior to transplantation when there is evidence of active or recent coccidioidomycosis. Several transplant centers in the endemic region provide universal antifungal prophylaxis for 6 months to 1 year after solid-organ and allogeneic stem cell transplantation, and lifelong universal prophylaxis has been advocated after lung transplantation to prevent the development of coccidioidomycosis after transplantation.

Cases of donor-transmitted coccidioidomycosis have been reported. Donors who are living or have lived in a coccidioidal endemic region should be screened serologically for coccidioidomycosis before transplantation and organ donation deferred if there is evidence of active infection.

Data on the use of antifungal agents for prophylaxis in other situations are limited. The administration of prophylactic antifungals is not recommended for HIV-1-infected persons who live in an endemic region. Most experts would administer a triazole antifungal to patients with a history of active coccidioidomycosis

or a positive coccidioidal serology in whom therapy with tumor necrosis factor- α antagonists or other biological response modifiers is being considered.

There are recent efforts to develop a vaccine for coccidioidomycosis, and a live avirulent product has demonstrated promising results in a canine model. Future studies will determine if this is a viable strategy for preventing or ameliorating coccidioidal infection in humans.

FURTHER READING

- GALGIANI JN et al: 2016 Infectious Diseases Society of America (IDSA) clinical practice guideline for the treatment of coccidioidomycosis. *Clin Infect Dis* 63:e112, 2016.
- GORRIS ME et al: Expansion of coccidioidomycosis endemic regions in the United States in response to climate change. *Geohealth* 3:308, 2019.
- SHUBITZ LF et al: Δ cps1 vaccine protects dogs against experimentally induced coccidioidomycosis. *Vaccine* 39:6894, 2021.
- TAYLOR JW, BARKER BM: The endozoan, small-mammal reservoir hypothesis and the life cycle of *Coccidioides* species. *Med Mycol* 57:S16, 2019.
- TROUNG CN et al: Universal lifelong fungal prophylaxis and risk of coccidioidomycosis in lung transplant recipients living in an endemic area. *Clin Infect Dis* 74:1966, 2021.

220

Blastomycosis

Gregory M. Gauthier, Bruce S. Klein



DEFINITION

Blastomycosis is a pyogranulomatous disease that follows the inhalation of *Blastomyces* conidia or hyphal fragments. Typically, *Blastomyces* causes pulmonary infection; however, a subset of patients will have disseminated disease that involves organs such as the skin, bone, brain, or genitourinary system. Blastomycosis is considered a primary fungal infection because it affects persons with either intact or impaired immune systems. A delay in diagnosis is common because blastomycosis mimics other diseases such as bacterial pneumonia, tuberculosis, and malignancy. Diagnosis involves culture- and nonculture-based tests. Amphotericin B formulations and triazoles are the drugs of choice for treatment.

ETIOLOGY

Blastomyces is a species complex comprising *B. dermatitidis*, *B. gilchristii*, *B. helicus*, *B. percursorus*, *B. emzantsi*, *B. silverae*, and *B. parvus*. *B. silverae* and *B. parvus* are not known to commonly infect humans. *Blastomyces* species exhibit thermal dimorphism, which involves the ability to convert between hyphal and yeast morphologies in response to temperature. In the soil (22–25°C), *Blastomyces* grows as septate hyphae that produce infectious conidia. Among the *Blastomyces* species, *B. helicus* is unique because its hyphae grow in a coiled pattern and it does not sporulate under in vitro growth conditions. In organs and tissues (37°C), *Blastomyces* grows as a pathogenic yeast (Fig. 220-1) that elicits pyogranulomatous inflammation. The yeast form of all *Blastomyces* species grows as broad-based budding yeast cells, with subtle differences in size among the different species (4–29 μ m).

EPIDEMIOLOGY

Although the majority of *Blastomyces* infections occur in North America, blastomycosis is a systemic fungal infection of global importance, with infections also occurring in Africa and Asia. In the United States, the traditional geographic range for *Blastomyces* includes the Mississippi



FIGURE 220-1 *Blastomyces* yeast at 37°C, with broad-based budding between mother and daughter cells (arrow). Bar = 10 μ m. (Gregory M. Gauthier, MD, MS.)

and Ohio River basins, the St. Lawrence River basin, states bordering the Great Lakes, and southeastern states. In Canada, the traditional geographic range includes the provinces of Saskatchewan, Manitoba, Ontario, and Quebec. In North America, *B. dermatitidis* is located throughout the traditional geographic range. *B. gilchristii* is geographically restricted to Minnesota, Wisconsin, Canada, and areas along the St. Lawrence River. *B. dermatitidis* and *B. gilchristii* are thought to have diverged 1.9 million years ago during the Pleistocene epoch, with *B. gilchristii* restricted to formerly glaciated areas. *B. dermatitidis* is found in glaciated and nonglaciated areas. In the environment, *B. dermatitidis* and *B. gilchristii* are not uniformly distributed; rather, they grow in ecologic niches often referred to as *microfoci*, which are characterized by acidic, sandy soils that are found near water and that contain decaying organic matter such as vegetation or wood. In upstate New York State, blastomycosis is an emerging pathogen in the Capitol District and upper Susquehanna River Subbasin with *B. dermatitidis* (88.4%) more common than *B. gilchristii* (11.6%). In Canada, blastomycosis is an emerging pathogen in the province of Saskatchewan. *B. helicus* infections have been reported in the western United States (California, Montana, Idaho, Colorado, Nebraska, Texas) and Canada (Saskatchewan, Alberta); their ecologic niche has yet to be defined. The geographic range and ecologic niche for *B. parvus* and *B. silvaeae* are unknown.

Outside of North America, blastomycosis has been reported in Africa (>100 cases), India (<10 cases), and Israel. On the basis of morphologic analysis, nearly all clinical isolates of *Blastomyces* in Africa were originally thought to be *B. dermatitidis*. However, molecular phylogenetic analysis of human clinical isolates has demonstrated that multiple *Blastomyces* species exist in Africa, including *B. dermatitidis*, *B. gilchristii*, *B. percursorus*, and *B. emzantsi*. A combination of internal transcribed spacer (ITS) sequencing, multilocus sequence typing (MLST), and whole genome sequencing was used to identify a new species, *B. emzantsi*, and to differentiate *B. percursorus* from other *Blastomyces* species. MLST has identified a *B. dermatitidis* isolate from Rwanda and *B. gilchristii* from Zimbabwe and South Africa. Analysis of 20 isolates from South Africa collected over a 40-year period identified them as either *B. emzantsi* or *B. percursorus*. The geographic distribution and ecologic niche of the four *Blastomyces* species in Africa are unknown. In India, there have been fewer than 10 autochthonous cases of blastomycosis, with the majority identified by morphologic analysis. One autochthonous case (caused by *B. percursorus*) with molecular confirmation has been reported from Israel.

Epidemiologic information about blastomycosis derives primarily from passive laboratory surveillance, retrospective studies, and

outbreak investigations. The lack of sensitive skin testing and serologic testing, along with difficulty in isolating *Blastomyces* from the environment by culture or molecular methods, has limited an in-depth epidemiologic understanding of blastomycosis. In North America, blastomycosis is reportable in five U.S. states (Minnesota, Wisconsin, Michigan, Arkansas, and Louisiana) and two Canadian provinces (Manitoba, Ontario). The annual incidence of blastomycosis in the traditional endemic area ranges from 0.11 to 2.17 cases/100,000 persons. In older persons (Medicare beneficiaries, 1999–2008), the nationwide annual incidence of blastomycosis was 0.7/100,000, with the highest rates in the midwestern and southern regions of the United States. Analysis of Healthcare Cost and Utilization Project (HCUP) data estimated that 11,776 persons were hospitalized for blastomycosis in the United States from 2010 through 2020, with the majority of patients from midwestern (58.8%) and southern (31.4%) states. In certain places, such as Vilas County, Wisconsin, and Kenora, Ontario, blastomycosis is hyperendemic, with annual incidence rates ranging from 40 to 117 cases/100,000 persons. Incidence data likely underestimate the true burden of infection because they are limited to persons with clinically apparent infection. Patients with asymptomatic or subclinical infections are undercounted.

Most *Blastomyces* infections are sporadic and can occur in either rural or urban areas. There have been at least 20 outbreaks of blastomycosis in the United States since the mid-1950s. Wisconsin, Minnesota, and North Carolina have had multiple outbreaks. The majority of outbreaks have been in rural areas, but several have occurred in urban settings. Activities associated with outbreaks include construction (of homes, cabins, factories, and roads), excavation of dirt, participation in water sports (canoeing, tubing on a river, and fishing), and exposure to a community compost pile or to beaver dams. *Blastomyces* infection is typically acquired from disturbed soil, which liberates infectious particles that are then inhaled into the lungs.

An investigation of a blastomycosis outbreak in Marathon County, Wisconsin (2009–2010), found that 45% of patients were of Hmong ethnicity. A retrospective study from the Marshfield Clinic in Wisconsin (1999–2014) found that 14.4% of patients with blastomycosis were of Asian ethnicity—a figure higher than was anticipated given that <2.5% of the population within the catchment area is Asian, including a large Hmong population. These findings suggest that persons of Hmong ethnicity have an increased risk of acquiring blastomycosis. A combination of whole genome sequencing and immunologic analyses indicated that polymorphisms in the interleukin 6 (IL-6) gene in the Hmong population result in decreased IL-6 production, which in turn impairs development of IL-17-producing CD4⁺ T lymphocytes. IL-17 is a critical cytokine for recruitment and activation of innate immune cells such as neutrophils and macrophages active against *Blastomyces*. Thus, alterations in IL-6 production may be responsible for the increased risk of blastomycosis in the Hmong population. Although data are limited, persons of Hmong ethnicity do not appear to be at increased risk for disseminated blastomycosis. Increased incidence rates of blastomycosis have also been reported in indigenous people of Canada and the United States. Compared with Caucasians, Asian and indigenous persons with blastomycosis tend to have fewer underlying medical conditions and to be younger.

■ PATHOGENESIS

A defining feature of the *Blastomyces* species complex is the ability to respond to shifts in temperature by switching between hyphal and yeast forms. In the soil, *Blastomyces* grows as mold cells with hyphae that produce conidia. Hyphal growth promotes environmental survival, genetic diversity through mating, and production of infectious conidia that facilitate transmission of *Blastomyces* from the environment to mammals, including humans. At 37°C (the core temperature of mammals), *Blastomyces* hyphae and conidia convert into pathogenic yeast that upregulate yeast phase-specific virulence factors and downregulate host immune defenses, thereby facilitating infection. Virulence traits that *Blastomyces* shares with *Histoplasma*, *Coccidioides*, *Sporothrix*, and *Paracoccidioides* are thermotolerance at 37°C, intracellular survival, and capacity to cause infection in persons with either

healthy or impaired immune defenses. Although *Emergomyces* and *Talaromyces marneffei* (formerly *Penicillium marneffei*) exhibit thermal dimorphism, growth as yeast at 37°C, and intracellular survival, these dimorphic fungi tend to cause infection primarily in immunocompromised persons.

The morphologic switch from hyphae to yeast at 37°C is driven chiefly by temperature and is coupled with the uptake of exogenous cysteine. Cysteine uptake is required to complete the transition to the yeast form because it helps restart mitochondrial respiration, which ceases during the morphologic switch. Over the past two decades, knowledge about the genetic mechanisms that promote the temperature-dependent transition between hyphae and yeast has substantially increased. The discovery of dimorphism-regulating kinase 1 (*DRK1*), which encodes a group III hybrid histidine kinase that is part of the high-osmolarity glycerol (HOG) signaling pathway, provided genetic proof that the transition to yeast is essential for virulence of the thermally dimorphic fungi. Disruption of *DRK1* by gene deletion or RNA interference resulted in *Blastomyces* cells that grew as hyphae at 37°C instead of yeast. Although viable at 37°C, these cells had altered cell-wall composition, failed to upregulate the *Blastomyces* adhesin 1 (*BAD1*, formerly *WI-1*) virulence factor, and were avirulent in a mouse model of lethal pulmonary infection. Subsequent studies of *Histoplasma* and *Talaromyces* demonstrated that the function of *DRK1* is conserved with regard to thermal dimorphism and virulence.

The temperature-dependent transition in the other direction—from yeast to hyphae—is regulated in part by a GATA-transcription factor encoded by siderophore biosynthesis repressor in *Blastomyces* (*SREB*), which influences neutral lipid metabolism. In addition, sensing of chitin by *NGT1* and *NGT2* N-acetylglucosamine transporters accelerates the conversion to hyphae following a drop in temperature from 37°C to 22°C. These two mechanisms are conserved in *Histoplasma capsulatum*.

As a primary fungal pathogen, *Blastomyces* is one of the few fungi that can infect immunocompetent persons. In its yeast form, *Blastomyces* evades and modulates immune defenses. Following disruption of soil, conidia that are aerosolized and inhaled into the lungs are phagocytosed by pulmonary macrophages, in which a subset of the conidia germinate as yeast and replicate during the early phases of infection. *Blastomyces* is also capable of replicating outside of macrophages. Upon conversion to the yeast phase, an essential virulence factor encoded by *BAD1* is upregulated. *BAD1* encodes a multifunctional 120-KDa cell-surface protein that facilitates yeast adherence to lung epithelial cells via interaction with heparin sulfate, attachment to host immune cells by binding to CR3 and CD14 complement receptors, and downregulation of tumor necrosis factor alpha (TNF-α) in macrophages and neutrophils. In addition, the *BAD1* protein impairs activation of CD4+ T lymphocytes, thereby decreasing the production of IL-17 and interferon gamma (IFN-γ). In vivo transcriptional profiling of *B. dermatitidis* yeast during pulmonary infection demonstrated that *BAD1* is the most highly upregulated gene. Deletion of *BAD1* renders *B. dermatitidis* avirulent in a murine model of pulmonary infection. Thus, *BAD1* is essential for virulence in *B. dermatitidis* and likely in *B. gilchristii* as well. In contrast, *BAD1* is absent from the sequenced genomes of *B. helicus*, *B. parvus*, *B. silverae*, *B. percursorus*, and *B. emzantsii*.

Additional factors that contribute to the virulence of *Blastomyces* yeast include relative resistance to oxidative stress, upregulation of catalase and superoxide dismutase during infection, active uptake of zinc by a *PRA1*-encoded zincophore and transmembrane transporter (*ZRT1*), and cleavage of granulocyte-macrophage colony-stimulating factor by dipeptidyl peptidase IVA, which blocks activation of innate immune cells (macrophages, neutrophils) and their recruitment to the lung.

APPROACH TO THE PATIENT

Blastomycosis

On the basis of outbreak investigations, it is estimated that 50% of persons exposed to *Blastomyces* develop symptomatic infection after a 3-week to 3-month incubation period. The relatively long incubation period means that patients can be diagnosed with

blastomycosis throughout the year. Blastomycosis has been referred to as the “the great pretender” because it can mimic infectious and noninfectious diseases. Blastomycotic pneumonia clinically and radiographically resembles community-acquired bacterial pneumonia, viral pneumonia, tuberculosis, and lung cancer. Patients often receive two or three courses of antibiotics before pulmonary blastomycosis is diagnosed. Without fungal stain and culture, cutaneous lesions can mimic skin cancer, sarcoidosis, and pyoderma gangrenosum. Rarely, blastomycosis can mimic laryngeal cancer. The most important aspect of the approach to a patient with a compatible illness is the consideration of *Blastomyces* as an etiologic agent in the differential diagnosis. This awareness facilitates early diagnosis and treatment, enhancing the potential for improved clinical outcomes. Clinical clues to blastomycosis, especially in persons who reside in or visit endemic regions, include pneumonia that does not improve with antibiotic treatment, pneumonia with extrapulmonary manifestations (e.g., skin lesions, osteomyelitis, central nervous system [CNS] involvement), and skin ulcers that do not respond to standard therapy. Blastomycosis should also be considered in persons from (or visited) an endemic area who have unexplained respiratory failure or acute respiratory distress syndrome (ARDS). In addition, a detailed exposure history can elevate blastomycosis in the differential diagnosis; approximately 50–60% of patients will have environmental risk factors for blastomycosis. This history should also include inquiries about a pet or family member with blastomycosis; these factors have been reported in 7.7–10% and 4–9% of patients, respectively.

CLINICAL MANIFESTATIONS

Pulmonary Blastomycosis Pulmonary manifestations occur in 69–93% of patients with symptomatic blastomycosis and are the most common clinical feature of infection. Signs and symptoms can include fever, chills, productive or nonproductive cough, shortness of breath, hemoptysis, malaise, and decreased appetite. Pulmonary blastomycosis also can manifest as asymptomatic infection, a brief influenza-like illness, acute pneumonia, chronic pneumonia, or ARDS. Radiographic findings in the lungs include lobar consolidation, a mass lesion, interstitial infiltrates, nodule(s), a miliary pattern, cavitary disease, and diffuse involvement of multiple lobes. Hilar adenopathy, pleural effusion, and empyema are uncommon. No distinctive features differentiate blastomycosis from other pulmonary diseases. Diabetes, receipt of a solid organ transplant, immunosuppression, and multilobar pneumonia are risk factors for severe pulmonary blastomycosis. Approximately 4–15% of patients with pulmonary blastomycosis develop ARDS, which is characterized by a fulminant course and high mortality rates ranging from 40 to 89% in most studies. The mortality rate in ARDS is increased when the diagnosis is delayed.

Disseminated Blastomycosis Disseminated blastomycosis occurs in 15–48% of patients and has the potential to involve nearly any organ in the body. The most common site of dissemination is the skin, in which the infection can manifest as papules, ulcers, verrucous lesions, or abscesses. The second most common site is bone, with consequent osteomyelitis characterized by bone pain, soft tissue swelling, soft tissue abscess, and sinus tract formation. Typically, a single bone is involved; however, multifocal osteomyelitis can occur. The most common sites for osteomyelitis include the spine, long bones, and ribs. Dissemination to the CNS (e.g., manifesting as meningitis, an abscess, or a mass lesion), the larynx, or the genitourinary system (e.g., to the prostate or epididymis) occurs in fewer than 10%; the majority of the affected patients have concomitant involvement of other organs, such as the lung or the skin.

Factors that influence dissemination include the infecting *Blastomyces* species, the duration of pulmonary symptoms, and concomitant AIDS. Multiple studies from Wisconsin, a state in which *B. dermatitidis* and *B. gilchristii* are endemic, have demonstrated that *B. dermatitidis* is more likely to cause disseminated infection (31.4–47.8% of cases), whereas *B. gilchristii* tends to remain localized to the lung (90.7–92.2%). Surprisingly, immunosuppression has only a minimal

influence on dissemination, an observation suggesting that *Blastomyces* virulence factors have a greater impact than host immune defenses. The frequency of disseminated blastomycosis among solid organ transplant recipients, persons receiving cancer chemotherapy, and patients undergoing pharmacologic immunosuppression is similar to that among patients with intact immune systems. Although patients treated with TNF- α antagonists are considered at risk for blastomycosis, the clinical manifestations and frequency of disseminated disease are unknown in this group because of a paucity of published data. Persons with AIDS and CD4⁺ T lymphocyte counts of <100/ μ L are an exception: they are at increased risk for CNS dissemination. Blastomycosis in pregnancy is uncommon, is typically diagnosed in the second or third trimester (91%), and most frequently manifests as pneumonia (74%) or disseminated infection (48%). Transmission to the neonate by either the transplacental route or aspiration of infected vaginal secretions is rare. Persons infected with *B. helicus* can have localized pulmonary infection or disseminated disease; they are typically immunosuppressed (e.g., as a result of solid organ transplantation, chemotherapy, HIV infection, or lupus) and have a high mortality rate (71.4% in seven patients). In contrast to *B. dermatitidis* and *B. gilchristii*, *B. helicus* commonly causes fungemia. Infections with *B. percursor* and *B. emzantsi* are often of long duration (persisting for 4 weeks to 5 years) and can involve the lungs or become disseminated (skin, bone, brain).

■ DIAGNOSIS

Timely diagnosis of blastomycosis requires a high degree of clinical suspicion because its clinical and radiographic presentations mimic more common etiologies, such as community-acquired pneumonia, malignancy, and tuberculosis. Laboratory findings such as leukocytosis, mild anemia, increased C-reactive protein level, and elevated erythrocyte sedimentation rate are nonspecific. Once suspected, the diagnosis of blastomycosis is straightforward and involves microscopic examination of stained specimens, fungal culture, and antigen testing. The poor sensitivity of complement fixation (9%) and immunodiffusion (28%) renders serologic testing diagnostically dispensable. However, a recently developed serologic test designed to detect antibodies to BAD1 has a sensitivity of 87% and a specificity of 94–99%.

A presumptive diagnosis of blastomycosis can be made by staining of clinical specimens and looking for broad-based budding yeast with a doubly refractile cell wall. Along with the broad-based budding pattern, yeast size (4–29 μ m) allows *Blastomyces* to be distinguished from other fungi. An exception is *B. helicus*, which has the potential to be confused with *Histoplasma* because of its small-sized yeast. Respiratory tract specimens such as sputum, tracheal aspirate, and bronchoalveolar (BAL) fluid can be stained with calcofluor, 10% potassium hydroxide, or Papanicolaou stain. Purulent drainage can be stained in a similar manner. The sensitivity of staining of respiratory samples ranges from 50 to 90%. Tissue samples for histopathology should be stained with Gomori methenamine silver or periodic acid–Schiff stain and assessed for pyogranulomatous inflammation and broad-based budding yeast. Traditional stains, such as Gram's stain or hematoxylin and eosin, do not permit optimal visualization of *Blastomyces* yeast.

Growth of *Blastomyces* in cultures of respiratory tissue or body fluid samples provides a definitive diagnosis of blastomycosis but typically requires 5–28 days of incubation. Special media such as Sabouraud dextrose, potato dextrose, and brain–heart infusion are required because *Blastomyces* does not grow well on standard bacteriologic media. Most clinical microbiology laboratories incubate fungal cultures at 25–30°C, a temperature that results in hyphal growth of *Blastomyces*. Unfortunately, *Blastomyces* hyphae are not morphologically distinct enough to confirm diagnosis. Thus, fungal identification and diagnosis are commonly confirmed via chemiluminescent DNA probe or, less commonly, via conversion to yeast upon incubation at 37°C. Diagnosis can also be confirmed by polymerase chain reaction. Neither the chemiluminescent DNA probe nor morphologic analysis of yeast by light microscopy differentiates among the different species of *Blastomyces*. Moreover, some species, such as *B. emzantsi*, are difficult to convert to yeast at 37°C. The species of *Blastomyces* is not typically determined in clinical labs because DNA sequencing is required.

An antigen test that detects a conserved galactomannan component in the *Blastomyces* cell wall has supplanted serologic testing. This test can be performed on urine, blood, BAL fluid, and cerebrospinal fluid. The sensitivity of the antigen test is 85–93% for urine and 57–82% for serum. Infection burden appears to influence test sensitivity, with a lower burden of infection resulting in reduced sensitivity. The antigen test can detect *B. dermatitidis*, *B. gilchristii*, and *B. helicus*; however, its utility for detection of other *Blastomyces* species is unknown. Cross-reactions in the antigen test occur during infection with other dimorphic fungi, including *H. capsulatum* (96%), *Paracoccidioides* species (100%), and *T. marneffe* (70%). Among these, only *H. capsulatum* is found in the same endemic region as *Blastomyces*. Rare cross-reactions can occur with *Aspergillus* and *Cryptococcus* infections. Antigen levels in urine and blood decline with successful treatment, and their measurement can be used to monitor the response to antifungal therapy.

TREATMENT

Blastomycosis

Guidelines for the treatment of blastomycosis have been published by the Infectious Diseases Society of America (2008), the American Thoracic Society (2011), the American Society of Transplantation (2019), and European Confederation of Medical Mycology (2021). Although there are isolated reports of self-limited pulmonary blastomycosis, there are no criteria to determine which patients will experience a resolution of infection. Thus, treatment is recommended for all patients with blastomycosis in order to prevent progressive infection, respiratory failure, and disseminated disease. Antifungal selection is influenced by immune status, CNS involvement, pregnancy, medical comorbidities (e.g., congestive heart failure, prolonged QT interval), and drug–drug interactions. Antifungal drugs active against *Blastomyces* include amphotericin B (AmB) formulations and triazoles. The minimal amount of beta-(1,3)-glucan in the *Blastomyces* yeast cell wall renders echinocandins ineffective, and they should not be used to treat blastomycosis. Hematologic, hepatic, and renal function should be assessed prior to initiation of antifungal therapy, and possible drug–drug interactions should be evaluated. In addition, patients should be educated about proper administration of triazole antifungals. For example, itraconazole capsules require an acidic gastric environment for optimal absorption and should be taken with food and an acidic beverage to improve bioavailability; they cannot be used by persons taking antacids, H₂ antagonists, or proton pump inhibitors. In contrast, itraconazole solution can be given to patients receiving gastric acid–lowering therapies and should be taken without food.

Treatment for blastomycosis is summarized in [Table 220-1](#). For immunocompetent patients with pulmonary or disseminated blastomycosis of mild or moderate severity (e.g., treatable in the outpatient setting), itraconazole therapy for 6 months is recommended. For severe blastomycosis (e.g., that requiring hospitalization), induction therapy with lipid AmB for 7–14 days (or until clinical improvement), followed by itraconazole treatment for 6–12 months, is recommended. Although not well studied, combination antifungal therapy with lipid AmB and itraconazole (or another azole) can be considered for patients with severe pulmonary blastomycosis. In patients with ARDS, prednisone can be considered; however, the benefits of steroids are unclear. Osteomyelitis due to blastomycosis requires at least 12 months of antifungal therapy, and some patients may require surgical debridement. For blastomycosis involving the CNS, lipid AmB is administered for 4–6 weeks and is followed by treatment with voriconazole, itraconazole, or fluconazole for at least 12 months. Although fluconazole has excellent CNS penetration, its minimum inhibitory concentration (MIC) against *B. dermatitidis* and *B. gilchristii* is higher than that of either itraconazole or voriconazole. Emerging data suggest that isavuconazonium sulfate can be used for treatment of CNS blastomycosis.

Immunosuppressed patients should be treated with 7–14 days of lipid AmB followed by 12 months of itraconazole. For patients

TABLE 220-1 Treatment of Blastomycosis

PATIENT POPULATION	SEVERITY OF INFECTION	SITE OF INFECTION	THERAPY
Immunocompetent	Mild to moderate ^a	Lung	Itraconazole for 6–12 months ^b
		Disseminated	Itraconazole for 6–12 months ^b (≥12 months for osteomyelitis)
	Severe ^c	CNS	Lipid AmB (5 mg/kg daily ^{d,e} for 4–6 weeks) followed by voriconazole (200–400 mg bid), itraconazole ^b or fluconazole (800 mg daily) for at least 12 months of treatment
		Lung	Lipid AmB (3–5 mg/kg daily ^{e,f} for 7–14 days) followed by itraconazole ^b for 6–12 months
		Disseminated	Lipid AmB (3–5 mg/kg daily ^{e,f} for 7–14 days) followed by itraconazole ^b for 12 months of treatment (≥12 months for osteomyelitis)
Immunocompromised	Any severity	CNS	Lipid AmB (5 mg/kg daily ^{d,e} for 4–6 weeks) followed by voriconazole (200–400 mg bid), itraconazole ^b or fluconazole (800 mg daily) for at least 1 year of treatment ^g
		Lung or disseminated	Lipid AmB (3–5 mg/kg daily ^{e,f} for 7–14 days) followed by itraconazole ^b for 12 months ^g
Pregnant ^h	Any severity	Any site	Lipid AmB (3–5 mg/kg daily ^{e,f} for 6–8 weeks), with avoidance of azole antifungals

^aMild to moderate infections can typically be managed in the outpatient setting. ^bA loading dose of 200 mg PO tid for 3 days followed by 200 mg PO daily or bid, with dosing based on serum itraconazole levels. The goal for levels of total itraconazole (i.e., itraconazole plus hydroxyitraconazole) is 1–5 µg/mL. Liquid itraconazole has greater bioavailability than the capsule formulation. Liquid itraconazole and oral capsules are administered differently (see text for details). Serum itraconazole levels should be measured after steady state has been reached (2 weeks). Because of the drug's long half-life, blood for serum itraconazole determinations can be drawn regardless of the time of administration. In contrast, serum drug levels for voriconazole, posaconazole, and isavuconazole should be measured before a dose is administered when steady state has been reached (~1 week). ^cSevere blastomycosis requires hospitalization on a medical ward, an intermediate care unit, or an intensive care unit. ^dLipid amphotericin B (AmB) is the preferred formulation because it has the best central nervous system (CNS) penetration among AmB formulations. For patients with CNS blastomycosis that results in neurologic dysfunction, surgical intervention should be considered. ^eFor patients with CNS blastomycosis, severe pulmonary blastomycosis, or severe disseminated blastomycosis, combination therapy with lipid AmB plus a triazole antifungal can be considered; however, this combination has not been formally studied. For patients with acute respiratory distress syndrome, adjunctive steroid therapy with prednisone (40–60 mg daily for 1–2 weeks) can be considered; however, the benefit of steroid administration is not well defined. ^fIf lipid AmB is not available, then AmB deoxycholate (0.7–1.0 mg/kg daily) can be substituted; however, this formulation is associated with higher rates of nephrotoxicity and infusion reactions than lipid AmB. ^gConsider lifelong suppression with itraconazole (200 mg daily) if immunosuppression cannot be reversed. This decision should be made on a case-by-case basis; not all immunosuppressed patients require lifelong suppressive therapy. In addition, lifelong antifungal suppression can be considered in patients who experience relapse after appropriate therapy. ^hAll women of childbearing age should undergo pregnancy testing before initiation of therapy.

requiring irreversible immunosuppression, indefinite suppressive azole therapy may be needed; however, in light of the heterogeneity of this patient population, a decision about suppressive therapy should be made on a case-by-case basis. The majority of solid organ transplant recipients do not require lifelong suppression because rates of relapse are low when treatment guidelines are followed.

For pregnant women, lipid AmB treatment for 6–8 weeks is recommended because, unlike the triazole antifungals, lipid AmB is not teratogenic. Fluconazole can cause craniofacial, skeletal, and cardiac defects in the developing fetus (Antley-Bixler-like syndrome); voriconazole and posaconazole also can result in skeletal abnormalities. Itraconazole increases the risk of spontaneous abortion. Before starting antifungal therapy, women of childbearing age with blastomycosis should have a pregnancy test.

Voriconazole, posaconazole, and isavuconazonium sulfate have potent activity against *B. dermatitidis* and *B. gilchristii* and can be considered as alternatives for persons who cannot tolerate itraconazole. These agents, along with itraconazole and AmB, also exhibit good activity against newly identified species of *Blastomyces*, such as *B. helicus*, *B. percursor*, and *B. emzantsi*. Fluconazole MICs against *B. percursor* and *B. emzantsi* are higher than those of other triazoles. Moreover, fluconazole appears to have poor activity against *B. helicus*, *B. parvus*, and *B. silverae*.

PROGNOSIS

Mortality rates for blastomycosis range from 5 to 13%; most deaths are associated with respiratory failure due to ARDS. Analysis of 11,776 hospital admissions for blastomycosis from 2010 through 2020 (HCUP data) demonstrated a 7.9% mortality rate, with increased mortality in patients who were older or who had underlying chronic obstructive pulmonary disease or concomitant malignancy (solid or hematologic). The vast majority of patients who recover from pulmonary blastomycosis do not experience long-term loss of pulmonary function. Cutaneous blastomycosis typically results in scarring.

PREVENTION

Prevention of blastomycosis is challenging because most infections are sporadic and unpredictably acquired from the environment. However, substantial progress has been made in understanding vaccine-mediated immunity conferred by a live, attenuated vaccine strain that

is deficient in BAD1. When injected subcutaneously into mice, the *B. dermatitidis* BAD1-null vaccine strain induces sterilizing immunity by activating T_H17 lymphocytes to protect against lethal pulmonary challenge. Major antigenic components of the vaccine identified to date include calnexin and *Blastomyces* endoglucanase-2, which is conserved in other pathogenic fungi, including *Histoplasma capsulatum*, *Coccidioides* species, *Aspergillus* species, *Fonsecaea pedrosoi*, and *Pseudogymnoascus destructans*. Neither the BAD1-null attenuated vaccine nor recombinant antigen-based vaccines are commercially available.

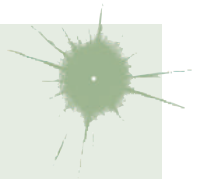
FURTHER READING

- BENEDICT K et al: Blastomycosis-associated hospitalizations, United States 2010–2020. *J Fungi* 9:867, 2023.
- LIMPER AH et al: An official American Thoracic Society statement: Treatment of fungal infections in adult pulmonary and critical care patients. *Am J Respir Crit Care Med* 183:96, 2011.
- SMITH D et al: Clinical testing guidance for coccidioidomycosis, histoplasmosis, and blastomycosis in patients with community-acquired pneumonia for primary and urgent care providers. *Clin Infect Dis* 78:1559, 2024.

221

Cryptococcosis

Arturo Casadevall, Shmuel Shoham



DEFINITION AND ETIOLOGY

Cryptococcus, a genus of yeast-like fungi, is the etiologic agent of cryptococcosis. In 2022, the World Health Organization (WHO) declared *Cryptococcus neoformans* as a critical priority pathogen. Until recently, cryptococcal strains were separated into two species. However, genome sequencing studies have now revealed tremendous diversity among isolates previously assigned to each species, leading to the proposal that each of the prior species classifications includes numerous new species. Hence, *C. neoformans* and *C. gattii* are now considered as

1704 species complexes. However, for clinical purposes, these species complexes cause indistinguishable disease referred to as cryptococcosis. Consequently, this chapter will continue to use the nomenclature *C. neoformans* and *C. gattii* with the understanding that these terms refer to species complexes.

■ EPIDEMIOLOGY

Cryptococcosis was first described in the 1890s but remained rare until the mid-twentieth century, when advances in diagnosis and increases in the number of immunosuppressed individuals markedly raised its reported prevalence. Although serologic evidence of cryptococcal infection is common among immunocompetent individuals, cryptococcal disease (cryptococcosis) is rare in the absence of impaired immunity. Conditions associated with high risk for *C. neoformans* infection include hematologic malignancies, receipt of solid-organ transplants, advanced liver disease, diabetes mellitus, illnesses necessitating glucocorticoid therapy, and advanced HIV infection with CD4⁺ T lymphocyte counts of <200/μL. In contrast, *C. gattii*-related disease is not generally associated with specific immune deficits and often occurs in immunocompetent individuals but is associated with autoantibodies to granulocyte-macrophage colony-stimulating factor.

Cryptococcal infection is acquired from the environment. *C. neoformans* and *C. gattii* species complexes inhabit different ecological niches. *C. neoformans* is frequently found in soils contaminated with avian excreta and can be recovered from shaded and humid soils contaminated with pigeon droppings. In contrast, *C. gattii* is not found in bird feces. Instead, it inhabits a variety of arboreal species, including eucalyptus trees. *C. neoformans* strains are found throughout the world; however, var. *grubii* (serotype A) strains are far more common than var. *neoformans* (serotype D) strains among both clinical and environmental isolates. *C. gattii* was thought to be largely limited to tropical regions until an outbreak of cryptococcosis caused by a new serotype B strain began in Vancouver in 1999. This outbreak has extended into the United States, and *C. gattii* infections are being encountered increasingly in several states in the Pacific Northwest.

The global burden of cryptococcosis was estimated in 2012 at ~1 million cases, with >600,000 deaths annually, although the prevalence of this disease has declined since then with the greater availability of antiretroviral therapy (ART) for HIV. Since the onset of the HIV pandemic in the early 1980s, the overwhelming majority of cryptococcosis cases have occurred in patients with AIDS (Chap. 208). To comprehend the impact of HIV infection on the epidemiology of cryptococcosis, it is instructive to note that in the early 1990s there were >1000 cases of cryptococcal meningitis each year in New York City—a figure far exceeding that for all cases of bacterial meningitis. With the advent of effective ART, the incidence of AIDS-related cryptococcosis has been sharply reduced among treated individuals. Therefore, most cases of cryptococcosis now occur in resource-limited regions of the world. The disease remains distressingly common in regions where ART is not readily available (e.g., parts of Africa and Asia); in these regions, up to one-third of patients with AIDS have cryptococcosis. Among HIV-infected persons, those with a decreased percentage of memory B cells expressing IgM may be at greater risk for cryptococcosis. In wealthier countries, most cryptococcosis cases are in people without HIV.

■ PATHOGENESIS

Cryptococcal infection is almost always acquired by inhalation of aerosolized particles. The exact nature of the infectious particles is not known; the two leading candidate forms are small desiccated yeast cells and basidiospores. Occasionally, infection is acquired via direct inoculation of the skin. Little is known about the pathogenesis of initial infection. Serologic studies have shown that cryptococcal infection is acquired in childhood, but it is not known whether the initial infection is symptomatic. Given that cryptococcal infection is common while disease is rare, the consensus is that pulmonary defense mechanisms in immunologically intact individuals are highly effective at containing this fungus. It is not clear whether initial infection leads to a state of immunity or whether most individuals are subject throughout life to frequent and recurrent infections that resolve without clinical disease.

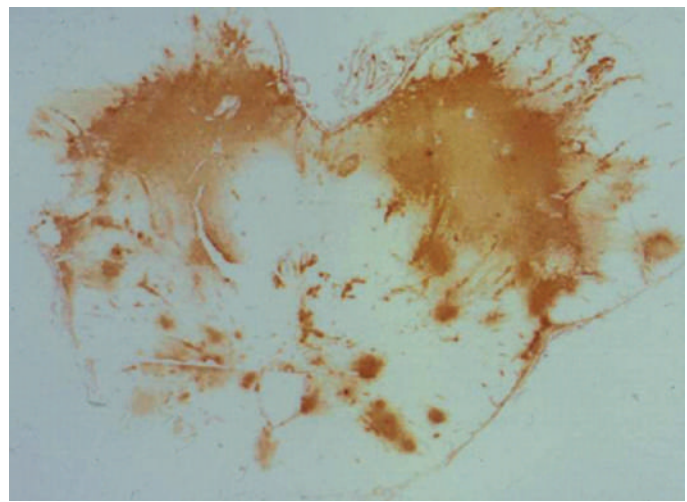


FIGURE 221-1 Cryptococcal antigen in human brain tissue, as revealed by immunohistochemical staining. Brown areas show polysaccharide deposits in the midbrain of a patient who died of cryptococcal meningitis. (Reproduced with permission from SC Lee et al: Immunohistochemical localization of capsular polysaccharide antigen in the central nervous system cells in cryptococcal meningoencephalitis. *Am J Pathol* 148:1267, 1996.)

However, some human cryptococcal infections lead to a state of latency in which viable organisms are harbored for prolonged periods, possibly in granulomas. Thus, the inhalation of cryptococcal cells and/or spores can be followed by either clearance or establishment of the latent state. The consequences of prolonged harboring of cryptococcal cells in the lung are not known, but evidence from animal studies indicates that the organisms' prolonged presence could alter the immunologic milieu in the lung and predispose to allergic airway disease.

Cryptococcosis usually presents clinically as pulmonary disease and/or chronic meningoencephalitis. The mechanisms by which the fungus undergoes extrapulmonary dissemination and enters the central nervous system (CNS) remain poorly understood. Current evidence suggests that both direct fungal-cell migration across the endothelium and fungal-cell carriage inside macrophages as “Trojan horse” invaders can occur. *Cryptococcus* species have well-defined virulence factors that include the expression of the polysaccharide capsule, the ability to make melanin, and the elaboration of enzymes (e.g., phospholipase and urease) that enhance the survival of fungal cells in tissue. Among these virulence factors, the capsule and melanin production have been most extensively studied. The cryptococcal capsule is antiphagocytic, and the capsular polysaccharide has been associated with numerous deleterious effects on host immune function. Cryptococcal infections can elicit little or no tissue inflammatory response. The immune dysfunction seen in cryptococcosis has been attributed to the release of copious amounts of capsular polysaccharide into tissues, where it probably interferes with local immune responses (Fig. 221-1). In clinical practice, the capsular polysaccharide is the antigen that is measured as a diagnostic marker of cryptococcal infection.

APPROACH TO THE PATIENT

Cryptococcosis

Cryptococcosis should be included in the differential diagnosis when any patient presents with findings suggestive of chronic pulmonary or CNS infection. Concern about cryptococcosis is heightened by a history of headache and neurologic symptoms in a patient with an underlying immunosuppressive disorder such as advanced HIV infection, malignancy, immunosuppressive use, or solid-organ transplantation. Evaluation of cerebrospinal fluid (CSF) is critical for diagnosis of CNS disease and should include measurements of CSF pressure, protein, glucose, cell count, Gram stain, cultures, and a cryptococcal antigen assay.

CLINICAL MANIFESTATIONS

The clinical manifestations reflect the site of infection. The spectrum of disease consists predominantly of meningoencephalitis and pneumonia, but skin and soft tissue infections also occur and cryptococcosis can affect any tissue or organ. Symptoms may arise from fungal tissue invasion and from an overactive immune response. In CNS disease, symptoms may reflect elevated intracranial pressure (ICP). CNS involvement usually presents as signs and symptoms of chronic meningitis, such as headache, fever, lethargy, sensory deficits, memory deficits, cranial nerve paresis, vision deficits, and meningismus. Cryptococcal meningitis differs from bacterial meningitis in that many patients present with symptoms of several weeks in duration. In addition, classic characteristics of meningeal irritation, such as meningismus, may be absent. Indolent cases can present as subacute dementia or depression. Meningeal cryptococcosis can lead to sudden catastrophic vision loss.

Pulmonary cryptococcosis usually presents as cough, increased sputum production, and chest pain. Radiographic findings include nodules, infiltrates, and masses. *C. gattii* pneumonia can present with granulomatous masses known as *cryptococcomas*. Fever develops in a minority of cases. Pulmonary cryptococcosis can follow an indolent course, and many cases probably do not come to clinical attention. In fact, many cases are discovered incidentally during the workup of an abnormal chest radiograph obtained for other purposes. Pulmonary cryptococcosis can be associated with antecedent diseases such as malignancy, diabetes, and tuberculosis.

Skin lesions are common in patients with disseminated cryptococcosis and can be highly variable, including papules, plaques, purpura, vesicles, tumor-like lesions, and rashes. The spectrum of cryptococcosis in HIV-infected patients is so varied and has changed so much since the advent of ART that a distinction between HIV-related and HIV-unrelated cryptococcosis is no longer pertinent. In highly immunosuppressed patients, the lesions of cutaneous cryptococcosis often resemble those of molluscum contagiosum (Fig. 221-2; Chap. 201).

DIAGNOSIS

A diagnosis of cryptococcosis requires the demonstration of cryptococcal cells or antigen in normally sterile tissues. Visualization of encapsulated fungal cells in CSF mixed with India ink remains a useful rapid diagnostic technique. Cryptococcal cells can also be visualized on Gram stain. However, these staining techniques may yield negative results in patients with a low fungal burden or be misread as positive by inexperienced operators. Cultures of CSF and blood that are positive for cryptococcal cells are diagnostic for cryptococcosis. CSF examination usually reveals evidence of chronic meningitis with mononuclear

cell pleocytosis and increased protein levels. A particularly useful test is cryptococcal antigen (CRAG) detection in CSF and blood. The assay is based on serologic detection of cryptococcal polysaccharide and is both sensitive and specific. A major advance in recent years was the introduction of rapid point-of-care CRAGs that provide results in minutes. A positive CRAG test provides strong presumptive evidence for cryptococcosis; however, because the result is often negative in pulmonary cryptococcosis, the test is less useful for diagnosing pulmonary disease and is of only limited usefulness in monitoring the response to therapy. Polymerase chain reaction is available and effective for the diagnosis of cryptococcosis, but physicians should seek confirmation of positive or negative tests with independent methods such as CRAG and culture.

In areas of Africa where there is a high prevalence of HIV infection, routine screening of blood for CRAG in HIV-infected patients with low CD4+ T lymphocyte counts may identify individuals at high risk of cryptococcal disease who are candidates for antifungal therapy. CRAG screening has shown that a significant proportion of HIV-infected patients hospitalized with pneumonia in Thailand harbor cryptococcal infection. Inexpensive point-of-care CRAG tests in the form of lateral flow assays provide rapid and accurate information and are of great diagnostic benefit in resource-limited regions.

TREATMENT

Cryptococcosis

When the infection is nonsevere and limited to the lungs or other non-CNS sites, treatment is typically with fluconazole. While pulmonary cryptococcosis in an immunocompetent host sometimes resolves without therapy, given the propensity of *Cryptococcus* species to disseminate from the lung, the inability to gauge the host's immune status precisely, and the availability of low-toxicity therapy in the form of fluconazole, the current recommendation for nonsevere pulmonary cryptococcosis in an immunocompetent individual is fluconazole 400 mg daily for 6–12 months. Fluconazole can cause drug interactions, QT interval prolongation, and liver dysfunction (especially at higher doses), and the dose should be adjusted for renal function. In general, extrapulmonary cryptococcosis without CNS involvement requires less intensive therapy, with the caveat that morbidity and death are associated with meningeal involvement. Thus, the decision to categorize cryptococcosis as “extrapulmonary without CNS involvement” should be made only after evaluation of the CSF reveals no evidence of cryptococcal infection.

When cryptococcosis is severe or involves the CNS, treatment involves induction, consolidation, and maintenance phases. The induction phase ideally includes a combination of intravenous amphotericin B (AmB) and oral flucytosine (5-FC). If feasible, lipid formulations of AmB should be used because those have less toxicity than deoxycholate AmB. The recommended daily doses are liposomal AmB 3–5 mg/kg, AmB lipid complex 5 mg/kg, and deoxycholate AmB 0.7–1 mg/kg. The dose of 5-FC is 25 mg/kg four times daily. 5-FC can cause bone marrow suppression, and the dose should be adjusted for renal function. The duration of induction therapy is a minimum of 2 weeks but may require extension to 4–6 weeks when there is poor initial response to therapy, neurologic complications, or brain cryptococcomas and in people without HIV. Cryptococcal cells with negative cultures may be seen on CSF staining, and elevated cryptococcal antigen may be present in CSF and blood, long after clearance of cultures. These do not indicate treatment failure.

For people with HIV, WHO recommends several alternative approaches to induction. These include a single dose of liposomal AmB 10 mg/kg given along with 14 days of 5-FC at 25 mg/kg four times daily and fluconazole 1200 mg daily. If liposomal AmB is not available, a 7-day course of AmB deoxycholate 1 mg/kg per day can be used in its place. If AmB is not available, an all-oral 14-day induction regimen of fluconazole 1200 mg daily and 5-FC 25 mg/kg four times daily can be used. The consolidation phase is typically with fluconazole 400–800 mg daily for 8 weeks. This is followed by fluconazole maintenance at 200–400 mg daily for a year or longer.



FIGURE 221-2 Disseminated fungal infection. A liver transplant recipient developed six cutaneous lesions similar to the one shown. Biopsy and serum antigen testing demonstrated *Cryptococcus*. Important features of the lesion include a benign-appearing fleshy papule with central umbilication resembling molluscum contagiosum. (Photo courtesy of Dr. Lindsey Baden; with permission.)

The goal of maintenance is to prevent recurrent disease, and its duration depends on the immune status of the patient. Individuals who remain deeply immunocompromised, such as those with HIV whose CD4+ T lymphocyte counts remain $<200/\mu\text{L}$ or have not attained control of HIV with ART may need to remain on fluconazole indefinitely. Newer triazoles like voriconazole, posaconazole, and isavuconazole are highly active against cryptococcal strains and appear to be clinically effective, but clinical experience with these agents is limited. An oral formulation (cochleated) of AmB is in development. The echinocandins are not effective against *Cryptococcus* species. Addition of a short course of interferon γ to antifungal therapy in patients with HIV infection increases clearance of cryptococci from the CSF. Antifungal drug resistance has not been a major problem with cryptococcal strains, but there are increasing reports of drug-resistant strains, including some emerging during the prolonged therapy needed for cryptococcosis. Hence, cryptococcosis that is refractory to antifungal therapy should prompt an investigation into the susceptibility of the clinical isolates in question.

Cryptococcal meningoencephalitis may be associated with increased ICP, which can damage the brain and cranial nerves. CSF pressure should be measured and increased ICP diligently managed in such patients. The mechanism leading to increased ICP is often excess CSF fluid. Neurologic symptoms, including headache, blurred vision, cranial nerve abnormalities, and altered mental status, are typical clues to increased ICP. Management requires reduction of pressure by repeated therapeutic lumbar punctures and the placement of shunts. Neither mannitol nor acetazolamide is effective. Glucocorticoids are not helpful unless the elevated ICP is from a brain lesion with associated mass effect and edema.

Several clinical syndromes in cryptococcosis are driven by an overactive immune response. In people with HIV disease who are treated with ART, an immune reconstitution inflammatory syndrome (IRIS) occurs when immunity rebounds in the setting of treated cryptococcosis (or an undiagnosed asymptomatic infection). Similar syndromes can occur in transplant recipients whose immunosuppressive regimens have been reduced to help control the infection and even in non-immunocompromised people with infection that had appeared to be resolving. The immune response triggers an inflammatory reaction that can be difficult to distinguish from a relapsing infection. Symptoms can include fevers, headache, lymphadenopathy, and pulmonary, CNS, and cutaneous manifestations. Administration of prophylactic dexamethasone in HIV-associated cryptococcosis is not recommended because it has been associated with reduced fungal clearance and increased mortality. The approach to patients that have already developed IRIS and the related immune-driven syndromes must attempt to balance resurgent immunity against immune-mediated damage. Management is individualized and can involve the use of glucocorticoids to reduce inflammation. Apart from the difficulties in distinguishing these inflammatory syndromes from cryptococcal relapse, their management is complex because the cause is often triggered by the desirable outcome of improving immunity, which is important in controlling cryptococcal infection and preventing relapses.

A major consideration for clinicians treating symptomatic AIDS-related cryptococcosis is when to begin ART, which can trigger rebounding immunity. Current recommendations are to start ART 4–6 weeks after initiating antifungal therapy. Screening with serum CRAG is recommended for asymptomatic people with HIV and CD4+ T lymphocyte counts of <100 – $200/\mu\text{L}$. Positive tests should be followed up with careful evaluation for active disease including CSF analysis. For those with antigenemia but no evidence of active disease, a preemptive fluconazole regimen is recommended with 14 days of induction dosing at 800 mg daily and up to 1200 mg daily when fungal burden is high (e.g., CRAG titers exceed 1:80) followed by consolidation and maintenance dosing. Initiation of ART should start 2 weeks after initiation of antifungal therapy in such circumstances.

■ PROGNOSIS AND COMPLICATIONS

Even with antifungal therapy, cryptococcosis is associated with high rates of morbidity and death. For most patients with cryptococcosis, the most important prognostic factors are the extent and the duration of the underlying immunologic deficits that predisposed them to develop the disease. Cryptococcosis is often curable with antifungal therapy in individuals with no apparent immunologic dysfunction, but in patients with irreversible immunosuppression, the best that can be hoped for is that antifungal therapy will induce remission, which can then be maintained with lifelong suppressive therapy. Before the advent of ART, the median overall survival period for AIDS patients with cryptococcosis was <1 year. Cryptococcosis in patients with underlying neoplastic disease has a particularly poor prognosis. For CNS cryptococcosis, poor prognostic markers are a CSF assay positive for yeast cells on initial India ink examination (evidence of a heavy fungal burden), high CSF pressure, low CSF glucose levels, low CSF pleocytosis ($<2/\mu\text{L}$), recovery of yeast cells from extraneural sites, absence of antibody to capsular polysaccharide, a CSF or serum cryptococcal antigen level of $\geq 1:32$, and concomitant glucocorticoid therapy or hematologic malignancy. A response to treatment does not guarantee cure since relapse of cryptococcosis is common even among patients with relatively intact immune systems and immune reconstitution syndromes can occur in patients who had been improving with antifungal therapy. Complications of CNS cryptococcosis include cranial nerve deficits, vision and hearing loss, and cognitive impairment.

■ PREVENTION

No vaccine is available for cryptococcosis. Primary prophylaxis with fluconazole 100–200 mg daily is an approach in high-risk HIV patients (e.g., CD4+ T lymphocyte count $<100/\mu\text{L}$) and can be used in places where cryptococcal antigen testing is not readily available. Since ART raises the CD4+ T lymphocyte count, it constitutes an immunologic form of prophylaxis.

■ FURTHER READING

- ALANIO A: Dormancy in *Cryptococcus neoformans*: 60 years of accumulating evidence. *J Clin Invest* 130:3353, 2020.
- BOYER-CHAMMARD T et al: Recent advances in managing HIV-associated cryptococcal meningitis. *F1000Res* 8:F1000 Faculty Rev-743, 2019.
- KWON-CHUNG KJ et al: The case for adopting the “species complex” nomenclature for the etiologic agents of cryptococcosis. *mSphere* 2:e00357, 2017.
- ROBERTSON EJ et al: *Cryptococcus neoformans* ex vivo capsule size is associated with intracranial pressure and host immune response in HIV-associated cryptococcal meningitis. *J Infect Dis* 209:74, 2014.
- SSEBAMBULIDDE K et al: Treatment recommendations for non-HIV associated cryptococcal meningoencephalitis including management of post-infectious inflammatory response syndrome. *Front Neurol* 13:994396, 2022.
- TUGUME L et al: Cryptococcal meningitis. *Nat Rev Dis Primers* 9:62, 2023.

222

Candidiasis

Michail S. Lionakis, Shakti Singh,
Ashraf S. Ibrahim, John E. Edwards, Jr.



The genus *Candida* encompasses >150 species, only a few of which cause disease in humans. With rare exceptions (although the exceptions are increasing in number), the human pathogens are *C. albicans*, *C. guilliermondii* (recently revised to *Meyerozyma guilliermondii*), *C. krusei* (recently revised to *Pichia kudriavzevii*), *C. parapsilosis*,

C. tropicalis, *C. lusitanae* (recently revised to *Clavispora lusitanae*), *C. dubliniensis*, *C. glabrata* (recently revised to *Nakaseomyces glabratus*), and the emerging, multidrug-resistant *C. auris*, which has been responsible for several outbreaks in health care facilities in recent years. Ubiquitous in nature, they inhabit the gastrointestinal tract (including the mouth and oropharynx), the female genital tract, and the skin in the majority of healthy persons. Although cases of candidiasis have been described since antiquity in debilitated patients, the advent of *Candida* species as common human pathogens dates to the introduction of modern therapeutic approaches that suppress normal host defense mechanisms. Of those advances, the most important are the use of antibacterial agents that alter the normal human microbiota and allow nonbacterial species to become more prevalent in the commensal flora, the use of indwelling intravenous catheters, and the use of cytotoxic, immunosuppressive treatments for malignant and autoimmune disorders. With the introduction of antifungal agents, the causes of *Candida* infections shifted from an almost complete dominance of *C. albicans* to the common involvement of *C. glabrata* and the other species listed above. The non-*albicans* species now account for approximately half of all cases of candidemia and hematogenously disseminated candidiasis. Recognition of this change is clinically important since the various species differ in susceptibility and are increasingly resistant to the newer antifungal agents.

Candida is a small, thin-walled, ovoid yeast that measures 4–6 µm in diameter and reproduces by budding. Organisms of this genus occur in three forms in tissue: blastospores, pseudohyphae, and hyphae. *Candida* grows readily on simple media; lysis centrifugation enhances its recovery from blood. Species are identified by biochemical testing (currently with automated devices) or on special agar (e.g., CHROMagar).

■ EPIDEMIOLOGY

Candida are present in humans as commensals, in animals, in foods, and on inanimate objects. In developed countries, where contemporary medical therapeutics are commonly used, *Candida* species are now among the most common nosocomial pathogens. In the United States, these species are among the four most common pathogens isolated from the blood of hospitalized patients. In fact, in a recent point-prevalence study in the United States, *Candida* species were the most common organisms infecting the bloodstream of hospitalized patients. In regions where advanced medical care is not readily available, mucocutaneous *Candida* infections, such as thrush, are more common than deep-organ infections, which rarely occur. However, the incidence of deep-organ candidiasis has been increasing steadily as advances in health care—such as therapy with broad-spectrum antibiotics, more aggressive treatment of cancer, and the use of immunosuppression for sustaining organ transplants—have been implemented. In aggregate, the global incidence of infections due to *Candida* species has risen over the past few decades.

C. auris is an emerging species of *Candida* that has spread rapidly in recent years to >50 countries and is a major public health concern. It was designated an urgent threat in the 2019 Centers for Disease Control and Prevention (CDC) Antimicrobial Resistance Threat Report, and it was included within the critical priority group in the 2022 World Health Organization (WHO) fungal priority pathogens list. This concern stems from its occurrence in health care facilities, its ability to adhere to and persist long term on inanimate objects (in hospitals) and the human skin despite decolonization efforts, its association with substantial mortality, its propensity for misidentification as other *Candida* species, the incomplete understanding of its environmental reservoirs, and its multidrug resistance to the current antifungal therapeutic armamentarium, with some *C. auris* strains being resistant to all antifungal drug classes currently available for treatment. *C. auris* (*auris* meaning *ear* in Latin) was first identified in 2009 from the ear drainage of a patient with an ear infection in Japan. However, subsequent retrospective analysis of *Candida* strain collections identified the earliest known *C. auris* strain to date back to 1996 in South Korea. Notably, whole genome sequencing analysis of *C. auris* strains from South Asia, East Asia, South America, South Africa, and Iran found that although

strains within each geographic region are closely related to each other, they are distinct compared to strains from other geographic regions. These findings indicate that *C. auris* emerged independently in multiple geographic locations around the same time; the epidemiologic reasons for this emergence remain poorly understood but may relate to the increasing use of antifungal drugs and climate change.

The presence of a central venous catheter and/or other invasive medical devices and recent residence in nursing homes are major risk factors for *C. auris* colonization and infection. Screening of selected patients who are in a hospital or nursing home where *C. auris* has been cultured and are at risk for dissemination from a colonization site may help in implementing effective infection control measures. Hand hygiene with an alcohol-based hand sanitizer is recommended when hands are not visibly soiled, in which case washing with soap and water is preferred. Identifying the source of contamination, if possible, and using an Environmental Protection Agency (EPA)-registered hospital-grade disinfectant effective against *Clostridioides difficile* spores are desirable. If a patient develops an invasive or bloodstream infection, it is recommended that the health care facility informs the CDC, or a similar agency in other countries, and adheres to recommendations for infection control, including isolation of patients (contact or enhanced barrier precautions), use of proper personal protective coverings, enforcement of hospital environment hygiene, and communicating with other health care facilities if the patient is being transferred.

■ PATHOGENESIS

In the most severe form of *Candida* infection, the organisms disseminate hematogenously and form microabscesses and small macroabscesses in major organs. Although the exact mechanism is not known, *Candida* probably enters the bloodstream from mucosal surfaces after growing to large numbers as a consequence of bacterial suppression by antibacterial drugs and breaches in the integrity of the mucosal barrier; alternatively, in some instances, the organism may enter the bloodstream from the skin via central venous catheters. A change from the blastospore stage to the pseudohyphal and hyphal stages is generally considered integral to *Candida*'s penetration into tissue. However, *C. glabrata* and *C. auris* can cause life-threatening infection, even though they do not transform into pseudohyphae or hyphae. Adherence to both epithelial and endothelial cells is thought to be the first step in invasion and infection; several adhesins have been identified as well as a mucosal toxin, candidalysin. Biofilm formation also is considered important in pathogenesis. Numerous reviews of cases of hematogenously disseminated candidiasis have identified the predisposing factors or conditions associated with disease (Table 222-1).

Several genes that are involved in the pathogenesis of other *Candida* species—such as those responsible for biofilm formation, proteinases, lipases, phospholipases, hydrolases, adhesins, secreted aspartyl proteases, and transporters involved in azole resistance—are also present in *C. auris*. Unlike other *Candida* species, several *C. auris* strains exhibit aggregate-forming properties in vivo, which may enable immune evasion. In addition, *C. auris* shows a unique tolerance to high temperature and saline concentrations and can grow optimally at up to 42°C and in a 10% saline concentration, making it possible to exist and persist in harsh environments. Furthermore, *C. auris* has significant affinity for

TABLE 222-1 Well-Recognized Factors and Conditions Predisposing to Hematogenously Disseminated Candidiasis

Antibacterial agents	Abdominal and thoracic surgery
Indwelling intravenous catheters	Cytotoxic chemotherapy
Hyperalimentation fluids	Immunosuppressive agents for organ transplantation
Indwelling urinary catheters	Respirators
Parenteral glucocorticoids	Myeloperoxidase deficiency
Severe burns	Neutropenia
CARD9 deficiency (central nervous system)	Low birth weight (neonates)
	Diabetes

abiotic surfaces such as plastic materials and medical devices, as well as human skin and nasal and ear cavities, which may account for its persistent colonization capabilities. The *C. auris*-specific adhesin surface colonization factor 1 (SCF1) was recently identified as a key fungal factor governing biofilm formation, colonization of skin and medical devices, and virulence during invasive infection.

Innate immunity is the most important defense mechanism against hematogenously disseminated candidiasis, and the neutrophil is the most potent component of this defense. Macrophages also play an important host defense role. On the other hand, interleukin (IL) 17-producing innate and adaptive lymphoid cells contribute significantly to defense against mucocutaneous candidiasis as evidenced by several monogenic disorders of IL-17 receptor signaling that manifest with *chronic mucocutaneous candidiasis* (CMC) (see “Clinical Manifestations,” below). Although many immunocompetent individuals have antibodies to *Candida*, the role of these antibodies in defense against the organism is not clear. Multiple genetic polymorphisms in host immune-related genes that predispose to both disseminated and focal candidiasis have been identified and may contribute to patient susceptibility.

CLINICAL MANIFESTATIONS

Mucocutaneous Candidiasis Thrush is characterized by white, adherent, painless, discrete or confluent patches in the mouth, on the tongue, or in the esophagus, occasionally with fissuring at the corners of the mouth. This form of disease caused by *Candida* can also occur at points of contact with dentures (called “denture sore mouth”). Organisms are identifiable in gram-stained scrapings from lesions. The occurrence of thrush in a young, otherwise healthy-appearing person should prompt an investigation for underlying HIV infection. More commonly, thrush is seen as a nonspecific manifestation of severe debilitating illness. Vulvovaginal candidiasis is accompanied by pruritus, pain, and vaginal discharge, which is usually thin but may contain whitish “curds” in severe cases. In contrast to oral thrush, HIV is not considered a major risk factor for vulvovaginal candidiasis. Instead, many women who receive antibiotics, particularly β -lactams, may develop vulvovaginal candidiasis. A subset of patients with recurrent vulvovaginitis may have a deficiency in the surface expression of Dectin-1, encoded by *CLEC7A*, a major recognition factor for β -glucan on the surface of *Candida* and/or in the downstream adaptor molecule CARD9, which ultimately increases the propensity for recurrent mucocutaneous (including vaginal) infections.

Other *Candida* skin infections include *paronychia*, a painful swelling at the nail-skin interface; *onychomycosis*, a fungal nail infection rarely caused by this genus; *intertrigo*, an erythematous irritation with redness and pustules in the skin folds; *balanitis*, an erythematous-pustular infection of the glans penis; *erosio interdigitalis blastomycetica*, an infection between the digits of the hands or toes; *folliculitis*, with pustules developing most frequently in the area of the beard; *perianal candidiasis*, a pruritic, erythematous, pustular infection surrounding the anus; *mastitis*; and *diaper rash*, a common erythematous, pustular perineal infection in infants. *Generalized disseminated cutaneous candidiasis*, another form of infection that occurs primarily in infants, is characterized by widespread eruptions over the trunk, thorax, and extremities. The diagnostic macronodular lesions of hematogenously disseminated candidiasis (Fig. 222-1) indicate a high probability of dissemination to multiple organs as well as the skin. While the lesions are seen predominantly in immunocompromised patients treated with cytotoxic drugs, they may also develop in patients without neutropenia.

CMC is a heterogeneous infection of the hair, nails, skin, and mucous membranes that persists despite intermittent antifungal therapy. The onset of disease usually comes in infancy or within the first two decades of life, but in rare cases, it occurs in later life. The condition may be mild and limited to a specific area of the skin or nails, or it may take a severely disfiguring form (*Candida* granuloma) characterized by exophytic outgrowths on the skin. CMC is usually associated with specific immunologic dysfunction; most frequently reported is a failure of lymphocytes to secrete or respond to type-17 cytokines following stimulation by *Candida* antigens in vitro. A subset of the affected



FIGURE 222-1 Macronodular skin lesions associated with hematogenously disseminated candidiasis. *Candida* organisms are usually but not always visible on histopathologic examination. The fungi grow when a portion of the biopsied specimen is cultured. Therefore, for optimal identification, both histopathology and culture should be performed. (Image courtesy of Dr. Noah Craft and the Victor Newcomer collection at UCLA, archived by Logical Images, Inc.; with permission.)

patients has mutations in the IL-17 receptors IL-17RA or IL-17RC, its adaptor molecule ACT1 (TRAF3IP2), or, more often, in STAT1.

Approximately half of patients with CMC have associated endocrine abnormalities either in the setting of gain-of-function mutations in *STAT1* or in the context of autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED) syndrome. This syndrome is due to mutations in the autoimmune regulator (*AIRE*) gene and is most prevalent among Finns, Iranian Jews, and Sardinians. Conditions that usually follow the onset of the disease include hypoparathyroidism, adrenal insufficiency, autoimmune thyroiditis, autoimmune hepatitis, autoimmune pneumonitis, alopecia, pernicious anemia, intestinal malabsorption, and primary hypogonadism. In addition, dental enamel dysplasia, vitiligo, nail dystrophy, asplenia, and calcification of the brain and tympanic membranes may occur. Patients with CMC rarely develop hematogenously disseminated candidiasis, reflecting their intact neutrophil function.

Deeply Invasive Candidiasis Deeply invasive *Candida* infections may or may not be due to hematogenous seeding. Deep esophageal infection may result from penetration by organisms from superficial esophageal erosions; joint or deep-wound infection from contiguous spread of organisms from the skin; kidney infection from catheter-initiated ascending spread of organisms through the urinary tract; infection of intraabdominal organs and the peritoneum from perforation of the gastrointestinal tract; and gallbladder infection from retrograde migration of organisms from the gastrointestinal tract into the biliary drainage system.

However, more commonly, deeply invasive candidiasis results from hematogenous seeding of various organs as a complication of candidemia. Once the organism gains access to the intravascular compartment (either from the gastrointestinal tract or, less often, from the skin through the site of an indwelling intravascular catheter), it may spread hematogenously to a variety of deep organs. The brain, chorioretina (Fig. 222-2), heart, and kidneys are most commonly infected and the liver and spleen are less commonly affected in nonneutropenic hosts (but most often involved in neutropenic patients). In fact, nearly any organ can become involved, including the endocrine glands, pancreas, heart valves (native or prosthetic), skeletal muscle, joints (native or prosthetic), bones, and meninges. *Candida* organisms can also spread hematogenously to the skin and cause classic macronodular lesions (Fig. 222-1). Frequently, painful muscular involvement is evident beneath the area of affected skin. Chorioretinal involvement and skin involvement are highly significant since both findings are associated with a very high probability of abscess formation in multiple



FIGURE 222-2 Hematogenous *Candida* endophthalmitis. A classic off-white lesion projecting from the chorioretina into the vitreous causes the surrounding haze. The lesion is composed primarily of inflammatory cells rather than organisms. Lesions of this type may progress to cause extensive vitreal inflammation and eventual loss of the eye. Partial vitrectomy, combined with IV and possibly intravitreal antifungal therapy, may be helpful in controlling the lesions. (Image courtesy of Dr. Gary Holland; with permission.)

deep organs as a result of generalized hematogenous seeding. Ocular involvement (Fig. 222-2) may require specific treatment (e.g., partial vitrectomy or intraocular injection of antifungal agents) to prevent permanent blindness. An ocular examination is indicated for patients with candidemia, whether or not they have ocular manifestations. *C. auris* invasive infections are similar to those of other *Candida* species and are most frequently associated with recent surgical procedures, immunosuppression, invasive devices such as catheters or various support or drainage tubes, and extended hospital stays. In the majority of invasive infections, *C. auris* has been isolated from the blood, but invasion of the kidney or spleen, and its recovery from cerebrospinal, bile, peritoneal, and pleural fluids demonstrate its invasiveness and dissemination potential. *C. auris*-associated candidemia can be life-threatening, with a crude mortality rate of 30–60%.

■ DIAGNOSIS

The diagnosis of *Candida* infection is established by visualization of pseudohyphae or hyphae on wet mount (saline and 10% KOH), tissue Gram stain, periodic acid–Schiff stain, or methenamine silver stain in the presence of inflammation. Absence of organisms on hematoxylin–eosin staining does not reliably exclude *Candida* infection. The most challenging aspect of diagnosis is determining which patients with *Candida* isolates have hematogenously disseminated

candidiasis. For instance, recovery of *Candida* from sputum, urine, or peritoneal catheters may indicate mere colonization rather than deep-seated infection, and *Candida* isolation from the blood of patients with indwelling intravascular catheters may reflect inconsequential seeding of the blood from or growth of the organisms on the catheter. Despite extensive research into both antigen and antibody detection systems, there is currently no widely available and validated diagnostic test to distinguish patients with inconsequential seeding of the blood from those whose positive blood cultures represent hematogenous dissemination to multiple organs. Many studies have examined the utility of the β -glucan test; at present, its greatest utility is its negative predictive value (~90%). Meanwhile, the presence of ocular or macronodular skin lesions is highly suggestive of widespread infection of multiple deep organs. Despite extensive diagnostic tests for hematogenous dissemination, such as polymerase chain reaction and T2 technology, no test is fully validated or widely available at present. Matrix-assisted laser desorption–ionization–time-of-flight mass spectrometry (MALDI-TOF MS) is now being used extensively for detection and speciation and is useful for the correct diagnosis of *C. auris*.

C. auris can be misdiagnosed in the microbiology laboratory, often leading to inappropriate treatment and delay in the implementation of appropriate infection control measures. Preliminary testing by culturing the fungus and examination of colony morphology may help in the initial identification, but this must be confirmed with more advanced diagnostic methods. For example, features such as budding yeast morphology, absence of hyphal growth or germ tubes, and growth at 40–42°C (unlike other *Candida* species) on CHROMagar that may appear white, pink, red, or purple should raise suspicion for *C. auris* (Fig. 222-3).

Several advanced molecular techniques accurately identify *C. auris* strains and therefore are being used for the follow-up testing and confirmation of the specimens that failed to be identified by traditional methods. MALDI-TOF equipment with upgraded libraries, such as Bruker Biotyper MALDI-TOF (CA System library version claim 4 or research use only [RUO] libraries versions 2014 [5627] and more recent), and using the bioMérieux VITEK MALDI-TOF MS (IVD v3.2 or RUO libraries with Saramis Ver 4.14 database and Saccharomycetaceae update), are the most common methods of *C. auris* identification. Other supplemental MALDI-TOF databases, such as MicrobeNet, which include additional *C. auris* strains from the four phylogenetic clades (i.e., South Asian, East Asian, South American, and South African) also can be used for the identification of *C. auris* strains. Sequencing of the D1–D2 region of the 28S rDNA or the internal transcribed region (ITS) of rDNA can also correctly identify *C. auris*. Recently, an automated, qualitative nucleic acid multiplex in vitro diagnostic test by GenMark called ePlex Blood Culture Identification Fungal Pathogen (BCID-FP) Panel was approved by the U.S. Food and Drug Administration for *C. auris* testing. Also, several polymerase chain reaction–based detection methods have been reported to identify

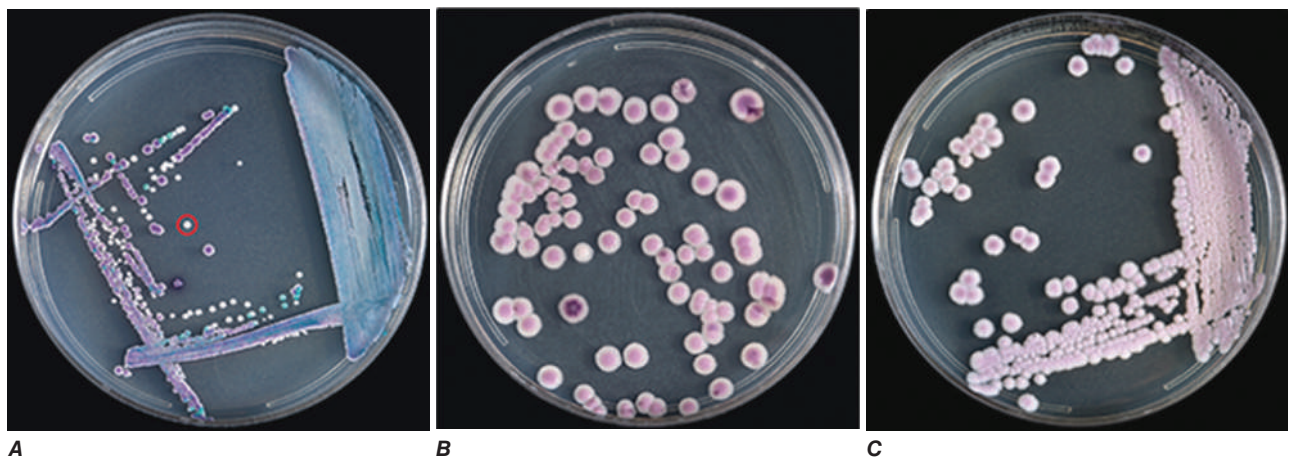


FIGURE 222-3 *C. auris* colony morphology and color on CHROMagar plates. **A.** *Candida* mixed culture: culture of *C. glabrata* (purple), *C. tropicalis* (navy blue), and *C. auris* (white, circled in red). **B.** *C. auris* showing multiple colony morphologies. **C.** *C. auris* after Salt SAB Dulcitol Broth enrichment. (From CDC: Identification of *Candida auris*. Available at: <https://www.cdc.gov/fungal/candida-auris/identification.html>.)

TABLE 222-2 Typical Decision-Making Steps in the Diagnosis of *C. auris*

NO.	METHOD	DATABASE/SOFTWARE	INITIAL FINDING	CONFIRMATION
1.	Bruker Biotyper MALDI-TOF	RUO libraries	<i>C. auris</i>	<i>C. auris</i>
		CA System library	<i>C. auris</i>	<i>C. auris</i>
2.	bioMérieux VITEK MS MALDI-TOF	RUO library	<i>C. auris</i>	<i>C. auris</i>
		IVD library (v3.2)	<i>C. auris</i>	<i>C. auris</i>
		Older IVD libraries	<i>C. haemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>C. lusitaniae</i>	<i>C. auris</i> possible: Needs further workup
3.	VITEK 2 YST	Software version 8.01	No identification	<i>C. auris</i> possible: Needs further workup
			<i>C. auris</i>	<i>C. auris</i> confirmed
			<i>C. haemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>C. duobushaemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
		Older versions	<i>C. haemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>C. duobushaemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
4.	API 20C		<i>Rhodotorula glutinis</i> , if characteristic red color absent	<i>C. auris</i> possible: Needs further workup
			<i>C. sake</i>	<i>C. auris</i> possible: Needs further workup
			<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
5.	API ID 32C		<i>C. intermedia</i>	<i>C. auris</i> possible: Needs further workup
			<i>C. sake</i>	<i>C. auris</i> possible: Needs further workup
			<i>Saccharomyces kluyveri</i>	<i>C. auris</i> possible: Needs further workup
6.	BD Phoenix		<i>C. catenulata</i>	<i>C. auris</i> possible: Needs further workup
			<i>C. haemulonii</i>	<i>C. auris</i> possible: Needs further workup
			<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
7.	MicroScan		<i>C. lusitaniae</i>	No hyphal growth present Can rule out <i>C. lusitaniae</i> , <i>C. guilliermondii</i> , and <i>C. parapsilosis</i> . <i>C. auris</i> possible: Needs further workup
			<i>C. guilliermondii</i>	
			<i>C. parapsilosis</i>	
			<i>C. lusitaniae</i>	Hyphal growth present Possibly <i>C. lusitaniae</i> , <i>C. guilliermondii</i> , <i>C. parapsilosis</i> , or <i>C. auris</i> : Needs further workup
			<i>C. guilliermondii</i>	
			<i>C. parapsilosis</i>	
			<i>C. famata</i>	<i>C. auris</i> possible: Needs further workup
8.	RapID Yeast Plus		<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
			<i>C. parapsilosis</i> → Test on corneal agar	No hyphal growth present Can rule out <i>C. parapsilosis</i> . <i>C. auris</i> possible: Needs further workup
9.	GenMark ePlex BCID-FP Panel		Hyphal growth present	Possibly <i>C. parapsilosis</i> or <i>C. auris</i> : Needs further workup
			<i>Candida</i> spp. not identified	<i>C. auris</i> possible: Needs further workup
9.	GenMark ePlex BCID-FP Panel		<i>C. auris</i>	<i>C. auris</i> confirmed

Abbreviations: IVD, in vitro diagnostic; RUO, research use only.

Source: Adapted from CDC: Identification of *Candida auris*. Available at: https://www.cdc.gov/fungal/candida-auris/pdf/Testing-algorithm_by-Method_508.pdf.

C. auris in various specimens. Table 222-2 outlines the typical decision-making steps in the diagnosis of *C. auris* by using different methods. A suspicious *C. auris* specimen is usually sent to a regional reference laboratory for further testing and confirmation of *C. auris*.

TREATMENT

Candida Infections

MUCOCUTANEOUS CANDIDA INFECTION

The treatment of mucocutaneous candidiasis is summarized in Table 222-3.

CANDIDEMIA AND SUSPECTED HEMATOGENOUSLY DISSEMINATED CANDIDIASIS

All patients with candidemia are treated with a systemic antifungal agent. A certain percentage of patients, including many of those who have candidemia associated with an indwelling intravascular catheter, probably have “benign” candidemia rather than

deep-organ seeding. However, because there is no reliable way to distinguish benign candidemia from deep-organ infection, and because antifungal drugs less toxic than amphotericin B are available, antifungal treatment for candidemia—with or without clinical evidence of deep-organ involvement—has become the standard of practice. In addition, if an indwelling intravascular catheter is

TABLE 222-3 Treatment of Mucocutaneous Candidal Infections

DISEASE	PREFERRED TREATMENT	ALTERNATIVES
Cutaneous	Topical azole	Topical nystatin
Vulvovaginal	Oral fluconazole (150 mg) or ibrexafungerp (300 mg twice daily for 1 day) or azole cream or suppository	Nystatin suppository
Oral (thrush)	Fluconazole tablets (100–200 mg/d)	Clotrimazole troches, nystatin
Esophageal	Fluconazole tablets (100–200 mg/d) or itraconazole solution (200 mg/d)	Caspofungin, micafungin, or amphotericin B

TABLE 222-4 Agents for the Treatment of Disseminated Candidiasis

AGENT	ROUTE OF ADMINISTRATION	DOSE ^a	COMMENT
Amphotericin B deoxycholate	IV only	0.5–1.0 mg/kg daily	Mostly replaced by lipid formulations
Amphotericin B lipid formulations			Not approved as primary therapy by the U.S. Food and Drug Administration, but used commonly because they are less toxic than amphotericin B deoxycholate
Liposomal (AmBisome, Abelcet)	IV only	3.0–5.0 mg/kg daily	Associated with frequent infusion reactions
Lipid complex (ABLC)	IV only	3.0–5.0 mg/kg daily	
Colloidal dispersion (ABCD)	IV only	3.0–5.0 mg/kg daily	
Azoles ^b			
Posaconazole	IV and oral	300 mg/d (IV) 200 mg tid (oral)	Approved for prophylaxis
Fluconazole	IV and oral	400 mg/d	Most commonly used
Voriconazole	IV and oral	400 mg/d	Multiple drug interactions, visual hallucinations, fluorosis, phototoxicity Approved for candidemia in nonneutropenic patients
Echinocandins			Broad spectrum against <i>Candida</i> species; approved for disseminated candidiasis; less toxic than amphotericin B formulations
Caspofungin	IV only	50 mg/d	
Anidulafungin	IV only	100 mg/d	
Micafungin	IV only	100 mg/d	
Rezafungin	IV only	400 mg loading dose, 200 mg once weekly thereafter	

^aFor loading doses and adjustments in renal failure, see Pappas PG et al: Clinical practice guidelines for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. Clin Infect Dis 62:e1, 2016. The recommended duration of therapy is 2 weeks beyond the last positive blood culture and the resolution of signs and symptoms of infection. ^bAlthough ketoconazole is approved for the treatment of disseminated candidiasis, it has been replaced by the newer agents listed in this table. Posaconazole has been approved for prophylaxis in neutropenic patients and for oropharyngeal candidiasis.

present, it is best to remove or replace the device whenever feasible. Moreover, an infectious disease consultation is recommended as it has been associated with improved patient outcomes.

The drugs used for the treatment of candidemia and suspected disseminated candidiasis are listed in Table 222-4. Various lipid formulations of amphotericin B, four echinocandins, the azoles fluconazole and voriconazole, and in some instances, the newer triazole posaconazole are used; no agent within a given class has been clearly identified as superior to the others. Most institutions choose an agent from each class on the basis of their own specific microbial epidemiology, strategies to minimize toxicities, and cost considerations. An echinocandin is the first choice of treatment. The U.S. Food and Drug Administration (FDA) recently approved the new-generation echinocandin rezafungin for the treatment of candidemia and invasive candidiasis in adult patients; rezafungin has a prolonged half-life, which allows for once-weekly 200-mg dosing following a 400-mg front-loading dose. Echinocandin treatment continues until sensitivities or speciation is determined. In stable patients, many centers then switch to fluconazole if a sensitive strain is identified and there is no evidence of hematogenous dissemination. For hemodynamically unstable or neutropenic patients, initial treatment with echinocandins is warranted, and once the clinical response has been assessed and the pathogen specifically identified, the regimen can be altered according to the sensitivities. At present, the vast majority of *C. albicans* isolates are sensitive to fluconazole. Isolates of *C. glabrata* and *C. krusei* are less sensitive to fluconazole and more sensitive to polyenes and echinocandins. *C. parapsilosis* is less sensitive to echinocandins in vitro; however, this lesser sensitivity is considered clinically insignificant. Posaconazole has been approved for prophylaxis, including against *Candida*, in neutropenic patients. Itraconazole is rarely used for *Candida* nowadays, and isavuconazole is not recommended for this indication.

Antifungal drug resistance is one of the hallmarks of *C. auris* infections. Some *C. auris* strains have multidrug resistance with elevated minimal inhibitory concentrations (MICs) to all three major antifungal classes—azoles, echinocandins, and polyenes—resulting in limited treatment options. A CDC study reported antifungal resistance in *C. auris* strains obtained from 54 patients

in India, Pakistan, South Africa, and Venezuela: 93% were resistant to fluconazole, 35% to amphotericin B, and 7% to echinocandins; 41% of the tested strains were resistant to two antifungal classes, and, alarmingly, 4% of the tested strains were resistant to all three classes of antifungal drugs. A 2023 CDC report indicated that rates of *C. auris* echinocandin resistance tripled in 2021 compared to the prior 2 years. Almost all *C. auris* strains that have been identified have elevated MICs for fluconazole with variable susceptibilities to other triazoles (Table 222-5), associated with mutations in *ERG11*-encoded lanosterol demethylase and/or overexpression of drug transporters/efflux pumps.

Due to the high rates of azole resistance among *C. auris* strains, the use of echinocandins is recommended as first-line therapy for *C. auris* infection. By contrast, the CDC discourages the use of antifungal drugs for the treatment of colonization of *C. auris* in the absence of invasive or bloodstream infection. A history of patient travel or residence in a health care or nursing facility with a known

TABLE 222-5 Typical MICs of Available Antifungal Drugs for *C. auris*

DRUG	TENTATIVE RESISTANCE BREAKPOINTS ^a	MIC RANGE, μ g/mL		
		MIC	MIC ₅₀	MIC ₉₀
Amphotericin B	≥ 2	0.06–8	0.5–1	2–4
Fluconazole	≥ 32	0.12– ≥ 64	≥ 64	≥ 64
Itraconazole	N/A	0.032–2	0.06–0.5	0.25–1
Voriconazole	N/A	0.032–16	0.5–2	2–8
Posaconazole	N/A	0.015–16	0.016–0.5	0.125–2
Isavuconazole	N/A	0.015–4	0.125–0.25	0.5–2
Caspofungin	≥ 2	0.03–16	0.25–1	1–2
Anidulafungin	≥ 4	0.015–16	0.125–0.5	0.5–1
Micafungin	≥ 4	0.015–8	0.125–0.25	0.25–2

^aTentative resistance breakpoints per Centers for Disease Control and Prevention (CDC) (www.cdc.gov/fungal/candida-auris/c-auris-antifungal.html).

Abbreviations: MIC, minimum inhibitory concentration; N/A, not available.

Source: Adapted from CDC: Antifungal susceptibility testing and interpretation. Available at: www.cdc.gov/fungal/candida-auris/c-auris-antifungal.html.

TABLE 222-6 List of CDC-Recommended Echinocandin Doses for the Treatment of *C. auris* Infections

DRUG	ADULTS	CHILDREN (>2 MONTHS)	INFANTS (<2 MONTHS)
Caspofungin	Loading dose 70 mg IV, then 50 mg IV daily	Loading dose 70 mg/m ² per day IV, then 50 mg/m ² per day IV	25 mg/m ² per day IV
Anidulafungin	Loading dose 200 mg IV, then 100 mg IV daily	Not approved for use in children	Not approved for use in children
Micafungin	100 mg IV daily	2 mg/kg per day IV with option to increase to 4 mg/kg per day IV in children at least 40 kg	10 mg/kg per day IV

Abbreviation: CDC, Centers for Disease Control and Prevention.
Source: Adapted from CDC: Treatment and management of infections and colonization. Available at: www.cdc.gov/fungal/candida-auris/c-auris-treatment.html.

outbreak of *C. auris* infection, as well as drug susceptibility data of identified strains, act as a guide for the effective choice of treatment of invasive and bloodstream infections. *C. auris* is known to develop antibiotic resistance during treatment. Therefore, the emergence of antifungal resistance should be closely monitored with follow-up cultures and repeat susceptibility testing. Antibiotic stewardship should be implemented to ameliorate the risk of development of drug resistance. Patients may remain colonized with *C. auris* during or after the successful treatment of invasive *C. auris* infection. Therefore, infection control measures should be implemented throughout patient care. Table 222-6 outlines CDC-recommended echinocandin doses for the initial antifungal treatment for *C. auris* infections.

In cases of echinocandin resistance, liposomal amphotericin B (5 mg/kg per day) can be considered. For neonates and infants (<2 months old), amphotericin B deoxycholate (1 mg/kg per day) treatment can be initiated. If this fails, liposomal amphotericin B (5 mg/kg per day) can be given. In very severe cases, if all treatment options fail, echinocandins per CDC recommendations can be given (Table 222-6). Other considerations for *C. auris* infection management can be referenced from the 2016 Infectious Diseases Society of America (IDSA) Clinical Practice Guideline for the Management of Candidiasis.

Some generalizations exist regarding the management of specific *Candida* infections. Recovery of *Candida* from sputum is almost never indicative of underlying pulmonary candidiasis and does not by itself warrant antifungal treatment. Similarly, *Candida* in the urine of a patient with an indwelling bladder catheter may represent colonization only, rather than bladder or kidney infection. However, the threshold for systemic treatment is lower in general in severely ill patients in this category since it is impossible to distinguish colonization from lower or upper urinary tract infection. If the isolate is *C. albicans*, most clinicians use oral fluconazole rather than a bladder washout with amphotericin B, which was more commonly used in the past. Although echinocandins are poorly excreted into the urine, they may be an option, especially for non-*albicans* isolates. The doses and duration are the same as for disseminated candidiasis. The significance of the recovery of *Candida* from abdominal drains in post-operative patients is unclear, but again, the threshold for treatment is generally low because most of the affected patients have been subjected to risk factors predisposing them to disseminated candidiasis. In addition, there has been a considerable increase in the recognition and diagnosis of intraabdominal candidiasis.

Removal of the infected valve and long-term antifungal administration constitute appropriate treatment for *Candida* endocarditis. Although definitive studies are not available, patients usually are treated for weeks with a systemic antifungal agent (Table 222-4) and then given chronic suppressive therapy for months or years (sometimes indefinitely) with an oral azole (usually fluconazole at 400–800 mg/d).

Hematogenous *Candida* endophthalmitis is a special problem requiring ophthalmologic consultation. When lesions are expanding

or are threatening the macula, an IV polyene combined with flucytosine (25 mg/kg four times daily) has been the regimen of choice, although comparative studies with other regimens have not yet been reported. As more data on the newer triazoles (e.g., voriconazole) and the echinocandins become available, new strategies involving these agents are developing, although it is important to note that echinocandins exhibit low penetration in ocular tissue. Of paramount importance is the decision to perform a partial vitrectomy. This procedure debulks the infection and can preserve sight, which may otherwise be lost due to vitreal scarring. All patients with candidemia should undergo ophthalmologic examination because of the relatively high frequency of this ocular complication (up to 15–20% in some case series). This examination can detect a developing eye lesion early in its course; in addition, identification of a lesion signifies a probability of ~90% of deep-organ abscesses and may prompt prolongation of therapy for candidemia beyond the recommended 2 weeks after the last positive blood culture. Although the basis for the consensus is a very small data set, the recommended treatment for *Candida* meningitis is a polyene (Table 222-4) plus flucytosine (25 mg/kg four times daily). Development of *Candida* meningo-encephalitis in an otherwise immunocompetent individual should raise suspicion for deficiency in the C-type lectin receptor adaptor molecule CARD9 and should prompt genetic testing to rule out this monogenic disorder. Successful treatment of *Candida*-infected prosthetic material (e.g., an artificial joint) nearly always requires removal of the infected material followed by long-term administration of an antifungal agent selected on the basis of the isolate’s sensitivity and the logistics of administration.

■ PROPHYLAXIS

The use of antifungal agents to prevent *Candida* infections has been controversial, but some general principles have emerged. Most centers administer prophylactic fluconazole (400 mg/d) to recipients of allogeneic hematopoietic stem cell transplantation. High-risk liver transplant recipients are also given fluconazole prophylaxis in most centers. The use of prophylaxis for neutropenic patients has varied considerably from center to center; many centers that elect to give prophylaxis to this population use either fluconazole (200–400 mg/d) or a lipid formulation of amphotericin B (AmBisome, 1–2 mg/d). Caspofungin (50 mg/d) also has been used. Posaconazole (200 mg three times daily) has been approved by the FDA for prophylaxis in neutropenic patients; it is gaining in popularity and may replace fluconazole in settings when mold activity is desired.

Prophylaxis is sometimes given to surgical patients at very high risk for candidiasis. The widespread use of prophylaxis for nearly all patients in general surgical or medical intensive care units is not—and should not be—a common practice for three reasons: (1) the incidence of disseminated candidiasis is relatively low, (2) the cost–benefit ratio is suboptimal, and (3) increased resistance with widespread prophylaxis is a valid concern.

Prophylaxis for oropharyngeal or esophageal candidiasis in HIV-infected patients is not recommended unless there are frequent recurrences.

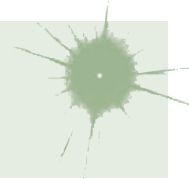
■ FURTHER READING

LIONAKIS MS, EDWARDS JE JR: *Candida* species, in *Mandell, Douglas, and Bennett’s Principles of Infectious Diseases*, 10th ed. Blaser MJ et al (eds). Philadelphia, Elsevier, 2025.
PAPPAS PG et al: Invasive candidiasis. *Nat Rev Dis Primers* 62:e1, 2018.
PECHACEK J, LIONAKIS MS: Host defense mechanisms against *Candida auris*. *Expert Rev Anti Infect Ther* 21:1087, 2023.
PROCTOR DM et al: Integrated genomic, epidemiologic investigation of *Candida auris* skin colonization in a skilled nursing facility. *Nat Med* 27:1401, 2021.
SANTANA DJ et al: A *Candida auris*-specific adhesin, Scf1, governs surface association, colonization, and virulence. *Science* 381:1461, 2023.
TSAI SV et al: Burden of candidemia in the United States, 2017. *Clin Infect Dis* 71:e449, 2020.

223

Aspergillosis

David W. Denning



Aspergillosis is the collective term used to describe all disease entities caused by any one of ~50 pathogenic and allergenic species of *Aspergillus*. Only those species that grow at 37°C can cause invasive infection, although some species without this ability can cause allergic syndromes. Each common pathogenic species is actually a complex of many species (many of them cryptic) but is referred to as a single species here for simplicity. *A. fumigatus* is responsible for most cases of invasive aspergillosis, >95% of cases of chronic aspergillosis, and most allergic syndromes. *A. flavus* is more prevalent in some hospitals and causes a higher proportion of cases of sinus infections, cutaneous infections, and keratitis than *A. fumigatus*. *A. niger* can cause invasive infection but more commonly colonizes the respiratory tract and causes external otitis. *A. terreus* causes only invasive disease, usually with a poor prognosis. *A. nidulans* occasionally causes invasive infection, primarily in patients with chronic granulomatous disease (CGD).

■ EPIDEMIOLOGY AND ECOLOGY

Aspergillus has a worldwide distribution, most commonly growing in decomposing plant materials (i.e., compost) and in bedding. This hyaline (nonpigmented), septate, branching mold produces vast numbers of conidia (spores) on stalks above the surface of mycelial growth. *Aspergilli* are found in indoor and outdoor air, on surfaces, and in water from surface reservoirs. Daily exposures vary from a few to many millions of conidia; high numbers of conidia are encountered in hay barns and other very dusty environments. The required size of the infecting inoculum is uncertain; however, only intense exposures (e.g., during construction work, handling of moldy bark or hay, or composting) are demonstrated to cause disease—acute community-acquired pulmonary aspergillosis—in healthy immunocompetent individuals or exacerbations of COPD. Allergic syndromes are exacerbated by continuous antigenic exposure arising from sinus or airway colonization or from nail infection. High-efficiency particulate air (HEPA) filtration is often protective against infection; thus, HEPA filters should be installed and monitored for efficiency in operating rooms, burn units and in areas of the hospital that house high-risk patients.

The incubation period of invasive aspergillosis after exposure is highly variable, extending in documented cases from 2 to 90 days. Thus, community acquisition of an infecting strain frequently manifests as invasive infection during hospitalization, although nosocomial acquisition is also common. Outbreaks usually are directly related to a contaminated air source or construction in the hospital.

Global aspergillosis incidence and prevalence have been estimated (Table 223-1). The frequency of different manifestations of aspergillosis varies considerably with geographic location; most notably, chronic granulomatous sinusitis is rare outside the Middle East and India. Fungal (mycotic) keratitis is particularly common in Southeast Asia but occurs globally. Chronic pulmonary aspergillosis follows pulmonary tuberculosis in ~6–13% of treated cases and also mimics pulmonary tuberculosis as smear-negative or “clinically diagnosed” tuberculosis. *Aspergillus* onychomycosis, usually of the toenail, has been reported in as low as <1% and as high as 35% of cases of onychomycosis and is more common in diabetes.

■ RISK FACTORS AND PATHOGENESIS

The primary risk factors for invasive aspergillosis are profound neutropenia, glucocorticoid use, and underlying respiratory disease; risk increases with longer duration of these conditions. Higher doses of glucocorticoids increase the risk of both acquisition of invasive aspergillosis and death from the infection. Neutrophil and/or phagocyte dysfunction also is an important risk factor, as evidenced by aspergillosis in CGD, advanced HIV infection, and relapsed leukemia. Invasive aspergillosis is increasingly recognized (if actively sought) in medical intensive care units (2–5%), those with severe influenza (8–25%) and

TABLE 223-1 Disease Frequency and Diagnostic Sensitivity for Different Manifestations of Aspergillosis

PARAMETER	TYPE OF DISEASE		
	INVASIVE	CHRONIC	ALLERGIC
Incidence/100,000 ^a	27.6	23.6	? ^b
Prevalence/100,000 ^a	—	55.4	286 ^c
Global burden ^a	~2,116,000	~6,141,000	~11,690,000
Mortality rate without treatment ^a	~100%	~50%	<1%
Respiratory Diagnostic Sensitivity^d			
Culture ^e	√	√-√√ ^e	√-√√ ^e
Microscopy	√	√	√
Antigen	√√√	√√	?
Real-time PCR	√√	√√	√√
Blood Diagnostic Sensitivity^d			
Culture	x	x	x
Antigen	√√	√	√
β-D-Glucan	√√	√	?
Real-time PCR	√√	x	x
IgG antibody ^f	√√ ^h	√√√	√√√
IgE antibody	x	√√	√√√√

^aDenning DW: Global incidence and mortality of severe fungal disease. *Lancet Infect Dis* 24:e428, 2024. ^bAllergic fungal disease can develop at any age, usually in adulthood; the annual frequency with which it occurs is not known. ^cAllergic bronchopulmonary aspergillosis and severe asthma with fungal sensitization. ^dKey for sensitivity: 1 check = limited (as the text indicates, 10–30% for culture); 2 checks = higher; 3 checks = >80%; and 4 checks = ~95%. ^eMultiple specimens, fungal media, and high-volume fungal culture increase sensitivity. ^fHigh cross-reactivity between *A. fumigatus* and other *Aspergillus* spp. ^gIgG is a useful assay in subacute invasive and nonneutropenic patients but may not be positive on initial testing.

Abbreviation: PCR, polymerase chain reaction.

severe COVID-19 (~10%), and patients admitted to hospital with chronic obstructive pulmonary disease (COPD; 1.3–3.9%). Temporary abrogation of protective responses from glucocorticoid use or compensatory anti-inflammatory response syndrome is a significant risk factor. Many patients have some evidence of prior pulmonary disease—typically, a history of pneumonia or COPD. Many new immunomodulating agents, such as infliximab and ibrutinib, increase the risk of invasive aspergillosis, as does severe liver disease.

Patients with chronic pulmonary aspergillosis have a wide spectrum of underlying pulmonary disease, including current and prior tuberculosis, sarcoidosis, prior pneumothorax, or COPD. These patients are apparently immunocompetent, but natural killer and/or interleukin 12 or gamma interferon production defects are common. Their inflammatory immune (T_H1-like) response is suboptimal, and persistent inflammation is typical. Glucocorticoids accelerate disease progression.

Allergic bronchopulmonary aspergillosis (ABPA) usually complicates asthma and cystic fibrosis. Many genetic associations indicate a strong basis for the development of a T_H2-like and “allergic” response to *A. fumigatus*. Remarkably, high-dose glucocorticoid treatment for exacerbations of ABPA almost never leads to invasive aspergillosis. Fungal, and especially *Aspergillus*, sensitization is especially common in those with poorly controlled asthma. Environmental exposure and *Aspergillus* sensitization drive some COPD exacerbations. Most patients with *Aspergillus* bronchitis have bronchiectasis, with or without cystic fibrosis, including heterozygous *CFTR* mutations.

 Different genetic traits are associated with invasive, chronic, and allergic aspergillosis; the majority of people probably are not at risk for aspergillosis. Multiple gene variants appear to be necessary for susceptibility to each form of aspergillosis.

■ CLINICAL FEATURES AND APPROACH TO THE PATIENT

(Table 223-2)

Invasive Pulmonary Aspergillosis Both the frequency of invasive disease and the pace of its progression increase with greater

TABLE 223-2 Major Manifestations of Aspergillosis

ORGAN	TYPE OF DISEASE			
	INVASIVE (ACUTE AND SUBACUTE)	CHRONIC	SAPROPHYTIC	ALLERGIC
Lung	Angioinvasive (in neutropenia), nonangioinvasive, granulomatous	Chronic cavitary, chronic fibrosing, bronchitis, <i>Aspergillus</i> nodule	Aspergilloma (single), airway colonization	Allergic bronchopulmonary, severe asthma with fungal sensitization, hypersensitivity pneumonitis
Sinus	Acute invasive	Chronic invasive, chronic granulomatous	Maxillary fungal ball	Allergic fungal sinusitis, eosinophilic fungal rhinosinusitis
Brain	Abscess, hemorrhagic infarction, meningitis	Granulomatous, meningitis	None	None
Skin	Acute disseminated, locally invasive (trauma, burns, IV access)	External otitis, onychomycosis	None	None
Heart	Endocarditis (native or prosthetic), pericarditis	None	None	None
Eye	Keratitis, endophthalmitis	None	None	None described

degrees of immunocompromise. Invasive aspergillosis is arbitrarily classified as acute and subacute, with courses of ≤ 1 month and 1–3 months, respectively. More than 80% of cases of invasive aspergillosis involve the lungs, and most are community acquired. The most common clinical features are no symptoms at all, fever, cough (sometimes productive), nondescript chest discomfort, trivial hemoptysis, and shortness of breath. Although the fever often responds to glucocorticoids, the disease progresses. In ventilated patients, screening for *Aspergillus* antigen on tracheobronchial lavage fluid is necessary for diagnosis as radiology is not distinctive. The keys to early diagnosis in at-risk patients are a high index of suspicion, screening for circulating antigen (in leukemia), and urgent computed tomography (CT) of the thorax. Invasive aspergillosis is one of the most common diagnostic errors revealed at autopsy.

Invasive Sinusitis The sinuses are involved in 5–10% of cases of invasive aspergillosis, especially affecting patients with leukemia and recipients of hematopoietic stem cell transplants. In addition to fever, the most common features are nasal or facial discomfort, blocked nose, and nasal discharge (sometimes bloody). Endoscopic examination of the nose reveals pale, dusky or necrotic-looking tissue in any location. CT or magnetic resonance imaging (MRI) of the sinuses is essential but does not distinguish invasive *Aspergillus* sinusitis from preexisting allergic or bacterial sinusitis early in the disease process.

Tracheobronchitis Occasionally, only the airways are infected by *Aspergillus*. The resulting manifestations seen on bronchoscopy range from acute or chronic bronchitis to ulcerative or pseudomembranous tracheobronchitis. These entities are particularly common among lung transplant recipients and patients on artificial ventilation. Obstruction with mucous plugs may occur and is called obstructing bronchial aspergillosis in immunocompromised patients and mucous impaction in other patients, such as those with ABPA.

Aspergillus Bronchitis Recurrent chest infections that only partially improve with antibiotic treatment and are associated with significant breathlessness or coughing up of thick sputum plugs are typical features of *Aspergillus* bronchitis. Patients are not significantly immunocompromised and usually have bronchiectasis or cystic fibrosis. Occasional patients present with respiratory failure because of airway obstruction with mucus. Concurrent bacterial bronchitis is common. The diagnosis rests on recurrent detection of *Aspergillus* in the airway by microscopy, culture, or polymerase chain reaction (PCR). *Aspergillus* IgG is usually detectable.

Disseminated Aspergillosis In the most severely immunocompromised patients, *Aspergillus* disseminates from the lungs to multiple organs—most often to the brain but also to the skin, thyroid, bone, kidney, liver, gastrointestinal tract, eye (endophthalmitis), and heart valve. Aside from cutaneous lesions, the most common features are gradual clinical deterioration over 1–3 days, with low-grade fever and features of mild sepsis, and nonspecific abnormalities in laboratory tests. In most cases, at least one localization becomes apparent before death. Blood cultures are almost always negative.

Cerebral Aspergillosis Hematogenous dissemination to the brain is a devastating complication of invasive aspergillosis. Single or multiple lesions may develop. In acute disease, hemorrhagic infarction is most typical, and cerebral abscess is common. Rarer manifestations include meningitis, mycotic aneurysm, and cerebral granuloma (mimicking a brain tumor). Local spread from cranial sinuses also occurs. Postoperative infection develops rarely and is exacerbated by glucocorticoids, which are often given after neurosurgery. The presentation can be either acute or subacute, with mood changes, focal signs, seizures, and decline in mental status. MRI is the most useful immediate investigation; unenhanced CT of the brain is usually nonspecific, and contrast is often contraindicated because of poor renal function. Cerebral aspergillosis is disproportionately common in those on ibritinib.

Endocarditis Most cases of *Aspergillus* endocarditis are prosthetic-valve infections resulting from contamination during surgery. Native-valve disease is reported, especially as a feature of disseminated infection and in persons using illicit IV drugs. Culture-negative endocarditis with large vegetations is the most common presentation; embolectomy occasionally reveals the diagnosis.

Cutaneous Aspergillosis Dissemination of *Aspergillus* occasionally results in cutaneous features, usually an erythematous or purplish nontender area that progresses to a necrotic eschar. Direct invasion of the skin occurs in neutropenic patients at the site of IV catheter insertion and in burn patients. Surgical, burn, and trauma wounds may become infected with *Aspergillus* (especially *A. flavus*).

Chronic Pulmonary Aspergillosis The hallmark of chronic cavitary pulmonary aspergillosis (Fig. 223-1) is one or more pulmonary cavities expanding over a period of months or years in association with pulmonary symptoms and systemic manifestations such as fatigue and weight loss. Often mistaken initially for tuberculosis, >90% of chronic cavitary pulmonary aspergillosis cases occur in patients with prior pulmonary disease (e.g., current or past tuberculosis or nontuberculous mycobacterial infection, sarcoidosis, rheumatoid lung disease, pneumothorax, bullae) or lung surgery. The onset is insidious, and systemic features (weight loss, fatigue) may be more prominent than pulmonary symptoms. On CT scan, an irregular internal cavity surface and thickened cavity walls are typical and indicative of disease activity. Irregular material, fluid level, and a well-formed fungal ball are seen in some cavities. Multiple cavities are more common than a single cavity, and most cavities are in the upper lobes. Pleural thickening and pericavitary infiltrates are typical and most obvious if a positron emission tomography scan has been done as part of the workup. Chronic cavitary pulmonary aspergillosis is usually caused by *A. fumigatus*, but *A. niger* has been implicated, particularly in diabetic patients, as have other species. IgG antibodies to *Aspergillus* are detectable in ~90% of patients with chronic cavitary pulmonary aspergillosis. Some patients have concurrent infections with mycobacteria and/or other bacterial pathogens. The most significant complication is life-threatening hemoptysis, which may be the presenting manifestation. If untreated, chronic cavitary pulmonary aspergillosis typically

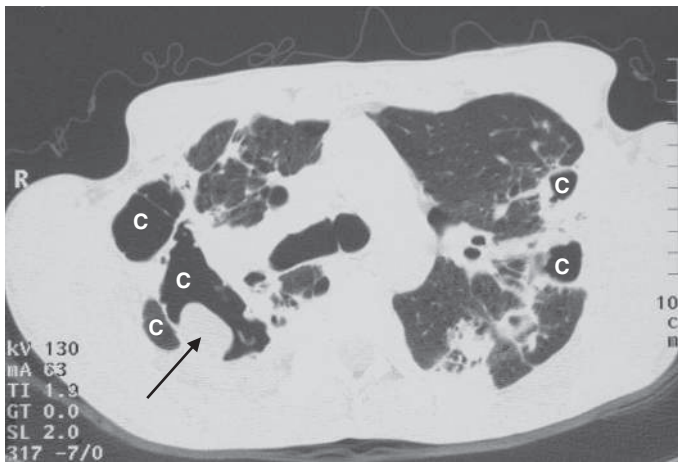


FIGURE 223-1 Computed tomography (CT) scan image of the chest in a patient with long-standing bilateral chronic cavitary pulmonary aspergillosis. This patient had a history of several bilateral pneumothoraces and had required bilateral pleurodesis in 1990. CT then demonstrated multiple bullae, and sputum cultures grew *A. fumigatus*. The patient had initially weakly and later strongly positive serum IgG *Aspergillus* antibody tests. This scan (2003) shows a mixture of thick- and thin-walled cavities in both lungs (each marked with C), with a probable fungal ball (black arrow) protruding into the large cavity on the patient's right side (R). There is also considerable pleural thickening bilaterally.

progresses (sometimes relatively rapidly) to unilateral or upper-lobe fibrosis. This end-stage entity is termed *chronic fibrosing pulmonary aspergillosis* (destroyed lung).

Aspergilloma Aspergilloma (fungal ball) is a late manifestation of chronic cavitary pulmonary aspergillosis, but some patients are asymptomatic. The inside of a pulmonary cavity allows fungal growth that peels off, forming the layers of the fungal ball. Signs and symptoms associated with single (simple) aspergillomas are minor, including cough (sometimes productive), hemoptysis, wheezing, and mild fatigue. More significant signs and symptoms are associated with chronic cavitary pulmonary aspergillosis and should be treated as such. About 10% of fungal balls resolve spontaneously (by being coughed up), but the cavity may still be infected and the patient symptomatic.

Aspergillus Nodule A recently recognized form of chronic pulmonary aspergillosis is the *Aspergillus* nodule, which may resemble early lung carcinoma and may cavitate. Nodules may be single or multiple and 5–50 mm in diameter. Larger mass lesions are rarely seen. Nodules are usually avid on positron emission tomography. IgG antibodies to *Aspergillus* are detectable in ~65% of patients with an *Aspergillus* nodule.

Chronic Aspergillus Sinusitis Three entities are subsumed under this broad designation: fungal ball of the sinus, chronic invasive sinusitis, and chronic granulomatous sinusitis. Fungal ball of the sinus is limited to the maxillary sinus (except in rare cases involving the sphenoid sinus) in which the sinus cavity is filled with a fungal ball. Maxillary disease is associated with prior upper-jaw root canal work and chronic (bacterial) sinusitis. About 90% of CT scans show focal hyperattenuation related to concretions; on MRI scans, the T2-weighted signal is decreased, whereas it is increased in bacterial sinusitis. Removal of the fungal ball is curative. No tissue invasion is demonstrable histologically or radiologically.

In contrast, chronic invasive sinusitis is a slowly destructive process that most commonly affects the ethmoid and sphenoid sinuses. Patients are usually but not always immunocompromised to some degree (e.g., as a result of diabetes or HIV infection). Imaging of the cranial sinuses shows opacification of one or more sinuses, local bone destruction, and invasion of local structures. The differential diagnosis is wide, including other infections. Apart from a history of chronic nasal discharge and blockage, loss of the sense of smell, and persistent headache, the usual presenting features are related to local involvement of critical structures. The orbital apex syndrome (blindness and proptosis) is characteristic. Facial swelling, cavernous sinus thrombosis,

carotid artery occlusion, and pituitary fossa and brain and skull-base invasion are complications.

Chronic granulomatous sinusitis due to *Aspergillus* is most commonly seen in the Middle East and India and is often caused by *A. flavus*. It typically presents late, with facial swelling and unilateral proptosis. The prominent granulomatous reaction histologically distinguishes this disease from chronic invasive sinusitis, in which tissue necrosis with a low-grade mixed-cell infiltrate is typical. IgG antibodies to *A. flavus* are usually detectable.

Allergic Bronchopulmonary Aspergillosis In almost all cases, ABPA represents a hypersensitivity reaction to *A. fumigatus*; other causes are other aspergilli and other fungi. ABPA occurs in ~2.5% of patients with asthma who are referred to secondary care, although it may be less common in the United States and more common in those from the Indian subcontinent. In cystic fibrosis, up to 15% of teenagers are affected. Episodes of bronchial obstruction with mucous plugs leading to coughing fits, “pneumonia,” consolidation, and breathlessness are typical. Many patients report coughing up thick sputum casts, often brown in color. Eosinophilia commonly develops before systemic glucocorticoids are given. The cardinal diagnostic test is detection of *Aspergillus*-specific IgE (or a positive skin-prick test in response to *A. fumigatus* extract) together with an elevated serum level of total IgE (usually >1000 IU/mL). The presence of hyperattenuated mucus in airways is highly specific. Bronchiectasis is characteristic, and a few patients develop chronic cavitary pulmonary aspergillosis.

Severe Asthma with Fungal Sensitization (SAFS) Many adults with severe asthma do not fulfill the criteria for ABPA and yet are allergic to fungi. Although *A. fumigatus* is a common allergen, numerous other fungi (e.g., *Cladosporium* and *Alternaria* species) are implicated by skin-prick testing and/or specific IgE testing. Serum total IgE concentrations are <1000 IU/mL, and bronchial-wall thickening is common. ABPA and SAFS are collectively referred to as *fungal asthma*.

Allergic Fungal Rhinosinusitis Like the lungs, the sinuses manifest allergic responses to *Aspergillus* and other fungi. The affected patients present with chronic (i.e., perennial) sinusitis that is relatively unresponsive to antibiotics. Many of these patients have nasal polyps, and all have congested nasal mucosae and sinuses full of mucoid material. The histologic hallmarks of allergic fungal sinusitis are local eosinophilia and Charcot-Leyden crystals. Removal of abnormal mucus and polyps, with local and occasionally systemic administration of glucocorticoids, usually leads to resolution. Persistent or recurrent signs and symptoms may require more extensive surgery (ethmoidectomy) and sometimes oral antifungal therapy. Recurrence is common, often after another bacterial or viral infection.

Superficial Aspergillosis *Aspergillus* can cause keratitis, onychomycosis, and otitis externa. The first can be difficult to diagnose early enough to save the patient's sight. Natamycin (5%) eye drops are the optimal therapy for fungal keratitis, often with surgery. Otitis externa usually resolves with debridement and local application of antifungal agents.

■ DIAGNOSIS

Several techniques are required to establish the diagnosis of any form of aspergillosis with confidence (Table 223-1).

Acute Invasive Aspergillosis Patients with acute invasive aspergillosis have a relatively heavy load of fungus in the affected organ; thus, antigen detection, PCR, microscopy, culture, and/or histopathology usually confirm the diagnosis. However, the pace of progression leaves only a narrow window for making the diagnosis without losing the patient, and some invasive procedures are not possible because of coagulopathy, respiratory compromise, and other factors. Many cases of invasive aspergillosis are missed clinically and are diagnosed only at autopsy. Histologic examination of affected tissue reveals either infarction, with invasion of blood vessels by many fungal hyphae, or acute necrosis, with limited inflammation and fewer hyphae. *Aspergillus* hyphae are hyaline, narrow, and septate, with branching at 45° in infected tissue. Hyphae can be seen in cytology or microscopy

1716 preparations, which therefore provide a rapid means of presumptive diagnosis.

One *Aspergillus* antigen test relies on detection of galactomannan release from *Aspergillus* organisms during growth, the other a novel protein antigen. Respiratory sample antigen detection is more sensitive than serum and is critical in the intensive care unit patient in whom radiology is nonspecific. Positive serum antigen results usually precede clinical or radiologic features by several days. The sensitivity of antigen detection is reduced by antifungal therapy.

A positive culture supports the diagnosis, given that multiple other (rarer) fungi can mimic *Aspergillus* species histologically, but only 10–30% of patients with invasive aspergillosis have a positive culture. Bacterial agar is less sensitive than fungal media for culture; thus, if physicians do not request fungal culture, the diagnosis may be missed. High-volume fungal cultures enhance yield. A positive culture may represent noninvasive forms of aspergillosis or airway colonization. Both antigen detection and real-time PCR are faster and much more sensitive than culture of respiratory samples and blood.

Definitive confirmation of a diagnosis of invasive aspergillosis requires (1) a positive culture of a sample taken directly from an ordinarily sterile site (e.g., a brain abscess) or (2) positive results of both histologic testing and culture (or molecular confirmation of *Aspergillus* spp.) of a sample taken from an affected organ (e.g., sinuses or skin). Most diagnoses of invasive aspergillosis are inferred from fewer data, including the presence of the *halo sign* on a thoracic CT scan—a localized ground-glass appearance representing hemorrhagic infarction surrounds a nodule or consolidation. Halo signs are present for ~7 days early in the course of infection in neutropenic patients and are a good prognostic feature, reflecting an early diagnosis. Nodules with halo signs are a feature of COVID-19 itself; invasive aspergillosis diagnosis requires supportive evidence. Other characteristic radiologic features of invasive pulmonary aspergillosis include nodules and pleural-based infarction or cavitation, but nonspecific consolidation is common (Fig. 223-2).

Chronic Aspergillosis For chronic aspergillosis, *Aspergillus* antibody testing combined with characteristic imaging is sufficient for the diagnosis. Biopsy of *Aspergillus* nodules reveals hyphae surrounded by cells of chronic inflammation and sometimes granulomas. Antibody titers fall slowly with successful therapy. Cultures are infrequently positive but are important in checking for azole resistance. PCR of sputum is often strongly positive. Some patients with chronic pulmonary aspergillosis often have elevated titers of total and *Aspergillus*-specific IgE.

ABPA, SAFS, and Allergic Aspergillus Sinusitis ABPA and SAFS are diagnosed serologically with elevated specific and total serum IgE levels or with skin-prick tests. Allergic *Aspergillus* sinusitis is usually diagnosed histologically, accompanied by *Aspergillus* IgE antibody.

TREATMENT

Aspergillosis

Antifungal drugs active against *Aspergillus* include voriconazole, itraconazole, posaconazole, isavuconazole, caspofungin, micafungin, and amphotericin B (AmB). Possible interactions with other drugs must be considered before azoles are prescribed. In addition, plasma itraconazole and voriconazole concentrations vary substantially from one patient to another, and many authorities recommend monitoring levels to ensure that drug concentrations are adequate but not excessive. Initial IV administration is preferred for acute invasive aspergillosis and oral administration for all other diseases that require antifungal therapy. Current recommendations are shown in Table 223-3.

Voriconazole, isavuconazole, and posaconazole are the preferred agents for invasive aspergillosis; caspofungin, micafungin, and lipid-associated AmB are second-line agents. AmB is not active against *A. terreus* or *A. nidulans*; multi-azole resistance in *A. fumigatus* is present in 3–20% of isolates but is increasing, especially in Southeast Asia; and *A. niger* is resistant to isavuconazole. An infectious disease consultation is advised for patients with invasive disease, given the



A



B

FIGURE 223-2 Markedly different appearances of invasive aspergillosis on computed tomography scan of the thorax. **A.** Patient with myelodysplasia and moderate neutropenia showing small right-sided nodules with minimal surrounding ground glass and a separate area of ground glass only on the left laterally. **B.** Patient with multiple myeloma undergoing intensive chemotherapy with corticosteroids showing bilateral areas of consolidation and some nonspecific atelectasis with probable ground glass surrounding the right-sided lesion. The anterior component of the left-sided lesion is demarcated by the fissure.

complexity of management. Immune reconstitution can complicate recovery. The duration of therapy for invasive aspergillosis varies from ~3 months to several years, depending on the patient's immune status and response to therapy. Relapse occurs if the response is sub-optimal and immune reconstitution is not complete.

Voriconazole is currently the preferred oral agent for chronic aspergillosis with itraconazole or posaconazole as substitutes when failure, emergence of resistance, or adverse events occur. Because chronic cavitary pulmonary aspergillosis responds slowly, therapy for at least 12 months is necessary, and disease control may require years of treatment, whereas the duration of treatment for other forms of chronic and allergic aspergillosis requires case-by-case evaluation. Glucocorticoids should be avoided in chronic cavitary pulmonary aspergillosis unless covered by adequate antifungal therapy; mortality is increased by 240% with corticosteroids. Acute exacerbations of ABPA respond well to voriconazole, itraconazole, or a short course of glucocorticoids—long-term azole therapy usually helps minimize corticosteroid exposure and maintain remission. Antifungal response in *Aspergillus* bronchitis is gratifying, but relapse after 4 months of therapy is common.

Resistance in *A. fumigatus* to one or more azoles, although uncommon, is increasingly found globally. Resistance may be derived from azole fungicide use for crops. Resistance arising from multiple mechanisms may develop during long-term treatment, and a positive culture during antifungal therapy is an indication for susceptibility testing.

Surgical treatment is important in several forms of aspergillosis, including fungal ball of the sinus and single aspergillomas, in which

TABLE 223-3 Treatment of Aspergillosis^a

INDICATION	PRIMARY TREATMENT	PRECAUTIONS	SECONDARY TREATMENT	COMMENTS
Invasive disease ^b	Voriconazole, isavuconazole, posaconazole	Drug interactions (especially with voriconazole) ^c	AmB, caspofungin, micafungin	As primary therapy, azoles have a 20% higher response rate than AmB. Therapeutic drug monitoring is recommended for voriconazole and isavuconazole in ICU patients.
Prophylaxis	Posaconazole tablet, itraconazole solution SUBA-itraconazole	Vincristine, cyclophosphamide interaction	Micafungin, aerosolized AmB	Some centers monitor plasma levels of itraconazole and posaconazole.
Single aspergilloma	Surgical resection	Multicavity disease: poor outcome of surgery, medical therapy preferable	Itraconazole, voriconazole, intracavity AmB	Single large cavities with an aspergilloma are best resected. Relapse reduced by pre- and perioperative antifungal therapy.
Chronic pulmonary disease ^b	Voriconazole, itraconazole	Poor absorption of itraconazole capsules with proton pump inhibitors or H ₂ blockers	Posaconazole, IV AmB, IV micafungin	Resistance may emerge during treatment, especially if plasma drug levels are low. Therapeutic azole drug monitoring recommended.
ABPA/SAFS ("fungal asthma")	Itraconazole	Some glucocorticoid interactions, including with inhaled formulations	Voriconazole, posaconazole, inhaled AmB	Long-term therapy is helpful in most cases. No evidence indicates whether therapy modifies progression to bronchiectasis/fibrosis.

^aFor information on duration of therapy and drug resistance in certain *Aspergillus* species, see text. ^bAn infectious disease consultation is appropriate for these patients.

^cOnline drug-interaction resource: www.aspergillus.org.uk/content/antifungal-drug-interactions.

Note: After loading doses, the oral dose is usually 200 mg bid for voriconazole and itraconazole, 65 mg bid for SUBA-itraconazole, 300 mg qd for posaconazole tablets, and 200 mg qd for isavuconazole. The IV dose of voriconazole for adults is 6 mg/kg twice at 12-h intervals (loading doses) followed by 4 mg/kg q12h; a larger dose is required for children and teenagers; a lower dose may be safer for persons >70 years of age. Plasma monitoring is helpful in optimizing the dosage. The IV dose of isavuconazole is 200 mg tid for 2 days (loading dose) followed by 200 mg qd. Caspofungin is given as a single loading dose of 70 mg and then at 50 mg/d; some authorities use 70 mg/d for patients weighing >80 kg, and lower doses are required with hepatic dysfunction. Micafungin is given as 50 mg/d for prophylaxis and as at least 150 mg/d for treatment; this drug has not yet been approved by the U.S. Food and Drug Administration (FDA) for this indication. AmB deoxycholate is given at a daily dose of 1 mg/kg if tolerated. Several strategies are available for minimizing renal dysfunction. Lipid-associated AmB is given at 3 mg/kg (AmBisome) or 5 mg/kg (Abelcet). Different regimens are available for aerosolized AmB, but none is FDA approved. Other considerations that may alter dose selection or route include age; concomitant medications; renal, hepatic, or intestinal dysfunction; and drug tolerability.

Abbreviations: ABPA, allergic bronchopulmonary aspergillosis; AmB, amphotericin B; SAFS, severe asthma with fungal sensitization ICU, intensive care unit.

surgery is curative; invasive aspergillosis involving bone, heart valve, sinuses, and proximal areas of the lung (to avoid catastrophic hemoptysis); brain abscess; keratitis; and endophthalmitis. In allergic fungal sinusitis, removal of abnormal mucus and polyps, with local and occasionally systemic glucocorticoid treatment, usually leads to resolution. Persistent or recurrent signs and symptoms may require more extensive surgery (ethmoidectomy) and possibly antifungal therapy. Surgery is problematic in chronic cavitary pulmonary aspergillosis, often resulting in serious complications. Bronchial artery embolization is preferred for problematic hemoptysis.

■ PROPHYLAXIS

In situations in which moderate or high risk is predicted (e.g., after induction therapy for acute myeloid leukemia), the need for antifungal prophylaxis for superficial and systemic candidiasis and for invasive aspergillosis is generally accepted. Fluconazole is commonly used in these situations but has no activity against *Aspergillus* species. Itraconazole solution or super bioavailability (SUBA)-itraconazole capsules provide enough bioavailability for modest efficacy, the latter with fewer adverse events. Posaconazole tablets are more effective in reducing infection rates and the need for empirical antifungal therapy. Some data support the use of IV micafungin in those with azole contraindications. No prophylactic regimen is completely successful.

■ OUTCOME

Invasive aspergillosis is curable if immune reconstitution occurs, whereas allergic and chronic forms are not. The mortality rate for invasive aspergillosis is 30–70% if the infection is treated but is 100% if the diagnosis is missed. Cerebral aspergillosis, *Aspergillus* endocarditis, and bilateral extensive invasive pulmonary aspergillosis have very poor outcomes, as does invasive infection in persons with late-stage AIDS, relapsed uncontrolled leukemia or liver failure.

For those with chronic cavitary pulmonary aspergillosis, ~70% deteriorate clinically without antifungal therapy. The mortality rate for chronic cavitary pulmonary aspergillosis is 15–20% in the first year, ~40% over 5 years, and 50–60% over 10 years if the patient is actively treated with antifungal agents. Antifungal therapy fails in ~30% of recipients and still more often if azole resistance is present.

Both ABPA and SAFS patients respond to glucocorticoids and antifungal therapy; ~60% respond to itraconazole and ~80% to

voriconazole and posaconazole (if tolerated). Inhaled amB is effective in and tolerated by ~15% of patients. Long-term antifungal therapy often allows oral glucocorticoids to be stopped and inhaled glucocorticoid dose. Relapse after discontinuation is common.

■ FURTHER READING

- DENNING DW, PEAVAYI LT: Poorly controlled asthma: Easy wins and future prospects for addressing fungal allergy. *Allergol Int* 72:493, 2023.
- KANG N et al: Clinical characteristics and treatment outcomes of pathologically confirmed *Aspergillus* nodules. *J Clin Med* 9:2185, 2020.
- LI Z, DENNING DW: The impact of corticosteroids on the outcome of fungal disease: A systematic review and meta-analysis. *Curr Fungal Infect Rep* 17:54, 2023.
- SEHGAL IS et al: Efficacy of 12-months oral itraconazole versus 6-months oral itraconazole to prevent relapses of chronic pulmonary aspergillosis: An open-label, randomised controlled trial in India. *Lancet Infect Dis* 22:1052, 2022.
- ULLMAN AJ et al: Diagnosis and management of *Aspergillus* diseases: Executive summary of the 2017 ESCMID-ECMM-ERS guideline. *Clin Microbiol Infect* 24:e1ee38, 2018.

224

Mucormycosis

Brad Spellberg, Ashraf S. Ibrahim

Mucormycosis represents a group of life-threatening infections caused by fungi of the order Mucorales of the subphylum Mucoromycotina. Mucormycosis is highly invasive and relentlessly progressive, resulting in higher rates of morbidity and mortality than many other infections. The mortality rates from mucormycosis have declined in recent years as a result of early initiation of more effective antifungal therapies. However, mortality remains high overall, often driven by progression of the underlying predisposing condition.

TABLE 224-1 Taxonomy of Fungi Causing Mucormycosis (Subphylum Mucoromycotina, Order Mucorales)

FAMILY	GENUS (SPECIES LISTED FOR SOME)
Mucoraceae	<i>Rhizopus oryzae</i> <i>Rhizopus delemar</i> <i>Rhizopus microsporus</i> <i>Rhizomucor</i> <i>Mucor</i> <i>Actinomucor</i>
Lichtheimiaceae	<i>Lichtheimia</i> (formerly <i>Mycocladius</i> , formerly <i>Absidia</i>)
Cunninghamellaceae	<i>Cunninghamella</i>
Thamnidaceae	<i>Cokeromyces</i>
Mortierellaceae	<i>Mortierella</i>
Saksenaceae	<i>Saksenaea</i> <i>Apophysomyces</i>
Syncephalastraceae	<i>Syncephalastrum</i>

■ ETIOLOGY

The fungal order Mucorales consists of seven families that are known to cause mucormycosis (Table 224-1). *Rhizopus oryzae* and *R. delemar* (both in the family Mucoraceae) are by far the most common causes of mucormycosis in the Western Hemisphere. Less frequently isolated species of the Mucoraceae that cause a similar spectrum of infections include *R. microsporus*, *Rhizomucor pusillus*, *Lichtheimia corymbifera* (formerly *Absidia corymbifera*), *Apophysomyces elegans*, and *Mucor* species. Increasing numbers of cases of mucormycosis due to infection by mold in the family Cunninghamellaceae have also been reported, particularly in highly immunocompromised patients. Other Mucorales can be the major cause of disease in certain geographic areas (e.g., *A. elegans* in India and *M. irregularis* in China), or in outbreaks following natural disasters (e.g., *A. trapeziformis* outbreak following Joplin tornado). Only rare case reports have demonstrated the ability of fungi in the remaining families of the Mucorales to cause mucormycosis.

■ PATHOGENESIS

The Mucorales are ubiquitous environmental fungi to which humans are constantly exposed. These fungi cause infection primarily in patients with uncontrolled diabetes, defects in phagocytic function (e.g., neutropenia or exposure to high doses of corticosteroids), and/or elevated levels of free iron, which supports fungal growth in serum and tissues. In the past, iron-overloaded patients with end-stage renal failure who were treated with deferoxamine had a high risk of developing rapidly fatal disseminated mucormycosis; deferoxamine is an iron chelator for the human host, but it serves as a fungal xenosiderophore, directly delivering iron to the Mucorales. Furthermore, patients with diabetic ketoacidosis (DKA) are at high risk of developing rhinocerebral mucormycosis. The acidosis causes dissociation of iron from sequestering proteins, resulting in enhanced fungal survival and virulence. The ketoacid β -hydroxybutyrate also increases expression of host and fungal receptors that result in fungal adherence and penetration into tissues.

Nevertheless, the majority of diabetic patients who present with mucormycosis are not acidotic, and, even absent acidosis, hyperglycemia directly contributes to the risk of mucormycosis by at least four likely mechanisms: (1) hyperglycation of iron-sequestering proteins, disrupting normal iron sequestration; (2) upregulation of a mammalian cell receptor (GRP78) that binds to Mucorales, enabling tissue penetration (due to both a direct effect of hyperglycemia and increasing levels of free iron); (3) induction of poorly characterized defects in phagocytic function; and (4) enhanced expression of Coth, a Mucorales-specific protein that mediates host cell invasion by binding to GRP78 (due to hyperglycemia and the resulting free iron).

■ EPIDEMIOLOGY

Mucormycosis typically occurs in patients with diabetes mellitus, solid-organ or hematopoietic stem cell transplantation (HSCT), prolonged

neutropenia or corticosteroid use, or malignancy. During the pandemic, mucormycosis occurred commonly as a co-infection in patients with COVID-19 severe enough to warrant hospitalization. Such infections were heavily described in low- and middle-income countries (LMICs) and in India in particular. However, the incidence of mucormycosis was higher in India than most other countries even before the pandemic. Hence, it is unclear if the virus itself somehow predisposed to fungal superinfection or if the routine use of high-dose corticosteroid therapy for COVID-19, possibly combined with high rates of diabetes mellitus in affected populations, was the primary driver of such infections.

As mentioned, the majority of diabetic patients are not acidotic on presentation with mucormycosis. Furthermore, patients often have no previously recognized history of diabetes mellitus when they present with mucormycosis. In these instances, presentation with mucormycosis may result in the first clinical recognition of hyperglycemia, which often has been unmasked by recent glucocorticoid use. Thus, a high index of suspicion of mucormycosis must be maintained, even in the absence of a known history of diabetes, if hyperglycemia is present and patients present with sinusitis, particularly with possible orbital extension.

In patients undergoing HSCT, mucormycosis develops at least as commonly during nonneutropenic as during neutropenic periods, probably because of glucocorticoid treatment of graft-versus-host disease. Mucormycosis can occur as isolated cutaneous or subcutaneous infection in immunologically normal individuals after traumatic implantation of soil or vegetation (e.g., due to natural disasters, motor vehicle accidents, or severe injuries in war zones) or in nosocomial settings via direct access through intravenous catheters, subcutaneous injections, or maceration of the skin by a moist dressing.

Patients receiving antifungal prophylaxis with either itraconazole or voriconazole may be at increased risk of mucormycosis. These patients typically present with disseminated mucormycosis, the most lethal form of disease. Breakthrough mucormycosis also has been described in patients receiving posaconazole, isavuconazole, or echinocandin prophylaxis.

■ CLINICAL MANIFESTATIONS

Mucormycosis presentations are often as one of five well-defined clinical syndromes: rhino-orbital-cerebral, pulmonary, cutaneous, gastrointestinal, and disseminated disease. However, infection of any body site can occur. Patients with specific defects in host defense tend to develop specific syndromes. For example, patients with diabetes mellitus and/or DKA typically develop the rhino-orbital-cerebral form and much more rarely develop pulmonary or disseminated disease. In contrast, pulmonary mucormycosis occurs most commonly in leukemic patients who are receiving chemotherapy and in patients undergoing HSCT.

Rhino-Orbital-Cerebral Disease Rhino-orbital-cerebral mucormycosis continues to be the most common form of the disease worldwide. Most cases occur in patients with diabetes, although such cases are also described in the transplantation setting, often along with glucocorticoid-induced diabetes mellitus. The initial symptoms of rhino-orbital-cerebral mucormycosis are nonspecific and include eye or facial pain and facial numbness followed by the onset of conjunctival suffusion and swelling, and blurry vision. In contrast to the acute, bright red, periocular skin manifestations typical of acute bacterial orbital cellulitis, periorbital skin in patients with rhino-orbital-cerebral mucormycosis may take on a more dusky, subacute appearance. Fever may be absent in up to half of cases. White blood cell counts are typically elevated as long as the patient has functioning bone marrow. If untreated, infection usually spreads from the ethmoid sinus to the orbit, resulting in compromise of extraocular muscle function and proptosis, typically with chemosis. From the orbit, the fungus can spread contiguously or hematogenously to the frontal lobe of the brain and/or via venous drainage to the cavernous sinus. Onset of signs and symptoms in the contralateral eye, with resulting bilateral proptosis, chemosis, vision loss, and ophthalmoplegia, is ominous, suggesting the development of cavernous sinus thrombosis.

Upon visual inspection, infected tissue often has a normal appearance during the earliest stages of fungal spread, which can make diagnosis difficult; blind biopsies of normal-appearing sinus tissue are warranted when suspicion for mucormycosis is high. Tissue then progresses through an erythematous phase, with or without edema, before the onset of a violaceous appearance and finally the development of a black necrotic eschar. Infection can sometimes extend from the sinuses into the mouth and produce painful necrotic ulcerations of the hard palate, but this is a late finding that suggests extensive, well-established infection.

One common misperception about mucormycosis is that it is always rapidly progressive. In fact, the rate of progression is extremely variable and is possibly dependent on the immune status of the patient, the infectious inoculum, and the causative *Mucorales* species, some of which are more virulent and/or have faster growth rates than others. Patients may go from initial symptoms to death in days; alternatively, it can take months or even a year or more for lethal progression to occur.

Pulmonary Disease Pulmonary mucormycosis is the second most common manifestation. Symptoms include dyspnea, cough, and chest pain; fever is often but not invariably present. Angioinvasion results in necrosis, cavitation, and/or hemoptysis. Lobar consolidation, isolated masses, nodular disease, cavities, or wedge-shaped infarcts may be seen on chest radiography. Chest computed tomography (CT) is the best method for determining the extent of pulmonary mucormycosis and may demonstrate evidence of infection before it is seen on chest x-ray. In the setting of cancer, where mucormycosis may be difficult to differentiate from aspergillosis, the presence of ≥ 10 pulmonary nodules, pleural effusion, or concomitant sinusitis make mucormycosis more likely. It is important to distinguish mucormycosis from aspergillosis because treatments for these infections may differ. Indeed, voriconazole—the first-line treatment for aspergillosis—exacerbates mucormycosis in mouse and fly models of infection. Isavuconazole was noninferior to voriconazole for the treatment of aspergillosis in a randomized controlled trial and also has activity against *Mucorales*. Hence if there is doubt about whether infection is caused by a septated mold (e.g., *Aspergillus*) or a *Mucorales*, inclusion of isavuconazole in a treatment regimen is a reasonable consideration. Consideration must also be given to the possibility of dual infection with both a septated mold and *Mucorales*; dual infection is not infrequently encountered in highly compromised patients.

Cutaneous Disease Cutaneous mucormycosis may result from external implantation of the fungus or from hematogenous dissemination. External implantation-related infection has been described in the setting of soil exposure from trauma (e.g., in a motor vehicle accident, a natural disaster, or combat-related injuries), penetrating injury with plant material (e.g., a thorn), injections of medications (e.g., insulin), catheter insertion, contamination of surgical dressings, and use of tape to secure endotracheal tubes. Cutaneous disease can be highly invasive, penetrating into muscle, fascia, and even bone. Necrotizing fasciitis caused by mucormycosis carries a mortality rate approaching 80%. Necrotic cutaneous lesions in the setting of hematogenous dissemination also are associated with an extremely high mortality rate. However, with prompt, aggressive surgical debridement, isolated, nonnecrotizing cutaneous mucormycosis has a favorable prognosis and a low mortality rate.

Gastrointestinal Disease In the past, gastrointestinal mucormycosis occurred primarily in premature neonates in association with disseminated disease and necrotizing enterocolitis. However, there has been a marked increase in case reports describing adults with neutropenia, glucocorticoid use, or other immunocompromising conditions. In addition, gastrointestinal disease has been reported as a nosocomial process following administration of medications mixed with

contaminated wooden applicator sticks. Nonspecific abdominal pain and distention associated with nausea and vomiting are the most common symptoms. Gastrointestinal bleeding is common, and fungating masses may be seen in the stomach at endoscopy. The disease may progress to visceral perforation, with extremely high mortality rates.

Disseminated and Miscellaneous Forms of Disease Hematogenously disseminated mucormycosis may originate from any primary site of infection. The most common site of dissemination is the brain, but metastatic lesions may also be found in any other organ. Mortality rates for widely disseminated mucormycosis exceed 90%; however, these high rates are likely to be due in part to the underlying predisposing condition leading to the infection and the inability to surgically remove the infected foci.

Mucormycosis may affect any body site, including bones, mediastinum, trachea, kidneys, peritoneum (in association with dialysis), scalp (causing a kerion), and even isolated infection of teeth.

■ DIAGNOSIS

A high index of suspicion is required for diagnosis of mucormycosis. Unfortunately, autopsy series have shown that up to half of cases are diagnosed only postmortem. Because the *Mucorales* are environmental isolates, definitive diagnosis requires a positive culture from a sterile site (e.g., a needle aspirate, a tissue biopsy specimen, or pleural fluid) or histopathologic evidence of invasive mucormycosis. A probable diagnosis of mucormycosis can be established by culture from a non-sterile site (e.g., sputum or bronchoalveolar lavage) or the detection of *Mucorales* on the surface of histopathology samples (without visualization of evidence of invasion) when a patient has appropriate risk factors as well as clinical and radiographic evidence of disease. In such cases, given the urgency of administering therapy early, the patient should be treated while confirmation of the diagnosis is awaited.

Biopsy with histopathologic examination remains the most sensitive and specific modality for definitive diagnosis (Fig. 224-1). Biopsy reveals characteristic wide (≥ 6 - to 30- μ m), thick-walled, ribbon-like, aseptate hyphal elements that branch at right angles. Other fungi, including *Aspergillus*, *Fusarium*, and *Scedosporium* species, have septa, are thinner, and branch at acute angles. However, artificial septa may result from folding of tissue during processing (which may also alter the appearance of the angle of branching), which can make *Mucorales* appear to have septa. Thus, the width and the ribbon-like form of the fungus are likely the most reliable features distinguishing mucormycosis from other pathogenic molds. The *Mucorales* are visualized most effectively with periodic acid–Schiff or hematoxylin and eosin; in contrast to many other fungi, methenamine silver may not result in optimal staining. While histopathology can identify the *Mucorales*, species can be identified only by culture. Several studies showed that polymerase chain reaction (PCR) of *Mucorales*-specific targets is useful in diagnosing mucormycosis. However, the U.S. Food and Drug Administration (FDA) has not approved any of these PCR-based assays for this purpose.

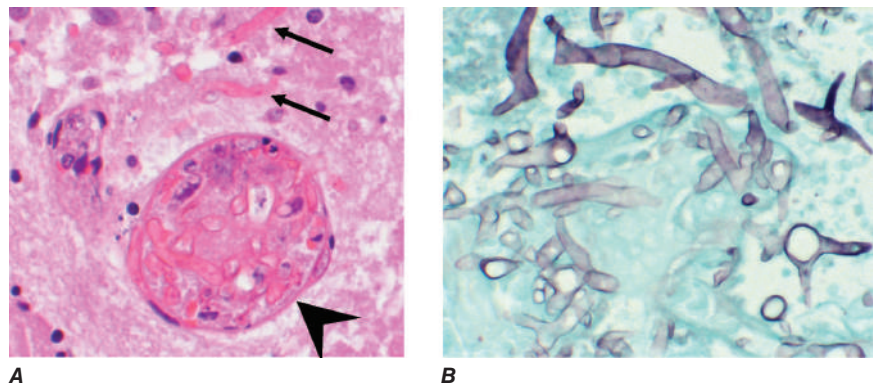


FIGURE 224-1 Histopathology sections of *Rhizopus delemar* in infected brain. **A.** Broad, ribbon-like, nonseptate hyphae in the parenchyma (arrows) and a thrombosed blood vessel with extensive intravascular hyphae (arrowhead) (hematoxylin and eosin). **B.** Extensive, broad, ribbon-like hyphae invading the parenchyma (Gomori methenamine silver).

Unfortunately, cultures are positive in fewer than half of cases of mucormycosis. Nevertheless, the Mucorales are not fastidious organisms and tend to grow quickly (i.e., within 48–96 h) on culture media. The likely explanation for the low sensitivity of culture is that the Mucorales form long filamentous structures that are killed by tissue homogenization—the standard method for preparing tissue cultures in the clinical microbiology laboratory. Therefore, the laboratory should be advised when a diagnosis of mucormycosis is suspected, and the tissue should be cut into sections and placed in the center of culture dishes rather than homogenized. Because there is also substantial variability among isolates in optimal growth temperature, growth at both room temperature and 37°C is advisable.

Imaging techniques often yield subtle findings that underestimate the extent of disease. For example, the most common finding on CT or magnetic resonance imaging (MRI) of the head or sinuses of a patient with rhino-orbital mucormycosis is sinusitis that is indistinguishable from bacterial sinusitis. While sinusitis is almost always seen on CT scans in patients with the rhino-orbital-cerebral disease, erosion through the sinus bones and into the orbit is rarely seen on CT even when it is clinically present. MRI is more sensitive (~80%) for detecting orbital and central nervous system (CNS) disease than is CT. High-risk patients should always undergo endoscopy and/or surgical exploration, with biopsy of the areas of suspected infection. If mucormycosis is suspected, initial empirical therapy with a polyene antifungal agent should be initiated while the diagnosis is being confirmed.

■ DIFFERENTIAL DIAGNOSIS

Other mold infections, including aspergillosis, scedosporiosis, fusariosis, and infections caused by the dematiaceous fungi (brown-pigmented soil organisms), can cause clinical syndromes identical to mucormycosis. Histopathologic examination usually allows distinction of the Mucorales from these other organisms, and a positive culture permits definitive species identification. As stated above, it is important to distinguish the Mucorales from these other fungi, as the preferred antifungal treatments differ (i.e., polyenes for the Mucorales vs expanded-spectrum triazoles for most septate molds). The entomophthoromycoses caused by *Basidiobolus* and *Conidiobolus* also can cause identical clinical syndromes. These fungi cannot be readily distinguished from the Mucorales by histopathology but can be reliably distinguished by culture. Fortunately, entomophthoromycoses are uncommon in developed countries and can be treated with polyenes; it is not urgent to distinguish them from mucormycosis.

In a patient with sinusitis and proptosis, orbital cellulitis and cavernous sinus thrombosis caused by bacterial pathogens (most commonly *Staphylococcus aureus*, but also streptococcal and gram-negative species) must be excluded. *Klebsiella rhinoscleromatis* is a rare cause of an indolent facial rhinoscleroma syndrome that may appear similar to mucormycosis. Finally, the Tolosa-Hunt syndrome is characterized by painful ophthalmoplegia, ptosis, headache, and cavernous sinus inflammation; biopsies and clinical follow-up may be needed to distinguish the Tolosa-Hunt syndrome from mucormycosis; there is a lack of progression of the former entity.

TREATMENT

Mucormycosis

GENERAL PRINCIPLES

Optimizing the chances for successful treatment of mucormycosis requires four steps: (1) early initiation of therapy; (2) surgical debridement, when possible; (3) rapid reversal of underlying predisposing risk factors, if possible; and (4) proceeding to treat underlying malignancy, if present, without waiting to complete antifungal therapy first.

Early initiation of antifungal therapy requires maintaining a high index of suspicion for at-risk patients. Multiple studies have found that earlier initiation of polyene-based therapy improves survival of patients with mucormycosis. Because the disease can present subtly at first and because confirmation of the diagnosis can take days,

therapy often must be started empirically before the diagnosis is established. When there is a reasonable suspicion of mucormycosis, clinicians should not hesitate to initiate therapy with a lipid polyene as soon as possible, since the toxicity of lipid polyenes (unlike that of amphotericin B [AmB] deoxycholate) is rarely substantial after one or two doses.

Blood vessel thrombosis and resulting tissue necrosis during mucormycosis can result in poor penetration of antifungal agents to the site of infection. Therefore, debridement of all necrotic tissues can help eradicate the disease. Surgery has been found (by logistic regression and in multiple case series) to be an independent variable for favorable outcome in patients with mucormycosis. However, these data are confounded by the fact that sicker patients are often unable to tolerate surgical procedures. Thus, a moderated approach in which tissue is debrided when and to the extent it is safe to do so is advisable. Limited data from a retrospective study support the use of intraoperative frozen sections to delineate the margins of infected tissues, with sparing of tissues lacking evidence of infection.

Rapidly reversing hyperglycemia, acidosis, or iron overload, and lowering corticosteroid doses are important to achieving cure. Indeed, a recent study confirmed that resolution of acidosis in mice with DKA via the administration of sodium bicarbonate (used in the mice in lieu of insulin) improved survival. Administration of glucocorticoids predisposes animals to death from mucormycosis in experimental models. Similarly, iron administration to patients with active mucormycosis should be avoided, as iron exacerbates infection in experimental models. Blood transfusion typically results in some liberation of free iron due to hemolysis, so a conservative approach to red blood cell transfusions is advisable.

One of the most common errors made in management of mucormycosis is the belief that mucormycosis must be eradicated before an underlying malignancy can be treated. This belief can result in halting or delaying treatment for the underlying disease (e.g., chemotherapy or transplantation) until the mucormycosis is cured. Three fallacies belie this concern. First, mucormycosis will not be definitively eradicated until near-normal immunity is restored; the antifungals provide a holding action and are unlikely to be curative until the underlying disease is treated. Second, modern antifungals can halt progression of mucormycosis temporarily, enabling aggressive chemotherapy or transplantation to be administered to cure the underlying disease. Finally, the primary driver of death in such patients is typically progression of the underlying disease due to failure to treat it appropriately.

Initially, some consideration can be given to moderating the level of aggressiveness of the chemotherapy and resulting duration and depth of neutropenia. The aggressiveness of immune suppression and antifungal therapy can then be adjusted during the course of treatment in response to changes in clinical status. Nevertheless, chemotherapy should be given sufficiently aggressively to attempt cure of the underlying disease. These patients are extremely complex, and multidisciplinary, team-based care is advisable.

ANTIFUNGAL THERAPY

Primary therapy for mucormycosis should be based on a polyene antifungal agent (Table 224-2), except perhaps in mild localized infection (e.g., isolated suprafacial cutaneous infection) that has been mostly excised surgically in an immunocompetent patient. Lipid formulations of AmB are significantly less nephrotoxic than AmB deoxycholate, can be administered at higher doses, and are probably more effective for this purpose. Liposomal amphotericin B (LAmB) may be preferred to amphotericin B lipid complex (ABLC) for management of brain infection based on retrospective survival data and superior brain penetration; there is no clear efficacy advantage of either agent for infections outside the brain, although LAmB may be less nephrotoxic than ABLC (but is also considerably more expensive in many settings).

Starting dosages of 1 mg/kg per day for AmB deoxycholate and 5 mg/kg per day for LAmB and ABLC are commonly given to adults

TABLE 224-2 Antifungal Options for the Treatment of Mucormycosis^a

DRUG	RECOMMENDED DOSAGE	ADVANTAGES AND SUPPORTING STUDIES	DISADVANTAGES
First-Line Antifungal Therapy			
AmB deoxycholate	1.0–1.5 mg/kg once per day	>5 decades of clinical experience - Inexpensive - FDA approved for treatment of mucormycosis	- Highly toxic - Poor CNS penetration
LAmB	5–10 mg/kg once per day	- Less nephrotoxic than AmB deoxycholate - Better CNS penetration than AmB deoxycholate or ABLC - Better outcomes than with AmB deoxycholate in murine models and a retrospective clinical review	Expensive
ABLC	5 mg/kg once per day	- Less nephrotoxic than AmB deoxycholate - Murine and retrospective clinical data suggest benefit of combination therapy with echinocandins	- Expensive - Possibly less efficacious than LAmB for CNS infection
Second-Line/Salvage Option			
Isavuconazole	200 mg of isavuconazole (372 mg of isavuconazonium sulfate), load q8h × 6 followed by once-daily dosing	- Efficacy similar to that of LAmB in mouse models - FDA approved for treatment of mucormycosis - May be a rational empirical option when septate mold vs mucormycosis is not yet established	- Much less clinical experience - Clinical study supporting approval was small and historically controlled
Posaconazole	200 mg four times per day	- In vitro activity against the Mucorales, with lower MICs than isavuconazole - Retrospective data for salvage therapy in mucormycosis	- Substantially lower blood levels than isavuconazole - No data on initial therapy for mucormycosis, and no evidence for posaconazole in combination with other therapy - Experience limited, potential use for salvage therapy
Combination Therapy^b			
Echinocandin plus lipid polyene	Standard echinocandin doses	- Favorable toxicity profile - Synergistic in murine disseminated mucormycosis - Retrospective clinical data suggest superior outcomes for rhino-orbital-cerebral mucormycosis.	Limited clinical data on combination therapy
Lipid polyene plus azole (posaconazole or isavuconazole)	Standard doses	Favorable toxicity profile	Limited efficacy data, with no available evidence of superiority vs monotherapy
Triple therapy (lipid polyene plus echinocandin plus azole)	Standard doses	Maximal aggressiveness	Limited efficacy data, with no available evidence for superiority vs monotherapy or dual therapy

^aPrimary therapy should generally include a polyene. Non-polyene-based regimens may be appropriate for patients who refuse polyene therapy or for relatively immunocompetent patients with mild disease (e.g., isolated suprafacial cutaneous infection) that can be surgically eradicated. ^bProspective randomized trials are necessary to confirm the suggested benefit (from animal and small retrospective human studies) of combination therapy for mucormycosis. Dose escalation of any echinocandin is not recommended because of a paradoxical loss of benefit of combination therapy at echinocandin doses of ≥ 3 mg/kg per day.

Abbreviations: ABLC, AmB lipid complex; AmB, amphotericin B; CNS, central nervous system; FDA, U.S. Food and Drug Administration; LAmB, liposomal AmB; MIC, minimal inhibitory concentration.

Source: Reproduced with permission from B Spellberg et al: Recent advances in the management of mucormycosis: From bench to bedside. Clin Infect Dis 48:1743, 2008.

and children to treat mucormycosis. Dose escalation of LAmB to 7.5 or 10 mg/kg per day for CNS mucormycosis may be considered in light of the limited penetration of polyenes into the brain. A recent observational study failed to find superior outcomes of LAmB dosed at 10 mg/kg per day. However, very few patients with brain involvement were included, and the analysis was not stratified by site of infection. Therefore, 10 mg/kg per day may be reasonable to consider for treating aggressive brain infection but is unlikely of benefit, and will increase toxicity, for other patients. Because of autoinduction of metabolism, which results in paradoxically lower drug levels, there is no advantage to escalating the LAmB dose above 10 mg/kg per day, and doses of 5 mg/kg per day are probably adequate for nonbrain infections. ABLC dose escalation above 5 mg/kg per day is not advisable given the lack of relevant data and the drug's potential toxicity.

In multiple studies, various combinations of lipid polyenes (both ABLC and LAmB) plus echinocandins (e.g., caspofungin, micafungin, and anidulafungin) improved survival rates among mice with disseminated mucormycosis (including CNS disease). Furthermore, combination lipid polyene–echinocandin therapy was associated with significantly better outcomes than polyene monotherapy in a

retrospective clinical study involving patients with rhino-orbital-cerebral mucormycosis (including CNS disease). The effect of echinocandins appears to be to downmodulate the virulence of the fungus and reduce tissue necrosis and destruction from fungal invasion. On the basis of such data, some experts prefer combination lipid polyene–echinocandin therapy as a first-line option. However, at least one retrospective study did not find an advantage of any combination regimens (including polyene-azole, polyene-echinocandin, or others) in patients who primarily had malignancy as the underlying disease. Ultimately, definitive randomized controlled trials are needed to establish whether the combination is superior in efficacy to monotherapy for mucormycosis. When used, echinocandins should be administered at standard, FDA-approved doses, since dose escalation has resulted in paradoxical loss of efficacy in preclinical models.

In contrast to deferoxamine, the iron chelator deferasirox is fungicidal against clinical isolates of the Mucorales. In mice with DKA and disseminated mucormycosis, combination deferasirox–LAmB therapy resulted in synergistic improvement of survival rates and reduced the fungal burden in the brain. Unfortunately, a small, randomized, double-blind, phase 2 safety clinical trial of adjunctive

therapy with deferasirox (plus LAMB) documented excess mortality among patients treated with deferasirox. Of note, the study population included primarily patients with active malignancy, and few patients in the study had diabetes mellitus as their only risk factor. Deferasirox is therefore contraindicated as therapy in patients with active malignancy, but its role in patients who have diabetes mellitus without malignancy (the setting in which its preclinical efficacy was optimal) remains uncertain.

Posaconazole and isavuconazole are the only FDA-approved azoles with reliable in vitro activity against the Mucorales. However, there are limited data regarding the efficacy of posaconazole monotherapy for mucormycosis, and in contrast to polyene-echinocandin therapy, no data support the use of combination posaconazole-polyene regimens. Although the minimal inhibitory concentrations (MICs) of isavuconazole against the Mucorales are four- to eightfold higher than those of posaconazole, blood levels may be higher with standard isavuconazole dosing than with posaconazole. Isavuconazole is FDA-approved for the treatment of mucormycosis based on a small, historically controlled study. Given this limited dataset, many experts continue to think that lipid polyenes are first-line options and that isavuconazole, like posaconazole, is best reserved for oral transitional therapy in patients whose condition has substantially improved on polyene-based therapy, or for salvage therapy in patients who are intolerant of polyene-based regimens or whose infection is refractory to these regimens.

No studies in mice or humans have ever shown benefit of adding posaconazole to another antifungal agent for combination therapy of mucormycosis. In contrast, in mice, a combination of isavuconazole plus polyene regimens did prolong survival time and lowered tissue fungal burden compared to monotherapy. However, this combination has not been studied compared to monotherapy in people. Some experts use triple therapy with a polyene, echinocandin, and either posaconazole or isavuconazole for patients who have extensive disease or whose disease has progressed on prior therapy. Empirical, dual lipid polyene-azole therapy is a rational choice in a patient with likely invasive mold infections when septate molds and mucormycosis are both in the differential diagnosis and the etiologic agent has not yet been confirmed. Alternatively, initial therapy with isavuconazole monotherapy may be reasonable for a brief period of time in a stable patient if mucormycosis is felt to be possible but less likely than a septated mold infection.

The roles of recombinant cytokines and neutrophil transfusions in the primary treatment of mucormycosis are not clear, although it is intuitive that earlier recovery of neutrophil counts should improve survival rates. Limited data from uncontrolled case series have described the use of hyperbaric oxygen in centers with the appropriate technical expertise and facilities; its efficacy remains undefined. As mentioned previously, one study in mice with DKA found that administration of sodium bicarbonate improved survival from mucormycosis; however, because insulin was not administered to the mice, it is unclear whether the therapeutic effect is clinically relevant.

In general, antifungal therapy for mucormycosis should be continued until resolution of clinical signs and symptoms of infection and resolution of underlying immunosuppression. However, after a week or two of daily therapy in a patient who is clinically improving, it is reasonable to consider switching to thrice-weekly lipid polyene doses—with ultimate weaning down to twice-weekly doses—for maintenance therapy. For patients with mucormycosis who are receiving immunosuppressive medications, secondary antifungal prophylaxis is typically continued for as long as the immunosuppressive regimen is administered. Transitioning to azoles for chronic suppression is a reasonable alternative to continuing polyene therapy in this setting, with re-initiation of polyenes during periods of deep neutropenia.

One common source of error in the long-term management of mucormycosis is follow-up radiology. Analysis of data from the DEFEAT Mucor study indicated that early radiographic progression (within the first 2 weeks) did not predict long-term mortality.

Changing the therapeutic plan based on early radiographic changes can result in therapeutic errors. For example, it is common for CNS Mucorales to cavitate in the brain parenchyma over time. This does not necessarily reflect therapeutic failure, but rather may reflect increased immune reactivity to the fungus, particularly in patients recovering from neutropenia or with removal of immune suppression. Thus, it may not be advisable to obtain serial radiographic studies in the short-term, and if such studies are obtained, caution should be used in reacting to their results. Greater emphasis should be placed on clinical response, particularly within the first 2–4 weeks after initiation of therapy.

■ PROGNOSIS

Over the past two decades, the prognosis of mucormycosis has substantially improved with aggressive antifungal therapy. Even CNS infection is often successfully treated. As mentioned, the key driver of outcome may be control of the patient's predisposing condition.

■ FURTHER READING

- ALQARIHI A et al: Mucormycosis in 2023: An update on pathogenesis and management. *Front Cell Infect Microbiol* 13:1254919, 2023.
- BALDIN C, IBRAHIM AS: Molecular mechanisms of mucormycosis: The bitter and the sweet. *PLoS Pathog* 13:e1006408, 2017.
- CORNELY O et al: Global guideline for the diagnosis and management of mucormycosis: An initiative of the ECMM in cooperation with ESCMID/EFISG. *Lancet Infect Dis* 19:e405, 2019.
- DANION F et al: What is new in pulmonary mucormycosis? *J Fungi (Basel)* 9:307, 2023.
- PETTRIKOS G et al: Epidemiology of mucormycosis in Europe. *Clin Microbiol Infect* 20:67, 2014.

225 Superficial Fungal Infections

Joseph Pechacek, Carol A. Kauffman,
Michail S. Lionakis

Fungal infections of the skin and skin structures are caused by molds and yeasts that do not invade deeper tissues but rather cause disease merely by inhabiting the superficial layers of skin, hair follicles, and nails. These agents cause the most common human fungal infections but only rarely cause serious infections.

■ MALASSEZIA INFECTIONS

Etiologic Agents, Epidemiology, and Pathogenesis *Malassezia* species, primarily *M. furfur* and *M. pachydermatis*, are lipophilic yeasts that generally cause only minor skin infections but, on occasion, can cause invasive infection. *Malassezia* species are part of the indigenous human microbiota found in the stratum corneum of the back, chest, scalp, and face—areas rich in sebaceous glands. The organisms do not invade below the stratum corneum and generally elicit little inflammation. Cutaneous interleukin (IL)-17 signaling controls *Malassezia* species in the skin, but excessive, *Malassezia*-induced IL-17 responses have been associated with exacerbating atopic inflammation.

Clinical Manifestations *Malassezia* species cause tinea versicolor (also called *pityriasis versicolor*), folliculitis, and seborrheic dermatitis. Tinea versicolor presents as flat round scaly patches of hypo- or hyperpigmented skin on the neck, chest, or upper arms. The lesions are usually asymptomatic but can be pruritic. They can be mistaken

for vitiligo, but the latter is not scaly. Folliculitis occurs on the back and chest and mimics bacterial folliculitis. Seborrheic dermatitis manifests as erythematous pruritic scaly lesions in the eyebrows, moustache, nasolabial folds, and scalp (dandruff). Seborrheic dermatitis can be severe in patients with AIDS. Fungemia and disseminated infection occur rarely with *Malassezia* species, and they almost always occur in premature neonates receiving parenteral lipid nutrition preparations.

Diagnosis *Malassezia* infections are diagnosed clinically in most cases. If scrapings are collected on a microscope slide on which a drop of potassium hydroxide has been placed, a mixture of budding, bottle-shaped yeasts and short septate hyphae is seen. To culture *Malassezia* from patients with suspected disseminated infection, sterile olive oil must be added to the medium.

Treatment and Prognosis Topical creams and lotions, including selenium sulfide shampoo, ketoconazole shampoo or cream, and terbinafine cream, are effective in treating *Malassezia* infections and are usually given for 2 weeks. Mild topical steroid creams are sometimes used to treat seborrheic dermatitis. For extensive disease, oral itraconazole or fluconazole (200 mg/day) can be used for 5–7 days. The rare cases of fungemia caused by *Malassezia* species are treated with amphotericin B (AmB) or an azole, such as voriconazole, prompt removal of central catheters, and discontinuation of parenteral lipid infusions. *Malassezia* skin infections are benign and self-limited, although recurrences are common. The outcome of systemic infection depends on the host's underlying conditions, but most infected neonates do well.

■ MUCOCUTANEOUS CANDIDA INFECTIONS

Candidiasis is discussed in [Chap. 222](#), and this section briefly describes mucocutaneous manifestations of candidiasis.

Etiologic Agents, Epidemiology, and Pathogenesis The most common species responsible for mucocutaneous candidiasis—e.g., oral thrush, esophageal candidiasis, vulvovaginal candidiasis (VVC)—is *Candida albicans*. *Candida* species are commensal organisms residing primarily in the oral mucosa, gastrointestinal tract, and genitourinary tract with the skin being colonized in certain settings, such as after antibiotic use in the intensive care unit or by certain species such as *C. parapsilosis* or *C. auris*. *C. auris* is an emerging multidrug-resistant species with a propensity for long-term colonization of the human skin, health care environments, and medical devices and for causing nosocomial outbreaks. *C. auris* grows avidly in human sweat and colonizes deep skin layers including hair follicles. IL-17-producing lymphoid cells at mucocutaneous barrier sites such as T_H17 cells, $\gamma\delta$ T cells, and innate lymphoid cells mediate control against *Candida*. Risk factors for oral thrush include inherited and pharmacologic defects of IL-17 signaling, AIDS, and conditions with excessive mucosal interferon γ responses such as autoimmune polyendocrine syndrome type 1. Antibiotic use, diabetes, and pregnancy are major risk factors for VVC. Skin candidiasis is uncommon and most often seen in inborn errors of immunity that cause chronic mucocutaneous candidiasis, primarily *STAT1* gain-of-function.

Diagnosis Diagnosis of oral thrush is generally clinical with white mucosal plaques that are removed with gentle scraping. Culture reveals the responsible *Candida* species. Biopsy of affected tissue may be needed to differentiate thrush from mimics such as leukoplakia. Biopsy of affected oral or esophageal tissue shows yeasts, pseudohyphae, and hyphae invading into the epithelium; observing only yeast forms superficial to epithelial tissue usually represents solely colonization. VVC is diagnosed with culture and microscopy of vaginal secretions collected via swabs. Skin and nail candidiasis is diagnosed by skin biopsy or nail clippings with culture and microscopy.

Treatment and Prognosis Mild oral thrush may be treated with topical agents such as nystatin suspension or clotrimazole troches, but moderate and severe disease or esophageal involvement should be treated with systemic, primarily azole therapy. VVC may be treated with topical or oral azoles. Newer options for VVC include the

triterpenoid antifungal ibrexafungerp, which inhibits β -glucan, and oteseconazole, a long-acting tetrazole that cannot be used in patients of reproductive potential owing to potential fetal ocular toxicity.

■ DERMATOPHYTE INFECTIONS

Etiologic Agents, Epidemiology, and Pathogenesis The molds that cause skin infections in humans include the genera *Trichophyton*, *Microsporum*, and *Epidermophyton*. These organisms, which are not components of the normal skin microbiota, can live within the keratinized structures of the skin—hence the term *dermatophytes*. Dermatophytes occur worldwide, and infections with these organisms are extremely common with an estimated ~1 billion people affected globally. Some organisms cause disease only in humans and can be transmitted by person-to-person contact and by fomites, such as hairbrushes or wet floors, that have been contaminated by infected individuals. Several species cause infections in cats and dogs and can readily be transmitted from these animals to humans, and others are spread from contact with soil. The characteristic ring shape of cutaneous lesions is the result of the organisms' outward growth in a centrifugal pattern in the stratum corneum. Fungal nail invasion usually occurs through the lateral or superficial nail plates and then spreads throughout the nail; when hair shafts are invaded, the organisms can be found either within the shaft or surrounding it. Symptoms are caused by the inflammatory reaction elicited by fungal antigens and not by tissue invasion. Dermatophyte infections are restricted by IL-17 responses and occur more often in males; progesterone can inhibit dermatophyte growth.

Clinical Manifestations Dermatophyte infection of the skin is often called *ringworm*. This term is confusing because worms are not involved. *Tinea*, the Latin word for *worm*, describes the serpentine nature of the skin lesions. *Tinea* is a less confusing term and can be used with the name of the body part affected—e.g., *tinea capitis* (head), *tinea pedis* (feet), *tinea corporis* (body), *tinea cruris* (crotch), and *tinea unguium* (nails, more often termed *onychomycosis*).

Tinea capitis occurs most commonly in 3- to 7-year-old children. Children with *tinea capitis* usually present with well-demarcated scaly patches in which hair shafts are broken off right above the skin; alopecia can result. *Tinea corporis* manifests as well-demarcated, annular, pruritic, scaly lesions that undergo central clearing. Usually, one or several small lesions are present; however, in some patients, *tinea corporis* can involve much of the trunk. The rash should be differentiated from contact dermatitis, eczema, and psoriasis. *Tinea cruris* is seen almost exclusively in men. The perineal rash is erythematous and pustular, has a discrete scaly border, is without satellite lesions, and is usually pruritic. The rash must be differentiated from intertriginous candidiasis, erythrasma, and psoriasis.

Tinea pedis also is also more common among men. It usually starts in the web spaces of the toes; peeling, maceration, and pruritus are followed by development of a scaly pruritic rash along the lateral and plantar surfaces of the feet. Hyperkeratosis of the soles of the feet often ensues. *Tinea pedis* has been implicated in lower-extremity cellulitis, as streptococci and staphylococci can gain entrance to the tissues through fissures between the toes. Onychomycosis affects toenails more often than fingernails and is most common among persons who have *tinea pedis*. The nail becomes thickened and discolored and may crumble; onycholysis almost always occurs. Onychomycosis is more common in older adults and in persons with vascular disease, diabetes mellitus, and nail trauma. Fungal infection must be differentiated from psoriasis, which can mimic onychomycosis but usually has associated skin lesions.

Diagnosis Many dermatophyte infections are diagnosed by their clinical appearance. If the diagnosis is in doubt, scrapings should be taken from the edge of a lesion with a scalpel blade, transferred to a slide to which a drop of potassium hydroxide is added, and examined under a microscope for the presence of hyphae. Cultures are indicated if an outbreak is suspected or the patient does not respond to therapy.

Treatment and Prognosis Dermatophyte infections usually respond to topical therapy with azoles or terbinafine. Lotions or sprays

TABLE 225-1 Suggested Oral Treatment for Extensive Tinea Infections and Onychomycosis

ANTIFUNGAL AGENT	SUGGESTED DOSAGE	COMMENTS
Extensive Tinea Infection		
Terbinafine	250 mg/day for 1–2 weeks	Adverse reactions minimal with short treatment period
Itraconazole ^a	200 mg/day for 1–2 weeks	Adverse reactions minimal with short treatment period except for drug interactions
Onychomycosis		
Terbinafine	250 mg/day for 3 months	Slightly superior to itraconazole; monitor for hepatotoxicity
Itraconazole ^a	200 mg/day for 3 months or 200 mg twice daily for 1 week each month for 3 months	Drug interactions frequent; monitor for hepatotoxicity; rarely causes hypokalemia, hypertension, edema; use with caution in patients with congestive heart failure Other triazoles such as posaconazole may be considered as alternative therapy

^aItraconazole capsules require food and gastric acid for absorption, whereas itraconazole solution is taken on an empty stomach. The newer SUBA-itraconazole formulation exhibits improved oral bioavailability and reduced interpatient variability.

are easier than creams to apply to large or hairy areas. Particularly for tinea cruris, the affected area should be kept as dry as possible. When patients have extensive skin lesions, oral itraconazole or terbinafine can hasten resolution (Table 225-1). Terbinafine interacts with fewer drugs than itraconazole and is generally the first-line systemic agent.

Onychomycosis generally does not respond to topical therapy, although efinaconazole topical solution applied to the affected nail for as long as a year has been beneficial in several trials. Itraconazole and terbinafine both accumulate in the nail plate and can treat onychomycosis (Table 225-1). The major decision to be made regarding therapy is whether the extent of nail involvement justifies the use of systemic antifungal agents that may have adverse effects, may interact with other drugs, and are costly. Relapses of tinea cruris and tinea pedis are common and should be treated as early as possible with topical creams to avoid development of more extensive disease. Relapses of onychomycosis follow treatment in 25–30% of cases.

■ SPOROTRICHOSIS

Etiologic Agent, Epidemiology, and Pathogenesis *Sporothrix schenckii* complex comprises six closely related organisms; *S. schenckii* and *S. brasiliensis* are the species that cause most human infections. *S. globosa* has grown in importance in Asia. *Sporothrix* species are relatively intolerant to heat and are found worldwide in sphagnum moss, decaying vegetation, and soil. Sporotrichosis most commonly affects persons who participate in outdoor activities such as landscaping, gardening, and tree farming. Infected animals—such as dogs, armadillos, and most commonly cats—can transmit *S. schenckii* to humans. A large ongoing outbreak of sporotrichosis in Brazil caused by *S. brasiliensis* has been traced to cats, which are highly susceptible to this infection. Sporotrichosis is primarily a localized infection of skin and subcutaneous tissues that follows traumatic conidial inoculation. Osteoarticular sporotrichosis is uncommon, occurring most often in middle-aged men who abuse alcohol, and pulmonary sporotrichosis occurs almost exclusively in persons with chronic obstructive pulmonary disease following fungal inhalation. Dissemination occurs almost entirely in markedly immunocompromised patients, especially those with AIDS.

Clinical Manifestations and Differential Diagnosis Days or weeks after inoculation, a papule develops at the site and then usually ulcerates but is not very painful. Similar lesions develop sequentially along the lymphatic channels proximal to the original lesion (Fig. 225-1). Some patients develop a fixed cutaneous lesion that can be verrucous



FIGURE 225-1 Several nodular lesions in a lymphangitic spread pattern that developed on the patient's arm after traumatic inoculation. Cultures from the biopsied lesion (circle) yielded growth of *Sporothrix schenckii*.

or ulcerative and that remains localized without lymphatic extension. The differential diagnosis of lymphocutaneous sporotrichosis includes nocardiosis (especially by *Nocardia brasiliensis*), tularemia, nontuberculous mycobacterial infection (especially by *Mycobacterium marinum*), and leishmaniasis. Osteoarticular sporotrichosis is usually monoarticular and can present as chronic synovitis or septic arthritis. Pulmonary sporotrichosis presents with cavitory or occasionally multifocal noncavitory pneumonia and must be differentiated from tuberculosis and other fungal pneumonias. Numerous ulcerated skin lesions, with or without spread to visceral organs (including the central nervous system [CNS]), are characteristic of disseminated sporotrichosis.

Diagnosis *S. schenckii* usually grows readily as a mold on Sabouraud's agar when material from a cutaneous lesion biopsy is incubated at room temperature. Histopathologic examination of biopsy material shows a mixed granulomatous and pyogenic reaction, and tiny oval or cigar-shaped yeasts sometimes can be seen with special stains.

Treatment and Prognosis Guidelines for the management of the various forms of sporotrichosis have been published by the Infectious Diseases Society of America (Table 225-2). Itraconazole is the drug of choice for cutaneous and lymphocutaneous sporotrichosis. Fluconazole is less effective, voriconazole is ineffective, and posaconazole has been used successfully in a small number of patients. Saturated solution of potassium iodide (SSKI) continues to be used for lymphocutaneous infection because of its low cost; however, SSKI is poorly

TABLE 225-2 Suggested Treatment for Sporotrichosis

DISEASE	FIRST-LINE THERAPY	ALTERNATIVES/COMMENTS
Cutaneous, lymphocutaneous	Itraconazole, 200 mg/day until 2–4 weeks after lesions resolve	SSKI, increasing doses ^a Terbinafine, 500 mg twice daily
Pulmonary, osteoarticular	Itraconazole, 200 mg twice daily for 12 months	Lipid AmB ^b for severe pulmonary disease until stable; then itraconazole
Disseminated, central nervous system	Lipid AmB ^b for 4–6 weeks	Itraconazole, 200 mg twice daily after AmB for 12 months Patients with AIDS: itraconazole maintenance, 200 mg/day until CD4+ T cell count is >200/μL for ≥12 months

^aThe starting dosage is 5–10 drops three times daily in water or juice. The dosage is increased weekly by 10 drops per dose, as tolerated, up to 40–50 drops three times daily. ^bThe dose of lipid AmB is 3–5 mg/kg daily; the higher dose should be used when the central nervous system is involved.

Abbreviations: AmB, amphotericin B; SSKI, saturated solution of potassium iodide.

tolerated because of adverse reactions, including metallic taste, salivary gland swelling, rash, and fever. High-dose terbinafine may be effective for lymphocutaneous infection. Treatment for lymphocutaneous sporotrichosis is continued for 2–4 weeks after all lesions have resolved, usually for a total of 3–6 months. The success rate for treatment of lymphocutaneous sporotrichosis is 90–100%.

Pulmonary and osteoarticular forms of sporotrichosis are treated with itraconazole for at least 1 year. Severe pulmonary infection and disseminated sporotrichosis, including that involving the CNS, should be treated initially with amphotericin B (AmB), with a switch to itraconazole after improvement has been noted. Lifelong suppressive therapy with itraconazole is often required for patients with AIDS. These forms of sporotrichosis respond poorly to antifungal therapy.

■ EUMYCETOMA AND CHROMOBLASTOMYCOSIS

Etiologic Agents, Epidemiology, and Pathogenesis Dematiaceous or brown-black fungi, the common soil organisms that cause phaeohyphomycoses, contain melanin, which causes the hyphae and conidia to be darkly pigmented. The term *phaeohyphomycosis* is used to describe any infection with a pigmented mold. This definition encompasses two cutaneous syndromes discussed herein—eumycetoma and chromoblastomycosis—and all other types of invasive infections caused by these organisms, which are discussed in [Chap. 226](#). It is important to note that eumycetomas can be caused by hyaline molds and by brown-black molds and that only ~50% of all mycetomas are due to fungi. Actinomycetes cause the remainder ([Chap. 180](#)).

The most common cause of eumycetoma is *Madurella* species. *Fonsecaea*, *Phialophora*, and *Cladophialophora* species are responsible for most cases of chromoblastomycosis and exhibit a pathogenic muriform phase. These infections are most often seen in resource-limited rural settings and affect mostly adult men. Inoculation is via minor, often unnoticed, trauma with soil or plant material contamination. Although exposure to these pathogens is common, the development of clinical disease is infrequent. Eumycetoma is found predominantly in tropical and subtropical areas with most cases reported from Sudan, Mexico, and India. Neutrophils mediate early inflammation around the hyphae-containing grains, and macrophages form granulomas at the site of infection.

Chromoblastomycosis is also prevalent in tropical and subtropical regions, with most cases reported from Latin America, Madagascar, and China. A broader geographic distribution is seen with certain species such as *Cladophialophora carrionii*, which thrives in semiarid climates and is a chief cause of chromoblastomycosis in Australia. Following traumatic inoculation, transformation of the fungus into thick-walled, multiseptated, muriform cells ensues. These cells can evade phagocytosis, resulting in chronic disease development.

Clinical Manifestations and Differential Diagnosis Eumycetoma—commonly called Madura foot—is a chronic cutaneous and subcutaneous infection that usually occurs on the lower extremities and manifests with swelling, sinus tract development, and appearance of grains that are actually fungal colonies discharged from the sinus tract. As the infection progresses, adjacent fascia and bony structures may become involved. The disease is indolent and disfiguring, progressing slowly over years. Complications include fractures of infected bone and bacterial superinfection.

Chromoblastomycosis is an indolent subcutaneous infection characterized by nodular, verrucous, or plaque-like painless lesions that occur predominantly on the lower extremities and grow slowly over months to years. There is hardly ever extension to adjacent structures, as is seen with eumycetoma. Long-term consequences include bacterial superinfection, chronic lymphedema, and (rarely) the development of squamous cell carcinoma.

Diagnosis A tentative clinical diagnosis of mycetoma can be made when a patient presents with a lesion characterized by swelling, sinus tracts, and grains. Histopathologic examination and culture are necessary to confirm that the etiologic agent is a mold and not an actinomycete. In chromoblastomycosis, the diagnosis rests on the histologic

demonstration of muriform cells or “sclerotic bodies” in tissues; culture establishes the causative pigmented mold. In culture-negative cases, PCR-based detection of the 28S ribosomal subunit or the rRNA internal transcribed spacer (ITS) may assist with identification.

Treatment and Prognosis Treatment of eumycetoma and chromoblastomycosis involves both surgical excision of lesions and antifungal therapy. Surgical removal of lesions is most effective if performed before extensive spread has occurred and is strongly recommended as a mainstay of treatment for both syndromes. In chromoblastomycosis, cryosurgery, heat therapy, and laser therapy have been used with variable success. Optimal antifungal therapy may vary depending on the causative fungal agent. Eumycetoma has been treated with itraconazole, voriconazole, posaconazole, and less commonly terbinafine with variable success rates. Long-term antifungal therapy is required, and cost or tolerability of these agents may limit their use. Itraconazole, terbinafine, and flucytosine have been used to treat chromoblastomycosis, again with variable success and with cost or tolerability limiting long-term treatment. Muriform cells may respond to antifungal therapy differently from hyphal forms, so caution is needed in interpreting in vitro susceptibility results. Species identification may be useful in determining optimal therapy, as different dematiaceous molds can differ in their susceptibilities to azoles, AmB, and flucytosine.

■ FURTHER READING

BONIFAZ A, TIRADO-SANCHEZ A: Cutaneous disseminated and extracutaneous sporotrichosis: Current status of a complex disease. *J Fungi* 3:6, 2017.

HAY R: Dermatophytosis (ringworm) and other superficial mycoses, in *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*, 10th ed. Blaser MJ et al (eds). Philadelphia, Elsevier, 2025.

HOSPENTHAL D: Uncommon fungi and related species, in *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*, 10th ed. Blaser MJ et al (eds). Philadelphia, Elsevier, 2025.

226 Uncommon Disseminated Fungal Infections

Joseph Pechacek, Carol A. Kauffman,
Michail S. Lionakis

■ ENDEMIC MYCOSES (DIMORPHIC FUNGI)

Dimorphic fungi—so named for their ability to convert between mold and yeast forms (or spherules or adiaspores in the case of *Coccidioides* and *Emmonsia crescens*, respectively) in a temperature-dependent manner—are found in discrete environmental niches as molds that produce conidia, which are the infectious form. In infected tissues and at temperatures of >35°C, they are found in the yeast form. The most common invasive endemic mycoses in the United States—histoplasmosis, coccidioidomycosis, and blastomycosis—are discussed in [Chaps. 218–220](#), respectively, and sporotrichosis is discussed in [Chap. 225](#).

■ PARACOCIDIIDOMYCOSIS

Etiologic Agent, Epidemiology, and Pathogenesis The species constituting the *Paracoccidioides brasiliensis* complex and the less frequently reported *Paracoccidioides lutzii* are thermally dimorphic fungi found in humid areas of Central and South America. The highest incidence of *Paracoccidioides* infections is seen in Brazil, Colombia, Venezuela, and Ecuador although infections have been identified as far north as Mexico. Most cases are found in an ecological niche

characterized by high rainfall and humidity and the presence of tropical or subtropical forests. A striking male-to-female ratio ranges from 14:1 to as high as 70:1 in various reports; this observation is largely accounted for by chronic disease, as the acute form of the infection lacks this stark male predominance. Most patients are middle-aged or elderly men from rural areas. Paracoccidioidomycosis develops after the inhalation of aerosolized conidia encountered in the environment into the lungs. In this early stage, neutrophils and macrophages play an important role in controlling the infection and preventing dissemination. Granulomas may be found in the lung and at sites of extrapulmonary dissemination in infected patients. Disease rarely develops at the time of the initial infection but appears years later, presumably after reactivation of a latent infection. Inborn errors of immunity affecting the interleukin (IL)-12/interferon (IFN)- γ signaling axis predispose to severe, disseminated paracoccidioidomycosis.

Clinical Manifestations The majority of patients exposed to *Paracoccidioides* develop asymptomatic pulmonary disease, which is contained and does not become clinically apparent. When clinical disease develops, two major syndromes are associated with paracoccidioidomycosis: the acute or juvenile form and the chronic or adult form. The acute form is uncommon, develops typically in persons <30 years old, and manifests primarily as disseminated infection of the reticuloendothelial system. The interval from initial fungal exposure to developing the acute form of paracoccidioidomycosis varies from as early as 45 days to as late as several months. Immunocompromised individuals also can develop this type of rapidly progressive disease. Peripheral eosinophilia is often seen in this form of the disease. The chronic form of paracoccidioidomycosis accounts for ~90% of cases and predominantly affects older men. The primary

manifestations are progressive pulmonary disease, primarily in the lower lobes, with fibrosis and ulcerative and nodular mucocutaneous lesions that occur primarily in mucous membranes of the upper respiratory tract and that must be differentiated from leishmaniasis ([Chap. 233](#)) and squamous cell carcinoma ([Chap. 82](#)). Metastatic foci of paracoccidioidomycosis may also occur via hematogenous or lymphatic spread, most often involving the central nervous system (CNS), skin, or adrenal glands, which can lead to the development of adrenal insufficiency.

Diagnosis The diagnosis is established by growth of the mold form of *Paracoccidioides* in culture at room temperature. A presumptive diagnosis can be made by detection of the distinctive thick-walled yeast in tissue biopsy specimens, which appears as a central yeast with multiple daughter cells attached circumferentially via narrow-necked buds and is classically described as a mariner's wheel. Serologic testing also can aid in diagnosis, although it should be noted that certain available tests may not detect *P. lutzii*.

Treatment and Prognosis Itraconazole is the treatment of choice for paracoccidioidomycosis ([Table 226-1](#)) owing to its longer-standing experience in this disease, although voriconazole and posaconazole also are active in vitro and have been effective in small patient cohorts. Sulfonamides have been used for years and are the least costly agents; however, the response is slower and the relapse rate higher. Seriously ill patients, such as those with respiratory failure, should be initially treated with amphotericin B (AmB) with subsequent azole therapy following clinical improvement. Patients with paracoccidioidomycosis have an excellent response to therapy, regardless of the causative species, but pulmonary fibrosis can be progressive in those with chronic disease.

TABLE 226-1 Suggested Treatment for Uncommon Disseminated Fungal Infections

DISEASE	FIRST-LINE THERAPY	ALTERNATIVES/COMMENTS
Paracoccidioidomycosis		
Chronic (adult form)	Itraconazole ^a , 100–200 mg/day for 6–12 months	Voriconazole ^a , 200 mg twice daily for 6–12 months Posaconazole ^a , 300 mg/day for 6–12 months TMP-SMX, 160/800 mg twice daily for 12–36 months
Acute (juvenile form)	Lipid AmB ^b until improvement	Itraconazole ^a , 200 mg twice daily after AmB for 12 months Voriconazole or posaconazole at doses noted above may be used
Talaromycosis		
Mild or moderate	Itraconazole ^a , 200 mg twice daily for 12 weeks	Voriconazole ^a , 200 mg twice daily Posaconazole ^a 300 mg/day
Severe	Lipid AmB ^b until improvement	Itraconazole ^a , 200 mg twice daily after AmB for 12 weeks Voriconazole or posaconazole may be considered as an alternative
Maintenance therapy (AIDS)	Itraconazole, 200 mg/day until CD4+ T cell count is >100/ μ L for $\geq$ 6 months	
Emergomycosis (Emmonsiosis)		
Mild or moderate	Itraconazole ^a 200 mg twice daily for 12 weeks	Voriconazole ^a , 200 mg twice daily Posaconazole ^a 300 mg/day
Severe	Lipid AmB ^b until improvement	Itraconazole ^a , 200 mg twice daily after AmB for 12 weeks Voriconazole or posaconazole may be considered as an alternative
Phaeohyphomycosis	Voriconazole ^a , 200 mg twice daily Itraconazole ^a , 200 mg twice daily Posaconazole ^a , 300 mg/day	Lipid AmB may be effective against some mold species
Fusariosis	Voriconazole ^a , 200–300 mg twice daily Lipid AmB, 5 mg/kg/day Posaconazole ^a , 300 mg/day	Lipid AmB plus voriconazole or posaconazole is used by some physicians for initial therapy. Investigational olorofim or fosmanogepix may be considered if available.
Scedosporiosis and lomentosporiosis	Voriconazole ^a , 200–300 mg twice daily Posaconazole ^a , 300 mg/day	Not susceptible to AmB. <i>Lomentospora prolificans</i> is resistant to almost all commercially available antifungal drugs. Investigational olorofim or fosmanogepix may be considered if available.
Trichosporonosis	Voriconazole ^a , 200–300 mg twice daily	Posaconazole ^a , 300 mg/day

^aDosing should be adjusted for appropriate drug trough levels. ^bThe dose of lipid AmB is 3–5 mg/kg daily; the higher dose should be used when the central nervous system is involved.

Abbreviations: AmB, amphotericin B; TMP-SMX, trimethoprim-sulfamethoxazole.

■ TALAROMYCOSIS

Etiologic Agent, Epidemiology, and Pathogenesis *Talaromyces marneffe* (formerly *Penicillium marneffe*) is a thermally dimorphic fungus that is endemic in the soil in certain areas of Vietnam, Thailand, and other southeastern Asian countries. The epidemiology of talaromycosis (formerly *penicilliosis*) is linked to bamboo rats that are infected with the fungus but rarely manifest disease. The disease occurs most often among persons living in rural areas in which the rats are found, but there is no evidence for transmission of the infection directly from rats to humans. Accidental laboratory inoculation has also been reported. Infection is rare in immunocompetent hosts, and most cases are reported in persons who have advanced AIDS. Other immune deficiencies associated with a higher risk of developing talaromycosis include corticosteroid use, solid organ transplantation, idiopathic CD4 lymphocytopenia, inborn errors of immunity affecting the IL-12/IFN- γ signaling axis, and neutralizing autoantibodies against IFN- γ . Infection is presumed to result from the inhalation of conidia from the environment though a definitive environmental reservoir that has yet to be identified. The organism converts to the yeast phase in the lungs and then spreads hematogenously throughout the reticuloendothelial system. IFN- γ -primed monocytes/macrophages and neutrophils both play a role in control of *Talaromyces* via fungal phagocytosis and killing.

Clinical Manifestations The clinical manifestations of talaromycosis mimic those of disseminated histoplasmosis and include fever, fatigue, weight loss, dyspnea, lymphadenopathy, hepatosplenomegaly, and mucocutaneous lesions, which appear as papules that often umbilicate and resemble molluscum contagiosum (Chap. 201). Patients with mild talaromycosis may present with isolated skin lesions.

Diagnosis Talaromycosis is diagnosed by culture of *T. marneffe* from blood or from biopsy samples of skin, bone marrow, or lymph node. The organism usually grows within 1 week as a mold producing a distinctive red pigment that diffuses into the agar. Histopathologic examination of tissues, blood smears, or material from skin lesions shows oval or elliptical yeast-like organisms that reproduce via bipolar fission (i.e., appearing as two elliptical yeasts separated by a central septation) and can quickly establish a presumptive diagnosis. Intramacrophage yeast-like organisms can also be seen on tissue or lymph node sections or bone marrow aspirates. *T. marneffe*-specific antigen and polymerase chain reaction (PCR)-based assays are under investigation.

Treatment and Prognosis Disseminated talaromycosis is usually fatal if not treated. With treatment, the mortality rate is ~10%. For mild or moderate infection, itraconazole is the drug of choice. Voriconazole also can be used, and posaconazole has good in vitro activity and has been successfully used in few case reports. Severe infection should be treated with AmB until clinical improvement occurs; then therapy can be changed to an effective triazole (Table 226-1). For patients with AIDS, suppressive therapy with itraconazole is recommended until the CD4⁺ T cell count has been >100 cells/ μ L for at least 6 months. Primary prophylaxis may be considered in certain patients with AIDS in endemic areas who are unable to receive effective antiretroviral therapy.

■ EMERGOMYCOSIS (DISSEMINATED EMMONSIOSIS)

Etiologic Agents, Epidemiology, and Pathogenesis As the name implies, *Emergomyces* (several members of which are formerly of the genus *Emmonsia*) is a genus of emerging thermally dimorphic fungi consisting of seven species, six of which have been shown to cause human disease. Among them, *E. pasteurianus* is the most geographically widespread species, and *E. africanus* has been the most studied given its relatively high incidence of infection in South Africa. Infection by *Emergomyces* species have now been reported in Europe, Asia (primarily in China and India), Africa (Uganda and South Africa), and North America (United States and Canada). Although *Emergomyces* has yet to be cultured from the environment and no animal

reservoir has been identified thus far, infection is thought to begin, as with other dimorphic fungi, through inhalation of conidia released by the mold form of the pathogen in the environment. The organism then undergoes temperature-dependent transformation into the invasive yeast form within the lungs and disseminates hematogenously to cause systemic disease in immunocompromised patients. Clinically apparent disease appears largely restricted to patients with impaired cellular immunity, including those with AIDS, hematologic malignancy, or solid organ transplantation, and recipients of immunosuppressive therapy such as corticosteroids.

Clinical Manifestations Pulmonary disease can manifest with parenchymal lesions and/or endobronchial lesions. Although isolated lung involvement has been described, the vast majority of patients with pulmonary disease have concomitant cutaneous involvement, which most frequently manifests with disseminated ulcers, papules, or nodules. Dissemination to the liver, bone marrow, and lymph nodes may also occur, whereas dissemination to the CNS is rare.

Diagnosis Diagnosis is made by culturing *Emergomyces* species from blood or tissue or by showing tissue invasion upon histologic examination. *Emergomyces* may grow from aerobic blood culture bottles within 7 days; the use of fungal isolator culture bottles may improve sensitivity. The yeast form appears on histopathology as intracellular and extracellular round or ovoid yeasts with narrow-based budding that may be misidentified as *Histoplasma capsulatum*; therefore, definitive identification is made on mold phase in culture. It is worth noting that the commercially available *Histoplasma* antigen test has a high degree of cross-reactivity with *Emergomyces* species.

Treatment and Prognosis Given the lack of clinical trials to guide optimal treatment of emergomycosis, treatment recommendations are currently based on expert opinion and largely mirror treatment recommendations for disseminated histoplasmosis. Immunosuppressed patients with emergomycosis should initiate treatment with AmB until clinical improvement occurs followed by therapy with a mold-active azole. Itraconazole is the azole most often used, although in vitro susceptibility testing results of voriconazole and posaconazole have been favorable. In case reports, itraconazole monotherapy has been used successfully for mild disease. Prolonged secondary prophylaxis may be considered in patients without a reversible cause of immunodeficiency.

■ ADIASPIROMYCOSIS

Adiaspiromycosis is a rare infection caused predominantly by the mold *Emmonsia crescens*, which is found in soil. Occupational dust exposure is a major risk factor. Human infections have been broadly reported including from the Americas, Europe, and Asia. The lungs are by far the most frequently affected organ, although cases of ocular and appendiceal disease have been described. The disease is defined by the presence of adiaspores in affected tissue, which results in a marked inflammatory response. After exposure to conidia via inhalation, ingestion, or mucosal contact with dust, the host mounts an immune response leading to the development of noncaseating granulomas. Once within host tissue and driven by thermal pressure, the conidia enlarge to thick-walled adiaspores measuring up to 500 μ m in diameter that are incapable of budding or replication. Clinical sequelae, including respiratory failure, are caused by the host's granulomatous inflammatory response in infected tissue. As *E. crescens* has not been cultured from human tissue with adiaspiromycosis, diagnosis depends upon histopathologic identification of adiaspores with surrounding granulomas, often with the assistance of molecular methods. The disease is often self-limiting. However, in severe disease, corticosteroids, with or without antifungal therapy, have been used successfully in several reports.

MOLD INFECTIONS

Molds produce conidia or spores, which cause infection via inhalation or traumatic skin inoculation. In infected tissues, filamentous structures termed *hyphae* form. The most common invasive mold infections—aspergillosis and mucormycosis—are discussed

in Chaps. 223 and 224, respectively. Dematiaceous or brown-black molds cause phaeohyphomycoses that can be either localized or disseminated infections, the latter mostly in immunocompromised patients. Three genera of hyaline (nonpigmented) molds, *Fusarium*, *Scedosporium*, and *Lomentospora*, have become prominent pathogens among immunocompromised patients. Infections with these fungi mimic aspergillosis in their clinical manifestations and their histopathologic appearance in tissues.

■ PHAEOHYPHOMYCOSIS

Dematiaceous or brown-black fungi cause phaeohyphomycoses. The term *phaeohyphomycosis* is used to describe any infection with a melanin-producing fungus with filamentous growth. Melanin is a virulence factor for all pigmented molds. Most dematiaceous fungi cause localized subcutaneous infections after direct inoculation. Disseminated infections and serious focal visceral infections can occur in immunocompetent individuals but are much more often seen in immunocompromised patients. Two specific cutaneous phaeohyphomycoses—*eumycetoma* and *chromoblastomycosis*—are discussed in Chap. 225.

Etiologic Agents, Epidemiology, and Pathogenesis A large number of genera and species of pigmented molds can cause human infection. Most are found in the soil or on plants, and some cause economically important plant diseases. *Alternaria*, *Exophiala*, *Curvularia*, *Cladophialophora*, *Bipolaris*, and *Wangiella* species are among the more common pigmented molds reported to cause human infection. Infections with dematiaceous molds are acquired by traumatic inoculation into the eye or through the skin, by inhalation, or by injection of contaminated medication. In 2012, *Exserohilum* species caused a large outbreak in the United States of severe—and in some patients fatal—infections of the CNS following the injection of methylprednisolone contaminated with this fungus. Subcutaneous infections follow traumatic inoculation and are frequently caused by *Curvularia*, *Exophiala*, and *Phialophora*, among other genera. Several organisms, specifically *Cladophialophora bantiana* and *Rhinocladiella mackenziei*, are neurotropic and likely to cause CNS infection. When a patient is immunocompromised or when a pigmented mold is injected directly into a deep structure, these organisms become opportunistic pathogens, invading blood vessels and mimicking better-known opportunistic infections, such as aspergillosis.

Disseminated phaeohyphomycosis may occur in patients with solid organ or hematopoietic stem cell transplantation or after corticosteroid use. It has also been described in patients with autosomal recessive deficiency in CARD9, an intracellular adaptor protein that relays fungus-sensing signals from multiple cell-surface C-type lectin receptors (CLRs). A recent report indicated that ~70% of putatively immunocompetent patients who developed disseminated and brain phaeohyphomycosis carried deleterious variants in *CLEC7A*, the gene that encodes for the CARD9-coupled CLR Dectin-1, which recognizes fungal cell wall β -glucan. Dectin-1 and CARD9 protect against this infection by mediating the production of proinflammatory cytokines in response to fungal infection, promoting macrophage fungal killing.

Clinical Manifestations Dematiaceous molds are the most common cause of allergic fungal sinusitis and a less common cause of invasive fungal sinusitis. Keratitis occurs with traumatic corneal inoculation. Even in many immunocompromised patients, inoculation through the skin generally produces only localized nodular lesions at the entry site. Other immunocompromised patients develop pneumonia, brain abscess, or disseminated infection. Patients with brain abscess often have no other signs of dissemination at the time of diagnosis and typically present with an indolent course. Certain species, such as *Cladophialophora bantiana*, exhibit a propensity for intraventricular spread within the CNS. In the *Exserohilum* outbreak mentioned above, epidural injections of fungus-contaminated glucocorticoids led to meningitis, basilar stroke, epidural abscess and phlegmon, vertebral osteomyelitis, and arachnoiditis.

Diagnosis The specific diagnosis of infection with a pigmented mold is established by growth of the organism in culture. Histologically,

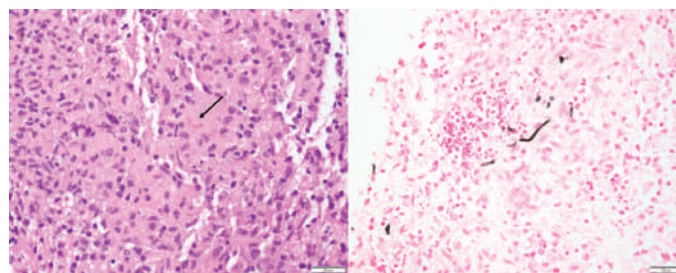


FIGURE 226-1 Histologic stains revealing pigmented mold from brain biopsy of a man with multifocal brain lesions. Cultures grew *Cladophialophora bantiana*. Images show tissue stained with hematoxylin and eosin (left panel) showing golden-brown pigmented hyphae (arrow) within multinucleated giant cells and associated neutrophilic infiltrate and Fontana-Masson stain (right panel) highlighting the melanin found in the cell wall of this organism. Scale bars, 20 μ m. (Images taken by Dr. Stefania Pittaluga, NCI.)

observation of melanized hyphal forms appearing golden-brown on biopsy specimens stained with hematoxylin and eosin supports the diagnosis of phaeohyphomycosis. Specialized melanin stains, such as the Fontana-Masson stain, highlight melanized fungal structures in infected tissues and provide important diagnostic support (Fig. 226-1). PCR assays are increasingly used in the diagnosis of infection due to dematiaceous molds but are available only through fungal reference laboratories.

Treatment and Prognosis Surgical debridement is vital for the treatment of phaeohyphomycosis in addition to antifungal treatment, which may need to be prolonged, especially in patients with CNS involvement. The choice of antifungal agent to treat disseminated and focal visceral infections with brown-black molds is based on the location and extent of the infection, in vitro test results, and clinical experience with the specific infecting organism. AmB is not effective against many of these organisms but has been used successfully against some species (Table 226-1). Itraconazole, voriconazole, or posaconazole can be used in the treatment of localized infections. Voriconazole has been the preferred azole when infections involve the CNS because it reaches adequate concentrations at that anatomic site; recent reports support posaconazole as an alternative azole. For certain pigmented molds, such as *C. bantiana*, flucytosine may have a role in addition to AmB or an azole. Disseminated and focal visceral infections, especially those involving the CNS, are associated with high mortality rates.

■ FUSARIOSIS

Etiologic Agent, Epidemiology, and Pathogenesis *Fusarium* species, which are found worldwide in soil and on plants, have emerged as major opportunistic pathogens in markedly immunocompromised patients. Strains of the *F. solani* complex are most commonly responsible for human infections, followed by *F. oxysporum*, *F. proliferatum*, and other species. *Fusarium* are molds with distinctive banana-shaped macroconidia. While *Fusarium* keratitis and onychomycosis can occur in immunocompetent patients, disseminated disease develops in immunodeficient patients. An outbreak of severe *Fusarium* keratitis among soft contact lens wearers was traced back to a particular brand of contact lens solution and individual contact lens cases that had been contaminated with this mold. In 2022 and 2023, two separate outbreaks of *F. solani* meningitis were reported in previously healthy young women who had received epidural anesthesia for plastic surgery procedures at two private clinics in Mexico. The case-fatality rate was ~50%.

Disseminated infection is reported most often in patients who have a hematologic malignancy, are neutropenic, have received hematopoietic stem cell or solid organ transplantation or corticosteroids, or have severe burn wounds. Neutrophils are the most critical cells for protection against fusariosis; hence the correlation with profound and prolonged neutropenia and susceptibility to disseminated infection. Neutrophil-derived reactive oxygen species are partly responsible for effective host defense against *Fusarium*. Patients with chronic granulomatous disease (CGD), who carry deleterious mutations in the NADPH oxidase complex that impair the phagocyte oxidative burst,

are at risk for fusariosis, albeit with far lower frequency compared with aspergillosis. Several studies have shown that the prognosis of fusariosis in neutropenic patients is significantly improved in those who recovered their neutrophil counts relative to those who remained persistently neutropenic.

Exposure to *Fusarium* can be through inhalation of conidia or direct inoculation via trauma, but disseminated disease can also occur secondary to pre-existing local disease, such as sinusitis or onychomycosis. For example, the presence of *Fusarium* onychomycosis preceding allogeneic hematopoietic stem cell transplantation has been associated with a higher risk of developing disseminated fusariosis post-transplantation.

Clinical Manifestations In immunocompetent persons, *Fusarium* species cause localized infections of various organs. These organisms are a common cause of fungal keratitis, which can extend into the anterior chamber of the eye, cause loss of vision, and require corneal transplantation. Onychomycosis due to *Fusarium* species, while a potential source of discomfort in immunocompetent patients, is a source of subsequent hematogenous dissemination and should be aggressively sought and treated in neutropenic patients. In profoundly immunocompromised patients, fusariosis is angioinvasive, and clinical manifestations mimic those of aspergillosis. Pulmonary infection is characterized by multiple nodular lesions that are best identified on CT scan. Fusariosis typically presents with persistent fever in neutropenic patients despite the administration of broad-spectrum antibiotics. Disseminated fusariosis differs from disseminated aspergillosis in that skin lesions are extremely common with fusariosis and may develop shortly after or even concurrently with the development of fever; the lesions are nodular or necrotic, are usually painful, and may appear over time in different locations (Fig. 226-2).

Diagnosis The diagnostic approach usually includes both documentation of the growth of *Fusarium* species from involved tissue and demonstration of invasion by histopathologic techniques that show acute angle septate hyphae in tissues. The organism is difficult to differentiate from *Aspergillus* species in tissues; therefore, identification with culture is imperative. An extremely helpful diagnostic clue is growth in blood cultures, which are positive in as many as 50% of patients with disseminated fusariosis. A positive blood culture may lead to rapid identification; subsequent rapid growth on agar reveals the characteristic banana-shaped macroconidia of *Fusarium*. Further identification of the species and antimicrobial susceptibility testing, generally in a reference laboratory, may aid in choosing the most appropriate therapy.

Treatment and Prognosis *Fusarium* species are resistant to many antifungal agents. A lipid formulation of AmB, voriconazole, or posaconazole is recommended. Many physicians use both a lipid

formulation of AmB and either voriconazole or posaconazole because susceptibility information is not standardized and is not always predictive of clinical response, bearing in mind that the prognosis is largely dictated by the degree of immune restoration. *F. solani* complex is frequently resistant to all azoles, while other species may have lower minimum inhibitory concentrations (MICs) to voriconazole and posaconazole. AmB is the most active antifungal drug against *Fusarium*; however, the MICs are relatively high for these organisms compared with other molds (Table 226-1). Mortality rates for disseminated fusariosis can be as high as 85%, with survival dependent on reversal of neutropenia. With the improved antifungal therapy now available, mortality rates have fallen to ~50%. However, if neutropenia persists, the mortality rate approaches 100%.

Two newer investigational antifungal agents show promise for the treatment of fusariosis. Fosmanogepix, which inhibits fungal Gwt1 and glycosylphosphatidylinositol-mannoprotein synthesis and anchoring, exhibits in vitro activity against *F. solani*, including against strains that are resistant to all available antifungals; it has been reported to successfully treat a few patients with meningeal and disseminated fusariosis. Olorofim, which inhibits fungal dihydroorotate dehydrogenase and pyrimidine synthesis, has shown favorable, but strain-specific, activity against *F. solani* complex in vitro.

■ SCEDOSPORIOSIS AND LOMENTOSPORIOSIS

Etiologic Agent, Epidemiology, and Pathogenesis *Scedosporium apiospermum* complex, which is composed of several related species, is reported more often as a cause of human infection than *Lomentospora prolificans*, formerly *Scedosporium prolificans*, but both are major emerging opportunistic pathogens in immunocompromised hosts and can cause pneumonia, disseminated infection, and brain abscess. Organisms of the *S. apiospermum* complex are found worldwide in temperate climates in tidal flats, swamps, ponds, manure, and soil. *L. prolificans* is also found in soil but is more geographically restricted with a greater representation of reported infections from Australia and Spain. Infection occurs predominantly through inhalation of conidia, but direct traumatic inoculation through the skin or the eye can also occur. *S. apiospermum* infections of the lungs and CNS have been described in putatively immunocompetent patients following near-drowning accidents, likely owing to the associated heavy burden of fungal exposure from contaminated water.

Clinical Manifestations Among immunocompetent persons, *Scedosporium* and *Lomentospora* species are a prominent cause of eumycetoma (Chap. 225). Keratitis resulting from accidental corneal inoculation is a sight-threatening infection. Traumatic inoculation during accidents may also lead to deep-seated infections, such as osteo-articular infections including septic arthritis, in immunocompetent patients. In patients who have hematologic malignancies, especially those with acute leukemia and prolonged neutropenia, recipients of solid organ or hematopoietic stem cell transplants, CGD patients, and patients receiving corticosteroids, these organisms are angioinvasive and can cause pneumonia and extrapulmonary dissemination, with a propensity for CNS spread. Pulmonary infection mimics that of aspergillosis both clinically and in appearance on CT scan: nodules, cavities, and lobar infiltrates are common radiographic findings. Contiguous spread from the lung to adjacent anatomic sites, such as the vertebrae, has rarely been described. Disseminated infection often involves the heart, brain, and many other organs. Skin lesions are not as common or as painful as those in patients with fusariosis.

Diagnosis Diagnosis depends on the growth of *Scedosporium* or *Lomentospora* species from infected tissue and the histologic demonstration of acute-angle septate hyphae invading tissues. Culture evidence is essential because these molds are difficult to morphologically differentiate from *Aspergillus* in tissues. Demonstration of tissue invasion is essential because these ubiquitous environmental molds can be mere contaminants or colonizers in the respiratory tract, especially in persons with bronchiectasis. As with *Fusarium*, *L. prolificans* can grow in blood cultures, but *S. apiospermum* usually does not.



FIGURE 226-2 Painful necrotic foot lesion that developed over a week in a woman with acute leukemia who had been neutropenic for 2 months. *Fusarium* species grew from a punch biopsy. (Courtesy of Dr. Nessrine Ktaich.)

Treatment and Prognosis *Scedosporium* and *Lomentospora* species are intrinsically resistant to AmB, echinocandins, and some azoles. Voriconazole has been the agent of choice for *S. apiospermum*, and posaconazole has been increasingly used for this infection (Table 226-1). *L. prolificans* is resistant in vitro to almost every commercially available antifungal agent; the addition of agents such as terbinafine to a voriconazole regimen has been attempted because in vitro data suggest possible synergy between these two agents against some strains of *L. prolificans*. Mortality rates for invasive *S. apiospermum* infection are ~50%, but those for invasive *L. prolificans* infection remain as high as 85–100%. The novel antifungal agents fosmanogepix and olorofim have shown in vitro activity against both *S. apiospermum* and *L. prolificans* and have demonstrated efficacy in preclinical mouse models of these infections. Ongoing clinical trials for these agents will help determine their clinical efficacy in infected patients.

YEAST INFECTIONS

In addition to the far more common human pathogenic yeasts *Candida* and *Cryptococcus*, which are discussed in [Chaps. 221](#) and [222](#) the yeast-like fungus *Trichosporon* has emerged as a significant opportunistic pathogen among immunocompromised patients.

■ TRICHOSPORONOSIS

Etiologic Agent, Epidemiology, and Pathogenesis The genus *Trichosporon* encompasses many species, some of which cause localized infection of hair and nails; white piedra is a superficial infection of the hair and scalp caused by *T. ovoides* that is characterized by the appearance of white nodules along the hair shaft. The most common species responsible for invasive infection is *Trichosporon asahii*, although other species also can cause disseminated disease. *Trichosporon* species grow as yeast-like colonies in vitro; in vivo, however, hyphae, pseudohyphae, and arthroconidia can be seen in addition to yeast forms. These yeasts are commonly found in soil, sewage, and water and in rare instances can colonize the human skin and the gastrointestinal and respiratory tracts. Most infections follow inhalation or entry via central venous catheters. The presence of foreign material is a risk factor for trichosporonosis, as both catheter-associated infections and prosthetic cardiac valve infections have been described. Systemic infection occurs almost exclusively in immunocompromised hosts, especially those who have hematologic malignancies, are neutropenic, have received a solid organ or hematopoietic stem cell transplantation, or are receiving corticosteroids. Severe neutropenia is a major risk factor for disseminated disease. Myeloid phagocytes and their oxidative burst are critical for protection against *Trichosporon*, as illustrated by the development of invasive trichosporonosis in patients with CARD9 deficiency or CGD.

Clinical Manifestations Disseminated trichosporonosis resembles invasive candidiasis, and fungemia is often the initial manifestation of infection typically presenting with fever. Pneumonia, skin lesions, and sepsis are common, although many patients can present without clinically apparent lung infection. The skin lesions in disseminated disease begin as papules or nodules surrounded by erythema and progress to central necrosis. Renal involvement is common in disseminated disease and may cause funguria and hematuria. A rare chronic form of infection mimics hepatosplenic candidiasis, now known as chronic disseminated candidiasis.

Diagnosis The diagnosis of systemic *Trichosporon* infection is established by growth of the organism from infected tissues or from blood. Histopathologic examination of a skin lesion showing a mixture of yeast forms, arthroconidia, pseudohyphae, and hyphae can lead to an early presumptive diagnosis of trichosporonosis. The presence of arthroconidia in particular can help differentiate trichosporonosis from candidemia. The serum cryptococcal antigen latex agglutination test may be positive in patients with disseminated trichosporonosis because *T. asahii* and *Cryptococcus neoformans* share polysaccharide antigens.

Treatment and Prognosis Rates of response to AmB have been disappointing, and many *Trichosporon* isolates are resistant to AmB in

vitro. Voriconazole has been the antifungal agent of choice (Table 226-1). The mortality rates for disseminated *Trichosporon* infection have been as high as 70% but are decreasing with the use of voriconazole; however, neutrophil recovery is vital in overcoming this infection, and mortality rates with persistent neutropenia remain very high despite antifungal therapy.

■ FURTHER READING

- DE ALMEIDA JUNIOR JN, HENNEQUIN C: Invasive *Trichosporon* infections: A systematic review on a re-emerging fungal pathogen. *Front Microbiol* 7:1629, 2016.
- HOSPENTHAL D: Uncommon fungi and related species in *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*, 10th ed, Blaser MJ et al (eds). Philadelphia, Elsevier, 2025.
- NEOH CF et al: Scedosporiosis and lomentosporiosis: Modern perspectives on these difficult-to-treat rare mold infections. *Clin Microbiol Rev* 37:e0000423, 2024.
- NUCCI M et al: Fusariosis. *Semin Respir Crit Care Med* 36:706, 2015.
- REVANKAR SG et al: A Mycoses Study Group international prospective study of phaeohyphomycosis: An analysis of 99 proven/probable cases. *Open Forum Infect Dis* 4:ofx200, 2017.

227

Pneumocystis Infections

Alison Morris, Henry Masur

■ DEFINITION AND DESCRIPTION

Pneumocystis is an opportunistic pathogen that is an important cause of pneumonia in immunocompromised hosts, particularly those with HIV infection ([Chap. 208](#)), organ transplants, or hematologic malignancies and those receiving high-dose glucocorticoids or certain immunosuppressive monoclonal antibodies. *Pneumocystis* was discovered in rodents in 1909 and was initially believed to be a protozoan. Because *Pneumocystis* cannot be cultured, understanding its biology has been limited, but molecular techniques have demonstrated that the organism is actually a fungus. Formerly known as *Pneumocystis carinii*, the species specific to humans has been renamed *Pneumocystis jirovecii*; the species that is specific to rats is *Pneumocystis carinii*; there are multiple variants of *Pneumocystis* that are specific to other animals.

■ EPIDEMIOLOGY

Pneumocystis jirovecii pneumonia (PCP) came to medical attention in the early 1950s when pathologists in Czechoslovakia recognized *Pneumocystis* in the alveolar exudates of infants involved in nursery outbreaks of interstitial pneumonia, outbreaks that had been described in Europe since the 1920s. Among adults, PCP was rarely recognized until the populations of immunosuppressed adults increased due to the development of immunosuppressive therapies for solid-organ transplantation, bone marrow transplantation, cancer, and autoimmune disorders, and until the development of better pulmonary diagnostic techniques such as bronchoscopy. In 1981, PCP was first reported in men who had sex with men and in intravenous (IV) drug users who had no obvious cause of immunosuppression. These cases were subsequently recognized as the first cases of what came to be known as the acquired immunodeficiency syndrome (AIDS) ([Chap. 208](#)).

The incidence of PCP increased dramatically as the AIDS epidemic grew: without PCP chemoprophylaxis or antiretroviral therapy (ART), 80–90% of patients with HIV/AIDS in North America and Western Europe ultimately developed one or more episodes of PCP. While its incidence declined with the introduction of anti-*Pneumocystis*

prophylaxis and durably effective ART, PCP has continued to be an important cause of AIDS-associated morbidity in the United States and Western Europe, particularly in individuals who do not know they are infected with HIV until they are profoundly immunosuppressed and in people living with HIV (PLWH) with CD4+ T lymphocyte counts of <200/ μ L who are not receiving ART or PCP prophylaxis.

PCP also develops in HIV-uninfected patients who are immunocompromised secondary to congenital immunodeficiencies, hematologic or malignant neoplasms, stem cell or solid-organ transplantation, and treatment with immunosuppressive medications. The incidence of PCP depends on the degree and duration of immunosuppression. PCP is increasingly reported among individuals receiving tumor necrosis factor α inhibitors and certain (but not all) immunosuppressive monoclonal antibodies for autoimmune, rheumatologic, or neoplastic diseases. In many health care systems, PCP occurs more often due to non-HIV-related immunosuppression than due to HIV infection.

While clinical disease due to *Pneumocystis* in immunocompetent hosts has not been clearly documented, studies have shown that *Pneumocystis* organisms can cause subclinical infection among children and adults who are not immunocompromised and can be associated with pulmonary pathology. The relevance of these organisms to acute or chronic clinical syndromes, such as chronic obstructive pulmonary disease (COPD), in immunocompetent patients is being investigated.

In some developing countries, the incidence of PCP among PLWH has been reported to be lower than that in more industrialized countries. This lower incidence may be due to competing mortality from infectious diseases such as tuberculosis and bacterial pneumonia, which typically occur before patients become immunosuppressed enough to develop PCP. Geographic variations in *Pneumocystis* exposure and underdiagnosis attributable to lack of diagnostic resources also may explain the apparent lower frequency of PCP in some countries.

■ PATHOGENESIS AND PATHOLOGY

Life Cycle and Transmission The life cycle of *Pneumocystis* likely involves both sexual and asexual reproduction. The organism exists as a trophic form, a cyst, and a precyst. Studies in rodents show that immunocompetent animals can serve as reservoirs for respiratory transmission of *P. carinii* (the infecting species in rats) to immunocompetent and immunosuppressed rats. Human *Pneumocystis* is thought to be transmitted by a respiratory route as well. *P. jirovecii*, like all pneumocystis species, is host-specific. Thus, humans are not infected, for example, by *P. carinii* (rodents) or *P. oryctolagi* (rabbits), but are only infected by *P. jirovecii*, which is from other humans.

Serologic and molecular studies have demonstrated that most humans are exposed to *P. jirovecii* and infected early in life. It was historically thought that *Pneumocystis* pneumonia usually developed from reactivation of latent infection. However, molecular evidence makes it clear that children and adults can develop PCP from primary infection or reinfection. The source of infection is thought to be either healthy or immunosuppressed individuals who themselves experienced recent infection or reinfection, or immunosuppressed persons with clinical PCP. Nosocomial outbreaks among immunosuppressed persons occur in inpatient and outpatient settings. The utility of droplet or airborne isolation for preventing transmission from patients with PCP to other immunosuppressed individuals has been debated; no clear evidence exists, although it seems prudent to isolate patients with active PCP from other immunosuppressed patients using at least droplet precautions.

Role of Immunity Defects in cellular and/or humoral immunity predispose to development of PCP. Such defects may be congenital, or they may be acquired as a result of HIV infection or treatment with certain immunosuppressive drugs such as glucocorticoids, fludarabine, temozolomide, temsirolimus, rituximab, or alemtuzumab. CD4+ T cells are critical in host defense against *Pneumocystis*. Among PLWH, the incidence of PCP is inversely related to the CD4+ T-cell count: at least 80% of cases occur at counts of <200/ μ L, and most cases actually develop at counts of <100/ μ L. HIV viral load is another factor that

predisposes patients to PCP. Clinicians must recognize that PCP can occur at CD4+ T-cell counts >200/ μ L in persons with HIV infection, but such occurrences are uncommon, especially with CD4+ T-cell counts substantially higher than 200/ μ L.

CD4+ T-cell counts are less useful in predicting the risk of PCP in patients who are immunosuppressed for reasons other than HIV infection. In these populations, a CD4+ T-cell count <200/ μ L is a sensitive indicator of susceptibility to PCP. However, a CD4+ T-cell count >200/ μ L in many populations does not imply protection.

Lung Pathology *Pneumocystis* has a unique tropism for the lung. Organisms are presumably inhaled into the alveolar space after being exhaled by another human. Clinically apparent pneumonia occurs only if an individual is immunocompromised. *Pneumocystis* proliferates in the lung, provoking a mononuclear cell response. The alveoli become filled with proteinaceous material, and alveolar damage results in increased alveolar-capillary injury and surfactant abnormalities. Stained lung sections typically show foamy, vacuolated alveolar exudates composed largely of viable and nonviable organisms (Fig. 227-1A). Interstitial edema and fibrosis may develop, and organisms can be seen in the alveolar space with silver or other stains. The organisms can also be seen when tissue is subjected to colorimetric or immunofluorescent staining (Fig. 227-1B-D).

■ CLINICAL FEATURES

Clinical Presentation PCP presents as acute or subacute pneumonia that may initially be characterized by a vague sense of dyspnea alone, but that subsequently manifests as fever and nonproductive cough with progressive shortness of breath. Patients may ultimately progress to respiratory failure and death. Extrapulmonary manifestations of PCP are rare, but can include involvement of almost any organ, most notably the lymph nodes, spleen, and liver.

Physical Examination, Oxygen Saturation, and Imaging The physical examination findings in PCP are nonspecific. Patients have decreased oxygen saturation—at rest or with exertion—that, without treatment, progresses to severe hypoxemia. Patients may initially have a normal chest examination and no adventitious sounds, but later develop diffuse rales and signs of consolidation.

Laboratory Findings The results of routine laboratory tests are nonspecific in PCP. Serum levels of lactate dehydrogenase (LDH) are often elevated as a result of pulmonary damage; however, a normal LDH level does not rule out PCP, nor is an elevated LDH value specific for PCP. The peripheral white blood cell count may be elevated in relation to the patient's baseline values, but the increase is usually modest and the baseline may have been below usual normal limits due to HIV infection. Hepatic and renal function are typically normal.

Radiographic Findings Although the initial chest radiograph may be normal when patients have mild symptoms, the classic radiographic appearance of symptomatic PCP consists of diffuse bilateral interstitial infiltrates that are perihilar and symmetric (Fig. 227-2A)—another finding that is not specific for PCP. The interstitial infiltrates can progress to alveolar filling (Fig. 227-2B). High-resolution chest computed tomography (CT) shows diffuse ground-glass opacities in virtually all patients with PCP, often before a routine chest radiograph becomes abnormal (Fig. 227-2C). A normal chest CT essentially rules out the diagnosis of PCP. Pneumatocoles and pneumothoraces are characteristic chest radiographic findings, especially in patients with HIV infection (Fig. 227-2D). A wide variety of atypical radiographic findings have been described, including asymmetric patterns, upper-lobe infiltrates, mediastinal adenopathy, nodules, cavities, and effusions.

■ DIAGNOSIS

The optimal sample for a specific microbiologic diagnostic examination depends on how ill the patient is and what resources are available. Before the 1990s, diagnoses of PCP were usually established by open lung biopsy; later, transbronchial lung biopsy was employed. Hematoxylin and eosin (H and E) staining of pulmonary tissue

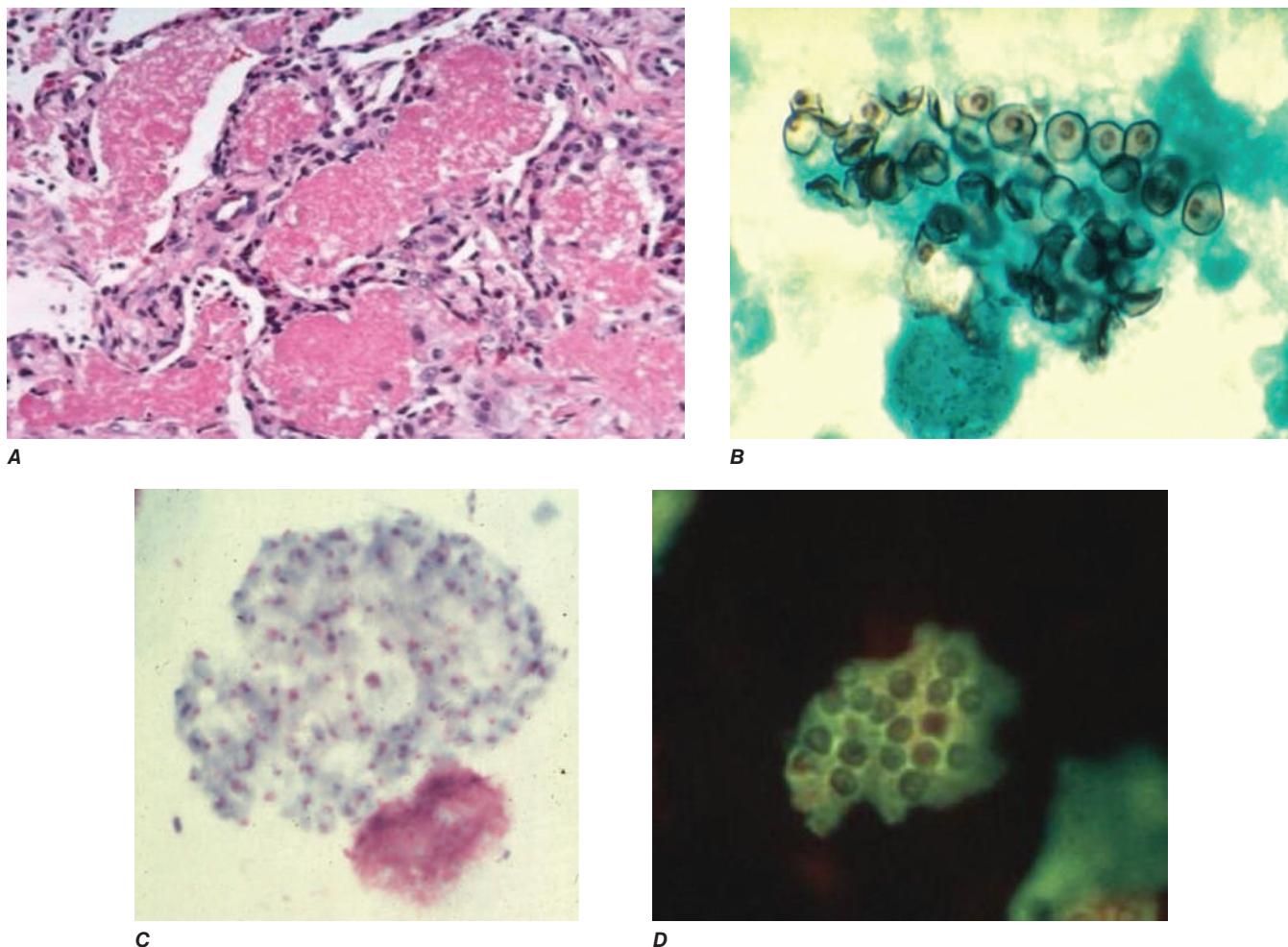


FIGURE 227-1 Direct microscopy of *Pneumocystis pneumonia*. **A.** Transbronchial lung biopsy stained with hematoxylin and eosin shows eosinophilic alveolar filling. **B.** Methenamine silver–stained bronchoalveolar lavage (BAL) fluid. **C.** Giemsa-stained BAL fluid. **D.** Immunofluorescent stain of BAL fluid.

demonstrates a foamy alveolar filling and a mononuclear interstitial infiltrate (Fig. 227-1A). This appearance is pathognomonic for PCP even though the organisms cannot be specifically identified with this H and E stain. Since human pneumocystis has never been successfully cultured, the diagnosis is typically established in lung tissue or pulmonary secretions by staining of the cyst and or trophozoite—e.g., with methenamine silver (Fig. 227-1B), toluidine blue O, or Giemsa (Fig. 227-1C)—or by staining with a specific immunofluorescent antibody (Fig. 227-1D).

Since the 1990s, bronchoalveolar lavage (BAL) has become the most common method for obtaining a sample of respiratory secretions in which to detect *Pneumocystis* organisms. The demonstration of organisms in BAL fluid is almost 100% sensitive and specific for PCP. The organisms are identified in pulmonary secretions with the specific stains indicated above for lung biopsy. While expectorated sputum or throat swabs have very low sensitivity, an induced sputum sample obtained and interpreted by an experienced provider can be highly sensitive and specific; however, the sensitivity is dependent on both the characteristics of the patient, the quality of the sputum sample, and the experience of the center conducting the test and is widely variable (55–90%).

Many laboratories offer polymerase chain reaction (PCR) testing of respiratory specimens for *Pneumocystis* in preference to direct microscopy of stained respiratory secretions. However, PCR tests are so sensitive that it is difficult to distinguish patients with colonization (i.e., those whose acute lung disease is due to some other process but who have low levels of *Pneumocystis* DNA in their lungs) from those with acute pneumonia due to *Pneumocystis*. The PCR cycle number is probably helpful, i.e., the lower the cycle number, the more DNA was

detected. However, PCR tests for PCP are not standardized and specimen collection can vary considerably resulting in variable quality of the specimens assessed. Therefore, there is no specific PCR threshold that can be deemed to be especially convincing. Such PCR tests on appropriate samples may be more useful for ruling out a diagnosis of PCP if they are negative than for definitively attributing the disease to *Pneumocystis* if they are positive.

There has been considerable interest in serologic tests, such as assays for (1→3)-β-D-glucan, a component of the fungal cell wall. Serum (1→3)-β-D-glucan levels are frequently elevated in patients with PCP. However, serum or BAL (1→3)-β-D-glucan levels are not perfectly sensitive or highly specific for PCP. There are increasing numbers of studies of serum PCR tests for *Pneumocystis*, but such serum tests are not yet useful for establishing the presence or absence of PCP.

■ COURSE AND PROGNOSIS

Untreated, PCP is invariably fatal. Patients with HIV infection often have an indolent course that may present early as mild exercise intolerance or chest tightness without fever or cough and a normal or nearly normal posterior–anterior chest radiograph. However, this process progresses over days, weeks, or even a few months to fever, cough, diffuse alveolar infiltrates, and profound hypoxemia. Some patients with HIV infection and most patients with other types of immunosuppression have more acute disease that progresses over a few days to respiratory failure. Rare patients also develop distributive shock due to PCP. A few unusual patients present with extrapulmonary manifestations in the skin or soft tissue, retina, brain, liver, kidney, or spleen. Extrapulmonary disease is nonspecific in presentation and can be diagnosed only by histology. When there is extrapulmonary clinical disease in a

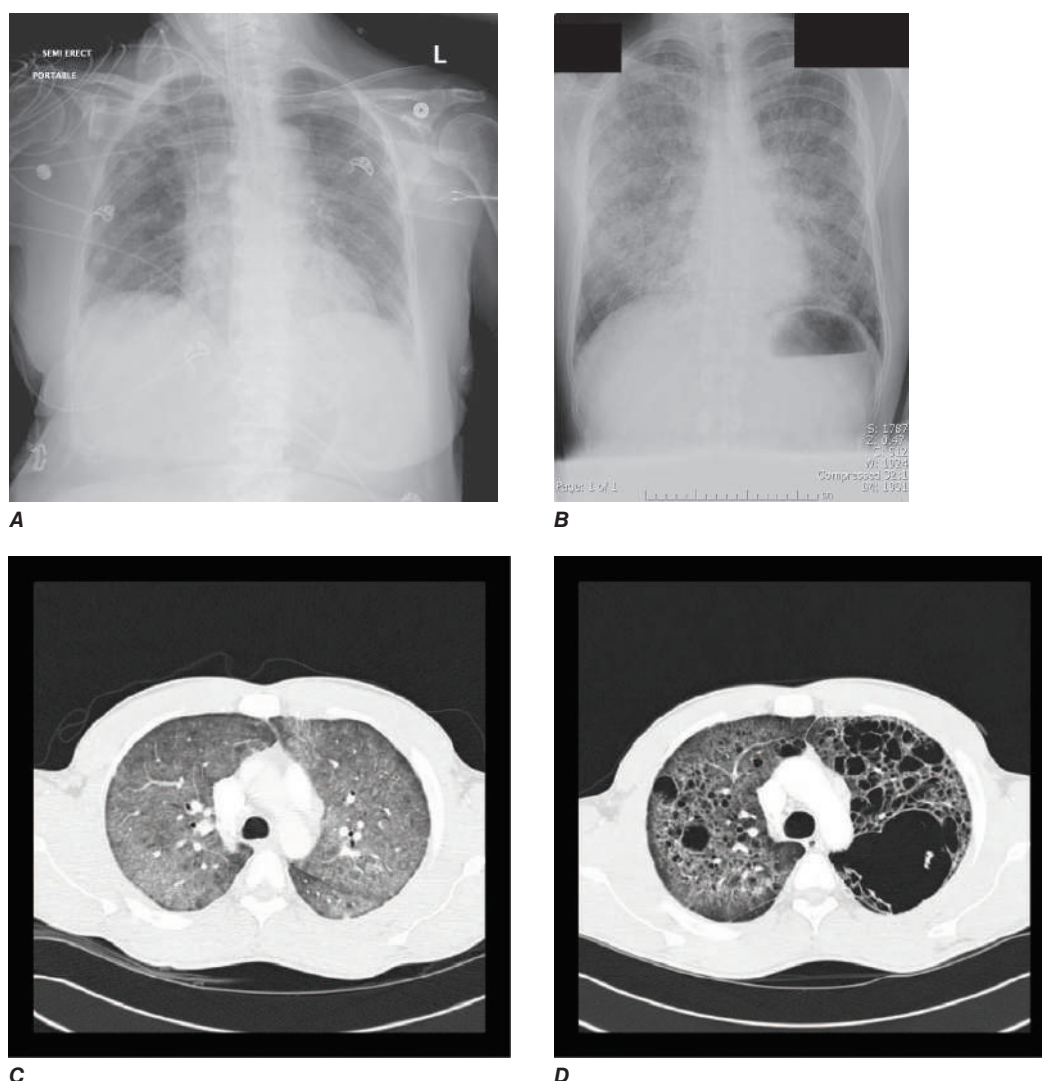


FIGURE 227-2 Radiographs in *Pneumocystis pneumonia*. **A.** Posterior–anterior chest radiograph showing symmetric interstitial infiltrates. **B.** Posterior–anterior chest radiograph showing symmetric alveolar infiltrates (courtesy of Alison Morris). **C.** Computed tomography (CT) image demonstrating symmetric interstitial infiltrates and ground-glass opacities. **D.** CT image showing symmetric interstitial infiltrates, ground-glass opacities, and pneumatoceles.

patient with PCP, the priority is to determine what other concurrent infectious or neoplastic process might be present, given the rarity of extrapulmonary pneumocystosis.

Factors that influence mortality risk of PCP include the patient's age and degree of immunosuppression as well as the presence of preexisting lung disease, the need for mechanical ventilation, and the development of a pneumothorax. With advances in supportive critical care, the prognosis for patients with PCP who require intubation and respiratory support has improved and now depends to a large extent on comorbidities and the prognosis of the underlying disease. Some patients do not respond to therapy for 4–8 days, and thus supportive care for a minimum of 10 days is a reasonable consideration if such support is compatible with the patient's wishes and the prognosis of comorbidities. In fact, if patients of any level of severity are treated with specific therapy but without corticosteroids, they often deteriorate during the first few days, presumably due to enhanced inflammation induced by dying organisms. Patients whose condition continues to deteriorate after 3 or 4 days or have not improved after 7–10 days should be reevaluated to determine whether other infectious processes are present (either having been missed on initial evaluation or having developed during treatment), whether initial anti-*Pneumocystis* treatment has failed, or whether noninfectious processes (e.g., congestive heart failure, pulmonary emboli, pulmonary hypertension, drug toxicity, or a neoplastic process) are causing pulmonary dysfunction.

TREATMENT

P. jirovecii Pneumonia

The treatment of choice for PCP is trimethoprim-sulfamethoxazole (TMP-SMX), given either IV or PO for 14 days to non-HIV-infected patients with mild disease and for 21 days to all other patients (Table 227-1). TMP-SMX, which interferes with the organism's folate metabolism, is at least as effective as alternative agents and is better tolerated. TMP-SMX can cause leukopenia, hepatitis, rash, fever, elevation of potassium and creatinine, and anaphylactic and anaphylactoid reactions. Patients with HIV infection have an unusually high incidence of hypersensitivity to TMP-SMX. Monitoring of serum drug levels is useful if renal function or toxicities are issues in order to enhance the likelihood that therapy will be effective and toxicity will be avoided. Maintenance of a 2-h post-dose serum sulfamethoxazole level of 100–150 µg/mL has been associated with a successful outcome. Resistance to TMP-SMX cannot be measured by organism growth inhibition in the laboratory because human *Pneumocystis* cannot be successfully cultured. However, mutations in the target gene for sulfamethoxazole that confer in vitro sulfa resistance when found in other organisms have been recognized in *Pneumocystis*. The clinical relevance of these mutations for the response to therapy is uncertain. Sulfadiazine plus pyrimethamine, an oral regimen more often used for treatment of toxoplasmosis, also is highly effective.

TABLE 227-1 Treatment of *Pneumocystis Pneumonia*^a

DRUG(S)	DOSE, ROUTE	ADVERSE EFFECTS
First-Choice Agent		
TMP-SMX	TMP (5 mg/kg) plus SMX (25 mg/kg) q6–8h PO or IV (i.e., 2 double-strength tablets tid or qid)	Fever, rash, cytopenias, hepatitis, hyperkalemia
Alternative Agents		
Atovaquone	750 mg bid PO	Rash, fever, hepatitis
Clindamycin plus Primaquine	300–450 mg q6h PO or 600 mg q6–8h IV 15–30 mg qd PO	Hemolysis (G6PD deficiency), methemoglobinemia, neutropenia, rash
Pentamidine	3–4 mg/kg qd IV	Hypotension, azotemia, cardiac arrhythmias (torsades des pointes), pancreatitis, dysglycemias, hypocalcemia, neutropenia, hepatitis
Adjunctive Agent		
Prednisone or methylprednisolone	40 mg bid × 5 d, 40 mg qd × 5 d, 20 mg qd × 11 d; PO or IV	Peptic ulcer disease, hyperglycemia, mood alteration, hypertension

^aTreatment can be administered for 14 days to non-HIV-infected patients with mild disease and for 21 days to all other patients.
Abbreviations: G6PD, glucose-6-phosphate dehydrogenase; TMP-SMX, trimethoprim-sulfamethoxazole.

Either intravenous pentamidine or the combination of clindamycin plus primaquine is an option for patients who cannot tolerate TMP-SMX and for patients in whom treatment with TMP-SMX appears to be failing. Pentamidine must be administered IV over at least 60 min to avoid potentially lethal hypotension. Adverse effects can be severe and irreversible and include renal dysfunction, dysglycemia (life-threatening hypoglycemia that can occur days or weeks after initial infusion and be followed by hyperglycemia), neutropenia, and torsades de pointes. Clindamycin plus primaquine is effective, but primaquine can be given only by the oral route—a disadvantage for patients who cannot ingest or absorb oral drugs and primaquine can cause hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency and can cause methemoglobinemia. Oral atovaquone is also a reasonable option for patients with mild disease who have no impediments to absorbing an oral drug that requires a high-fat meal for optimal absorption. There is some evidence for activity of echinocandins against the cyst form (but not the trophozoite form) of *Pneumocystis*, but the role for echinocandins as part of combination therapy is currently uncertain, and these drugs should never be used as single drug therapy for PCP.

A major advance in therapy for PCP was the recognition that glucocorticoids could improve survival rates among PLWH with moderate to severe disease (initial room air PO_2 <70 mmHg or alveolar–arterial oxygen gradient ≥ 35 mmHg). Glucocorticoids appear to reduce the pulmonary inflammation that occurs after specific therapy is started and organisms begin to die, eliciting inflammation. Therapy with glucocorticoids should be the standard of care for patients with HIV infection and moderate to severe PCP. Therapy with glucocorticosteroids is also probably effective for patients with other immunodeficiencies who have moderate to severe PCP. This treatment should be started for moderate or severe disease when therapy for PCP is initiated, even if the diagnosis is suspected but has not yet been confirmed. If PLWH or HIV-uninfected patients are receiving high-dose glucocorticoids when they develop PCP, there are theoretical advantages to decreasing the steroid dose to improve immune function, but there is no convincing evidence on which to base any specific strategy.

No definitive trials have identified the best therapeutic algorithm for patients in whom TMP-SMX treatment for PCP is failing. If no other treatable infectious or noninfectious processes are detected and pulmonary dysfunction appears to be due to PCP alone, many

authorities would switch from TMP-SMX to either IV pentamidine or IV clindamycin plus oral primaquine. Clindamycin-primaquine is certainly less toxic than IV pentamidine. Some authorities would add the second drug or drug combination to TMP-SMX rather than switching regimens. If patients are not already receiving them, glucocorticoids should be added to the regimen; the dosage and regimen are usually chosen empirically and depend on what glucocorticoid regimen (if any) the patient was receiving when PCP therapy was begun.

For PLWH who present with PCP before the initiation of ART, ART should be started within the first 2 weeks of therapy for PCP in most situations. Immune reconstitution inflammatory syndrome (IRIS) can occur, and the decision to initiate ART thus requires considerable clinical judgement that factors in the severity of pneumonia, the response of PCP to therapy, and concurrent medical conditions.

PREVENTION

The most effective method for preventing PCP is to eliminate the cause of immunosuppression by withdrawing immunosuppressive therapy or treating the underlying cause (e.g., HIV infection). Patients who are susceptible to PCP benefit from chemoprophylaxis during the period of susceptibility. For patients with HIV infection, CD4+ T-cell counts are a reliable marker of susceptibility, and counts <200/ μ L are an indication to start or continue prophylaxis (Table 227-2).

For patients who are immunosuppressed as a result of factors other than HIV infection, CD4+ T-cell counts <200/ μ L are a plausible but not absolute marker of susceptibility. However, counts >200/ μ L are not a reliable marker of protection from PCP. For these patients who are immunosuppressed due to causes other than HIV infection, the period of susceptibility is usually estimated on the basis of experience with the underlying disease and the specific immunosuppressive regimen. Cessation of prophylaxis has been associated with clusters of cases in certain patient populations, such as solid-organ transplant recipients, where the period of susceptibility is not well-defined. Patients receiving a prolonged course of high-dose glucocorticoids

TABLE 227-2 Prophylaxis of *Pneumocystis Pneumonia*

DRUG(S)	DOSE, ROUTE	COMMENTS
First-Choice Agent		
TMP-SMX	1 tablet (double- or single-strength) qd PO	Incidence of hypersensitivity is high. Rechallenge for non-life-threatening hypersensitivity; consider dose-escalation protocol.
Alternative Agents		
Dapsone	50 mg bid or 100 mg qd PO	Hemolysis is associated with G6PD deficiency.
Dapsone plus Pyrimethamine plus Leucovorin	50 mg qd PO 50 mg weekly PO 25 mg weekly PO	Leucovorin ameliorates cytopenias due to pyrimethamine.
Dapsone plus Pyrimethamine plus Leucovorin	200 mg weekly PO 75 mg weekly PO 25 mg weekly PO	Leucovorin ameliorates cytopenias due to pyrimethamine.
Pentamidine	300 mg monthly via Respigard II nebulizer	Aerosol may cause bronchospasm. Pentamidine is probably less effective than TMP-SMX or dapsone regimens.
Atovaquone	1500 mg qd PO	Requires fatty meal for optimal absorption.

Abbreviations: G6PD, glucose-6-phosphate dehydrogenase; TMP-SMX, trimethoprim-sulfamethoxazole.

Section 17 Protozoal and Helminthic Infections: General Considerations

228 Introduction to Parasitic Infections

Sharon L. Reed, Charles E. Davis



appear to be particularly susceptible to PCP. The glucocorticoid exposure threshold that warrants chemoprophylaxis is ill-defined, but such preventive therapy should be strongly considered for any patient who is receiving more than the equivalent of 20 mg of prednisone daily for 30 days or who is receiving glucocorticoids in conjunction with other immunosuppressive agents. Clinical experience also suggests that chemoprophylaxis is useful for patients receiving certain immunosuppressive agents (e.g., tumor necrosis factor inhibitors, antithymocyte globulin, rituximab, and alemtuzumab). The duration of such chemoprophylaxis is empirically estimated based on prior clinical experience and immunologic factors that would plausibly relate to immunity, such as CD4+ T-cell counts, recognizing that such estimates are not precise.

TMP-SMX is the most effective prophylactic drug; few patients experience a PCP breakthrough when they are reliably taking a recommended TMP-SMX chemoprophylactic regimen. Several TMP-SMX regimens have been used successfully. Regimens of one single-strength or double-strength tablet daily are the regimens with which there is the most experience, but one double-strength tablet two or three times weekly also has been recommended for various PLWH and non-HIV-infected populations of patients.

For patients who cannot tolerate TMP-SMX (usually because of hypersensitivity or bone marrow suppression), alternative drugs include daily dapsone, weekly dapsone-pyrimethamine, atovaquone, and monthly aerosol pentamidine. Patients who develop hypersensitivity to TMP-SMX can sometimes tolerate the drug if a gradual dose-escalation protocol is used. Atovaquone is effective and well tolerated; however, this drug is available only as an oral preparation, and gastrointestinal absorption is unpredictable in patients with abnormal gastrointestinal motility or function. Aerosolized pentamidine is effective, but it is not as effective as TMP-SMX and may not provide protection in areas of the lung that are not well-ventilated. Dapsone cross-reacts with sulfonamides in a substantial fraction of patients and is rarely useful in patients with a history of life-threatening reactions to TMP-SMX.

■ FURTHER READING

- BUCHACZ K et al: Incidence of AIDS-defining opportunistic infections in a multicohort analysis of HIV-infected persons in the United States and Canada, 2000–2010. *J Infect Dis* 214:862, 2016.
- DEL CORPO O et al: Diagnostic accuracy of serum (1-3)- β -D-glucan for *Pneumocystis jirovecii* pneumonia: A systematic review and meta-analysis. *Clin Microbiol Infect* 26:1137, 2020.
- LÉCUYER R et al: Characteristics and prognosis factors of *Pneumocystis jirovecii* pneumonia according to underlying disease. *Chest* 165:1319, 2024.
- LE GAL S et al: *Pneumocystis* infection outbreaks in organ transplantation units in France: A nation-wide survey. *Clin Infect Dis* 70:2216, 2020.
- MA L et al: Genome analysis of three *Pneumocystis* species reveals adaptation mechanisms to life exclusively in mammalian hosts. *Nat Commun* 7:10740, 2016.
- PANEL ON OPPORTUNISTIC INFECTIONS IN HIV-INFECTED ADULTS AND ADOLESCENTS: Guidelines for the prevention and treatment of opportunistic infections in HIV-infected adults and adolescents: Recommendations from the Centers for Disease Control and Prevention, the National Institutes of Health, and the HIV Medicine Association of the Infectious Diseases Society of America. Available at <https://clinicalinfo.hiv.gov/en/guidelines/hiv-clinical-guidelines-adult-and-adolescent-opportunistic-infections/pneumocystis-0?view=full>. Accessed December 21, 2024.
- WILLS NK et al: The prevalence of laboratory-confirmed *Pneumocystis jirovecii* in HIV-infected adults in Africa: A systematic review and meta-analysis. *Med Mycol* 59:802, 2021.
- ZOLOPA A et al: Early antiretroviral therapy reduces AIDS progression/death in individuals with acute opportunistic infections: A multicenter randomized strategy trial. *PLoS One* 4:e5575, 2009.

The word *parasite* comes originally from the Greek *parasitos* (*para*, alongside of; and *sitos*, food), meaning someone who eats at another's table or lives at another's expense. Although the same is true of many bacteria and viruses, the designation *parasite* is reserved, by convention, for helminths and protozoa. These organisms are larger and more complex than bacteria, with a eukaryotic cell structure similar to that of human host cells. Historically, this similarity has made it difficult to find effective antiparasitic agents that do not cause unacceptable toxicity to human cells. Fortunately, intensive research and modern techniques have now provided suitable agents for safe and effective treatment of most parasitic infections. **See Chap. S12 for details on diagnostic procedures and Chap. 229 for details on treatment.**

Internal parasites of human beings are divided into two types: helminths (worms) and protozoa. *Helminths* are multicellular organisms that can often be seen with the naked eye (**Chap. 237**). There are two phyla: Platyhelminthes (flat worms) and Nematelminthes (roundworms). Both phyla include some genera that mature in the gastrointestinal tract and others that migrate through the tissue after ingestion or skin penetration. **Tables S12-1 and S12-2** present the helminthic genera, their definitive and intermediate hosts, their geographic distributions, and the parasitic stages in the human body.

The key to understanding which helminths use humans as definitive hosts is to remember that helminth ova develop into larvae, and larval stages develop into adults. Humans serve as the definitive host when they ingest helminth larvae, which develop into adults in the intestine and usually cause mild disease, often without any symptoms. (The exception is ingestion of the late-stage larvae of the somatic or tissue flukes, as shown in **Table S12-1**.) In contrast, if humans ingest helminth ova and serve as the intermediate host, the ova develop into larvae, which penetrate the intestine, migrate through the tissue, and invade organs where they mature into adults. Intermediate hosts with parasitic invasion of organs may experience severe disease.

Protozoa are microscopic single-celled organisms. Among the many differences between helminths and protozoans, the most important is the ability of protozoa (like bacteria) to multiply within the human body and cause overwhelming infections. A major mechanism promoting unrestrained growth is evasion of the host immune response either by antigenic variation (*Trypanosoma brucei*) or by survival inside host cells (e.g., *Plasmodium*, *Babesia*, *Cryptosporidium*, *Leishmania*, and *Toxoplasma*). In contrast, almost all helminths require stages in other hosts to complete their life cycles and multiply. As a result, except for *Strongyloides* and *Capillaria*, which can complete their life cycle in humans, increases in the burden of infection with helminths require repeated exogenous reinfections. Thus, permanent residents of endemic countries, who are exposed repeatedly, may have heavy severe infections, while most travelers with one or two exposures are unlikely to experience the full spectrum of chronic helminthic infections.

In contrast to helminthic infections, naïve patients with their first protozoal infection usually are the most severely affected because partial immunity often limits the number of parasites during recurrent infections. Protozoan replication to large numbers in the host also promotes the development of drug-resistant forms, especially in malaria (**Chap. 229**). Because protozoa belong to many different phyla, it is easier to understand the pathogenesis and management of protozoal infections when they are classified by the site of infection (intestinal protozoans, free-living amebae, and blood and tissue protozoans).

1736 (Table S12-3). Immunocompromised hosts are at risk of disseminated infection with several protozoa, including *Leishmania*, *Toxoplasma*, *Cryptosporidium*, and *Trypanosoma cruzi*, which are AIDS-defining illnesses. Among the helminths, *Strongyloides* can disseminate in immunocompromised individuals.

HELMINTHIC INFECTIONS

The Platyhelminthes (flatworms) are categorized as tapeworms (cestodes) and flukes (trematodes). Tapeworms are composed of a head or scolex bearing the holdfast organs and segments, which become gravid as they mature. Some tapeworms can reach lengths of many yards; the longest tapeworms develop in the intestine, where they rarely cause serious disease. In contrast, flukes are small leaf-shaped organisms whose size is not a measure of disease severity.

■ FLATWORMS

Cestodes Tapeworms cause either intestinal or somatic infection, depending on the species. *Intestinal* infections occur when the human host ingests larvae in the tissue of the intermediate host, whereas *somatic* infections occur when humans accidentally ingest ova excreted from the wild or domesticated definitive animal host.

INTESTINAL TAPEWORMS As shown in Table S12-1, humans acquire most intestinal tapeworms by eating the insufficiently cooked flesh of the intermediate host. Thus, *Taenia saginata* is commonly called the beef tapeworm, *Taenia solium* the pork tapeworm, and *Diphyllobothrium latum* the fish tapeworm. *Hymenolepis nana* is capable of completing its life cycle in the human intestine and is acquired by ingestion of infected grain beetles or of ova from infected humans or mice. None of these parasites causes significant damage, and infection is usually asymptomatic. There are two occasional exceptions. When people ingest *T. solium* ova from their own intestine or from another infected individual, it can cause somatic infection. *D. latum* avidly absorbs vitamin B₁₂ in the intestine and can cause pernicious anemia in 1–2% of infected Scandinavians with a genetic predisposition.

SOMATIC TAPEWORMS There are three major causes of somatic tapeworm infections. Two species of *Echinococcus* cause echinococcosis. *E. granulosus* is acquired by accidental ingestion of ova from dogs infected when fed the infected tissues of sheep or other animals by sheepherders or hunters. *E. multilocularis* is transmitted primarily in sub-Arctic areas when humans ingest ova from foxes, dogs, or cats that have been infected through consumption of the tissues of infected rodents. Both species cause hydatid cysts when the eggs hatch into larvae, penetrate the intestine, and migrate into the liver or lung. Ingested *T. solium* ova cause somatic disease (cysticercosis) when the larvae penetrate the intestine, migrate into tissue, and form cysts (*cysterci*), usually in the muscles or central nervous system (CNS).

Trematodes Flukes also cause both intestinal and somatic infections (Chap. 241 and Table S12-1). Most fluke infections are localized to Asia, Africa, Southeast Asia, or the Pacific islands. Infection with intestinal flukes is usually asymptomatic, although heavy infections sometimes cause abdominal discomfort and mucous diarrhea. Liver flukes and lung flukes cause somatic infections when humans ingest a larval form from an intermediate host. Adults develop in the intestine, migrate into adjacent tissues, and cause disease. The major liver flukes (*Clonorchis sinensis*, *Opisthorchis* spp., and *Fasciola hepatica*) are causes of recurrent bacterial cholangitis (due to obstruction) or portal hypertension and cirrhosis. Only *F. hepatica* can be acquired worldwide; it is especially common in sheep-raising areas, where the animals ingest water plants (e.g., watercress). The lung flukes (*Paragonimus* spp.) occur globally except in Europe; most lesions occur as pulmonary cysts, although occasional lesions develop in the CNS or the abdominal cavity.

The blood flukes cause schistosomiasis, one of the most common and serious parasitic infections (Chap. 241 and Table S12-1). The major species are *Schistosoma mansoni*, *S. haematobium*, and *S. japonicum*. All are transmitted to humans when free-swimming larvae exit an infected snail in freshwater and penetrate the skin. Swimmer's itch sometimes follows skin penetration but is usually of short duration.

The larvae then wander in the skin until they find a blood vessel and migrate to the target organ. *S. mansoni* and *S. japonicum* migrate to the mesentery vessels and eventually make their way to the liver, while *S. haematobium* targets the veins around the ureter and bladder. Extensive egg deposition by *S. mansoni* and *S. japonicum* and the immune reactions to the ova cause granuloma formation and, with many repeated exposures, portal vein obstruction and cirrhosis. The same process in the ureters and bladders during infection with *S. haematobium* eventually interferes with urine flow and leads to repeated urinary tract infections and kidney damage.

■ ROUNDWORMS

Nematodes Roundworms are nonsegmented bisexual organisms. The species that infect humans include intestinal and tissue groups. Humans may also acquire certain nonhuman mammalian roundworms that either can be limited to the skin or can migrate to tissues and cause serious disease (the larva migrans syndromes).

INTESTINAL ROUNDWORMS The major intestinal roundworms are *Ascaris lumbricoides*, *Necator americanus* (New World hookworms), *Ancylostoma duodenale* (Old World hookworms), *Trichuris trichiura* (whipworms), *Enterobius vermicularis* (pinworms), and *Strongyloides stercoralis*. Taken together, infections caused by intestinal roundworms are the most common infections in the world. *Ascaris*, hookworms, and *Trichuris* infect about 900 million individuals, and at least 100–370 million have strongyloidiasis. These infections are most common in resource-poor developing countries, especially where people defecate outside and/or human feces is used as fertilizer (“night soil”). Infection is transmitted either by ingestion of ova (*A. lumbricoides*, *T. trichiura*, and *E. vermicularis*) or by active penetration of the skin by larvae (hookworms and *S. stercoralis*) (Table S12-2).

Intestinal roundworms cause serious health problems in residents of endemic regions with poor sanitation, but travelers are at low risk of developing significant disease from most of these parasites. Intestinal blockage and malnutrition from heavy *Ascaris* infections and anemia from heavy hookworm infections are now restricted to areas of heavy endemicity. Except in the case of *Strongyloides* and *Capillaria*, which can reproduce in the body, multiple exposures over time are necessary for the development of severe disease. *Strongyloides* infection persists over decades and can disseminate when the immune system is compromised. Although *Capillaria* remains localized to the intestine, infections can become so heavy that protein-losing enteropathy and malnutrition cause serious disease.

The life cycles of *Ascaris* and the hookworms involve migration through the heart and lungs before development into adults in the intestine. In particular, *Ascaris* occasionally causes eosinophilic pneumonia (Loeffler's syndrome) during heavy infections. Pinworms are the most common causes of intestinal roundworm infection persisting in the United States and other developed countries. The anal and perineal itching caused by pinworm migration out of the anus and subsequent egg deposition is well known to families throughout the world.

TISSUE ROUNDWORMS The major diseases caused by tissue roundworms are filariasis, angiostrongyliasis, gnathostomiasis, and trichinellosis. By far, the most important globally is filariasis; the thread-like filarial worms infect an estimated 80 million individuals in tropical and subtropical areas of the world. Four filarial species cause three distinct diseases: lymphatic filariasis (*Wuchereria bancrofti* and *Brugia malayi*), river blindness (*Onchocercus volvulus*), and loiasis (*Loa loa*, the African eye worm). Humans, the major reservoir, acquire these infections from bites of infected arthropods (Table S12-2). The larvae develop into adults, which remain static in tissue: the lymphatics for lymphatic filariasis and subcutaneous tissue for *O. volvulus* and *L. loa*. After adults mate, next-stage larvae are produced, and their migration causes additional damage.

Repeated bouts of migrating larvae and blocking of the lymphatics by adults are necessary to establish the syndrome of lymphatic filariasis; thus, it is unusual for the short-term traveler (<3 months' residence in an endemic region) to develop significant disease. In river blindness, the larvae produced by adult *O. volvulus* migrate through the skin and

eye, causing skin damage and eventual blindness. Loiasis is a milder disease restricted to central and western Africa. Although both the adults and the larvae of *L. loa* migrate through the skin and eye, many infected individuals are asymptomatic, and the infection is often diagnosed only when an adult worm migrates across the subconjunctival tissue and is visible to the patient and the physician. Red lumps in the skin from heavy cutaneous migration are called *Calabar swellings*.

The other four major roundworm tissue infections are acquired by ingestion of larvae in undercooked food. The sources for trichinellosis are swine and other large mammals; for gnathostomiasis, freshwater fish and chicken; for ancylostomiasis, snails, fish, prawns, and crabs; and for Guinea worm, infected water fleas. Guinea worm infection (dracunculiasis, caused by *Dracunculus medinensis*) has been almost eradicated. *Trichinella spiralis* larvae penetrate the intestine and migrate widely, with a preference for skeletal tissue; the release of eosinophils and IgE causes muscle soreness and may cause palpebral swelling and other manifestations of generalized allergic reactions. *Angiostrongylus cantonensis* is the most common parasitic cause of eosinophilic meningitis. Ingested larvae penetrate the intestine and migrate to the brain and meninges, where they quickly die and attract massive numbers of eosinophils. Although complications can occur, most individuals recover spontaneously. *Gnathostoma spinigerum* larvae also penetrate the intestine and migrate, showing a preference for the skin, eyes, and meninges. Mechanical damage from the migration and inflammation produced by the resultant immune reaction can cause boil-like lesions on the skin, painful eye damage, and eosinophilic meningitis. Although eosinophilic meningitis caused by *G. spinigerum* is less common than that caused by *A. cantonensis*, it is often more severe and can result in paralysis or brain hemorrhage.

PROTOZOAL INFECTIONS

■ INTESTINAL PROTOZOA

Entamoeba histolytica is the one intestinal protozoan that causes invasive disease. This disease consists of dysentery or bloody diarrhea that must be differentiated from that due to bacteria such as *Salmonella*, *Campylobacter*, and *Shigella*. Although amebiasis usually has a slower onset with lower fever than these bacterial infections, *E. histolytica* can disseminate from the bloodstream to cause distant abscesses, particularly of the liver. The diagnosis cannot be made by identification of the characteristic cyst or trophozoites (Chap. 230), as they are identical to those of the noninvasive *E. dispar*, which is more common globally.

Cryptosporidium and *Giardia* are the most common water-borne protozoal infections. *Cryptosporidium* can cause major outbreaks because it is highly infectious and resistant to high levels of chlorine (Chap. 236). Without immune reconstitution, immunosuppressed patients, particularly those with AIDS, can develop severe, even fatal watery diarrhea. Infections caused by the remaining intestinal protozoans—*Giardia*, *Isospora*, *Cyclospora*, and microsporidia (Chap. 236)—have a much more indolent course, with intermittent diarrhea. Microsporidia, unique intracellular protozoa that form infectious spores, may cause limited gastrointestinal infection in immunocompetent hosts, but patients with AIDS can develop chronic diarrhea and wasting or disseminated infection to the biliary or respiratory tract.

■ FREE-LIVING AMOEBAS

The free-living amoebas *Acanthamoeba* and *Naegleria* are found worldwide in freshwater and brackish water (Chap. 230 and Table S12-3). Organisms of these two genera cause very different syndromes. In immunocompromised individuals, *Acanthamoeba* may cause invasive infection, with brain masses and skin lesions. However, all humans are susceptible to *Acanthamoeba* keratitis after trauma to the eye and exposure to contaminated water. In contrast, naeglerial meningitis, acquired in warm lakes or hot springs, causes sudden pyogenic and usually fatal meningitis. *Balamuthia*, reported only from the Americas, causes indolent meningoencephalitis, with both cerebrospinal fluid pleocytosis and a space-occupying lesion, in immunocompetent patients. Despite the availability of miltefosine, which is active in vitro against *Naegleria*, infection of the CNS is almost universally fatal.

Plasmodium and Babesia Malaria, caused by six species of *Plasmodium*, carries higher mortality rates than any other parasitic infection (Chap. 231). All species are transmitted in tropical and subtropical areas by female *Anopheles* mosquitoes. *Plasmodium falciparum* is most common in sub-Saharan Africa, where it causes more than 80% of malaria infections and 90% of malarial deaths. Infection with *P. falciparum* may be particularly severe because the organism can invade any erythrocyte, reaches very high parasite loads, damages organs by adhering to vascular epithelium, and is the most likely *Plasmodium* species to be resistant to antimalarial drugs. *Plasmodium vivax*, the dominant cause of malaria outside sub-Saharan Africa, reaches lower levels of parasitemia and exhibits less drug resistance because it invades only reticulocytes with Duffy antigen. Many Africans, especially in the western part of the continent, lack the Duffy blood group; consequently, *Plasmodium ovale*, another cause of milder malaria, can compete successfully with *P. vivax*. Both *P. vivax* and *P. ovale* produce persistent liver forms, which must be treated with primaquine (Chap. 229). Because malaria can cause a variety of symptoms ranging from acute fever to coma, this diagnosis must be considered in any traveler or immigrant from a malarial area. Recently, locally acquired *P. vivax* malaria has been detected in Florida and Texas and *P. falciparum* in Maryland; thus, malaria should be included in the differential diagnoses of unexplained fever even in patients who have never left the United States. *Babesia* also infects erythrocytes and may cause a nonspecific febrile illness or, in asplenic patients, severe infection. This parasite is carried by ixodid ticks and is geographically limited to the northeastern and midwestern United States, with only sporadic cases in Europe and other temperate areas.

Trypanosomes The three species of trypanosomes all have flagellated bloodstream forms, but they cause very different diseases. *T. cruzi*, the cause of Chagas disease, is transmitted in South and Central America in the feces of blood-sucking reduviid bugs (Chap. 234). After initial parasitemia, patients are often asymptomatic for years while the parasite multiplies intracellularly in muscle and ganglion cells. Although only a minority of patients go on to develop organ damage (megaesophagus and cardiomyopathy), all infected patients can spread the disease through transfusions, mother-to-child transmission, and organ transplants.

African trypanosomiasis is limited to sub-Saharan Africa, where it is transmitted by the bite of a tsetse fly. A history of a tsetse bite and the presence of a painful chancre are strong diagnostic clues (Chap. 234). Although the parasites causing this disease in western Africa (*Trypanosoma brucei gambiense*) and eastern Africa (*T. brucei rhodesiense*) look identical, they are genetically and clinically distinct. *T. b. gambiense* causes low-level parasitemia with cyclical fevers over months or years before CNS invasion, whereas *T. b. rhodesiense* causes high-level parasitemia, invades the CNS early on, and can lead to death within weeks of onset.

Leishmania Leishmaniasis is caused by more than 20 species of obligate intracellular protozoa transmitted by sandflies, which are present in almost 100 countries in tropical and temperate zones (Chap. 233). A wide spectrum of clinical symptoms result, ranging from self-healing, painless skin ulcers to mucocutaneous disease with destruction of the nose and palate to disseminated visceral leishmaniasis with hepatic and splenic involvement. The resulting disease depends on the infecting strain and the host immune response. Visceral leishmaniasis can present as an acute febrile illness, with the later development of hepatosplenomegaly, and is an AIDS-defining illness in HIV-infected patients. More than 90% of cases of visceral leishmaniasis occur in India, Bangladesh, Ethiopia, Sudan, and Brazil.

Toxoplasma *Toxoplasma gondii* is an obligate intracellular parasite that is found worldwide. Infection follows ingestion of oocysts in food or water contaminated by cat feces, ingestion of tissue cysts in undercooked meat, or transplacental transmission. After gastrointestinal invasion, tachyzoites can invade any nucleated cell and cause lifelong infection in most patients (Chap. 235). Clinical manifestations depend

1738 on the host's age and immune status at the time of infection. Congenital toxoplasmosis results from primary maternal infection; outcomes are most severe early in pregnancy and include visual, hearing, and cognitive impairments. Babies infected later in pregnancy may appear normal but can develop chorioretinitis decades later. Primary infection in immunocompetent hosts may be asymptomatic, may present as an infectious mononucleosis–like syndrome, or may manifest as chorioretinitis during outbreaks. During immunosuppression by AIDS or organ transplantation, reactivation of latent cerebral infection can be fatal unless diagnosed and treated early.

APPROACH TO THE PATIENT

Parasitic Infection

A thorough history and physical examination are the keys to diagnosis of any disease and particularly of parasitic infections. Because many of the more serious parasitic infections are uncommon in the United States, a travel history, particularly to developing nations, is a critical component. The longer the stay in an area endemic for significant parasitic infections, the greater the risk, even for healthy travelers. In addition, other factors increase the chance of acquiring these infections. Notably, immunocompromise greatly increases the likelihood of developing some of the more serious parasitic infections. Even healthy travelers with adventure itineraries, extensive travel to rural areas, or involvement in war zones or refugee camps are at increased risk. Immigrants from developing countries may seek care for symptoms or signs associated with parasitic infections.

Information on the patient's immunization history and adherence to appropriate malarial chemoprophylaxis is critical. The recent approval of the first parasitic vaccines against *P. falciparum* is very exciting, but it will be targeted initially only for children in high prevalence areas. Typhoid fever is much less likely to be the cause of prolonged fever in an immunized individual. Similarly,

hepatitis A or B is unlikely to be the cause of jaundice and fever in fully immunized patients. In this era of increasing drug resistance, even adherence to appropriate malarial chemoprophylaxis does not guarantee that fever is not malarial. Nevertheless, most travelers who acquire malaria have taken inadequate or no prophylaxis. Although these considerations do not prove that the symptoms are caused by parasites, they narrow the differential diagnosis.

There are many other important aspects of the history, including when symptoms began. Was the individual still in the endemic area at the time, or did the symptoms commence after return to the United States? If they started during travel, was any treatment received? Malaria must be the first consideration in a febrile patient returning from an endemic area. If the patient was well upon return from travel, the timing of symptom onset is a critical point. For example, if the chief manifestation is fever that began >10–14 days after departure from the endemic region, many tropical diseases can be ruled out, including dengue fever, chikungunya fever, and Zika virus infection. On the other hand, fever beginning several months or later after return makes malaria a likely diagnosis. Travelers' diarrhea, the most common complaint of travelers, is usually caused by bacteria or viruses and resolves in a short time with or without treatment. Travelers' diarrhea that persists for weeks is much more likely to be parasitic in origin.

Most patients who consult physicians after international travel either have troublesome symptoms or have been referred for symptoms or signs whose source was unclear to a referring caregiver. After a careful travel history including the individual's symptoms and the exact geographic zones visited, a thorough physical examination must be conducted. The symptoms, signs, and physical findings should help to establish possible diagnoses. [Table 228-1](#) breaks down the symptoms of major parasitic infections by organ system and geographic distribution, with comments on clinical and epidemiologic associations.

TABLE 228-1 Parasitic Infections, by Organ System and Signs/Symptoms^a

ORGAN SYSTEM, MAJOR SIGN(S)/SYMPTOM(S)	PARASITE(S)	GEOGRAPHIC DISTRIBUTION	COMMENTS
Skin			
Serpentine rash	Hookworm	Worldwide	Can cause anemia in heavy infections
	<i>Strongyloides</i>	Moist tropics and subtropics	Disseminated infection in immunocompromise
	<i>Toxocara</i> (animal roundworm)	Tropical and temperate zones	Cutaneous or visceral larva migrans
Itchy skin rash	<i>Onchocerca</i>	Mexico, Central/South America, Africa	Larvae detectable in skin snips and nodules
Painless ulcers	<i>Leishmania</i>	Tropics and subtropics	Amastigotes detectable in biopsies; may cause destructive mucocutaneous infection; AIDS-defining infection
Skin nodules	<i>Onchocerca</i>	Mexico, South America, Africa	Large nodules of adult worms
	<i>Loa loa</i> (African eye worm)	Western and central Africa	Migratory nodules
	<i>Gnathostoma</i>	Southeast Asia and China	Migratory nodules with eosinophilia
Painful nodules, especially involving feet	<i>Dracunculus</i> (Guinea worm)	Africa	Nearly eradicated
Central Nervous System			
Somnolence, seizures, coma	<i>Plasmodium falciparum</i>	Subtropics and tropics	Cerebral malaria, especially in children
	<i>Trypanosoma brucei rhodesiense</i>	Sub-Saharan eastern Africa	Painful chancre from tsetse fly bite; death in weeks to months
Space-occupying lesions, seizures	<i>Acanthamoeba</i>	Worldwide	Immunocompromised individuals
	<i>Balamuthia</i>	Americas	Indolent meningoencephalitis with brain mass
	<i>Toxoplasma</i>	Worldwide	Reactivation disease in immunocompromise; ring-enhancing lesions; AIDS-defining infection
	<i>Taenia solium</i>	Mexico, Central/South America, Africa	Cysticercosis; variable sized or calcified larval cysts on CT
	<i>Schistosoma japonicum</i>	Far East	Aberrant eggs can form brain or spinal cord masses.
	<i>Schistosoma mansoni</i>	Africa, Central/South America	Aberrant eggs can form brain or spinal cord masses.

(Continued)

TABLE 228-1 Parasitic Infections, by Organ System and Signs/Symptoms^a (Continued)

ORGAN SYSTEM, MAJOR SIGN(S)/SYMPTOM(S)	PARASITE(S)	GEOGRAPHIC DISTRIBUTION	COMMENTS
Pyogenic meningitis	<i>Naegleria</i>	Worldwide	Motile trophozoites in fresh cerebrospinal fluid; pyogenic; rapid death
Eosinophilic meningitis	<i>Angiostrongylus</i> (rat lung worm)	Southeast Asia, Pacific, Caribbean	Most common cause globally of eosinophilic meningitis; spontaneous resolution
	<i>Gnathostoma</i>	Southeast Asia and China	Migratory nodules
Eyes			
Painful corneal ulcers	<i>Acanthamoeba</i>	Worldwide	Freshwater and brackish water; corneal trauma; long-wear contact lenses
Corneal opacification	<i>Onchocerca</i>	Mexico, Central/South America, Africa	Immune response to microfilaria in cornea
Congenital or adult visual loss	<i>Toxoplasma</i>	Worldwide	Primary infection in pregnancy and normal hosts; reactivation infection in immunocompromised
Retinal mass	<i>Toxocara</i>	Worldwide	Ocular larva migrans
Visible roundworm in eye	<i>Onchocerca</i>	Mexico, Central/South America, Africa	Worms may cross eye during migration.
	<i>L. loa</i>	Western and central Africa	Worms may cross eye during migration.
Pain, possible vision loss	<i>Gnathostoma</i>	Southeast Asia and China	Migratory skin nodules, eosinophilia
Lungs			
Pulmonary nodule/abscess	<i>Paragonimus</i>	Far East, Africa, Americas	Ectopic migration to abdomen or central nervous system
Cough, transient infiltrates, eosinophilia	Migrating helminths	Worldwide	Loeffler's syndrome from migrating <i>Ascaris</i> , hookworm, <i>Strongyloides</i>
Heart			
Pulmonary edema	<i>P. falciparum</i> (complication)	Tropics and subtropics	End-organ damage from severe malaria
Cardiomegaly, arrhythmias	<i>Trypanosoma cruzi</i>	Mexico, Central/South America	Late amastigote infection of myocardium; AIDS-defining infection
Gastrointestinal Tract			
Hepatosplenomegaly	Malaria (multiple episodes)	Tropics and subtropics	Splenomegaly with anemia and recurrent fever are hallmarks of malaria.
	<i>S. mansoni</i>	Africa, Central/South America	Portal obstruction with cirrhosis and late varices
	<i>Leishmania donovani</i> complex	Tropics and subtropics	Visceral leishmaniasis; AIDS-defining infection
Hepatomegaly	<i>Entamoeba histolytica</i>	Tropics	Acute with fever, right-upper-quadrant pain; or chronic with enlarged liver; hypoechoic abscess(es) on ultrasound or CT
	<i>Echinococcus</i>	Sheep-raising areas	Characteristic cysts of liver > lung
	<i>Fasciola</i>	Sheep-raising areas	Eosinophilia
Cholangitis	<i>Clonorchis</i>	China, Southeast Asia	Recurrent cholangitis and late cholangiocarcinoma
	Microsporidia	Worldwide	AIDS
	<i>Cryptosporidium</i>	Worldwide	AIDS-defining infection
Bloody diarrhea	<i>E. histolytica</i>	Tropics	Less fever than in diarrhea of bacterial etiology
	<i>S. mansoni</i>	Africa, Central/South America	Only in heavy, acute infection with fever and eosinophilia
	<i>S. japonicum</i>	Far East	Only in heavy, acute infection
Watery diarrhea	<i>Cryptosporidium</i>	Worldwide	Severe in immunocompromised patients
	<i>Giardia</i>	Worldwide	Foul-smelling stool with steatorrhea
	<i>Isospora belli</i>	Worldwide	Fever, abdominal pain, chronic diarrhea
	Microsporidia	Worldwide	Chronic diarrhea with AIDS
	<i>Capillaria</i>	Southeast Asia, Egypt	Malabsorption, wasting
Passage of large roundworm (>6 cm)	<i>Ascaris</i>	Worldwide	Patients may confuse the roundworm with an earthworm.
Small roundworms visible around anus	Pinworm	Worldwide	Anal itching; eggs rarely detected by ova and parasite (O&P) exam
	<i>Trichuris</i>	Worldwide	Rectal prolapse with heavy infection in children
Passage of tapeworm segments	<i>T. solium</i> or <i>Taenia saginata</i>	Worldwide	Usual reason for seeking medical care
	<i>Diphyllobothrium latum</i>	Worldwide	Pernicious anemia in genetically predisposed Scandinavians
Genitourinary System			
Itchy discharge	<i>Trichomonas vaginalis</i>	Worldwide	Common sexually transmitted disease of both sexes
Hematuria	<i>Schistosoma haematobium</i>	Africa	Hematuria with negative cultures, urinary tract infections, and late bladder cancer

(Continued)

ORGAN SYSTEM, MAJOR SIGN(S)/SYMPTOM(S)	PARASITE(S)	GEOGRAPHIC DISTRIBUTION	COMMENTS
Muscular System			
Myalgias, myositis	<i>Trichinella</i>	Worldwide	Palpebral swelling; high-level eosinophilia
Bloodstream			
Fever without localizing symptoms	<i>Plasmodium</i>	Tropics and subtropics	Consider in any patient from a malarious area.
	<i>Babesia</i>	New England, United States	Geographically limited; worse with splenectomy
	<i>T. brucei rhodesiense</i> , <i>T. brucei gambiense</i>	Sub-Saharan Africa	Limited to tsetse fly range; painful chancre; adenopathy and cyclical fevers; early (<i>rhodesiense</i>) or late (<i>gambiense</i>) central nervous system involvement
	Filariae	Asia, India	Periodic fever with eosinophilia, adenolymphangitis, chronic lymphangitis
	<i>L. donovani</i> complex	Tropics and subtropics	Hepatosplenomegaly, fever, wasting; AIDS-defining infection

^aSee also text and Tables S12-1, S12-2, and S12-3 for vectors and routes of transmission.

FURTHER READING

BLACKBURN D et al: Outbreak of locally acquired mosquito-transmitted malaria—Florida and Texas, May–July 2023. *MMWR* 72:973, 2023.
CONRAD MD et al: Evolution of partial resistance to artemisinins in malaria parasites in Uganda. *N Engl J Med* 389:722, 2023.
DIAZ AV et al: Reaching the World Health Organization elimination targets for schistosomiasis: The importance of a One Health perspective. *Philos Trans R Soc Lond B Biol Sci* 378:20220274, 2023.
LOUKAS A et al: The yin and yang of human soil-transmitted helminths. *Int J Parasitol* 51:1243, 2021.
RUBIN EJ: Making the worm turn. *N Engl J Med* 388:1908, 2023.

parasitic diseases of humans are listed in subsequent chapters. Many of the agents discussed herein are approved by the U.S. Food and Drug Administration (FDA) but are considered investigational for the treatment of certain infections. Drugs marked in the text with an asterisk (*) are available through the Centers for Disease Control and Prevention (CDC) Drug Service (telephone: 404-639-3670; email: drugservice@cdc.gov). Drugs marked with a dagger (†) are available only through their manufacturers; contact information for these manufacturers may be available from the CDC.

Table 229-1 presents a brief overview of each agent (including some drugs that are covered in other chapters), along with major toxicities, spectrum of activity, and safety for use during pregnancy and lactation.

Albendazole Like all benzimidazoles, albendazole acts by selectively binding to free β-tubulin in nematodes, inhibiting the polymerization of tubulin and the microtubule-dependent uptake of glucose. Irreversible damage occurs in gastrointestinal (GI) cells of the nematodes, resulting in starvation, death, and expulsion by the host. This fundamental disruption of cellular metabolism offers treatment for a wide range of parasitic diseases.

Albendazole is poorly absorbed from the GI tract, a feature that is advantageous for the treatment of intestinal helminths but not for that of tissue helminth infections (e.g., hydatid disease and neurocysticercosis), which requires a sufficient amount of active drug to reach the site of infection. Administration with a high-fat meal (~40 g) increases the drug's absorption by up to fivefold. The metabolite albendazole sulfoxide is responsible for the drug's therapeutic effect outside the gut lumen. Albendazole sulfoxide crosses the blood-brain barrier, reaching a level significantly higher than that achieved in plasma. The high concentrations of albendazole sulfoxide attained in cerebrospinal fluid (CSF) may explain the efficacy of albendazole in the treatment of neurocysticercosis.

Albendazole is extensively metabolized in the liver, but there are few data regarding the drug's use in patients with hepatic disease. Single-dose albendazole therapy in humans is largely without side effects (overall frequency, ≤1%). More prolonged courses (e.g., as administered for cystic and alveolar echinococcal disease) have been associated with liver function abnormalities and bone marrow toxicity. Thus, when prolonged use is anticipated, the drug should be administered in treatment cycles of 28 days interrupted by 14-day intervals off therapy. Prolonged therapy with full-dose albendazole (800 mg/d) should be approached cautiously in patients also receiving drugs with known effects on the cytochrome P450 system.

Amodiaquine Amodiaquine has been widely used in the treatment of malaria for >60 years. Like chloroquine (the other major 4-aminoquinoline), amodiaquine is now of limited use because of the spread of resistance. Amodiaquine interferes with hemozoin formation through complexation with heme. It is rapidly absorbed and acts as a prodrug after oral administration; the principal plasma metabolite, monodesethylamodiaquine, is the predominant antimalarial agent. Amodiaquine

229

Agents Used to Treat Parasitic Infections

Thomas A. Moore



Parasitic infections continue to afflict more than half of the world's population and impose a substantial health burden, particularly in underdeveloped nations, where they are most prevalent. The reach of some parasitic diseases, including malaria, has expanded over the past few decades due to factors such as deforestation, population shifts, global warming, and other climatic events. Although there have been significant advances in vaccine development and vector control, chemotherapy remains the single most effective means of controlling parasitic infections. Efforts to combat the spread of some diseases are hindered by the development and spread of drug resistance, the limited introduction of new antiparasitic agents, the proliferation of counterfeit medications, profiteering, and, most recently, the widespread and unsupported use of antiparasitic agents to treat COVID-19, all of which have dramatically increased the cost of these once-affordable agents. However, there are good reasons to be optimistic. Ambitious global initiatives aimed at controlling or eliminating threats such as AIDS, tuberculosis, and malaria continue to demonstrate success. The ongoing efforts of multinational partnerships to address the substantial burden imposed by neglected tropical diseases have generated new effective antiparasitic agents. In addition, agents approved for other uses are being re-evaluated for antiparasitic efficacy, and some have been subsequently repurposed.

This chapter deals exclusively with the agents used to treat infections due to parasites. Specific treatment recommendations for the

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
4-Aminoquinolines Amodiaquine Chloroquine	Malaria ^b Malaria ^b	Agranulocytosis, hepatotoxicity <i>Occasional</i> : pruritus, nausea, vomiting, headache, hair depigmentation, exfoliative dermatitis, reversible corneal opacity <i>Rare</i> : irreversible retinal injury, nail discoloration, blood dyscrasias	No information Antacids and kaolin: reduced absorption of chloroquine Ampicillin: bioavailability reduced by chloroquine Cimetidine: increased serum levels of chloroquine Cyclosporine: serum levels increased by chloroquine	Not assigned Not assigned ^d	Yes ^c Yes ^c
Piperaquine	Malaria ^b	<i>Occasional</i> : GI disturbances	None reported	Not assigned	Yes
8-Aminoquinolines Primaquine	Malaria ^b	<i>Frequent</i> : hemolysis in patients with G6PD deficiency <i>Occasional</i> : methemoglobinemia, GI disturbances <i>Rare</i> : CNS symptoms	Quinacrine: potentiated toxicity of primaquine	Contraindicated	Yes
Tafenoquine	Malaria ^b	<i>Frequent</i> : hemolysis in patients with G6PD deficiency, mild GI upset <i>Occasional</i> : methemoglobinemia, headache	No information	Not assigned	Yes
Aminoalcohols Halofantrine	Malaria ^b	<i>Frequent</i> : abdominal pain, diarrhea <i>Occasional</i> : ECG disturbances (dose-related prolongation of QTc and PR interval), nausea, pruritus; contraindicated in persons who have cardiac disease or who have taken mefloquine in the preceding 3 weeks	Concomitant use of agents that prolong QTc interval contraindicated	C	No information
Lumefantrine	Malaria ^b	<i>Occasional</i> : nausea, vomiting, diarrhea, abdominal pain, anorexia, headache, dizziness	Plasma levels increased by darunavir and nevirapine, decreased by etravirine	Not assigned	No information
Aminoglycosides Paromomycin	Amebiasis, ^b infection with <i>Dientamoeba fragilis</i> , giardiasis, cryptosporidiosis, leishmaniasis	<i>Frequent</i> : GI disturbances (oral dosing only) <i>Occasional</i> : nephrotoxicity, ototoxicity, vestibular toxicity (parenteral dosing only)	No major interactions	Oral: B Parenteral: not assigned ^d	No information
Amphotericin B Amphotericin B deoxycholate Amphotec (InterMune) Amphotericin B lipid complex, ABLC (Abelcet) Amphotericin B, liposomal (AmBisome)	Leishmaniasis, ^e amebic meningoencephalitis	<i>Frequent</i> : fever, chills, hypokalemia, hypomagnesemia, nephrotoxicity <i>Occasional</i> : vomiting, dyspnea, hypotension	Antineoplastic agents: renal toxicity, bronchospasm, hypotension Glucocorticoids, ACTH, digitalis: hypokalemia Zidovudine: increased myelo- and nephrotoxicity	B	No information
Antimonials Meglumine antimoniate ^h	Leishmaniasis	<i>Frequent</i> : arthralgias/myalgias, pancreatitis, ECG changes (QT prolongation, T-wave flattening or inversion)	Antiarrhythmics and tricyclic antidepressants: increased risk of cardiotoxicity	Not assigned	No information
Artemisinin and derivatives Arteether Artemether Artesunate ^h Dihydroartemisinin	Malaria ^g	<i>Occasional</i> : neurotoxicity (ataxia, convulsions), nausea, vomiting, anorexia, contact dermatitis	No information Artemether levels decreased by darunavir, etravirine, and nevirapine Mefloquine: levels decreased and clearance accelerated by artesunate Mefloquine: increased absorption	Not assigned C C Not assigned	Yes ^c Yes ^c Yes ^c Yes ^c

(Continued)

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections (Continued)

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
Atovaquone	Malaria, ^b babesiosis	<i>Frequent:</i> nausea, vomiting <i>Occasional:</i> abdominal pain, headache	Plasma levels decreased by rifampin, tetracycline, atazanavir, efavirenz, lopinavir/ritonavir; bioavailability decreased by metoclopramide	C	No information
Azoles Fluconazole Itraconazole Ketoconazole	Leishmaniasis	<i>Serious:</i> hepatotoxicity <i>Rare:</i> exfoliative skin disorders, anaphylaxis	Warfarin, oral hypoglycemics, phenytoin, cyclosporine, theophylline, digoxin, dofetilide, quinidine, carbamazepine, rifabutin, busulfan, docetaxel, vinca alkaloids, pimozide, alprazolam, diazepam, midazolam, triazolam, verapamil, atorvastatin, cerivastatin, lovastatin, simvastatin, tacrolimus, sirolimus, indinavir, ritonavir, saquinavir, alfentanil, buspirone, methylprednisolone, trimetrexate: plasma levels increased by azoles Carbamazepine, phenobarbital, phenytoin, isoniazid, rifabutin, rifampin, antacids, H ₂ -receptor antagonists, proton pump inhibitors, nevirapine: decreased plasma levels of azoles Clarithromycin, erythromycin, indinavir, ritonavir: increased plasma levels of azoles	C	Yes
Benzimidazoles Albendazole	Ascariasis, capillariasis, clonorchiasis, cutaneous larva migrans, cysticercosis, ^b echinococcosis, ^b enterobiasis, eosinophilic enterocolitis, gnathostomiasis, hookworm, lymphatic filariasis, microsporidiosis, strongyloidiasis, trichinellosis, trichostrongyliasis, trichuriasis, visceral larva migrans	<i>Occasional:</i> nausea, vomiting, abdominal pain, headache, reversible alopecia, elevated aminotransferases <i>Rare:</i> leukopenia, rash	Dexamethasone, praziquantel: plasma level of albendazole sulfoxide increased by ~50%	C	Yes ^c
Mebendazole	Ascariasis, ^b capillariasis, eosinophilic enterocolitis, enterobiasis, ^b hookworm, ^b trichinellosis, trichostrongyliasis, trichuriasis, ^b visceral larva migrans	<i>Occasional:</i> diarrhea, abdominal pain, elevated aminotransferases <i>Rare:</i> agranulocytosis, thrombocytopenia, alopecia	Cimetidine: inhibited mebendazole metabolism	C	No information
Thiabendazole	Strongyloidiasis, ^b cutaneous larva migrans, ^b visceral larva migrans ^b	<i>Frequent:</i> anorexia, nausea, vomiting, diarrhea, headache, dizziness, asparagus-like urine odor <i>Occasional:</i> drowsiness, giddiness, crystalluria, elevated aminotransferases, psychosis <i>Rare:</i> hepatitis, seizures, angioneurotic edema, Stevens-Johnson syndrome, tinnitus	Theophylline: serum levels increased by thiabendazole	C	No information
Triclabendazole	Fascioliasis, paragonimiasis	<i>Occasional:</i> abdominal cramps, diarrhea, biliary colic, transient headache	No information	Not assigned	Yes
Benznidazole	Chagas disease	<i>Frequent:</i> rash, pruritus, nausea, leukopenia, paresthesias	No major interactions	Not assigned	No information
Clindamycin	Babesiosis, malaria, toxoplasmosis	<i>Occasional:</i> pseudomembranous colitis, abdominal pain, diarrhea, nausea/vomiting <i>Rare:</i> pruritus, skin rashes	No major interactions	B	Yes ^c
Diloxanide furoate	Amebiasis	<i>Frequent:</i> flatulence <i>Occasional:</i> nausea, vomiting, diarrhea <i>Rare:</i> pruritus	None reported	Contraindicated	No information

(Continued)

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections (Continued)

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
Eflornithine ^f (difluoromethylornithine, DFMO)	Trypanosomiasis	<i>Frequent:</i> pancytopenia <i>Occasional:</i> diarrhea, seizures <i>Rare:</i> transient hearing loss	No major interactions	Contraindicated	No information
Emetine and dehydroemetine	Amebiasis, fascioliasis	<i>Severe:</i> cardiotoxicity <i>Frequent:</i> pain at injection site <i>Occasional:</i> dizziness, headache, GI symptoms	None reported	X	No information
Folate antagonists					
Dihydrofolate reductase inhibitors					
Pyrimethamine	Malaria, ^b isosporiasis, toxoplasmosis ^b	<i>Occasional:</i> folate deficiency <i>Rare:</i> rash, seizures, severe skin reactions (toxic epidermal necrolysis, erythema multiforme, Stevens-Johnson syndrome)	Sulfonamides, proguanil, zidovudine: increased risk of bone marrow suppression when used concomitantly	C	Yes
Proguanil and chlorproguanil	Malaria	<i>Occasional:</i> urticaria <i>Rare:</i> hematuria, GI disturbances	Atazanavir, efavirenz, lopinavir/ritonavir: plasma levels of proguanil decreased	C	Yes
Trimethoprim	Cyclosporiasis, isosporiasis	Hyperkalemia, GI upset, mild stomatitis	Methotrexate: reduced clearance Warfarin: effect prolonged Phenytoin: hepatic metabolism increased	C	Yes
Dihydropteroate synthetase inhibitors: sulfonamides	Malaria, ^b toxoplasmosis ^b	<i>Frequent:</i> GI disturbances, allergic skin reactions, crystalluria <i>Rare:</i> severe skin reactions (toxic epidermal necrolysis, erythema multiforme, Stevens-Johnson syndrome), agranulocytosis, aplastic anemia, hypersensitivity of the respiratory tract, hepatitis, interstitial nephritis, hypoglycemia, aseptic meningitis	Thiazide diuretics: increased risk of thrombocytopenia in elderly patients Warfarin: effect prolonged by sulfonamides Methotrexate: levels increased by sulfonamides Phenytoin: metabolism impaired by sulfonamides Sulfonylureas: effect prolonged by sulfonamides	B	Yes
Sulfadiazine					
Sulfamethoxazole					
Sulfadoxine					
Dihydropteroate synthetase inhibitors: sulfones					
Dapsone	Leishmaniasis, malaria, toxoplasmosis	<i>Frequent:</i> rash, anorexia <i>Occasional:</i> hemolysis, methemoglobinemia, neuropathy, allergic dermatitis, anorexia, nausea, vomiting, tachycardia, headache, insomnia, psychosis, hepatitis <i>Rare:</i> agranulocytosis	Rifampin: lowered plasma levels of dapsone	C	Yes
Fumagillin	Microsporidiosis	<i>Rare:</i> neutropenia, thrombocytopenia	None reported	No information	No information
Furazolidone	Giardiasis	<i>Frequent:</i> nausea/vomiting, brown urine <i>Occasional:</i> rectal itching, headache <i>Rare:</i> hemolytic anemia, disulfiram-like reactions, MAO inhibitor interactions	Risk of hypertensive crisis when administered for >5 days with MAO inhibitors	C	No information
Iodoquinol	Amebiasis, ^b balantidiasis, <i>D. fragilis</i> infection	<i>Occasional:</i> headache, rash, pruritus, thyrotoxicosis, nausea, vomiting, abdominal pain, diarrhea <i>Rare:</i> optic neuritis, peripheral neuropathy, seizures, encephalopathy	No major interactions	C	No information
Lactones					
Ivermectin	Ascariasis, cutaneous larva migrans, gnathostomiasis, loiasis, lymphatic filariasis, onchocerciasis, ^b scabies, strongyloidiasis, ^b trichuriasis	<i>Occasional:</i> fever, pruritus, headache, myalgias <i>Rare:</i> hypotension	No major interactions	C	Yes ^c
Moxidectin	Onchocerciasis	<i>Occasional:</i> fever, pruritus, headache, myalgias <i>Rare:</i> orthostatic hypotension, elevated transaminases	No major interactions	C	Yes ^c

(Continued)

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections (Continued)

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
Macrolides					
Azithromycin	Babesiosis	<i>Occasional:</i> nausea, vomiting, diarrhea, abdominal pain <i>Rare:</i> angioedema, cholestatic jaundice	Cyclosporine and digoxin: levels increased by azithromycin Nelfinavir: increased levels of azithromycin	B	Yes
Spiramycin ^h	Toxoplasmosis	<i>Occasional:</i> GI disturbances, transient skin eruptions <i>Rare:</i> thrombocytopenia, QT prolongation in an infant, cholestatic hepatitis	No major interactions	Not assigned ^d	Yes ^c
Mefloquine	Malaria ^b	<i>Frequent:</i> lightheadedness, nausea, headache <i>Occasional:</i> confusion; nightmares; insomnia; visual disturbance; transient and clinically silent ECG abnormalities, including sinus bradycardia, sinus arrhythmia, first-degree AV block, prolongation of QTc interval, and abnormal T waves <i>Rare:</i> psychosis, convulsions, hypotension	Administration of halofantrine <3 weeks after mefloquine use may produce fatal QTc prolongation. Mefloquine may lower plasma levels of anticonvulsants. Levels are decreased and clearance is accelerated by artesunate. Mefloquine decreases plasma levels of ritonavir and possibly other protease inhibitors.	C	Yes
Melarsoprol ^f	Trypanosomiasis	<i>Frequent:</i> myocardial injury, encephalopathy, peripheral neuropathy, hypertension <i>Occasional:</i> G6PD-induced hemolysis, erythema nodosum leprosum <i>Rare:</i> hypotension	No major interactions	Not assigned	No information
Metrifonate	Schistosomiasis	<i>Frequent:</i> abdominal pain, nausea, vomiting, diarrhea, headache, vertigo, bronchospasm <i>Rare:</i> cholinergic symptoms	No major interactions	B	No
Miltefosine	Leishmaniasis, ^b primary amebic meningoencephalitis	<i>Frequent:</i> mild and transient (1–2 days) GI disturbances within first 2 weeks of therapy (resolve after treatment completion); motion sickness <i>Occasional:</i> reversible elevations of creatinine and aminotransferases	No major interactions	Not assigned	No information
Niclosamide	Intestinal cestode infections ^b	<i>Occasional:</i> nausea, vomiting, dizziness, pruritus	No major interactions	B	No information
Nifurtimox	Chagas disease	<i>Frequent:</i> nausea, vomiting, abdominal pain, insomnia, paresthesias, weakness, tremors <i>Rare:</i> seizures (all reversible and dose-related)	No major interactions	Not assigned	No information
Nitazoxanide	Cryptosporidiosis, ^b giardiasis ^b	<i>Occasional:</i> abdominal pain, diarrhea <i>Rare:</i> vomiting, headache	Increases plasma levels of highly protein-bound drugs (e.g., phenytoin, warfarin)	B	No information
Nitroimidazoles					
Metronidazole	Amebiasis, ^b balantidiasis, dracunculiasis, giardiasis ^b , trichomoniasis, ^b <i>D. fragilis</i> infection	<i>Frequent:</i> nausea, headache, anorexia, metallic aftertaste <i>Occasional:</i> vomiting, insomnia, vertigo, paresthesias, disulfiram-like effects <i>Rare:</i> seizures, peripheral neuropathy	Warfarin: effect enhanced by metronidazole Disulfiram: psychotic reaction Phenobarbital, phenytoin: accelerate elimination of metronidazole Lithium: serum levels elevated by metronidazole Cimetidine: prolonged half-life of metronidazole Oral solutions of antiretrovirals containing alcohol: disulfiram effect due to alcohol	B	Yes
Tinidazole	Amebiasis, ^b giardiasis, trichomoniasis	<i>Occasional:</i> nausea, vomiting, metallic taste	See metronidazole	C	Yes

(Continued)

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections (Continued)

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
Oxamniquine	Schistosomiasis	<i>Occasional:</i> dizziness, drowsiness, headache, orange urine, elevated aminotransferases <i>Rare:</i> seizures	No major interactions	C	No information
Pentamidine isethionate	Leishmaniasis, trypanosomiasis	<i>Frequent:</i> hypotension, hypoglycemia, pancreatitis, sterile abscesses at IM injection sites, GI disturbances, reversible renal failure <i>Occasional:</i> hepatotoxicity, cardiotoxicity, delirium <i>Rare:</i> anaphylaxis	No major interactions	C	No information
Piperazine and derivatives					
Piperazine	Ascariasis, enterobiasis	<i>Occasional:</i> nausea, vomiting, diarrhea, abdominal pain, headache <i>Rare:</i> neurotoxicity, seizures	None reported	C	No information
Diethylcarbamazine ^f	Lymphatic filariasis, loiasis, tropical pulmonary eosinophilia	<i>Frequent:</i> dose-related nausea, vomiting <i>Rare:</i> fever, chills, arthralgias, headache	None reported	Not assigned ^d	No information
Praziquantel	Clonorchiasis, ^b cysticercosis, diphyllbothriasis, hymenolepiasis, taeniasis, opisthorchiasis, intestinal trematodes, paragonimiasis, schistosomiasis ^b	<i>Frequent:</i> abdominal pain, diarrhea, dizziness, headache, malaise <i>Occasional:</i> fever, nausea <i>Rare:</i> pruritus, singultus	No major interactions	B	Yes
Pyrantel pamoate	Ascariasis, eosinophilic enterocolitis, enterobiasis, ^b hookworm, trichostrongyliasis	<i>Occasional:</i> GI disturbances, headache, dizziness, elevated aminotransferases	No major interactions	C	No information
Pyronaridine	Malaria	<i>Occasional:</i> headache, nausea	None reported to date	B	Yes
Quinacrine ^h	Giardiasis ^b	<i>Frequent:</i> headache, nausea, vomiting, bitter taste <i>Occasional:</i> yellow-orange discoloration of skin, sclerae, urine; begins after 1 week of treatment and lasts up to 4 months after drug discontinuation <i>Rare:</i> psychosis, exfoliative dermatitis, retinopathy, G6PD-induced hemolysis, exacerbation of psoriasis, disulfiram-like effects	Primaquine: toxicity potentiated by quinacrine	C	No information
Quinine and quinidine	Malaria, babesiosis	<i>Frequent:</i> cinchonism (tinnitus, high-tone deafness, headache, dysphoria, nausea, vomiting, abdominal pain, visual disturbances, postural hypotension), hyperinsulinemia resulting in life-threatening hypoglycemia <i>Occasional:</i> deafness, hemolytic anemia, arrhythmias, hypotension due to rapid IV infusion	Carbonic anhydrase inhibitors, thiazide diuretics: reduced renal elimination of quinidine Amiodarone, cimetidine: increased quinidine levels Nifedipine: decreased quinidine levels; quinidine slows metabolism of nifedipine Phenobarbital, phenytoin, rifampin: accelerated hepatic elimination of quinidine Verapamil: reduced hepatic clearance of quinidine Diltiazem: decreased clearance of quinidine	X	Yes ^c
Quinolones					
Ciprofloxacin	Cyclosporiasis, isosporiasis	<i>Occasional:</i> nausea, diarrhea, vomiting, abdominal pain/discomfort, headache, restlessness, rash <i>Rare:</i> myalgias/arthralgias, tendon rupture, CNS symptoms (nervousness, agitation, insomnia, anxiety, nightmares, or paranoia); convulsions	Probenecid: increased serum levels of ciprofloxacin Theophylline, warfarin: serum levels increased by ciprofloxacin	C	Yes

(Continued)

TABLE 229-1 Overview of Agents Used for the Treatment of Parasitic Infections (Continued)

DRUGS BY CLASS	PARASITIC INFECTION(S)	ADVERSE EFFECTS	MAJOR DRUG-DRUG INTERACTIONS	PREGNANCY CLASS ^a	BREAST MILK
Suramin ^f	Trypanosomiasis	<i>Frequent:</i> immediate: fever, urticaria, nausea, vomiting, hypotension; delayed (up to 24 h): exfoliative dermatitis, stomatitis, paresthesias, photophobia, renal dysfunction <i>Occasional:</i> nephrotoxicity, adrenal toxicity, optic atrophy, anaphylaxis	No major interactions	Not assigned	No information
Tetracyclines	Balantidiasis, <i>D. fragilis</i> infection, malaria; lymphatic filariasis (doxycycline)	<i>Frequent:</i> GI disturbances <i>Occasional:</i> photosensitivity dermatitis <i>Rare:</i> exfoliative dermatitis, esophagitis, hepatotoxicity	Warfarin: effect prolonged by tetracyclines	D	Yes

^aBased on U.S. Food and Drug Administration (FDA) pregnancy categories of A–D, X. ^bApproved by the FDA for this indication. ^cNot believed to be harmful. ^dUse in pregnancy is recommended by international organizations outside the United States. ^eOnly AmBisome has been approved by the FDA for this indication. ^fAvailable through the CDC. ^gOnly artemether (in combination with lumefantrine) and artesunate have been approved by the FDA for this indication. ^hAvailable through the manufacturer.

Abbreviations: ACTH, adrenocorticotropic hormone; AV, atrioventricular; CNS, central nervous system; ECG, electrocardiogram; G6PD, glucose 6-phosphate dehydrogenase; GI, gastrointestinal; MAO, monoamine oxidase.

and its metabolites are all excreted in urine, but there are no recommendations concerning dosage adjustment in patients with impaired renal function. Agranulocytosis and hepatotoxicity can develop with repeated use; therefore, this drug should not be used for prophylaxis. Indeed, because of its adverse effects and widespread resistance, amodiaquine is no longer in use in Europe or the United States, and it was dropped from malaria control programs as a single agent by the WHO in 1990; however, it remains effective in some areas when combined with other antimalarial drugs (e.g., artesunate, sulfadoxine-pyrimethamine).

Amphotericin B See Table 229-1 and Chap. 217.

Antimonials[†] Despite associated adverse reactions and the need for prolonged parenteral treatment, the pentavalent antimonial compounds (designated Sb^v) have remained the first-line therapy for all forms of leishmaniasis throughout the world, primarily because they are affordable and effective and have survived the test of time. Pentavalent antimonials are active only after bioreduction to the trivalent Sb(III) form, which inhibits trypanothione reductase, a critical enzyme involved in the oxidative stress management of *Leishmania* species. The fact that *Leishmania* species use trypanothione rather than glutathione (which is used by mammalian cells) may explain the parasite-specific activity of antimonials. The drugs are taken up by the reticuloendothelial system, and their activity against *Leishmania* species may be enhanced by this localization.

Resistance is a major problem in some areas. Although low-level unresponsiveness to Sb^v was identified in India in the 1970s, incremental increases in both the recommended daily dosage (to 20 mg/kg) and the duration of treatment (to 28 days) satisfactorily compensated for the growing resistance until around 1990. Since then, the capacity of Sb^v to induce long-term cure in patients with kala-azar who live in eastern India has steadily eroded. Co-infection with HIV impairs the treatment response.

Pentavalent antimonials are available in aqueous solution and are administered parenterally. Antimony appears to have two elimination phases. When the drug is administered IV, the mean half-life of the first phase is <2 h; the mean half-life of the terminal elimination phase is nearly 36 h. This slower phase may be due to conversion of pentavalent antimony to a trivalent form that is the likely cause of the side effects often seen with prolonged therapy.

In 2020, the global manufacturer of sodium stibogluconate notified the CDC that the product would be discontinued due to the inability to source the necessary raw materials. As a result, CDC ended its distribution program following expiration of the last lot in use. However, meglumine antimoniate is available from the manufacturer through the FDA (301-796-1400).

Artemisinin Derivatives* Artesunate, artemether, artemotil, and the parent compound artemisinin are sesquiterpene lactones derived from the wormwood plant *Artemisia annua*. These agents are at least

10-fold more potent in vivo than other antimalarial drugs and presently show no cross-resistance with known antimalarial drugs; thus, they have become first-line agents for the treatment of severe falciparum malaria. The artemisinin compounds are rapidly effective against the asexual blood forms of *Plasmodium* species but are not active against intrahepatic forms. With the exception of artesunate, artemisinin and its derivatives are highly lipid soluble and readily cross both host and parasite cell membranes. One factor that explains the drugs' highly selective toxicity against malaria is that parasitized erythrocytes concentrate artemisinin and its derivatives to concentrations 100-fold higher than those in uninfected erythrocytes. The antimalarial effect of these agents results primarily from the active metabolite dihydroartemisinin; in the presence of heme or molecular iron, the endoperoxide moiety of dihydroartemisinin decomposes, generating free radicals and other metabolites that damage parasite proteins. The compounds are available for oral, rectal, IV, or IM administration, depending on the derivative. Following FDA approval in May 2020, artesunate became commercially available in the United States.

Artemisinin and its derivatives are cleared rapidly from the circulation. Their short half-lives limit their value for prophylaxis and monotherapy. Side effects appear to be minor, although sinus bradycardia and transient first-degree heart block have been reported. Although seen in animal models, embryotoxicity and neurotoxicity have not been identified in humans despite active investigation. These agents should be used only in combination with another, longer-acting agent (e.g., artesunate-mefloquine, dihydroartemisinin-piperaquine). A combined formulation of artemether and lumefantrine is widely available for the treatment of acute uncomplicated falciparum malaria acquired in areas where *Plasmodium falciparum* is resistant to chloroquine and antifolates.

Atovaquone Atovaquone is a hydroxynaphthoquinone that exerts broad-spectrum antiprotozoal activity via selective inhibition of parasite mitochondrial electron transport. This agent exhibits potent activity against toxoplasmosis and babesiosis when used with pyrimethamine and azithromycin, respectively. Atovaquone possesses a novel mode of action against *Plasmodium* species, inhibiting the electron transport system at the level of the cytochrome bc1 complex. The drug is active against both the erythrocytic and the exoerythrocytic stages of *Plasmodium* species; however, because it does not eradicate hypnozoites from the liver, patients with *P. vivax* or *P. ovale* infections must be given radical prophylaxis.

Malarone is a fixed-dose combination of atovaquone and proguanil used for malaria prophylaxis as well as for the treatment of acute, uncomplicated *P. falciparum* malaria. Malarone has been shown to be effective in regions with multidrug-resistant *P. falciparum*. Resistance to atovaquone develops rapidly via mutations in the parasite's mitochondrial cytochrome b complex. However, the mutations result in sterility of female parasites; thus, atovaquone-resistant parasites cannot

be transmitted to another person. This situation may explain why clinical resistance has yet to be reported.

The bioavailability of atovaquone varies considerably. Absorption after a single oral dose is slow, increases two- to threefold with a fatty meal, and is dose-limited above 750 mg. The elimination half-life is increased in patients with moderate hepatic impairment. Because of the potential for drug accumulation, the use of atovaquone is generally contraindicated in persons with a creatinine clearance rate <30 mL/min. No dosage adjustments are needed in patients with mild to moderate renal impairment.

Azithromycin See Table 229-1 and Chap. 149.

Azoles See Table 229-1 and Chap. 217.

Benznidazole Introduced in 1971, this oral nitroimidazole derivative is used to treat Chagas disease, with cure rates of 80–90% recorded in acute infections. Benznidazole is believed to exert its trypanocidal effects by generating oxygen radicals to which the parasites are more sensitive than mammalian cells because of a relative deficiency in antioxidant enzymes. Benznidazole also appears to alter the balance between pro- and anti-inflammatory mediators by downregulating the synthesis of nitrite, interleukin (IL) 6, and IL-10 in macrophages. Benznidazole is highly lipophilic and readily absorbed. The drug is extensively metabolized; only 5% of the dose is excreted unchanged in the urine. Benznidazole is well tolerated; adverse effects are rare and usually manifest as GI upset or pruritic rash. Following FDA approval in 2018, this drug is now commercially available in the United States.

Chloroquine This 4-aminoquinoline has marked, rapid schizonticidal and gametocidal activity against blood forms of *P. ovale* and *P. malariae* and against susceptible strains of *P. vivax* and *P. falciparum*. It is not active against intrahepatic forms (*P. vivax* and *P. ovale*). Parasitized erythrocytes accumulate chloroquine in significantly greater concentrations than do normal erythrocytes. Chloroquine, a weak base, concentrates in the food vacuoles of intraerythrocytic parasites because of a relative pH gradient between the extracellular space and the acidic food vacuole. Once it enters the acidic food vacuole, chloroquine is rapidly converted to a membrane-impermeable protonated form and is trapped. Continued accumulation of chloroquine in the parasite's acidic food vacuoles results in drug levels that are 600-fold higher at this site than in plasma. The high accumulation of chloroquine results in an increase in pH within the food vacuole to a level above that required for the acid proteases' optimal activity, inhibiting parasite heme polymerase; as a result, the parasite is effectively killed with its own metabolic waste. Compared with susceptible strains, chloroquine-resistant plasmodia transport chloroquine out of intraparasitic compartments more rapidly and maintain lower chloroquine concentrations in their acid vesicles. Hydroxychloroquine, a congener of chloroquine, is equivalent to chloroquine in its antimalarial efficacy but is preferred to chloroquine for the treatment of autoimmune disorders because it produces less ocular toxicity when used in high doses.

Chloroquine is well absorbed. However, because it exhibits extensive tissue binding, a loading dose is required to yield effective plasma concentrations. A therapeutic drug level in plasma is reached 2–3 h after oral administration (the preferred route). Chloroquine can be administered IV, but excessively rapid parenteral administration can result in seizures and death from cardiovascular collapse. The mean half-life of chloroquine is 4 days, but the rate of excretion decreases as plasma levels decline, making once-weekly administration possible for prophylaxis in areas with sensitive strains. About one-half of the parent drug is excreted in urine, but the dose should not be reduced for persons with acute malaria and renal insufficiency.

Ciprofloxacin See Table 229-1 and Chap. 149.

Clindamycin See Table 229-1 and Chap. 149.

Dapsone See Table 229-1 and Chap. 188.

Dehydroemetine Emetine is an alkaloid derived from ipecac; dehydroemetine is synthetically derived from emetine and is considered less

toxic. Both agents are active against *Entamoeba histolytica* and appear to work by blocking peptide elongation and thus inhibiting protein synthesis. Emetine is rapidly absorbed after parenteral administration, rapidly distributed throughout the body, and slowly excreted in the urine in unchanged form. Both agents are contraindicated in patients with renal disease.

Diethylcarbamazine* A derivative of the antihelminthic agent piperazine with a long history of successful use, diethylcarbamazine (DEC) remains the treatment of choice for lymphatic filariasis and loiasis and has also been used for visceral larva migrans. Although piperazine itself has no antifilarial activity, the piperazine ring of DEC is essential for the drug's activity. DEC's mechanism of action remains to be fully defined. Proposed mechanisms include immobilization due to inhibition of parasite cholinergic muscle receptors, disruption of microtubule formation, and alteration of helminthic surface membranes resulting in enhanced killing by the host's immune system. DEC enhances adherence properties of eosinophils. The development of resistance under drug pressure (i.e., a progressive decrease in efficacy when the drug is used widely in human populations) has not been observed, although DEC has variable effects when administered to persons with filariasis. Monthly administration provides effective prophylaxis against both bancroftian filariasis and loiasis.

DEC is well absorbed after oral administration, with peak plasma concentrations reached within 1–2 h. No parenteral form is available. The drug is eliminated largely by renal excretion, with <5% found in feces. If more than one dose is to be administered to an individual with renal dysfunction, the dose should be reduced commensurate with the reduction in creatinine clearance rate. Alkalinization of the urine prevents renal excretion and increases the half-life of DEC. Use in patients with onchocerciasis can precipitate a Mazzotti reaction, with pruritus, fever, and arthralgias. Like other piperazines, DEC is active against *Ascaris* species. Patients co-infected with this nematode may expel live worms after treatment.

Diloxanide Furoate Diloxanide furoate, a substituted acetanilide, is a luminally active agent used to eradicate the cysts of *E. histolytica*. After ingestion, diloxanide furoate is hydrolyzed by enzymes in the lumen or mucosa of the intestine, releasing furoic acid and the ester diloxanide; the latter acts directly as an amebicide.

Diloxanide furoate is given alone to asymptomatic cyst passers. For patients with active amebic infections, diloxanide is generally administered in combination with a 5-nitroimidazole such as metronidazole or tinidazole. Diloxanide furoate is rapidly absorbed after oral administration. When coadministered with a 5-nitroimidazole, diloxanide levels peak within 1 h and disappear within 6 h. About 90% of an oral dose is excreted in the urine within 48 h, chiefly as the glucuronide metabolite. Diloxanide furoate is contraindicated in pregnant and breast-feeding women and in children <2 years of age.

Eflornithine* Eflornithine (difluoromethylornithine, or DFMO) is a fluorinated analogue of the amino acid ornithine. Although originally designed as an antineoplastic agent, eflornithine has proven effective against some trypanosomatids.

Eflornithine has specific activity against all stages of infection with *Trypanosoma brucei gambiense*; however, it is inactive against *T. b. rhodesiense*. The drug acts as an irreversible suicide inhibitor of ornithine decarboxylase, the first enzyme in the biosynthesis of the polyamines putrescine and spermidine. Polyamines are essential for the synthesis of trypanothione, an enzyme required for the maintenance of intracellular thiols in the correct redox state and for the removal of reactive oxygen metabolites. However, polyamines are also essential for cell division in eukaryotes, and ornithine decarboxylase is similar in trypanosomes and mammals. The selective antiparasitic activity of eflornithine is partly explained by the structure of the trypanosomal enzyme, which lacks a 36-amino-acid C-terminal sequence found on mammalian ornithine decarboxylase. This difference results in a lower turnover of ornithine decarboxylase and a more rapid decrease of polyamines in trypanosomes than in the mammalian host. The diminished effectiveness of eflornithine against *T. b. rhodesiense* appears to be due

to the parasite's ability to replace the inhibited enzyme more rapidly than *T. b. gambiense*.

Eflornithine is less toxic but more costly than conventional therapy. It can be administered IV or PO. The dose should be reduced in renal failure. Eflornithine readily crosses the blood-brain barrier; CSF levels are highest in persons with the most severe central nervous system (CNS) involvement.

Flubendazole This agent, a methylcarbamate benzimidazole, is highly active against a broad spectrum of gut nematodes and filaria. Its antihelminthic effect is similar to other benzimidazoles; however, it is also an effective inducer of reactive oxygen species and disrupts glucose transport and metabolism. It has limited solubility in water, so bioavailability is very low, but absorption increases if taken after meals, and is further enhanced when reconstituted as a cyclodextrin-based formulation. It is approved for human use in Europe but not in the United States.

Fosmidomycin Originally developed in the 1970s as an antibiotic, this *Streptomyces*-derived agent is an aminopropylphosphonic acid that exhibits antimalarial activity through inhibition of 1-deoxy-D-xylulose 5-phosphate reductoisomerase, an essential enzyme on the non-mevalonate pathway of isoprenoid biosynthesis, a pathway that is common in most eukaryotes but absent in humans. For >15 years, fosmidomycin has been evaluated alone and in combination with other antimalarials, and it has demonstrated both safety and efficacy. However, the drug is highly charged—resulting in a short plasma half-life (3.5 h)—causing high rates of recrudescence malaria, particularly in children. It is not currently available, but clinical studies are ongoing.

Fumagillin[†] Originally discovered as an antiangiogenic compound derived from the fungus *Aspergillus fumigatus*, fumagillin is a water-insoluble antibiotic that is active against microsporidia and is used topically to treat ocular infections due to *Encephalitozoon* species. When given systemically, fumagillin was effective but caused thrombocytopenia in all recipients in the second week of treatment; this side effect was readily reversed when administration of the drug was stopped. Fumagillin acts by binding to methionine aminopeptidase 2, thus inhibiting microsporidial replication by irreversibly blocking the active site.

Furazolidone This nitrofurant derivative is an effective alternative agent for the treatment of giardiasis and also exhibits activity against *Isoospora belli*. Because it is the only agent active against *Giardia* that is available in liquid form, it is most often used to treat young children. Furazolidone undergoes reductive activation in *Giardia lamblia* trophozoites—an event that, unlike the reductive activation of metronidazole, involves an NADH oxidase. The killing effect correlates with the toxicity of reduced products, which damage important cellular components, including DNA. Although furazolidone had been thought to be largely unabsorbed when administered orally, the occurrence of systemic adverse reactions indicates that this is not the case. More than 65% of the drug dose can be recovered from the urine as colored metabolites. Omeprazole reduces the oral bioavailability of furazolidone.

Furazolidone is a monoamine oxidase (MAO) inhibitor; thus, caution should be used in its concomitant administration with other drugs (especially indirectly acting sympathomimetic amines) and in the consumption of food and drink containing tyramine during treatment. However, hypertensive crises have not been reported in patients receiving furazolidone, and it has been suggested that—because furazolidone inhibits MAOs gradually over several days—the risks are small if treatment is limited to a 5-day course. Because hemolytic anemia can occur in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency and glutathione instability, furazolidone treatment is contraindicated in mothers who are breast-feeding and in neonates.

Halofantrine This 9-phenanthrenemethanol is one of three classes of arylaminoalcohols first identified as potential antimalarial agents by the World War II Malaria Chemotherapy Program. Its activity is believed to be similar to that of chloroquine, although it is an oral alternative for the treatment of malaria due to chloroquine-resistant *P. falciparum*.

Halofantrine is thought to share one or more mechanisms with the 4-aminoquinolines, forming a complex with ferriprotoporphyrin IX and interfering with the degradation of hemoglobin. It has been shown to bind to plasmeypsin, a hemoglobin-degrading enzyme unique to plasmodia.

Halofantrine exhibits erratic bioavailability, but its absorption is significantly enhanced when it is taken with a fatty meal. The elimination half-life of halofantrine is 1–2 days; it is excreted mainly in feces. Halofantrine is metabolized into *N*-debutyl-halofantrine by the cytochrome P450 enzyme CYP3A4. Grapefruit juice should be avoided during treatment because it increases both halofantrine's bioavailability and halofantrine-induced QT interval prolongation by inhibiting CYP3A4 at the enterocyte level. Halofantrine should not be given simultaneously with or <3 weeks after mefloquine because of the potential occurrence of a fatal prolongation of the QTc interval on electrocardiography.

Iodoquinol Iodoquinol (diiodohydroxyquin), a hydroxyquinoline, is an effective luminal agent for the treatment of amebiasis, balantidiasis, and infection with *Dientamoeba fragilis*. Its mechanism of action is unknown. It is poorly absorbed. Because the drug contains 64% organically bound iodine, it should be used with caution in patients with thyroid disease. Iodine dermatitis occurs occasionally during iodoquinol treatment. Protein-bound serum iodine levels may be increased during treatment and can interfere with certain tests of thyroid function. These effects may persist for as long as 6 months after discontinuation of therapy. Iodoquinol is contraindicated in patients with liver disease. Most serious are the reactions related to prolonged high-dose therapy (optic neuritis, peripheral neuropathy), which should not occur if the recommended dosage regimens are followed.

Ivermectin Ivermectin (22,23-dihydroavermectin) is a derivative of the macrocyclic lactone avermectin produced by the soil-dwelling actinomycete *Streptomyces avermitilis*. Ivermectin is active at low doses against a wide range of helminths and ectoparasites. It is the drug of choice for the treatment of onchocerciasis, strongyloidiasis, cutaneous larva migrans, and scabies. Ivermectin is highly active against microfilariae of the lymphatic filariases but has no macrofilaricidal activity. When ivermectin is used in combination with other agents such as DEC or albendazole for treatment of lymphatic filariasis, synergistic activity is seen. Although active against the intestinal helminths *Ascaris lumbricoides* and *Enterobius vermicularis*, ivermectin is only variably effective in trichuriasis and is ineffective against hookworms. Widespread use of ivermectin for treatment of intestinal nematode infections in sheep and goats has led to the emergence of drug resistance in veterinary practice; this development may portend problems in human medical use.

Data suggest that ivermectin acts by opening the neuromuscular membrane-associated, glutamate-dependent chloride channels. The influx of chloride ions results in hyperpolarization and muscle paralysis—particularly of the nematode pharynx, with consequent blockage of the oral ingestion of nutrients. As these chloride channels are present only in invertebrates, paralysis is seen only in the parasite.

Ivermectin is available for administration to humans only as an oral formulation. The drug is highly protein bound; it is almost completely excreted in feces. Both food and beer increase the bioavailability of ivermectin significantly. Ivermectin is distributed widely throughout the body; animal studies indicate that it accumulates at the highest concentration in adipose tissue and liver, with little accumulation in the brain. Few data exist to guide therapy in hosts with conditions that may influence drug pharmacokinetics.

Ivermectin is generally administered as a single dose of 150–200 µg/kg. In the absence of parasitic infection, the adverse effects of ivermectin in therapeutic doses are minimal. Adverse effects in patients with filarial infections include fever, myalgia, malaise, lightheadedness, and (occasionally) postural hypotension. The severity of such side effects is related to the intensity of parasite infection, with more symptoms in individuals with a heavy parasite burden. In onchocerciasis, skin edema, pruritus, and mild eye irritation may also occur. The adverse effects are generally self-limiting and only occasionally require

symptom-based treatment with antipyretics or antihistamines. More severe complications of ivermectin therapy for onchocerciasis include encephalopathy in patients heavily infected with *Loa loa*.

Lumefantrine Lumefantrine (benflumetol), a fluorene arylamino-alcohol derivative synthesized in the 1970s by the Chinese Academy of Military Medical Sciences (Beijing), has marked blood schizonticidal activity against a wide range of plasmodia. This agent conforms structurally and in mode of action to other arylaminoalcohols (quinine, mefloquine, and halofantrine). Lumefantrine exerts its antimalarial effect as a consequence of its interaction with heme, a degradation product of hemoglobin metabolism. Although its antimalarial activity is slower than that of the artemisinin-based drugs, the recrudescence rate with the recommended lumefantrine regimen is lower. The pharmacokinetic properties of lumefantrine are reminiscent of those of halofantrine, with variable oral bioavailability, considerable augmentation of oral bioavailability by concomitant fat intake, and a terminal elimination half-life of ~4–5 days in patients with malaria.

Artemether and lumefantrine have synergistic activity, and the combined formulation of artemether and lumefantrine is effective for the treatment of falciparum malaria in areas where *P. falciparum* is resistant to chloroquine and antifolates.

Mebendazole This benzimidazole is a broad-spectrum antiparasitic agent widely used to treat intestinal helminthiasis. Its mechanism of action is similar to that of albendazole; however, it is a more potent inhibitor of parasite malic dehydrogenase and exhibits a more specific and selective effect against intestinal nematodes than the other benzimidazoles.

Mebendazole is available only in oral form but is poorly absorbed from the GI tract; only 5–10% of a standard dose is measurable in plasma. The proportion absorbed from the GI tract is extensively metabolized in the liver. Metabolites appear in the urine and bile; impaired liver or biliary function results in higher plasma mebendazole levels in treated patients. No dose reduction is warranted in patients with renal function impairment. Because mebendazole is poorly absorbed, its incidence of side effects is low. Transient abdominal pain and diarrhea sometimes occur, usually in persons with massive parasite burdens.

Mefloquine Mefloquine is used for prophylaxis of chloroquine-resistant malaria; high doses can be used for treatment. Despite the development of drug-resistant strains of *P. falciparum* in parts of Africa and Southeast Asia, mefloquine remains an effective drug throughout most of the world. Cross-resistance of mefloquine with halofantrine and with quinine has been documented in limited areas. Like quinine and chloroquine, this quinoline is active only against the asexual erythrocytic stages of malarial parasites. Unlike quinine, however, mefloquine has a relatively poor affinity for DNA and, as a result, does not inhibit the synthesis of parasitic nucleic acids and proteins. Although both mefloquine and chloroquine inhibit hemozoin formation and heme degradation, mefloquine differs in that it forms a complex with heme that may be toxic to the parasite.

Mefloquine HCl is poorly water soluble and intensely irritating when given parenterally; thus, it is available only in tablet form. Its absorption is adversely affected by vomiting and diarrhea but is significantly enhanced when the drug is administered with or after food. About 98% of the drug binds to protein. Mefloquine is excreted mainly in the bile and feces; therefore, no dose adjustment is needed in persons with renal insufficiency. The drug and its main metabolite are not appreciably removed by hemodialysis. No dosage adjustments are indicated for the achievement of plasma concentrations in dialysis patients. Pharmacokinetic differences have been detected among various ethnic populations; however, these distinctions are of minor importance compared with host immune status and parasite susceptibility. In patients with impaired liver function, the elimination of mefloquine may be prolonged, leading to higher plasma levels.

Mefloquine should be used with caution by individuals participating in activities requiring alertness and fine-motor coordination because dizziness, vertigo, or tinnitus can develop and persist. If the drug is to be administered for a prolonged period, periodic evaluations are

recommended, including liver function tests and ophthalmic examinations. Sleep abnormalities (insomnia, abnormal dreams) have occasionally been reported. Psychosis and seizures occur rarely; mefloquine should not be prescribed to patients with neuropsychiatric conditions. The development of acute anxiety, depression, restlessness, or confusion may be considered prodromal to a more serious event, and the drug should be discontinued.

Concomitant use of quinine, quinidine, or drugs producing β -adrenergic blockade may cause significant electrocardiographic abnormalities or cardiac arrest. Halofantrine must not be given simultaneously with or <3 weeks after mefloquine because a potentially fatal prolongation of the QTc interval on electrocardiography may occur. No data exist on mefloquine use after halofantrine use. Administration of mefloquine with quinine or chloroquine may increase the risk of convulsions. Mefloquine may lower plasma levels of anticonvulsants. Caution should be exercised with concomitant antiretroviral therapy, because mefloquine has been shown to exert variable effects on ritonavir pharmacokinetics that are not explained by hepatic CYP3A4 activity or ritonavir protein binding. Vaccinations with attenuated live bacteria should be completed at least 3 days before the first dose of mefloquine.

Women of childbearing age who are traveling to areas where malaria is endemic should be warned against becoming pregnant and encouraged to practice contraception during malaria prophylaxis with mefloquine and for up to 3 months thereafter. However, in the case of unplanned pregnancy, use of mefloquine is not considered an indication for pregnancy termination. Analysis of prospectively monitored cases demonstrates a prevalence of birth defects and fetal loss comparable to background rates.

Melarsoprol Melarsoprol has been used since 1949 for the treatment of human African trypanosomiasis. This trivalent arsenical compound is indicated for the treatment of African trypanosomiasis with neurologic involvement and for the treatment of early disease that is resistant to suramin or pentamidine. Melarsoprol, like other drugs containing heavy metals, interacts with thiol groups of several different proteins; however, its antiparasitic effects appear to be more specific. Trypanothione reductase is a key enzyme involved in the oxidative stress management of both *Trypanosoma* and *Leishmania* species, helping to maintain an intracellular reducing environment by reduction of disulfide trypanothione to its dithiol derivative dihydrotrypanothione. Melarsoprol sequesters dihydrotrypanothione, depriving the parasite of its main sulfhydryl antioxidant, and inhibits trypanothione reductase, depriving the parasite of the essential enzyme system that is responsible for keeping trypanothione reduced. These effects are synergistic. The selectivity of arsenical action against trypanosomes is due at least in part to the greater melarsoprol affinity of reduced trypanothione than of other monothioles (e.g., cysteine) on which the mammalian host depends for maintenance of high thiol levels. Melarsoprol enters the parasite via an adenosine transporter; drug-resistant strains lack this transport system.

Melarsoprol is always administered IV. A small but therapeutically significant amount of the drug enters the CSF. The compound is excreted rapidly, with ~80% of the arsenic found in feces.

Melarsoprol is highly toxic. The most serious adverse reaction is reactive encephalopathy, which affects 6% of treated individuals and usually develops within 4 days of the start of therapy, with an average case–fatality rate of 50%. Glucocorticoids are administered with melarsoprol to prevent this development. Because melarsoprol is intensely irritating, care must be taken to avoid infiltration of the drug.

Metrifonate Metrifonate has selective activity against *Schistosoma haematobium*. This organophosphorous compound is a prodrug that is converted nonenzymatically to dichlorvos (2,2-dichlorovinyl dimethylphosphate, DDVP), a highly active chemical that irreversibly inhibits the acetylcholinesterase enzyme. Schistosomal cholinesterase is more susceptible to dichlorvos than is the corresponding human enzyme. The exact mechanism of action of metrifonate is uncertain, but the drug is believed to inhibit tegumental acetylcholine receptors that mediate glucose transport.

Metrifonate is administered in a series of three doses at 2-week intervals. After a single oral dose, metrifonate produces a 95% decrease in plasma cholinesterase activity within 6 h, with a fairly rapid return to normal. However, 2.5 months are required for erythrocyte cholinesterase levels to return to normal. Treated persons should not be exposed to neuromuscular blocking agents or organophosphate insecticides for at least 48 h after treatment.

Metronidazole and Other Nitroimidazoles See Table 229-1 and Chap. 149.

Miltefosine In the early 1990s, miltefosine (hexadecylphosphocholine), originally developed as an antineoplastic agent, was discovered to have significant antiproliferative activity against *Leishmania* species, *T. cruzi*, and *T. brucei* parasites in vitro and in experimental animal models. Miltefosine is the first oral drug that has proved to be highly effective and comparable to amphotericin B against visceral leishmaniasis in India, where antimonial-resistant cases are prevalent. Miltefosine is also effective in previously untreated visceral infections. Cure rates in cutaneous leishmaniasis are comparable to those obtained with antimony. Miltefosine is also effective against the free-living ameba *Naegleria fowleri*.

The activity of miltefosine is attributed to interaction with cell signal transduction pathways and inhibition of phospholipid and sterol biosynthesis. Resistance to miltefosine has not been observed clinically. The drug is readily absorbed from the GI tract, is widely distributed, and accumulates in several tissues. The efficacy of a 28-day treatment course in Indian visceral leishmaniasis is equivalent to that of amphotericin B therapy; however, it appears that a shortened course of 21 days may be equally efficacious.

General recommendations for the use of miltefosine are limited by the exclusion of specific groups from the published clinical trials: persons <12 or >65 years of age, persons with the most advanced disease, breast-feeding women, HIV-infected patients, and individuals with significant renal or hepatic insufficiency.

Moxidectin Like ivermectin, moxidectin is a macrocyclic lactone that is an effective antihelminthic. In 2018, the FDA approved its use for the treatment of onchocerciasis. The primary mode of action of moxidectin is believed to be like that of ivermectin; however, there are likely different binding sites, as suggested by the identification of ivermectin-resistant helminths that are susceptible to moxidectin. The drug is well tolerated, with most adverse effects attributed to death of microfilariae. Some adverse effects occurred more commonly compared with ivermectin, including orthostatic hypotension (5 vs 2%) and elevated transaminases (1 vs 0.6%). In clinical trials, no clinically significant differences in the pharmacokinetics were observed with age, gender, weight, or renal impairment. The effect of hepatic dysfunction is unknown.

Niclosamide⁺ Niclosamide is active against a wide variety of adult tapeworms but not against tissue cestodes. The drug uncouples oxidative phosphorylation in parasite mitochondria, thereby blocking the uptake of glucose by the intestinal tapeworm and resulting in the parasite's death. Niclosamide rapidly causes spastic paralysis of intestinal cestodes in vitro. Its use is limited by its side effects, the necessarily long duration of therapy, the recommended use of purgatives, and—most important—limited availability (i.e., on a named-patient basis from the manufacturer).

Niclosamide is poorly absorbed. Tablets are given on an empty stomach in the morning after a liquid meal the night before, and this dose is followed by another 1 h later. For treatment of hymenolepiasis, the drug is administered for 7 days. A second course is often prescribed. The scolex and proximal segments of the tapeworms are killed on contact with niclosamide and may be digested in the gut. However, disintegration of the adult tapeworm results in the release of viable ova, which theoretically can result in autoinfection. Although fears of the development of cysticercosis in patients with *Taenia solium* infections have proved unfounded, it is still recommended that a brisk purgative be given 2 h after the first dose.

Nifurtimox This nitrofurant compound is an inexpensive and effective oral agent for the treatment of acute Chagas disease. Trypanosomes lack catalase and have very low levels of peroxidase; as a result, they are very vulnerable to by-products of oxygen reduction. When nifurtimox is reduced in the trypanosome, a nitro anion radical is formed and undergoes autooxidation, resulting in the generation of the superoxide anion O_2^- , hydrogen peroxide (H_2O_2), hydroperoxyl radical (HO_2), and other highly reactive and cytotoxic molecules. Despite the abundance of catalases, peroxidases, and superoxide dismutases that neutralize these destructive radicals in mammalian cells, nifurtimox has a poor therapeutic index. Prolonged use is required, but the course may have to be interrupted because of drug toxicity, which develops in 40–70% of recipients. Nifurtimox is well absorbed and undergoes rapid and extensive biotransformation; <0.5% of the original drug is excreted in urine. In 2020, the FDA approved this agent for the treatment of Chagas disease in children; it is now commercially available.

Nitazoxanide Nitazoxanide is a 5-nitrothiazole compound used for the treatment of cryptosporidiosis and giardiasis; it is active against other intestinal protozoa as well. The drug is approved for use in children 1–11 years of age.

The antiprotozoal activity of nitazoxanide is believed to be due to interference with the pyruvate-ferredoxin oxidoreductase (PFOR) enzyme-dependent electron transfer reaction that is essential to anaerobic energy metabolism. Studies have shown that the PFOR enzyme from *G. lamblia* directly reduces nitazoxanide by transfer of electrons in the absence of ferredoxin. The DNA-derived PFOR protein sequence of *Cryptosporidium parvum* appears to be similar to that of *G. lamblia*. Interference with the PFOR enzyme-dependent electron transfer reaction may not be the only pathway by which nitazoxanide exerts antiprotozoal activity.

After oral administration, nitazoxanide is rapidly hydrolyzed to an active metabolite, tizoxanide (desacetyl-nitazoxanide). Tizoxanide then undergoes conjugation, primarily by glucuronidation. It is recommended that nitazoxanide be taken with food; however, no studies have been conducted to determine whether the pharmacokinetics of tizoxanide and tizoxanide glucuronide differ in fasted versus fed subjects. Tizoxanide is excreted in urine, bile, and feces, and tizoxanide glucuronide is excreted in urine and bile. The pharmacokinetics of nitazoxanide in patients with impaired hepatic and/or renal function have not been studied. Tizoxanide is highly bound to plasma protein (>99.9%). Therefore, caution should be used when administering this agent concurrently with other highly plasma protein-bound drugs that have narrow therapeutic indices, as competition for binding sites may occur.

Oxamniquine This tetrahydroquinoline derivative is an effective alternative agent for the treatment of *S. mansoni*, although susceptibility to this drug exhibits regional variation. Oxamniquine exhibits anticholinergic properties, but its primary mode of action seems to rely on ATP-dependent enzymatic drug activation generating an intermediate that alkylates essential macromolecules, including DNA. In treated adult schistosomes, oxamniquine produces marked tegumental alterations like those seen with praziquantel but that develop less rapidly, becoming evident 4–8 days after treatment.

Oxamniquine is administered orally as a single dose and is well absorbed. Food retards absorption and reduces bioavailability. About 70% of an administered dose is excreted in urine as a mixture of pharmacologically inactive metabolites. Patients should be warned that their urine might have an intense orange-red color. Side effects are uncommon and usually mild, although hallucinations and seizures have been reported.

Paromomycin (Aminosidine) First isolated in 1956, this aminoglycoside is an effective oral agent for the treatment of infections due to intestinal protozoa. Parenteral paromomycin appears to be effective against visceral leishmaniasis in India.

Paromomycin inhibits protozoan protein synthesis by binding to the 30S ribosomal RNA in the aminoacyl-tRNA site, causing misreading

of mRNA codons. Paromomycin is less active against *G. lamblia* than standard agents; however, like other aminoglycosides, paromomycin is poorly absorbed from the intestinal lumen, and the high levels of drug in the gut compensate for this relatively weak activity. If absorbed or administered systemically, paromomycin can cause ototoxicity and nephrotoxicity. However, systemic absorption is very limited, and toxicity should not be a concern in persons with normal kidneys. Topical formulations are not generally available.

Pentamidine Isethionate This diamidine is an effective alternative agent for some forms of leishmaniasis and trypanosomiasis. It is available for parenteral and aerosolized administration. Although its mechanism of action remains undefined, it is known to exert a wide range of effects, including interaction with trypanosomal kinetoplast DNA; interference with polyamine synthesis by a decrease in the activity of ornithine decarboxylase; and inhibition of RNA polymerase, topoisomerase, ribosomal function, and the synthesis of nucleic acids and proteins.

Pentamidine isethionate is well absorbed, highly tissue bound, and excreted slowly over several weeks, with an elimination half-life of 12 days. No steady-state plasma concentration is attained in persons given daily injections; the result is extensive accumulation of pentamidine in tissues, primarily the liver, kidney, adrenal gland, and spleen. Pentamidine does not penetrate well into the CNS. Pulmonary concentrations of pentamidine are increased when the drug is delivered in aerosolized form, but not when it is delivered systemically.

Rapid (<1-h) infusion of intravenous pentamidine often results in hypotension. Because electrolyte disturbances and mild to moderate nephrotoxicity occur commonly, pentamidine should be used with caution with other nephrotoxic agents. Pancreatitis and QT prolongation may also occur; cumulative damage to pancreatic islet cells may result in drug-induced diabetes mellitus. Similarly, hypoglycemia can develop, although much less commonly when pentamidine is given by the inhaled route.

Piperaquine This bisquinoline was synthesized in the 1960s and used widely for malaria control in China until resistance emerged. The development of artemisinin-based combination therapy led to its evaluation as a partner drug, and it is now combined with dihydroartemisinin. Piperaquine is highly lipophilic and has a prolonged half-life (~20 days), thus providing a period of posttreatment prophylaxis. The drug's mechanisms of action and resistance have not been well studied but are presumed to be similar to the other 4-aminoquinolines.

Piperazine The antihelminthic activity of piperazine is confined to ascariasis and enterobiasis. Piperazine acts as an agonist at extrasynaptic γ -aminobutyric acid (GABA) receptors, causing an influx of chloride ions in the nematode somatic musculature. Although the initial result is hyperpolarization of the muscle fibers, the ultimate effect is flaccid paralysis, leading to the expulsion of live worms. Patients should be warned, as this occurrence can be unsettling.

Praziquantel This heterocyclic pyrazinoisoquinoline derivative is highly active against a broad spectrum of trematodes and cestodes. It is the mainstay of treatment for schistosomiasis and is a critical part of community-based control programs.

All of the effects of praziquantel can be attributed either directly or indirectly to an alteration of intracellular calcium concentrations. Although the exact mechanism of action remains unclear, the major mechanism is disruption of the parasite tegument, causing tetanic contractures with loss of adherence to host tissues and, ultimately, disintegration or expulsion. Praziquantel induces changes in the antigenicity of the parasite by causing the exposure of concealed antigens. Praziquantel also produces alterations in schistosomal glucose metabolism, including decreases in glucose uptake, lactate release, glycogen content, and ATP levels.

Praziquantel exerts its parasitic effects directly and does not need to be metabolized to be effective. It is well absorbed but undergoes extensive first-pass hepatic clearance. Levels of the drug are increased when it is taken with food, particularly carbohydrates, or with cimetidine.

Serum levels are reduced by glucocorticoids, chloroquine, carbamazepine, and phenytoin. Praziquantel is completely metabolized in humans, with 80% of the dose recovered as metabolites in urine within 4 days. It is not known to what extent praziquantel crosses the placenta, but retrospective studies suggest that it is safe in pregnancy.

Patients with schistosomiasis who have heavy parasite burdens may develop abdominal discomfort, nausea, headache, dizziness, and drowsiness. Symptoms begin 30 min after ingestion, may require spasmolytics for relief, and usually disappear spontaneously after a few hours.

Primaquine Phosphate Primaquine, an 8-aminoquinoline, has a broad spectrum of activity against all stages of plasmodial development in humans but has been used most effectively for eradication of the hepatic stage of these parasites. Primaquine must be metabolized by the host to be effective. It is, in fact, rapidly metabolized; only a small fraction of the dose of the parent drug is excreted unchanged. Although the parasitocidal activity of the three oxidative metabolites remains unclear, they are believed to affect both pyrimidine synthesis and the mitochondrial electron transport chain. The metabolites appear to have significantly less antimalarial activity than primaquine; however, their hemolytic activity is greater than that of the parent drug.

Primaquine causes marked hypotension after parenteral administration and therefore is given only by the oral route. It is rapidly and almost completely absorbed from the GI tract. Patients should be tested for G6PD deficiency before they receive primaquine. The drug may induce the oxidation of hemoglobin into methemoglobin, regardless of the G6PD status of the patient. Primaquine is otherwise well tolerated.

Proguanil (Chloroguanide) Proguanil inhibits plasmodial dihydrofolate reductase and is used with atovaquone for oral treatment of uncomplicated malaria or with chloroquine for malaria prophylaxis in parts of Africa without widespread chloroquine-resistant *P. falciparum*.

Proguanil exerts its effect primarily by means of the metabolite cycloguanil, whose inhibition of dihydrofolate reductase in the parasite disrupts deoxythymidylate synthesis, thus interfering with a key pathway involved in the biosynthesis of pyrimidines required for nucleic acid replication. There are no clinical data indicating that folate supplementation diminishes drug efficacy; women of childbearing age for whom atovaquone/proguanil is prescribed should continue taking folate supplements to prevent neural tube birth defects.

Proguanil is extensively absorbed regardless of food intake. The drug is 75% protein bound. The main routes of elimination are hepatic biotransformation and renal excretion; 40–60% of the proguanil dose is excreted by the kidneys. Drug levels are increased and elimination is impaired in patients with hepatic insufficiency.

Pyrantel Pamoate Pyrantel is a tetrahydropyrimidine formulated as pamoate. This safe, well-tolerated, inexpensive drug is used to treat a variety of intestinal nematode infections but is ineffective in trichuriasis. Pyrantel pamoate is usually effective in a single dose. Its target is the nicotinic acetylcholine receptor on the surface of nematode somatic muscle. Pyrantel depolarizes the neuromuscular junction of the nematode, resulting in its irreversible paralysis and allowing the natural expulsion of the worm.

Pyrantel pamoate is poorly absorbed from the intestine; >85% of the dose is passed unaltered in feces. The absorbed portion is metabolized and excreted in urine. Piperazine is antagonistic to pyrantel pamoate and should not be used concomitantly.

Pyrantel pamoate has minimal toxicity at the oral doses used to treat intestinal helminthic infection. It is not recommended for pregnant women or for children <12 months old.

Pyrimethamine When combined with short-acting sulfonamides, this diaminopyrimidine is effective in malaria, toxoplasmosis, and isosporiasis. Unlike mammalian cells, the parasites that cause these infections cannot use preformed pyrimidines obtained through salvage pathways but rather rely completely on de novo pyrimidine synthesis, for which folate derivatives are essential cofactors. The efficacy of pyrimethamine is increasingly limited by the

development of resistant strains of *P. falciparum* and *P. vivax*. Single amino acid substitutions to parasite dihydrofolate reductase confer resistance to pyrimethamine by decreasing the enzyme's binding affinity for the drug.

Pyrimethamine is well absorbed; the drug is 87% bound to human plasma proteins. In healthy volunteers, drug concentrations remain at therapeutic levels for up to 2 weeks; drug levels are lower in patients with malaria.

At the usual dosage, pyrimethamine alone causes little toxicity except for occasional skin rashes and, more rarely, blood dyscrasias. Bone marrow suppression sometimes occurs at the higher doses used for toxoplasmosis; at these doses, the drug should be administered with folic acid.

Pyronaridine This potent antimalarial is a benzonaphthylidene derivative first synthesized by Chinese researchers in 1970. Like chloroquine, pyronaridine targets hemozoin formation, inhibiting the production of β -hemozoin by forming complexes with it, with consequent enhancement of hemozoin-induced hemolysis. However, this drug is more potent than chloroquine: for complete lysis, pyronaridine is required at only 1/100th of the concentration needed with chloroquine. It also inhibits glutathione-dependent heme degradation. Despite its similar mode of action, pyronaridine remains effective against chloroquine-resistant strains. When combined with artesunate, it is effective for the treatment of acute, uncomplicated infection caused by *P. falciparum* or *P. vivax* in areas of low transmission with evidence of artemisinin resistance.

Pyronaridine is readily absorbed, widely distributed throughout the body, metabolized by the liver, and excreted in urine and stool. Its use is contraindicated in patients with severe liver or kidney impairment. Pyronaridine inhibits both CYP2D6 and P-glycoprotein in vitro, and these effects may have clinical relevance for patients taking medications for cardiac disease (e.g., metoprolol and digoxin).

Quinacrine[†] Despite being one of the first antimalarials, the antiprotozoal mechanism of quinacrine has not been fully elucidated. It is generally considered to intercalate into DNA and thereby inhibit replication and transcription. The drug also inhibits adenosine uptake, ATP incorporation into RNA, and NADH oxidase—the same enzyme that activates furazolidone. The differing relative quinacrine uptake rate between human cells and *G. lamblia* may explain the selective toxicity of the drug. Resistance correlates with decreased drug uptake.

Quinacrine is rapidly absorbed from the intestinal tract and is widely distributed in body tissues. Alcohol is best avoided because of a disulfiram-like effect. Although its production was discontinued in 1992, quinacrine can be obtained commercially from compounding pharmacies.

Quinine and Quinidine When combined with another agent, the cinchona alkaloid quinine is effective for the oral treatment of both uncomplicated, chloroquine-resistant malaria and babesiosis. Quinine acts rapidly against the asexual blood stages of all forms of the human malaria parasites. For severe malaria, only quinidine (the dextroisomer of quinine) is available in the United States. Quinine concentrates in the acidic food vacuoles of *Plasmodium* species. The drug inhibits the nonenzymatic polymerization of the highly reactive, toxic heme molecule into the nontoxic polymer pigment hemozoin.

Quinine is readily absorbed when given orally. In patients with malaria, the elimination half-life of quinine increases according to the severity of the infection. However, toxicity is avoided by an increase in the concentration of plasma glycoproteins. The cinchona alkaloids are extensively metabolized, particularly by CYP3A4; only 20% of the dose is excreted unchanged in urine. The drug's metabolites are also excreted in urine and may be responsible for toxicity in patients with renal failure. Renal excretion of quinine is decreased when cimetidine is taken and increased when the urine is acidic. The drug readily crosses the placenta.

Quinidine is both more potent as an antimalarial and more toxic than quinine. Its use requires cardiac monitoring. Dose reduction is necessary in persons with severe renal impairment.

Spiramycin[†] This macrolide is used to treat acute toxoplasmosis in pregnancy and congenital toxoplasmosis. While the mechanism of action is similar to that of other macrolides, the efficacy of spiramycin in toxoplasmosis appears to stem from its rapid and extensive intracellular penetration, which results in macrophage drug concentrations 10–20 times greater than serum concentrations.

Spiramycin is rapidly and widely distributed throughout the body and reaches concentrations in the placenta up to five times those in serum. This agent is excreted mainly in bile. Indeed, in humans, the urinary excretion of active compounds represents only 20% of the administered dose.

Serious reactions to spiramycin are rare. Of the available macrolides, spiramycin appears to have the lowest risk of drug interactions. Complications of treatment are rare but, in neonates, can include life-threatening ventricular arrhythmias that disappear with drug discontinuation.

Spiramycin is not formally approved for use in the United States, but it is accessible through a compassionate use program for toxoplasmosis in pregnancy through the FDA (301-796-1400).

Sulfonamides See Table 229-1 and Chap. 149.

Suramin^{*} This derivative of urea is the drug of choice for the early stage of African trypanosomiasis. The drug is polyanionic and acts by forming stable complexes with proteins, thus inhibiting multiple enzymes essential to parasite energy metabolism. Suramin appears to inhibit all trypanosome glycolytic enzymes more effectively than it inhibits the corresponding host enzymes.

Suramin is parenterally administered. It binds to plasma proteins and persists at low levels for several weeks after infusion. Its metabolism is negligible. This drug does not penetrate the CNS.

Tafenoquine Tafenoquine is an 8-aminoquinoline with causal prophylactic activity. Its prolonged half-life (2–3 weeks) allows longer dosing intervals when the drug is used for prophylaxis. Tafenoquine has been well tolerated in clinical trials. When tafenoquine is taken with food, its absorption is increased by 50% and the most commonly reported adverse event—mild GI upset—is diminished. Like primaquine, tafenoquine is a potent oxidizing agent, causing hemolysis in patients with G6PD deficiency as well as methemoglobinemia. It has been commercially available since FDA approval in 2018.

Tetracyclines See Table 229-1 and Chap. 149.

Thiabendazole Discovered in 1961, thiabendazole remains one of the most potent of the numerous benzimidazole derivatives. However, its use has declined significantly because of a higher frequency of adverse effects than is seen with other, equally effective agents.

Thiabendazole is active against most intestinal nematodes that infect humans. Although the exact mechanism of its antihelminthic activity has not been fully elucidated, it is likely to be similar to that of other benzimidazole drugs: namely, inhibition of polymerization of parasite β -tubulin. The drug also inhibits the helminth-specific enzyme fumarate reductase. In animals, thiabendazole has anti-inflammatory, antipyretic, and analgesic effects, which may explain its usefulness in dracunculiasis and trichinellosis. Thiabendazole also suppresses egg and/or larval production by some nematodes and may inhibit the subsequent development of eggs or larvae passed in feces. Despite the emergence and global spread of thiabendazole-resistant trichostrongyliasis among sheep, there have been no reports of drug resistance in humans.

Thiabendazole is available in tablet form and as an oral suspension. The drug is rapidly absorbed from the GI tract but can also be absorbed through the skin. Thiabendazole should be taken after meals. This agent is extensively metabolized in the liver before ultimately being excreted; most of the dose is excreted within the first 24 h. The usual dose of thiabendazole is determined by the patient's weight, but some treatment regimens are parasite specific. No adjustments are recommended in patients with renal or hepatic failure; only cautious use is advised.

Coadministration of thiabendazole to patients taking theophylline can result in an increase in theophylline levels by >50%. Therefore,

Tinidazole This nitroimidazole is effective for the treatment of amebiasis, giardiasis, and trichomoniasis. Like metronidazole, tinidazole must undergo reductive activation by the parasite's metabolic system before it can act on protozoal targets. Tinidazole inhibits the synthesis of new DNA in the parasite and causes degradation of existing DNA. The reduced free-radical derivatives alkylate DNA, with consequent cytotoxic damage to the parasite. This damage appears to be produced by short-lived reduction intermediates, resulting in helix destabilization and strand breakage of DNA. The mechanism of action and side effects of tinidazole are similar to those of metronidazole, but adverse events appear to be less frequent and severe with tinidazole. In addition, the significantly longer half-life of tinidazole (>12 h) offers potential cure with a single dose.

Tribendimidine Tribendimidine, a diamidine derivative of aminophenylamidine amidantel, is a cholinergic agonist that is selective for the nicotinic acetylcholine receptors of nematode muscle. This novel agent has a broad spectrum of activity against a wide variety of helminths and is highly effective against food-borne trematodes, with a similar cure rate to praziquantel. Clinical trials have demonstrated efficacy of a single dose alone or in combination with other helminthics against soil-transmitted helminth infections. The drug is an L-type nicotinic acetylcholine receptor agonist and exhibits the same method of action as levamisole and pyrantel; therefore, it may not be effective in regions where resistance to these agents is widespread. It is not available in the United States.

Triclabendazole In contrast to other benzimidazoles, the antihelminthic activity of triclabendazole is highly specific for *Fasciola* and *Paragonimus* species, with little activity against nematodes, cestodes, and other trematodes. Triclabendazole is effective against all stages of *Fasciola* species. The active sulfoxide metabolite of triclabendazole binds to fluke tubulin by assuming a unique nonplanar configuration and disrupts microtubule-based processes. Resistance to triclabendazole in veterinary use has been reported in Australia and Europe; however, no resistance has been documented in humans.

Triclabendazole is rapidly absorbed after oral ingestion; administration with food enhances its absorption and shortens the elimination half-life of the active metabolite. Both the sulfoxide and the sulfone metabolites are highly protein bound (>99%). Treatment with triclabendazole is typically given in one or two doses. No clinical data are available regarding dose adjustment in renal or hepatic insufficiency; however, given the short course of therapy and extensive hepatic metabolism of triclabendazole, dose adjustment is unlikely to be necessary. No information exists on drug interactions.

Trimethoprim-Sulfamethoxazole See Table 229-1 and Chap. 149.

FURTHER READING

- DZIDUCH K et al: The current directions of searching for antiparasitic drugs. *Molecules* 27:1534, 2022.
- FEHINTOLA FA et al: Drug interactions in the treatment and chemoprophylaxis of malaria in HIV infected individuals in sub-Saharan Africa. *Curr Drug Metab* 12:51, 2011.
- KEISER J, HÄBERLI C: Evaluation of commercially available anthelmintics in laboratory models of human intestinal nematode infections. *ACS Infect Dis* 7:1177, 2021.
- KEISER J et al: Antiparasitic drugs for paediatrics: Systematic review, formulations, pharmacokinetics, safety, efficacy, and implications for control. *Parasitology* 138:1620, 2011.
- KELESIDIS T, FALAGAS ME: Substandard/counterfeit antimicrobial drugs. *Clin Microbiol Rev* 28:443, 2015.
- PINK R et al: Opportunities and challenges in antiparasitic drug discovery. *Nat Rev Drug Discov* 4:727, 2005.

230 Amebiasis and Infection with Free-Living Amebae

Rosa M. Andrade, Sharon L. Reed



AMEBIASIS

DEFINITION

Amebiasis is an infection caused by *Entamoeba histolytica*, an intestinal protozoan. Its spectrum of clinical syndromes ranges from asymptomatic colonization (90% of cases) to invasive amebiasis, which accounts for 10% of affected individuals. Invasive amebiasis frequently presents as intestinal colitis (dysentery or diarrhea) or as extraintestinal amebiasis, in which abscesses of the liver are more commonly found than involvement of the lungs or brain.

LIFE CYCLE AND TRANSMISSION

E. histolytica is acquired by ingestion of viable cysts from fecally contaminated water, food, or hands (Fig. 230-1). Food-borne exposure is the most prevalent form of transmission. It occurs when food handlers are shedding cysts or food is being grown with feces-contaminated soil, fertilizer, or water. Less common means of transmission include oral and anal sexual practices and, in rare instances, direct rectal inoculation through colonic irrigation devices. Motile trophozoites are released from cysts in the small intestine and, in most patients, remain as harmless commensals in the large bowel. After encystation, infectious cysts are shed in the stool and can survive for several weeks in a moist environment. In some patients, the trophozoites invade either the bowel mucosa, causing symptomatic colitis, or the bloodstream, causing distant abscesses of the liver, lungs, or brain. The trophozoites may not encyst in patients with active dysentery, and motile hematophagous trophozoites are frequently present in fresh stools. Trophozoites are rapidly killed by exposure to air or stomach acid and therefore cannot transmit infection.

EPIDEMIOLOGY

E. histolytica infection typically affects underdeveloped tropical regions with poor sanitation systems and hygiene, occurring often in children <5 years of age. This infection is widespread in the Indian subcontinent and Africa, parts of East Asia (Thailand), and Central and South America (Mexico and Colombia). According to the Global Burden of Disease 2016 study, amebiasis accounts for 26,748 all-age deaths, including 4567 children <5 years old.

In contrast, returning travelers, recent immigrants, men who have sex with men (MSM), military personnel, and inmates of institutions are the main groups at risk for amebiasis in developed countries. The GeoSentinel Surveillance Network, which encompasses information from tropical medicine clinics on six continents, showed that, among long-term travelers (trip duration, >6 months), diarrhea due to *E. histolytica* was among the most common diagnoses. In countries like Japan and Spain, amebiasis is emerging as a sexually transmitted disease, first reported in HIV-positive MSM. However, in Japan, *E. histolytica* infections, symptomatic or asymptomatic, seem to be steadily spreading among non-HIV men and women.

Worldwide, *E. histolytica* is the second most common cause of death related to parasitic infection (after malaria). Invasive colitis and liver abscesses are tenfold more common among men than among women; this difference has been attributed to a disparity in complement-mediated killing and effects of testosterone on the secretion of interferon γ . The wide spectrum of clinical disease caused by *Entamoeba* is due in part to the differences between the two major infecting species,

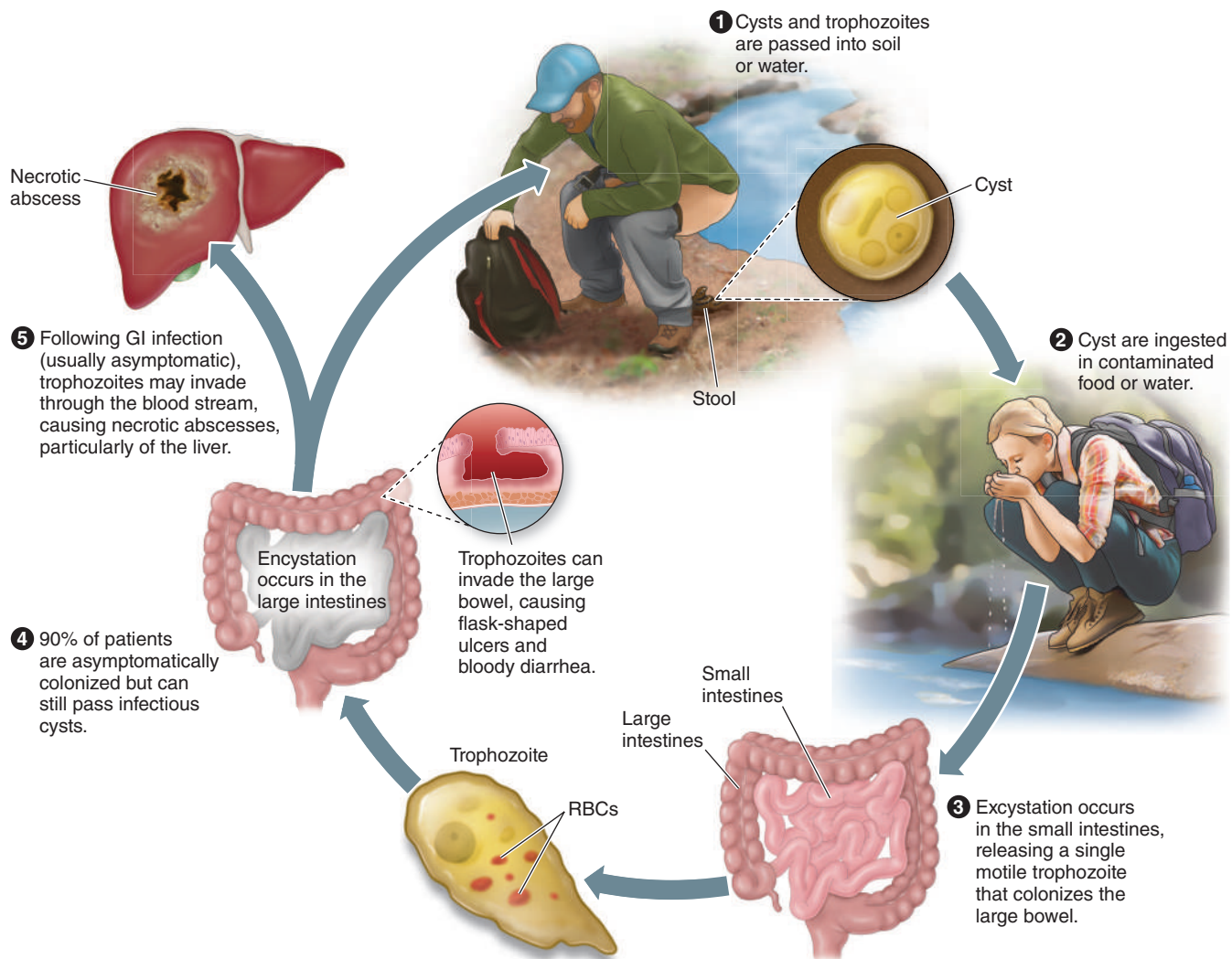


FIGURE 230-1 Life cycle of *Entamoeba histolytica*. GI, gastrointestinal; RBCs, red blood cells.

E. histolytica and *E. dispar*. *E. histolytica* has unique surface antigens, is genetically distinct, and possesses virulence properties that distinguish it from the morphologically identical *E. dispar*.

Most asymptomatic carriers, including MSM and patients with AIDS, harbor *E. dispar* and have self-limited infections. In this respect, *E. dispar* is dissimilar to other enteric pathogens such as *Cryptosporidium* and *Cystoisospora belli*, which can cause self-limited illnesses in immunocompetent hosts but devastating diarrhea in patients with AIDS. These observations indicate that *E. dispar* is incapable of causing invasive disease. Through genomic sequencing, new species of *Entamoeba* have been identified: *E. moshkovskii* and *E. bangladeshi*. These new species are microscopically indistinguishable from *E. histolytica*. Although *E. moshkovskii* causes diarrhea, weight loss, and colitis in mice, a prospective evaluation of children from the Mirpur community of Dhaka, Bangladesh, found that most children who had diarrheal diseases associated with *E. moshkovskii* were simultaneously infected with at least one other enteric pathogen. *E. bangladeshi* nov. sp., Bangladesh was first reported in 2012 in this Bangladeshi community; however, it has been isolated in South African subjects of all ages in recent years. Additional clinical and epidemiologic studies are needed to discern the true role of *E. bangladeshi* in the human host.

■ PATHOGENESIS AND PATHOLOGY

Both trophozoites and cysts are found in the intestinal lumen, but only trophozoites of *E. histolytica* invade tissue. The trophozoite is 20–60 μm in diameter and contains vacuoles and a nucleus with a characteristic central nucleolus. Trophozoites attach to colonic mucus and epithelial cells by Gal/GalNAc adherence lectin and release glycosidases and

proteases that cause degradation of mucus polymers. Extracellular cysteine proteinases degrade collagen, elastin, IgA, IgG, and the anaphylatoxins C3a and C5a. After disruption of the mucous layer, trophozoites damage the mucosa by contact-dependent and contact-independent cytotoxicity. The contact-dependent cytotoxicity is attributable to induction of apoptotic cell death; trophozoite-mediated cell death (ingestion of fragments of living cells); and lysis of inflammatory cells (neutrophils, monocytes, and lymphocytes), colonic cells, and hepatic cells through release of phospholipase A and pore-forming peptides. Contact-independent cytotoxicity follows production of inflammatory mediators, such as prostaglandin E₂, by trophozoites, ultimately leading to increased ion permeability of intercellular tight junctions.

E. histolytica trophozoites are constantly exposed to reactive oxygen and nitrogen species arising from their own metabolism and from the host during tissue invasion. The ability to resist reactive oxygen species or reactive nitrogen species such as nitric oxide or S-nitrosothiols (e.g., S-nitrosoglutathione [GSNO] and S-nitrosocysteine [CySNO]) is also a virulence factor. Since *E. histolytica* lacks glutathione and glutathione reductase, it relies on its thioredoxin–thioredoxin reductase system to prevent, regulate, and repair the damage caused by oxidative stress. This antioxidant system is versatile: it has the ability to reduce reactive nitrogen species and use an alternative electron donor, such as nicotinamide adenine dinucleotide. In addition, trophozoites preincubated with *E. coli* 055 are more resistant to hydrogen peroxide-mediated killing by using bacterially produced malate dehydrogenase and oxaloacetate for protection from oxidative stress. Metronidazole, the current standard of therapy for amebiasis, seems to exert its antiparasitic effect through inhibition of this antioxidant system.

Phagocytosis is a virulence factor that leads to a defective proliferation of *E. histolytica* if inhibited. Trophozoites use membrane-associated carbohydrate-binding proteins to phagocytose intestinal bacteria, especially gram-negative Enterobacteriaceae, for their nutrients. Interactions with commensal bacteria, such as *Escherichia coli*, can attenuate the virulence of *E. histolytica* by decreasing the expression of Gal/GalNAc lectin. In contrast, ingestion of enteropathogenic bacteria, such as enteropathogenic *E. coli* and *Shigella dysenteriae*, increases expression of the Gal/GalNAc lectin and enhances *E. histolytica* cysteine protease activity.

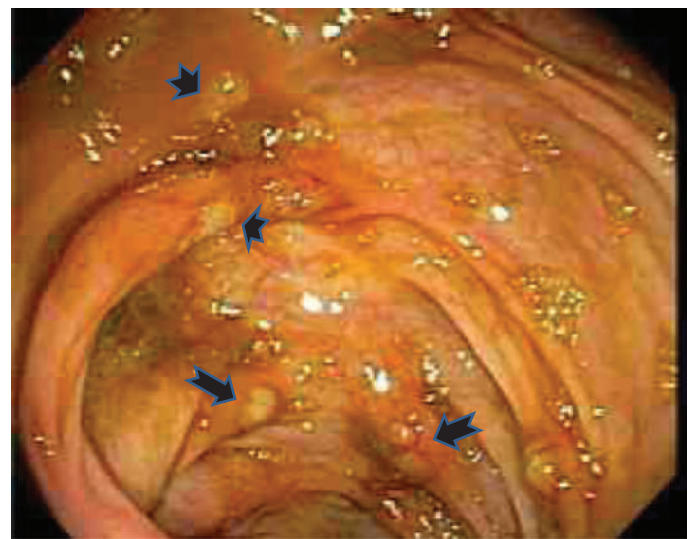
The interactions between *E. histolytica* and the gut microbiome can determine the severity of disease. During the first 2 years of life, the gut immune system and the microbiome mature rapidly. In one study, ~80% of children from the Bangladeshi community of Dhaka were found to be infected with *E. histolytica* by 2 years of age. Fecal anti-Gal/GalNAc lectin IgA was associated with protection from reinfection, while a high parasite burden in the first year of life was associated with the expansion of *Prevotella copri* in their gut microbiome and presence of diarrhea. In adults, the diversity of the gut microbiome can determine the pathogenesis of disease. As shown in a Japanese cohort through next-generation sequencing analysis, the gut microbiome in asymptomatic *E. histolytica* carriers was more homogeneous, with high abundance of *Ruminococcaceae*, *Coriobacteriaceae*, and *Clostridiaceae*, while these bacteria were absent or in low abundance in patients with invasive disease (symptomatic). These observations suggest the protective role of these bacteria.

Antimicrobial peptides, such as cathelicidins, are an important component of innate immunity and are induced by *E. histolytica* upon intestinal invasion in a mouse model. In this model, cecal cathelicidin-related antimicrobial peptide mRNA increased by >4-fold at 3 days and >100-fold at 7 days. However, *E. histolytica* remained resistant to cathelicidin-mediated killing, probably because the antimicrobial peptide was digested by amebic cysteine proteinases. Other in vitro studies have shown that α - and β -defensins may damage *E. histolytica* membrane integrity.

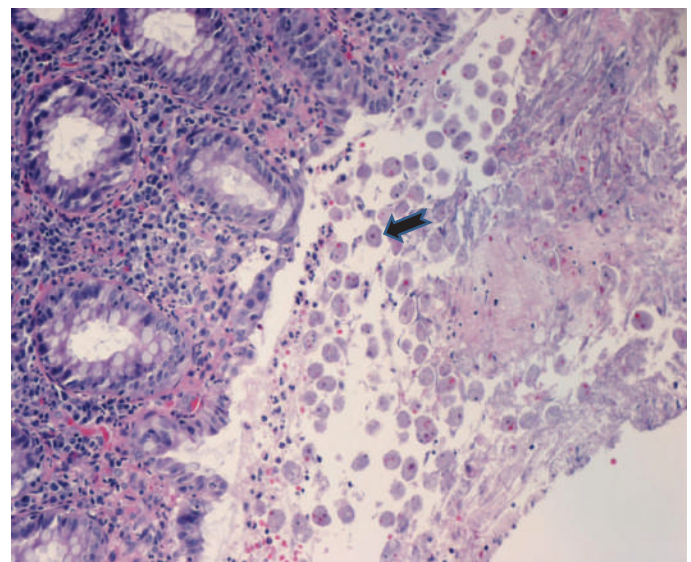
IgA plays a critical role in acquired immunity to *E. histolytica*. A study in Bangladeshi schoolchildren revealed that an intestinal IgA response to Gal/GalNAc reduced the risk of new *E. histolytica* infection by 64%. Serum IgG antibody is not protective; titers correlate with the duration of illness rather than with the severity of disease. Indeed, Bangladeshi children with a serum IgG response were more likely than those without such a response to develop new *E. histolytica* infection. In infants from this same Bangladeshi community, passive immunity conferred by maternal parasite-specific IgA via breastfeeding resulted in a 39% reduced risk of infection and a 64% reduced risk of diarrheal disease from *E. histolytica* during the first year of life. However, this protection appeared to be species-specific, with little or no protection conferred from infections with other species such as *E. dispar* or *E. bangladeshi*.

This Bangladeshi cohort has furthered our understanding of the genetic susceptibility factors associated with *E. histolytica* disease. Heterozygosity of the major histocompatibility complex (MHC) class II allele DQB1*0601 was found to protect against amebic intestinal disease, which supports the role of antigen processing and CD4+ T cells in resistance to amebiasis. Adipocyte leptin receptors (LEPRs) are expressed on intestinal epithelial cells, prevent apoptosis, promote tissue repair, and may decrease neutrophil infiltration. In this cohort, a single amino acid substitution (Q223R) in LEPRs nearly quadrupled the risk for amebic intestinal disease in children and increased the risk for amebic liver abscesses in adults. Similarly, variations in the locus of cAMP-responsive element modulator/cullin 2 (CREM/CUL2) may increase the risk for diarrhea in children who acquired *E. histolytica* within their first year of life. Interestingly, both genetic variations, Q223R and CREM, are overrepresented in this geographical region.

The earliest intestinal lesions are microulcerations of the mucosa of the cecum, sigmoid colon, or rectum that release erythrocytes, inflammatory cells, and epithelial cells. A colonoscopy reveals small ulcers with heaped-up margins and normal intervening mucosa (Fig. 230-2A). Submucosal extension of ulcerations under viable-appearing surface



A



B

FIGURE 230-2 Endoscopic and histopathologic features of intestinal amebiasis. **A.** Appearance of ulcers on colonoscopy (arrows). **B.** Inflammatory infiltrate and *Entamoeba histolytica* trophozoites (arrows) in invasive amebic colitis (hematoxylin and eosin). (Courtesy of the Department of Pathology and Gastroenterology, San Diego VA Medical Center.)

mucosa causes the classic “flask-shaped” ulcer containing trophozoites at the margins of dead and viable tissues. Although neutrophilic infiltrates may accompany early lesions in animals, human intestinal infection is marked by a paucity of inflammatory cells, probably in part because of the killing of neutrophils by trophozoites (Fig. 230-2B). Treated ulcers characteristically heal with little or no scarring. Occasionally, however, full-thickness necrosis and perforation occur.

Rarely, intestinal infection results in the formation of a mass lesion, or *ameboma*, in the bowel lumen. The overlying mucosa is usually thin and ulcerated, while other layers of the wall are thickened, edematous, and hemorrhagic; this condition results in exuberant formation of granulation tissue with little fibrous-tissue response.

Amebic liver abscesses are age- and gender-dependent. Men 30–60 years of age are most commonly infected at a rate 10–12 times higher than women in the same age group. Studies in animal models have demonstrated that testosterone may increase susceptibility to amebic liver abscess by modulating the secretion of interferon γ by natural killer T cells, which are activated through *E. histolytica* lipopeptidophosphoglycan present on the surface of ameba trophozoites. Liver

abscesses are always preceded by intestinal colonization, which may be asymptomatic. Blood vessels may be compromised early by wall lysis and thrombus formation. Trophozoites invade veins to reach the liver through the portal venous system. *E. histolytica* is resistant to complement-mediated lysis—a property critical to survival in the bloodstream. Inoculation of amebae into the portal system of hamsters results in an acute cellular infiltrate consisting predominantly of neutrophils. Later, the neutrophils are lysed by contact with amebae, and the release of neutrophil toxins may contribute to necrosis of hepatocytes. The liver parenchyma is replaced by necrotic material that is surrounded by a thin rim of congested liver tissue. Although the necrotic contents of a liver abscess are classically described as “anchovy paste,” the fluid is variable in color; it is composed of bacteriologically sterile granular debris with few or no cells. Amebae, if seen, tend to be found near the capsule of the abscess.

■ CLINICAL SYNDROMES

Intestinal Amebiasis The most common type of amebic infection is asymptomatic cyst passage (90% of patients). Even in highly endemic areas, most patients harbor *E. dispar*.

Symptomatic amebic colitis develops 2–6 weeks after the ingestion of infectious cysts. A gradual onset of lower abdominal pain and mild diarrhea is followed by malaise, weight loss, and diffuse lower abdominal or back pain. Cecal involvement may mimic acute appendicitis. Patients with full-blown dysentery may pass 10–12 stools per day. The stools contain little fecal material and consist mainly of blood and mucus. In contrast to those with bacterial diarrhea, fewer than 40% of patients with amebic dysentery are febrile. Virtually all patients have heme-positive stools.

More fulminant intestinal infection, with severe abdominal pain, high fever, and profuse diarrhea, is rare and occurs predominantly in children. Patients may develop toxic megacolon, in which there is severe bowel dilation with intramural air. Patients receiving glucocorticoids are at risk for severe amebiasis. The association between severe amebiasis complications and glucocorticoid therapy emphasizes the importance of excluding amebiasis when inflammatory bowel disease is suspected. An occasional patient presents with only an asymptomatic or tender abdominal mass caused by an ameboma, which is easily confused with cancer on barium studies. A positive serologic test or biopsy can prevent unnecessary surgery in this setting.

Environmental enteropathy (“impoverished gut”; blunted small-intestinal villi with lamina propria inflammation) is observed in tropical developing areas with endemic enteric infections, such as amebiasis. It is associated with functional gastrointestinal impairment causing malnutrition and stunted growth in children within the first 2 years of life. Bangladeshi children with symptomatic *E. histolytica* infections were 2.9 times more likely to be malnourished and 4.7 times more likely to be short for their age than were children without symptomatic infections. These factors affect their cognitive development and may be linked to loss of productivity in adulthood.

Amebic Liver Abscess Extraintestinal infection by *E. histolytica* most often involves the liver. Of travelers who develop an amebic liver abscess after leaving an endemic area, 95% do so within 5 months. Young patients with an amebic liver abscess are more likely than older patients to present in the acute phase with prominent symptoms of <10 days’ duration. Most patients are febrile and have right-upper-quadrant pain, which may be dull or pleuritic in nature and may radiate to the shoulder. Point tenderness over the liver and right-sided pleural effusion are common. Jaundice is rare. Although the initial site of infection is the colon, fewer than one-third of patients with an amebic abscess have active diarrhea. Older patients from endemic areas are more likely to have a subacute course lasting 6 months, with weight loss and hepatomegaly. About one-third of patients with chronic presentations are febrile. Thus, the clinical diagnosis of an amebic liver abscess may be difficult to establish because the symptoms and signs are often nonspecific. Since 10–15% of patients present only with fever, amebic liver abscess must be considered in the differential diagnosis of fever of unknown origin ([Chap. 22](#)).

Complications of Amebic Liver Abscess Pleuropulmonary involvement, which is reported in 20–30% of patients, is the most frequent complication of amebic liver abscess. Manifestations include sterile effusions, contiguous spread from the liver, and rupture into the pleural space. Sterile effusions and contiguous spread usually resolve with medical therapy, but frank rupture into the pleural space requires drainage. A hepatobronchial fistula may cause cough productive of large amounts of necrotic material that may contain amebae. This dramatic complication carries a good prognosis. Abscesses that rupture into the peritoneum may present as an indolent leak or an acute abdomen and require both percutaneous catheter drainage and medical therapy. Rupture into the pericardium, usually from abscesses of the left lobe of the liver, carries the gravest prognosis; it can occur during medical therapy and requires surgical drainage.

Involvement of Other Extraintestinal Sites The genitourinary tract may become involved by direct extension of amebiasis from the colon or by hematogenous spread of the infection. Painful genital ulcers, characterized by a punched-out appearance and profuse discharge, may develop secondary to extension from either the intestine or the liver. Both of these conditions respond well to medical therapy. Cerebral involvement has been reported in fewer than 0.1% of patients in large clinical series. Symptoms and prognosis depend on the size and location of the lesion.

■ DIAGNOSTIC TESTS

Laboratory Diagnosis Stool examinations, serologic tests, and noninvasive imaging of the liver remain the most important procedures in the diagnosis of amebiasis in most parts of the world. Fecal findings suggestive of amebic colitis include a positive test for heme, a paucity of neutrophils, and amebic cysts or trophozoites. The definitive diagnosis of amebic colitis is made by the demonstration of hematophagous trophozoites of *E. histolytica*. Because trophozoites are killed rapidly by water, drying, or barium, it is important to examine at least three fresh stool specimens. Examination of a combination of wet mounts, iodine-stained concentrates, and trichrome-stained preparations of fresh stool and concentrates for cysts or trophozoites confirms the diagnosis in 75–95% of cases. Cultures of amebae are more sensitive but are not routinely available. If stool examinations are negative, sigmoidoscopy with biopsy of the edge of ulcers may increase the yield, but this procedure is dangerous during fulminant colitis because of the risk of perforation. Trophozoites in a biopsy specimen from a colonic mass confirm the diagnosis of ameboma, but trophozoites are rare in liver aspirates because they are found in the abscess capsule and not in the readily aspirated necrotic center. Accurate diagnosis requires experience, since the trophozoites may be confused with neutrophils and the cysts must be differentiated morphologically from those of *Entamoeba hartmanni*, *Entamoeba coli*, and *Endolimax nana*, which do not cause clinical disease and do not warrant therapy. Unfortunately, the cysts of *E. histolytica* cannot be distinguished microscopically from those of *E. dispar*, *E. moshkovskii*, or *E. bangladeshi*. Therefore, the microscopic diagnosis of *E. histolytica* can be made only by the detection of *Entamoeba* trophozoites that have ingested erythrocytes.

Serology is an important addition to the methods used for parasitologic diagnosis of invasive amebiasis. Enzyme-linked immunosorbent assays are positive in >90% of cases with colitis, ameboma, or liver abscess. Positive results in conjunction with the appropriate clinical syndrome suggest active disease because serologic findings usually revert to negative within 6–12 months. Even in highly endemic areas such as South Africa, fewer than 10% of asymptomatic individuals have a positive amebic serology. The interpretation of the indirect hemagglutination test is difficult because titers may remain positive for as long as 10 years.

Early in infection, up to 10% of patients with acute amebic liver abscess may have negative serologic findings, which usually become positive within a week. In contrast to carriers of *E. dispar*, most asymptomatic carriers of *E. histolytica* develop antibodies. Thus, serologic tests are helpful in assessing the risk of invasive amebiasis in asymptomatic, cyst-passing individuals in nonendemic areas. Serologic tests

should also be performed in patients with ulcerative colitis before the institution of glucocorticoid therapy to prevent the development of severe colitis or toxic megacolon owing to unsuspected amebiasis.

More sensitive, specific, and rapid tests in stool are now in use in most developed countries. Enzyme immunoassays using monoclonal antibodies for the Gal/GalNAc lectin allow specific detection of *E. histolytica* in 96-well plates or a single cassette for rapid detection. Sensitivity is 87% and specificity >90% in stool and can also be used in liver abscess aspirates. The number of commercially available PCR tests has significantly expanded. Most are multiplex for GI protozoa alone or include major bacterial and viral pathogens, as well. The sensitivity and specificity approaches 100%, and results can be available in as little as an hour. Their cost precludes use in most developing countries.

Other amplification tests such as a loop-mediated isothermal amplification (LAMP) assay may be a potential alternative for direct detection of *E. histolytica* DNA in pus samples from amebic liver abscesses. LAMP is a relatively simple, rapid, and low-cost method of DNA amplification that could be a better alternative for diagnosis in developing countries.

Routine hematology and chemistry tests usually are not very helpful in the diagnosis of invasive amebiasis. About three-fourths of patients with an amebic liver abscess have leukocytosis (>10,000 cells/ μ L); this condition is particularly likely if symptoms are acute or complications have developed. Invasive amebiasis does not elicit eosinophilia. Anemia, if present, is usually multifactorial. Even with large liver abscesses, liver enzyme levels are normal or minimally elevated. The alkaline phosphatase level is most often elevated and may remain so for months. Aminotransferase elevations suggest acute disease or a complication.

Radiographic Studies Radiographic barium studies are potentially dangerous in acute amebic colitis. Amebomas are usually identified first by a barium enema, but biopsy is necessary for differentiation from carcinoma.

Radiographic techniques such as ultrasonography, CT, and MRI are all useful for detection of the round or oval hypoechoic cyst. More than 80% of patients who have had symptoms for >10 days have a single abscess of the right lobe of the liver (Fig. 230-3). Approximately 50% of patients who have had symptoms for <10 days have multiple abscesses. Findings associated with complications include large abscesses (>10 cm) in the superior part of the right lobe, which may rupture into the pleural space; multiple lesions, which must be differentiated from pyogenic abscesses; and lesions of the left lobe, which may rupture into the pericardium. Because abscesses resolve slowly and may increase in size despite a clinical response to therapy, frequent follow-up ultrasonography may prove confusing. Complete resolution of a liver abscess

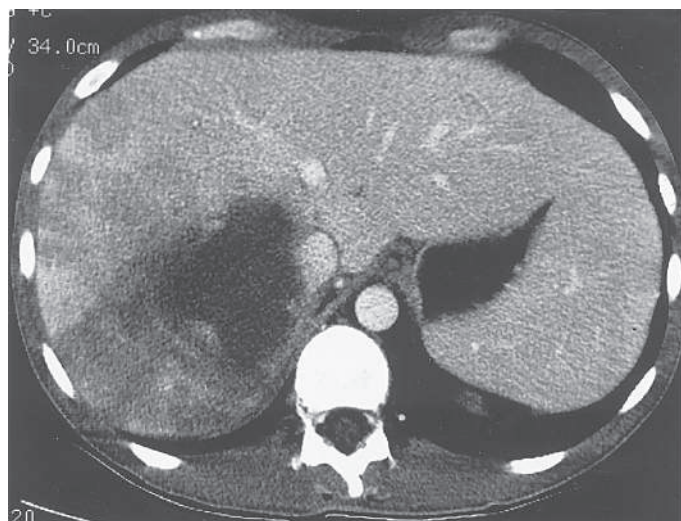


FIGURE 230-3 Abdominal CT scan of a large amebic abscess of the right lobe of the liver. (Courtesy of the Department of Radiology, UCSD Medical Center, San Diego; with permission.)

within 6 months can be anticipated in two-thirds of patients, but 10% may have persistent abnormalities for a year.

Differential Diagnosis The differential diagnosis of intestinal amebiasis includes bacterial diarrheas (Chap. 138) caused by *Campylobacter* (Chap. 169); enteroinvasive *Escherichia coli* (Chap. 166); and species of *Shigella* (Chap. 172), *Salmonella* (Chap. 171), and *Vibrio* (Chap. 173). Because the typical patient with amebic colitis has less prominent fever than in these other conditions as well as heme-positive stools with few neutrophils, correct diagnosis requires bacterial cultures, microscopic examination of stools, amebic serologic testing, and stool specific antigens or PCR. As has been mentioned, amebiasis must be ruled out in any patient thought to have inflammatory bowel disease.

Because of the variety of presenting signs and symptoms, amebic liver abscess can easily be confused with pulmonary or gallbladder disease or with any febrile illness with few localizing signs, such as malaria (Chap. 231) or typhoid fever (Chap. 171). The diagnosis should be considered in members of high-risk groups who have recently traveled outside the United States (Chap. 130) and in inmates of institutions. Once radiographic studies have identified an abscess in the liver, the most important differential diagnosis is between amebic and pyogenic abscess. Patients with pyogenic abscess typically are older and have a history of underlying bowel disease or recent surgery. Amebic serology is helpful, but aspiration of the abscess, with Gram's staining and culture of the material, may be required for differentiation of the two diseases.

TREATMENT

Amebiasis

INTESTINAL DISEASE (TABLE 230-1)

The drugs used to treat amebiasis can be classified according to their primary site of action. Luminal amebicides are poorly absorbed and reach high concentrations in the bowel, but their activity is limited to cysts and trophozoites close to the mucosa. Only two luminal drugs are available in the United States: iodoquinol and paromomycin. Indications for the use of luminal agents include eradication of cysts in patients with colitis or a liver abscess and treatment of asymptomatic carriers. The majority of asymptomatic individuals who pass cysts are colonized with *E. dispar*, which does not warrant specific therapy. However, it is prudent to treat asymptomatic individuals who pass cysts unless *E. dispar* colonization can be definitively demonstrated by specific antigen-detection tests.

Tissue amebicides reach high concentrations in the blood and tissue after oral or parenteral administration. The development of nitroimidazole compounds, especially metronidazole, was a major advance in the treatment of invasive amebiasis. Patients with amebic colitis should be treated with IV or oral metronidazole. Side effects include nausea, vomiting, abdominal discomfort, and a disulfiram-like reaction. Another, longer-acting imidazole compound, tinidazole, is likewise effective and is available in the United States. All patients should also receive a full course of therapy with

TABLE 230-1 Drug Therapy for Amebiasis

INDICATION	THERAPY
Asymptomatic carriage	Luminal agent: iodoquinol (650-mg tablets), 650 mg tid for 20 days; or paromomycin (250-mg tablets), 500 mg tid for 10 days
Acute colitis	Metronidazole (250- or 500-mg tablets), 750 mg PO or IV tid for 5–10 days; or tinidazole, 2 g/d PO for 3 days plus Luminal agent as above
Amebic liver abscess	Metronidazole, 750 mg PO or IV for 5–10 days; or tinidazole, 2 g PO once; or ornidazole,* 2 g PO once plus Luminal agent as above

*Not available in the United States.